Quant Interview God Notes

Probability, statistics, machine learning, and market making for quantitative interviews — with full proofs, worked examples, and a verified problem bank

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Contents

Acknowledgements

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Μηδεὶς ἀλόγιστος εἰσίτω

Preface

Who this is for. These notes are written for a strong student or early-career researcher preparing for quantitative-finance and AI-research interviews. Everything needed is developed in place: the document is self-contained given single-variable calculus, elementary linear algebra, and comfort with mathematical notation. Probability is built up from its working tools rather than from measure theory. If you are able to understand and answer all the questions and examples, you will undoubtedly pass just about any assessment/interview in quant or AI research.

The proof policy. Three kinds of statement appear, and the typography distinguishes them honestly:

- **Theorems, Propositions, Lemmas, Corollaries** are proved — either on the spot, or, where a problem in the Problem Bank asks for exactly that proof, by a deferral naming the problem (the full argument then appears in its solution).
- **Facts** are true results whose proofs lie beyond the scope of this document (the Central Limit Theorem, the spectral theorem, and similar); each is stated precisely, attributed inline where a specific source deserves credit, and never silently treated as proven here.
- **Empirical facts** are regularities of markets or computation (the square-root impact law, the variance risk premium); they are labelled as empirical and should be quoted as evidence, not as mathematics.

A fourth numbered environment, **Procedure**, appears occasionally: it is a recipe — steps to execute.

Provenance. Two small marks record where material comes from. A star ($\bigstar$) after a problem, an example, or a statement means that item reproduces, from memory, a question asked in a live 2026 interview (video or on-site); a diamond ($\blacklozenge$) means the same for a timed online assessment. The firms are deliberately not named. Every problem's answer was independently re-derived by two blind solvers before inclusion, and the expository material has been audited by independent recomputation.

Notation

One symbol keeps one meaning throughout. Estimators wear hats; matrices are capital letters.

$P(A)$, $P(A \mid B)$	probability; conditional probability
$\mathbb{E}[X]$, $\mathrm{Var}(X)$, $\mathrm{Cov}(X, Y)$, ρ	expectation, variance, covariance, correlation
$\mathbf{1}_A$	indicator of the event A
$X \sim \mathcal{N}(\mu, \sigma^2)$	normal with mean μ and *variance* σ^2
$\mathrm{U}(a, b)$, $\mathrm{Exp}(\lambda)$, $\mathrm{Bin}(n, p)$, $\mathrm{Geom}(p)$, $\mathrm{Poisson}(\lambda)$	uniform; exponential; binomial; geometric; Poisson
Φ, φ	standard normal CDF and density
iid	independent and identically distributed
$A^\top$, λ_i, tr, det	transpose, eigenvalues, trace, determinant
$\hat{\beta}$, $\hat{\sigma}^2$	estimators (hats)
$\|x\|$	Euclidean norm
i, N, p, $q = 1 - p$, $r = q/p$	random-walk state, absorbing target, step probabilities
log	natural logarithm ($\log_{10}$ written explicitly)
f', ∂, $\int$	differentiation and integration
$\bigstar$, $\blacklozenge$	provenance: reproduced from a live 2026 interview ($\bigstar$) or a timed online assessment ($\blacklozenge$); firms not named

Part I

Probability and Games of Chance

1 Expected Value for Trading Games

Expected value is the fair price of a random payoff, and the games in this section — coin patterns, dice markets, re-roll options, shuffled hats — are the standard vehicles by which quantitative interviews test whether a candidate can compute it quickly and without error. Three tools do almost all of the work: linearity of expectation, indicator variables, and conditioning. This section develops those tools formally, records the named distributions used throughout the document, and then works through the classic game families in turn: geometric waiting times, pattern waits, coupon collection, order statistics of dice, the option value of a re-roll, a conditioning trap, and the hat-check problem.

1.1 The Toolkit

This subsection states and proves the three results that dispose of most expected-value games: linearity of expectation, the indicator lemma, and the tower rule. Everything later in the section is an application of these three, so the proofs here repay careful reading.

Setup. Throughout this subsection, all random variables are discrete: each takes values in a countable subset of $\mathbb{R}$. Probabilities are written P, and $P(X = x, Y = y)$ denotes the joint probability mass function of a pair (X, Y) defined on a common probability space. For an event A, the indicator $\mathbf{1}_A$ is the random variable equal to 1 when A occurs and 0 otherwise.

Definition 1.1 (Expectation of a discrete random variable)**.** Let X be a random variable taking values in a countable set $\mathcal{X} \subset \mathbb{R}$. If $\sum_{x\in\mathcal{X}} |x|\, P(X = x) < \infty$, we call X integrable and define its expectation as

$$\mathbb{E}[X] = \sum_{x\in\mathcal{X}} x\, P(X = x).$$

Absolute convergence guarantees that the value does not depend on the order in which the elements of $\mathcal{X}$ are enumerated.

Theorem 1.2 (Linearity of expectation)**.** *Let X and Y be integrable random variables on a common discrete probability space, and let $a, b \in \mathbb{R}$. Then $aX + bY$ is integrable and*

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y].$$

No independence assumption is made: the identity holds for arbitrarily dependent X and Y.

Proof. We proceed by direct computation with the joint probability mass function, justifying every rearrangement by absolute convergence. Write $p(x, y) = P(X = x, Y = y)$, with (x, y) ranging over the countable joint range.

Step 1: integrability. By the triangle inequality, and since all terms below are nonnegative (so the sums may be rearranged and grouped freely),

$$\sum_{x,y} |ax + by|\, p(x, y) \;\le\; |a| \sum_{x,y} |x|\, p(x, y) + |b| \sum_{x,y} |y|\, p(x, y) \;=\; |a|\,\mathbb{E}|X| + |b|\,\mathbb{E}|Y| < \infty,$$

where the final equality uses $\sum_y p(x, y) = P(X = x)$ and $\sum_x p(x, y) = P(Y = y)$ (marginalisation). Hence $aX + bY$ is integrable.

Step 2: the identity. Since the double series now converges absolutely, we may split and regroup it:

$$\mathbb{E}[aX + bY] = \sum_{x,y}(ax + by)\, p(x, y) = a \sum_x x \sum_y p(x, y) + b \sum_y y \sum_x p(x, y) = a\, \mathbb{E}[X] + b\, \mathbb{E}[Y].$$

At no point did we use any relationship between X and Y beyond their being defined on the same space: only the marginal sums entered. $\square$

Example 1.3 (Sum of two dice)**.** Let X and Y be the faces of two fair six-sided dice, each with $\mathbb{E}[X] = \mathbb{E}[Y] = \frac{1+2+\cdots+6}{6} = 3.5$. By Theorem 1.2, $\mathbb{E}[X + Y] = 3.5 + 3.5 = 7$, with no need to enumerate the 36 joint outcomes and no need to know whether the dice are independent. The same one-line answer holds even for two dice glued together.

Lemma 1.4 (Indicator lemma)**.** *For any event A, the indicator satisfies $\mathbb{E}[\mathbf{1}_A] = P(A)$.*

Proof. We compute directly from Definition 1.1. The random variable $\mathbf{1}_A$ takes only the values 1 and 0, with probabilities $P(A)$ and $1 - P(A)$ respectively, so

$$\mathbb{E}[\mathbf{1}_A] = 1 \cdot P(A) + 0 \cdot (1 - P(A)) = P(A). \qquad \square$$

Corollary 1.5 (Counting via indicators)**.** *Let $A_1, \ldots, A_m$ be events, in any dependence structure whatsoever, and let $X = \sum_{i=1}^m \mathbf{1}_{A_i}$ count how many of them occur. Then*

$$\mathbb{E}[X] = \sum_{i=1}^m P(A_i).$$

Proof. We proceed by induction on m. The case $m = 1$ is Lemma 1.4. For the inductive step, write $X = \left(\sum_{i=1}^{m-1} \mathbf{1}_{A_i}\right) + \mathbf{1}_{A_m}$; both summands are integrable (they are bounded), so Theorem 1.2 applies and gives $\mathbb{E}[X] = \sum_{i=1}^{m-1} P(A_i) + P(A_m)$ by the inductive hypothesis and Lemma 1.4. $\square$

Example 1.6 (Aces without replacement)**.** Draw two cards from a standard 52-card deck without replacement, and let X count the aces drawn. The two draws are dependent, yet by symmetry each draw individually is uniform over the deck, so $P(\text{draw } i \text{ is an ace}) = \frac{4}{52} = \frac{1}{13}$ for $i = 1, 2$. Corollary 1.5 gives $\mathbb{E}[X] = \frac{2}{13} \approx 0.154$. The dependence between the draws, which would complicate any attempt to work with the distribution of X itself, is simply irrelevant to the expectation.

Trading application. Questions of the form "expected number of . . . " — fills across a set of resting quotes, winning trades in a day, fixed points of a shuffle — are solved by Corollary 1.5: the answer is the sum of the individual probabilities, whatever the dependence between the events. Most game questions in this section fall to writing the payoff as a sum of indicators before doing anything else.

Proposition 1.7 (Tower rule)**.** *Let X be integrable and let Y be a discrete random variable on the same space. For each y with $P(Y = y) > 0$, define the conditional expectation*

$$g(y) = \mathbb{E}[X \mid Y = y] = \sum_x x\, P(X = x \mid Y = y).$$

Then $\mathbb{E}[g(Y)] = \mathbb{E}[X]$; in the usual shorthand, $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$.

Proof. We expand the outer expectation and swap the order of summation, which is legitimate because the double series converges absolutely ($\sum_{x,y} |x| P(X = x, Y = y) = \mathbb{E}|X| < \infty$):

$$\mathbb{E}[g(Y)] = \sum_y P(Y = y) \sum_x x\, \frac{P(X = x, Y = y)}{P(Y = y)} = \sum_x x \sum_y P(X = x, Y = y).$$

The inner sum on the right is $P(X = x)$ by marginalisation, so the right-hand side equals $\sum_x x\, P(X = x) = \mathbb{E}[X]$ by Definition 1.1. $\square$

Example 1.8 (A two-stage payout)**.** A fair coin is flipped. On heads, the payout is the face of a fair die; on tails, the payout is a flat 2. Conditioning on the flip Y and applying Proposition 1.7,

$$\mathbb{E}[X] = \tfrac{1}{2} \cdot 3.5 + \tfrac{1}{2} \cdot 2 = 2.75,$$

where 3.5 is the die's mean from Example 1.3. No joint distribution ever needs to be written down.

Intuition. Linearity does the bookkeeping; conditioning does the dynamics. First-step analysis — the workhorse for every waiting-time game below — is nothing but the tower rule of Proposition 1.7 applied at the first move of a sequential game: condition on what the first step does, and let the structure of the game tell you what the conditional expectations are.

1.2 Named Distributions

Seven named distributions cover nearly every random quantity in this document. This subsection records their definitions and collects their means and variances, together with the one-line "hook" worth memorising for each, in Table 1. Two of the table's entries are proved immediately in the flow of this section: the Bernoulli variance below, and the geometric mean as Theorem 1.13 in the next subsection.

Setup. For an integrable random variable X with mean $\mu = \mathbb{E}[X]$ and integrable square, the variance is

$$\mathrm{Var}(X) = \mathbb{E}\big[(X-\mu)^2\big],$$

and the standard deviation is its square root. The reader should verify, by expanding the square and applying Theorem 1.2, the shortcut form

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mu^2. \tag{1.1}$$

Definition 1.9 (The seven named distributions)**.** Let $p \in [0,1]$ ($p \in (0,1]$ for the geometric), $n \geq 1$ an integer, $\lambda > 0$, $a < b$ reals, $\mu \in \mathbb{R}$, and $\sigma > 0$.

- **Bernoulli(p):** $X \in \{0,1\}$ with $P(X=1) = p$ — a single yes/no trial.
- **Binomial** $\mathrm{Bin}(n,p)$**:** the number of successes in n iid Bernoulli(p) trials; $P(X=k) = \binom{n}{k}p^k(1-p)^{n-k}$ for $0 \leq k \leq n$.
- **Geometric** $\mathrm{Geom}(p)$**:** the number of iid Bernoulli(p) trials up to and including the first success; $P(X=k) = (1-p)^{k-1}p$ for $k \geq 1$. (This document always counts trials, not failures.)
- **Poisson(λ):** $P(X=k) = e^{-\lambda}\lambda^k/k!$ for integers $k \geq 0$.
- **Uniform** $\mathrm{U}(a,b)$**:** continuous with density $\frac{1}{b-a}$ on (a,b).
- **Exponential** $\mathrm{Exp}(\lambda)$**:** continuous with density $\lambda e^{-\lambda x}$ for $x > 0$; the parameter λ is a rate.
- **Normal** $\mathcal{N}(\mu,\sigma^2)$**:** continuous with density $\frac{1}{\sigma\sqrt{2\pi}}\exp\big(-\frac{(x-\mu)^2}{2\sigma^2}\big)$; the second parameter is the variance.

Proposition 1.10 (Bernoulli mean and variance)**.** *If $X \sim$ Bernoulli(p), then $\mathbb{E}[X] = p$ and $\mathrm{Var}(X) = p(1-p)$.*

Proof. We compute both moments directly. The mean is $\mathbb{E}[X] = 1 \cdot p + 0 \cdot (1-p) = p$, which is Lemma 1.4 in disguise (X is the indicator of its own success event). Because X takes only the values 0 and 1, we have $X^2 = X$ pointwise, hence $\mathbb{E}[X^2] = p$. By the shortcut (1.1),

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1-p). \qquad \square$$

Distribution	$\mathbb{E}$	Var	**Hook**
Bernoulli(p)	p	$p(1-p)$	the atom of everything
Bin(n, p)	np	$np(1-p)$	$\approx \mathcal{N}$ for $np(1-p)$ large
Geom(p)	$1/p$	$(1-p)/p^2$	memoryless
Poisson(λ)	λ	λ	rare-event limit; sums add
U(a, b)	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\mathbb{E}[U^k] = \frac{1}{k+1}$ on $(0, 1)$
Exp(λ)	$1/\lambda$	$1/\lambda^2$	memoryless; $\mathbb{E}[X^n] = n!/\lambda^n$
$\mathcal{N}(\mu, \sigma^2)$	μ	σ^2	sums stay normal; $\mathbb{E}[(X-\mu)^4] = 3\sigma^4$

Table 1: The seven named distributions: mean, variance, and the one-line hook worth carrying into an interview.

Example 1.11 (Variance of a directional bet)**.** A strategy wins a fixed unit with probability $p = 0.55$ and loses nothing otherwise, so the per-trade outcome is Bernoulli(0.55). By Proposition 1.10, its variance is $0.55 \times 0.45 = 0.2475$ — barely below the maximum 0.25 attained at $p = 0.5$. Modest edges leave the per-trial variance essentially at its fair-coin worst case, which is why single trials tell you almost nothing and sample sizes matter.

Remark 1.12 (Where the table is proved)*.* The Bernoulli row is Proposition 1.10. The binomial mean np follows in one line from Corollary 1.5, since a Bin(n, p) count is a sum of n success indicators each with probability p. The geometric mean $1/p$ is proved as Theorem 1.13 below. The remaining entries are standard; they are quoted here without proof and re-derived where the document needs them.

1.3 Geometric Waiting Times

How long until the first success? This is the simplest waiting-time game, it introduces first-step analysis — the tower rule of Proposition 1.7 applied at the first move — and every later waiting-time result in this section reduces to it or generalises it.

Setup. Let $X_1, X_2, \dots$ be iid Bernoulli(p) trials with $0 < p \le 1$, and let

$$T = \min\{\, n \ge 1 : X_n = 1 \,\}$$

be the number of trials up to and including the first success. Since the first success occurs at trial n exactly when the first $n-1$ trials fail and the n-th succeeds, $P(T = n) = (1-p)^{n-1}p$; that is, $T \sim$ Geom(p) in the convention of Definition 1.9.

Theorem 1.13 (Expected wait for the first success)**.** *With the setup above,* $\mathbb{E}[T] = 1/p$*.*

Proof. We proceed by first-step analysis: condition on the outcome of the first trial and apply the tower rule (Proposition 1.7). Figure 1 displays the decomposition.

Step 1: T is integrable. If $p = 1$ then $T \equiv 1$ and there is nothing to prove, so assume $p < 1$. The series $\sum_{n \ge 1} n(1-p)^{n-1}p$ converges by the ratio test (the ratio of consecutive terms tends to $1 - p < 1$), so $\mathbb{E}[T] < \infty$. Finiteness is what makes the algebra of Step 2 legitimate.

Step 2: the recursion. On the event $\{X_1 = 1\}$, which has probability p, we have $T = 1$. On $\{X_1 = 0\}$, one trial has been spent, and the remaining trials $X_2, X_3, \dots$ are again iid Bernoulli(p), so the residual wait $T - 1$ has the same distribution as T; hence $\mathbb{E}[T \mid X_1 = 0] = 1 + \mathbb{E}[T]$. The tower rule gives

$$\mathbb{E}[T] = p \cdot 1 + (1-p)\big(1 + \mathbb{E}[T]\big) = 1 + (1-p)\,\mathbb{E}[T]. \tag{1.2}$$

Since $\mathbb{E}[T]$ is finite, we may subtract $(1-p)\mathbb{E}[T]$ from both sides of (1.2) to get $p\,\mathbb{E}[T] = 1$, i.e. $\mathbb{E}[T] = 1/p$. □

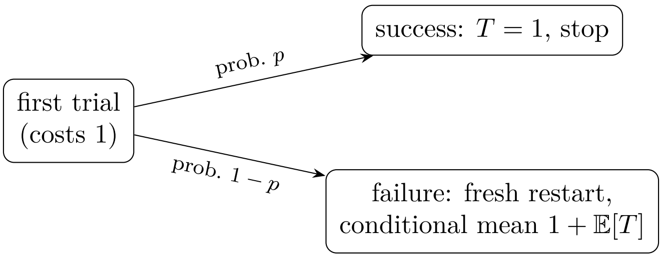


Figure 1: The first-step-analysis tree behind Theorem 1.13. Weighting the two branches by their probabilities yields the recursion (1.2).

Example 1.14 (Coins and dice)**.** For a fair coin, $p = \frac{1}{2}$ and the expected number of flips to the first head is $\mathbb{E}[T] = 2$. For a fair die waiting for the first six, $p = \frac{1}{6}$ and $\mathbb{E}[T] = 6$. As a check on the recursion (1.2) with $p = \frac{1}{2}$: the claimed value 2 satisfies $2 = 1 + \frac{1}{2} \cdot 2$, as it must.

Proposition 1.15 (Memorylessness of the geometric distribution)**.** *With the setup above, assume $p < 1$. Then for all integers $m, n \geq 0$,*

$$P(T > m + n \mid T > m) = P(T > n).$$

Proof. We compute the tail probabilities explicitly. The event $\{T > k\}$ occurs exactly when the first k trials all fail, so $P(T > k) = (1 - p)^k$ by independence. Since $\{T > m + n\} \subset \{T > m\}$,

$$P(T > m + n \mid T > m) = \frac{P(T > m + n)}{P(T > m)} = \frac{(1 - p)^{m+n}}{(1 - p)^m} = (1 - p)^n = P(T > n). \qquad \square$$

Example 1.16 (Failed flips carry no information)**.** For the fair coin, given that no head has appeared in the first two flips, the probability that none appears in the next two is $P(T > 4 \mid T > 2) = (\frac{1}{2})^2 = \frac{1}{4}$, exactly the unconditional $P(T > 2)$. Two wasted flips leave the forecast for the future untouched.

Intuition. The recursion (1.2) works because the process is memoryless: after a failure you are back exactly where you started, one flip poorer. Proposition 1.15 is the precise statement of "back where you started", and $\mathbb{E}[T \mid X_1 = 0] = 1 + \mathbb{E}[T]$ is its expectation-level shadow.

1.4 Pattern Waiting Times: HH versus HT

Any two fixed consecutive flips of a fair coin are HH with probability $\frac{1}{4}$ and HT with probability $\frac{1}{4}$ — yet the expected wait for the first HH differs from the expected wait for the first HT. This subsection resolves the apparent paradox by first-step analysis on a two-state automaton, the same technique as in Theorem 1.13 but with more than one "restart" state.

Setup. A fair coin is flipped repeatedly; flips are iid with $P(\mathrm{H}) = P(\mathrm{T}) = \frac{1}{2}$. Let T_{HH} be the number of flips until the last two flips first read HH, and T_{HT} likewise for HT. For each pattern we track progress with two states: S_0 (no useful progress: either no flips yet, or the last flip was not H) and S_1 (the last flip was H). Write E_0 and E_1 for the expected number of further flips to complete the pattern from S_0 and S_1 respectively; each game starts in S_0.

Theorem 1.17 (Expected pattern waits, HH versus HT)**.** *With the setup above,*

$$\mathbb{E}[T_{\mathrm{HH}}] = 6 \qquad \textit{and} \qquad \mathbb{E}[T_{\mathrm{HT}}] = 4.$$

Proof. We proceed by first-step analysis on the automata of Figure 2, one flip at a time.

Step 1: all expectations are finite. Partition the flips into disjoint blocks of two. Whatever the past, each block independently equals the target pattern with probability $\frac{1}{4}$, so if the pattern has not appeared by flip $2k$ then in particular none of the first k blocks equalled it: $P(T > 2k) \le (\frac{3}{4})^k$ for either pattern. Geometrically decaying tails give finite expectations, so the linear equations below may be solved by ordinary algebra.

Step 2: the HH system. From S_0, one flip is spent; heads (probability $\frac{1}{2}$) moves to S_1, tails stays in S_0. From S_1, heads completes HH; tails destroys all progress and returns to S_0. Hence

$$E_0 = 1 + \tfrac{1}{2}E_1 + \tfrac{1}{2}E_0, \qquad E_1 = 1 + \tfrac{1}{2}\cdot 0 + \tfrac{1}{2}E_0. \tag{1.3}$$

The first equation of (1.3) rearranges to $E_0 = 2 + E_1$; substituting into the second gives $E_1 = 1 + \frac{1}{2}(2 + E_1)$, so $E_1 = 4$ and $E_0 = 6$. Since the game starts in S_0, $\mathbb{E}[T_{\mathrm{HH}}] = 6$.

Step 3: the HT system. From S_0 the transitions are as before. From S_1, however, tails completes HT, while heads — the failure — leaves the process still holding an H, i.e. still in S_1. This is the crucial structural difference: for HT, failure does not destroy progress. Hence

$$E_0 = 1 + \tfrac{1}{2}E_1 + \tfrac{1}{2}E_0, \qquad E_1 = 1 + \tfrac{1}{2}E_1 + \tfrac{1}{2}\cdot 0. \tag{1.4}$$

The second equation of (1.4) gives $E_1 = 2$; the first then gives $\frac{1}{2}E_0 = 1 + \frac{1}{2}\cdot 2 = 2$, so $E_0 = 4$ and $\mathbb{E}[T_{\mathrm{HT}}] = 4$. $\square$

(Problem P1.10 asks you to reproduce this proof from scratch.)

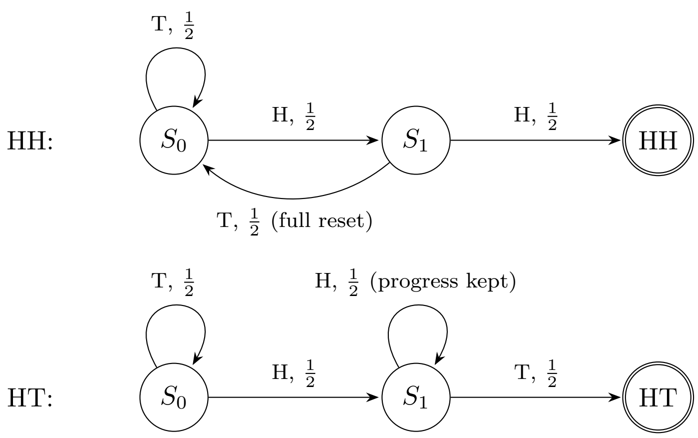


Figure 2: State-transition automata for the waits in Theorem 1.17. The two machines differ only in the failure edge out of S_1: for HH it returns to S_0 (all progress lost), while for HT it loops at S_1 (an extra head still leaves an H on the board).

Example 1.18 (HT as two geometric stages)**.** The HT wait admits an independent numeric check. Because HT progress never resets, the wait splits cleanly: first wait for an H, then — holding an H forever after, since further heads keep the H alive — wait for a T. Each stage is a first-success wait with $p = \frac{1}{2}$, worth 2 flips by Theorem 1.13, so $\mathbb{E}[T_{\mathrm{HT}}] = 2 + 2 = 4$, matching Theorem 1.17. No such decomposition exists for HH, precisely because its failure edge destroys the first stage's work.

Intuition. Self-overlapping patterns such as HH are slower to arrive: a failure at the final step destroys all progress and returns the automaton to S_0, while HT's failure edge preserves progress. The same mathematics governs stop-hunting sequences and losing streaks — "bad" events that reset you entirely are costlier than their probability alone suggests.

1.5 Coupon Collector

How many draws are needed to see every type at least once? The answer combines the two main tools so far — the geometric wait of Theorem 1.13 for each new type, and linearity (Theorem 1.2) to add the stages — and is a fixture of both interviews and applied sampling questions.

Setup. Draws are iid and uniform over n equally likely types. Let T be the number of draws until every type has appeared at least once, and write $H_n = \sum_{k=1}^{n} \frac{1}{k}$ for the n-th harmonic number.

Theorem 1.19 (Coupon collector)**.** *With the setup above,* $\mathbb{E}[T] = nH_n$.

Proof. We decompose the collection into stages and apply linearity. For $j = 0, 1, \ldots, n-1$, let T_j be the number of draws made while exactly j distinct types have been seen, so that $T = \sum_{j=0}^{n-1} T_j$. While exactly j types have been seen, each fresh draw is a new type with probability $\frac{n-j}{n}$, independently of everything drawn so far; the stage therefore ends at a first success of probability $\frac{n-j}{n}$, i.e. $T_j \sim \mathrm{Geom}(\frac{n-j}{n})$, and Theorem 1.13 gives $\mathbb{E}[T_j] = \frac{n}{n-j}$. By Theorem 1.2 — which requires no independence between the stages —

$$\mathbb{E}[T] = \sum_{j=0}^{n-1} \frac{n}{n-j} = n \sum_{k=1}^{n} \frac{1}{k} = nH_n,$$

where the middle step substitutes $k = n - j$. □

(Problem P1.13 asks you to reproduce this proof from scratch.)

Example 1.20 (All six faces of a die)**.** For a fair die, $n = 6$ and

$$\mathbb{E}[T] = 6\Big(1 + \tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{4} + \tfrac{1}{5} + \tfrac{1}{6}\Big) = 6 \cdot \tfrac{49}{20} = \tfrac{147}{10} = 14.7.$$

The final face alone costs 6 expected rolls — the last stage is the whole first-six wait of Example 1.14 — while the first face is free after one roll.

1.6 Order Statistics of Dice

"Make me a market on the max of two dice" is a staple prompt: it tests whether a candidate reaches for the CDF rather than guessing symmetry. This subsection computes the maximum by CDF differencing and gets the minimum for free by linearity (Theorem 1.2).

Setup. Let X and Y be independent, each uniform on $\{1, \ldots, 6\}$, and set $M = \max(X, Y)$ and $L = \min(X, Y)$.

Proposition 1.21 (Max and min of two dice)**.** *With the setup above, for* $k = 1, \ldots, 6$,

$$P(M \le k) = \Big(\frac{k}{6}\Big)^2, \qquad P(M = k) = \frac{2k-1}{36},$$

and consequently

$$\mathbb{E}[M] = \frac{161}{36} \approx 4.47, \qquad \mathbb{E}[L] = \frac{91}{36} \approx 2.53.$$

Proof. We proceed by CDF differencing. The maximum is at most k exactly when both dice are, so independence gives $P(M \le k) = P(X \le k)\, P(Y \le k) = (k/6)^2$. Differencing consecutive CDF values yields the probability mass function:

$$P(M = k) = P(M \le k) - P(M \le k-1) = \frac{k^2 - (k-1)^2}{36} = \frac{2k-1}{36}.$$

The expectation is then

$$\mathbb{E}[M] = \sum_{k=1}^{6} k \cdot \frac{2k-1}{36} = \frac{2\sum_{k=1}^{6} k^2 - \sum_{k=1}^{6} k}{36} = \frac{2 \cdot 91 - 21}{36} = \frac{161}{36}.$$

For the minimum, observe the pointwise identity $L + M = X + Y$: on every outcome, the min and the max are the two rolled values in some order. Theorem 1.2 then gives

$$\mathbb{E}[L] = \mathbb{E}[X] + \mathbb{E}[Y] - \mathbb{E}[M] = 7 - \frac{161}{36} = \frac{91}{36},$$

using $\mathbb{E}[X + Y] = 7$ from Example 1.3. $\square$

(Problem P1.20 runs the same CDF-differencing pipeline on the maximum of four dice.)

Example 1.22 (Sanity checks on the max distribution)**.** The masses $\frac{2k-1}{36}$ for $k = 1, \ldots, 6$ are $\frac{1}{36}, \frac{3}{36}, \frac{5}{36}, \frac{7}{36}, \frac{9}{36}, \frac{11}{36}$; their numerators are the first six odd numbers, summing to $6^2 = 36$, so the pmf sums to 1. The mass has visibly shifted upward — $P(M = 6) = \frac{11}{36} \approx 0.31$ against $P(M = 1) = \frac{1}{36}$ — which is why $\mathbb{E}[M] \approx 4.47$ sits well above the single-die mean 3.5.

Remark 1.23 (Median of five dice)*.* The median (third-smallest) of five iid dice has expectation exactly 3.5, by symmetry: the face swap $k \mapsto 7 - k$ preserves each die's distribution and maps the third-smallest of the five values to the third-largest, which among five values is again the median. Hence the median $\tilde{M}$ and $7 - \tilde{M}$ have the same distribution, so $\mathbb{E}[\tilde{M}] = 7 - \mathbb{E}[\tilde{M}]$, i.e. $\mathbb{E}[\tilde{M}] = 3.5$.

Interview trap. "Make me a market on the max of two dice" — the naive fair value 3.5 ignores that the max is biased up. Before quoting, always ask: is the payoff a *mean*, an *extreme*, or a *median* of the randomness? Extremes shift the fair value (Proposition 1.21); medians resist outliers (Remark 1.23).

1.7 Option Value of a Re-Roll

A die roll with the right to discard and try again is the smallest possible optimal-stopping problem, and it prices an option with nothing but the tower rule. The values 4.25 and $\frac{14}{3}$ below are standard interview answers worth knowing cold.

Setup. A fair die is rolled and the player is paid the face value in dollars. Before the game, the player is granted the right to discard the current roll and re-roll, usable at most k times in total; a discarded roll is gone forever. A fresh roll X is uniform on $\{1, \ldots, 6\}$ with $\mathbb{E}[X] = 3.5$. Let V_k denote the value of the game — the expected payoff under optimal play — with k re-rolls remaining and no roll yet showing, so that $V_0 = 3.5$.

Proposition 1.24 (Optimal re-roll policies and values)**.** *With the setup above:*

(i) With one re-roll, the optimal policy is to keep a first roll in $\{4, 5, 6\}$ *and re-roll* $\{1, 2, 3\}$*, and* $V_1 = \frac{17}{4} = 4.25$.

(ii) With two re-rolls, the optimal policy is to keep only $\{5, 6\}$ *on the first roll, then play as in (i); and* $V_2 = \frac{14}{3} \approx 4.67$.

Proof. We proceed by backward induction on the number of re-rolls remaining. With $k \geq 1$ re-rolls remaining and a face v showing, the player chooses between stopping (worth v) and discarding, which restarts the game with $k - 1$ re-rolls and is worth V_{k-1}; optimal play takes the larger, so the pre-roll value satisfies the dynamic-programming recursion

$$V_k = \mathbb{E}\big[\max(X, V_{k-1})\big], \qquad V_0 = 3.5, \tag{1.5}$$

by the tower rule (Proposition 1.7) conditioning on the face rolled.

One re-roll. A face v beats the continuation exactly when $v > V_0 = 3.5$, i.e. $v \in \{4, 5, 6\}$, whose average is 5. Evaluating (1.5),

$$V_1 = \tfrac{3}{6} \cdot 5 + \tfrac{3}{6} \cdot 3.5 = 2.5 + 1.75 = 4.25 = \tfrac{17}{4}.$$

Two re-rolls. Now the continuation is worth $V_1 = 4.25$, so only $v \in \{5, 6\}$ (average 5.5) is kept — in particular a 4, kept in the one-re-roll game, is discarded here because $4 < 4.25$. Evaluating (1.5) again,

$$V_2 = \tfrac{2}{6} \cdot 5.5 + \tfrac{4}{6} \cdot 4.25 = \tfrac{11}{6} + \tfrac{17}{6} = \tfrac{28}{6} = \tfrac{14}{3} \approx 4.67. \qquad \square$$

Example 1.25 (What the option itself is worth)**.** Subtracting the no-option value $V_0 = 3.5$ from the results of Proposition 1.24: a single re-roll right is worth $\frac{17}{4} - \frac{7}{2} = \frac{3}{4} = \$0.75$; the two-re-roll right is worth $\frac{14}{3} - \frac{7}{2} = \frac{7}{6} \approx \$1.17$. The second re-roll adds $\frac{14}{3} - \frac{17}{4} = \frac{5}{12} \approx \$0.42$ — total option value grows with each extra chance, though at a diminishing rate.

Intuition. The re-roll is an *option*: worth 0.75 here, and its value grows with the number of chances and with the dispersion of the underlying outcomes. Optionality on a volatile underlying is worth more — the deepest fact in derivatives, visible in a dice game.

1.8 Conditioning Traps

Conditioning on a property of an entire sequence is not the same as altering the mechanism that generates each step. The following die problem is the canonical illustration, and nearly every first instinct about it is wrong.

Setup. A fair die is rolled repeatedly until the first 6 appears. Let T be the number of rolls (including the final 6), and let A be the event that every roll in the sequence — including the final 6 — came up even.

Proposition 1.26 (Conditioned wait: first 6 given all rolls even)**.** *With the setup above, $P(A) = \frac{1}{4}$ and*

$$\mathbb{E}[T \mid A] = \frac{3}{2}.$$

Proof. We compute the conditional distribution of T directly from the joint probabilities. The event $\{T = n\} \cap A$ requires the first $n - 1$ rolls to land in $\{2, 4\}$ (probability $\frac{2}{6} = \frac{1}{3}$ each) and roll n to be a 6 (probability $\frac{1}{6}$), so by independence

$$P(T = n,\ A) = \Big(\frac{1}{3}\Big)^{n-1} \frac{1}{6}, \qquad n \geq 1. \tag{1.6}$$

Summing the geometric series (1.6) over n gives

$$P(A) = \frac{1/6}{1 - 1/3} = \frac{1}{4}.$$

Dividing (1.6) by $P(A)$ yields the conditional probability mass function

$$P(T = n \mid A) = 4 \cdot \Big(\frac{1}{3}\Big)^{n-1} \frac{1}{6} = \frac{2}{3} \Big(\frac{1}{3}\Big)^{n-1},$$

which is exactly the $\text{Geom}(\frac{2}{3})$ pmf of Definition 1.9. By Theorem 1.13, $\mathbb{E}[T \mid A] = \frac{1}{2/3} = \frac{3}{2}$. Alternatively, summing the series directly: $\sum_{n \geq 1} n \cdot \frac{2}{3} (\frac{1}{3})^{n-1} = \frac{2}{3} \cdot \frac{1}{(1-1/3)^2} = \frac{2}{3} \cdot \frac{9}{4} = \frac{3}{2}$. $\square$

Example 1.27 (The first-roll check)**.** From the conditional pmf in the proof of Proposition 1.26, $P(T = 1 \mid A) = \frac{2}{3}$, versus $\frac{1}{3}$ under the naive three-face model discussed below — the conditioning visibly favours stopping immediately, consistent with a mean of $\frac{3}{2} < 3$. As a bound check, the answer must lie between 1 and the naive 3, and $\frac{3}{2}$ does.

Interview trap. The instant answer is 3: "conditioning on all-even reduces the die to the three faces $\{2, 4, 6\}$, a first-success wait with $p = \frac{1}{3}$." That reasoning is wrong — it describes a genuinely three-faced die, a different experiment. Conditioning on "all rolls even" reweights *whole sequences*, not the die's faces: long sequences are penalised because each extra roll must dodge the odd faces, so the conditional law tilts sharply toward short sequences and the mean drops to $\frac{3}{2}$. Trigger to remember: conditioning reweights whole sequences, not just the die's faces.

1.9 Hat-Check Fixed Points

A uniformly random shuffle hands n hats back to their n owners; how many owners get their own hat? The count has mean 1 and variance 1 for every n — a striking constancy that indicator variables (Corollary 1.5) explain in a few lines, and that foreshadows a Poisson limit.

Setup. Let τ be a uniformly random permutation of $\{1, \dots, n\}$, $n \geq 1$, and let

$$X = \#\{\, i : \tau(i) = i \,\}$$

be its number of fixed points — the number of owners who receive their own hat.

Theorem 1.28 (Hat-check moments)**.** *With the setup above,* $\mathbb{E}[X] = 1$ *for every* $n \geq 1$*, and* $\operatorname{Var}(X) = 1$ *exactly, for every* $n \geq 2$*.*

Proof. We use indicators and linearity, computing the second moment through pair indicators. Write $X = \sum_{i=1}^{n} \mathbf{1}_{A_i}$ with $A_i = \{\tau(i) = i\}$.

Mean. The permutations fixing i number $(n-1)!$, so $P(A_i) = \frac{(n-1)!}{n!} = \frac{1}{n}$, and Corollary 1.5 gives $\mathbb{E}[X] = n \cdot \frac{1}{n} = 1$ — for every n, with no independence needed.

Second moment. Expanding the square of the indicator sum, and using that an indicator squared is itself while a product of indicators is the indicator of the intersection,

$$X^2 = \sum_{i=1}^{n} \mathbf{1}_{A_i} + \sum_{i \neq j} \mathbf{1}_{A_i \cap A_j}.$$

For $i \neq j$ (this is where $n \geq 2$ enters), the permutations fixing both i and j number $(n-2)!$, so $P(A_i \cap A_j) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$. Taking expectations with Theorem 1.2 and Lemma 1.4, the $n(n-1)$ ordered pairs contribute

$$\mathbb{E}[X^2] = n \cdot \frac{1}{n} + n(n-1) \cdot \frac{1}{n(n-1)} = 1 + 1 = 2.$$

By the variance shortcut (1.1), $\operatorname{Var}(X) = 2 - 1^2 = 1$. $\square$

Example 1.29 (All six permutations of three hats)**.** For $n = 3$ the six permutations split as: the identity (3 fixed points), three transpositions (1 fixed point each), and two 3-cycles (0 fixed points). Direct averaging:

$$\mathbb{E}[X] = \frac{3+1+1+1+0+0}{6} = 1, \qquad \mathbb{E}[X^2] = \frac{9+1+1+1+0+0}{6} = 2,$$

so $\operatorname{Var}(X) = 2 - 1 = 1$, confirming Theorem 1.28 by brute force.

Fact 1.30 (Poisson limit of fixed points)**.** As $n \to \infty$, the number of fixed points of a uniformly random permutation of n items converges in distribution to Poisson(1) — consistent with Theorem 1.28, since a Poisson(λ) variable has mean and variance both λ (Table 1), here $\lambda = 1$. The underlying matching problem goes back to de Montmort (1713). A proof is beyond our scope.

Intuition. Each indicator $\mathbf{1}_{A_i}$ is a rare event of probability $\frac{1}{n}$, and the pairwise dependence is weak — from the proof of Theorem 1.28, $P(A_i \cap A_j) = \frac{1}{n(n-1)}$ barely exceeds the independent value $\frac{1}{n^2}$. Many weakly dependent rare events with total mean 1 is exactly the regime in which Poisson behaviour emerges, which is why the limit in the fact above should feel inevitable rather than surprising.

Practice: problems P1.1–P1.20 in §20, solutions in §21.

2 Bayesian Updating and Reading Flow

Bayesian updating is the arithmetic of changing one's mind: a prior belief meets evidence and becomes a posterior. This section proves Bayes' theorem from the definition of conditional probability, recasts it in the odds form that turns interview questions into one-line multiplications, and extends it to streams of conditionally independent evidence. The same three results explain how a market maker reads order flow, which is where the section ends.

2.1 Conditional Probability and Bayes' Theorem

Everything in this section rests on a single definition — conditional probability — and on the theorem that follows from it in two lines. We state both carefully, because every later formula in the section is an algebraic rearrangement of them.

Setup. Throughout this section we work on a probability space with probability measure P; every set named below is an event. We write H for a hypothesis under consideration, H^c for its complement, and E for observed evidence. Conditioning on an event requires that event to have positive probability, and each statement below lists exactly the positivity it needs.

Definition 2.1 (Conditional probability)**.** Let A and B be events with $P(B) > 0$. The *conditional probability of A given B* is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Conditioning renormalises: among the outcomes in which B occurs, the conditional probability measures the fraction that also lie in A. Multiplying both sides of the definition by $P(B)$ gives the *multiplication rule*

$$P(A \cap B) = P(A \mid B)\, P(B), \tag{2.1}$$

valid whenever $P(B) > 0$.

Lemma 2.2 (Law of total probability)**.** *Let $B_1, \ldots, B_n$ be events that partition the sample space (they are pairwise disjoint and their union is the whole space), with $P(B_i) > 0$ for every i. Then, for any event A,*

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)\, P(B_i).$$

Proof. We decompose A along the partition and apply additivity. Since the B_i cover the sample space and are pairwise disjoint, the sets $A \cap B_1, \ldots, A \cap B_n$ are pairwise disjoint and their union is A. Additivity of P gives $P(A) = \sum_{i=1}^{n} P(A \cap B_i)$, and by the multiplication rule (2.1) each term equals $P(A \mid B_i)\, P(B_i)$. □

Theorem 2.3 (Bayes' theorem)**.** *Let H and E be events with $P(H) > 0$ and $P(E) > 0$. Then*

$$P(H \mid E) = \frac{P(E \mid H)\, P(H)}{P(E)}. \tag{2.2}$$

If moreover $0 < P(H) < 1$, the denominator expands over the partition $\{H, H^c\}$:

$$P(H \mid E) = \frac{P(E \mid H)\, P(H)}{P(E \mid H)\, P(H) + P(E \mid H^c)\, P(H^c)}. \tag{2.3}$$

Proof. We compute the probability of the intersection $H \cap E$ in two ways. By the multiplication rule (2.1) with $B = E$ we have $P(H \cap E) = P(H \mid E)\, P(E)$; with $B = H$ we have $P(H \cap E) = P(E \mid H)\, P(H)$. Equating the two expressions and dividing by $P(E) > 0$ yields (2.2). For (2.3), note that when $0 < P(H) < 1$ the events H and H^c partition the sample space and both have positive probability, so Lemma 2.2 gives $P(E) = P(E \mid H)\,P(H) + P(E \mid H^c)\,P(H^c)$; substituting this into the denominator of (2.2) completes the proof. □

In Bayesian language, $P(H)$ is the *prior*, $P(H \mid E)$ the *posterior*, $P(E \mid H)$ the *likelihood* of the evidence under the hypothesis, and $P(E)$ the *normalising constant*. The theorem inverts a conditional: it converts the easy direction, "probability of the evidence given the hypothesis," into the direction one actually wants, "probability of the hypothesis given the evidence."

Example 2.4 (Rare-disease test)**.** A disease has prevalence 1% in the population. A test has sensitivity 99% (it is positive on 99% of the sick) and specificity 95% (it is negative on 95% of the healthy, so its false-positive rate is 5%). A randomly chosen person tests positive; we compute the probability that the person has the disease.

Let D be the event "has the disease" and let $+$ denote a positive test, so that $P(D) = 0.01$, $P(+ \mid D) = 0.99$, and $P(+ \mid D^c) = 0.05$. By (2.3),

$$\begin{aligned} P(D \mid +) &= \frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.05 \times 0.99} = \frac{0.0099}{0.0099 + 0.0495} \\ &= \frac{0.0099}{0.0594} = \frac{1}{6} \approx 16.7\%. \end{aligned}$$

Despite a 99%-sensitive test, a positive result leaves only a one-in-six chance of disease: the false positives generated by the enormous healthy pool (joint mass 0.0495) outweigh the true positives (joint mass 0.0099) five to one. Figure 3 displays the computation as a probability tree.

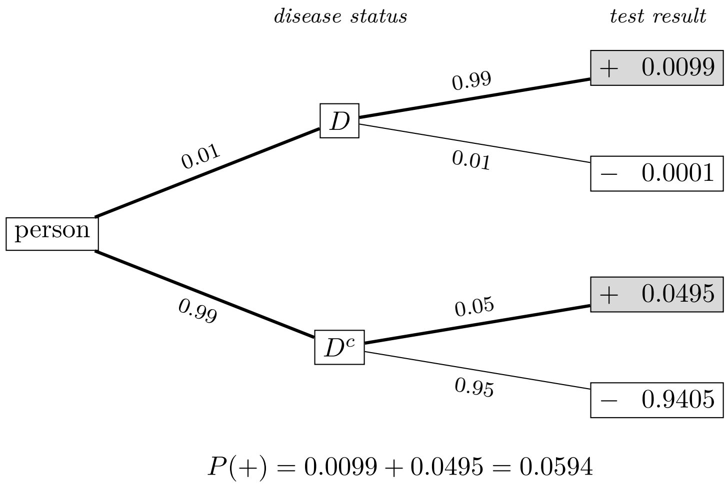


Figure 3: Probability tree for the rare-disease test of Example 2.4. Edge labels are conditional probabilities; each leaf shows the joint probability of its path, obtained by multiplying along the edges. The two thickened paths (shaded leaves) are the only routes to a positive test, so $P(+) = 0.0594$, and the posterior $P(D \mid +) = 0.0099/0.0594 = 1/6$ is the share of that shaded mass carried by the disease branch.

Intuition. The tree in Figure 3 shows why an accurate test can still leave a small posterior. A positive result can be manufactured in two ways: a sick person (rare — branch mass 0.01) triggering a true positive, or a healthy person (almost everyone — branch mass 0.99) triggering a false alarm. The posterior is a contest between the two joint masses, 0.0099 against 0.0495, and the healthy branch wins five to one — which is exactly the statement that the odds of disease given a positive test are 1:5.

2.2 The Odds Form: the Fast Recipe

Interviews rarely leave time to assemble the denominator of (2.3). The odds form removes it: dividing Bayes' theorem for H by Bayes' theorem for H^c cancels $P(E)$, leaving a recipe — posterior odds equal prior odds times likelihood ratio — that can be executed mentally.

Setup. As before, H is a hypothesis with $0 < P(H) < 1$ and E is evidence with $P(E) > 0$. The next definition introduces the two ratios the recipe runs on.

Definition 2.5 (Odds and likelihood ratio)**.** Let H be an event with $0 < P(H) < 1$. The *odds* of H are

$$\mathrm{O}(H) = \frac{P(H)}{P(H^c)},$$

and, for evidence E with $P(E) > 0$, the *posterior odds* are $\mathrm{O}(H \mid E) = P(H \mid E)/P(H^c \mid E)$ (defined when the denominator is positive). If in addition $P(E \mid H^c) > 0$, the *likelihood ratio* of E for H against H^c is

$$\mathrm{LR}(E) = \frac{P(E \mid H)}{P(E \mid H^c)}.$$

Odds and probabilities carry the same information. Dividing the identity $P(H \mid E)+P(H^c \mid E) = 1$ by $P(H^c \mid E)$ gives $\mathrm{O}(H \mid E) + 1 = 1/P(H^c \mid E)$, and therefore

$$P(H \mid E) = \frac{\mathrm{O}(H \mid E)}{1 + \mathrm{O}(H \mid E)}. \tag{2.4}$$

The same conversion applies to prior odds: odds of 1:5, say, mean a probability of $(1/5)/(1+1/5) = 1/6$.

Corollary 2.6 (Odds form of Bayes' theorem)**.** *Let H be an event with $0 < P(H) < 1$ and let E be evidence with $P(E) > 0$ and $P(E \mid H^c) > 0$. Then*

$$\underbrace{\frac{P(H \mid E)}{P(H^c \mid E)}}_{\textit{posterior odds}} = \underbrace{\frac{P(H)}{P(H^c)}}_{\textit{prior odds}} \times \underbrace{\frac{P(E \mid H)}{P(E \mid H^c)}}_{\textit{likelihood ratio}}, \tag{2.5}$$

that is, $\mathrm{O}(H \mid E) = \mathrm{O}(H) \times \mathrm{LR}(E)$*.*

Proof. We divide one instance of Bayes' theorem by another. Theorem 2.3 applied to H gives $P(H \mid E) = P(E \mid H)\, P(H)/P(E)$; applied to H^c (which has positive probability since $P(H) < 1$) it gives $P(H^c \mid E) = P(E \mid H^c)\, P(H^c)/P(E)$. The second quantity is positive because $P(E \mid H^c) > 0$ and $P(H^c) > 0$, so we may divide the first equation by the second. The common factor $P(E)$ cancels, leaving exactly (2.5). □

Example 2.7 (Rare-disease test, odds form)**.** We redo Example 2.4 in two lines. The prior odds of disease are $0.01/0.99 = 1/99$, and the likelihood ratio of a positive test is $0.99/0.05 = 19.8$. By (2.5), the posterior odds are

$$\frac{1}{99} \times 19.8 = \frac{19.8}{99} = \frac{1}{5},$$

and by (2.4) the posterior probability is $(1/5)/(6/5) = 1/6$ — the same answer, with no denominator bookkeeping.

Intuition. The odds form is fast because the normalising constant $P(E)$ — the entire denominator of (2.3) — divides out. What remains says that evidence acts on beliefs multiplicatively: the likelihood ratio is the exchange rate at which an observation converts prior odds into posterior odds. A problem that looks like heavy conditioning collapses to one multiplication and one conversion via (2.4).

2.3 Sequential Evidence and Reading Flow

Evidence usually arrives in a stream — repeated coin flips, repeated test results, repeated fills against a standing quote. When the pieces are conditionally independent given each hypothesis, the odds form of Corollary 2.6 iterates: each new piece multiplies the running odds by its own likelihood ratio. We first make the independence assumption explicit, since it is the hypothesis on which the whole multiplication rests.

Setup. As before, H is a hypothesis with $0 < P(H) < 1$. The evidence now consists of n events $E_1, \ldots, E_n$; think of E_i as "the i-th observation came out a particular way." The following definition pins down the independence we need.

Definition 2.8 (Conditional independence given an event)**.** Let B be an event with $P(B) > 0$. Events $E_1, \ldots, E_n$ are *conditionally independent given* B if for every subset $S \subseteq \{1, \ldots, n\}$,

$$P\Big(\bigcap_{i \in S} E_i \;\Big|\; B\Big) = \prod_{i \in S} P(E_i \mid B).$$

Theorem 2.9 (Multiplication of likelihood ratios)**.** *Let H be an event with $0 < P(H) < 1$, and let $E_1, \ldots, E_n$ be events that are conditionally independent given H and also conditionally independent given H^c, with $P(E_i \mid H^c) > 0$ for every i. Write $E = E_1 \cap \cdots \cap E_n$ for the compound evidence. Then $P(E) > 0$ and*

$$\mathrm{O}(H \mid E) = \mathrm{O}(H) \times \prod_{i=1}^{n} \mathrm{LR}(E_i), \qquad \textit{where } \mathrm{LR}(E_i) = \frac{P(E_i \mid H)}{P(E_i \mid H^c)}.$$

Proof. We apply the odds form to the compound event E and factor its likelihood ratio using conditional independence. First, E has positive probability: Definition 2.8 applied with $B = H^c$ and $S = \{1, \ldots, n\}$ gives $P(E \mid H^c) = \prod_{i=1}^{n} P(E_i \mid H^c) > 0$, and hence $P(E) \geq P(E \cap H^c) = P(E \mid H^c)\, P(H^c) > 0$ by the multiplication rule (2.1). The hypotheses of Corollary 2.6 are therefore satisfied for the pair (H, E), and it gives

$$\mathrm{O}(H \mid E) = \mathrm{O}(H) \times \frac{P(E \mid H)}{P(E \mid H^c)}.$$

Conditional independence given H factors the numerator as $P(E \mid H) = \prod_{i=1}^{n} P(E_i \mid H)$, and conditional independence given H^c factors the denominator as $P(E \mid H^c) = \prod_{i=1}^{n} P(E_i \mid H^c)$. The ratio of the two products is the product of the ratios, namely $\prod_{i=1}^{n} \mathrm{LR}(E_i)$. $\square$

Example 2.10 (The thousand coins)**.** A bag contains 1000 coins: 999 fair and one double-headed. A coin is drawn uniformly at random and flipped 10 times, and all 10 flips come up heads. We compute the probability that the drawn coin is the double-headed one.

Let H be the event that the double-headed coin was drawn, and let E_i be the event that the i-th flip is heads. The prior odds are

$$\mathrm{O}(H) = \frac{1/1000}{999/1000} = \frac{1}{999}.$$

Given H, every flip is heads with probability 1; given H^c, the flips are independent and each is heads with probability $1/2$. In both cases the E_i are conditionally independent given the hypothesis, so Theorem 2.9 applies with

$$\mathrm{LR}(E_i) = \frac{1}{1/2} = 2 \quad \text{for each } i, \qquad \prod_{i=1}^{10} \mathrm{LR}(E_i) = 2^{10} = 1024.$$

The posterior odds are therefore 1024/999, and by (2.4) the posterior probability is

$$P(H \mid E) = \frac{1024/999}{1 + 1024/999} = \frac{1024}{1024 + 999} = \frac{1024}{2023} \approx 50.6\%.$$

Despite ten straight heads, the answer is essentially a coin flip: the prior was roughly 1000:1 against, and the evidence factor $2^{10} = 1024$ only barely overcomes 999.

Remark 2.11 (Correlated evidence). Conditional independence is the load-bearing assumption of Theorem 2.9, and it must hold given *each* hypothesis separately — it neither implies nor follows from unconditional independence of the E_i. When the pieces of evidence share a common mechanism — for instance, several fills that are child slices of a single parent order — the factorisation fails, and multiplying likelihood ratios overstates the evidence.

To read order flow, instantiate the machinery: let H be the event that the counterparty currently trading is informed, and let E_i record the direction of the i-th aggressive fill — for instance, that it lifted the offer rather than hitting the bid. The prior odds come from the population of traders on the venue; each fill's likelihood ratio comes from how much more often informed flow trades in that direction than noise flow.

Trading application. A market maker is a Bayesian machine: every fill is evidence. The observation "the same counterparty lifted the offer three times in a row" carries a likelihood ratio strongly favouring *informed buyer* over *noise*, and by Theorem 2.9 sequential fills multiply likelihood ratios exactly as sequential tests do. The correct response to a posterior that has swung toward "informed" is to move the market up — raise both quotes — not to keep selling at the same price.

Interview trap. Base-rate neglect in reverse: with mostly-noise flow (a high base rate of uninformed traders), one aggressive fill is weak evidence; with thin, professional flow, the same fill is strong evidence. The *prior* on who trades on the venue changes the update — one reason the same quote logic behaves differently across exchanges.

Practice: problems P2.1–P2.6 in §20, solutions in §21.

3 Betting, Bankroll, Ruin and Markov Chains

This section develops the mathematics of repeated betting: the Kelly criterion for sizing a favorable bet, gambler's ruin for survival probabilities and game durations, first-step analysis as the general solving method behind both, stationary distributions as a first contact with the long-run behavior of Markov chains, and the St. Petersburg paradox as the canonical warning about infinite expectations. The same mathematics governs position sizing on a trading desk, the capital buffer of a strategy, and a market maker's inventory, which is why every topic here is a recurring interview theme.

3.1 The Kelly Criterion

A bettor with a genuine edge still faces a sizing decision: staking too little forgoes growth, while staking too much guarantees eventual ruin even though every individual bet has positive expected value. The Kelly criterion resolves the tension by choosing the stake that maximizes the long-run exponential growth rate of the bankroll.

Setup. Fix odds $b > 0$ and a win probability $p \in (0, 1)$, and write $q = 1 - p$. A bet at odds b-to-1 returns b units of profit per unit staked when it wins and loses the unit staked when it loses. A bettor starts with bankroll $W_0 > 0$ and repeatedly stakes the same fraction $f \in [0, 1)$ of the current bankroll on independent copies of this bet. We write R_k for the gross return (bankroll multiplier) of the k-th bet; the symbol r is reserved for the ratio q/p of the ruin subsections and is not used here.

Definition 3.1 (Fixed-fractional betting and expected log growth)**.** Let $R_1, R_2, \dots$ be iid random variables with

$$R_k = \begin{cases} 1 + bf & \text{with probability } p, \\ 1 - f & \text{with probability } q, \end{cases}$$

so that the bankroll after n bets is $W_n = W_0 \prod_{k=1}^n R_k$. The *expected log growth rate* of the strategy is

$$g(f) = \mathbb{E}\left[\log R_1\right] = p\log(1+bf) + q\log(1-f), \qquad f \in [0,1). \tag{3.1}$$

The next lemma explains why g is the right objective: it is the almost-sure exponential rate at which the bankroll grows or decays.

Lemma 3.2 (Almost-sure growth rate)**.** *Fix $f \in [0,1)$. Then*

$$\frac{1}{n}\log\frac{W_n}{W_0} \longrightarrow g(f) \quad \textit{almost surely as } n \to \infty.$$

Consequently $W_n \to \infty$ almost surely when $g(f) > 0$, and $W_n \to 0$ almost surely when $g(f) < 0$.

Proof. We appeal to the strong law of large numbers. Taking logarithms of the product,

$$\log\frac{W_n}{W_0} = \sum_{k=1}^n \log R_k,$$

a sum of iid random variables taking the two finite values $\log(1+bf)$ and $\log(1-f)$, hence with finite mean $g(f)$ by (3.1). The strong law gives $n^{-1}\sum_{k=1}^n \log R_k \to g(f)$ almost surely. On the event where the convergence holds, $\log W_n = \log W_0 + n\,(g(f) + o(1))$, which tends to $+\infty$ if $g(f) > 0$ and to $-\infty$ if $g(f) < 0$; exponentiating gives the two consequences. □

Maximizing long-run growth therefore means maximizing g over f. The following theorem describes the shape of g completely and locates its maximizer.

Theorem 3.3 (Kelly criterion)**.** *Let g be as in (3.1) with $b > 0$ and $p \in (0,1)$.*

(i) The function g is strictly concave on $[0,1)$, with $g(0) = 0$, $g'(0) = bp - q$, and $g(f) \to -\infty$ as $f \uparrow 1$.

(ii) If $bp - q > 0$, then g has a unique global maximizer on $[0,1)$, namely

$$f^* = \frac{bp - q}{b},$$

which lies in $(0,1)$, and $g(f^*) > 0$.

(iii) If $bp - q \le 0$, then g is strictly decreasing on $[0,1)$, so the unique maximizer is $f = 0$: the growth-optimal action is not to bet.

Proof. We proceed by elementary calculus: we compute two derivatives, establish strict concavity, and solve the first-order condition.

Step 1 (derivatives and concavity). On $[0,1)$ both logarithms in (3.1) have positive arguments, and

$$g'(f) = \frac{pb}{1+bf} - \frac{q}{1-f}, \qquad g''(f) = -\frac{pb^2}{(1+bf)^2} - \frac{q}{(1-f)^2}.$$

Both terms of g'' are strictly negative for every $f \in [0,1)$, so $g'' < 0$ and g is strictly concave. Evaluating at $f = 0$ gives $g(0) = p \log 1 + q \log 1 = 0$ and $g'(0) = pb - q$. As $f \uparrow 1$ the first term of g tends to the finite limit $p \log(1+b)$ while $q \log(1-f) \to -\infty$, so $g(f) \to -\infty$.

Step 2 (first-order condition). Setting $g'(f) = 0$ and clearing denominators,

$$pb(1-f) = q(1+bf) \iff pb - q = bf(p+q) = bf \iff f = \frac{pb-q}{b}.$$

Step 3 (case $bp - q > 0$). The stationary point $f^* = (bp-q)/b = p - q/b$ is positive by assumption and satisfies $f^* < p < 1$ because $q/b > 0$, so $f^* \in (0,1)$. Strict concavity makes g' strictly decreasing; since $g'(f^*) = 0$, we get $g' > 0$ on $[0, f^*)$ and $g' < 0$ on $(f^*, 1)$, so f^* is the unique global maximizer. Because $g(0) = 0$ and g is strictly increasing on $[0, f^*]$, we conclude $g(f^*) > 0$.

Step 4 (case $bp - q \leq 0$). Here $g'(0) = bp - q \leq 0$, and g' is strictly decreasing by Step 1, so $g'(f) < 0$ for every $f \in (0,1)$. Hence g is strictly decreasing on $[0,1)$ and its unique maximizer is $f = 0$. □

(Problem P3.1 asks you to reproduce this derivation from scratch.)

The quantity $g'(0) = bp - q$ is the expected profit per unit staked — the *edge* of the bet. The Kelly fraction is therefore "edge over odds": $f^* = (bp-q)/b$.

Corollary 3.4 (Even-money Kelly)**.** *For an even-money bet ($b = 1$) with $p > \frac{1}{2}$, the Kelly fraction is*

$$f^* = p - q = 2p - 1.$$

Proof. Substituting $b = 1$ into Theorem 3.3(ii) gives $f^* = (p-q)/1 = p - q = 2p - 1$. □

Example 3.5 (A 60/40 coin)**.** Consider an even-money bet won with probability $p = 0.6$, so $q = 0.4$ and $b = 1$. By Corollary 3.4, the Kelly fraction is $f^* = 2(0.6) - 1 = 0.2$: stake 20% of the bankroll on each bet. The optimal growth rate is

$$g(0.2) = 0.6 \log 1.2 + 0.4 \log 0.8 = 0.6(0.18232) + 0.4(-0.22314) \approx 0.02014,$$

so capital compounds by a factor $e^{0.02014} \approx 1.0203$ per bet — about 2.0% — and over 100 bets the typical growth factor is $e^{2.01} \approx 7.5$. Figure 4 plots this growth curve to scale.

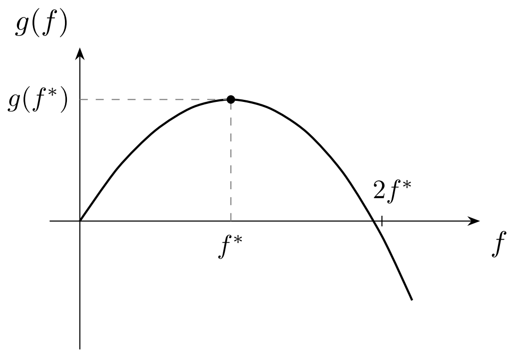


Figure 4: The expected log growth rate $g(f)$ of (3.1) for an even-money bet with $p = 0.6$, drawn to scale. As Theorem 3.3 asserts, the curve is strictly concave, leaves the origin with slope equal to the edge $bp - q = 0.2$, peaks at $f^* = 0.2$, and diverges to $-\infty$ as $f \to 1$. It crosses zero just below $2f^* = 0.4$ (exactly at $2f^*$ in the quadratic approximation of Proposition 3.6); betting beyond the crossing shrinks capital almost surely despite a positive per-bet expectation.

The next proposition makes precise three claims that practitioners quote constantly: half-Kelly keeps about three quarters of the growth, volatility scales linearly in the stake (so half the stake means a quarter of the variance), and growth turns negative at about twice the Kelly fraction.

Proposition 3.6 (Fractional Kelly, small-edge approximation)**.** *Consider an even-money bet* ($b = 1$) *with edge* $\varepsilon = p - q > 0$*, so that* $f^* = \varepsilon$ *by Corollary 3.4. Let* $\tilde{g}(f) = \varepsilon f - f^2/2$*.*

(i) The polynomial $\tilde{g}$ *is the second-order Taylor polynomial of* g *at* $f = 0$*.*

(ii) The polynomial $\tilde{g}$ *is uniquely maximized at* $f = \varepsilon = f^*$*, with* $\tilde{g}(f^*) = \varepsilon^2/2$*, and for every* $c \geq 0$*,*

$$\tilde{g}\left(cf^*\right) = c\left(2 - c\right)\tilde{g}\left(f^*\right).$$

In particular $\tilde{g}(f^*/2) = \frac{3}{4}\,\tilde{g}(f^*)$ *and* $\tilde{g}(2f^*) = 0$*.*

(iii) The variance of the per-bet log return at fraction f *is* $4pq\,f^2 + O(f^3)$ *as* $f \downarrow 0$*. Hence, to leading order, betting* cf^* *scales the log-return volatility by* c *and the variance by* c^2*.*

Consequently, to second order in the edge, half-Kelly earns 75% *of the maximal growth at half the volatility (a quarter of the variance), and the growth rate vanishes at twice the Kelly fraction and is negative beyond it.*

Proof. We compute Taylor expansions directly.

Part (i). From the proof of Theorem 3.3 with $b = 1$: $g(0) = 0$, $g'(0) = p - q = \varepsilon$, and $g''(0) = -p - q = -1$. The second-order Taylor polynomial at 0 is therefore $\varepsilon f - f^2/2 = \tilde{g}(f)$.

Part (ii). The polynomial $\tilde{g}$ is a downward parabola with $\tilde{g}'(f) = \varepsilon - f$, so its unique maximizer is $f = \varepsilon$ with value $\varepsilon^2 - \varepsilon^2/2 = \varepsilon^2/2$. Substituting $f = c\varepsilon$,

$$\tilde{g}(c\varepsilon) = \varepsilon(c\varepsilon) - \frac{(c\varepsilon)^2}{2} = \varepsilon^2\left(c - \frac{c^2}{2}\right) = c(2-c)\,\frac{\varepsilon^2}{2} = c(2-c)\,\tilde{g}(f^*).$$

Evaluating at $c = \frac{1}{2}$ gives the factor $\frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}$, and at $c = 2$ the factor $2 \cdot 0 = 0$.

Part (iii). Write L for the per-bet log return, so $L = \log(1 + f)$ with probability p and $L = \log(1 - f)$ with probability q. Expanding, $\log(1 \pm f) = \pm f + O(f^2)$, hence $\mathbb{E}[L] = \varepsilon f + O(f^2)$ and $\mathbb{E}[L^2] = f^2 + O(f^3)$, giving

$$\mathrm{Var}(L) = \mathbb{E}[L^2] - (\mathbb{E}[L])^2 = \left(1 - \varepsilon^2\right) f^2 + O(f^3) = 4pq\,f^2 + O(f^3),$$

where the last equality uses $1 - \varepsilon^2 = (p+q)^2 - (p-q)^2 = 4pq$. Replacing f by cf multiplies the leading term by c^2, which is the claimed scaling of variance; taking square roots gives the scaling of volatility. $\square$

Example 3.7 (Half-Kelly and over-betting on the 60/40 coin)**.** Continue Example 3.5 ($p = 0.6$, $b = 1$, $f^* = 0.2$, $g(f^*) \approx 0.02014$). The exact numbers match Proposition 3.6 closely.

- *Half-Kelly.* At $f = 0.1$: $g(0.1) = 0.6\log 1.1 + 0.4\log 0.9 \approx 0.01504$, which is 74.7% of $g(f^*)$ — the predicted three quarters. The exact per-bet log-return standard deviation is 0.0983 at $f = 0.1$ versus 0.1986 at $f = 0.2$ (half), and the variances are 0.00966 versus 0.03946 (a quarter).
- *Twice Kelly.* At $f = 2f^* = 0.4$: $g(0.4) = 0.6\log 1.4 + 0.4\log 0.6 \approx -0.00245 < 0$. The exact zero crossing sits at $f \approx 0.389$, just below $2f^*$, exactly as Figure 4 shows. Meanwhile the expected gross return per bet at $f = 0.4$ is $1 + 0.4(p - q) = 1.08$, so the *expected* bankroll grows by 8% per bet even though, by Lemma 3.2, the actual bankroll tends to zero almost surely. The expectation is carried by vanishingly rare lucky paths.

Interview trap. Over-betting is catastrophic while under-betting is merely slow: the growth curve of Figure 4 falls off a cliff to the right of f^* and turns negative beyond roughly $2f^*$. In particular, "the expected value is positive, so bet everything" fails twice over — at f near 1 a single loss is nearly fatal ($g \to -\infty$), and even moderate over-betting produces $g(f) < 0$, which by Lemma 3.2 means almost-sure ruin despite an exponentially growing expected bankroll.

Remark 3.8 (Why practitioners bet less than full Kelly). Three reasons, all standard and all worth stating in an interview.

(1) The win probability p is *estimated*, not known. Plugging an overestimated edge into $f^* = (bp - q)/b$ means betting beyond the true optimum, and Proposition 3.6 shows the penalty is asymmetric: beyond about $2f^*$ the true growth rate is negative.
(2) Log-optimality ignores drawdown pain: a full-Kelly bettor routinely halves the bankroll along the way. Half-Kelly keeps about 75% of the growth at half the volatility — a quarter of the variance — by Proposition 3.6.
(3) Edges correlate: simultaneous "independent" bets often are not independent, so the effective aggregate fraction at risk exceeds what per-bet Kelly arithmetic suggests.

Trading application. A live prediction-market strategy sized positions fixed-fractionally with hard per-market caps — fractional-Kelly logic under parameter uncertainty. That is the shape of a strong sizing answer in an interview: quote Kelly as the theoretical anchor, apply an explicit haircut for estimation error and correlated exposures, and impose hard caps because a model's fitted probability is never the true p.

3.2 First-Step Analysis as a General Method

A single method solves coin-pattern races (HH before TH and their relatives), gambler's ruin, and most "expected time until ..." games: condition on the first move and let the Markov property restart the problem. This subsection names the method once, so that the proofs of §3.3 can invoke it; the worked example shows the companion trick of collapsing symmetric states.

Setup. Consider a time-homogeneous Markov chain on a finite state space, together with a target quantity defined per state: the probability of reaching one absorbing set before another, or the expected number of steps until absorption.

Method (first-step analysis).

(1) *Symmetry first.* Collapse equivalent states before writing any equations: states whose transition structure relative to the target is identical can share a single unknown. This is the biggest practical simplification.
(2) Define one unknown per (collapsed) state — a probability h_i or an expected time m_i.
(3) Write one equation per non-absorbing state by conditioning on the first move: the law of total probability (or total expectation) over the one-step transitions, with the Markov property guaranteeing that after the first step the process is a fresh copy started from the new state. Expected-time equations carry an extra "+1" for the step just taken.
(4) Impose boundary conditions at absorbing states: probabilities 0 or 1, expected times 0.
(5) Solve the resulting linear system.

For expected-time versions the unknowns must be finite for the equations to be valid; finiteness holds whenever an absorbing state is reachable from every state, by the block argument proved as Lemma 3.10 below.

Example 3.9 (Ant on a cube: state collapse)**.** An ant stands on one corner of a cube and walks along edges, choosing among the three incident edges uniformly at random at each step. We compute the expected number of steps to reach the opposite corner.

Collapse. Identify the cube's vertices with binary strings in $\{0,1\}^3$ and the target with the string differing from the start in all three coordinates. Each step flips one uniformly chosen coordinate. If the ant is at Hamming distance d from the target, then d of the three flips decrease the distance and $3 - d$ increase it, so the distance process is itself a Markov chain with

$$P(d \to d-1) = \frac{d}{3}, \qquad P(d \to d+1) = \frac{3-d}{3},$$

and the eight vertices collapse to the four states $d \in \{0, 1, 2, 3\}$. Let m_d denote the expected number of steps to reach $d = 0$ from distance d; the unknowns are finite by the argument of Lemma 3.10.

First-step equations. Conditioning on the first move from each transient state, with boundary condition $m_0 = 0$:

$$m_3 = 1 + m_2, \qquad m_2 = 1 + \tfrac{1}{3}m_3 + \tfrac{2}{3}m_1, \qquad m_1 = 1 + \tfrac{2}{3}m_2 + \tfrac{1}{3} \cdot 0.$$

Solve. Substituting the first equation into the second, $m_2 = 1+\frac{1}{3}(1+m_2)+\frac{2}{3}m_1$, so $\frac{2}{3}m_2 = \frac{4}{3}+\frac{2}{3}m_1$, giving $m_2 = 2+m_1$. The third equation becomes $m_1 = 1+\frac{2}{3}(2+m_1) = \frac{7}{3}+\frac{2}{3}m_1$, so $\frac{1}{3}m_1 = \frac{7}{3}$ and $m_1 = 7$. Back-substituting, $m_2 = 9$ and $m_3 = 10$: the ant needs 10 steps on average. Symmetry reduced an eight-state system to three equations solved in four lines.

Intuition. Hunting for the collapse is step zero of every such problem: the cube's 8 states become 4, a deuce game's score lattice becomes 3 states (lead +1, level, lead −1), and the gambler's-ruin chain is already collapsed. Writing equations before finding the symmetry wastes time and invites algebra errors.

3.3 Gambler's Ruin

Gambler's ruin is the canonical absorption problem and a standing metaphor in trading: the state is a capital buffer or a market maker's inventory, and the absorbing barriers are bankruptcy and a profit target. This subsection proves the three formulas worth knowing cold — the fair-game hitting probability i/N, the fair-game expected duration $i(N - i)$, and the biased-game hitting probability — as applications of the first-step method of §3.2.

Setup. Fix an integer $N \geq 1$ and $p \in (0, 1)$, and set $q = 1 - p$ and $r = q/p$. Let $(S_n)_{n \geq 0}$ be the random walk on $\{0, 1, \ldots, N\}$ started at $S_0 = i$ that, from any interior state, steps up by 1 with probability p and down by 1 with probability q, independently at each step; the states 0 and N are absorbing. Figure 5 shows the chain. Define the absorption time $T = \min\{n \geq 0 : S_n \in \{0, N\}\}$, the hitting probability $h_i = P(S_T = N \mid S_0 = i)$, and the expected duration $D_i = \mathbb{E}[T \mid S_0 = i]$.

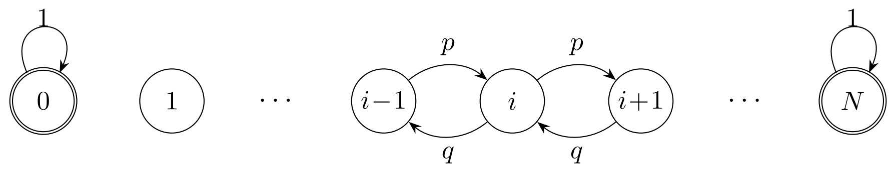


Figure 5: The gambler's-ruin random walk on $\{0, 1, \ldots, N\}$. From an interior state i the walk moves up with probability p and down with probability $q = 1 - p$ (arrows shown around the generic state i); the boundary states 0 and N are absorbing (double circles, self-loop probability 1).

Before solving for h_i and D_i we verify that the game actually ends, and quickly enough for expected durations to be finite — the hypothesis the first-step method needs.

Lemma 3.10 (Absorption is certain and has finite mean time)**.** *For every starting state $i \in \{0, \ldots, N\}$,*

$$P(T < \infty \mid S_0 = i) = 1 \qquad \textit{and} \qquad \mathbb{E}[T \mid S_0 = i] < \infty.$$

Proof. We use a block argument. Partition time into consecutive blocks of N steps. Suppose the walk is not yet absorbed at the start of a block, so its state lies in $\{1, \ldots, N-1\}$. With probability at least p^N all N steps of the block go up; starting from a state ≥ 1, a run of N consecutive up-moves would carry the walk past level N, which is impossible without first hitting the absorbing state N. Hence absorption occurs within the block with probability at least $\delta = p^N > 0$, and this bound holds for every block regardless of the past. Iterating,

$$P(T > kN) \le (1-\delta)^k \longrightarrow 0 \quad (k \to \infty),$$

so $T < \infty$ almost surely. For the mean, sum tail probabilities: each n with $kN \le n < (k+1)N$ satisfies $P(T > n) \le P(T > kN) \le (1-\delta)^k$, so

$$\mathbb{E}[T] = \sum_{n \ge 0} P(T > n) \le \sum_{k \ge 0} N(1-\delta)^k = \frac{N}{\delta} < \infty. \qquad \square$$

Theorem 3.11 (Gambler's ruin, fair case)**.** *Suppose $p = \frac{1}{2}$. Then for every $i \in \{0, \dots, N\}$,*

$$h_i = P(\textit{reach } N \textit{ before } 0 \mid S_0 = i) = \frac{i}{N}.$$

Proof. We proceed by first-step analysis (§3.2). Conditioning on the first step from an interior state i, the law of total probability and the Markov property give

$$h_i = \tfrac{1}{2}\, h_{i+1} + \tfrac{1}{2}\, h_{i-1} \qquad (1 \le i \le N-1), \qquad h_0 = 0, \quad h_N = 1. \tag{3.2}$$

Relation (3.2) says that h is *discrete harmonic*: each interior value is the average of its two neighbors. Rearranging,

$$h_{i+1} - h_i = h_i - h_{i-1} \qquad (1 \le i \le N-1),$$

so all successive increments are equal to a common value d, and telescoping from the left boundary gives $h_i = h_0 + i\,d = i\,d$. The right boundary condition $h_N = Nd = 1$ forces $d = 1/N$, hence $h_i = i/N$. Both boundary conditions hold, and the argument derived this form for *every* solution of (3.2), so the solution is unique — the discrete counterpart of the fact that harmonic functions on an interval are affine. $\square$

Theorem 3.12 (Expected duration, fair case)**.** *Suppose $p = \frac{1}{2}$. Then for every $i \in \{0, \dots, N\}$,*

$$D_i = \mathbb{E}[T \mid S_0 = i] = i\,(N - i).$$

Proof. We again proceed by first-step analysis; Lemma 3.10 guarantees the unknowns D_i are finite, so the following equations are valid. Conditioning on the first step (which costs one step and restarts the walk from the new state),

$$D_i = 1 + \tfrac{1}{2}\, D_{i+1} + \tfrac{1}{2}\, D_{i-1} \qquad (1 \le i \le N-1), \qquad D_0 = D_N = 0. \tag{3.3}$$

Multiplying by 2 and rearranging puts (3.3) in the form of a second-order linear recurrence with constant inhomogeneous term:

$$D_{i+1} - 2D_i + D_{i-1} = -2.$$

Homogeneous part. The characteristic equation of $D_{i+1} - 2D_i + D_{i-1} = 0$ is $x^2 - 2x + 1 = (x-1)^2 = 0$, a double root at $x = 1$, so the homogeneous solutions are exactly the affine sequences $\alpha + \beta i$.

Particular solution. Since constants and linear terms already solve the homogeneous equation, we try a quadratic: substituting $D_i = \gamma i^2$ gives

$$\gamma \left[(i+1)^2 - 2i^2 + (i-1)^2 \right] = 2\gamma = -2,$$

so $\gamma = -1$ and $-i^2$ is a particular solution.

General solution and boundaries. Every solution of the recurrence has the form $D_i = \alpha + \beta i - i^2$. The boundary condition $D_0 = 0$ forces $\alpha = 0$, and $D_N = \beta N - N^2 = 0$ forces $\beta = N$. Hence $D_i = Ni - i^2 = i(N-i)$. Uniqueness holds because the difference of two solutions of (3.3) solves the homogeneous equation with zero boundary values, i.e. is affine and vanishes at 0 and N, hence is identically zero. $\square$

(Problem P3.3 asks you to reproduce this proof and that of Theorem 3.11 from scratch.)

Example 3.13 (A fair game to 100)**.** Two players flip a fair coin for one unit per flip; one starts with 40 units, the other with 60, and they play until someone is broke, so $i = 40$ and $N = 100$. By Theorem 3.11 the poorer player finishes with everything with probability $h_{40} = 40/100 = 0.40$ — in a fair game, the probability of winning the pot equals the fraction of the total capital contributed. By Theorem 3.12 the game lasts $D_{40} = 40 \times 60 = 2400$ flips on average, far longer than the 100 units at stake might suggest.

Theorem 3.14 (Gambler's ruin, biased case)**.** *Suppose $p \neq \frac{1}{2}$ and set $r = q/p$. Then for every $i \in \{0, \dots, N\}$,*

$$h_i = P(\textit{reach } N \textit{ before } 0 \mid S_0 = i) = \frac{1 - r^{\,i}}{1 - r^{\,N}}.$$

Proof. We use first-step analysis followed by the standard ansatz for linear recurrences. Conditioning on the first step from an interior state,

$$h_i = p\,h_{i+1} + q\,h_{i-1} \qquad (1 \le i \le N-1), \qquad h_0 = 0, \quad h_N = 1. \tag{3.4}$$

Ansatz. Seek geometric solutions $h_i = x^i$. Substituting into (3.4) and dividing by x^{i-1} gives the characteristic equation

$$px^2 - x + q = 0.$$

The two roots multiply to q/p and sum to $1/p$; the reader should verify that $x = 1$ and $x = q/p = r$ satisfy both, so these are the roots, and they are distinct precisely because $p \neq q$. The general solution of (3.4) is therefore

$$h_i = \alpha \cdot 1^i + \beta\, r^i = \alpha + \beta\, r^i.$$

Boundary conditions. The condition $h_0 = 0$ gives $\alpha + \beta = 0$, so $\alpha = -\beta$, and $h_N = 1$ gives $\beta(r^N - 1) = 1$, so $\beta = 1/(r^N - 1)$. Hence

$$h_i = \frac{r^i - 1}{r^N - 1} = \frac{1 - r^i}{1 - r^N}.$$

Uniqueness follows as in Theorem 3.12: the difference of two solutions of (3.4) is of the form $\alpha + \beta r^i$ with zero values at $i = 0$ and $i = N$, which forces $\alpha = \beta = 0$. ☐

(Problem P3.4 asks you to reproduce this derivation and push it to the no-upper-target limit of Corollary 3.16.)

Proposition 3.15 (The fair case as a limit)**.** *Fix i and N. As $p \to \frac{1}{2}$ (equivalently $r \to 1$),*

$$\frac{1 - r^{\,i}}{1 - r^{\,N}} \longrightarrow \frac{i}{N},$$

recovering Theorem 3.11.

Proof. We apply l'Hôpital's rule in the variable r at $r = 1$: numerator and denominator both vanish there and both are differentiable, so

$$\lim_{r \to 1} \frac{1 - r^i}{1 - r^N} = \lim_{r \to 1} \frac{-i\, r^{\,i-1}}{-N\, r^{\,N-1}} = \frac{i}{N}.$$

Alternatively, factoring geometric sums, $(1 - r^i)/(1 - r^N) = (1 + r + \cdots + r^{\,i-1})/(1 + r + \cdots + r^{\,N-1})$, whose numerator and denominator tend to i and N respectively. ☐

Corollary 3.16 (Ruin with no upper target)**.** *Suppose the walk has no upper absorbing barrier (the adversary is infinitely rich, or the strategy simply never stops). Starting from $i \ge 1$:*

(i) if $p > \frac{1}{2}$ (so $r < 1$), the probability of ever hitting 0 is $r^{\,i}$, decaying exponentially in the buffer i;

(ii) if $p \le \frac{1}{2}$, ruin is certain.

Proof. We compute the limit of the two-barrier formulas as $N \to \infty$. Writing T_0 and T_N for the hitting times of 0 and N, the events $\{T_0 < T_N\}$ increase to $\{T_0 < \infty\}$ as $N \to \infty$: if the walk ever hits 0, it visits only finitely many levels beforehand, so $T_0 < T_N$ for all N large enough. By continuity of probability from below, $P(T_0 < \infty) = \lim_{N\to\infty}(1 - h_i)$.

For $p > \frac{1}{2}$ we have $r < 1$, so $r^N \to 0$ and, by Theorem 3.14, $1 - h_i \to 1 - (1 - r^i) = r^i$.

For $p < \frac{1}{2}$ we have $r > 1$; dividing numerator and denominator by r^N,

$$1 - h_i = \frac{r^i - r^N}{1 - r^N} = \frac{r^{\,i-N} - 1}{r^{-N} - 1} \longrightarrow \frac{0 - 1}{0 - 1} = 1.$$

For $p = \frac{1}{2}$, Theorem 3.11 gives $1 - h_i = 1 - i/N \to 1$. □

Example 3.17 (Small edge, large consequence)**.** Two computations show how violently a small bias moves the answer.

- *Two-barrier.* Take $p = 0.48$, $i = 10$, $N = 20$ — a nearly fair game, starting halfway. Then $r = 0.52/0.48 = 13/12$, and $\log(13/12) \approx 0.0800$, so $r^{10} \approx e^{0.800} \approx 2.226$ and $r^{20} \approx 4.957$. By Theorem 3.14,
$$h_{10} = \frac{1 - 2.226}{1 - 4.957} = \frac{1.226}{3.957} \approx 0.310.$$
A two-point deficit in win rate turns a 50% shot (Theorem 3.11 with $i/N = \frac{1}{2}$; also the $p \to \frac{1}{2}$ limit of Proposition 3.15) into a 31% shot.
- *No upper target.* A strategy wins each unit-stake round with probability $p = 0.55$ and never stops. With a buffer of $i = 10$ units, $r = 0.45/0.55 = 9/11$ and $\log(9/11) \approx -0.2007$, so by Corollary 3.16 the lifetime ruin probability is $(9/11)^{10} \approx e^{-2.007} \approx 0.134$. Doubling the buffer to $i = 20$ squares this: $(9/11)^{20} \approx 0.018$, below 2%. Capitalization buys exponential safety.

Intuition. A small negative edge is lethal over many trials: with $p < \frac{1}{2}$ and a distant target, the probability of reaching it decays exponentially and ruin probability tends to 1, by Theorem 3.14 and Corollary 3.16. This is why a market maker obsesses over a fraction of a tick of edge — compounding acts on the *sign* of the edge — and why paying the spread as a taker is a tax that must be beaten before anything else compounds in the right direction.

Trading application. The ruin walk doubles as a capitalization model for a trading operation: the state is the capital buffer in units of per-trade risk, p is the per-trade win probability, and Corollary 3.16 says the lifetime ruin probability of a positive-edge operation is $r^{\,i}$. A desk therefore sizes its buffer so that $r^{\,i}$ sits below its ruin tolerance — and treats anything that flips the sign of the edge (fees, spread paid, adverse selection) as an existential threat rather than a cost line.

3.4 Markov Chains and Stationary Distributions

Absorption questions ask where a chain ends; stationary distributions ask how a chain that never ends shares its time among states. This first contact covers the definition and the two-state chain, which is solvable in closed form and already carries the main intuition.

Setup. Let $(X_n)_{n\ge 0}$ be a time-homogeneous Markov chain on the finite state space $\{1, \dots, K\}$. Its one-step transition matrix is written $A = (A_{jk})$ with $A_{jk} = P(X_{n+1} = k \mid X_n = j)$; each row of A is nonnegative and sums to 1. We write A rather than the customary P because the symbol P is reserved for probability throughout this document. Distributions over states are row vectors, so that if X_n has distribution μ then X_{n+1} has distribution μA.

Definition 3.18 (Stationary distribution)**.** A row vector $\pi = (\pi_1, \dots, \pi_K)$ is a *stationary distribution* of the chain if

$$\pi_j \geq 0 \quad (1 \leq j \leq K), \qquad \sum_{j=1}^{K} \pi_j = 1, \qquad \pi A = \pi.$$

If X_0 has distribution π, then by induction X_n has distribution π for every n: the chain is in equilibrium.

Proposition 3.19 (Two-state stationary distribution)**.** *Consider the two-state chain that switches from state* 1 *to state* 2 *with probability* a *and from state* 2 *to state* 1 *with probability* b *(staying put otherwise), i.e.*

$$A = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}, \qquad 0 \leq a, b \leq 1, \quad a + b > 0.$$

Within this subsection a *and* b *denote these switch probabilities; the odds symbol* b *of §3.1 is not used here. The chain has the unique stationary distribution*

$$\pi = \left(\frac{b}{a+b}, \ \frac{a}{a+b} \right).$$

Proof. We solve the defining linear system directly. The first coordinate of $\pi A = \pi$ reads

$$\pi_1(1-a) + \pi_2 \, b = \pi_1 \iff a\,\pi_1 = b\,\pi_2,$$

the *balance equation*: in equilibrium, probability flow from 1 to 2 equals flow from 2 to 1. The second coordinate reads $\pi_1 a + \pi_2(1-b) = \pi_2$, which rearranges to the same equation $a\pi_1 = b\pi_2$ (as it must, since the rows of A sum to one). The system therefore reduces to

$$a\,\pi_1 = b\,\pi_2, \qquad \pi_1 + \pi_2 = 1.$$

Substituting $\pi_2 = 1 - \pi_1$ into the balance equation gives $a\pi_1 = b(1-\pi_1)$, so $(a+b)\pi_1 = b$ and, since $a + b > 0$,

$$\pi_1 = \frac{b}{a+b}, \qquad \pi_2 = \frac{a}{a+b}.$$

Both entries are nonnegative and sum to one, and every step was an equivalence, so this is the unique stationary distribution. ☐

Example 3.20 (A two-regime order-flow model)**.** Model a venue's order flow as quiet (state 1) or busy (state 2) period by period, switching from quiet to busy with probability $a = 0.05$ and from busy to quiet with probability $b = 0.20$. By Proposition 3.19,

$$\pi = \left(\frac{0.20}{0.25}, \ \frac{0.05}{0.25} \right) = (0.8, \ 0.2) :$$

the venue is quiet 80% of the time in the long run. Verification by direct multiplication: the first entry of πA is $0.8 \times 0.95 + 0.2 \times 0.20 = 0.76 + 0.04 = 0.80$ and the second is $0.8 \times 0.05 + 0.2 \times 0.80 = 0.04 + 0.16 = 0.20$, so $\pi A = \pi$ as required.

Intuition. Time is shared inversely to how easily each state is left. When $a, b > 0$, a visit to state 1 lasts Geom(a) steps (mean $1/a$) and a visit to state 2 lasts Geom(b) steps (mean $1/b$), and the chain alternates visits; the long-run fraction of time in state 1 is therefore

$$\frac{1/a}{1/a + 1/b} = \frac{b}{a+b},$$

matching Proposition 3.19. Sticky states accumulate probability.

3.5 St. Petersburg and Infinite Expected Value

The St. Petersburg game is the standard cautionary tale about treating expected value as a price: its expectation is infinite, yet nobody pays more than about twenty dollars to play. This subsection computes the divergence, proves the bankroll-capped value, and names the classical resolutions.

Setup. A fair coin is flipped until the first head. If the first head occurs on flip n (an event of probability 2^{-n}, $n = 1, 2, \ldots$), the player is paid $Y = 2^n$ dollars.

Proposition 3.21 (St. Petersburg divergence)**.** *The expected payoff of the St. Petersburg game is infinite:*

$$\mathbb{E}[Y] = \sum_{n=1}^{\infty} 2^{-n} \cdot 2^n = \sum_{n=1}^{\infty} 1 = \infty.$$

Proof. We compute the defining series of the expectation of a discrete nonnegative random variable. The event $\{Y = 2^n\}$ has probability 2^{-n}, so the n-th term of $\sum_n 2^n P(Y = 2^n)$ equals $2^n \cdot 2^{-n} = 1$. The partial sums are m after m terms and grow without bound, so the series of nonnegative terms diverges and $\mathbb{E}[Y] = \infty$. $\square$

Despite the infinite expectation, essentially nobody will pay more than about \$20 to play once. The sharpest quantitative resolution is that no real counterparty can pay unbounded amounts: capping the payoff at the counterparty's bankroll collapses the value to a small number.

Proposition 3.22 (Capped St. Petersburg)**.** *Fix an integer $M \geq 1$ and suppose the payoff is capped at 2^M dollars, i.e. the player receives $Y_M = \min(Y, 2^M)$. Then*

$$\mathbb{E}[Y_M] = M + 1.$$

Proof. We split the expectation at the cap. For $n \leq M$ the payoff is 2^n, and for $n > M$ it is 2^M:

$$\mathbb{E}[Y_M] = \sum_{n=1}^{M} 2^{-n} \cdot 2^n + 2^M \sum_{n=M+1}^{\infty} 2^{-n} = \underbrace{M}_{\text{one dollar per level}} + 2^M \cdot 2^{-M} = M + 1,$$

where the tail sum is geometric: $\sum_{n=M+1}^{\infty} 2^{-n} = 2^{-M}$. Each uncapped level contributes exactly one dollar of expected value, and the entire capped tail contributes one more. $\square$

(Problem P3.9 asks you to reproduce both computations with a different cap.)

Example 3.23 (A 2^{30} cap)**.** Suppose the counterparty can pay at most 2^{30} dollars — about \$1.07 billion, a very rich casino. Proposition 3.22 with $M = 30$ gives

$$\mathbb{E}[Y_{30}] = 30 + 1 = 31$$

dollars: one dollar of expected value per level up to the cap, plus one dollar for the capped tail. An infinite expectation backed by a billion-dollar bankroll is worth \$31.

Remark 3.24 (Three resolutions)*.* Three classical resolutions of the paradox are worth naming.

(1) *Log utility (Bernoulli, 1738).* A log-utility agent values the uncapped game by $\mathbb{E}[\log_2 Y] = \sum_{n \geq 1} n\, 2^{-n} = 2$ doublings, a certainty equivalent of $2^2 = 4$ dollars — finite and small. This is the same concavity that drives the Kelly criterion of §3.1.

(2) *Finite counterparty bankroll.* No counterparty pays unbounded sums; Proposition 3.22 shows a cap at 2^M collapses the value to $M + 1$ dollars.

(3) *Variance and one-shot reasoning.* The payoff distribution is so heavy tailed that the expectation is carried by astronomically rare outcomes: a single play pays at most \$8 with probability 7/8, and with no finite mean (uncapped) sample averages never stabilize over any feasible number of plays.

Interview trap. Quoting the infinite expectation as a fair price is the trap. *Price* is not *expected value* when distributions are wild — managing the gap between the two is the whole point of risk management. In an interview, the standard follow-up "how much would you pay to play once?" is answered with a small number in the low tens of dollars, justified by the three resolutions of Remark 3.24, never with "infinity".

Practice: problems P3.1–P3.10 in §20, solutions in §21.

Part II

Statistics and Regression

4 Statistics I: Estimators and Inference

Estimation is where probability meets data: a trading signal's mean return, a strategy's volatility, a Sharpe ratio — each is a number computed from a finite sample and therefore a random variable in its own right, with a bias, a variance, and a sampling distribution. This section builds the estimator's error budget (bias, variance, mean squared error), corrects the most famous estimation bias (Bessel's correction), states what the central limit theorem does and does not license, turns hypothesis testing into an explicit procedure with a fully worked coin example, and closes with the standard error of the Sharpe ratio — the single formula that most reliably deflates backtest overconfidence.

4.1 Bias, Variance, and Mean Squared Error

Every estimate carries two kinds of error: a systematic offset (bias) and sampling noise (variance). The mean squared error adds them in exactly one way, and that decomposition is the reason a deliberately biased estimator can beat an unbiased one — the fact that underlies ridge regression and shrinkage throughout statistics and quantitative finance.

Setup. Let $X_1, \ldots, X_n$ be data drawn from a distribution that depends on an unknown parameter $\theta \in \mathbb{R}$. All probabilities and expectations in this subsection are taken under the distribution indexed by the true value θ, which is a fixed (non-random) number.

Definition 4.1 (Estimator, bias, and mean squared error)**.** An *estimator* of θ is a function $\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n)$ of the data alone (it may not depend on θ), used as a guess for θ. Its *bias* is

$$\mathrm{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta,$$

and the estimator is called *unbiased* when $\mathrm{Bias}(\hat{\theta}) = 0$ for every value of θ. Its *mean squared error* is

$$\mathrm{MSE}(\hat{\theta}) = \mathbb{E}\big[(\hat{\theta} - \theta)^2\big].$$

Theorem 4.2 (Bias–variance decomposition of MSE)**.** *For any estimator $\hat{\theta}$ of θ with $\mathbb{E}[\hat{\theta}^2] < \infty$,*

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Bias}(\hat{\theta})^2 + \mathrm{Var}(\hat{\theta}). \tag{4.1}$$

Proof. We expand the square after adding and subtracting the mean of the estimator. Write $m = \mathbb{E}[\hat{\theta}]$, so that

$$(\hat{\theta} - \theta)^2 = \big((\hat{\theta} - m) + (m - \theta)\big)^2 = (\hat{\theta} - m)^2 + 2(\hat{\theta} - m)(m - \theta) + (m - \theta)^2.$$

Now take expectations term by term (linearity). The first term gives $\mathbb{E}[(\hat{\theta}-m)^2] = \mathrm{Var}(\hat{\theta})$ by the definition of variance. In the cross term, the factor $m-\theta$ is a deterministic number, so

$$\mathbb{E}[2(\hat{\theta}-m)(m-\theta)] = 2(m-\theta)\,\mathbb{E}[\hat{\theta}-m] = 2(m-\theta)\cdot 0 = 0.$$

The last term is the constant $(m-\theta)^2 = \mathrm{Bias}(\hat{\theta})^2$. Summing the three terms yields (4.1). □

Example 4.3 (Shrinkage can dominate on MSE)**.** Let $X_1, \ldots, X_{25}$ be iid with mean $\mu = 1$ and variance $\sigma^2 = 4$; here the parameter is $\theta = \mu$. Compare two estimators of μ.

Unbiased estimator. The sample mean $\bar{X} = \frac{1}{25}\sum_{i=1}^{25} X_i$ has $\mathbb{E}[\bar{X}] = 1$ (bias 0) and $\mathrm{Var}(\bar{X}) = \sigma^2/n = 4/25 = 0.16$. By (4.1), $\mathrm{MSE}(\bar{X}) = 0 + 0.16 = 0.16$.

Shrunk estimator. Take $\tilde{\mu} = 0.9\,\bar{X}$. Then $\mathbb{E}[\tilde{\mu}] = 0.9$, so the bias is $0.9-1 = -0.1$ and $\mathrm{Bias}^2 = 0.01$; the variance is $0.9^2 \times 0.16 = 0.1296$. By (4.1),

$$\mathrm{MSE}(\tilde{\mu}) = 0.01 + 0.1296 = 0.1396 < 0.16 = \mathrm{MSE}(\bar{X}).$$

The deliberately biased estimator wins: shrinking toward zero costs 0.01 in squared bias but saves 0.0304 in variance. The comparison depends on the true parameter — if instead $\mu = 4$, the squared bias becomes $(0.4)^2 = 0.16$ and $\mathrm{MSE}(\tilde{\mu}) = 0.16 + 0.1296 = 0.2896 > 0.16$, so shrinkage helps when the true parameter is small relative to the sampling noise.

Intuition. Unbiasedness is a constraint, not a virtue in itself. An unbiased estimator is forced to track the data wherever it goes, and pays for that fidelity in variance. Trading a little bias for a large variance reduction lowers total error, and Example 4.3 shows the exchange rate explicitly. Ridge regression and shrinkage estimators institutionalise exactly this trade and can dominate unbiased alternatives on MSE.

4.2 Bessel's Correction

The most common variance estimator divides by $n-1$ rather than n, and interviewers routinely ask why. The answer is a two-line expectation computation resting on one algebraic identity, which we isolate as a lemma because it recurs across statistics (it is the finite-sample ancestor of the analysis-of-variance decomposition).

Setup. Let $X_1, \ldots, X_n$ be iid random variables with unknown mean μ and unknown variance $\sigma^2 < \infty$, and let $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ denote the sample mean. Following universal convention we write s^2 (rather than a hatted symbol) for the Bessel-corrected sample variance defined in Theorem 4.5 below, and $\hat{\sigma}^2_{\mathrm{ML}}$ for the divide-by-n estimator.

Lemma 4.4 (Decomposition around the sample mean)**.** *For any realisation of the data,*

$$\sum_{i=1}^{n}(X_i-\mu)^2 = \sum_{i=1}^{n}(X_i-\bar{X})^2 + n(\bar{X}-\mu)^2. \tag{4.2}$$

Proof. We add and subtract the sample mean inside each squared term and show that the cross term vanishes. Writing $X_i - \mu = (X_i - \bar{X}) + (\bar{X}-\mu)$ and expanding the square,

$$\sum_{i=1}^{n}(X_i-\mu)^2 = \sum_{i=1}^{n}(X_i-\bar{X})^2 + 2(\bar{X}-\mu)\sum_{i=1}^{n}(X_i-\bar{X}) + n(\bar{X}-\mu)^2,$$

where the factor $\bar{X}-\mu$ came out of the middle sum because it does not depend on i. The middle sum is

$$\sum_{i=1}^{n}(X_i-\bar{X}) = \sum_{i=1}^{n} X_i - n\bar{X} = n\bar{X} - n\bar{X} = 0$$

by the definition of $\bar{X}$, so the cross term vanishes and (4.2) follows. □

Theorem 4.5 (Bessel's correction)**.** *Under the setup above,*

$$\mathbb{E}\Big[\sum_{i=1}^{n}(X_i - \bar{X})^2\Big] = (n-1)\,\sigma^2,$$

and therefore the sample variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

is an unbiased estimator of σ^2.

Proof. We take expectations on both sides of the identity (4.2) and evaluate each term from the iid assumptions. On the left, each summand has $\mathbb{E}[(X_i - \mu)^2] = \sigma^2$ by the definition of variance, so the left side has expectation $n\sigma^2$. For the final term on the right, the sample mean satisfies $\mathbb{E}[\bar{X}] = \mu$ (linearity) and, by independence,

$$\operatorname{Var}(\bar{X}) = \frac{1}{n^2}\sum_{i=1}^{n}\operatorname{Var}(X_i) = \frac{\sigma^2}{n},$$

so $\mathbb{E}[n(\bar{X} - \mu)^2] = n\operatorname{Var}(\bar{X}) = \sigma^2$. Substituting both evaluations into the expectation of (4.2) gives

$$n\sigma^2 = \mathbb{E}\Big[\sum_{i=1}^{n}(X_i - \bar{X})^2\Big] + \sigma^2,$$

and rearranging yields $\mathbb{E}\big[\sum_i(X_i - \bar{X})^2\big] = (n-1)\sigma^2$. Dividing by $n-1$ shows $\mathbb{E}[s^2] = \sigma^2$. □

(Problem P4.3 asks you to reproduce this proof from scratch.)

Example 4.6 (Bessel's correction on a small sample)**.** Take the sample $\{1, 2, 6\}$, so $n = 3$ and $\bar{X} = 3$. The deviations from $\bar{X}$ are $-2, -1, 3$, with squared deviations $4, 1, 9$ summing to 14. The Bessel-corrected estimate is $s^2 = 14/2 = 7$, while the divide-by-n estimate is $\hat{\sigma}^2_{\mathrm{ML}} = 14/3 \approx 4.67$.

The systematic direction of the gap is visible in expectation. Suppose the true variance is $\sigma^2 = 10$ and $n = 5$. Theorem 4.5 gives $\mathbb{E}\big[\sum_i(X_i - \bar{X})^2\big] = 4 \times 10 = 40$, so

$$\mathbb{E}[\hat{\sigma}^2_{\mathrm{ML}}] = \frac{40}{5} = 8 = \frac{n-1}{n}\,\sigma^2,$$

an underestimate of the true variance by 20%, whereas $\mathbb{E}[s^2] = 40/4 = 10$ exactly.

Intuition. The sample mean chases the data: because $\bar{X}$ is computed from the same sample it is being compared against, deviations from $\bar{X}$ are systematically smaller than deviations from the true mean μ — by exactly one degree of freedom. The identity (4.2) makes this precise: the discarded term $n(\bar{X} - \mu)^2$ carries expectation exactly σ^2, one summand's worth of variance.

Remark 4.7 (The MLE divisor and the regression generalisation)*.* The maximum-likelihood estimator of σ^2 for a normal sample is $\hat{\sigma}^2_{\mathrm{ML}}$, which divides by n and is therefore biased low by the factor $(n-1)/n$, as Example 4.6 quantifies. Regression generalises Bessel's correction: with p fitted parameters, the unbiased estimator of the residual variance is $\hat{\sigma}^2 = \mathrm{RSS}/(n-p)$, with RSS the residual sum of squares; each fitted parameter absorbs one degree of freedom, just as fitting $\bar{X}$ absorbed one here.

4.3 Sampling Distributions: the Central Limit Theorem and Student's t

Inference needs the distribution of an estimator, not just its first two moments. The central limit theorem supplies a universal normal approximation for averages; Student's t corrects that approximation when the variance itself must be estimated. Both are stated here without proof — what matters for practice is knowing exactly what they assert and, just as importantly, what they do not.

Setup. Let $X_1, X_2, \dots$ be iid random variables with mean μ and finite variance $\sigma^2 > 0$, and write $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ for the sample mean of the first n observations. As throughout, Φ denotes the standard normal CDF, and s^2 is the Bessel-corrected sample variance of Theorem 4.5.

Fact 4.8 (Central Limit Theorem)**.** Under the setup above, the standardised sample mean converges in distribution to a standard normal: for every real x,

$$P\left(\frac{\sqrt{n}\,(\bar{X}_n - \mu)}{\sigma} \le x\right) \longrightarrow \Phi(x) \qquad \text{as } n \to \infty.$$

A proof is beyond our scope.

Example 4.9 (Normal approximation to a fair-coin count)**.** Let K be the number of heads in 100 flips of a fair coin, so $K \sim \mathrm{Bin}(100, 1/2)$ is the sum of 100 iid Bernoulli variables, each with mean $1/2$ and variance $1/4$. The count has mean $100 \times 1/2 = 50$ and variance $100 \times 1/4 = 25$, so Fact 4.8 licenses the approximation

$$K \approx \mathcal{N}(50,\, 25), \qquad \text{standard deviation 5.}$$

This approximation is the engine of the hypothesis test in Example 4.13 below.

Remark 4.10 (What the CLT does not say)*.* Fact 4.8 is a limit statement about the centre of the distribution, and three misreadings of it are common enough to deserve explicit warnings.

(i) *Nothing about a fixed small n.* The theorem asserts a limit; it attaches no accuracy guarantee to any particular finite sample size.

(ii) *Nothing about the far tails at moderate n.* Fat-tailed summands converge slowly in the tails — exactly where risk lives. A normal approximation can be excellent near the mean and badly wrong at the 4-sigma quantile simultaneously.

(iii) *Nothing for dependent or infinite-variance data.* Both hypotheses (independence, finite variance) are load-bearing, and financial returns violate both in stress: dependence spikes and tails thicken precisely when the approximation is being leaned on hardest.

Fact 4.11 (Student's t and the estimated standard deviation (Student, 1908))**.** Let $X_1, \dots, X_n$ be iid $\mathcal{N}(\mu, \sigma^2)$. If the true σ were known, the statistic $\sqrt{n}\,(\bar{X}_n - \mu)/\sigma$ would be exactly standard normal. Replacing σ by its estimate s changes the distribution: the statistic

$$T = \frac{\sqrt{n}\,(\bar{X}_n - \mu)}{s}$$

follows Student's t distribution with $n-1$ degrees of freedom, written t_{n-1}, whose tails are wider than the standard normal's. The widening reflects the extra uncertainty from estimating σ, matters materially for small n, and vanishes in the limit: t_{n-1} converges to $\mathcal{N}(0,1)$ as $n \to \infty$. A proof is beyond our scope.

4.4 Hypothesis Testing and Confidence Intervals

A hypothesis test is not an inspiration but a checklist: fix the null, fix the statistic, know the statistic's null distribution, and compute how surprising the data would be if the null were true. This subsection states the checklist as a formal procedure, works it end to end on a coin, and proves the duality that makes confidence intervals and tests two views of the same object.

Setup. The data are $X_1, \dots, X_n$; a *null hypothesis* H_0 is a claim that fully determines the distribution of a chosen statistic; and $\alpha \in (0,1)$ is a *significance level* fixed before seeing the data (conventionally $\alpha = 0.05$). To avoid a symbol collision we reserve the letter p for p-values in this subsection and write θ for unknown parameters.

Definition 4.12 (Hypothesis-testing procedure)**.** A hypothesis test at significance level α consists of the following steps.

(i) State the null hypothesis H_0.
(ii) Choose a test statistic T, a function of the data.
(iii) Derive the distribution of T under H_0.
(iv) Compute the *p-value*: the probability, under H_0, of a value of T at least as extreme as the one observed. For a two-sided test whose null distribution is symmetric about zero, with observed value t_{obs}, this is $p = P_{H_0}(|T| \geq |t_{\text{obs}}|)$.
(v) Reject H_0 if $p \leq \alpha$; otherwise do not reject.

Equivalently, step (v) rejects exactly when T falls in a *rejection region* chosen to have probability α under H_0 (Figure 6).

Example 4.13 (Sixty heads in one hundred flips)**.** A coin is flipped 100 times and lands heads 60 times. Is it fair? We run Definition 4.12 step by step at $\alpha = 0.05$.

(i) The null hypothesis is that the coin is fair: the heads count satisfies $K \sim \text{Bin}(100, 1/2)$.
(ii) The statistic is the standardised count $z = (K - 50)/5$.
(iii) By Example 4.9, under H_0 the count is approximately $\mathcal{N}(50, 25)$, so z is approximately standard normal.
(iv) The observed value is $z = (60 - 50)/5 = 2$, and the two-sided p-value is

$$p = 2(1 - \Phi(2)) = 2 \times 0.02275 \approx 0.045.$$

(v) Since $0.045 < 0.05$, reject fairness at the 5% level — barely. Equivalently, the observed $z = 2$ lies just beyond the critical value 1.96, inside the shaded rejection region of Figure 6.

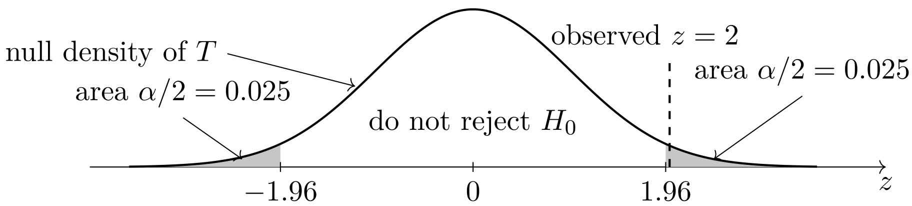


Figure 6: Two-sided rejection region at level $\alpha = 0.05$ under the null density of the test statistic. The shaded tails each carry probability $\alpha/2 = 0.025$ and begin at the critical values ± 1.96. The observed statistic $z = 2$ of Example 4.13 falls just inside the right tail, matching the two-sided p-value of about $0.045 < 0.05$: a rejection, but a marginal one.

Step (iii) of Definition 4.12 is where Fact 4.11 earns its place: when the statistic standardises by an *estimated* rather than known standard deviation, its null distribution is t_{n-1} rather than normal — the z-test becomes a t-test, the critical values widen, and the difference matters for small n.

Proposition 4.14 (Confidence-interval–test duality)**.** *Let $\{P_\theta : \theta \in \Theta\}$ be a family of distributions for the data X, and suppose that for each $\theta_0 \in \Theta$ there is a test of $H_0 : \theta = \theta_0$ with acceptance region $A(\theta_0)$ (the set of data values for which the test does not reject) of level α, meaning $P_{\theta_0}(X \notin A(\theta_0)) \leq \alpha$. Define*

$$C(X) = \{\theta_0 \in \Theta : X \in A(\theta_0)\},$$

the set of null values the data do not reject. Then $C(X)$ is a $(1 - \alpha)$-confidence set:

$$P_\theta(\theta \in C(X)) \geq 1 - \alpha \qquad \textit{for every } \theta \in \Theta.$$

Conversely, if $C(X)$ is any $(1 - \alpha)$-confidence set, the test that rejects $H_0 : \theta = \theta_0$ exactly when $\theta_0 \notin C(X)$ has level α.

Proof. The proof is a direct verification from the definitions of level and coverage. For the first claim, fix $\theta \in \Theta$. By the construction of C, the event $\{\theta \in C(X)\}$ is exactly the event $\{X \in A(\theta)\}$, so

$$P_\theta(\theta \in C(X)) = P_\theta(X \in A(\theta)) \geq 1 - \alpha,$$

where the inequality is the level condition applied at $\theta_0 = \theta$. For the converse, fix θ_0 and compute the type-I error of the proposed test:

$$P_{\theta_0}(\text{reject}) = P_{\theta_0}(\theta_0 \notin C(X)) = 1 - P_{\theta_0}(\theta_0 \in C(X)) \leq 1 - (1-\alpha) = \alpha,$$

where the inequality is the coverage guarantee applied at $\theta = \theta_0$. □

In the standard configuration $\alpha = 0.05$, Proposition 4.14 says: *the* 95% *confidence interval is exactly the set of null values not rejected at the* 5% *level.*

Example 4.15 (Confidence interval for the coin's bias)**.** Continue Example 4.13 and write θ for the coin's heads-probability, estimated by $\hat{\theta} = 60/100 = 0.6$. The Wald standard error is

$$\mathrm{SE}(\hat{\theta}) = \sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}} = \sqrt{\frac{0.6 \times 0.4}{100}} \approx 0.049,$$

so the 95% Wald interval is

$$0.6 \pm 1.96 \times 0.049 = [0.504,\ 0.696].$$

The fair value $\theta_0 = 0.5$ lies just outside the interval, which by Proposition 4.14 is the same statement as rejection at the 5% level — and indeed the Wald test gives $z = 0.1/0.049 \approx 2.04$ with two-sided $p \approx 0.041$, in agreement with the marginal rejection ($p \approx 0.045$) found in Example 4.13. The small numerical gap arises because Example 4.13 standardised by the null standard error $\sqrt{0.5 \times 0.5/100} = 0.05$ while the Wald interval uses 0.049; the duality is exact when the interval inverts the same family of tests, and here both variants reject, both barely.

4.5 The Sharpe-Ratio Standard Error (the Humility Formula)

A Sharpe ratio computed from data is an estimate, and like every estimate it deserves a standard error. The formula below is short, its consequences are brutal, and quoting it accurately is one of the fastest ways to demonstrate statistical maturity when discussing backtests.

Setup. Let $r_1, \ldots, r_n$ be a strategy's returns, assumed iid at a fixed observation frequency (say daily), with mean μ and standard deviation $\sigma > 0$. The (per-period) Sharpe ratio is $\mathrm{SR} = \mu/\sigma$, estimated by $\widehat{\mathrm{SR}} = \hat{\mu}/\hat{\sigma}$ with the sample mean and sample standard deviation; annualised quantities multiply the daily ones by $\sqrt{252}$.

Fact 4.16 (Standard error of the Sharpe ratio (Lo, 2002))**.** Under the iid setup above, assuming additionally that the returns are normally distributed, the estimated Sharpe ratio has approximate standard error

$$\mathrm{SE}\left(\widehat{\mathrm{SR}}\right) \approx \sqrt{\frac{1 + \mathrm{SR}^2/2}{n}}, \tag{4.3}$$

with SR and n both measured at the observation frequency. The result follows from a delta-method computation; a proof is beyond our scope.

Example 4.17 (One year of daily data)**.** Take one year of daily returns, $n = 252$, from a strategy whose true annual Sharpe ratio is 2; the daily Sharpe ratio is $2/\sqrt{252} \approx 0.126$. By (4.3), the daily-frequency standard error is

$$\mathrm{SE}\left(\widehat{\mathrm{SR}}\right) \approx \sqrt{\frac{1 + 0.126^2/2}{252}} = \sqrt{\frac{1.0079}{252}} \approx 0.0632,$$

and annualising (multiplying by $\sqrt{252} \approx 15.87$) gives an annualised standard error of $0.0632 \times 15.87 \approx 1.0$. *One year of daily data measures an annualised Sharpe ratio to about* ± 1.

Trading application. Reporting a Sharpe ratio of 2.3 versus 2.1 off a year of daily returns is reporting noise: the difference of 0.2 is a fifth of one standard error. Equation (4.3) is the humility check to run before believing any backtest headline, comparing two strategies on short samples, or reallocating between books on the strength of a single year.

Interview trap. "Your signal's t-statistic is 2.5 over 1000 days — do you trade it?" The required first answer is *multiple testing*: if 100 signals were tried, the expected maximum $|t|$ under the null is well above 2.5, so a best-of-many 2.5 is unremarkable. The correction is Bonferroni-style thinking or the Deflated Sharpe Ratio. Only after that come the follow-ups: look-ahead bias and leakage, overlapping observations (which shrink the effective n), heteroskedasticity, and finally costs and capacity.

Practice: problems P4.1–P4.12 in §20, solutions in §21.

5 Regression and Regularisation

Linear regression is the workhorse of quantitative research: it prices hedges, evaluates signals, and underlies almost every factor model. This section develops ordinary least squares and its projection geometry, the Gauss–Markov optimality theorem and what happens when its assumptions fail, the two classical regularisers (ridge and lasso), the inverse-regression identity, weighted least squares, and a dart-throwing problem that ties squared and absolute loss to the mean and the median.

5.1 Ordinary Least Squares

This subsection derives the least-squares estimator from first principles, establishes its interpretation as an orthogonal projection, and specialises it to the one-regressor formulas that every interview expects on demand.

Setup. Throughout this subsection, the data consist of a response vector $y \in \mathbb{R}^n$ and a design matrix $X \in \mathbb{R}^{n\times p}$ of full column rank $p \le n$, treated as fixed. The working model is

$$y = X\beta + \varepsilon, \qquad \mathbb{E}[\varepsilon] = 0, \qquad \mathrm{Var}(\varepsilon) = \sigma^2 I,$$

with $\beta \in \mathbb{R}^p$ unknown. The least-squares criterion is $L(\beta) = \|y - X\beta\|^2$, and $\mathrm{col}(X) \subset \mathbb{R}^n$ denotes the span of the columns of X.

Theorem 5.1 (Normal equations and the OLS estimator)**.** *The criterion $L(\beta) = \|y - X\beta\|^2$ has a unique minimiser, characterised by the normal equations*

$$X^\top X\beta = X^\top y, \tag{5.1}$$

and given explicitly by

$$\hat\beta_{\mathrm{OLS}} = (X^\top X)^{-1}X^\top y. \tag{5.2}$$

Proof. We compute the gradient and Hessian of L and use strict convexity. Expanding the squared norm,

$$L(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X\beta,$$

so the gradient is $\nabla_\beta L = -2X^\top y + 2X^\top X\beta$ and the Hessian is $\nabla^2_\beta L = 2X^\top X$. For any $u \neq 0$ we have $u^\top X^\top Xu = \|Xu\|^2 > 0$, because full column rank makes $Xu \neq 0$; hence $X^\top X$ is positive definite, so it is invertible and L is strictly convex. Setting the gradient to zero yields (5.1), whose unique solution is (5.2); by strict convexity this stationary point is the unique global minimiser. □

Theorem 5.2 (Projection interpretation and hat-matrix properties)**.** *Define the hat matrix $H = X(X^\top X)^{-1}X^\top$, the fitted vector $\hat{y} = X\hat{\beta}_{\mathrm{OLS}} = Hy$, and the residual vector $\hat{\varepsilon} = y - \hat{y}$. Then:*

(i) $X^\top\hat{\varepsilon} = 0$*: the residual is orthogonal to every column of X;*

(ii) $H^\top = H$ *and* $H^2 = H$*: the hat matrix is a symmetric idempotent, i.e. an orthogonal projector;*

(iii) $\mathrm{tr}(H) = p$*;*

(iv) $\hat{y}$ *is the orthogonal projection of y onto* $\mathrm{col}(X)$*: for every $v \in \mathrm{col}(X)$,*

$$\|y - v\|^2 = \|y - \hat{y}\|^2 + \|\hat{y} - v\|^2 \;\geq\; \|y - \hat{y}\|^2.$$

Proof. We verify each claim directly from the closed form (5.2), using the cyclic property of the trace for (iii).

(i) Substituting the definitions, $X^\top\hat{\varepsilon} = X^\top y - X^\top X(X^\top X)^{-1}X^\top y = X^\top y - X^\top y = 0$.

(ii) Symmetry follows by transposing: $H^\top = X\big((X^\top X)^{-1}\big)^\top X^\top = H$, because $X^\top X$ is symmetric and so is its inverse. Idempotence follows by cancellation:

$$H^2 = X(X^\top X)^{-1}\,(X^\top X)\,(X^\top X)^{-1}X^\top = H.$$

(iii) By the cyclic property of the trace, $\mathrm{tr}(H) = \mathrm{tr}\big[(X^\top X)^{-1}X^\top X\big] = \mathrm{tr}(I_p) = p$.

(iv) The fitted vector satisfies $\hat{y} = X\hat{\beta}_{\mathrm{OLS}} \in \mathrm{col}(X)$. Any $v \in \mathrm{col}(X)$ can be written $v = Xb$, and then $\hat{\varepsilon}^\top(\hat{y} - v) = \hat{\varepsilon}^\top X(\hat{\beta}_{\mathrm{OLS}} - b) = 0$ by (i). Hence the decomposition $y - v = \hat{\varepsilon} + (\hat{y} - v)$ is orthogonal, and the Pythagorean identity gives the display. ☐

Figure 7 shows the geometry: the fitted vector is the foot of the perpendicular dropped from y onto the plane $\mathrm{col}(X)$.

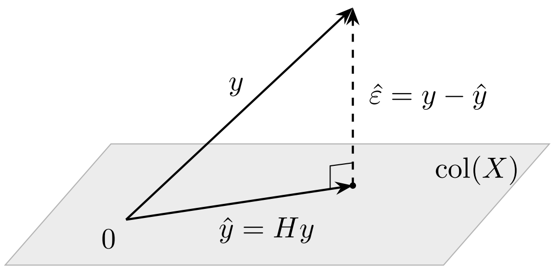


Figure 7: Ordinary least squares as orthogonal projection. The fitted vector $\hat{y} = Hy$ is the point of $\mathrm{col}(X)$ closest to y; the residual $\hat{\varepsilon}$ is perpendicular to every vector in $\mathrm{col}(X)$ (Theorem 5.2).

Intuition. Residuals are orthogonal to every regressor — that one identity generates most regression facts: the Pythagorean decomposition of sums of squares, the vanishing of cross terms in variance calculations, and the proof of Theorem 5.6. The trace identity $\mathrm{tr}(H) = p$ is read as "degrees of freedom used": fitting p coefficients consumes p dimensions of the data, which is why residual variance is estimated with divisor $n - p$.

The one-regressor specialisation below is worth having cold: interviews expect it instantly and build follow-ups on it.

Setup (univariate case). Suppose the regression has an intercept and a single regressor: observations (x_i, y_i) for $i = 1, \dots, n$, model $y_i = \alpha + \beta x_i + \varepsilon_i$. Write $\bar{x}, \bar{y}$ for the sample means, and

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2, \qquad S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \qquad S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2,$$

assumed positive. The sample standard deviations are $\hat{\sigma}_x = \sqrt{S_{xx}/n}$ and $\hat{\sigma}_y = \sqrt{S_{yy}/n}$, the sample correlation is $\hat{\rho} = S_{xy}/\sqrt{S_{xx}S_{yy}}$, and the coefficient of determination is $R^2 = 1 - \mathrm{RSS}/\mathrm{TSS}$, where $\mathrm{RSS} = \sum_i \hat{\varepsilon}_i^2$ and $\mathrm{TSS} = S_{yy}$.

Corollary 5.3 (Univariate slope and R-squared)**.** *In the univariate regression with intercept,*

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} = \hat{\rho}\,\frac{\hat{\sigma}_y}{\hat{\sigma}_x}, \qquad \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}, \qquad R^2 = \hat{\rho}^{\,2}. \tag{5.3}$$

Equivalently, the slope is the ratio of sample covariance to sample variance of the regressor.

Proof. We specialise the normal equations (5.1) to the design whose columns are the all-ones vector and x. The first normal equation reads $\sum_i (y_i - \alpha - \beta x_i) = 0$, which gives $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$. Substituting this into the second normal equation $\sum_i x_i(y_i - \alpha - \beta x_i) = 0$ yields

$$\sum_i x_i\big[(y_i - \bar{y}) - \beta(x_i - \bar{x})\big] = 0.$$

Because $\sum_i (y_i - \bar{y}) = \sum_i (x_i - \bar{x}) = 0$, replacing the factor x_i by $x_i - \bar{x}$ changes nothing, and the equation becomes $S_{xy} = \beta S_{xx}$, i.e. $\hat{\beta} = S_{xy}/S_{xx}$. Dividing numerator and denominator by n and factoring out standard deviations gives $\hat{\beta} = \hat{\rho}\,\hat{\sigma}_y/\hat{\sigma}_x$.

For R^2, the fitted deviations are $\hat{y}_i - \bar{y} = \hat{\beta}(x_i - \bar{x})$, so the explained sum of squares is $\mathrm{ESS} = \hat{\beta}^2 S_{xx} = S_{xy}^2/S_{xx}$. By Theorem 5.2(i) the residual vector is orthogonal to both columns of the design, hence to the fitted deviations, so the cross term in $\mathrm{TSS} = \sum_i \big[\hat{\varepsilon}_i + (\hat{y}_i - \bar{y})\big]^2$ vanishes and $\mathrm{TSS} = \mathrm{RSS} + \mathrm{ESS}$. Therefore

$$R^2 = \frac{\mathrm{ESS}}{\mathrm{TSS}} = \frac{S_{xy}^2}{S_{xx}S_{yy}} = \hat{\rho}^{\,2}. \qquad \square$$

(Problem P5.2 asks you to reproduce the derivations of this subsection — estimator, orthogonality, hat-matrix properties, and the univariate reduction — from scratch.)

Example 5.4 (A three-point regression, fully worked)**.** Take $n = 3$ observations $x = (1, 2, 3)$, $y = (1, 2, 2)$, and fit $y_i = \alpha + \beta x_i + \varepsilon_i$. The design and its Gram matrix are

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \qquad X^\top X = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}, \qquad (X^\top X)^{-1} = \frac{1}{6}\begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}, \qquad X^\top y = \begin{pmatrix} 5 \\ 11 \end{pmatrix},$$

using $\det(X^\top X) = 42 - 36 = 6$. Hence

$$\hat{\beta}_{\mathrm{OLS}} = \frac{1}{6}\begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}\begin{pmatrix} 5 \\ 11 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 2/3 \\ 1/2 \end{pmatrix}:$$

intercept $2/3$, slope $1/2$. The univariate formula (5.3) confirms this: $\bar{x} = 2$, $\bar{y} = 5/3$, $S_{xy} = 1$, $S_{xx} = 2$, so $\hat{\beta} = 1/2$ and $\hat{\alpha} = 5/3 - 2/2 = 2/3$. The fitted values are $(7/6,\ 5/3,\ 13/6)$ and the residuals $\hat{\varepsilon} = (-1/6,\ 1/3,\ -1/6)$. Orthogonality checks out: $\sum_i \hat{\varepsilon}_i = 0$ and $\sum_i x_i\hat{\varepsilon}_i = -1/6 + 4/6 - 3/6 = 0$. The leverages (diagonal of H) are $h_{11} = 5/6$, $h_{22} = 1/3$, $h_{33} = 5/6$, summing to $2 = p$ as Theorem 5.2(iii) requires. Finally $S_{yy} = 2/3$ and $\mathrm{RSS} = 1/6$, so $R^2 = 1 - (1/6)/(2/3) = 3/4$, matching $\hat{\rho}^{\,2} = S_{xy}^2/(S_{xx}S_{yy}) = 1/(2 \cdot 2/3) = 3/4$.

5.2 The Gauss–Markov Theorem

Having derived the OLS estimator, we now ask in what sense it is optimal. The Gauss–Markov theorem gives the classical answer: under four assumptions — none of which is normality — OLS has the smallest variance among all linear unbiased estimators. The proof is short and appears in interviews verbatim; the follow-up questions about broken assumptions appear even more often.

Setup. As before, $y = X\beta + \varepsilon$ with $X \in \mathbb{R}^{n\times p}$. The four Gauss–Markov assumptions, conditional on X, are:

(GM1) *Linearity / correct specification:* the model $y = X\beta + \varepsilon$ holds.
(GM2) *Full column rank:* $\operatorname{rank}(X) = p$.
(GM3) *Strict exogeneity:* $\mathbb{E}[\varepsilon \mid X] = 0$.
(GM4) *Spherical errors:* $\operatorname{Var}(\varepsilon \mid X) = \sigma^2 I$ — homoskedastic *and* uncorrelated.

Normality of the errors is *not* assumed. All expectations and variances below are conditional on X; we suppress the conditioning in the notation.

Definition 5.5 (Linear estimator and BLUE)**.** An estimator $\tilde\beta$ of β is *linear* if $\tilde\beta = Cy$ for some matrix $C \in \mathbb{R}^{p\times n}$ that may depend on X but not on y, and *unbiased* if $\mathbb{E}[\tilde\beta] = \beta$ for every $\beta \in \mathbb{R}^p$. A linear unbiased estimator $\hat\beta$ is *BLUE* (best linear unbiased estimator) if, for every competing linear unbiased $\tilde\beta$, the matrix $\operatorname{Var}(\tilde\beta) - \operatorname{Var}(\hat\beta)$ is positive semidefinite — equivalently, $\operatorname{Var}(\ell^\top\tilde\beta) \ge \operatorname{Var}(\ell^\top\hat\beta)$ for every linear combination $\ell \in \mathbb{R}^p$.

Theorem 5.6 (Gauss–Markov)**.** *Under (GM1)–(GM4), the OLS estimator $\hat\beta_{\mathrm{OLS}}$ is linear and unbiased, and it is BLUE: every linear unbiased estimator $\tilde\beta = Cy$ satisfies*

$$\operatorname{Var}(\tilde\beta) - \operatorname{Var}(\hat\beta_{\mathrm{OLS}}) = \sigma^2\Delta\Delta^\top \succeq 0, \qquad \textit{where } \Delta = C - (X^\top X)^{-1}X^\top,$$

with equality if and only if $\tilde\beta = \hat\beta_{\mathrm{OLS}}$.

Proof. We decompose an arbitrary competitor as OLS plus a slack matrix that is orthogonal to the design, and show the slack can only add variance.

First, OLS is linear with $C_0 = (X^\top X)^{-1}X^\top$, and it is unbiased because $C_0X = I$ and (GM3) give $\mathbb{E}[\hat\beta_{\mathrm{OLS}}] = C_0X\beta + C_0\,\mathbb{E}[\varepsilon] = \beta$. Its variance is

$$\operatorname{Var}(\hat\beta_{\mathrm{OLS}}) = C_0\ \operatorname{Var}(\varepsilon)\,C_0^\top = \sigma^2 C_0C_0^\top = \sigma^2(X^\top X)^{-1},$$

using (GM4) and $C_0C_0^\top = (X^\top X)^{-1}X^\top X(X^\top X)^{-1} = (X^\top X)^{-1}$.

Now let $\tilde\beta = Cy$ be any linear estimator that is unbiased for every β. Then $\mathbb{E}[\tilde\beta] = CX\beta$ must equal β for all $\beta \in \mathbb{R}^p$, which forces $CX = I$. Write $C = C_0 + \Delta$; subtracting $C_0X = I$ from $CX = I$ gives

$$\Delta X = 0.$$

By (GM4), $\operatorname{Var}(\tilde\beta) = \sigma^2 CC^\top$, and expanding,

$$CC^\top = (C_0 + \Delta)(C_0 + \Delta)^\top = C_0C_0^\top + C_0\Delta^\top + \Delta C_0^\top + \Delta\Delta^\top.$$

The cross terms vanish because they contain ΔX: indeed $\Delta C_0^\top = \Delta X(X^\top X)^{-1} = 0$, and $C_0\Delta^\top$ is its transpose. Hence

$$\operatorname{Var}(\tilde\beta) = \sigma^2\big[(X^\top X)^{-1} + \Delta\Delta^\top\big] = \operatorname{Var}(\hat\beta_{\mathrm{OLS}}) + \sigma^2\Delta\Delta^\top.$$

The matrix $\Delta\Delta^\top$ is positive semidefinite, since $u^\top\Delta\Delta^\top u = \|\Delta^\top u\|^2 \ge 0$ for every u. Equality of the variance matrices holds if and only if $\Delta\Delta^\top = 0$, i.e. $\Delta = 0$, i.e. $\tilde\beta = \hat\beta_{\mathrm{OLS}}$. The scalar statement follows by sandwiching: $\ell^\top\big[\operatorname{Var}(\tilde\beta) - \operatorname{Var}(\hat\beta_{\mathrm{OLS}})\big]\ell \ge 0$ for every ℓ. □

(Problem P5.5 asks you to reproduce this proof from scratch.)

Intuition. One idea carries the whole proof: *any competitor equals OLS plus a piece orthogonal to the design, and that piece only adds variance.* Unbiasedness pins down the estimator's action on $\operatorname{col}(X)$ exactly; the only remaining freedom is slack in the orthogonal directions, which cannot help and strictly hurts unless it is zero.

Example 5.7 (Estimating a mean: which weights are best?)**.** The simplest instance of Theorem 5.6 takes $X = (1, 1, \ldots, 1)^\top$ and scalar $\beta = \mu$, so the model is $y_i = \mu + \varepsilon_i$ with uncorrelated errors of variance σ^2. Every linear estimator is a weighting $\tilde{\mu} = \sum_i w_i y_i$; unbiasedness for all μ forces $\sum_i w_i = 1$, and then $\mathrm{Var}(\tilde{\mu}) = \sigma^2 \sum_i w_i^2$. The OLS estimator here is the sample mean ($w_i = 1/n$). With $n = 4$ and $\sigma^2 = 1$: equal weights give variance $4 \times (1/4)^2 = 0.25$, while the unequal weighting $(0.4, 0.3, 0.2, 0.1)$ — still unbiased, since the weights sum to 1 — gives $0.16 + 0.09 + 0.04 + 0.01 = 0.30$, strictly worse. The Cauchy–Schwarz inequality shows equal weights are optimal in general: $n \sum_i w_i^2 \geq \big(\sum_i w_i\big)^2 = 1$, so $\mathrm{Var}(\tilde{\mu}) \geq \sigma^2/n$ with equality only at $w_i = 1/n$.

Remark 5.8 (What breaks, what it costs, what fixes it)*.* Interviews rarely stop at the theorem; the follow-up is which assumption a given desk failure violates and what to do. The summary table:

Broken	**Consequence**	**Fix**
exogeneity (omitted variable, leakage)	bias + inconsistency (fatal)	add variable, IV, redesign
heteroskedasticity / autocorrelation	inefficient + wrong SEs	GLS/WLS (then BLUE); robust SEs
near-multicollinearity	variance explodes	ridge, drop/combine features
non-normality	only exact t/F affected	CLT asymptotics, bootstrap

Two gradations matter. A broken (GM3) poisons the estimate itself and no amount of data cures it. Broken (GM4) leaves OLS unbiased and consistent but inefficient, with wrong standard errors — an inference problem, not an estimation catastrophe. Near-multicollinearity violates nothing formally (only exact rank deficiency violates (GM2)) but makes $(X^\top X)^{-1}$ huge, so coefficient variances explode. Non-normality breaks none of (GM1)–(GM4): only exact finite-sample t and F distributions are lost.

The most dangerous violation — exogeneity — has a formula every candidate should be able to derive on demand.

Setup. Suppose the true model involves two centred regressors, $y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$ with $\mathbb{E}[\varepsilon \mid x_1, x_2] = 0$, but the analyst regresses y on x_1 alone. Write $\widehat{\mathrm{Cov}}(x_1, x_2) = \frac{1}{n} \sum_i x_{1i} x_{2i}$ and $\widehat{\mathrm{Var}}(x_1) = \frac{1}{n} \sum_i x_{1i}^2 > 0$ for the sample moments of the fixed regressors.

Proposition 5.9 (Omitted-variable bias)**.** *The short-regression slope $\hat{\beta}_1 = \sum_i x_{1i} y_i / \sum_i x_{1i}^2$ satisfies*

$$\mathbb{E}[\hat{\beta}_1] = \beta_1 + \beta_2 \, \frac{\widehat{\mathrm{Cov}}(x_1, x_2)}{\widehat{\mathrm{Var}}(x_1)}. \tag{5.4}$$

The bias is the omitted coefficient times the regression slope of the omitted variable on the included one; it vanishes only if $\beta_2 = 0$ or the regressors are uncorrelated in sample.

Proof. We substitute the true model into the estimator and take expectations. Substituting $y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$,

$$\hat{\beta}_1 = \beta_1 + \beta_2 \, \frac{\sum_i x_{1i} x_{2i}}{\sum_i x_{1i}^2} + \frac{\sum_i x_{1i} \varepsilon_i}{\sum_i x_{1i}^2}.$$

Conditional on the regressors, the last term has expectation zero by exogeneity of the *true* model, and the middle ratio equals $\widehat{\mathrm{Cov}}(x_1, x_2)/\widehat{\mathrm{Var}}(x_1)$ after dividing through by n. Taking expectations gives (5.4). □

Example 5.10 (Phantom alpha, numerically)**.** Suppose the truth is $\beta_1 = 0.2$ (a weak genuine signal) and $\beta_2 = 1.5$ (a strong market effect), and in sample $\widehat{\mathrm{Cov}}(x_1, x_2)/\widehat{\mathrm{Var}}(x_1) = 0.4$. By (5.4), the short regression reports

$$\mathbb{E}[\hat{\beta}_1] = 0.2 + 1.5 \times 0.4 = 0.8$$

— four times the true coefficient. Three quarters of the reported "alpha" is the market loading in disguise, and collecting more data makes the estimate more precise around 0.8, not closer to 0.2.

Trading application. Omit market beta from a return regression while the signal under test correlates with the market, and the backtest rediscovers beta and calls it alpha — the classic way (5.4) bites in practice. The standard cure is residualisation: a systematic equity desk residualises returns against risk-model factors before evaluating signals, precisely so that whatever survives is orthogonal to the omitted-variable directions the risk model spans.

Interview trap. "You just proved OLS is best — so why does ridge regularly beat it out of sample?" Gauss–Markov crowns OLS best *among linear unbiased estimators.* Ridge is biased, so it competes outside that class, and the mean-squared-error decomposition MSE = $\text{bias}^2 + \text{variance}$ lets it win: a small bias can buy a large variance reduction (Proposition 5.14 makes this exact). Under normality, OLS is best among *all* unbiased estimators — it attains the Cramér–Rao bound and equals the MLE — but it still loses to biased competitors on MSE.

5.3 Ridge Regression

Ridge regression adds an L2 penalty to the least-squares criterion. This buys three things, each proved below: an estimator that exists for *every* design (the mechanical fix for multicollinearity), a transparent shrinkage action, and — the punchline the trap above promised — a strictly smaller mean-squared error than OLS for a well-chosen penalty.

Setup. The data are $y \in \mathbb{R}^n$ and $X \in \mathbb{R}^{n\times p}$; full column rank is *not* assumed in this subsection (that is the point). For a penalty level $\lambda > 0$, the ridge criterion is

$$L_\lambda(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|_2^2,$$

and we write $\hat\beta_\lambda$ for its minimiser. Because λ is reserved for the penalty throughout this section, the eigenvalues of the symmetric positive-semidefinite matrix $X^\top X$ are written $\gamma_1 \geq \gamma_2 \geq \cdots \geq \gamma_p \geq 0$ (renaming the document-wide λ_i).

Theorem 5.11 (Ridge estimator and guaranteed invertibility)**.** *For every $\lambda > 0$ and every design X — including rank-deficient designs and $p > n$ — the criterion L_λ is strictly convex with unique minimiser*

$$\hat\beta_\lambda = (X^\top X + \lambda I)^{-1}X^\top y, \tag{5.5}$$

the matrix $X^\top X + \lambda I$ being invertible because its eigenvalues are $\gamma_i + \lambda \geq \lambda > 0$.

Proof. We repeat the gradient–Hessian argument of Theorem 5.1 with the penalised criterion. The gradient is $\nabla_\beta L_\lambda = -2X^\top y + 2(X^\top X + \lambda I)\beta$ and the Hessian is $2(X^\top X + \lambda I)$. By the spectral theorem, the symmetric matrix $X^\top X$ has an orthonormal eigenbasis with eigenvalues $\gamma_i \geq 0$ (nonnegative because $u^\top X^\top X u = \|Xu\|^2 \geq 0$); adding λI shifts every eigenvalue to $\gamma_i + \lambda \geq \lambda > 0$. Hence the Hessian is positive definite regardless of the rank of X: the criterion is strictly convex, the matrix in (5.5) is invertible, and the unique stationary point (5.5) is the unique global minimiser. □

Proposition 5.12 (Orthonormal shrinkage)**.** *If the design is orthonormal, $X^\top X = I$, then*

$$\hat\beta_\lambda = \frac{\hat\beta_{\text{OLS}}}{1+\lambda}:$$

every coefficient is shrunk toward zero by the common factor $1/(1+\lambda)$, and no coefficient is set exactly to zero unless its OLS value is already zero.

Proof. We substitute $X^\top X = I$ into (5.5): the estimator becomes $\hat\beta_\lambda = (I+\lambda I)^{-1}X^\top y = (1+\lambda)^{-1}X^\top y$, and in the orthonormal case (5.2) reduces to $\hat\beta_{\text{OLS}} = X^\top y$. The factor $1/(1+\lambda)$ lies in $(0,1)$ for $\lambda > 0$ and multiplies each coordinate, so a coordinate vanishes exactly when its OLS value does. □

Example 5.13 (Shrinkage, numerically)**.** With an orthonormal design, an OLS coefficient of 0.9 and penalty $\lambda = 0.2$ give the ridge coefficient $0.9/1.2 = 0.75$. Doubling the penalty to $\lambda = 0.4$ gives $0.9/1.4 \approx 0.643$: more penalty, more shrinkage, but never an exact zero.

Proposition 5.14 (Some penalty strictly lowers MSE, orthonormal case)**.** *Assume the orthonormal design $X^\top X = I$ and errors with mean zero and variance $\sigma^2 I$, and let* $\mathrm{MSE}(\lambda) = \mathbb{E}\,\|\hat\beta_\lambda - \beta\|^2$. *Then*

$$\mathrm{MSE}(\lambda) = \frac{\lambda^2\|\beta\|^2 + p\sigma^2}{(1+\lambda)^2}, \tag{5.6}$$

so $\mathrm{MSE}(0) = p\sigma^2$ *recovers OLS and* $\mathrm{MSE}'(0) = -2p\sigma^2 < 0$*: some $\lambda > 0$ strictly beats OLS. If $\beta \neq 0$, the MSE is uniquely minimised at*

$$\lambda^* = \frac{p\sigma^2}{\|\beta\|^2}, \qquad \mathrm{MSE}(\lambda^*) < \mathrm{MSE}(0);$$

if $\beta = 0$, the MSE is strictly decreasing in λ, so any $\lambda > 0$ beats OLS.

Proof. We compute the bias–variance decomposition of $\hat\beta_\lambda$ and optimise over λ by elementary calculus. In the orthonormal case $\hat\beta_{\mathrm{OLS}} = X^\top y = \beta + X^\top\varepsilon$, with $\mathbb{E}[X^\top\varepsilon] = 0$ and $\mathrm{Var}(X^\top\varepsilon) = \sigma^2 X^\top X = \sigma^2 I$. By Proposition 5.12, $\hat\beta_\lambda = \hat\beta_{\mathrm{OLS}}/(1+\lambda)$, so

$$\mathbb{E}[\hat\beta_\lambda] - \beta = \frac{\beta}{1+\lambda} - \beta = -\frac{\lambda}{1+\lambda}\,\beta, \qquad \mathrm{Var}(\hat\beta_\lambda) = \frac{\sigma^2}{(1+\lambda)^2}\,I.$$

Summing squared bias and total variance,

$$\mathrm{MSE}(\lambda) = \left\|\tfrac{\lambda}{1+\lambda}\beta\right\|^2 + \operatorname{tr}\mathrm{Var}(\hat\beta_\lambda) = \frac{\lambda^2\|\beta\|^2 + p\sigma^2}{(1+\lambda)^2},$$

which is (5.6). Differentiating with the quotient rule and simplifying,

$$\mathrm{MSE}'(\lambda) = \frac{2(\lambda\|\beta\|^2 - p\sigma^2)}{(1+\lambda)^3}.$$

At $\lambda = 0$ this equals $-2p\sigma^2 < 0$, so the MSE initially decreases and some positive penalty strictly improves on OLS. If $\beta \neq 0$, the derivative is negative on $(0, \lambda^*)$ and positive on (λ^*, ∞) with $\lambda^* = p\sigma^2/\|\beta\|^2$, so λ^* is the unique minimiser and $\mathrm{MSE}(\lambda^*) < \mathrm{MSE}(0)$ because the MSE strictly decreases on $(0, \lambda^*]$. If $\beta = 0$, the derivative is negative for all $\lambda > 0$. □

Example 5.15 (Halving the MSE)**.** Take $p = 2$, $\beta = (1,1)^\top$, $\sigma^2 = 1$ in the orthonormal setting. Then OLS has $\mathrm{MSE}(0) = p\sigma^2 = 2$, the optimal penalty is $\lambda^* = 2/2 = 1$, and (5.6) gives

$$\mathrm{MSE}(1) = \frac{1\cdot 2 + 2}{(1+1)^2} = 1$$

— half the OLS error, split as squared bias $\frac14\|\beta\|^2 = 0.5$ plus variance $\frac14\,p\sigma^2 = 0.5$. The accepted small bias bought a variance reduction three times its size.

Fact 5.16 (Ridge improvement for general designs (Hoerl–Kennard, 1970))**.** The orthonormal assumption is a convenience, not the substance: for an arbitrary full-column-rank design and spherical errors, there always exists $\lambda > 0$ such that $\mathbb{E}\|\hat\beta_\lambda - \beta\|^2 < \mathbb{E}\|\hat\beta_{\mathrm{OLS}} - \beta\|^2$; the same one-sided derivative argument goes through eigenvalue by eigenvalue of $X^\top X$. A full proof is beyond our scope.

Remark 5.17 (Bayesian interpretation). Place the prior $\beta \sim \mathcal{N}(0, \tau^2 I)$ on the coefficients and take the Gaussian likelihood $y \mid \beta \sim \mathcal{N}(X\beta, \sigma^2 I)$. Minus twice the log-posterior is, up to constants, $\|y - X\beta\|^2/\sigma^2 + \|\beta\|^2/\tau^2$; multiplying through by σ^2 shows the MAP estimate is exactly ridge with $\lambda = \sigma^2/\tau^2$. A tight prior (small τ) means heavy shrinkage; an uninformative prior ($\tau \to \infty$) recovers OLS.

5.4 The Lasso

Replacing the squared-norm penalty by an absolute-value penalty changes the character of the solution entirely: instead of shrinking every coefficient a little, the lasso sets some coefficients exactly to zero. This subsection proves the soft-thresholding formula that makes the mechanism explicit and gives the geometric picture behind it.

Setup. For $\lambda > 0$ the lasso criterion is

$$L(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|_1, \qquad \|\beta\|_1 = \sum_{j=1}^{p} |\beta_j|.$$

The criterion is convex but not differentiable at coordinates equal to zero, and it has no closed-form minimiser in general. In the orthonormal case $X^\top X = I$, however, writing $z = X^\top y = \hat\beta_{\text{OLS}}$ and expanding the squared norm,

$$L(\beta) = \|y\|^2 + \sum_{j=1}^{p}\Big[\beta_j^2 - 2\beta_j z_j + \lambda|\beta_j|\Big] = \text{const} + \sum_{j=1}^{p}\Big[(\beta_j - z_j)^2 + \lambda|\beta_j|\Big],$$

so the problem separates into p one-dimensional problems. We write $(a)_+ = \max(a, 0)$ and $\operatorname{sign}(z) \in \{-1, 0, 1\}$ for the sign function.

Theorem 5.18 (Soft-thresholding, orthonormal design)**.** *For any $z \in \mathbb{R}$ and $\lambda > 0$, the function $\psi_z(b) = (b - z)^2 + \lambda|b|$ has the unique minimiser*

$$b^*(z) = \operatorname{sign}(z)\left(|z| - \tfrac{\lambda}{2}\right)_+. \tag{5.7}$$

Consequently, in the orthonormal case the lasso solution is coordinatewise soft-thresholding of OLS:

$$\hat\beta_j^{\text{lasso}} = \operatorname{sign}(\hat\beta_j^{\text{OLS}})\left(|\hat\beta_j^{\text{OLS}}| - \tfrac{\lambda}{2}\right)_+,$$

and every OLS coefficient with $|\hat\beta_j^{\text{OLS}}| \le \lambda/2$ is set exactly to zero — the lasso performs variable selection.

Proof. We argue by case analysis on the sign of the minimiser, using strict convexity to guarantee uniqueness. The function ψ_z is strictly convex, being the sum of the strictly convex quadratic $(b - z)^2$ and the convex $\lambda|b|$; hence it has exactly one minimiser, and any point satisfying a local optimality condition is that minimiser. Three mutually exclusive possibilities cover it.

Case $b^* > 0$. On $(0, \infty)$ the function is differentiable with $\psi_z'(b) = 2(b - z) + \lambda$, so a stationary point requires $b = z - \lambda/2$, which is positive precisely when $z > \lambda/2$.

Case $b^* < 0$. Symmetrically, $\psi_z'(b) = 2(b - z) - \lambda$ on $(-\infty, 0)$, giving $b = z + \lambda/2$, which is negative precisely when $z < -\lambda/2$.

Case $b^* = 0$. The one-sided derivatives at zero are $\psi_z'(0^+) = -2z + \lambda$ and $\psi_z'(0^-) = -2z - \lambda$. A convex function is minimised at 0 if and only if the left derivative is ≤ 0 and the right derivative is ≥ 0, i.e.

$$-2z - \lambda \le 0 \le -2z + \lambda \iff |z| \le \tfrac{\lambda}{2}.$$

The three conditions $z > \lambda/2$, $z < -\lambda/2$, $|z| \le \lambda/2$ partition $\mathbb{R}$, and in each case the minimiser matches (5.7). The coordinatewise claim follows because the separated criterion is minimised by minimising each coordinate's term. □

(Problem P5.8 asks you to reproduce this derivation from scratch.)

Example 5.19 (Soft-thresholding in action)**.** Take $\lambda = 0.8$, so the threshold is $\lambda/2 = 0.4$, and orthonormal-design OLS coefficients $(1.6,\ 0.5,\ -0.2)$. Formula (5.7) gives

$$1.6 \mapsto 1.6 - 0.4 = 1.2, \qquad 0.5 \mapsto 0.5 - 0.4 = 0.1, \qquad -0.2 \mapsto 0 \quad (|-0.2| \le 0.4).$$

The third coefficient is *exactly* zero: the variable is dropped from the model. Ridge at any penalty would have returned three nonzero coefficients (Proposition 5.12).

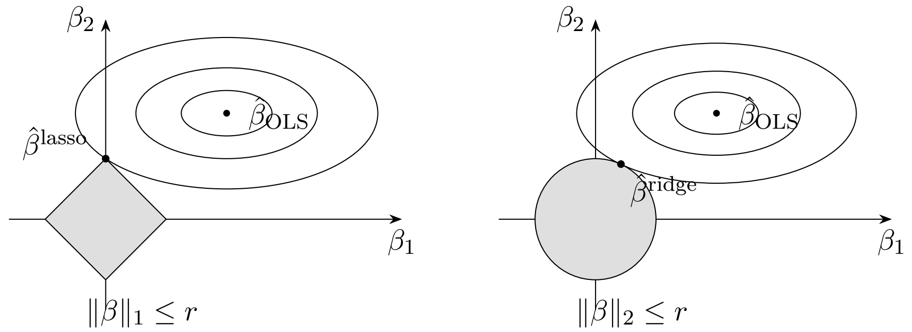


Figure 8: Constrained view of the two penalties for $p = 2$: minimise the quadratic loss (elliptical level sets centred at $\hat\beta_{\text{OLS}}$) subject to a norm budget r. Left: the L1 ball is a diamond with corners on the axes; the level sets typically first touch at a corner, where a coordinate is exactly zero. Right: the L2 ball is round, the contact point is generically off-axis, and both coordinates stay nonzero.

Intuition. Figure 8 is the standard geometric picture. The penalised problems are equivalent to minimising $\|y - X\beta\|^2$ subject to $\|\beta\|_1 \le r$ (lasso) or $\|\beta\|_2 \le r$ (ridge) for a budget r matched to λ. The elliptical level sets of the loss expand from $\hat\beta_{\text{OLS}}$ until they first touch the constraint set. The L1 diamond presents corners that sit exactly on the coordinate axes, and the expanding ellipse typically lands on a corner — exact zeros. The L2 ball has no corners, so the touching point generically has all coordinates nonzero: ridge shrinks, lasso selects.

Remark 5.20 (Bayesian view)*.* Just as ridge is the MAP estimate under a Gaussian prior (Remark 5.17), the lasso is the MAP estimate under independent Laplace (double-exponential) priors on the coefficients: minus the log of the Laplace density is proportional to $|\beta_j|$, producing the L1 penalty. The Laplace prior's sharp peak at zero is the Bayesian face of variable selection.

5.5 Inverse Regression, Beta and Hedging

Regressing Y on X and regressing X on Y do not give reciprocal slopes, and confusing the two is a classic interview trap with direct hedging consequences. The identity that governs the relationship follows in one line from Corollary 5.3.

Setup. Let the pair (X, Y) have standard deviations $\sigma_X, \sigma_Y > 0$ and correlation ρ (population moments, or sample moments used consistently). By Corollary 5.3, the slope of the regression of Y on X and the slope of the regression of X on Y are

$$B = \rho\,\frac{\sigma_Y}{\sigma_X}, \qquad D = \rho\,\frac{\sigma_X}{\sigma_Y}.$$

Proposition 5.21 (Inverse-regression identity)**.** *The two slopes satisfy*

$$BD = \rho^2 \le 1.$$

Consequently, when $\rho \ne 0$,

$$D = \frac{\rho^2}{B} \ne \frac{1}{B} \quad \textit{unless } |\rho| = 1:$$

the naive reciprocal $1/B$ overstates the true inverse slope by the factor $1/\rho^2$.

Proof. We multiply the two slope formulas: the standard-deviation ratios cancel, leaving $BD = \rho^2$, and $\rho^2 \le 1$ is the Cauchy–Schwarz inequality for correlations. When $\rho \neq 0$, both slopes are nonzero and division gives $D = \rho^2/B$; this equals $1/B$ precisely when $\rho^2 = 1$, i.e. when the variables are perfectly linearly related, in which case both regression lines coincide. □

Example 5.22 (The factor-of-four error)**.** Take $\rho = 0.5$ and $\sigma_X = \sigma_Y$. Then $B = D = 0.5$, while the naive inversion quotes $1/B = 2$ — overstating the true slope $D = 0.5$ by the factor $1/\rho^2 = 4$. The two correct regression lines are far from being mirror images of each other.

Intuition. Regressing Y on X minimises *vertical* distances to the line; regressing X on Y minimises *horizontal* ones. These are different projections, and each slope is attenuated toward zero by a factor ρ relative to the standard-deviation ratio: noise in the regressor always pulls the fitted slope toward flatness, in whichever direction the regression is run.

Trading application. Pair trading: the hedge ratio depends on which asset is regressed on which. Taking the slope from the convenient regression and inverting it systematically over- or under-hedges the book by the factor $1/\rho^2$; with a typical pair correlation of 0.7, that is a factor of two.

Remark 5.23 (Beta versus correlation)*.* The regression slope $\beta = \rho\,\sigma_Y/\sigma_X$ mixes two ingredients that should be kept mentally separate. Correlation is normalised: it measures direction and tightness of the linear relationship and is dimensionless. Beta additionally carries *relative volatility*: with $\rho \approx 1$ but Y twice as volatile as X, the beta is 2. Conversely a large beta can coexist with a weak correlation if σ_Y/σ_X is large — the relationship is loose, but when Y moves, it moves a lot.

Interview trap. "Which do you hedge with — beta or correlation?" The *beta*: hedging neutralises profit-and-loss magnitude, and magnitude lives in the ratio σ_Y/σ_X. Correlation tells you whether a hedge is worth doing at all (how much residual variance will remain); beta tells you how large to make it. Sizing a hedge by correlation alone ignores the volatility ratio and mis-sizes whenever the two assets have different volatilities.

5.6 Weighted Least Squares and Whitening

When error variances differ across observations, assumption (GM4) fails and OLS, while still unbiased, is no longer efficient. The remedy is to whiten: rescale each observation by its noise level and run OLS on the transformed data. The optimality of the result is an immediate corollary of Gauss–Markov, which is the point of this short subsection.

Setup. Consider $y = X\beta + \varepsilon$ with X of full column rank, $\mathbb{E}[\varepsilon \mid X] = 0$, and *heteroskedastic but uncorrelated* errors: $\mathrm{Var}(\varepsilon \mid X) = \mathrm{diag}(\sigma_1^2, \dots, \sigma_n^2)$ with all $\sigma_i^2 > 0$ known. Let $W = \mathrm{diag}(1/\sigma_1, \dots, 1/\sigma_n)$, and define the whitened data $\tilde{y} = Wy$, $\tilde{X} = WX$: row i of the whole observation is divided by σ_i. The weighted least squares (WLS) estimator is

$$\hat{\beta}_{\mathrm{WLS}} = \arg\min_{\beta} \sum_{i=1}^{n} \frac{\left(y_i - x_i^\top \beta\right)^2}{\sigma_i^2},$$

where $x_i^\top$ is the i-th row of X — OLS on the whitened data, with weights $w_i = 1/\sigma_i^2$.

Theorem 5.24 (WLS is BLUE after whitening)**.** *Under the setup above, the whitened model $\tilde{y} = \tilde{X}\beta + \tilde{\varepsilon}$ with $\tilde{\varepsilon} = W\varepsilon$ satisfies (GM1)–(GM4) with error variance* 1*, and*

$$\hat{\beta}_{\mathrm{WLS}} = (\tilde{X}^\top \tilde{X})^{-1} \tilde{X}^\top \tilde{y}$$

is BLUE among all estimators linear in y and unbiased for β.

Proof. We transform to a homoskedastic model and invoke Theorem 5.6. Multiplying the model by W gives $\tilde{y} = \tilde{X}\beta + \tilde{\varepsilon}$, which is (GM1) for the whitened data. The matrix W is invertible, so

$\tilde{X} = WX$ retains full column rank (GM2). Exogeneity passes through: $\mathbb{E}[\tilde{\varepsilon} \mid X] = W\,\mathbb{E}[\varepsilon \mid X] = 0$ (GM3). Finally

$$\operatorname{Var}(\tilde{\varepsilon} \mid X) = W \operatorname{diag}(\sigma_1^2, \dots, \sigma_n^2)\, W^\top = I,$$

which is (GM4) with unit variance. The whitened least-squares criterion is

$$\|\tilde{y} - \tilde{X}\beta\|^2 = \sum_{i=1}^{n} \frac{(y_i - x_i^\top \beta)^2}{\sigma_i^2},$$

so its minimiser is exactly $\hat{\beta}_{\mathrm{WLS}}$, with closed form given by Theorem 5.1 applied to $(\tilde{y}, \tilde{X})$. Theorem 5.6 now says this estimator is BLUE among estimators linear in $\tilde{y}$ and unbiased. Because W is invertible, an estimator is linear in $\tilde{y}$ if and only if it is linear in $y = W^{-1}\tilde{y}$, so the two competitor classes coincide, and the BLUE property holds in the original model as claimed. □

Example 5.25 (Precision weighting two measurements)**.** Measure the same unknown μ twice with independent noise: $x_1 = \mu + \varepsilon_1$ with $\operatorname{sd}(\varepsilon_1) = 1$, and $x_2 = \mu + \varepsilon_2$ with $\operatorname{sd}(\varepsilon_2) = 3$. The WLS weights are proportional to $1/\sigma_i^2$, i.e. 1 and 1/9, so

$$\hat{\mu} = \frac{9x_1 + x_2}{10}, \qquad \operatorname{Var}(\hat{\mu}) = \frac{81 \cdot 1 + 1 \cdot 9}{100} = 0.9.$$

Even the noisy measurement helps: variance 0.9 beats the 1.0 of using the good measurement alone. Equal weighting, by contrast, gives $\operatorname{Var}\big((x_1 + x_2)/2\big) = (1 + 9)/4 = 2.5$ — worse than discarding the noisy measurement altogether. Weighting by precision is not a refinement; it is the difference between the noisy observation helping and hurting.

Trading application. Market data are heteroskedastic by regime: residual variance in a returns regression can be several times larger in high-volatility periods. Unweighted OLS lets those crisis observations dominate the fit, because the criterion counts squared errors and crises produce the largest ones. Down-weighting by inverse variance restores efficiency, and heteroskedasticity-robust standard errors repair the inference even when the weights are left alone (Remark 5.8).

Remark 5.26 (From WLS to GLS). Nothing in the proof of Theorem 5.24 needs the covariance to be diagonal. With a general known $\operatorname{Var}(\varepsilon \mid X) = \sigma^2\Omega$, Ω positive definite, whitening with $\Omega^{-1/2}$ in place of W produces a model satisfying (GM1)–(GM4), and the same argument shows the generalised least squares (GLS) estimator $(X^\top\Omega^{-1}X)^{-1}X^\top\Omega^{-1}y$ is BLUE — covering autocorrelated as well as heteroskedastic errors.

Remark 5.27 (Residual weighting is boosting, not GLS). A superficially similar move points in the opposite direction. Fitting a model A, then training a second model B with sample weights $w_i = |y_i - \hat{y}_{A,i}|$, makes B specialise on A's errors — the *boosting* idea in disguise. Safe names for the scheme: boosting-style reweighting, residual fitting, cascading. Avoid calling it "stacking", which properly means training a meta-learner on the models' *outputs*. Distinguish the two weightings sharply: GLS *down*-weights noisy observations to gain efficiency for one model; residual weighting *up*-weights hard observations so a second model specialises. One fights noise, the other chases signal — and the residual-weighting scheme backfires exactly when the "hard" points are hard because they are noise.

5.7 The Dart Problem: Squared Loss, Absolute Loss and Where to Aim

A single physical puzzle packages the deepest fact of this section — that the choice of loss function decides *what* is being estimated. Squared loss points at the mean, absolute loss at the median; a dart game with Gaussian versus Laplace scatter turns the abstraction into an aiming decision, complete with a bifurcation.

Setup. A dart aimed at the point $a \in \mathbb{R}$ lands at $a + Z$, where the scatter Z has density proportional to $\exp\big(-|z|^p/(2\sigma^p)\big)$: the Gaussian case is $p = 2$ (standard deviation σ) and the

Laplace case is $p = 1$. Two targets sit at $x_1 < x_2$ with separation $d = x_2 - x_1$ and midpoint $m = (x_1 + x_2)/2$. The expected score is proportional to the landing density evaluated at the targets,

$$S(a) = f(x_1 - a) + f(x_2 - a),$$

where f is the scatter density; the question is which aim a maximises S. Before the game itself, we record the two estimation lemmas the interview follow-up is really about.

Lemma 5.28 (Squared loss selects the mean)**.** *Let X be a random variable with $\mathbb{E}[X^2] < \infty$. The function $c \mapsto \mathbb{E}[(X - c)^2]$ is minimised uniquely at $c = \mathbb{E}[X]$, with minimum value* $\mathrm{Var}(X)$.

Proof. We complete the square around the mean. Adding and subtracting $\mathbb{E}[X]$,

$$\mathbb{E}[(X - c)^2] = \mathbb{E}[(X - \mathbb{E}[X])^2] + \big(\mathbb{E}[X] - c\big)^2 = \mathrm{Var}(X) + \big(\mathbb{E}[X] - c\big)^2,$$

the cross term vanishing because $\mathbb{E}[X - \mathbb{E}[X]] = 0$. The second term is nonnegative and zero only at $c = \mathbb{E}[X]$. ☐

Lemma 5.29 (Absolute loss selects the median)**.** *Let X have $\mathbb{E}|X| < \infty$ and CDF F.*

(i) If F is continuous, then $g(c) = \mathbb{E}|X - c|$ is convex and is minimised exactly at the points c with $F(c) = 1/2$, i.e. at the medians of X.

(ii) For the two-point empirical distribution placing mass $1/2$ at each of $x_1 < x_2$, the function g is constant, equal to $d/2$, on the whole segment $[x_1, x_2]$, and strictly larger outside: every *point of the segment is an optimal aim under absolute loss.*

Proof. We differentiate under the integral sign for (i) and compute directly for (ii).

(i) Convexity holds because $c \mapsto |x - c|$ is convex for each fixed x and expectation preserves convexity. Splitting the expectation at c,

$$g(c) = \int_{-\infty}^{c} (c - x)\, dF(x) + \int_{c}^{\infty} (x - c)\, dF(x),$$

and differentiating each term (the boundary contributions vanish because the integrand is zero at $x = c$),

$$g'(c) = F(c) - \big(1 - F(c)\big) = 2F(c) - 1.$$

A convex function is minimised exactly where its derivative changes sign from ≤ 0 to ≥ 0; since F is nondecreasing and continuous, that set is precisely $\{c : F(c) = 1/2\}$, which is nonempty by the intermediate value theorem.

(ii) Here $g(c) = \frac{1}{2}\big(|x_1 - c| + |x_2 - c|\big)$. For $c \in [x_1, x_2]$ the two distances sum to the fixed separation: $g(c) = \frac{1}{2}\big[(c - x_1) + (x_2 - c)\big] = d/2$. For $c < x_1$, both distances grow by $x_1 - c$ compared to $c = x_1$, giving $g(c) = d/2 + (x_1 - c) > d/2$; the case $c > x_2$ is symmetric. ☐

With the estimation story in hand — L2 points at the mean, L1 at the median — we can analyse the game, where the answers are close cousins of the lemmas but not identical to them.

Proposition 5.30 (Gaussian dart and the pitchfork at $d = 2\sigma$)**.** *Let $p = 2$, so that*

$$S(a) = e^{-(a - x_1)^2/(2\sigma^2)} + e^{-(a - x_2)^2/(2\sigma^2)},$$

and set $\kappa = d^2/(4\sigma^2)$. Then:

(i) the midpoint m is a critical point of S for every separation d;

(ii) if $d \leq 2\sigma$ (i.e. $\kappa \leq 1$), the midpoint is the unique global maximiser: aim at the mean of the targets;

(iii) if $d > 2\sigma$ (i.e. $\kappa > 1$), the midpoint is a local minimum*, and S has exactly two global maximisers $m \pm u^*$, where $u^* = (d/2)v^*$ and $v^* \in (0,1)$ is the unique positive solution of*

$$\operatorname{artanh}(v) = \kappa v. \tag{5.8}$$

The optimal aims lie strictly between the midpoint and the targets and approach the targets as $d/\sigma \to \infty$.

Proof. We centre coordinates at the midpoint, characterise all critical points by a transcendental equation, and count its roots; the technique is the standard normal-form analysis of a pitchfork bifurcation.

Step 1 (reduction). Substituting $u = a - m$ places the targets at $\pm d/2$:

$$S(u) = e^{-(u+d/2)^2/(2\sigma^2)} + e^{-(u-d/2)^2/(2\sigma^2)},$$

an even, positive, smooth function of u with $S(u) \to 0$ as $|u| \to \infty$.

Step 2 (critical points lie between the targets). Differentiating,

$$S'(u) = -\frac{1}{\sigma^2}\Big[(u + \tfrac{d}{2})\, e^{-(u+d/2)^2/(2\sigma^2)} + (u - \tfrac{d}{2})\, e^{-(u-d/2)^2/(2\sigma^2)}\Big].$$

At $u = 0$ the two bracketed terms are $(d/2)e^{-d^2/(8\sigma^2)}$ and $-(d/2)e^{-d^2/(8\sigma^2)}$, which cancel: the midpoint is critical for every d, proving (i). For $u \geq d/2$, both factors $u + d/2 > 0$ and $u - d/2 \geq 0$ are nonnegative and the first is strictly positive, so $S'(u) < 0$; by evenness $S'(u) > 0$ for $u \leq -d/2$. Hence every critical point satisfies $|u| < d/2$: candidate aims lie strictly between the targets.

Step 3 (the transcendental characterisation). For $|u| < d/2$, setting $S'(u) = 0$ and moving one term across,

$$\big(\tfrac{d}{2} + u\big)\, e^{-(u+d/2)^2/(2\sigma^2)} = \big(\tfrac{d}{2} - u\big)\, e^{-(u-d/2)^2/(2\sigma^2)},$$

where both prefactors are positive. Dividing and using $(u + d/2)^2 - (u - d/2)^2 = 2du$,

$$\frac{d/2 + u}{d/2 - u} = \exp\Big(\frac{du}{\sigma^2}\Big).$$

Taking logarithms, halving, and substituting $v = 2u/d \in (-1, 1)$ with the identity $\operatorname{artanh}(v) = \frac{1}{2}\log\frac{1+v}{1-v}$ turns the condition into exactly (5.8), $\operatorname{artanh}(v) = \kappa v$.

Step 4 (counting roots). Let $h(v) = \operatorname{artanh}(v) - \kappa v$ on $(-1, 1)$; h is odd with $h(0) = 0$, and $h'(v) = \frac{1}{1-v^2} - \kappa$, where $1/(1 - v^2)$ increases strictly from 1 at $v = 0$ to $+\infty$ as $v \to 1^-$. If $\kappa \leq 1$, then $h'(v) \geq 1 - \kappa \geq 0$ with equality at most at $v = 0$, so h is strictly increasing and $v = 0$ is its only zero: the midpoint is the only critical point. If $\kappa > 1$, then $h' < 0$ near 0 and $h' > 0$ near 1, with h' strictly increasing; hence h strictly decreases to a negative minimum and then strictly increases to $+\infty$, so it has exactly one zero $v^* \in (0, 1)$, plus the symmetric $-v^*$: three critical points $0, \pm u^*$ with $u^* = (d/2)v^*$.

Step 5 (classification). Because S is continuous, positive, and vanishes at infinity, its supremum is attained, necessarily at a critical point. For $\kappa \leq 1$ the only critical point is $u = 0$, proving (ii). For $\kappa > 1$, differentiating once more and evaluating at the midpoint gives

$$S''(0) = \frac{2}{\sigma^2}\,(\kappa - 1)\, e^{-d^2/(8\sigma^2)} > 0,$$

so the midpoint is a strict local minimum, and the maximum is attained at the remaining critical points $\pm u^*$, which carry equal values by evenness: exactly two global maximisers, strictly between midpoint and targets by Step 2. Finally, as $\kappa \to \infty$ the solution of (5.8) satisfies $v^* \to 1$ (for fixed $v < 1$, κv eventually exceeds $\operatorname{artanh}(v)$), so the optimal aims approach the targets. □

(Problem P5.16 asks you to reproduce the core of this analysis.)

Example 5.31 (Aiming numbers for a supercritical separation)**.** Take $\sigma = 1$ and targets at 0 and 2.5, so $d = 2.5 > 2\sigma$ and $\kappa = 2.5^2/4 = 1.5625$. Solving (5.8) numerically: at $v = 0.88$, $\operatorname{artanh}(0.88) = 1.3758$ against $\kappa v = 1.3750$, so $v^* \approx 0.880$ and $u^* = 1.25 \times 0.880 = 1.10$. The optimal aims are $m \pm u^* = 1.25 \pm 1.10$, i.e. 0.15 and 2.35 — about 0.15 *inside* each target. The scores confirm the ranking:

$$S(0.15) = e^{-0.15^2/2} + e^{-2.35^2/2} \approx 0.9888 + 0.0632 = 1.0520,$$

against $S(0) = 1 + e^{-3.125} \approx 1.0439$ at a target and $S(1.25) = 2e^{-0.78125} \approx 0.9157$ at the midpoint. Aim slightly inside a target: better than the target itself, and much better than the middle. For a subcritical separation, say $d = 1.5$ ($\kappa = 0.5625 \leq 1$), part (ii) applies and the unique optimum is the midpoint 0.75.

Proposition 5.32 (Laplace dart: aim at a target)**.** *Let $p = 1$ and measure distances in units of the Laplace scale, so a target at distance r contributes e^{-r} and*

$$S(a) = e^{-|a-x_1|} + e^{-|a-x_2|}.$$

Then S is maximised exactly at the two targets $a \in \{x_1, x_2\}$, and the endpoint beats the midpoint by

$$S(x_1) - S(m) = \left(1 - e^{-d/2}\right)^2 > 0 \quad \textit{for every } d > 0. \tag{5.9}$$

Proof. We show monotonicity outside the segment $[x_1, x_2]$ and strict convexity inside it; the maximum of a convex function on an interval sits at an endpoint. Without loss of generality $x_1 = 0$ and $x_2 = d$.

For $a \leq 0$, both distances shrink as a increases: $S(a) = e^a + e^{a-d} = e^a(1 + e^{-d})$, strictly increasing, so $S(a) < S(0)$ for $a < 0$; symmetrically $S(a) < S(d)$ for $a > d$.

For $a \in [0, d]$, the score is $S(a) = e^{-a} + e^{a-d}$, with $S''(a) = e^{-a} + e^{a-d} > 0$: strictly convex. A strictly convex function on $[0, d]$ satisfies $S(a) < \max\{S(0), S(d)\}$ for every interior a, and here the endpoint values are equal, $S(0) = S(d) = 1 + e^{-d}$. Hence the global maximisers are exactly the two targets. The interior stationary point, from $S'(a) = -e^{-a} + e^{a-d} = 0$, is the midpoint $a = d/2$, where $S(d/2) = 2e^{-d/2}$ — the *worst* aim on the segment. The gap is

$$S(0) - S(d/2) = 1 + e^{-d} - 2e^{-d/2} = \left(1 - e^{-d/2}\right)^2,$$

a perfect square, strictly positive for all $d > 0$. ☐

(Problem P5.17 asks you to reproduce both the median characterisation of Lemma 5.29 and this endpoint computation.)

Example 5.33 (Laplace aiming numbers)**.** With separation $d = 2$ scale units, aiming at a target scores $S(x_1) = 1 + e^{-2} \approx 1.1353$, while the midpoint scores $S(m) = 2e^{-1} \approx 0.7358$. The gap is $(1 - e^{-1})^2 \approx 0.632^2 \approx 0.400$, matching (5.9): with Laplace tails, committing to one target beats hedging between them by a wide margin.

Intuition. The estimation lemmas and the game agree in spirit but not in letter, and the contrast is the insight. Under absolute loss, *estimation* is indifferent across the whole segment between the targets (Lemma 5.29(ii)); the Laplace *game* flips to the extreme points, because maximising a density sum rewards concentration — the Laplace peak is sharp, so being exactly on one target buys more than being mediocre for both. The Gaussian's flat top around its centre rewards compromise, which is why the midpoint (the mean, per Lemma 5.28) wins — but only until the targets separate past $d = 2\sigma$, where the compromise collapses in a pitchfork and the optimum breaks symmetry toward one target.

Interview trap. "Does this remind you of anything in regression?" The expected mapping: **L2 loss** $\leftrightarrow$ conditional *mean*, outlier-sensitive (the square in the Gaussian exponent is the squared loss of OLS); **L1 loss** $\leftrightarrow$ conditional *median*, robust (quantile regression at the median). Then keep *losses* distinct from *penalties*: losses (L2 versus L1 on the residuals) decide what functional you predict — mean versus median; penalties (ridge versus lasso on the coefficients) decide how estimates behave — shrinkage versus selection. Conflating the two families because both carry the labels "L1/L2" is a known trap.

Practice: problems P5.1–P5.17 in §20, solutions in §21.

Part III

Linear Algebra and Calculus

6 Linear Algebra, PCA and Covariance

6.1 Symmetric Matrices, Eigenvalues and Positive Semidefiniteness

This subsection assembles the spectral toolkit on which the rest of the section runs: the diagonalisation of symmetric matrices, the notion of positive semidefiniteness, and the trace and determinant identities that let a hand-computed eigensystem be checked in seconds. Interviewers use exactly these facts as rapid-fire warm-ups before the PCA derivation.

Setup. Throughout this subsection, vectors $x, y \in \mathbb{R}^p$ are columns, $A \in \mathbb{R}^{p \times p}$ is a symmetric matrix ($A = A^\top$), and $\|x\| = \sqrt{x^\top x}$ denotes the Euclidean norm. The standard basis vectors are $e_1, \ldots, e_p$.

Definition 6.1 (Quadratic form; positive semidefiniteness)**.** The *quadratic form* of a symmetric matrix A is the map $x \mapsto x^\top A x = \sum_{i,j} A_{ij} x_i x_j$. The matrix A is *positive semidefinite* (PSD) if $x^\top A x \geq 0$ for every $x \in \mathbb{R}^p$, and *positive definite* (PD) if $x^\top A x > 0$ for every $x \neq 0$.

Fact 6.2 (Spectral theorem)**.** Every symmetric matrix $A \in \mathbb{R}^{p \times p}$ has p real eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$ (listed with multiplicity) and an orthonormal basis of corresponding eigenvectors $q_1, \ldots, q_p$. Equivalently,

$$A = Q \Lambda Q^\top, \qquad Q = (q_1 \mid \cdots \mid q_p), \quad Q^\top Q = I, \quad \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_p).$$

This is the structural fact the entire section leans on. A proof is beyond our scope.

Proposition 6.3 (Trace and determinant read off the eigenvalues)**.** *For a symmetric matrix A with eigenvalues $\lambda_1, \ldots, \lambda_p$,*

$$\operatorname{tr} A = \sum_{i=1}^{p} \lambda_i, \qquad \det A = \prod_{i=1}^{p} \lambda_i.$$

Proof. We exploit the invariance of trace and determinant under the factorisation of Fact 6.2. First, for any matrices B, C of compatible sizes, $\operatorname{tr}(BC) = \sum_i \sum_j B_{ij} C_{ji} = \sum_j \sum_i C_{ji} B_{ij} = \operatorname{tr}(CB)$. Applying this with $B = Q\Lambda$ and $C = Q^\top$ gives

$$\operatorname{tr} A = \operatorname{tr}(Q \Lambda Q^\top) = \operatorname{tr}(\Lambda Q^\top Q) = \operatorname{tr} \Lambda = \sum_i \lambda_i.$$

For the determinant, multiplicativity gives $\det A = \det Q \cdot \det \Lambda \cdot \det Q^\top = (\det Q)^2 \det \Lambda$, and $Q^\top Q = I$ forces $(\det Q)^2 = 1$, so $\det A = \det \Lambda = \prod_i \lambda_i$. $\square$

Intuition. These two identities are the fastest sanity checks available: after any hand computation of an eigensystem, confirm that the eigenvalues sum to the trace and multiply to the determinant. A mismatch localises the arithmetic error immediately.

Theorem 6.4 (PSD equivalences)**.** *Let A be symmetric with eigenvalues $\lambda_1, \dots, \lambda_p$. Then A is positive semidefinite if and only if $\lambda_i \geq 0$ for all i, and positive definite if and only if $\lambda_i > 0$ for all i.*

Proof. We expand the quadratic form in the eigenbasis furnished by Fact 6.2. Write $x = \sum_i c_i q_i$ with $c_i = q_i^\top x$; orthonormality gives $\|x\|^2 = \sum_i c_i^2$ and

$$x^\top A x = \sum_{i=1}^{p} \lambda_i c_i^2. \tag{6.1}$$

If every $\lambda_i \geq 0$, then (6.1) is a sum of nonnegative terms, so $x^\top A x \geq 0$ for all x. Conversely, if A is PSD, take $x = q_j$: then $c_i = \mathbf{1}_{\{i=j\}}$ and $q_j^\top A q_j = \lambda_j \geq 0$. For the definite case: if every $\lambda_i > 0$ and $x \neq 0$, some $c_j \neq 0$, so the sum in (6.1) is strictly positive; conversely $x = q_j$ gives $\lambda_j = q_j^\top A q_j > 0$. □

Intuition. In the eigenbasis the quadratic form is a weighted sum of squared coordinates, each weighted by an eigenvalue. For a PD matrix the level set $\{x : x^\top A x = 1\}$ is an ellipsoid whose axes point along the eigenvectors, with semi-axis length $1/\sqrt{\lambda_i}$ along q_i: large eigenvalues squeeze the ellipsoid, small ones stretch it.

Example 6.5 (The two-by-two staple)**.** Consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The characteristic polynomial is $(2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$, so $\lambda_1 = 3$ and $\lambda_2 = 1$, with eigenvectors $q_1 = (1,1)^\top/\sqrt{2}$ and $q_2 = (1,-1)^\top/\sqrt{2}$: indeed $A(1,1)^\top = (3,3)^\top$ and $A(1,-1)^\top = (1,-1)^\top$. Both eigenvalues are positive, so A is positive definite by Theorem 6.4. The checks of Proposition 6.3 pass: $\operatorname{tr} A = 4 = 3 + 1$ and $\det A = 4 - 1 = 3 = 3 \cdot 1$. A shortcut worth memorising: a symmetric 2×2 matrix with equal diagonal entries d and off-diagonal c has eigenvalues $d \pm c$ with eigenvectors $(1, \pm 1)^\top$.

6.2 Covariance Matrices Are Positive Semidefinite

The single most quoted one-liner of quant linear algebra: every covariance matrix is PSD, because a quadratic form in a covariance matrix is a variance. This subsection proves it, then explains why matrices *estimated* entrywise in practice can nevertheless fail the property, and what to do about it.

Setup. Let $Z = (Z_1, \dots, Z_p)^\top$ be a random vector with finite second moments, mean vector $\mu = \mathbb{E}[Z]$, and covariance matrix

$$\Sigma = \mathbb{E}\big[(Z - \mu)(Z - \mu)^\top\big], \qquad \Sigma_{ij} = \operatorname{Cov}(Z_i, Z_j).$$

Theorem 6.6 (Covariance matrices are PSD)**.** *Every covariance matrix Σ is symmetric positive semidefinite.*

Proof. We evaluate the quadratic form and recognise it as a variance. Symmetry is immediate from $\operatorname{Cov}(Z_i, Z_j) = \operatorname{Cov}(Z_j, Z_i)$. For any fixed $x \in \mathbb{R}^p$, linearity of expectation lets the (deterministic) vector x pass inside:

$$x^\top \Sigma x = \mathbb{E}\big[x^\top (Z - \mu)(Z - \mu)^\top x\big] = \mathbb{E}\big[\big(x^\top (Z - \mu)\big)^2\big] = \operatorname{Var}(x^\top Z) \geq 0.$$

By Theorem 6.4, all eigenvalues of Σ are nonnegative. □

Interview trap. "Prove that a covariance matrix is positive semidefinite" is a standard interview opener, and the expected answer is exactly the one-line identity $x^\top \Sigma x = \mathrm{Var}(x^\top Z) \geq 0$: the quadratic form is the variance of the portfolio with weights x, and variances cannot be negative. Produce it without hesitation; attempts to argue entrywise (for example from the signs or sizes of individual correlations) stall.

Example 6.7 (Portfolio-variance check)**.** Let $Z = (Z_1, Z_2)^\top$ with $\mathrm{Var}(Z_1) = \mathrm{Var}(Z_2) = 2$ and $\mathrm{Cov}(Z_1, Z_2) = 1$, so that Σ is the matrix of Example 6.5. The spread portfolio $x = (1, -1)^\top$ has

$$\mathrm{Var}(Z_1 - Z_2) = x^\top \Sigma x = 2 - 2(1) + 2 = 2 \geq 0.$$

The eigen route agrees: x is proportional to the eigenvector q_2, so $x^\top \Sigma x = \lambda_2 \|x\|^2 = 1 \cdot 2 = 2$.

Remark 6.8 (Why a sample covariance can fail PSD in practice). Theorem 6.6 covers the population covariance, and also any sample covariance computed from a *single* complete panel: with a centred data matrix X (rows are observations), the estimate $\hat{\Sigma} = \frac{1}{n} X^\top X$ satisfies $x^\top \hat{\Sigma} x = \|Xx\|^2/n \geq 0$. The guarantee breaks when the entries are estimated *pairwise on different samples* — asynchronous data across assets, or pairwise-complete observations — because then no single dataset generates all the entries at once: the single-quadratic-form structure underlying the proof is gone, and the assembled matrix can have negative eigenvalues. Three standard repairs:

1. *Shrinkage / regularisation:* blend the estimate toward a PSD target, in the simplest form by adding a small positive multiple of the identity;
2. *Eigenvalue clipping:* floor the negative eigenvalues at zero and renormalise (for a correlation matrix, rescale the diagonal back to one);
3. *Factor structure:* impose a low-rank-plus-diagonal model, which is PSD by construction.

Note the rhyme with ridge regression: adding a positive multiple of I is the same medicine, there applied to a near-singular $X^\top X$.

Interview trap. Feasibility itself can be the question: with $\rho_{AB} = 0.9$ and $\rho_{AC} = -0.9$, no joint distribution allows $\rho_{BC} = 0.8$ — the PSD constraint forces $\rho_{BC} \in [-1, -0.62]$. A risk report can still display such a matrix if its entries were estimated pairwise, which is precisely the failure mode of Remark 6.8. Problem P6.4 works this through in full.

6.3 Weighted Inner Products and A-Orthogonality

Ordinary orthogonality is orthogonality with respect to the identity; timed online assessments ♦ probe the general weighted version, in which a symmetric positive definite matrix reweights the whole geometry. This subsection defines the weighted inner product, derives the matrix criterion for maps that preserve it, and classifies the standard candidates for a concrete diagonal weight matrix.

Setup. Let $A \in \mathbb{R}^{p \times p}$ be symmetric positive definite (Definition 6.1), and let $x, y \in \mathbb{R}^p$.

Definition 6.9 (A-inner product; A-orthogonal matrix)**.** The *A-inner product* of x and y is

$$\langle x, y \rangle_A = x^\top A y.$$

A matrix $Q \in \mathbb{R}^{p \times p}$ is *A-orthogonal* if it preserves this inner product: $\langle Qx, Qy \rangle_A = \langle x, y \rangle_A$ for all $x, y \in \mathbb{R}^p$. The reader should verify, using the symmetry and positive definiteness of A, that $\langle \cdot, \cdot \rangle_A$ is a genuine inner product; the choice $A = I$ recovers the Euclidean case.

Proposition 6.10 (Matrix criterion for A-orthogonality)**.** *A matrix Q is A-orthogonal if and only if*

$$Q^\top A Q = A.$$

In particular, for $A = I$ the criterion reads $Q^\top Q = I$: ordinary orthogonality.

Proof. We reduce equality of bilinear forms to equality of matrices by testing on standard basis vectors. For any x, y,

$$\langle Qx, Qy\rangle_A = (Qx)^\top A(Qy) = x^\top (Q^\top AQ)y.$$

If $Q^\top AQ = A$, the right side equals $x^\top Ay = \langle x, y\rangle_A$ for all x, y. Conversely, suppose $x^\top (Q^\top AQ)y = x^\top Ay$ for all x, y; choosing $x = e_i$ and $y = e_j$ reads off the (i, j) entry, $(Q^\top AQ)_{ij} = A_{ij}$, for every i, j, hence $Q^\top AQ = A$. □

Example 6.11 (A-orthogonality for weights $2, 3, 5$ ♦)**.** Let $A = \mathrm{diag}(2, 3, 5)$, whose diagonal weights are pairwise distinct. We test four natural candidates against the criterion of Proposition 6.10.

1. *Permutation swapping $e_1 \leftrightarrow e_3$:* here $Q^\top AQ = \mathrm{diag}(5, 3, 2) \neq A$. Not A-orthogonal — a permutation *reorders* the weights, and can only succeed when the swapped weights are equal.
2. *Sign flip $Q = \mathrm{diag}(1, -1, 1)$:* here $Q^\top AQ = \mathrm{diag}(1^2 \cdot 2,\ (-1)^2 \cdot 3,\ 1^2 \cdot 5) = A$. This one is A-orthogonal: signs square away, so axis reflections are A-orthogonal for *any* diagonal A.
3. *Permutation swapping $e_1 \leftrightarrow e_2$:* here $Q^\top AQ = \mathrm{diag}(3, 2, 5) \neq A$. Not A-orthogonal.
4. *Rotation by angle t in the x–y plane:* the upper 2×2 block of $Q^\top AQ$ is $R_t^\top\,\mathrm{diag}(2, 3)R_t$, whose off-diagonal entry is $(3-2)\sin t\cos t = \sin t\cos t \neq 0$ for generic t; even at $t = \pi/2$ the block becomes $\mathrm{diag}(3, 2) \neq \mathrm{diag}(2, 3)$. Not A-orthogonal — a rotation *mixes* unequal weights, and preserves $\langle\cdot,\cdot\rangle_A$ only on blocks where A is a multiple of the identity.

Conclusion: with distinct diagonal weights, axis sign flips qualify, while axis permutations and in-plane rotations do not. *(Problem P6.10 asks you to reproduce this classification from scratch.)*

Intuition. An A-orthogonal map is a rigid motion of the geometry in which direction i carries weight A_{ii}. Equal weights everywhere (A proportional to I) leave the full rotation group available; unequal weights break the rotational symmetry, and the only surviving axis symmetries are the reflections that keep each axis in place.

6.4 Principal Component Analysis

Principal component analysis answers one question: along which direction does centred data vary the most? The answer is a variational problem solved completely by the spectral decomposition of Fact 6.2. Interviews ask for the derivation, not the recipe — this subsection gives the rigorous argument and the explained-variance bookkeeping that follows from it.

Setup. Let $X \in \mathbb{R}^{n\times p}$ be a data matrix whose n rows are observations of p variables, with each column *centred* (summing to zero). The sample covariance matrix is

$$\hat\Sigma = \frac{1}{n}X^\top X$$

(the $1/(n-1)$ convention rescales every entry by the same factor and changes no direction; we return to it in Proposition 6.18). For a unit vector $w \in \mathbb{R}^p$, the projected data $Xw \in \mathbb{R}^n$ has sample variance $\frac{1}{n}\|Xw\|^2 = w^\top\hat\Sigma w$, so "the direction of maximal variance" means the maximiser of the quadratic form over the unit sphere. The theorem is stated for an arbitrary symmetric matrix Σ and applied with $\Sigma = \hat\Sigma$.

Theorem 6.12 (Variational characterisation of the first principal direction)**.** *Let $\Sigma \in \mathbb{R}^{p\times p}$ be symmetric with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_p$ and orthonormal eigenvectors $q_1, \ldots, q_p$ as in Fact 6.2. Then*

$$\max_{\|w\|=1} w^\top\Sigma w = \lambda_1, \quad \textit{attained at } w = q_1, \qquad \min_{\|w\|=1} w^\top\Sigma w = \lambda_p, \quad \textit{attained at } w = q_p.$$

If $\lambda_1 > \lambda_2$, the maximiser is unique up to sign.

Proof. We expand in the eigenbasis, as in the proof of Theorem 6.4. Write $w = \sum_i c_i q_i$; the constraint $\|w\| = 1$ reads $\sum_i c_i^2 = 1$, and by (6.1), $w^\top \Sigma w = \sum_i \lambda_i c_i^2$. The right side is a weighted average of the eigenvalues with nonnegative weights c_i^2 summing to one, hence

$$w^\top \Sigma w = \sum_i \lambda_i c_i^2 \le \lambda_1 \sum_i c_i^2 = \lambda_1,$$

with equality if and only if all the weight sits on indices i with $\lambda_i = \lambda_1$. The choice $w = q_1$ attains the bound, proving the maximum; the minimum statement is identical with the inequality reversed and $w = q_p$. If $\lambda_1 > \lambda_2$, equality forces $c_1^2 = 1$ and $c_i = 0$ for $i \ge 2$, that is $w = \pm q_1$. ☐

Remark 6.13 (The Lagrangian route, made honest). The derivation usually quoted in interviews maximises $w^\top \Sigma w$ subject to $w^\top w = 1$ via the Lagrangian $L(w, \lambda) = w^\top \Sigma w - \lambda(w^\top w - 1)$: setting $\nabla_w L = 2\Sigma w - 2\lambda w = 0$ yields $\Sigma w = \lambda w$, so the stationary points are exactly the eigenvectors, the multiplier is the eigenvalue, and the objective at an eigenvector equals that eigenvalue. This is worth reproducing fluently — but stationarity alone does not say *which* eigenvector is the maximiser. The eigenbasis expansion in the proof of Theorem 6.12 is what certifies that the top eigenvector wins.

Definition 6.14 (Principal directions and components)**.** Let $\hat{\Sigma}$ be the sample covariance of a centred data matrix X, with eigenvalues sorted $\lambda_1 \ge \cdots \ge \lambda_p$ and orthonormal eigenvectors $q_1, \dots, q_p$. The vectors q_i are the *principal directions* of the data; the projected series $Xq_i \in \mathbb{R}^n$ is the *i-th principal component* (PC i).

Corollary 6.15 (Explained variance)**.** *The sample variance of the i-th principal component equals λ_i, the total sample variance of the data equals* $\operatorname{tr} \hat{\Sigma} = \sum_{j=1}^p \lambda_j$*, and the fraction of total variance explained by PC i is*

$$\frac{\lambda_i}{\sum_{j=1}^p \lambda_j}.$$

Proof. The sample variance of Xq_i is $q_i^\top \hat{\Sigma} q_i = \lambda_i q_i^\top q_i = \lambda_i$, using the eigenvector property and $\|q_i\| = 1$. The total sample variance of the data is the sum of the column variances of X, which is exactly the sum of the diagonal entries $\operatorname{tr} \hat{\Sigma}$; by Proposition 6.3 this equals $\sum_j \lambda_j$. The ratio of the two gives the displayed fraction. ☐

Example 6.16 (PCA of the two-by-two staple)**.** Suppose two centred variables have sample covariance matrix $\hat{\Sigma} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, the matrix of Example 6.5. The first principal direction is $q_1 = (1, 1)^\top / \sqrt{2}$ with variance $\lambda_1 = 3$; the second is $q_2 = (1, -1)^\top / \sqrt{2}$ with variance $\lambda_2 = 1$. By Corollary 6.15, PC 1 explains $3/(3 + 1) = 75\%$ of total variance and PC 2 the remaining 25%; the check $\operatorname{tr} \hat{\Sigma} = 4 = 3 + 1$ passes. Figure 9 shows a data cloud with this covariance structure and its principal axes. *(Problem P6.5 asks you to reproduce this derivation from scratch; Problem P6.6 applies it to an equicorrelated covariance matrix.)*

Interview trap. Principal component analysis is defined on the *covariance*, so the data must be centred first. Without centring, $X^\top X$ equals the centred scatter plus the rank-one term $n\bar{x}\bar{x}^\top$ built from the mean vector $\bar{x}$; when the mean sits far from the origin this term dominates, and the leading "principal direction" chases the direction of the mean rather than the direction of maximal variance, with an inflated explained-variance share.

6.5 Computation via the SVD

In practice principal directions are not computed by eigendecomposing a covariance matrix: they are read off the singular value decomposition of the data matrix itself. This subsection proves the link and records the numerical reason for preferring it.

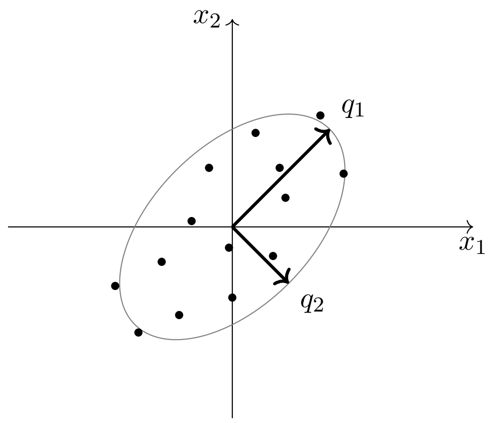


Figure 9: A centred data cloud with covariance as in Example 6.16. The principal axes point along the eigenvectors $q_1 = (1,1)^\top/\sqrt{2}$ and $q_2 = (1,-1)^\top/\sqrt{2}$ of the sample covariance (each eigenvector is defined only up to sign), with arrow lengths proportional to $\sqrt{\lambda_1} = \sqrt{3}$ and $\sqrt{\lambda_2} = 1$; the gray curve is the one-standard-deviation ellipse.

Setup. Let $X \in \mathbb{R}^{n\times p}$ be a centred data matrix with $n \geq p$.

Fact 6.17 (Singular value decomposition)**.** Every matrix $X \in \mathbb{R}^{n\times p}$ with $n \geq p$ admits a thin factorisation

$$X = USV^\top, \qquad U \in \mathbb{R}^{n\times p},\ U^\top U = I_p, \qquad V \in \mathbb{R}^{p\times p} \text{ orthogonal}, \qquad S = \operatorname{diag}(s_1, \dots, s_p)$$

with singular values $s_1 \geq s_2 \geq \cdots \geq s_p \geq 0$. A proof is beyond our scope.

Proposition 6.18 (The SVD–PCA link)**.** *Let $X = USV^\top$ be as in Fact 6.17. Then*

$$X^\top X = VS^2V^\top,$$

so the columns $v_1, \dots, v_p$ of V are the eigenvectors of $X^\top X$ with eigenvalues $s_1^2, \dots, s_p^2$: the principal directions of the data are the columns of V, and the sample variance along v_i is s_i^2/n under the $1/n$ covariance convention and $s_i^2/(n-1)$ under the $1/(n-1)$ convention. The explained-variance fraction of PC i is $s_i^2/\sum_j s_j^2$ under either convention.

Proof. We substitute the factorisation and cancel:

$$X^\top X = (USV^\top)^\top(USV^\top) = VS\,(U^\top U)\,SV^\top = VS^2V^\top,$$

using $U^\top U = I_p$ and the symmetry of the diagonal factor S. Since V is orthogonal and S^2 is diagonal with nonnegative entries, this *is* a spectral decomposition of $X^\top X$ in the sense of Fact 6.2: the columns of V are orthonormal eigenvectors and the eigenvalues are s_i^2. Dividing by n gives $\hat{\Sigma} = V(S^2/n)V^\top$ — the same eigenvectors, with eigenvalues s_i^2/n, which by Corollary 6.15 are the sample variances along the principal directions; the $1/(n-1)$ variant differs only by the scalar $n/(n-1)$. In the explained-variance ratio of Corollary 6.15 the normalisation cancels, leaving $s_i^2/\sum_j s_j^2$. $\square$

Remark 6.19 (Pick one convention). The two normalisations $1/n$ and $1/(n-1)$ produce identical directions and identical explained-variance shares; only the per-direction variance figures differ, by the factor $n/(n-1)$. Either is defensible — what is not defensible is mixing them within one computation. Pick one and stay consistent.

Example 6.20 (Variances from singular values)**.** Suppose $n = 5$ observations of $p = 2$ centred variables give singular values $s_1 = 4$ and $s_2 = 2$. Under the $1/(n-1)$ convention the sample variances along the principal directions are $16/4 = 4$ and $4/4 = 1$, and PC 1 explains $4/(4+1) = 80\%$ of total variance — equivalently $s_1^2/(s_1^2+s_2^2) = 16/20 = 80\%$, where the normalisation has cancelled as promised by Proposition 6.18. *(Problem P6.7 asks you to reproduce this proof and computation from scratch.)*

Remark 6.21 (Conditioning: decompose X, not $X^\top X$). Given a numerical library offering both routines, compute the SVD of X rather than the eigendecomposition of $X^\top X$. Forming the product explicitly squares the condition number, $\kappa(X^\top X) = \kappa(X)^2$, so the directions belonging to small singular values drown in floating-point error, and rounding can even return small negative eigenvalues for a matrix that is PSD in exact arithmetic (compare Remark 6.8). The SVD works on X directly, is backward stable, and never materialises the $p \times p$ product.

6.6 Orthogonalising Correlated Signals (Gram–Schmidt)

A recurring desk problem: a candidate signal overlaps an existing one, and only the part of the candidate *not* already captured should be kept. The answer is a single Gram–Schmidt step — project and keep the residual — and it is simultaneously an ordinary least squares regression.

Setup. Two signals recorded over n periods are vectors $a, b \in \mathbb{R}^n$ (entry a_t is the first signal's value in period t), with $a \neq 0$. The Euclidean inner product is $\langle a, b\rangle = \sum_{t=1}^n a_t b_t$, and $\mathbf{1} \in \mathbb{R}^n$ denotes the all-ones vector. A signal is *demeaned* if its entries sum to zero, that is $\langle a, \mathbf{1}\rangle = 0$; for demeaned signals the sample covariance is $\langle a, b\rangle/n$ (or the $1/(n-1)$ variant, Remark 6.19), so zero inner product is the same statement as zero sample correlation.

Proposition 6.22 (Gram–Schmidt residual)**.** *Define the projection coefficient and residual*

$$\hat\beta = \frac{\langle a, b\rangle}{\langle a, a\rangle}, \qquad b_\perp = b - \hat\beta\, a. \tag{6.2}$$

Then:

(i) the residual is orthogonal to a: $\langle a, b_\perp\rangle = 0$;
(ii) norms split (Pythagoras): $\|b\|^2 = \hat\beta^2\|a\|^2 + \|b_\perp\|^2$;
(iii) if a and b are both demeaned, then $b_\perp$ is demeaned and (when $b_\perp \neq 0$) the sample correlation between a and $b_\perp$ is zero;
(iv) the coefficient $\hat\beta$ is the OLS slope of the regression of b on a without intercept, and $b_\perp$ is its residual vector.

Proof. We argue by direct computation. (i) By bilinearity and (6.2),

$$\langle a, b_\perp\rangle = \langle a, b\rangle - \hat\beta\langle a, a\rangle = \langle a, b\rangle - \langle a, b\rangle = 0.$$

(ii) Expanding $b = \hat\beta a + b_\perp$ and using (i) to kill the cross term,

$$\|b\|^2 = \hat\beta^2\|a\|^2 + 2\hat\beta\langle a, b_\perp\rangle + \|b_\perp\|^2 = \hat\beta^2\|a\|^2 + \|b_\perp\|^2.$$

(iii) If $\langle a, \mathbf{1}\rangle = \langle b, \mathbf{1}\rangle = 0$, then $\langle b_\perp, \mathbf{1}\rangle = \langle b, \mathbf{1}\rangle - \hat\beta\langle a, \mathbf{1}\rangle = 0$, so $b_\perp$ is demeaned; the sample covariance of a and $b_\perp$ is then proportional to $\langle a, b_\perp\rangle = 0$, and dividing by the (nonzero) sample standard deviations gives sample correlation zero. (iv) The least squares problem $\min_\beta \|b - \beta a\|^2$ has derivative $-2\langle a, b - \beta a\rangle$, which vanishes at $\beta = \langle a, b\rangle/\langle a, a\rangle = \hat\beta$; the second derivative $2\langle a, a\rangle > 0$ confirms a minimum. The residual vector $b - \hat\beta a$ is exactly $b_\perp$, and (i) is the familiar OLS orthogonality of residuals to regressors. □

Figure 10 shows the geometry: the projection $\hat\beta a$ and the residual $b_\perp$ form the two legs of a right triangle with hypotenuse b, which is precisely statement (ii).

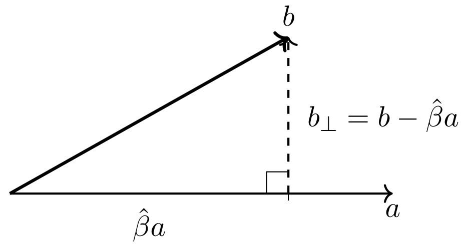


Figure 10: The Gram–Schmidt step of Proposition 6.22: the candidate signal b decomposes into its projection $\hat{\beta} a$ onto the incumbent signal a and the orthogonal residual $b_\perp$. Pythagoras splits $\|b\|^2 = \hat{\beta}^2 \|a\|^2 + \|b_\perp\|^2$.

Remark 6.23 (Demeaning caveat). Orthogonality equals zero correlation *only for demeaned signals*. In general the sample covariance is $\frac{1}{n}\langle a, b\rangle - \bar{a}\bar{b}$ with $\bar{a}, \bar{b}$ the sample means, so with nonzero means two orthogonal vectors can be correlated and two uncorrelated signals can fail to be orthogonal. Demean first — or, equivalently, include an intercept in the regression of (iv), after which the residuals are orthogonal to the constant column as well.

Example 6.24 (Orthogonalising a candidate signal)**.** Let a and b be demeaned with unit sample variance and sample correlation 0.8, so that $\hat{\beta} = 0.8$ and $b_\perp = b - 0.8\,a$ by (6.2). The residual's sample variance is

$$1 - 2(0.8)(0.8) + (0.8)^2(1) = 1 - 1.28 + 0.64 = 0.36 = 1 - 0.8^2,$$

the general pattern $1 - \rho^2$ for unit-variance signals with correlation ρ. In words: 64% of the candidate's variance was already contained in the incumbent, and only 36% of it is new information. *(Problem P6.8 asks you to reproduce this computation from scratch.)*

Trading application. Suppose a desk runs a production model whose output cannot be modified — an untouchable black box — and a candidate signal overlaps it. Orthogonalise the demeaned candidate against the demeaned incumbent output via (6.2): what survives is the candidate's *incremental* information. The combined book then aggregates cleanly — by Proposition 6.22(iii) the two components are uncorrelated, so variances add with no covariance term — and combined risk stops double-counting the shared exposure.

6.7 Beta Neutralisation (Market-Neutral Signals)

The Gram–Schmidt step of Proposition 6.22 applied in the *cross-section* strips market exposure out of a trading signal. The recipe rests on one empirical fact about equity return covariances and one line of algebra.

Setup. Consider a universe of p assets. A cross-sectional signal is now a vector $b \in \mathbb{R}^p$ whose i-th entry scores asset i (note the change of arena: entries index assets, not time periods). Let $m \in \mathbb{R}^p$ denote the first principal direction of the asset-return covariance matrix, estimated as in Definition 6.14 (in practice via Proposition 6.18).

Fact 6.25 (PC1 as the market mode, empirical)**.** For broad equity universes, the first principal direction of the return covariance matrix empirically has same-sign weights on every asset: it is the *market mode*, the direction in which the whole market moves together. This is an empirical regularity, not a theorem, although positively correlated toy models reproduce it exactly — for an equicorrelated covariance with correlation $\rho > 0$, PC1 is exactly the equal-weight vector (Problem P6.6 derives this).

Corollary 6.26 (Beta-neutral projection)**.** *The projected signal*

$$b' = b - \frac{\langle b, m\rangle}{\langle m, m\rangle}\, m$$

satisfies $\langle b', m\rangle = 0$*: the neutralised signal carries no exposure along the market mode. The neutrality holds by construction, not by a regression estimate.*

Proof. This is Proposition 6.22(i) with the incumbent vector taken to be m and the candidate taken to be b. □

Example 6.27 (Neutralising a three-asset signal)**.** Take the market mode $m = (1, 1, 1)^\top$ and the raw signal $b = (2, -1, 1)^\top$. The two inner products are $\langle b, m\rangle = 2 - 1 + 1 = 2$ and $\langle m, m\rangle = 3$, so

$$b' = b - \tfrac{2}{3}\,(1, 1, 1)^\top = \left(\tfrac{4}{3},\ -\tfrac{5}{3},\ \tfrac{1}{3}\right)^\top.$$

The check passes: $\langle b', m\rangle = \frac{4}{3} - \frac{5}{3} + \frac{1}{3} = 0$, that is, the neutralised weights sum to zero. The position is long–short balanced against the equal-weight market portfolio: a broad move lifting all three assets equally produces zero profit or loss to first order. *(Problem P6.9 asks you to reproduce and extend this computation.)*

Trading application. The full recipe for a market-neutral signal: (1) run PCA on the asset returns (Proposition 6.18); (2) inspect the first principal direction's weights — same-sign loadings identify it as the market mode per Fact 6.25; (3) project the signal off it by Corollary 6.26. The result is beta-neutral by construction. When m has exactly equal weights the projection reduces to cross-sectional demeaning of the signal; an estimated PC1 instead removes a loading-weighted average, giving each name credit for its actual market sensitivity.

Practice: problems P6.1–P6.10 in §20, solutions in §21.

7 Calculus for Quant Interviews

Assessments test calculus at two speeds. Reflex questions ask for a derivative or a standard antiderivative and are graded on speed; reasoning questions ask whether an improper integral converges, what a moment equals, or why a root-finder stalls, and are graded on the one-sentence justification that accompanies the verdict. This section builds the toolkit in that order: derivative fluency, integration technique, the convergence decision, the bridge from integrals to moments, Taylor expansions, and Newton's method. Subsections sourced from real assessments carry the company tag in their title.

7.1 The Derivative Toolkit

Differentiation questions reward fluency: the rules and the core table below should come without visible thought, because they are the substrate for every optimisation, sensitivity, and Taylor question later in the section. The one identity worth proving is the logistic derivative, which turns every "sensitivity of a price to news" question into a two-second multiplication.

Throughout this subsection, u, v, and g denote real functions differentiable wherever the rules below are applied, and a denotes a positive constant.

Fact 7.1 (Differentiation rules and the core table)**.** Let u and v be differentiable at x, with $v(x) \neq 0$ where v appears in a denominator, and let f be differentiable at $g(x)$. Then

$$(uv)' = u'v + uv', \qquad \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}, \qquad \frac{d}{dx} f(g(x)) = f'(g(x))\, g'(x).$$

The core table, for n real and $a > 0$:

$$(x^n)' = nx^{n-1}, \qquad (e^{ax})' = ae^{ax}, \qquad (\ln x)' = \frac{1}{x}, \qquad (a^x)' = a^x \ln a,$$

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x, \qquad (\sqrt{x}\,)' = \frac{1}{2\sqrt{x}}.$$

These are standard first-course results; we use them without proof.

In an interview, narrate the chain rule from the outside in: "the derivative of the outside, evaluated at the inside, times the derivative of the inside." The narration prevents the classic slip of forgetting the inner factor.

The logistic (sigmoid) function maps log-odds to a probability, and it is the standard model of a prediction-market contract price responding to news. In this subsection only, the symbol p denotes this price function (elsewhere in the document p is a probability parameter).

Proposition 7.2 (Logistic derivative identity)**.** *Let $p(x) = (1 + e^{-x})^{-1}$ for $x \in \mathbb{R}$. Then*

$$p'(x) = p(x)(1 - p(x)) \qquad \textit{for all } x,$$

and the derivative attains its maximum value $\frac{1}{4}$ at $x = 0$, where $p = \frac{1}{2}$.

Proof. We differentiate directly with the chain rule and then recognise the factors. Writing $p(x) = (1 + e^{-x})^{-1}$,

$$p'(x) = -(1 + e^{-x})^{-2} \cdot (-e^{-x}) = \frac{e^{-x}}{(1 + e^{-x})^2} = \frac{1}{1 + e^{-x}} \cdot \frac{e^{-x}}{1 + e^{-x}}.$$

The first factor is $p(x)$. For the second, note

$$1 - p(x) = 1 - \frac{1}{1 + e^{-x}} = \frac{e^{-x}}{1 + e^{-x}},$$

which proves the identity. For the maximum, set $s = p(x)$; as x ranges over $\mathbb{R}$, the value s ranges over the open interval $(0, 1)$, and $p'(x) = s(1 - s)$. The map $s \mapsto s(1 - s)$ is a downward parabola with vertex at $s = \frac{1}{2}$, where it equals $\frac{1}{4}$. Since p is continuous and strictly increasing with $p(0) = \frac{1}{2}$, the maximum sensitivity $\frac{1}{4}$ is attained exactly at $x = 0$. □

(Problem P7.3 asks you to reproduce this proof from scratch.)

Example 7.3 (Sensitivity at eighty cents)**.** A contract trades at $p = 0.80$. By Proposition 7.2, the price moves at rate $p(1 - p) = 0.80 \times 0.20 = 0.16$ price units per unit of log-odds. At $p = 0.50$ the rate would be $0.50 \times 0.50 = 0.25$, the maximum possible: a 50-cent contract moves most per unit of news.

Trading application. Any "sensitivity of price to log-odds" question reduces to: compute p, multiply $p(1-p)$. The qualitative fact that sensitivity peaks at $p = \frac{1}{2}$ is itself a common follow-up: contracts near even odds are where news moves prices fastest.

The optimisation recipe. Interview optimisation questions are graded in three parts. (1) Set $f' = 0$ and solve. (2) *Justify* that the critical point is a maximum: by the sign of the second derivative, by a sign change of f', or by boundary behaviour (if $f \to 0$ at both ends of the domain and there is a single interior critical point, that point is the maximum). (3) Report both the maximiser x^* and the optimal value $f(x^*)$.

Example 7.4 (The recipe on a decaying objective)**.** Maximise $f(x) = xe^{-x}$ over $x > 0$. Step 1: the product rule gives $f'(x) = (1-x)e^{-x}$, which vanishes only at $x^* = 1$. Step 2: the factor e^{-x} is positive, so f' changes sign from positive to negative at $x = 1$; alternatively, $f(x) \to 0$ as $x \downarrow 0$ and as $x \to \infty$, and $x = 1$ is the only interior critical point, so it is the maximum. Step 3: the maximiser is $x^* = 1$ with value $f(1) = e^{-1} \approx 0.368$.

Interview trap. The justification sentence in step (2) is graded. A stated critical point without the one-line argument for *why* it is a maximum, or an answer that reports x^* but not $f(x^*)$, leaves points on the table even when the calculus is flawless.

7.2 Integration: Substitution, Parts, Symmetry

Two named integrals anchor a large share of quant calculus: the Gaussian integral and the Gamma integral. This subsection proves both and then organises the three techniques — substitution, parts, symmetry — that evaluate almost everything an assessment asks.

Setup. Throughout this subsection x, u and t are real variables and n is a nonnegative integer. Integrals over unbounded ranges are improper Riemann integrals, evaluated as limits of integrals over bounded ranges (the convergence theory is developed in §7.3).

Theorem 7.5 (Gaussian integral)**.**

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}.$$

Proof. We use the polar-coordinates squaring trick. First, the integral converges: for $|x| \geq 1$ we have $x^2 \geq |x|$, hence $e^{-x^2} \leq e^{-|x|}$, and $\int_{-\infty}^{\infty} e^{-|x|}dx = 2$ is finite. Let I denote the integral; the integrand is positive, so $I > 0$. Since the name of the integration variable is immaterial,

$$I^2 = \Big(\int_{-\infty}^{\infty} e^{-x^2}dx\Big)\Big(\int_{-\infty}^{\infty} e^{-y^2}dy\Big) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy,$$

where writing the product as a double integral over the plane is justified by Tonelli's theorem for nonnegative integrands. Now change to polar coordinates with radius s, $x = s\cos\theta$, $y = s\sin\theta$ with area element $s\,ds\,d\theta$:

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-s^2}\,s\,ds\,d\theta.$$

The factor s is exactly the derivative factor a substitution wants: with $u = s^2$, $du = 2s\,ds$, the inner integral is $\frac{1}{2}\int_0^{\infty} e^{-u}du = \frac{1}{2}$. Hence $I^2 = 2\pi \cdot \frac{1}{2} = \pi$, and since $I > 0$ we conclude $I = \sqrt{\pi}$. □

Example 7.6 (Normalising the standard normal density)**.** The substitution $x = \sqrt{2}\,u$, $dx = \sqrt{2}\,du$, converts Theorem 7.5 into the normalisation of the standard normal density:

$$\int_{-\infty}^{\infty} e^{-x^2/2}\,dx = \sqrt{2}\int_{-\infty}^{\infty} e^{-u^2}\,du = \sqrt{2\pi} \approx 2.5066,$$

so $\varphi(x) = (2\pi)^{-1/2}e^{-x^2/2}$ integrates to exactly 1. Every "find the constant that makes this a density" question with a Gaussian kernel is this computation with different constants.

Theorem 7.7 (Gamma integral at the integers)**.** *For every integer $n \geq 0$,*

$$\int_0^{\infty} x^n e^{-x}\,dx = n!$$

(in Gamma-function notation, $\Gamma(n+1) = n!$).

Proof. We proceed by induction on n, with integration by parts supplying the inductive step.

Base case. For $n = 0$, $\int_0^\infty e^{-x} dx = \lim_{t\to\infty} (1 - e^{-t}) = 1 = 0!$.

Inductive step. Assume $\int_0^\infty x^n e^{-x} dx = n!$. Integrating by parts on $[0, t]$ with $u = x^{n+1}$ and $dv = e^{-x} dx$ (so $du = (n+1)x^n dx$, $v = -e^{-x}$),

$$\int_0^t x^{n+1} e^{-x}\, dx = \big[-x^{n+1}e^{-x}\big]_0^t + (n+1)\int_0^t x^n e^{-x}\, dx.$$

The boundary term vanishes in the limit: from the exponential series, $e^t \geq t^{n+2}/(n+2)!$ for $t > 0$, so $t^{n+1}e^{-t} \leq (n+2)!/t \to 0$ as $t \to \infty$. Letting $t \to \infty$ and applying the inductive hypothesis,

$$\int_0^\infty x^{n+1} e^{-x}\, dx = (n+1)\cdot n! = (n+1)!,$$

which completes the induction. □

Example 7.8 (Gamma integral in action)**.** Theorem 7.7 with $n = 3$ gives

$$\int_0^\infty x^3 e^{-x}\, dx = 3! = 6$$

in one line — no triple integration by parts required. Check by a single parts step: $\int_0^\infty x^3 e^{-x} dx = 3\int_0^\infty x^2 e^{-x} dx = 3 \cdot 2! = 6$. A rescaled variant follows from the substitution $u = 2x$:

$$\int_0^\infty x^2 e^{-2x}\, dx = \frac{1}{2^3}\int_0^\infty u^2 e^{-u}\, du = \frac{2!}{2^3} = 0.25,$$

the pattern developed systematically in Proposition 7.23.

Substitution: spot the g' factor. If the integrand contains $g(x)$ *and* a constant multiple of $g'(x)$, substitute $u = g(x)$. The screaming tell is a factor $x\,dx$ next to anything built from x^2. Two habits keep definite integrals honest: transform the limits immediately, and never return to the x variable.

Example 7.9 (The substitution reflex)**.** Consider $\int_0^\infty xe^{-x^2}\, dx$. The factor $x\,dx$ beside e^{-x^2} calls for $u = x^2$, $du = 2x\,dx$; the limits $x = 0$ and $x = \infty$ become $u = 0$ and $u = \infty$:

$$\int_0^\infty xe^{-x^2}\, dx = \frac{1}{2}\int_0^\infty e^{-u}\, du = \frac{1}{2},$$

where the final integral is the $n = 0$ case of Theorem 7.7.

Parts: u is what dies when differentiated. In $\int u\,dv$, choose as u the factor that simplifies under differentiation. The LIATE mnemonic orders the candidates: log, inverse trig, algebraic, trig, exponential — $\ln x$ is always u, and a power of x against e^x or $\cos x$ is u. The prototype is the antiderivative of the logarithm: with $u = \ln x$ and $dv = dx$,

$$\int \ln x\, dx = x\ln x - \int x \cdot \frac{1}{x}\, dx = x\ln x - x + C.$$

(Problem P7.5 asks you to reproduce this computation.) The reflex classics, each verifiable in ten seconds by differentiating the right-hand side:

$$\int x\cos x\, dx = x\sin x + \cos x + C, \qquad \int xe^x\, dx = (x-1)e^x + C,$$

$$\int x^2 e^x\, dx = e^x(x^2 - 2x + 2) + C, \qquad \int x\ln x\, dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C.$$

For $\int x^n e^x dx$ and its relatives, tabular integration (repeated parts laid out in a table) takes n passes for degree n.

Proposition 7.10 (Vanishing of odd integrals)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be odd, meaning $f(-x) = -f(x)$ for all x, and absolutely integrable, meaning $\int_{-\infty}^{\infty} |f(x)|\,dx < \infty$. Then*

$$\int_{-\infty}^{\infty} f(x)\,dx = 0.$$

The same conclusion holds for an odd integrable f on a bounded symmetric interval $[-a, a]$.

Proof. We substitute $x \mapsto -x$ on the negative half-line. With $u = -x$,

$$\int_{-\infty}^{0} f(x)\,dx = \int_{\infty}^{0} f(-u)\,(-du) = \int_{0}^{\infty} f(-u)\,du = -\int_{0}^{\infty} f(u)\,du,$$

using oddness in the last step. Absolute integrability guarantees that both half-line integrals are finite numbers, so their sum is legitimately

$$\int_{-\infty}^{\infty} f(x)\,dx = -\int_{0}^{\infty} f(u)\,du + \int_{0}^{\infty} f(x)\,dx = 0.$$

On a bounded interval the same substitution applies and finiteness is automatic. ☐

Example 7.11 (Symmetry with the caveat)**.** Consider $\int_{-\infty}^{\infty} \sin(2x)\,e^{-x^2}\,dx$. The integrand is odd times even, hence odd, and it is dominated in absolute value by e^{-x^2}, whose integral is $\sqrt{\pi}$ by Theorem 7.5; Proposition 7.10 therefore gives the value 0. The absolute-integrability hypothesis is not decoration: "odd, so zero" alone would equally "prove" $\int_{-\infty}^{\infty} x\,dx = 0$, yet that integral diverges (each half-line piece is infinite; only the symmetric principal value is zero). In an interview, state the caveat explicitly — it is the difference between a right answer and a right argument.

Interview trap. After *any* antiderivative, differentiate the result back — a ten-second check that catches most sign errors. And a positive integrand can never produce a negative or zero answer: if it does, something upstream is wrong. Both checks return in the improper integrals below.

7.3 Improper Integrals: the Convergence Decision

Converge-or-diverge questions are a reported assessment topic ♦, and the expected deliverable is a verdict plus a one-line justification — that is *full* credit. This subsection defines convergence, proves the tests that decide it (the two p-tests, direct comparison, limit comparison, and the logarithmic refinements), and ends with the decision procedure in speaking order.

Definition 7.12 (Improper integral and convergence)**.** Let f be continuous on $[a, \infty)$. The improper integral $\int_a^{\infty} f(x)\,dx$ *converges* to L if $\lim_{t\to\infty} \int_a^t f(x)\,dx = L$ exists finitely, and *diverges* otherwise. Similarly, if f is continuous on $(a, b]$ but unbounded near a, then $\int_a^b f(x)\,dx = \lim_{t\downarrow a} \int_t^b f(x)\,dx$ when the limit exists finitely. An integral whose domain contains several *problem points* — infinite endpoints, finite endpoints where the integrand is unbounded, and interior singularities — is split into pieces containing exactly one problem point each; the integral converges if and only if *every* piece converges.

Rule one is built into the definition: before any test, list every problem point — $\pm\infty$, endpoint blow-ups, *and* interior singularities — and split there. Missing an interior singularity is the classic fatal error; it reappears in the trap at the end of this subsection.

Proposition 7.13 (p-test at infinity)**.** *For real p, the integral $\int_1^{\infty} x^{-p}\,dx$ converges if and only if $p > 1$, in which case its value is $\frac{1}{p-1}$.*

Proof. We evaluate the antiderivative directly. For $p \neq 1$, an antiderivative of x^{-p} is $x^{1-p}/(1-p)$, so

$$\int_1^t x^{-p}\,dx = \frac{t^{1-p}-1}{1-p}.$$

If $p > 1$ then $1-p < 0$, so $t^{1-p} \to 0$ as $t \to \infty$ and the limit is $\frac{-1}{1-p} = \frac{1}{p-1}$, finite. If $p < 1$ then $t^{1-p} \to \infty$ and the integral diverges. In the boundary case $p = 1$, $\int_1^t x^{-1}dx = \ln t \to \infty$: logarithmic divergence. □

Proposition 7.14 (p-test at zero)**.** *For real p, the integral $\int_0^1 x^{-p}\,dx$ converges if and only if $p < 1$, in which case its value is $\frac{1}{1-p}$.*

Proof. We evaluate the same antiderivative at the other problem point. For $p \neq 1$,

$$\int_t^1 x^{-p}\,dx = \frac{1-t^{1-p}}{1-p}.$$

As $t \downarrow 0$, the term t^{1-p} tends to 0 precisely when $1-p > 0$, giving the value $\frac{1}{1-p}$; when $p > 1$ it blows up. At $p = 1$, $\int_t^1 x^{-1}dx = -\ln t \to \infty$: again logarithmic divergence. □

Intuition. At infinity the integrand must decay *fast*; at zero a *mild* blow-up is affordable. The function $1/x$ fails both tests — it is log-divergent at each end — and it is the exact boundary case of each: whenever $p = 1$ appears, the verdict is logarithmic divergence.

Example 7.15 (The p-tests with values)**.** The tail integral $\int_1^\infty x^{-3/2}dx$ has $p = \frac{3}{2} > 1$: it converges, with value $\frac{1}{3/2-1} = 2$ by Proposition 7.13. The endpoint integral $\int_0^1 x^{-1/2}dx$ has $p = \frac{1}{2} < 1$: it converges, with value $\frac{1}{1-1/2} = 2$ by Proposition 7.14.

Proposition 7.16 (Direct comparison test)**.** *Let f and g be continuous on $[a,\infty)$ with $0 \leq f(x) \leq g(x)$ for all $x \geq a$.*

(i) If $\int_a^\infty g$ converges, then $\int_a^\infty f$ converges, and $\int_a^\infty f \leq \int_a^\infty g$.
(ii) If $\int_a^\infty f$ diverges, then $\int_a^\infty g$ diverges.

The same statements hold on an interval whose problem point is a finite endpoint.

Proof. We use monotone boundedness. Define $F(t) = \int_a^t f(x)\,dx$. Since $f \geq 0$, the function F is nondecreasing in t. For every t,

$$F(t) \leq \int_a^t g(x)\,dx \leq \int_a^\infty g(x)\,dx,$$

the second inequality because $g \geq 0$ makes $t \mapsto \int_a^t g$ nondecreasing with the improper integral as its limit. A nondecreasing function that is bounded above converges, as $t \to \infty$, to its supremum, which is finite; hence $\int_a^\infty f = \lim_{t\to\infty} F(t)$ exists and is at most $\int_a^\infty g$. Statement (ii) is the contrapositive of (i). At a finite problem point the argument is identical with t tending to that point. □

Figure 11 shows the geometry: the region under f sits inside the region under g, so a finite total area for g forces a finite total area for f.

Example 7.17 (Comparison from below)**.** Consider $\int_1^\infty \frac{2+\sin x}{x}\,dx$. Since $\sin x \geq -1$, the integrand satisfies $\frac{2+\sin x}{x} \geq \frac{1}{x} \geq 0$ on $[1,\infty)$, and $\int_1^\infty \frac{dx}{x}$ diverges by Proposition 7.13 with $p = 1$. Part (ii) of Proposition 7.16 gives divergence.

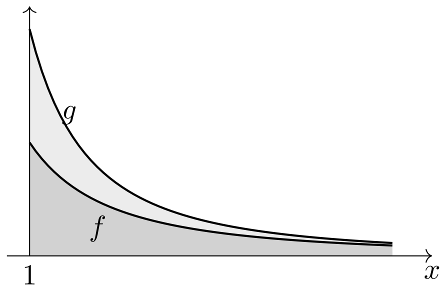


Figure 11: The direct comparison test (Proposition 7.16). Here $f(x) = \frac{1}{x(x+1)}$ (dark region) is dominated by $g(x) = x^{-2}$; the light band is the gap between the two graphs. The area under f is contained in the area under g, so a finite area under g forces a finite area under f.

Corollary 7.18 (Exponentials beat polynomials)**.** *For every real $k \geq 0$, every rate $\lambda > 0$, and every $a > 0$, both*

$$\int_a^\infty x^k e^{-\lambda x}\, dx \qquad \text{and} \qquad \int_a^\infty x^k e^{-\lambda x^2}\, dx$$

converge.

Proof. We compare against a pure exponential. Choose an integer $j > k$. For $x > 0$ the exponential series gives $e^{\lambda x/2} \geq (\lambda x/2)^j/j!$, hence

$$x^k e^{-\lambda x/2} \leq j!\,(2/\lambda)^j\, x^{k-j} \longrightarrow 0 \quad \text{as } x \to \infty,$$

so there is a constant B with $x^k e^{-\lambda x/2} \leq B$ for all $x \geq a$. Therefore

$$x^k e^{-\lambda x} = \left(x^k e^{-\lambda x/2}\right) e^{-\lambda x/2} \leq B\, e^{-\lambda x/2}, \qquad \int_a^\infty e^{-\lambda x/2}\, dx = \frac{2}{\lambda} e^{-\lambda a/2} < \infty,$$

and Proposition 7.16(i) gives convergence of the first integral. For the second, on $x \geq \max(a, 1)$ we have $\lambda x^2 \geq \lambda x$, so $x^k e^{-\lambda x^2} \leq x^k e^{-\lambda x}$ and the first case applies; the remaining piece over $[a, \max(a, 1)]$ is a proper integral of a continuous function. The displayed bound also shows the pointwise fact that a polynomial times $e^{-\lambda x}$ or $e^{-\lambda x^2}$ tends to zero at infinity; we reuse this when discarding boundary terms in integration by parts. □

Proposition 7.19 (Limit comparison test)**.** *Let f and g be continuous on $[a, \infty)$ with $f \geq 0$ and $g > 0$, and suppose*

$$\lim_{x \to \infty} \frac{f(x)}{g(x)} = c \in (0, \infty).$$

Then $\int_a^\infty f$ and $\int_a^\infty g$ either both converge or both diverge.

Proof. We run the epsilon argument for the limit and hand the result to the direct comparison test. Apply the definition of the limit with $\varepsilon = c/2 > 0$: there exists $M \geq a$ such that $|f(x)/g(x) - c| < c/2$ for all $x \geq M$. Multiplying through by $g(x) > 0$,

$$\frac{c}{2}\, g(x) \;<\; f(x) \;<\; \frac{3c}{2}\, g(x) \qquad \text{for all } x \geq M. \tag{7.1}$$

Each of the two improper integrals splits at M into a proper integral over $[a, M]$ (a continuous function on a compact interval, hence finite) plus a tail over $[M, \infty)$, so each converges if and only if its tail converges.

Suppose $\int_M^\infty g$ converges. Then so does $\int_M^\infty \frac{3c}{2} g$ (a constant multiple), and the right-hand inequality in (7.1) together with Proposition 7.16(i) gives convergence of $\int_M^\infty f$.

Suppose instead $\int_M^\infty g$ diverges. Then $\int_M^\infty \frac{c}{2} g$ diverges (a positive constant multiple of a divergent integral of a nonnegative function), and the left-hand inequality in (7.1) together with Proposition 7.16(ii) gives divergence of $\int_M^\infty f$.

The two cases together prove the claim. The same argument, run as x tends to a finite problem point, gives the analogous statement there. $\square$

In practice the limit comparison test is applied by replacing the integrand with its *asymptotic shape* — the simplest function with the same leading behaviour at the problem point — and it is usually faster than hunting for a pointwise domination.

Example 7.20 (Asymptotic shape in action)**.** (a) For $\int_1^\infty \frac{dx}{x(x+1)}$: at infinity the integrand behaves like x^{-2}, and indeed $\frac{1/(x(x+1))}{1/x^2} = \frac{x}{x+1} \to 1 \in (0, \infty)$. Since $\int_1^\infty x^{-2} dx$ converges ($p = 2$), Proposition 7.19 gives convergence. Here the value is also available: partial fractions give $\frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1}$, so

$$\int_1^T \frac{dx}{x(x+1)} = \ln T - \ln(T+1) + \ln 2 \longrightarrow \ln 2 \approx 0.693.$$

(b) For $\int_1^\infty \frac{x}{1+x^2}\, dx$: the asymptotic shape is $1/x$, and $\frac{x/(1+x^2)}{1/x} = \frac{x^2}{1+x^2} \to 1$. Since $\int_1^\infty \frac{dx}{x}$ diverges ($p = 1$), the integral diverges.

Proposition 7.21 (Logarithmic refinements)**.** *The integral $\int_2^\infty \frac{dx}{x \ln x}$ diverges, while $\int_2^\infty \frac{dx}{x(\ln x)^2}$ converges, with value $\frac{1}{\ln 2}$.*

Proof. We substitute $u = \ln x$, $du = dx/x$, which transforms both integrals into p-integrals; the limits $x = 2$ and $x = t$ become $u = \ln 2$ and $u = \ln t$. For the first integral,

$$\int_2^t \frac{dx}{x \ln x} = \int_{\ln 2}^{\ln t} \frac{du}{u} = \ln \ln t - \ln \ln 2 \longrightarrow \infty,$$

so it diverges (doubly logarithmically slowly). For the second,

$$\int_2^t \frac{dx}{x(\ln x)^2} = \int_{\ln 2}^{\ln t} \frac{du}{u^2} = \frac{1}{\ln 2} - \frac{1}{\ln t} \longrightarrow \frac{1}{\ln 2} \approx 1.4427.$$

$\square$

Intuition. The substitution $u = \ln x$ is the general device: it turns every $\int^\infty \frac{dx}{x(\ln x)^q}$ into the p-test with exponent q. The borderline $p = 1$ integrand $1/x$ thus acquires a whole ladder of refinements, and the first rung $1/(x \ln x)$ still diverges.

Interview trap. Three standing errors. (1) "The integrand tends to 0" proves *nothing*: $1/x$ tends to zero and diverges — the question is always *how fast*. (2) Never apply the fundamental theorem of calculus across an *interior* singularity. The canonical wrong answer is $\int_{-1}^1 x^{-2} dx = [-x^{-1}]_{-1}^1 = -2$: the integrand blows up at the interior point 0, so one must split there, and each half diverges (Proposition 7.14 with $p = 2$ after shifting the endpoint); moreover a positive integrand can never produce -2 — the positive-integrand check of §7.2. (3) Whenever the exponent $p = 1$ appears, the answer is logarithmic divergence.

The decision procedure, in speaking order. List the problem points and split. For a tail at infinity: compare to x^{-p} (need $p > 1$) or spot exponential decay (Corollary 7.18). For a finite blow-up at a: compare to $(x - a)^{-p}$ (need $p < 1$). For a messy integrand: limit-compare against its asymptotic shape (Proposition 7.19). If the integral converges and a value is asked: substitution, parts, partial fractions, symmetry, or a named result (Theorems 7.5 and 7.7). For a converge-or-diverge question, the verdict plus a one-line justification is full credit.

7.4 Moments as Integrals: the Probability Bridge

Moment computations are improper integrals in probability clothing, and the named integrals of §7.2 do all the heavy lifting. This subsection states the transfer principle (LOTUS) and derives the moment formulas — exponential, normal, uniform — that assessments expect on demand.

Throughout, X denotes a continuous random variable with density f_X (nonnegative, integrating to 1), and k and n denote nonnegative integers.

Fact 7.22 (Law of the unconscious statistician (LOTUS))**.** Let g be a function for which the integral below converges absolutely, that is, $\int |g(x)|\, f_X(x)\, dx < \infty$. Then

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx.$$

There is no need to find the density of $g(X)$ first. A full proof requires measure-theoretic machinery and is beyond our scope.

Proposition 7.23 (Exponential moments)**.** *Let $X \sim \mathrm{Exp}(\lambda)$ with rate $\lambda > 0$, that is, with density $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Then for every integer $n \geq 0$,*

$$\mathbb{E}[X^n] = \frac{n!}{\lambda^n}.$$

In particular $\mathbb{E}[X] = 1/\lambda$, $\mathbb{E}[X^2] = 2/\lambda^2$ (so $\mathrm{Var}(X) = 1/\lambda^2$), and $\mathbb{E}[X^3] = 6/\lambda^3$.

Proof. We substitute the integral into the Gamma form. By Fact 7.22 (the integrand is nonnegative, and finiteness follows from Corollary 7.18),

$$\mathbb{E}[X^n] = \int_0^{\infty} x^n \lambda e^{-\lambda x}\, dx.$$

Substituting $u = \lambda x$ (so $x = u/\lambda$, $dx = du/\lambda$, limits unchanged in form),

$$\mathbb{E}[X^n] = \int_0^{\infty} \frac{u^n}{\lambda^n}\, \lambda\, e^{-u}\, \frac{du}{\lambda} = \frac{1}{\lambda^n} \int_0^{\infty} u^n e^{-u}\, du = \frac{n!}{\lambda^n},$$

the last equality by Theorem 7.7. The variance follows: $\mathrm{Var}(X) = 2/\lambda^2 - (1/\lambda)^2 = 1/\lambda^2$. □

Example 7.24 (Exponential moments at rate two)**.** Let $X \sim \mathrm{Exp}(2)$. Then $\mathbb{E}[X] = 1/2 = 0.5$, $\mathbb{E}[X^2] = 2/4 = 0.5$, $\mathrm{Var}(X) = 0.5 - 0.25 = 0.25$, and $\mathbb{E}[X^3] = 6/8 = 0.75$.

Proposition 7.25 (Normal moments up to order four)**.** *Let $X \sim \mathcal{N}(0, \sigma^2)$ with density $\varphi_\sigma(x) = (2\pi\sigma^2)^{-1/2} e^{-x^2/(2\sigma^2)}$.*

(i) Every odd moment vanishes: $\mathbb{E}[X^k] = 0$ for odd k.
(ii) $\mathbb{E}[X^2] = \sigma^2$.
(iii) $\mathbb{E}[X^4] = 3\sigma^4$; equivalently, the kurtosis $\mathbb{E}[X^4]/\sigma^4$ equals 3.

Proof. Part (i) is symmetry: for odd k the integrand $x^k \varphi_\sigma(x)$ is an odd function (odd power times even density), and it is absolutely integrable by Corollary 7.18, so Proposition 7.10 gives 0.

For the even moments we use an integration-by-parts reduction. The density satisfies $\varphi_\sigma'(x) = -(x/\sigma^2)\, \varphi_\sigma(x)$, so $-\sigma^2 \varphi_\sigma$ is an antiderivative of $x\, \varphi_\sigma(x)$. For $k \geq 2$, write $x^k \varphi_\sigma = x^{k-1} \cdot (x\, \varphi_\sigma)$ and integrate by parts on $[-t, t]$ with $u = x^{k-1}$, $dv = x\, \varphi_\sigma(x)\, dx$:

$$\int_{-t}^{t} x^k \varphi_\sigma(x)\, dx = \Big[-\sigma^2 x^{k-1} \varphi_\sigma(x) \Big]_{-t}^{t} + (k-1)\, \sigma^2 \int_{-t}^{t} x^{k-2} \varphi_\sigma(x)\, dx.$$

The boundary term is a polynomial times a Gaussian and tends to zero as $t \to \infty$ (the pointwise bound in the proof of Corollary 7.18). Letting $t \to \infty$,

$$\mathbb{E}[X^k] = (k-1)\,\sigma^2\,\mathbb{E}[X^{k-2}] \qquad (k \geq 2). \tag{7.2}$$

The base of the reduction is $\mathbb{E}[X^0] = \int \varphi_\sigma = 1$: substituting $x = \sigma u$ reduces this to Example 7.6. Applying (7.2) with $k = 2$ and then $k = 4$:

$$\mathbb{E}[X^2] = 1 \cdot \sigma^2 \cdot 1 = \sigma^2, \qquad \mathbb{E}[X^4] = 3\sigma^2 \cdot \mathbb{E}[X^2] = 3\sigma^4.$$

The kurtosis is $3\sigma^4/\sigma^4 = 3$. □

(Problem P7.11 asks you to reproduce the fourth-moment derivation.)

Example 7.26 (Fourth moment of a return model)**.** Model a daily return as $X \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 0.02$. Then $\mathbb{E}[X^4] = 3\,(0.02)^4 = 4.8 \times 10^{-7}$, and the ratio $\mathbb{E}[X^4]/\sigma^4 = 3$ regardless of σ: under normality, kurtosis is always exactly 3.

Proposition 7.27 (Uniform moments)**.** *Let $U \sim \mathrm{U}(0,1)$. Then for every integer $k \geq 0$,*

$$\mathbb{E}[U^k] = \frac{1}{k+1}.$$

Proof. We integrate directly against the density, which is 1 on $[0,1]$: $\mathbb{E}[U^k] = \int_0^1 u^k\,du = \frac{1}{k+1}$. □

Example 7.28 (Variance of a squared uniform)**.** By Proposition 7.27, $\mathbb{E}[U^2] = \frac{1}{3}$ and $\mathbb{E}[U^4] = \frac{1}{5}$, so

$$\operatorname{Var}(U^2) = \mathbb{E}[U^4] - \left(\mathbb{E}[U^2]\right)^2 = \frac{1}{5} - \frac{1}{9} = \frac{4}{45} \approx 0.089.$$

Remark 7.29 (Normalisation questions)*.* Questions of the form “find c making $c\,h(x)$ a density” are improper-integral evaluations in probability clothing: first confirm $\int h$ converges (the tests of §7.3), then set $c = 1/\int h$. The evaluation is typically Theorem 7.7 or Theorem 7.5 after a substitution.

7.5 Taylor Expansions and Limits

Limit questions are usually expansion questions in disguise. The kit below covers nearly every limit an assessment asks; each entry is an instance of Taylor’s theorem, so we state them as facts.

Setup. Throughout this subsection x is a real variable, n a positive integer, and a, α fixed real constants; $O(x^k)$ as $x \to 0$ denotes a quantity bounded by a constant multiple of $|x|^k$ in a neighbourhood of 0.

Fact 7.30 (Standard expansions near zero (Taylor))**.** As $x \to 0$,

$$e^x = 1 + x + \frac{x^2}{2} + O(x^3), \qquad \ln(1+x) = x - \frac{x^2}{2} + O(x^3), \qquad (1+x)^\alpha = 1 + \alpha x + O(x^2),$$

$$\sin x = x + O(x^3), \qquad \cos x = 1 - \frac{x^2}{2} + O(x^4),$$

and, for fixed real a, $\left(1 + \frac{a}{n}\right)^n \to e^a$ as $n \to \infty$. Each expansion is an instance of Taylor’s theorem with remainder, and the compounding limit follows from the logarithm expansion by taking logs; we use all of them without proof.

Example 7.31 (A limit by expansion)**.** Compute $\lim_{x\to 0} \frac{1-\cos x}{x^2}$. By Fact 7.30, the numerator is $\frac{x^2}{2} + O(x^4)$, so the ratio is $\frac{1}{2} + O(x^2) \to \frac{1}{2}$. Numerical sanity check at $x = 0.1$: $(1-\cos 0.1)/0.01 = 0.49958$, already within 0.1 percent of the limit.

Interview trap. L'Hôpital's rule applies only to the indeterminate forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$ — check the form *before* differentiating, because applying the rule to a non-indeterminate form silently produces nonsense. And reach for the expansion first: it is usually faster and shows more structure (for instance the next-order term, which L'Hôpital hides).

7.6 Newton's Method: Convergence Rates

Root-finding convergence rates are a reported assessment topic ♦, and the graded distinction is between a *simple* root, where Newton's method is quadratic, and a *multiple* root, where it silently degrades to linear. This subsection proves both rates, using the Taylor expansions of §7.5 as the engine.

Throughout, f is a real function, r denotes a root ($f(r) = 0$), and $e_k = x_k - r$ denotes the error of the k-th iterate.

Definition 7.32 (Newton iteration)**.** Given a differentiable function f and a starting point x_0, the Newton iteration is

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}, \qquad k = 0, 1, 2, \ldots,$$

defined as long as $f'(x_k) \neq 0$. Geometrically, x_{k+1} is the point where the tangent line to the graph at $(x_k, f(x_k))$ crosses the horizontal axis; Figure 12 shows two steps.

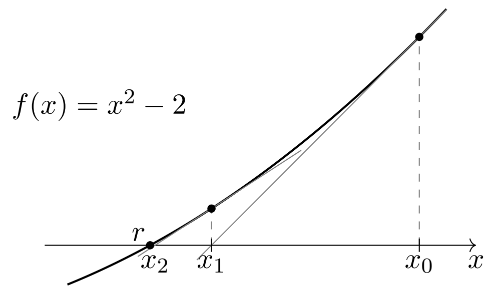


Figure 12: Two Newton steps on $f(x) = x^2 - 2$ from $x_0 = 2.6$. Each gray tangent line is followed to its axis crossing: $x_1 \approx 1.685$, then $x_2 \approx 1.436$. The filled dot on the axis marks the root $r = \sqrt{2} \approx 1.414$; after two steps the iterate almost coincides with it.

Theorem 7.33 (Quadratic convergence at a simple root)**.** *Let f be twice continuously differentiable on an open interval containing r, with $f(r) = 0$ and $f'(r) \neq 0$ (a* simple *root). Then there exist $\delta > 0$ and $C \geq 0$ such that for every starting point x_0 with $|x_0 - r| \leq \delta$, the Newton iterates are defined, remain within δ of r, converge to r, and satisfy*

$$|x_{k+1} - r| \leq C\,|x_k - r|^2 \qquad \textit{for all } k.$$

Consequently, once the error is small, the number of correct digits roughly doubles at every step.

Proof. We apply Taylor's theorem with Lagrange remainder and then trap the iterates.

Step 1: a safe interval. Since f' is continuous and $f'(r) \neq 0$, choose $\delta_0 > 0$ such that $I = [r - \delta_0,\, r + \delta_0]$ lies in the domain and $|f'(x)| \geq |f'(r)|/2 > 0$ for all $x \in I$. Set $K = \max_{x \in I} |f''(x)|$ and $C = K/|f'(r)|$.

Step 2: the one-step error identity. Suppose $x_k \in I$. Expanding f about x_k and evaluating at r, Taylor's theorem with Lagrange remainder gives, for some ξ_k between r and x_k,

$$0 = f(r) = f(x_k) - f'(x_k)\, e_k + \tfrac{1}{2} f''(\xi_k)\, e_k^2.$$

Dividing by $f'(x_k) \neq 0$ and rearranging, the Newton update satisfies

$$x_{k+1} - r \;=\; e_k - \frac{f(x_k)}{f'(x_k)} \;=\; \frac{f''(\xi_k)}{2f'(x_k)}\, e_k^2, \qquad \text{hence} \qquad |e_{k+1}| \le \frac{K}{2 \cdot |f'(r)|/2}\, e_k^2 = C\, e_k^2. \quad (7.3)$$

Step 3: trapping and convergence. If $K = 0$ then f'' vanishes on I, so by (7.3) the very first step lands exactly on r. Otherwise set $\delta = \min(\delta_0,\, 1/(2C))$. If $|e_k| \le \delta$, then by (7.3)

$$|e_{k+1}| \le C\,|e_k| \cdot |e_k| \le C\delta\,|e_k| \le \tfrac{1}{2}|e_k| \le \delta,$$

so the next iterate stays in the safe interval and the error at least halves. By induction $|e_k| \le 2^{-k}|e_0| \to 0$, and the quadratic bound in (7.3) holds at every step. For the digit count, take base-10 logarithms in (7.3): $-\log_{10}|e_{k+1}| \ge 2\big(-\log_{10}|e_k|\big) - \log_{10} C$, so the number of correct digits roughly doubles per step once it dominates $\log_{10} C$. □

Example 7.34 (Square root of two: digits doubling)**.** Take $f(x) = x^2 - 2$, whose positive root is $r = \sqrt{2} = 1.4142136$. The iteration is $x_{k+1} = x_k - (x_k^2 - 2)/(2x_k) = (x_k + 2/x_k)/2$. From $x_0 = 1.5$:

$$x_1 = 1.4166667, \qquad x_2 = 1.4142157,$$

with errors $|e_0| = 8.58 \times 10^{-2}$, $|e_1| = 2.45 \times 10^{-3}$, $|e_2| = 2.1 \times 10^{-6}$: roughly 1, then 3, then 6 correct digits — doubling per step. The ratios $|e_1|/e_0^2 = 0.33$ and $|e_2|/e_1^2 = 0.35$ approach $|f''(r)/(2f'(r))| = 1/(2\sqrt{2}) \approx 0.354$, exactly as the error identity (7.3) predicts.

Proposition 7.35 (Linear convergence at a multiple root)**.** *Fix an integer $m \ge 2$ and consider the model function $f(x) = (x - r)^m$. For every $x_k \neq r$, the Newton iterate satisfies*

$$x_{k+1} - r = \Big(1 - \frac{1}{m}\Big)(x_k - r):$$

convergence is only linear*, with the error multiplied by $\frac{m-1}{m}$ at every step (at a double root, the error merely halves). The modified iteration $x_{k+1} = x_k - m\, f(x_k)/f'(x_k)$ reaches r in a single step for this model.*

Proof. We compute the update directly. With $f(x) = (x - r)^m$ we have $f'(x) = m(x - r)^{m-1}$, so for $x_k \neq r$,

$$\frac{f(x_k)}{f'(x_k)} = \frac{(x_k - r)^m}{m(x_k - r)^{m-1}} = \frac{x_k - r}{m}, \qquad x_{k+1} - r = (x_k - r) - \frac{x_k - r}{m} = \Big(1 - \frac{1}{m}\Big)(x_k - r).$$

For the modified step, the subtracted quantity is $m \cdot (x_k - r)/m = x_k - r$, so $x_{k+1} = x_k - (x_k - r) = r$ exactly. □

Remark 7.36 (General multiplicity). A root of multiplicity m means $f(x) = (x - r)^m h(x)$ with h continuously differentiable and $h(r) \neq 0$. The model of Proposition 7.35 captures the leading behaviour: near such a root the error ratio tends to $\frac{m-1}{m}$, and when m is known, the modified step $x - mf/f'$ restores quadratic convergence. The practical diagnostic runs in reverse: unexpectedly slow, linear-looking Newton convergence is itself evidence of sitting on a multiple root.

Example 7.37 (Double root: errors halve)**.** Take $f(x) = (x - 1)^2$ from $x_0 = 2$. The iterates are $2,\ 1.5,\ 1.25,\ 1.125, \ldots$ with errors $1,\ 0.5,\ 0.25,\ 0.125$: the ratio is exactly $\frac{1}{2} = \frac{m-1}{m}$ for $m = 2$, as Proposition 7.35 predicts. The modified step with $m = 2$ jumps home immediately: $x_1 = 2 - 2\, f(2)/f'(2) = 2 - 2 \cdot \frac{1}{2} = 1$.

Interview trap. "Newton's method always converges quadratically" is the textbook overstatement: quadratic convergence requires a *simple* root and a sufficiently good start. When a function has several roots, the slowest is the one of *highest multiplicity.* Assessment instance ♦: for $f(x) = \tanh(x^4 - x^2)$, the roots are where $x^4 - x^2 = x^2(x - 1)(x + 1) = 0$, namely $x = 0$ and $x = \pm 1$. Near zero, $\tanh z = z + O(z^3)$ gives $f(x) = -x^2 + O(x^4)$: a double root, where Newton merely halves the error (Proposition 7.35) while it converges quadratically at the simple roots ± 1 (Theorem 7.33). The slowest root is therefore the one of smallest absolute value.

Practice: problems P7.1–P7.17 in §20, solutions in §21.

Part IV

Machine Learning and Deep Learning

8 ML Theory: Trees, Ensembles, Bias–Variance

This section develops the three results that anchor machine-learning theory questions in quant interviews: the decomposition of prediction error into bias, variance, and irreducible noise; the variance of an average of correlated predictors; and the reading of gradient boosting as gradient descent in function space. Decision trees are the running example throughout, because the two great ensemble designs — random forests and gradient boosting — are engineering responses to the two reducible terms of the first decomposition.

8.1 The Bias–Variance Decomposition

Almost every qualitative model-selection question — why a deep tree overfits, why averaging helps, why a deliberately biased estimator can beat an unbiased one — reduces to a single identity for the expected squared prediction error. This subsection states and proves that identity, with the cross-term cancellations argued in full, since those cancellations are precisely what an interviewer probes.

Setup. Fix a test point x_0. The data-generating model is

$$y = f(x_0) + \varepsilon,$$

where f is an unknown deterministic function, and the test noise ε satisfies $\mathbb{E}[\varepsilon] = 0$ and $\mathrm{Var}(\varepsilon) = \sigma^2$. A training set D is drawn at random (from whatever process generates training data), independently of ε, and a learning algorithm maps D to a fitted predictor $\hat{f}_D$. We abbreviate $\hat{f} := \hat{f}_D(x_0)$, a random variable whose randomness comes entirely from D. All expectations below are over the joint randomness of D and ε.

Definition 8.1 (Bias, variance, and irreducible error at a test point)**.** In the setup above, the *bias* of the predictor at x_0 is

$$\mathrm{Bias}(\hat{f}) = \mathbb{E}[\hat{f}] - f(x_0),$$

the systematic offset of the average fitted prediction from the truth; the *variance* is

$$\mathrm{Var}(\hat{f}) = \mathbb{E}\big[(\hat{f} - \mathbb{E}[\hat{f}])^2\big],$$

the fluctuation of the fitted prediction across training sets; and the noise variance σ^2 is called the *irreducible error*, since no predictor — however trained — can remove it.

Theorem 8.2 (Bias–variance decomposition)**.** *In the setup above, the expected squared prediction error at x_0 satisfies*

$$\mathbb{E}\big[(y - \hat{f})^2\big] = \mathrm{Bias}(\hat{f})^2 + \mathrm{Var}(\hat{f}) + \sigma^2. \tag{8.1}$$

Proof. We proceed by centering: we add and subtract $\mathbb{E}[\hat{f}]$ inside the square and expand the resulting three-term sum. Write

$$y - \hat{f} = \underbrace{\varepsilon}_{A} + \underbrace{(f(x_0) - \mathbb{E}[\hat{f}])}_{B} + \underbrace{(\mathbb{E}[\hat{f}] - \hat{f})}_{C},$$

where A is the test noise, B is a deterministic constant (it involves no randomness at all), and C is a mean-zero function of the training set: indeed $\mathbb{E}[C] = \mathbb{E}[\hat{f}] - \mathbb{E}[\hat{f}] = 0$. Expanding the square,

$$(y - \hat{f})^2 = A^2 + B^2 + C^2 + 2AB + 2AC + 2BC.$$

We take expectations term by term. The three squared terms give exactly the three terms of (8.1):

$$\mathbb{E}[A^2] = \mathrm{Var}(\varepsilon) = \sigma^2, \qquad B^2 = \big(f(x_0) - \mathbb{E}[\hat{f}]\big)^2 = \mathrm{Bias}(\hat{f})^2, \qquad \mathbb{E}[C^2] = \mathrm{Var}(\hat{f}),$$

where the middle equality holds because squaring removes the sign difference between B and $\mathrm{Bias}(\hat{f}) = -B$. It remains to show that each cross term vanishes, and the three cancellations have two distinct causes.

- $\mathbb{E}[AB] = B\,\mathbb{E}[\varepsilon] = 0$: the constant B factors out, and the noise has mean zero.
- $\mathbb{E}[AC] = \mathbb{E}[A]\,\mathbb{E}[C] = 0$: here $A = \varepsilon$ depends only on the test noise while C depends only on the training set D, and the two are independent by assumption, so the expectation factorizes; each factor is zero.
- $\mathbb{E}[BC] = B\,\mathbb{E}[C] = 0$: the constant factors out, and C has mean zero by construction — this is exactly what centering at $\mathbb{E}[\hat{f}]$ buys.

Summing the six expectations yields (8.1). □

(Problem P8.1 asks you to reproduce this proof from scratch.)

Example 8.3 (A shrinkage estimator beats the unbiased one)**.** Take a location model: the true value is $f(x_0) = 1$, the noise variance is $\sigma^2 = 1$, and the training set consists of $n = 4$ independent noisy observations of $f(x_0)$, with sample mean $\bar{y}$. Consider the family of shrinkage predictors $\hat{f}_\alpha = \alpha\bar{y}$ for $\alpha \in [0, 1]$. Then $\mathbb{E}[\hat{f}_\alpha] = \alpha$, so

$$\mathrm{Bias}(\hat{f}_\alpha)^2 = (1-\alpha)^2, \qquad \mathrm{Var}(\hat{f}_\alpha) = \alpha^2 \cdot \frac{\sigma^2}{n} = \frac{\alpha^2}{4},$$

and by Theorem 8.2 the risk is $R(\alpha) = (1-\alpha)^2 + \alpha^2/4 + 1$. Three instructive values:

$$R(1) = 0 + 0.25 + 1 = 1.25, \qquad R(0.8) = 0.04 + 0.16 + 1 = 1.20, \qquad R(0) = 1 + 0 + 1 = 2.$$

Setting $R'(\alpha) = -2(1-\alpha) + \alpha/2 = 0$ gives $\alpha^* = 0.8$: the optimal predictor deliberately accepts squared bias 0.04 to save 0.09 of variance, and beats the unbiased choice $\alpha = 1$ outright. The constant predictor $\alpha = 0$ shows the opposite failure: zero variance, but crippling bias. The irreducible $\sigma^2 = 1$ is untouched by every choice.

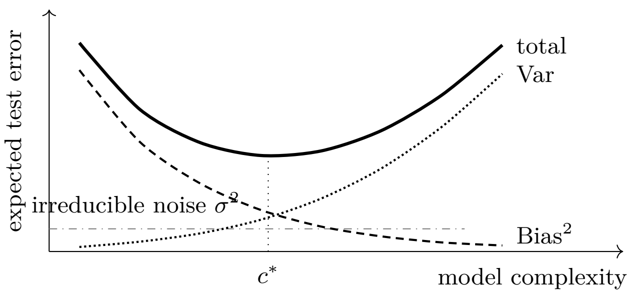


Figure 13: The bias–variance trade-off of Theorem 8.2 as model complexity grows: squared bias falls (dashed), variance rises (dotted), the irreducible noise σ^2 is a constant floor (dash-dotted), and their sum — the expected test error (solid) — is U-shaped, minimized at an interior complexity c^*.

Intuition. Figure 13 is the canonical picture behind (8.1). A more flexible model class tracks f more closely on average (bias falls) but its fit swings more from one training set to the next (variance rises); the noise floor never moves. Every regularisation device in this section — tree depth limits, averaging, shrinkage, early stopping — is a way of walking along this curve toward c^*.

Interview trap. When an interviewer asks for this derivation, the graded step is the cross-term argument, not the expansion. Asserting "the cross terms vanish" without a reason loses the points; the two reasons are distinct and must both be named: the terms containing ε die because the test noise is *independent of the training randomness* and has mean zero, while the remaining term dies because $\hat{f} - \mathbb{E}[\hat{f}]$ is *mean zero by centering* against a deterministic factor.

8.2 Decision Trees and Split Criteria

A decision tree grows by greedily choosing, at each node, the split that most reduces an impurity measure of the node's label distribution. Interviews test the two standard impurity measures on small numeric cases, and the bias–variance behaviour of trees of different depths — the property that motivates both ensemble designs of the next two subsections.

Setup. Consider a classification node holding n training samples distributed over K classes, with p_k the proportion of the node's samples in class k, so $\sum_{k=1}^{K} p_k = 1$. A candidate split partitions the node's samples into a left child L and a right child R holding n_L and n_R samples respectively, with $n_L + n_R = n$.

Definition 8.4 (Gini impurity, entropy, and impurity decrease)**.** The *Gini impurity* of the node is

$$G = 1 - \sum_{k=1}^{K} p_k^2,$$

and its *entropy*, measured in bits, is

$$H = -\sum_{k=1}^{K} p_k \log_2 p_k,$$

with the convention $0 \log_2 0 = 0$. (Entropy in natural-log units differs only by the constant factor $\log 2$ and ranks all splits identically.) For an impurity measure $I \in \{G, H\}$, the *impurity decrease* of the candidate split is the parent impurity minus the sample-weighted child impurity,

$$\Delta I = I_{\text{parent}} - \frac{n_L}{n} I_L - \frac{n_R}{n} I_R,$$

and the tree greedily selects the split maximizing ΔI; when $I = H$ the quantity ΔH is called the *information gain*. For regression trees the impurity of a node is the mean squared deviation of the node's targets from the node mean, and splits maximize the same weighted decrease (MSE reduction).

Example 8.5 (A ten-sample split)**.** A node holds 10 samples, 5 positive and 5 negative, and a candidate split sends 5 samples left (4 positive, 1 negative) and 5 right (1 positive, 4 negative); Figure 14 shows the split. The parent has $p = (0.5, 0.5)$, so

$$G_{\text{parent}} = 1 - 0.5^2 - 0.5^2 = 0.5, \qquad H_{\text{parent}} = -2 \cdot 0.5 \log_2 0.5 = 1 \text{ bit}.$$

Each child has proportions $(0.8, 0.2)$, hence

$$G_{\text{child}} = 1 - 0.8^2 - 0.2^2 = 1 - 0.64 - 0.04 = 0.32,$$

$$H_{\text{child}} = -0.8\log_2 0.8 - 0.2\log_2 0.2 \approx 0.8 \times 0.322 + 0.2 \times 2.322 \approx 0.722 \text{ bits}.$$

Both children hold half the samples, so the weighted child impurities equal the single-child values, and

$$\Delta G = 0.5 - 0.32 = 0.18, \qquad \Delta H = 1 - 0.722 = 0.278 \text{ bits}.$$

Both criteria would accept this split over any candidate with a smaller decrease.

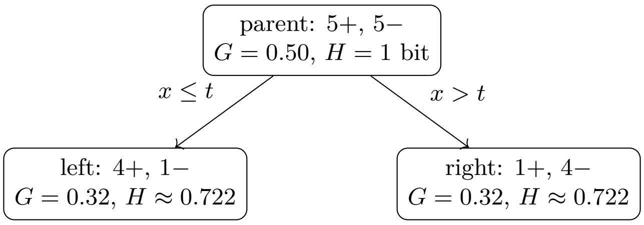


Figure 14: The split of Example 8.5: a threshold t on one feature partitions a mixed node into two purer children, for an impurity decrease of $\Delta G = 0.18$ (Gini) or $\Delta H \approx 0.278$ bits (information gain).

Intuition. Depth is the tree's complexity dial in the sense of Figure 13. A deep tree carves feature space finely and fits the training data almost exactly: low bias, *high variance* — a slightly different training set produces very different splits. A shallow tree is the reverse: stable but crude, high bias and low variance. Trees are therefore the canonical high-variance base learner, and this single property dictates both ensemble designs that follow: averaging many deep trees to cut variance (§8.3), or adding many shallow trees sequentially to cut bias (§8.4).

8.3 Bagging and Random Forests

Bagging fits many deep trees on bootstrap resamples of the training data and averages their predictions. Whether — and how much — averaging helps is settled by one covariance computation, whose limit explains the defining design choice of random forests.

Setup. At a fixed test point, let $X_1, \dots, X_N$ be the predictions of N models (for instance N trees fit on N bootstrap resamples), and let $\bar{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$ be the ensemble prediction. Assume the predictions are exchangeable with common variance $\mathrm{Var}(X_i) = v$ and common pairwise correlation ρ, so that $\mathrm{Cov}(X_i, X_j) = \rho v$ for all $i \neq j$. We write v for the variance of an individual model's prediction — a different quantity from the irreducible noise σ^2 of §8.1, so it gets its own symbol.

Proposition 8.6 (Ensemble variance and the correlation floor)**.** *In the setup above,*

$$\mathrm{Var}(\bar{X}) = \rho v + \frac{1-\rho}{N}\, v. \tag{8.2}$$

For $\rho < 1$ this is strictly decreasing in N, and for any fixed $\rho \in [0, 1)$,

$$\lim_{N\to\infty} \mathrm{Var}(\bar{X}) = \rho v.$$

Proof. We compute the variance of the sum by bilinearity of covariance and then rescale. Expanding,

$$\mathrm{Var}\Big(\sum_{i=1}^{N} X_i\Big) = \sum_{i=1}^{N} \mathrm{Var}(X_i) + \sum_{i\neq j} \mathrm{Cov}(X_i, X_j) = Nv + N(N-1)\,\rho v,$$

since the diagonal contributes N variance terms and there are exactly $N(N-1)$ ordered pairs (i,j) with $i \neq j$, each contributing ρv. Dividing by N^2,

$$\operatorname{Var}(\bar{X}) = \frac{v}{N} + \frac{N-1}{N}\,\rho v = \frac{v}{N} + \rho v - \frac{\rho v}{N} = \rho v + \frac{1-\rho}{N}\,v,$$

which is (8.2). Monotonicity is immediate: the first term does not depend on N, and for $\rho < 1$ the coefficient $(1-\rho)v > 0$ multiplies the strictly decreasing sequence $1/N$. The same observation gives the limit: the second term tends to zero, and the first term ρv survives. □

(Problem P8.4 asks you to reproduce this computation from scratch.)

Example 8.7 (The floor in numbers)**.** Take $v = 1$ and $\rho = 0.2$. By (8.2),

$$\operatorname{Var}(\bar{X}) = \begin{cases} 1.00 & N = 1, \\ 0.2 + 0.8/10 = 0.28 & N = 10, \\ 0.2 + 0.8/100 = 0.208 & N = 100, \end{cases}$$

with limit 0.2: the first ten trees do almost all the work, and no number of additional trees crosses the floor $\rho v = 0.2$. Now compare two stylised design choices. Plain bagging with highly correlated trees, say $\rho = 0.5$ and $v = 1$, has floor 0.5. Decorrelating the trees at the price of making each one individually worse — say $\rho = 0.2$ but $v = 1.2$ — gives floor $\rho v = 0.24$, and already at $N = 100$ the decorrelated ensemble achieves $0.24 + 0.96/100 \approx 0.25$, half the correlated ensemble's 0.505. Attacking ρ beats polishing the $1/N$ term.

Intuition. Correlation between learners is the floor on ensemble variance: as $N \to \infty$ the $1/N$ term in (8.2) dies and ρv survives, so averaging can never remove the shared component of the trees' errors. Bagging alone leaves trees highly correlated — fit on overlapping resamples, they keep chasing the same strong features near the root. This is why random forests additionally sample a random subset of features at *every split*: the restriction forces different trees to use different features, attacking ρ itself — which averaging cannot — at the acceptable price of slightly worse individual trees, as Example 8.7 quantifies.

8.4 Boosting and Gradient Boosting

The second ensemble design runs in the opposite direction: instead of averaging deep trees fit in parallel, boosting adds *shallow* trees *sequentially*, each fit to what the current ensemble still gets wrong. Gradient boosting (Friedman, 2001) makes this precise: the procedure is gradient descent in function space, and for squared loss the descent direction is literally the residual vector.

Setup. Fix a training sample $(x_1, y_1), \ldots, (x_n, y_n)$ and a per-sample loss $\ell(y, u)$, differentiable in its second argument (the prediction u). A candidate model F enters the training objective only through its values at the training points, so we identify F with the vector $\big(F(x_1), \ldots, F(x_n)\big) \in \mathbb{R}^n$ and write the empirical risk as

$$R(F) = \sum_{i=1}^{n} \ell\big(y_i, F(x_i)\big).$$

Base learners h are drawn from a class $\mathcal{H}$ of weak models, typically shallow trees, and $\eta \in (0,1]$ is a learning rate.

Definition 8.8 (Gradient boosting)**.** Initialize F_0 (for instance the constant minimizing R). At each round $t = 1, 2, \ldots$, compute the *pseudo-residuals*

$$g_i^{(t)} = -\left.\frac{\partial\, \ell(y_i, u)}{\partial u}\right|_{u = F_{t-1}(x_i)}, \qquad i = 1, \ldots, n,$$

fit a base learner $h_t \in \mathcal{H}$ to the targets $g^{(t)} = (g_1^{(t)}, \dots, g_n^{(t)})$ by least squares (a projection of the direction $g^{(t)}$ onto the base class), and update

$$F_t = F_{t-1} + \eta\, h_t. \tag{8.3}$$

Proposition 8.9 (Residuals are the negative gradient under squared loss)**.** *Take the squared loss $\ell(y,u) = \frac{1}{2}(y-u)^2$. Then the pseudo-residuals of Definition 8.8 are the ordinary residuals,*

$$g_i^{(t)} = y_i - F_{t-1}(x_i), \qquad i = 1, \dots, n,$$

and the vector $g^{(t)}$ equals $-\nabla R$ evaluated at the current value vector $(F_{t-1}(x_1), \dots, F_{t-1}(x_n))$. Consequently, "fit the next tree to the residuals" is steepest descent on R over $\mathbb{R}^n$, with each step projected onto the base class $\mathcal{H}$ and scaled by η.

Proof. We differentiate directly. For the squared loss,

$$\frac{\partial\, \ell(y,u)}{\partial u} = \frac{\partial}{\partial u}\, \tfrac{1}{2}(y-u)^2 = -(y-u),$$

so the pseudo-residual at sample i is

$$g_i^{(t)} = -\big[-(y_i - u)\big]_{u=F_{t-1}(x_i)} = y_i - F_{t-1}(x_i),$$

the residual. Because R is a sum of per-sample terms, each depending on a single coordinate $F(x_i)$, the gradient of R with respect to the value vector is the vector of per-sample derivatives, $\big(\partial\ell(y_i,u)/\partial u\big|_{u=F(x_i)}\big)_{i=1}^n$; its negative at F_{t-1} is exactly $g^{(t)}$. The update (8.3) changes the value vector by $\eta\,\big(h_t(x_i)\big)_{i=1}^n$, where h_t is the least-squares approximation to $g^{(t)}$ within $\mathcal{H}$ — a gradient step of length η taken in the closest direction the base class can represent. □

(Problem P8.6 asks you to reproduce this derivation from scratch.)

Example 8.10 (Two rounds of boosting on one observation)**.** Take a single training observation with $y = 8$, squared loss, initial model $F_0 = 0$, learning rate $\eta = 0.25$, and base learners rich enough to fit any target exactly (with $n = 1$, a single constant leaf suffices). By Proposition 8.9, each round fits the current residual $r_t = y - F_t$:

$$r_0 = 8, \quad F_1 = 0 + 0.25 \times 8 = 2; \qquad r_1 = 6, \quad F_2 = 2 + 0.25 \times 6 = 3.5; \qquad r_2 = 4.5.$$

In general each update (8.3) replaces the residual by $(1-\eta)$ times itself, so $r_t = 8 \times 0.75^t$ and $F_t = 8\,(1 - 0.75^t)$: geometric convergence to the target, which is exactly gradient descent with step size η on the quadratic $R(F) = \frac{1}{2}(8-F)^2$. With $\eta = 1$ one round would fit exactly — the point of shrinking η is that real base learners must also generalize away from the training points, and small steps plus early stopping keep the ensemble from fitting noise.

Intuition. Boosting attacks *bias*: each round fits precisely what the current ensemble still misses, so weak, shallow, high-bias trees suffice as base learners. Its variance is controlled by the shrinkage η, early stopping on a validation set, tree depth, and L1/L2 penalties on the leaf weights — the regularisation toolkit implemented in XGBoost and LightGBM.

Interview trap. "Would you rather have one very deep tree or many small ones?" The graded answer runs through the decomposition (8.1). One deep tree overfits: low bias, unchecked variance. *Many deep* trees averaged — a random forest — keep the low bias and cut the variance toward the floor of Proposition 8.6. *Many shallow* trees added sequentially — gradient boosting — start from high bias and remove it round by round (Proposition 8.9). The choice between the ensembles is which error component must be attacked, plus the tuning budget: forests are nearly tuning-free, boosting buys lower bias at the cost of the knobs listed above.

Trading application. Feature guidance for LightGBM and tree models generally, as posed in a live quantitative interview ★. Good tree features are monotone-informative and interaction-revealing raw quantities with low-cardinality encodings. Because a tree splits on thresholds, its splits are invariant to monotone transforms of a feature: scaling and normalisation are wasted effort. What trees do *not* get for free are relations *across* columns — engineer those explicitly as differences and ratios (spreads, imbalances, a price relative to its moving average). Two standing dangers: high-cardinality categorical features invite the tree to memorise identities rather than learn structure, and every feature must be constructed *point-in-time* — time-safe, using only information available at prediction time — or the backtest silently leaks the future.

Interview trap. "Give five examples of regularisation" expects breadth across both model worlds, not five variants of one idea. For neural networks: L1/L2 weight penalties, dropout, early stopping, data augmentation. For trees: limits on depth and leaf count, minimum child samples, row and feature subsampling, L1/L2 penalties on leaf weights, and early stopping on boosting rounds. An answer drawn entirely from one column signals a narrow toolkit.

Practice: problems P8.1–P8.7 in §20, solutions in §21.

9 Deep Learning

Deep-learning questions in quant interviews concentrate on a small set of mechanisms: how gradients are actually computed and why they vanish with depth, why attention scores carry a $1/\sqrt{d_k}$ factor, where the transformer's cost profile breaks down on high-frequency data, and a layer of engineering judgment about the numerical computing stack. The first two admit genuine mathematics and are proved below; the rest are engineering facts and are presented honestly as such, in structured remarks and applications.

9.1 Backpropagation and Vanishing Gradients

Backpropagation is the chain rule organised efficiently: gradients are computed layer by layer, back to front, and intermediate sensitivities are reused so that the entire gradient costs a small constant times one forward pass. This subsection works the mechanics explicitly on a two-layer scalar network, then isolates what goes wrong with depth — vanishing gradients — and why each standard architectural fix restores a usable gradient path.

Setup. Throughout this section the symbol σ denotes the logistic (sigmoid) activation function

$$\sigma(z) = \frac{1}{1+e^{-z}},$$

not a standard deviation; where a standard deviation is needed in this section it is written $\mathrm{Var}(\cdot)^{1/2}$. A *scalar chain network* of depth L maps an input $a_0 \in \mathbb{R}$ through

$$z_\ell = w_\ell \, a_{\ell-1}, \qquad a_\ell = \sigma(z_\ell), \qquad \ell = 1, \ldots, L,$$

where $w_1, \ldots, w_L$ are scalar weights, z_ℓ is the *pre-activation* of layer ℓ, and a_ℓ its *activation*. For the worked example we take depth $L = 2$ and attach the squared loss $\mathcal{L} = \frac{1}{2}(a_2 - y)^2$ with a fixed target y.

Lemma 9.1 (Sigmoid derivative bound)**.** *The sigmoid satisfies $\sigma'(z) = \sigma(z)\,(1 - \sigma(z))$ for every $z \in \mathbb{R}$, and consequently*

$$0 < \sigma'(z) \le \tfrac{1}{4},$$

with equality exactly at $z = 0$.

Proof. We differentiate directly and then complete the square. Writing $\sigma(z) = (1+e^{-z})^{-1}$ and applying the chain rule,

$$\sigma'(z) = \frac{e^{-z}}{(1+e^{-z})^2} = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}} = \sigma(z)\,(1-\sigma(z)).$$

Now set $u = \sigma(z) \in (0,1)$. Completing the square gives

$$u(1-u) = \tfrac{1}{4} - \left(u - \tfrac{1}{2}\right)^2 \le \tfrac{1}{4},$$

with equality if and only if $u = \frac{1}{2}$, that is, at $z = 0$; positivity holds because $u \in (0,1)$. $\square$

The next example is the whole of backpropagation in miniature: every partial derivative is written out, and Figure 15 displays the computational graph with the gradient flowing backward along it.

Example 9.2 (Backpropagation through a two-layer network)**.** Take input $a_0 = 1$, target $y = 0$, and weights $w_1 = 0$, $w_2 = 1$ in the depth-two network of the Setup.

Forward pass (left to right in Figure 15): the network evaluates

$$z_1 = w_1 a_0 = 0, \quad a_1 = \sigma(0) = 0.5, \quad z_2 = w_2 a_1 = 0.5, \quad a_2 = \sigma(0.5) = 0.6225,$$

and the loss is $\mathcal{L} = \frac{1}{2}(0.6225 - 0)^2 = 0.1937$. Every intermediate value is stored.

Backward pass (right to left): the chain rule is applied from the loss backward, one local factor at a time. Each partial is written out in full:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial a_2} &= a_2 - y = 0.6225,\\
\frac{\partial a_2}{\partial z_2} &= \sigma'(z_2) = a_2(1-a_2) = 0.6225 \times 0.3775 = 0.2350,\\
\delta_2 := \frac{\partial \mathcal{L}}{\partial z_2} &= \frac{\partial \mathcal{L}}{\partial a_2} \cdot \frac{\partial a_2}{\partial z_2} = 0.6225 \times 0.2350 = 0.1463,\\
\frac{\partial \mathcal{L}}{\partial w_2} &= \delta_2 \cdot \frac{\partial z_2}{\partial w_2} = \delta_2\, a_1 = 0.1463 \times 0.5 = 0.0731,\\
\frac{\partial \mathcal{L}}{\partial a_1} &= \delta_2 \cdot \frac{\partial z_2}{\partial a_1} = \delta_2\, w_2 = 0.1463 \times 1 = 0.1463,\\
\frac{\partial a_1}{\partial z_1} &= \sigma'(z_1) = a_1(1-a_1) = 0.5 \times 0.5 = 0.25,\\
\delta_1 := \frac{\partial \mathcal{L}}{\partial z_1} &= \frac{\partial \mathcal{L}}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1} = 0.1463 \times 0.25 = 0.0366,\\
\frac{\partial \mathcal{L}}{\partial w_1} &= \delta_1 \cdot \frac{\partial z_1}{\partial w_1} = \delta_1\, a_0 = 0.0366 \times 1 = 0.0366.
\end{aligned}$$

Two observations carry the whole subject. First, the sensitivity δ_2 was computed *once* and used *twice* — for the weight gradient $\partial\mathcal{L}/\partial w_2$ and to continue the backward sweep to a_1; this reuse of intermediates is what makes backpropagation cheap. Second, propagating one layer further back multiplied the sensitivity by $w_2\,\sigma'(z_1) = 0.25$, so $\delta_1 = 0.25\,\delta_2$: a single backward step through a sigmoid shrank the signal by a factor of four. Proposition 9.3 shows how this attenuation compounds with depth.

Intuition. Backpropagation is not a new differentiation rule; it is bookkeeping. The chain rule is organised back to front so that each layer's sensitivity δ_ℓ is computed once and reused by every parameter feeding that layer, which is why the full gradient costs a small constant times one forward pass. Differentiating naively — one perturbed forward pass per parameter, as finite differences would — pays the forward-pass cost once per parameter instead of once in total.

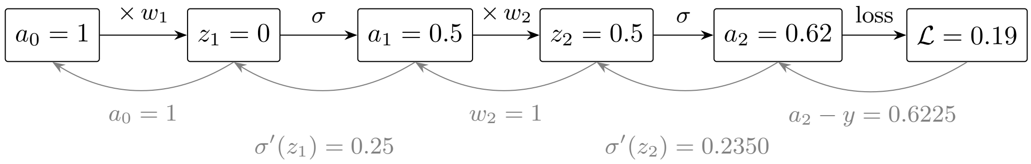


Figure 15: The computational graph of Example 9.2 (node values rounded to two decimals; the final edge applies the squared loss $\frac{1}{2}(a_2 - y)^2$). The forward pass (black, left to right) stores every intermediate value; the backward pass (gray, right to left) multiplies the running sensitivity by one local factor per edge. Weight gradients branch off the running product: $\partial\mathcal{L}/\partial w_2 = \delta_2\, a_1$ and $\partial\mathcal{L}/\partial w_1 = \delta_1\, a_0$.

Proposition 9.3 (Vanishing gradients in a deep sigmoid chain)**.** *Consider the scalar chain network of the Setup with depth $L \geq 1$ and $|a_0| \leq 1$. Then*

$$\frac{\partial a_L}{\partial w_1} = \Big[\prod_{\ell=2}^{L} w_\ell\, \sigma'(z_\ell)\Big]\, \sigma'(z_1)\, a_0, \tag{9.1}$$

a product containing one derivative factor $\sigma'(z_\ell)$ for every layer. If moreover $|w_\ell| \leq 1$ for all ℓ, then

$$\Big|\frac{\partial a_L}{\partial w_1}\Big| \leq 4^{-L} :$$

the gradient reaching the first layer decays exponentially in the depth.

Proof. We prove (9.1) by induction on the depth L. For the base case $L = 1$ we have $a_1 = \sigma(w_1 a_0)$, so the chain rule gives $\partial a_1/\partial w_1 = \sigma'(z_1)\, a_0$, which is (9.1) with the empty product equal to 1. For the inductive step, suppose (9.1) holds at depth $L - 1$. Since $a_L = \sigma(w_L\, a_{L-1})$ and the weight w_L does not depend on w_1, the chain rule gives

$$\frac{\partial a_L}{\partial w_1} = \sigma'(z_L)\, w_L\, \frac{\partial a_{L-1}}{\partial w_1},$$

and substituting the inductive hypothesis extends the product by the factor $w_L\, \sigma'(z_L)$, which is exactly (9.1) at depth L.

For the bound, Lemma 9.1 gives $|\sigma'(z_\ell)| \leq \frac{1}{4}$ for each of the L sigmoid factors in (9.1), while $|w_\ell| \leq 1$ for the $L - 1$ weight factors and $|a_0| \leq 1$. Taking absolute values and multiplying the bounds yields $|\partial a_L/\partial w_1| \leq (\frac{1}{4})^L = 4^{-L}$. □

(Problem P9.1 asks you to reproduce this derivation and bound from scratch.)

Example 9.4 (A depth-twenty chain)**.** For depth $L = 20$ the bound of Proposition 9.3 gives

$$\Big|\frac{\partial a_{20}}{\partial w_1}\Big| \leq 4^{-20} = 2^{-40} \approx 9.1 \times 10^{-13}.$$

For mental arithmetic: $2^{40} = (2^{10})^4 = 1024^4 \approx (10^3)^4 = 10^{12}$, so the bound is roughly 10^{-12}. At that scale the first layer receives essentially no learning signal at any learning rate for which the last layers remain stable: the early layers are frozen while the late layers train.

Intuition. Nothing in the mechanism is specific to the sigmoid. Any deep composition whose local derivative factors are uniformly below 1 in magnitude — say bounded by $c < 1$ — has gradients decaying like c^L, and factors uniformly above 1 produce the mirror image, exploding gradients. Saturating activations manufacture small factors by construction: their derivative is near zero on most of the input range, so deep products of such factors collapse exponentially.

Remark 9.5 (Architectural fixes and the gradient path each one preserves). Every standard fix attacks a specific factor of the product (9.1); naming *which* factor is what an interviewer listens for.

- *ReLU.* With $\mathrm{ReLU}(z) = \max(z, 0)$ the derivative is exactly 1 wherever the unit is active ($z > 0$), so the per-layer shrink factor of at most $\frac{1}{4}$ becomes 1: an active path passes the gradient through undiminished.
- *Residual connections.* A residual block computes $h + f(h)$, whose derivative $1 + f'(h)$ contains an identity term. Composed over depth, there is a gradient path built purely from identity terms — additive rather than multiplicative — so signal reaches early layers without traversing the full product of derivatives.
- *LSTM/GRU gating.* The cell state is updated *additively* through gates; when the forget gate sits near 1, gradient propagation through time is a near-identity, sum-like highway rather than a product of Jacobians at every step.
- *Batch normalisation.* Normalising pre-activations keeps each z_ℓ in the non-saturated region of the activation, so the factors $\sigma'(z_\ell)$ are not driven toward 0 in the first place.

9.2 Self-Attention

Scaled dot-product attention is the most frequently requested formula in modern deep-learning interviews, and the follow-up is nearly always the same: why divide by $\sqrt{d_k}$? The complete answer is a one-line variance computation (Proposition 9.7) combined with an understanding of softmax saturation — the same pathology as the saturated sigmoid factors behind Proposition 9.3.

Setup. Fix a sequence length T and head dimensions d_k (queries and keys) and d_v (values). The inputs are matrices $Q, K \in \mathbb{R}^{T\times d_k}$ and $V \in \mathbb{R}^{T\times d_v}$, whose rows q_i, k_j, v_j are the query, key, and value vectors of individual tokens. For a vector $z \in \mathbb{R}^T$ the softmax is defined componentwise by

$$\mathrm{softmax}(z)_i = \frac{e^{z_i}}{\sum_{j=1}^{T} e^{z_j}},$$

and applied to a matrix it acts row by row. The resulting attention weights are written α_{ij}.

Definition 9.6 (Scaled dot-product attention)**.** Given Q, K, V as in the Setup, *scaled dot-product attention* is the map

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}}\Big)\, V, \tag{9.2}$$

with the softmax applied row-wise. The (i, j) entry of $QK^\top$ is the *score* $q_i \cdot k_j$, measuring the match between query i and key j; row i of the output is the convex combination $\sum_{j=1}^{T} \alpha_{ij}\, v_j$ of the value vectors, with weights α_{ij} given by the softmax of the scaled scores.

Proposition 9.7 (Variance of an unscaled attention score)**.** *Let $q, k \in \mathbb{R}^{d_k}$, and suppose the $2d_k$ components $q_1, \dots, q_{d_k}, k_1, \dots, k_{d_k}$ are jointly independent, each with mean 0 and variance 1. Then the unscaled score $s = q \cdot k = \sum_{i=1}^{d_k} q_i k_i$ satisfies*

$$\mathbb{E}[s] = 0, \qquad \mathrm{Var}(s) = d_k.$$

Proof. We compute directly, using independence to factor every mixed moment. Each term has mean $\mathbb{E}[q_i k_i] = \mathbb{E}[q_i]\,\mathbb{E}[k_i] = 0$, so $\mathbb{E}[s] = 0$ by linearity of expectation. Each term's variance is

$$\mathrm{Var}(q_i k_i) = \mathbb{E}[q_i^2 k_i^2] - \big(\mathbb{E}[q_i k_i]\big)^2 = \mathbb{E}[q_i^2]\,\mathbb{E}[k_i^2] - 0 = 1 \cdot 1 = 1,$$

where the second equality uses independence of q_i and k_i and the fact that mean 0 and variance 1 force $\mathbb{E}[q_i^2] = \mathbb{E}[k_i^2] = 1$. For $i \neq j$ the terms are uncorrelated:

$$\mathrm{Cov}(q_i k_i,\ q_j k_j) = \mathbb{E}[q_i k_i q_j k_j] - 0 = \mathbb{E}[q_i]\,\mathbb{E}[k_i]\,\mathbb{E}[q_j]\,\mathbb{E}[k_j] = 0,$$

again by joint independence. The variance of a sum of uncorrelated terms is the sum of their variances, which is d_k. □

(Problem P9.2 asks you to reproduce this computation.)

Corollary 9.8 (Unit variance under $\sqrt{d_k}$ scaling)**.** *In the setting of Proposition 9.7, the scaled score $s/\sqrt{d_k}$ has mean* 0 *and variance* 1*, for every head dimension d_k.*

Proof. The claim follows from the scaling rule for variance. With $c = 1/\sqrt{d_k}$, linearity gives $\mathbb{E}[cs] = 0$, and $\mathrm{Var}(cs) = c^2\,\mathrm{Var}(s) = d_k/d_k = 1$ by Proposition 9.7. □

Unscaled scores therefore grow like $\sqrt{d_k}$ in standard deviation, while the scaled scores stay at unit scale regardless of head dimension. The next example shows what the unscaled growth does to the softmax.

Example 9.9 (Saturation at $d_k = 64$)**.** Take $d_k = 64$, a standard head dimension. By Proposition 9.7, unscaled scores have standard deviation $\sqrt{64} = 8$, so the *gap* between two competing logits — the difference of two independent scores — has standard deviation $8\sqrt{2} \approx 11.3$. For two logits the softmax reduces to a sigmoid of the gap: with $\Delta = z_1 - z_2$,

$$\mathrm{softmax}(z_1, z_2)_1 = \frac{e^{z_1}}{e^{z_1} + e^{z_2}} = \frac{1}{1 + e^{-\Delta}} = \sigma(\Delta),$$

the sigmoid of the Setup of §9.1. At a typical unscaled gap $\Delta \approx 11$,

$$\sigma(11) = \frac{1}{1 + e^{-11}} \approx 1 - 1.7 \times 10^{-5},$$

essentially one-hot: at initialisation, attention freezes onto single random tokens. After dividing by $\sqrt{d_k} = 8$, the gap has standard deviation $\sqrt{2} \approx 1.4$, and $\sigma(1.4) \approx 0.80$: weights like $(0.80, 0.20)$, a diffuse distribution that training can still reshape. Figure 16 shows both regimes on the softmax curve.

Remark 9.10 (Softmax saturation kills the gradient). The softmax Jacobian is $\partial\alpha_i/\partial z_j = \alpha_i(\mathbf{1}\{i = j\} - \alpha_j)$. At a near-one-hot output every entry is close to 0: for each pair (i, j), either $\alpha_i \approx 0$ or $\mathbf{1}\{i = j\} - \alpha_j \approx 0$ (diagonal entries $\alpha_i(1 - \alpha_i)$ die because α_i is near 0 or 1; off-diagonal entries $-\alpha_i\alpha_j$ die because at most one weight is non-negligible). Almost no gradient flows back to the scores, so the attention pattern cannot learn its way out of its random initial assignment — the same mechanism as the vanished sigmoid factors of Proposition 9.3, where a saturated squashing function contributes a near-zero derivative factor. Dividing by $\sqrt{d_k}$ keeps the score variance at 1 for every head dimension (Corollary 9.8), which keeps the softmax in its sensitive regime at initialisation.

9.3 Transformer Limitations (Tick-Data Question)

A recurring interview prompt asks why one should not run a transformer directly on raw tick data. A precise answer separates attention's memory cost from its compute cost, and training cost from streaming-inference cost. These are engineering facts about implementations rather than theorems, and they are stated as such. Throughout this subsection T denotes the attention sequence length, as in §9.2 (the symbol L already denotes network depth in this section).

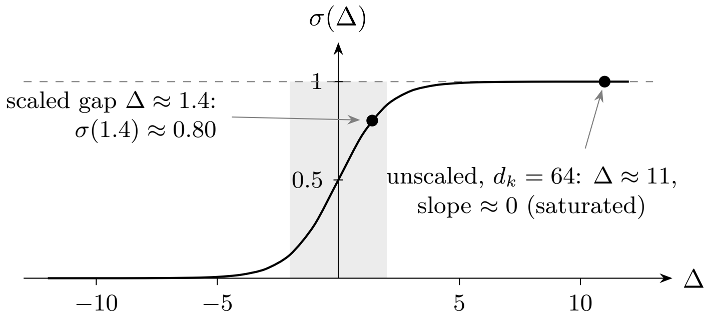


Figure 16: Softmax saturation for two competing logits: the larger logit's weight is $\sigma(\Delta)$, the sigmoid of the gap Δ (Example 9.9). The shaded band $|\Delta| \le 2$ is the sensitive regime where the curve has usable slope. Scaled scores (unit variance, Corollary 9.8) place typical gaps inside the band; unscaled scores at $d_k = 64$ place them near $\Delta \approx 11$, deep in the flat tail where the gradient through the softmax is essentially zero.

Remark 9.11 (What exactly is quadratic). Naive attention materialises the full $T \times T$ score matrix $QK^\top$ of (9.2): both memory and compute are $O(T^2)$ per head per layer. FlashAttention removes the memory problem, not the compute problem: it streams tiles of the score matrix through fast on-chip memory without ever materialising the whole matrix, reducing attention memory to $O(T)$ — but every query–key pair is still computed, so *compute remains* $O(T^2)$. The scale is easy to underestimate: a liquid instrument prints on the order of 5×10^5 ticks per day, so a full-day window means $T^2 = 2.5 \times 10^{11}$ score entries per head per layer — roughly 500 GB at two bytes per entry if materialised, far beyond any single accelerator's memory.

Interview trap. Claiming that "FlashAttention makes attention linear" is a half-truth that interviewers probe. FlashAttention linearises *memory* only; compute stays quadratic in the sequence length. Be precise about which resource each statement refers to, and state both halves.

Trading application. For tick data the quadratic profile is the wrong shape twice over. Training must attend over years of history at $O(T^2)$ compute per sequence; and streaming inference re-attends over a growing window each time a new tick arrives, so the per-tick cost grows with the session. The standard alternatives, in the order worth naming:

- *State-space models.* S4 trains in $O(T \log T)$ and Mamba in $O(T)$; both run recurrently at inference — a fixed-size state updated in $O(1)$ per step — which is the right computational shape for live trading.
- *Linear and sparse attention* variants, which cut the quadratic cost by approximating or restricting the score matrix.
- *The pragmatic first answer:* aggregate ticks into bars or engineered features before any sequence model sees the data, shrinking T by orders of magnitude while injecting domain knowledge.

9.4 "The Model Worked 2005–2018, Then Died"

Another open-ended standard: a systematic strategy performed well for over a decade, then stopped working. The question tests whether a candidate organises hypotheses systematically rather than guessing a single cause; the expected structure is the checklist below.

Remark 9.12 (Five failure modes of a dead strategy). Five distinct hypotheses explain a strategy that worked and then died, and a complete answer names all five:

1. *Non-stationarity / regime change* — the statistical relationship the model exploits weakened as the market changed.

2. *Overfitting all along* — the historical performance was in-sample luck; no real edge ever existed.
3. *Alpha decay* — the edge was real but has been arbitraged away as other participants found it.
4. *Look-ahead bias* — the backtest used information not available at the time, so the historical performance was never achievable live.
5. *Structural change* — regulation, new participants, or market-structure shifts broke the mechanism the model relied on.

Trading application. In an interview, a strong opening response commits to a diagnostic plan rather than a verdict: suspect regime change or overfitting first (Remark 9.12, items 1 and 2), check feature stationarity over time, re-run walk-forward validation, and test whether the model over-relies on patterns with no causal story.

9.5 numpy, PyTorch and GPUs

Numerical-stack questions ★ test whether a candidate understands why the standard tools are fast, one level below the API. The three standard questions — why numpy beats lists, why PyTorch exists at all, and what a GPU actually buys — are collected here as engineering facts.

Remark 9.13 (Why numpy arrays beat Python lists). A Python list of floats is an array of *pointers* to individually heap-allocated Python objects: every element is boxed with a type tag and a reference count, the elements live wherever the allocator placed them, and every operation passes through the interpreter's dynamic dispatch — the interpreter loop pays that dispatch cost on every single element. A numpy array is one *contiguous, typed* block of memory, and a vectorised operation on it runs as a compiled C loop with SIMD instructions over cache-friendly memory: no per-element boxing, no pointer chasing, no per-element reference counting, and no per-element interpreter dispatch. The speedup is a memory-layout and compiled-code story, not a parallelism story.

Remark 9.14 (Why PyTorch does not simply use numpy). Tensors carry three things numpy arrays lack. First, *autograd*: every operation records a node in the computational graph — exactly the structure drawn in Figure 15 — so the backward pass of Example 9.2 runs automatically over arbitrary programs. Second, *device placement*: the same tensor API executes on GPU through CUDA, with asynchronous kernel launches. Third, kernels *fused and optimised* for deep-learning workloads and shapes. The numpy library, by contrast, is CPU-only, records no graph, and executes synchronously.

Fact 9.15 (GPU versus CPU speedup, empirical)**.** For large parallel workloads — big matrix multiplications above all — a GPU delivers on the order of 10–100$\times$ over a CPU. The advantage shrinks or disappears for small, branchy, or memory-bound workloads, and host–device transfers over PCIe can consume the entire gain. These are measured engineering regularities, not derivable constants.

Interview trap. Quoting "10–100$\times$" without qualification. The expected answer volunteers the caveats unprompted: the range holds for large parallel workloads such as big matrix multiplications, and it shrinks or dies when the workload is small, branchy, or memory-bound — or when PCIe transfer time dominates the computation it feeds.

Practice: problems P9.1–P9.11 in §20, solutions in §21.

Part V

Trading, Portfolios, Market Making, and Market Structure

10 Sharpe, Loss Probabilities and Model Combination

This section connects the single most quoted performance statistic in systematic trading — the Sharpe ratio — to the quantity a risk manager actually cares about: the probability of losing money over a given horizon. We first derive the exact loss probability under a Gaussian model and establish how the Sharpe ratio scales with the observation horizon. We then work through a complete assessment-style loss calculation and audit every assumption behind it, quantify the model risk with Cantelli's distribution-free bound, and finish with the algebra of combining strategies: how variances add, why uncorrelated legs earn a $\sqrt{2}$ Sharpe improvement, and how to decorrelate a new model from an existing black box.

10.1 The Sharpe Ratio and the Probability of Loss

This subsection derives the identity $P(R < 0) = \Phi(-\mathrm{SR})$ and the square-root-of-time scaling law for the Sharpe ratio. Together they answer the two most common warm-up questions on strategy risk: "how often does a Sharpe-s strategy lose money?" and "why does the same strategy look so much worse on a monthly statement than on an annual one?"

Setup. Let the annual P&L of a strategy be a random variable R (measured in currency units; the scale plays no role) with

$$R \sim \mathcal{N}(\mu, \sigma^2), \qquad \mu \in \mathbb{R}, \quad \sigma > 0,$$

where μ is the annual mean and σ the annual volatility. Recall that Φ denotes the standard normal CDF and that a standardised normal variable $Z = (R - \mu)/\sigma$ satisfies $Z \sim \mathcal{N}(0, 1)$.

Definition 10.1 (Sharpe ratio)**.** The (annualised) *Sharpe ratio* of the strategy is

$$\mathrm{SR} = \frac{\mu}{\sigma}.$$

Throughout this document the ratio is taken directly as mean over volatility, with no risk-free adjustment; interview problems intend this convention unless they state a risk-free rate explicitly.

Theorem 10.2 (Loss probability under normality)**.** *Let $R \sim \mathcal{N}(\mu, \sigma^2)$ with $\sigma > 0$ and $\mathrm{SR} = \mu/\sigma$. Then*

$$P(R < 0) = \Phi(-\mathrm{SR}).$$

Proof. We standardise. Set $Z = (R - \mu)/\sigma$, so that $Z \sim \mathcal{N}(0, 1)$. Dividing by the positive constant σ preserves the inequality, so the events coincide:

$$\{R < 0\} = \Big\{\frac{R - \mu}{\sigma} < \frac{0 - \mu}{\sigma}\Big\} = \{Z < -\mathrm{SR}\}.$$

Taking probabilities and using the definition of the standard normal CDF gives $P(R < 0) = P(Z < -\mathrm{SR}) = \Phi(-\mathrm{SR})$. □

Setup for time scaling. Now let the strategy's P&L over each of T consecutive unit periods be $X_1, \dots, X_T$, assumed iid with per-period mean μ_1 and per-period standard deviation $\sigma_1 > 0$, and write $S_T = X_1 + \cdots + X_T$ for the cumulative P&L. The per-period Sharpe ratio is $\mathrm{SR}_1 = \mu_1/\sigma_1$.

Proposition 10.3 (Sharpe time-scaling under iid returns)**.** *Under the setup above,*

$$\mathbb{E}[S_T] = T\mu_1 \qquad \textit{and} \qquad \sqrt{\mathrm{Var}(S_T)} = \sigma_1\sqrt{T},$$

and hence the horizon-T Sharpe ratio is

$$\mathrm{SR}_T \;=\; \frac{\mathbb{E}[S_T]}{\sqrt{\mathrm{Var}(S_T)}} \;=\; \mathrm{SR}_1\sqrt{T}.$$

In particular, taking the month as the unit period and the year as $T = 12$ months, $\mathrm{SR}_{\mathrm{annual}} = \sqrt{12}\,\mathrm{SR}_{\mathrm{monthly}}$*, so a strategy with annual Sharpe ratio* SR *has monthly Sharpe ratio* $\mathrm{SR}_M = \mathrm{SR}/\sqrt{12}$*; and over a horizon of T years,* $\mathrm{SR}_T = \mathrm{SR}_{\mathrm{annual}}\sqrt{T}$*.*

Proof. We compute the mean by linearity of expectation and the variance by expanding the square. Linearity gives

$$\mathbb{E}[S_T] = \sum_{t=1}^{T} \mathbb{E}[X_t] = T\mu_1,$$

which holds with no independence assumption. For the variance, write $\bar{X}_t = X_t - \mu_1$ for the centred increments, so that $S_T - T\mu_1 = \sum_t \bar{X}_t$, and expand:

$$\begin{aligned}\mathrm{Var}(S_T) = \mathbb{E}\Big[\Big(\sum_{t=1}^{T} \bar{X}_t\Big)^2\Big] &= \sum_{t=1}^{T} \mathbb{E}[\bar{X}_t^2] + \sum_{s\neq t} \mathbb{E}[\bar{X}_s\bar{X}_t] \\ &= T\sigma_1^2 + \sum_{s\neq t} \mathrm{Cov}(X_s, X_t).\end{aligned}$$

Independence makes every cross term vanish, leaving $\mathrm{Var}(S_T) = T\sigma_1^2$ and $\sqrt{\mathrm{Var}(S_T)} = \sigma_1\sqrt{T}$. Dividing,

$$\mathrm{SR}_T = \frac{T\mu_1}{\sigma_1\sqrt{T}} = \sqrt{T}\,\frac{\mu_1}{\sigma_1} = \mathrm{SR}_1\sqrt{T}. \qquad \square$$

Example 10.4 (Loss frequencies at Sharpe one)**.** Consider a strategy with annual Sharpe ratio 1 and normally distributed, iid returns. By Theorem 10.2, the probability of a losing *year* is

$$\Phi(-1) \approx 15.9\%.$$

By Proposition 10.3, the monthly Sharpe ratio is $1/\sqrt{12} \approx 0.289$, so the probability of a losing *month* is

$$\Phi(-0.289) \approx 38.6\%.$$

The same strategy loses money in roughly one year in six but in nearly two months in five — nothing about the strategy has changed, only the clock on which it is read.

Intuition. Shrinking the horizon shrinks the drift linearly but the noise only as the square root, so the signal-to-noise ratio of the P&L degrades like $\sqrt{T}$. At short horizons the drift is invisible inside the noise and the loss frequency drifts toward $1/2$; as $T \to \infty$ it tends to 0. Watching a good strategy at too fine a time scale makes it feel like a coin flip.

Interview arithmetic in this area runs through a handful of standard normal lower-tail values. Table 2 collects the four worth memorising.

k	1	2	3	4
$\Phi(-k)$	15.9%	2.28%	0.135%	3.2×10^{-5}

Table 2: Normal lower-tail anchors (memorise).

10.2 A Worked Loss Probability and Its Assumption Audit

The identity of Theorem 10.2 turns a Sharpe ratio into a loss probability in one line, and assessment questions exploit exactly that. The example below is a real assessment question; the audit that follows it is the part of the answer that separates a complete response from a mere calculation.

Example 10.5 (Loss probability at Sharpe two ★)**.** A strategy has an annual Sharpe ratio of 2 and annual P&L volatility of \$50M. What is the probability of losing at least \$100M in a year?

Solution. Work in units of \$M and model the annual P&L as $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\sigma = 50$. Definition 10.1 recovers the mean from the two given numbers:

$$\mu = \mathrm{SR} \times \sigma = 2 \times 50 = 100.$$

A loss of at least \$100M is the event $\{X \leq -100\}$. Standardising as in the proof of Theorem 10.2, the threshold sits at

$$z = \frac{-100 - 100}{50} = -4$$

standard deviations, so by Table 2

$$P(X \leq -100) = \Phi(-4) \approx 3.2 \times 10^{-5}.$$

Figure 17 locates this threshold in the standard normal density.

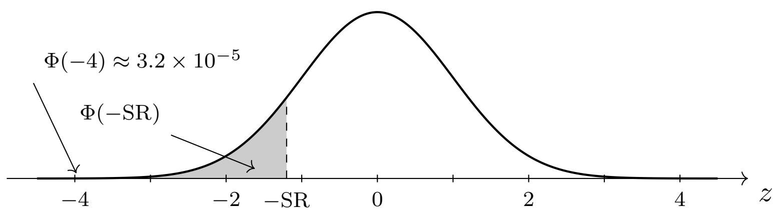


Figure 17: The standard normal density with the lower tail beyond $-\mathrm{SR}$ shaded (the threshold is drawn at $z = -1.2$ so the region is visible; for a Sharpe-1 strategy it would sit at -1). By Theorem 10.2 the shaded mass $\Phi(-\mathrm{SR})$ is the probability of a losing period. The mark at $z = -4$ locates the \$100M loss threshold of Example 10.5: essentially no Gaussian mass lies beyond it, which is exactly why the assumption audit of Remark 10.7 — and the distribution-free bound of Theorem 10.8 — matter.

The number $\Phi(-4)$ is only as good as the model behind it, and an interviewer will press on exactly that. One of the assumptions concerns what "losing \$100M" means: terminal P&L at year end, or the running P&L ever touching $-\$100$M during the year. The following classical result quantifies the difference.

Fact 10.6 (Reflection bound for path losses)**.** Let $(W_t)_{t \geq 0}$ be a driftless Brownian motion, $\sigma > 0$, and $a > 0$. By the reflection principle (classical, going back to Bachelier, 1900), the driftless P&L path σW_t satisfies

$$P\Big(\min_{0 \leq t \leq T} \sigma W_t \leq -a\Big) \;=\; 2\,P(\sigma W_T \leq -a) \;=\; 2\,\Phi\Big(\frac{-a}{\sigma\sqrt{T}}\Big):$$

the path probability is *exactly* twice the terminal one. Adding a positive drift $\mu > 0$ pushes the path probability strictly *above* twice the terminal probability: conditional on the running

P&L touching the loss level $-a$, the upward drift makes finishing below it less likely than $1/2$, so the path-to-terminal ratio exceeds two and can be arbitrarily large — the path-versus-terminal correction is even larger than the reflection factor of two. The exact formula for the drifted process $X_t = \mu t + \sigma W_t$ is

$$P\Big(\min_{0\le t\le T} X_t \le -a\Big) \;=\; \Phi\Big(\frac{-a-\mu T}{\sigma\sqrt{T}}\Big) \;+\; e^{-2\mu a/\sigma^2}\,\Phi\Big(\frac{-a+\mu T}{\sigma\sqrt{T}}\Big),$$

whose first term is the terminal probability $P(X_T \le -a)$. As a consistency check, at $\mu = 0$ the exponential factor equals 1 and both normal terms reduce to $\Phi(-a/(\sigma\sqrt{T}))$, so the formula collapses to exactly the driftless identity above. Proofs are beyond our scope.

Remark 10.7 (Demolishing the assumptions)*.* A complete answer to Example 10.5 names its assumptions and states how each one fails in reality:

1. *Normality.* Real P&L distributions are fat-tailed; the Gaussian badly underprices four-sigma events, so the true tail probability is higher than $\Phi(-4)$.
2. *Independence across time.* Returns exhibit autocorrelation in their scale — volatility clustering — so the iid aggregation implicit in the annual figure understates the tails.
3. *Constant volatility.* Volatility moves in regimes. Even if the P&L is conditionally normal within each regime, a scale mixture of normals is unconditionally fat-tailed.
4. *Stationarity of the Sharpe ratio itself.* The quoted $\mathrm{SR} = 2$ is an estimate, typically a selected one (the best of many strategies), and edges decay; the true mean is likely lower, moving the loss threshold closer than four sigmas.
5. *Path versus terminal.* If "losing \$100M" means the drawdown ever touching $-$\$100M during the year, the probability is strictly larger — for a driftless process exactly twice the terminal probability by Fact 10.6, and *more* than twice it under the positive drift of the example.

Every correction pushes in the same direction: the real loss probability is *higher* than $\Phi(-4)$.

10.3 Cantelli's Inequality: the Distribution-Free Bound

If the Gaussian answer of Example 10.5 is the optimistic end of the model spectrum, the pessimistic end is the best bound that uses *no* distributional assumption at all — only the mean and the variance. That bound is Cantelli's inequality, the one-sided sharpening of Chebyshev, and the gap between the two ends is the model-risk conversation the interviewer is fishing for.

Setup. Let X be any random variable with mean μ and finite variance $\sigma^2 \in (0,\infty)$; no other assumption is made on its distribution. Fix $k > 0$.

Theorem 10.8 (Cantelli's inequality, one-sided Chebyshev)**.** *Under the setup above,*

$$P(X-\mu \le -k\sigma) \;\le\; \frac{1}{1+k^2}.$$

Proof. We apply Markov's inequality to a shifted square and then optimise the shift. Let $Y = \mu - X$, so that $\mathbb{E}[Y] = 0$, $\mathrm{Var}(Y) = \sigma^2$, and the event of interest is $\{Y \ge k\sigma\}$.

Fix any $u \ge 0$. On the event $\{Y \ge k\sigma\}$ we have $Y + u \ge k\sigma + u > 0$, and squaring both sides of this inequality between positive quantities gives $(Y+u)^2 \ge (k\sigma+u)^2$. The variable $(Y+u)^2$ is nonnegative, so Markov's inequality applies:

$$P(Y \ge k\sigma) \;\le\; P\big((Y+u)^2 \ge (k\sigma+u)^2\big) \;\le\; \frac{\mathbb{E}[(Y+u)^2]}{(k\sigma+u)^2} \;=\; \frac{\sigma^2+u^2}{(k\sigma+u)^2}, \tag{10.1}$$

where the last equality expands $\mathbb{E}[(Y+u)^2] = \mathbb{E}[Y^2] + 2u\,\mathbb{E}[Y] + u^2 = \sigma^2 + u^2$, using $\mathbb{E}[Y] = 0$.

The bound (10.1) holds for every $u \geq 0$, so we may minimise the right-hand side over u. Differentiating with respect to u,

$$\frac{d}{du}\frac{\sigma^2+u^2}{(k\sigma+u)^2} = \frac{2u(k\sigma+u)^2 - 2(k\sigma+u)(\sigma^2+u^2)}{(k\sigma+u)^4} = \frac{2\,(uk\sigma-\sigma^2)}{(k\sigma+u)^3},$$

which is negative for $u < \sigma/k$ and positive for $u > \sigma/k$; the minimiser is $u^* = \sigma/k$. Substituting,

$$\frac{\sigma^2+\sigma^2/k^2}{(k\sigma+\sigma/k)^2} = \frac{\sigma^2\,(k^2+1)/k^2}{\sigma^2\,(k^2+1)^2/k^2} = \frac{1}{1+k^2}. \qquad \square$$

(Problem P10.3 asks you to reproduce this proof from scratch; it also shows the bound is tight, by exhibiting a two-point distribution that attains it, so no better distribution-free bound exists.)

Example 10.9 (The distribution-free ceiling at four sigma)**.** Return to the numbers of Example 10.5: annual mean P&L \$100M, volatility \$50M, distribution now *unknown.* Losing at least \$100M means $X \leq -100 = \mu - 4\sigma$, a $k = 4$ shortfall, so Theorem 10.8 gives

$$P(X-\mu \leq -4\sigma) \;\leq\; \frac{1}{1+16} = \frac{1}{17} \approx 5.9\%.$$

The Gaussian model gave $\Phi(-4) \approx 3.2 \times 10^{-5}$; the distribution-free ceiling is larger by over three orders of magnitude (a factor of roughly two thousand).

Trading application. The gap between 3.2×10^{-5} and 5.9% *is* the model-risk conversation. Neither number is "the answer": the Gaussian figure is the best case of one specific thin-tailed model, the Cantelli figure is the worst case over all distributions with that mean and volatility, and every number in between corresponds to a choice of tail assumption. In an interview, naming that gap — and saying which way reality leans, per Remark 10.7 — is the point of the question.

10.4 Combining Models: $\mathrm{Var}(A+B)$ and the Root-Two Rule

A desk rarely runs one strategy. This subsection establishes the variance algebra of running two strategies side by side and proves the classic diversification result: two uncorrelated legs of equal Sharpe and equal risk combine to a Sharpe $\sqrt{2}$ times as large.

Setup. In this section, A and B denote the P&Ls of two strategies over a common period — random variables with finite, strictly positive variances, *not* matrices (the document-wide matrix notation is suspended here; no matrix appears in this section). Write $\sigma_A^2 = \mathrm{Var}(A)$, $\sigma_B^2 = \mathrm{Var}(B)$, and $\rho = \mathrm{Corr}(A,B) = \mathrm{Cov}(A,B)/(\sigma_A\sigma_B)$.

Proposition 10.10 (Variance of a sum)**.** *For any random variables A and B with finite variances,*

$$\mathrm{Var}(A+B) \;=\; \mathrm{Var}(A) + \mathrm{Var}(B) + 2\,\mathrm{Cov}(A,B).$$

Proof. We expand the square and apply linearity of expectation. Write $\bar{A} = A - \mathbb{E}[A]$ and $\bar{B} = B - \mathbb{E}[B]$ for the centred variables, and note $A + B - \mathbb{E}[A+B] = \bar{A} + \bar{B}$ by linearity. Then

$$\begin{aligned}\mathrm{Var}(A+B) = \mathbb{E}\big[(\bar{A}+\bar{B})^2\big] &= \mathbb{E}[\bar{A}^2] + 2\,\mathbb{E}[\bar{A}\bar{B}] + \mathbb{E}[\bar{B}^2] \\ &= \mathrm{Var}(A) + 2\,\mathrm{Cov}(A,B) + \mathrm{Var}(B),\end{aligned}$$

where the middle equality expands the square inside the expectation and the last one applies the definitions of variance and covariance term by term. $\square$

Intuition. The covariance term is the whole game. Positive correlation makes risk superadditive: the two legs are partly the same bet, and running both buys redundancy. Negative correlation is diversification in its strongest form — hedging — and the combined risk can fall below that of either leg alone. Zero correlation is the neutral case in which risks add in quadrature, and it is where the cleanest Sharpe arithmetic lives.

Theorem 10.11 (Root-two Sharpe combination)**.** *Let A and B be uncorrelated ($\rho = 0$) with a common mean $\mu > 0$ and a common standard deviation $\sigma > 0$ — equal Sharpe ratios* $\mathrm{SR}_1 = \mu/\sigma$ and *equal volatilities. Then the combined book $A + B$ has Sharpe ratio*

$$\mathrm{SR}_{A+B} \;=\; \sqrt{2}\,\mathrm{SR}_1.$$

Proof. We compute the mean and the variance of the sum directly. Linearity of expectation gives $\mathbb{E}[A + B] = 2\mu$. Proposition 10.10 with $\mathrm{Cov}(A, B) = 0$ gives $\mathrm{Var}(A + B) = \sigma^2 + \sigma^2 = 2\sigma^2$, hence a combined volatility of $\sigma\sqrt{2}$. Dividing,

$$\mathrm{SR}_{A+B} = \frac{2\mu}{\sigma\sqrt{2}} = \sqrt{2}\,\frac{\mu}{\sigma} = \sqrt{2}\,\mathrm{SR}_1. \qquad \square$$

Corollary 10.12 (Quadrature rule for equal-risk uncorrelated legs)**.** *Let $A_1, \ldots, A_N$ be pairwise uncorrelated, each with mean $\mu > 0$ and standard deviation $\sigma > 0$. Then the book $A_1 + \cdots + A_N$ has Sharpe ratio*

$$\mathrm{SR}_N \;=\; \sqrt{N}\,\mathrm{SR}_1 \;=\; \sqrt{\textstyle\sum_{i=1}^{N} \mathrm{SR}_i^2},$$

the last equality holding because all the SR_i *equal* SR_1*.*

Proof. We repeat the argument of Theorem 10.11 with N summands. The mean is $N\mu$ by linearity. Expanding the square of the centred sum as in the proof of Proposition 10.3, all $N(N - 1)$ cross terms are covariances of distinct legs and vanish by pairwise uncorrelatedness, so the variance is $N\sigma^2$ and the volatility $\sigma\sqrt{N}$. Hence $\mathrm{SR}_N = N\mu/(\sigma\sqrt{N}) = \sqrt{N}\,\mu/\sigma$, and $\sqrt{N}\,\mathrm{SR}_1 = \sqrt{N\,\mathrm{SR}_1^2} = \sqrt{\sum_i \mathrm{SR}_i^2}$. $\square$

Example 10.13 (Two legs at three correlations)**.** Strategies A and B each have annual mean P&L \$20M and annual volatility \$20M (Sharpe 1 each). Working in \$M, Proposition 10.10 gives $\mathrm{Var}(A + B) = 400 + 400 + 2\rho \cdot 400 = 800(1 + \rho)$, while the combined mean is always 40.

- At $\rho = 0.6$: variance 1280, volatility $\sqrt{1280} \approx$ \$35.8M, Sharpe $40/35.8 \approx 1.12$ — redundancy, a modest gain over a single leg.
- At $\rho = 0$: variance 800, volatility $\approx$ \$28.3M, Sharpe $40/28.3 \approx 1.41 = \sqrt{2}$, confirming Theorem 10.11.
- At $\rho = -0.6$: variance 320, volatility $\approx$ \$17.9M — *below either leg's* \$20M — and Sharpe $40/17.9 \approx 2.24$: hedging with positive carry.

Remark 10.14 (Risk-balanced weights, not a raw sum)*.* For uncorrelated legs with *equal* Sharpe ratios but *unequal* volatilities, the quadrature formula $\mathrm{SR} = \sqrt{\sum_i \mathrm{SR}_i^2} = \sqrt{N}\,\mathrm{SR}_1$ requires risk-balanced weights, not a raw sum. A raw dollar sum weights each leg by its volatility: a leg with tiny volatility contributes almost nothing to the book's P&L, while a high-volatility leg dominates the book's risk. Scaling each leg to a common volatility (weights proportional to $1/\sigma_i$) restores the symmetric setting of Corollary 10.12, and only then does the quadrature formula hold. The equal-Sharpe hypothesis is essential: with unequal Sharpe ratios, the equal-risk book earns $\big(\sum_i \mathrm{SR}_i\big)/\sqrt{N}$, which by the Cauchy–Schwarz inequality is strictly below $\sqrt{\sum_i \mathrm{SR}_i^2}$ unless all the SR_i are equal; attaining the quadrature value in general requires allocating risk in proportion to each leg's Sharpe ratio — the mean–variance weights $w_i \propto \mu_i/\sigma_i^2$ — which coincide with $1/\sigma_i$ weights only in the equal-Sharpe case.

10.5 Decorrelating a New Model from a Black Box

Corollary 10.12 makes low correlation between models directly valuable, which raises a practical question: a desk already runs a profitable production model whose internals cannot be inspected, and a newly built model correlates substantially with it — how should the new model be decorrelated? This subsection gives the exact one-line answer, two alternatives, and the interview trap lurking next to them.

Setup. Let A denote the output (or P&L) of the production black-box model — observable, but with hidden internals — and let B denote the output of the new model. Both are random variables with finite variances, $\mathrm{Var}(A) > 0$, and $\rho = \mathrm{Corr}(A, B)$. The new model is trained on features $X_1, \ldots, X_p$, and its training loss is denoted $\mathcal{L}$.

Proposition 10.15 (Exact decorrelation by regression residual)**.** *Set*

$$\beta^* = \frac{\mathrm{Cov}(A, B)}{\mathrm{Var}(A)} \qquad \textit{and} \qquad B_\perp = B - \beta^* A.$$

Then $\mathrm{Cov}(B_\perp, A) = 0$*, and* β^* *is the unique coefficient with this property. Moreover*

$$\mathrm{Var}(B_\perp) \;=\; \mathrm{Var}(B) - \frac{\mathrm{Cov}(A, B)^2}{\mathrm{Var}(A)} \;=\; (1 - \rho^2)\,\mathrm{Var}(B).$$

Proof. We use bilinearity of covariance. For any coefficient β,

$$\mathrm{Cov}(B - \beta A,\, A) = \mathrm{Cov}(B, A) - \beta\,\mathrm{Var}(A),$$

which vanishes if and only if $\beta = \mathrm{Cov}(A, B)/\,\mathrm{Var}(A) = \beta^*$; this proves both the decorrelation and the uniqueness. For the variance, apply Proposition 10.10 to the sum $B + (-\beta^* A)$:

$$\mathrm{Var}(B_\perp) = \mathrm{Var}(B) + (\beta^*)^2\,\mathrm{Var}(A) - 2\beta^*\,\mathrm{Cov}(A, B) = \mathrm{Var}(B) - \frac{\mathrm{Cov}(A, B)^2}{\mathrm{Var}(A)},$$

substituting β^* and simplifying. Writing $\mathrm{Cov}(A, B)^2 = \rho^2\,\mathrm{Var}(A)\,\mathrm{Var}(B)$ gives the final form. □

Example 10.16 (Orthogonalising at correlation 0.7)**.** Normalise both model outputs to unit variance and suppose $\rho = 0.7$. Then $\beta^* = \mathrm{Cov}(A, B) = 0.7$, the residual model is $B_\perp = B - 0.7A$, and by Proposition 10.15

$$\mathrm{Var}(B_\perp) = 1 - 0.7^2 = 0.51,$$

a standard deviation of about 0.71. Nearly half of the new model's output variance was redundant exposure to the black box; what remains is exactly its incremental information, and by Theorem 10.11 it is the uncorrelated part that improves the book's Sharpe.

Three levers are available, in decreasing order of exactness:

1. *Orthogonalise the outputs.* Regress B on A and trade the residual $B_\perp$ of Proposition 10.15 — exact, and one line of algebra.
2. *Feature filtering.* Drop features X_j with high $\mathrm{Corr}(X_j, A)$ before training — crude, for the reasons the trap below makes precise.
3. *A correlation penalty in the loss.* Train B with $\mathcal{L} = \mathrm{MSE} + \lambda\,\mathrm{Corr}(A, B)^2$ — a continuous trade-off between standalone accuracy and orthogonality, tuned through λ.

Interview trap. "I dropped every feature correlated with A, then retrained. Thoughts?" Two independent failures: (i) the procedure discards genuinely predictive features — a feature can correlate with A's output and still carry incremental information A does not use; (ii) pairwise feature correlations do *not* bound the correlation of the trained model's combined output with A, so the procedure does not even achieve its stated goal. The cleaner answers are the exact one — orthogonalise the output — or the tunable one — constrain the loss.

Practice: problems P10.1–P10.6 in §20, solutions in §21.

11 Portfolio Theory: Mean–Variance Optimisation, Diversification, and the Capital Market Line

This section extends the two-strategy variance algebra of §10.4 — Proposition 10.10 and Theorem 10.11 — to n assets held in weights, following Markowitz (1952). It derives the minimum-variance and mean–variance-efficient portfolios by Lagrangian, shows that the efficient frontier is a hyperbola spanned by two funds, adds a risk-free asset to obtain the tangency (maximum-Sharpe) portfolio and the capital market line, reads the capital asset pricing model off the tangency algebra, decomposes portfolio risk into Euler contributions (risk parity), and closes with the reason the optimiser fails when fed estimated inputs. The $r_f = 0$ convention of Definition 10.1 is kept until §11.4, where a risk-free rate enters. Two running examples recur: a two-asset pair with volatilities 10% and 20% and means 5% and 10%, and a hand-invertible three-asset universe with means $(4\%, 7\%, 8\%)$ and $r_f = 2\%$, introduced in §11.2.

11.1 Portfolios, Weights and the Two-Asset Diversification Formula

The warm-up block at every quant fund is "write the variance of a two-asset portfolio", "why does diversification work", and the sharper follow-up "asset B has the same mean and twice the volatility of asset A — should you hold any of it?". This subsection fixes the notation for the whole section, proves the portfolio mean and variance formulas (the quadratic form of Theorem 6.6; the n-fold form of Proposition 10.10), and settles the two-asset case in closed form, including the degenerate lines at $\rho = \pm 1$.

Setup. There are $n \geq 2$ assets with one-period return vector $R = (R_1, \ldots, R_n)^\top$, each R_i having finite second moments. The mean vector is $\mu = \mathbb{E}[R] \in \mathbb{R}^n$ — a *vector* throughout this section, unlike the scalar μ of §10. The covariance matrix is $\Sigma \in \mathbb{R}^{n \times n}$ with entries

$$\Sigma_{ij} = \mathrm{Cov}(R_i, R_j) = \rho_{ij}\,\sigma_i\sigma_j, \qquad \sigma_i = \sqrt{\Sigma_{ii}} > 0,$$

where ρ_{ij} is the correlation of R_i and R_j (so $\rho_{ii} = 1$). By Theorem 6.6, Σ is symmetric positive semidefinite. *Standing assumption:* wherever Σ^{-1} appears, Σ is positive definite (PD) — by Definition 6.1 and the identity $x^\top \Sigma x = \mathrm{Var}(x^\top R)$ displayed in the proof of Theorem 6.6, this says that no nonzero combination $x^\top R$ of the assets is riskless. It excludes pairs with $|\rho_{ij}| = 1$ and redundant assets; for $n = 2$ it is the condition $\det \Sigma = \sigma_1^2\sigma_2^2(1 - \rho_{12}^2) > 0$. Remark 6.8 explains why an *estimated* Σ can fail even semidefiniteness. The all-ones vector is $\mathbf{1} \in \mathbb{R}^n$, and e_i is the i-th standard basis vector. The asset count is n, not N, because N is the random-walk target of the notation table and the leg count of Corollary 10.12.

Symbol notice for this section. The scalars A, B, C, D are reserved for the frontier constants of §11.3 (§10.4 used A, B for P&L random variables and §6 used A for a generic matrix; both were local there, and a generic positive-definite matrix is written G here). The letter m is a target mean (§12 uses it for a mid-price). Lagrange multipliers are θ_1, θ_2, because λ is reserved for eigenvalues, which §11.7 needs; in that subsection the multipliers of sign constraints form a vector ν, a shrinkage intensity is ζ, and a perturbation of the mean vector is δ. Risk aversion is γ. The mixing coefficient between two frontier portfolios is τ, so that α is reserved for Jensen's alpha of §11.5; a common Sharpe ratio is s_0. The risk-free rate r_f is set up formally in §11.4; before that it appears only in an informal arbitrage remark in the proof of Proposition 11.3(iv) and in Example 11.4. The symbol β is the population CAPM beta of §11.5, whose sample estimate is the $\hat{\beta}$ of §5.

Definition 11.1 (Portfolio, weights and portfolio return)**.** A *portfolio* is a weight vector $w \in \mathbb{R}^n$, entry w_i being the fraction of capital placed in asset i. It is *fully invested* if $\mathbf{1}^\top w = 1$, and *long-only* if additionally $w_i \geq 0$ for every i; short positions $w_i < 0$ are allowed unless "long-only" is stated. Its return is the random variable

$$R_w = w^\top R = \sum_{i=1}^{n} w_i R_i,$$

itself the return on capital when the portfolio is fully invested. The *Sharpe ratio* of a fully invested portfolio is

$$\mathrm{SR}(w) = \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}},$$

the $r_f = 0$ convention of Definition 10.1; Definition 11.20 in §11.4 generalises it to a risk-free rate.

Proposition 11.2 (Mean and variance of a portfolio)**.** *For every* $w \in \mathbb{R}^n$*,*

$$\mathbb{E}[R_w] = w^\top \mu \qquad \text{and} \qquad \mathrm{Var}(R_w) = w^\top \Sigma w,$$

and, written out,

$$\mathrm{Var}(R_w) = \sum_{i=1}^{n} w_i^2 \sigma_i^2 + \sum_{i \neq j} w_i w_j \, \rho_{ij} \sigma_i \sigma_j. \tag{11.1}$$

In particular $\mathrm{Var}(R_w) \geq 0$ *for every* w*.*

Proof. We use linearity of expectation for the mean and the quadratic-form identity of Theorem 6.6 for the variance. Linearity gives $\mathbb{E}[R_w] = \sum_i w_i \mathbb{E}[R_i] = \sum_i w_i \mu_i = w^\top \mu$. The identity $\mathrm{Var}(w^\top R) = w^\top \Sigma w$ is the display in the proof of Theorem 6.6 with $Z = R$ and $x = w$, and nonnegativity is the same display. Writing $w^\top \Sigma w = \sum_{i,j} w_i w_j \Sigma_{ij}$ and splitting the double sum into its diagonal terms ($i = j$, where $\Sigma_{ii} = \sigma_i^2$) and off-diagonal terms ($i \neq j$, where $\Sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$) gives (11.1). ☐

The case $n = 2$ is Proposition 10.10 applied to the two summands $w_1 R_1$ and $w_2 R_2$. Example 6.7 evaluates the same quadratic form for the spread $x = (1, -1)^\top$, a vector that does not sum to one: the formula does not need the budget constraint; only the Sharpe ratio of Definition 11.1, defined for fully invested portfolios, does.

Setup (two assets). Let $n = 2$ with volatilities $\sigma_1, \sigma_2 > 0$ and correlation $\rho = \rho_{12} \in [-1, 1]$. A fully invested portfolio is $(w, 1 - w)$ for a single real number w, and we write $\sigma_p^2(w)$ for its variance and $\mu_p(w) = w\mu_1 + (1 - w)\mu_2$ for its mean.

Proposition 11.3 (Two-asset variance, the minimum-variance mix, and the degenerate lines)**.** *In the two-asset setup:*

(i) The variance is

$$\sigma_p^2(w) = w^2\sigma_1^2 + (1 - w)^2\sigma_2^2 + 2w(1 - w)\, \rho\, \sigma_1\sigma_2, \tag{11.2}$$

a quadratic in w *whose leading coefficient* $V = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$ *equals* $\mathrm{Var}(R_1 - R_2)$ *and is strictly positive when* $|\rho| < 1$*. For fixed* $w \in (0, 1)$*, the variance is strictly increasing in* ρ*.*

(ii) If $|\rho| < 1$*, the variance has the unique minimiser*

$$w^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}, \qquad \sigma_p^2(w^*) = \frac{\sigma_1^2\sigma_2^2(1 - \rho^2)}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}.$$

(iii) Label the assets so that $\sigma_1 \leq \sigma_2$*. The higher-volatility asset receives positive weight (*$w^* \in (0, 1)$*) if and only if* $\rho < \sigma_1/\sigma_2$*. For* $\rho \geq \sigma_1/\sigma_2$ *the long-only minimum is asset* 1 *alone, while the unconstrained minimiser shorts asset* 2 *(or holds none of it, at equality).*

(iv) At $\rho = +1$, $\sigma_p(w) = |w\sigma_1 + (1-w)\sigma_2|$, which vanishes at the short position $w = \sigma_2/(\sigma_2 - \sigma_1)$ when $\sigma_1 \neq \sigma_2$. At $\rho = -1$, $\sigma_p(w) = |w\sigma_1 - (1-w)\sigma_2|$, which vanishes at the long-only weight $w = \sigma_2/(\sigma_1 + \sigma_2)$. In the ($\sigma$, mean) plane the attainable set (shorts allowed) is, at both degenerate correlations, a V with vertex on the $\sigma = 0$ axis: at $\rho = +1$ the vertex is the short position $w = \sigma_2/(\sigma_2 - \sigma_1)$, which lies outside $[0, 1]$, so the long-only mixes trace the segment joining the two asset points; at $\rho = -1$ the vertex is the long-only weight $w = \sigma_2/(\sigma_1 + \sigma_2)$ and lies between the asset points. When $\sigma_1 = \sigma_2$ and $\rho = +1$ the attainable set is the vertical line $\sigma = \sigma_1$. Every $|\rho| < 1$ traces a curve between the two degenerate cases (Figure 18).

Proof. We specialise Proposition 11.2 to two assets, then locate the minimum by differentiating a quadratic, and treat the degenerate correlations by completing the square.

Part (i). Formula (11.1) with $w_1 = w$, $w_2 = 1 - w$ has one diagonal term per asset and two equal off-diagonal terms, giving (11.2). Collecting powers of w,

$$\sigma_p^2(w) = Vw^2 - 2cw + \sigma_2^2, \qquad V = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2, \quad c = \sigma_2^2 - \rho\sigma_1\sigma_2. \tag{11.3}$$

Proposition 10.10 applied to R_1 and $-R_2$ gives $\operatorname{Var}(R_1 - R_2) = \sigma_1^2 + \sigma_2^2 - 2\operatorname{Cov}(R_1, R_2) = V$, and $V = (\sigma_1 - \sigma_2)^2 + 2(1-\rho)\sigma_1\sigma_2 > 0$ when $|\rho| < 1$ (the first summand is nonnegative and the second is strictly positive). The coefficient of ρ in (11.2) is $2w(1-w)\sigma_1\sigma_2$, positive for $w \in (0, 1)$, so the variance increases strictly with ρ there.

Part (ii). We differentiate (11.3): $\frac{d}{dw}\sigma_p^2(w) = 2Vw - 2c$, which vanishes at the single point $w^* = c/V$; the second derivative is $2V = 2\operatorname{Var}(R_1 - R_2) > 0$, so the stationary point is the strict global minimum of the quadratic. Writing c/V out gives the stated w^*. Substituting back, $\sigma_p^2(w^*) = \sigma_2^2 - c^2/V = (V\sigma_2^2 - c^2)/V$, and the numerator collapses in two lines:

$$\begin{aligned} V\sigma_2^2 - c^2 &= \sigma_1^2\sigma_2^2 + \sigma_2^4 - 2\rho\sigma_1\sigma_2^3 - (\sigma_2^4 - 2\rho\sigma_1\sigma_2^3 + \rho^2\sigma_1^2\sigma_2^2) \\ &= \sigma_1^2\sigma_2^2(1 - \rho^2). \end{aligned}$$

Part (iii). Since $V > 0$, $w^* < 1$ if and only if $c < V$, i.e. $\sigma_2^2 - \rho\sigma_1\sigma_2 < \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$, i.e. $\rho\sigma_1\sigma_2 < \sigma_1^2$, i.e. $\rho < \sigma_1/\sigma_2$. Likewise $w^* > 0$ if and only if $c > 0$, i.e. $\rho < \sigma_2/\sigma_1$; with $\sigma_1 \leq \sigma_2$ we have $\sigma_1/\sigma_2 \leq 1 \leq \sigma_2/\sigma_1$, so the second condition is automatic once the first holds. When $\rho \geq \sigma_1/\sigma_2$, we have $w^* \geq 1$, so $1 - w^* \leq 0$: the unconstrained minimiser holds a nonpositive position in asset 2. On the long-only range $[0, 1]$ the convex quadratic is decreasing on $[0, w^*] \supseteq [0, 1]$, so its minimum over $[0, 1]$ is at $w = 1$, asset 1 alone.

Part (iv). We complete the square. At $\rho = +1$, (11.2) reads $w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\sigma_1\sigma_2 = \big(w\sigma_1 + (1-w)\sigma_2\big)^2$; at $\rho = -1$ the cross term changes sign and the square is $\big(w\sigma_1 - (1-w)\sigma_2\big)^2$. Taking square roots gives the two absolute values. The first vanishes when $w(\sigma_1 - \sigma_2) = -\sigma_2$, i.e. $w = \sigma_2/(\sigma_2 - \sigma_1)$, which exceeds 1 when $\sigma_1 < \sigma_2$ and is negative when $\sigma_1 > \sigma_2$ (a short position either way); the second vanishes when $w\sigma_1 = (1-w)\sigma_2$, i.e. $w = \sigma_2/(\sigma_1 + \sigma_2) \in (0, 1)$. The mean $\mu_p(w)$ is affine in w and $\sigma_p(w)$ is the absolute value of an affine function of w with nonzero slope ($\sigma_1 - \sigma_2$ at $\rho = +1$ when $\sigma_1 \neq \sigma_2$; $\sigma_1 + \sigma_2$ at $\rho = -1$), so the image of $w \mapsto (\sigma_p(w), \mu_p(w))$ is piecewise linear with a single fold at the zero of the affine function, i.e. a V with vertex on the vertical axis in both cases. The fold lies outside $[0, 1]$ at $\rho = +1$, so the long-only mixes stay on one arm, the segment through the two asset points ($w = 1$ and $w = 0$); it lies inside $(0, 1)$ at $\rho = -1$. If $\sigma_1 = \sigma_2$ and $\rho = +1$ the affine function is the constant σ_1, giving the vertical line. At either value $\det\Sigma = \sigma_1^2\sigma_2^2(1 - \rho^2) = 0$, so Σ is singular, consistent with the standing assumption: a riskless combination exists, and if a risk-free rate r_f is available, any such combination whose mean differs from r_f is an arbitrage. □

Example 11.4 (Ten and twenty percent volatility, six correlations)**.** Take $\sigma_1 = 10\%$, $\sigma_2 = 20\%$ and $\mu = (5\%, 10\%)^\top$. First the equal-weight portfolio at $\rho = 0.3$, by Proposition 11.2: the mean is $0.5(0.05) + 0.5(0.10) = 7.5\%$, the plain average of the two means; the variance is

$$0.25(0.01) + 0.25(0.04) + 2(0.25)(0.3)(0.02) = 0.0025 + 0.0100 + 0.0030 = 0.0155,$$

a volatility of $\sqrt{0.0155} = 12.45\%$, against the weight-average of the volatilities, $0.5(10\%) + 0.5(20\%) = 15\%$. The mean is exactly the weighted average; the volatility is strictly below it.

Next the minimum-variance mix of Proposition 11.3 at several correlations, using $V = 0.05 - 0.04\rho$ and $c = 0.04 - 0.02\rho$:

- $\rho = 0.3$: $w^* = (0.04 - 0.006)/(0.05 - 0.012) = 0.034/0.038 = 0.8947$; minimum variance $0.0004 \times 0.91/0.038 = 0.009579$, volatility 9.79% — below asset 1's 10% — with mean $0.8947(5\%) + 0.1053(10\%) = 5.53\%$.
- $\rho = 0$: $w^* = 0.04/0.05 = 0.8$; variance $0.0004/0.05 = 0.008$, volatility 8.94%, mean 6.0%.
- $\rho = 0.5 = \sigma_1/\sigma_2$ exactly: $w^* = 0.03/0.03 = 1$, the threshold of part (iii); asset 1 alone.
- $\rho = 0.6$: $w^* = 0.028/0.026 = 1.077 > 1$; the unconstrained minimiser shorts the volatile asset (7.7% short, volatility 9.92%), and the long-only minimum is asset 1 alone at 10%.
- $\rho = -1$: part (iv) gives $w = 0.20/0.30 = 2/3$, volatility 0, mean $\frac{2}{3}(5\%) + \frac{1}{3}(10\%) = 6.67\%$; absence of arbitrage would force $r_f = 6.67\%$.
- $\rho = +1$: $w = 0.20/0.10 = 2$, i.e. long 200% of asset 1 and short 100% of asset 2, volatility 0, mean $2(5\%) - 10\% = 0$. This is the vertex of the V of part (iv), which sits at the origin: the long-only mixes lie on the arm mean $= 0.5\,\sigma$ through the two asset points, and positions beyond $w = 2$ trace the reflected arm mean $= -0.5\,\sigma$.

Two readings. The minimum-variance weight leans toward the lower-volatility asset and moves further toward it as ρ rises (0.80 at $\rho = 0$, 0.89 at $\rho = 0.3$, 1 at $\rho = 0.5$). Reducing variance moved the mean: 5.53% at $\rho = 0.3$ against 7.5% for the equal-weight mix — minimum variance and maximum Sharpe are different objectives, a point made precise in §11.4. The three-correlation pattern matches Example 10.13, the equal-volatility, equal-weight cousin of this computation.

Figure 18 draws the long-only mixes of Example 11.4 in the (σ, mean) plane at $\rho = 1$, $\rho = 0.3$ and $\rho = -1$.

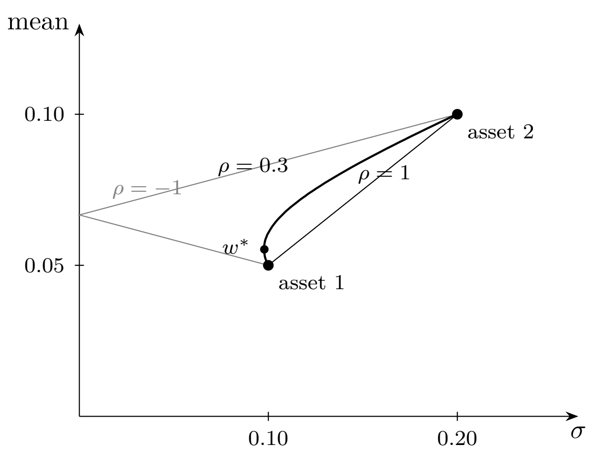


Figure 18: Long-only mixes of the two assets of Example 11.4 (volatilities 10% and 20%, means 5% and 10%) in the (σ, mean) plane. At $\rho = 1$ the long-only mixes lie on the segment joining the assets (the full attainable set is a V whose vertex, at the origin, needs a 200% position); at $\rho = 0.3$ they trace a curve bulging leftward, whose leftmost point w^* is the minimum-variance mix of Proposition 11.3(ii); at $\rho = -1$ they form a V touching the vertical axis at mean 6.67%, part (iv). Curves nest leftward as ρ falls. With two assets, every fully invested portfolio lies on its curve, so each curve is also the frontier of §11.3 for $n = 2$.

Intuition. The cross term is the only place correlation enters. The diagonal terms of (11.1) are what the assets contribute alone; the off-diagonal terms are what they contribute *together*, and they can be negative. Diversification is the observation that the off-diagonal terms are smaller than the diagonal ones whenever $\rho < 1$, so the whole is less volatile than the weighted sum of its parts — while the mean, being linear, is exactly the weighted sum of its parts.

Interview trap. Diversification does not lower expected return. Since $\mathbb{E}[R_w] = w^\top \mu$ is linear in w, a long-only mix has mean exactly the weighted average of the asset means (7.5% for the equal-weight mix in Example 11.4); only the volatility is subadditive, and only its cross term carries ρ. What moves the mean is the choice of weights, not the act of combining.

Interview trap. "Never hold an asset with the same mean and twice the volatility" is wrong whenever $\rho < \sigma_1/\sigma_2$: at $\rho = 0.3$ a 10.5% allocation to the 20%-volatility asset cuts the volatility from 10.0% to 9.79%. The correct test compares ρ with the volatility *ratio*, not one volatility with the other, and "diversification requires negative correlation" is equally false — $\rho = 0.3$ already diversifies. The companion arithmetic slip is quoting $w^* = \sigma_2^2/(\sigma_1^2+\sigma_2^2)$ as "the" minimum-variance weight: that is the $\rho = 0$ special case only (0.80 against 0.8947 at $\rho = 0.3$), and dropping the factor 2 on the cross term, or writing ρ where $\rho\sigma_1\sigma_2$ belongs, is the same error in different clothes; expanding $w^\top \Sigma w$ from (11.1) is safer than quoting a formula from memory.

11.2 The Global Minimum-Variance Portfolio and the Diversification Floor

"Derive the minimum-variance portfolio" is the standard Lagrangian derivation at quant funds, and "how does risk fall with the number of assets, and what stops it?" is the follow-up; AI-lab interviews ask the identical computation as ensemble variance. This subsection first proves one small lemma about positive-definite matrices that every later theorem reuses (the inverse is positive definite; Cauchy–Schwarz holds in the weighted inner product of Definition 6.9), then proves $w_g = \Sigma^{-1}\mathbf{1}/(\mathbf{1}^\top\Sigma^{-1}\mathbf{1})$ by Lagrangian together with a completing-the-square certificate — the certificate Remark 6.13 warns is missing from the naive Lagrangian — and shows that equal-weight diversification bottoms out at the average covariance, the formula of Proposition 8.6.

Setup. The covariance matrix Σ is positive definite (standing assumption of §11.1). The feasible set is $\{w \in \mathbb{R}^n : \mathbf{1}^\top w = 1\}$, an affine subspace. We write

$$A = \mathbf{1}^\top\Sigma^{-1}\mathbf{1},$$

a scalar shown positive in Lemma 11.5, in which G denotes a generic symmetric positive-definite matrix. The running three-asset universe, used through §11.5, is

$$\Sigma = 0.01 \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix},$$

in which every asset has volatility $\sqrt{0.02} = 14.14\%$, $\rho_{12} = \rho_{23} = 0.5$ and $\rho_{13} = 0$; its mean vector $\mu = (4\%, 7\%, 8\%)^\top$ is not needed until §11.3.

Lemma 11.5 (Inverse of a positive-definite matrix; weighted Cauchy–Schwarz)**.** *Let $G \in \mathbb{R}^{n\times n}$ be symmetric positive definite. Then:*

(i) The inverse G^{-1} exists, is symmetric, and is positive definite; in particular $x^\top G^{-1}x > 0$ for every $x \neq 0$.

(ii) For all $x, y \in \mathbb{R}^n$,

$$(x^\top Gy)^2 \ \le\ (x^\top Gx)(y^\top Gy),$$

with equality if and only if x and y are linearly dependent. This is the Cauchy–Schwarz inequality for the G-inner product $\langle x, y\rangle_G = x^\top Gy$ of Definition 6.9.

Proof. We argue directly from the definition of positive definiteness, without the spectral theorem.

Part (i). If $Gx = 0$ then $x^\top Gx = 0$, which forces $x = 0$ by positive definiteness; so G has trivial kernel and is invertible. Transposing $GG^{-1} = I$ gives $(G^{-1})^\top G^\top = I$, and $G^\top = G$, so $(G^{-1})^\top$ is also an inverse of G; inverses are unique, hence $(G^{-1})^\top = G^{-1}$. For $x \neq 0$ put $y = G^{-1}x$, which is nonzero because G^{-1} is invertible; then

$$x^\top G^{-1}x = (Gy)^\top y = y^\top G^\top y = y^\top Gy > 0.$$

Part (ii). We use the discriminant of a nonnegative quadratic. If $y = 0$ both sides vanish, so assume $y \neq 0$. For real t define

$$q(t) = (x - ty)^\top G(x - ty) = x^\top Gx - 2t\, x^\top Gy + t^2\, y^\top Gy,$$

using the symmetry of G to merge the two cross terms. Positive definiteness gives $q(t) \geq 0$ for every t, and the leading coefficient $y^\top Gy$ is strictly positive, so q is a genuine quadratic that never goes below zero: its discriminant is nonpositive,

$$4(x^\top Gy)^2 - 4(x^\top Gx)(y^\top Gy) \leq 0,$$

which is the inequality. If equality holds, the discriminant is zero and q has a real double root t_0 with $q(t_0) = 0$, i.e. $(x - t_0y)^\top G(x - t_0y) = 0$, so $x = t_0y$ by positive definiteness. Conversely, if $x = ty$ then both sides equal $t^2(y^\top Gy)^2$. □

The lemma is used with $G = \Sigma^{-1}$ to prove that the frontier constant D of §11.3 is positive, and with $G = \Sigma$ to prove the tangency theorem of §11.4.

Theorem 11.6 (Global minimum-variance portfolio)**.** *Under the setup above:*

(i) The problem $\min_w w^\top \Sigma w$ *subject to* $\mathbf{1}^\top w = 1$ *has the unique solution*

$$w_g = \frac{\Sigma^{-1}\mathbf{1}}{A}, \qquad A = \mathbf{1}^\top\Sigma^{-1}\mathbf{1} > 0, \tag{11.4}$$

with minimum variance $\sigma_g^2 = w_g^\top \Sigma w_g = 1/A$.

(ii) The minimum-variance portfolio satisfies $\Sigma w_g = (1/A)\,\mathbf{1}$*: every asset has the same covariance* $1/A$ *with* w_g*, equivalently — in the language of Definition 11.32 below — beta* 1 *to it.*

(iii) If $\Sigma = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)$*, then*

$$w_{g,i} = \frac{\sigma_i^{-2}}{\sum_{j=1}^n \sigma_j^{-2}}, \qquad \sigma_g^2 = \frac{1}{\sum_{j=1}^n \sigma_j^{-2}},$$

the inverse-variance (precision) weights; the minimum variance is the parallel-resistor combination of the variances and lies strictly below the smallest σ_j^2 *when* $n \geq 2$.

Proof. We proceed by the Lagrangian to locate the candidate, then certify optimality by completing the square — the honesty point of Remark 6.13, which observes that stationarity alone does not identify a minimum.

Gradient of the quadratic form. Differentiating $w^\top \Sigma w = \sum_{j,k} w_j \Sigma_{jk} w_k$ with respect to w_i picks up the terms with $j = i$ and the terms with $k = i$:

$$\frac{\partial}{\partial w_i}(w^\top \Sigma w) = \sum_k \Sigma_{ik} w_k + \sum_j w_j \Sigma_{ji} = 2(\Sigma w)_i, \tag{11.5}$$

using $\Sigma_{ji} = \Sigma_{ij}$. Positivity of A is Lemma 11.5(i) with $G = \Sigma$ and $x = \mathbf{1} \neq 0$.

Candidate. The Lagrangian is $L(w, \theta) = \frac{1}{2} w^\top \Sigma w - \theta(\mathbf{1}^\top w - 1)$. By (11.5), stationarity in w reads $\Sigma w = \theta\mathbf{1}$, so $w = \theta\Sigma^{-1}\mathbf{1}$; the budget constraint gives $1 = \theta\,\mathbf{1}^\top\Sigma^{-1}\mathbf{1} = \theta A$, hence $\theta = 1/A$ and $w = w_g$.

Certificate. Let w be any feasible portfolio and write $w = w_g + d$; feasibility of both w and w_g gives $\mathbf{1}^\top d = 0$. Expanding the quadratic form,

$$w^\top \Sigma w = w_g^\top \Sigma w_g + 2\, d^\top \Sigma w_g + d^\top \Sigma d.$$

By the stationarity equation, $\Sigma w_g = (1/A)\mathbf{1}$, so $d^\top \Sigma w_g = (1/A)\, \mathbf{1}^\top d = 0$, and

$$w^\top \Sigma w = w_g^\top \Sigma w_g + d^\top \Sigma d \;\geq\; w_g^\top \Sigma w_g,$$

with equality if and only if $d = 0$ by positive definiteness. This is strict convexity of the objective on the affine feasible set, and it is why stationarity suffices here, unlike on the sphere constraint of Theorem 6.12. The value is

$$w_g^\top \Sigma w_g = \frac{1}{A^2}\, \mathbf{1}^\top \Sigma^{-1} \Sigma \Sigma^{-1} \mathbf{1} = \frac{1}{A^2}\, \mathbf{1}^\top \Sigma^{-1} \mathbf{1} = \frac{1}{A}.$$

Part (ii) is the stationarity equation at $\theta = 1/A$. The covariance of asset i with R_{w_g} is $\mathrm{Cov}(R_i, w_g^\top R) = e_i^\top \Sigma w_g = 1/A$ by bilinearity, and dividing by $\mathrm{Var}(R_{w_g}) = 1/A$ gives a ratio of 1, which §11.5 names the beta of asset i to w_g.

Part (iii). With Σ diagonal, $\Sigma^{-1} = \mathrm{diag}(\sigma_1^{-2}, \dots, \sigma_n^{-2})$, so $\Sigma^{-1}\mathbf{1}$ has entries σ_i^{-2} and $A = \sum_j \sigma_j^{-2}$; substituting into (11.4) gives the weights and $\sigma_g^2 = 1/A$. Since every summand of A is positive, $A > \sigma_j^{-2}$ for each j when $n \geq 2$, i.e. $1/A < \sigma_j^2$. $\square$

Remark 11.7 (Precision weights are the best linear unbiased weights). Combining independent unbiased estimates $X_1, \dots, X_n$ of one quantity, with variances $\sigma_1^2, \dots, \sigma_n^2$, by a linear rule $\sum_i w_i X_i$ that stays unbiased ($\sum_i w_i = 1$) and has the smallest variance is the problem $\min \sum_i w_i^2 \sigma_i^2$ subject to $\mathbf{1}^\top w = 1$ — Theorem 11.6 with diagonal Σ, since independence makes every covariance vanish. Its solution is therefore the precision weights of part (iii), by the same certificate, and Example 5.7 is the equal-variance case in which they reduce to $1/n$.

Example 11.8 (The minimum-variance portfolio of the running universe)**.** For the running universe, $\det \Sigma = 0.01^3(8 - 2 - 2) = 4 \times 10^{-6}$ and

$$\Sigma^{-1} = 25 \begin{pmatrix} 3 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 3 \end{pmatrix};$$

the reader should verify by multiplying that $\Sigma\Sigma^{-1} = I$ (each diagonal entry is 0.25×4 and every off-diagonal entry cancels). Summing the rows, $\Sigma^{-1}\mathbf{1} = (50, 0, 50)^\top$, so $A = 100$ and by (11.4)

$$w_g = (0.5,\ 0,\ 0.5)^\top, \qquad \sigma_g^2 = 1/100, \qquad \sigma_g = 10\%,$$

against 14.14% for each asset alone. Part (ii) checks: $\Sigma w_g = 0.01(1, 1, 1)^\top = (1/A)\mathbf{1}$. Asset 2 receives zero weight because its covariance with the equal mix of assets 1 and 3 is $0.01(0.5 + 0.5) = 0.01$, exactly the variance of that mix: at the margin it adds nothing. Consistency with Proposition 11.3 on assets 1 and 3 alone ($\rho = 0$, equal volatilities): $w^* = 1/2$ and variance $0.02/2 = 0.01$.

Part (iii) in numbers: three uncorrelated assets with variances 0.01, 0.02, 0.04 give $\Sigma^{-1}\mathbf{1} = (100, 50, 25)^\top$, $A = 175$, $w_g = (4/7, 2/7, 1/7)^\top = (0.571, 0.286, 0.143)^\top$ and $\sigma_g^2 = 1/175 = 0.005714$, a volatility of 7.56%, below the best single asset's 10%.

Interview trap. Presenting the Lagrangian stationary point without the convexity certificate invites "why is it a minimum?". The answer is strict convexity on the affine feasible set, shown by the completing-the-square line in the proof of Theorem 11.6: any feasible deviation d adds $d^\top \Sigma d > 0$. A second slip is reading the result as "equal weights" or "equal variances": the minimum-variance portfolio makes every asset's *covariance with the portfolio* equal ($\Sigma w_g = (1/A)\mathbf{1}$), and asset 2 of Example 11.8 receives no weight at all. Inverse-variance weights $\propto 1/\sigma_i^2$ (part (iii), diagonal Σ) are not inverse-volatility weights $\propto 1/\sigma_i$, which equalise volatility budgets rather than variances; keep the exponents straight.

Setup (equal weights). Write $\bar{v} = \frac{1}{n}\sum_i \Sigma_{ii}$ for the average variance and $\bar{c} = \frac{1}{n(n-1)}\sum_{i\neq j}\Sigma_{ij}$ for the average covariance of the n assets present; both depend on n.

Definition 11.9 (Equal-weight portfolio and the equicorrelated model)**.** The *equal-weight* portfolio is $w = \frac{1}{n}\mathbf{1}$. The *equicorrelated model* is the covariance structure with a common variance $\Sigma_{ii} = v > 0$ for all i and a common correlation $\rho_{ij} = \rho$ for all $i \neq j$, i.e. $\Sigma = v\big[(1-\rho)I + \rho\,\mathbf{1}\mathbf{1}^\top\big]$.

Proposition 11.10 (Equal-weight diversification and the systematic floor)**.** *Under the setup above:*

(i) For any covariance matrix,

$$\operatorname{Var}\Big(\frac{1}{n}\mathbf{1}^\top R\Big) = \frac{\bar{v}}{n} + \Big(1 - \frac{1}{n}\Big)\bar{c},$$

so that $\operatorname{Var}(\frac{1}{n}\mathbf{1}^\top R) - \bar{c} \to 0$ *as* $n \to \infty$ *when* $\bar{v}$ *stays bounded.*

(ii) In the equicorrelated model of Definition 11.9, Σ *is positive definite if and only if* $-1/(n-1) < \rho < 1$*, and then*

$$\operatorname{Var}\Big(\frac{1}{n}\mathbf{1}^\top R\Big) = \rho v + \frac{1-\rho}{n}\,v,$$

which is (8.2) of Proposition 8.6 with $X_i = R_i$*. For* $\rho \geq 0$ *the floor* ρv *(volatility* $\sqrt{\rho v}$*) is the* systematic *(undiversifiable) variance and* $(1-\rho)v/n$ *the* idiosyncratic *part.*

(iii) In the equicorrelated model the equal-weight portfolio is the minimum-variance portfolio of Theorem 11.6, with $1/A = \rho v + (1-\rho)v/n$*.*

Proof. We expand the quadratic form at $w = \frac{1}{n}\mathbf{1}$ and, for the equicorrelated case, compute $\Sigma\mathbf{1}$ directly.

Part (i). Formula (11.1) at $w_i = 1/n$ gives

$$\frac{1}{n^2}\Big(\sum_i \Sigma_{ii} + \sum_{i\neq j}\Sigma_{ij}\Big) = \frac{1}{n^2}\big(n\bar{v} + n(n-1)\bar{c}\big) = \frac{\bar{v}}{n} + \frac{n-1}{n}\,\bar{c}.$$

The difference between this variance and $\bar{c}$ is $(\bar{v}-\bar{c})/n$, and $|\bar{c}| \leq \bar{v}$ because $|\Sigma_{ij}| \leq \sigma_i\sigma_j \leq \frac{1}{2}(\sigma_i^2+\sigma_j^2)$; so the difference tends to 0 when $\bar{v}$ stays bounded.

Part (ii). For d with $\mathbf{1}^\top d = 0$, $\Sigma d = v(1-\rho)d + v\rho\,\mathbf{1}(\mathbf{1}^\top d) = v(1-\rho)d$; and $\Sigma\mathbf{1} = v(1-\rho)\mathbf{1} + v\rho\,\mathbf{1}\,n = v(1+(n-1)\rho)\mathbf{1}$. Since $\mathbb{R}^n$ is spanned by $\mathbf{1}$ and the $(n-1)$-dimensional space $\{d : \mathbf{1}^\top d = 0\}$, the eigenvalues of Σ are $v(1-\rho)$ with multiplicity $n-1$ and $v(1+(n-1)\rho)$ once; by Theorem 6.4, Σ is positive definite if and only if both are positive, i.e. $\rho < 1$ and $\rho > -1/(n-1)$. Substituting $\bar{v} = v$ and $\bar{c} = \rho v$ into part (i) gives $v/n + (1-1/n)\rho v = \rho v + (1-\rho)v/n$, and comparing with (8.2) shows the two formulas are identical: an equal-weight portfolio of n correlated assets is an average of n correlated random variables, so the mathematics is the same.

Part (iii). From $\Sigma\mathbf{1} = v(1+(n-1)\rho)\mathbf{1}$ and invertibility, $\Sigma^{-1}\mathbf{1} = \mathbf{1}/\big(v(1+(n-1)\rho)\big)$, a multiple of $\mathbf{1}$. Hence $A = n/\big(v(1+(n-1)\rho)\big)$ and (11.4) gives $w_g = \frac{1}{n}\mathbf{1}$, with $1/A = v(1+(n-1)\rho)/n = \rho v + (1-\rho)v/n$. (The same vector is the equal-weight first principal direction of Fact 6.25 when $\rho > 0$, since $\mathbf{1}$ is the eigenvector of the largest eigenvalue $v(1+(n-1)\rho)$.) ☐

Example 11.11 (The diversification floor in numbers)**.** Take $\sigma = 20\%$ for every asset and $\rho = 0.3$, so $v = 0.04$ and $\rho v = 0.012$. By Proposition 11.10(ii) the equal-weight volatility $\sqrt{0.012 + 0.028/n}$ is

n	1	2	10	30	100	1000
volatility	20.0%	16.1%	12.2%	11.4%	11.1%	10.97%

with limit $\sqrt{0.3} \times 20\% = 10.95\%$. Ten names capture $(20 - 12.2)/(20 - 10.95) \approx 86\%$ of the achievable reduction; the next 990 names buy about 1.2 points; no n breaks below 10.95%. Example 8.7 ($v = 1$, $\rho = 0.2$) is the same table read as ensemble variance.

Intuition. Averaging kills the diagonal at rate $1/n$ and leaves the average off-diagonal entry standing. Whatever the assets share — the market, a sector, a common factor — survives in every one of them and therefore in their average; only what is specific to each asset cancels. The floor is the shared part, which is why ρv is called systematic or market risk and the $1/n$ term idiosyncratic risk. Note also that $w_g = \Sigma^{-1}\mathbf{1}/A$ never looks at μ: minimum variance is not maximum Sharpe, and Theorem 11.21(iii) in §11.4 shows exactly when that costs nothing.

Interview trap. "With enough assets all risk diversifies away" is false for $\rho > 0$: the limit is the average covariance, and going from 30 to 100 names removes only 0.3 points of volatility in Example 11.11. Averaging removes idiosyncratic variance at rate $1/n$ and leaves ρv intact — the random-forest floor of §8.3 in different clothes.

11.3 The Mean–Variance Frontier and Two-Fund Separation

"Sketch the efficient frontier and explain its shape" is a whiteboard staple; written tests ask for the variance minimisation with two constraints; and two-fund separation is the one-sentence theory question ("why does every investor hold the same two funds?"). Fixing a target mean m and minimising variance subject to two linear constraints produces the frontier. The solution is affine in m, which is the whole content of two-fund separation for the risky frontier (Merton, 1972; Black, 1972), and expressed through four constants the frontier is a hyperbola in the (σ, m) plane whose vertex is the minimum-variance portfolio of Theorem 11.6. Figure 19 is built here and completed in §11.4 with the capital market line.

Setup. The covariance matrix Σ is positive definite, and μ is assumed not to be a multiple of $\mathbf{1}$: otherwise every fully invested portfolio has the same mean and the frontier collapses to the single point w_g. A target mean is a real number m.

Definition 11.12 (Frontier portfolio, efficient frontier and the frontier constants)**.** The *frontier constants* are the scalars

$$A = \mathbf{1}^\top\Sigma^{-1}\mathbf{1}, \qquad B = \mathbf{1}^\top\Sigma^{-1}\mu, \qquad C = \mu^\top\Sigma^{-1}\mu, \qquad D = AC - B^2, \tag{11.6}$$

local to this section as announced in §11.1. For a target mean m, a *frontier portfolio* is a minimiser of $w^\top\Sigma w$ over $\{w : \mathbf{1}^\top w = 1,\ \mu^\top w = m\}$, and the *mean–variance frontier* is the set of frontier portfolios over all m, drawn as the set of their (σ, m) pairs. The *efficient frontier* is the subset with $m \geq B/A$ (the threshold is identified in Corollary 11.17).

Lemma 11.13 (Signs of the frontier constants)**.** *The constant A is positive; the constant C is positive whenever $\mu \neq 0$; and $D \geq 0$ always, with $D > 0$ exactly when μ and $\mathbf{1}$ are linearly independent — in particular under the setup above.*

Proof. We read all three claims off Lemma 11.5 with $G = \Sigma^{-1}$, which is symmetric positive definite by part (i) of that lemma. The constants A and C are the quadratic form of Σ^{-1} at $\mathbf{1} \neq 0$ and at μ, hence positive when the vector is nonzero. The constant D is the Gram determinant of $\mathbf{1}$ and μ in the Σ^{-1}-inner product, $D = \langle \mathbf{1}, \mathbf{1}\rangle_{\Sigma^{-1}}\langle \mu, \mu\rangle_{\Sigma^{-1}} - \langle \mathbf{1}, \mu\rangle_{\Sigma^{-1}}^2$, which is nonnegative by Lemma 11.5(ii) with $x = \mathbf{1}$, $y = \mu$, and zero if and only if the two vectors are linearly dependent. ☐

Theorem 11.14 (Minimum variance at a target mean)**.** *Under Definition 11.12, for each $m \in \mathbb{R}$ the problem $\min_w w^\top\Sigma w$ subject to $\mathbf{1}^\top w = 1$ and $\mu^\top w = m$ has the unique solution*

$$w(m) = \theta_1\,\Sigma^{-1}\mathbf{1} + \theta_2\,\Sigma^{-1}\mu, \qquad \theta_1 = \frac{C - Bm}{D}, \quad \theta_2 = \frac{Am - B}{D},$$

equivalently

$$w(m) = g + h\,m, \qquad g = \frac{C\,\Sigma^{-1}\mathbf{1} - B\,\Sigma^{-1}\mu}{D}, \quad h = \frac{A\,\Sigma^{-1}\mu - B\,\Sigma^{-1}\mathbf{1}}{D}, \tag{11.7}$$

where the fixed vectors g, h satisfy $\mathbf{1}^\top g = 1$, $\mathbf{1}^\top h = 0$, $\mu^\top g = 0$, $\mu^\top h = 1$. The minimum variance is

$$\sigma^2(m) = w(m)^\top \Sigma w(m) = \frac{Am^2 - 2Bm + C}{D}. \tag{11.8}$$

At $m = B/A$ the solution is w_g and the variance is $1/A$.

Proof. We proceed by the Lagrangian with two multipliers and certify optimality by completing the square, exactly as in Theorem 11.6.

Candidate. With $L(w, \theta_1, \theta_2) = \frac{1}{2} w^\top \Sigma w - \theta_1(\mathbf{1}^\top w - 1) - \theta_2(\mu^\top w - m)$, the gradient formula (11.5) makes stationarity read

$$\Sigma w = \theta_1 \mathbf{1} + \theta_2 \mu, \qquad \text{i.e.} \qquad w = \theta_1 \Sigma^{-1}\mathbf{1} + \theta_2 \Sigma^{-1}\mu. \tag{11.9}$$

Imposing the two constraints on this w and using (11.6),

$$\mathbf{1}^\top w = A\theta_1 + B\theta_2 = 1, \qquad \mu^\top w = B\theta_1 + C\theta_2 = m,$$

a 2×2 linear system in (θ_1, θ_2) with determinant $AC - B^2 = D > 0$ by Lemma 11.13. Cramer's rule gives $\theta_1 = (C \cdot 1 - B \cdot m)/D$ and $\theta_2 = (A \cdot m - B \cdot 1)/D$. Collecting the terms in m yields (11.7); the four identities for g and h are direct: $\mathbf{1}^\top g = (CA - B^2)/D = 1$, $\mathbf{1}^\top h = (AB - BA)/D = 0$, $\mu^\top g = (CB - BC)/D = 0$, $\mu^\top h = (AC - B^2)/D = 1$.

Certificate. Let w be any portfolio satisfying both constraints and write $w = w(m) + d$. Then $\mathbf{1}^\top d = 0$ and $\mu^\top d = 0$, so by (11.9)

$$d^\top \Sigma w(m) = \theta_1 \, \mathbf{1}^\top d + \theta_2 \, \mu^\top d = 0,$$

hence

$$w^\top \Sigma w = w(m)^\top \Sigma w(m) + d^\top \Sigma d \;\geq\; w(m)^\top \Sigma w(m),$$

with equality if and only if $d = 0$ by positive definiteness. So $w(m)$ is the unique minimiser.

Variance. Using (11.9) once more,

$$w(m)^\top \Sigma w(m) = w(m)^\top(\theta_1 \mathbf{1} + \theta_2 \mu) = \theta_1 + \theta_2 m = \frac{C - Bm + Am^2 - Bm}{D},$$

which is (11.8). At $m = B/A$: $\theta_2 = 0$ and $\theta_1 = (C - B^2/A)/D = (AC - B^2)/(AD) = 1/A$, so $w(B/A) = \Sigma^{-1}\mathbf{1}/A = w_g$ and $\sigma^2(B/A) = 1/A$. ☐

Corollary 11.15 (Frontier membership from the first-order condition)**.** *If a fully invested portfolio w satisfies $\Sigma w = \theta_1 \mathbf{1} + \theta_2 \mu$ for some scalars θ_1, θ_2, then w is the frontier portfolio at its own mean: $w = w(\mu^\top w)$.*

Proof. We repeat the completing-the-square certificate. Put $m = \mu^\top w$ and let w' be any portfolio with $\mathbf{1}^\top w' = 1$ and $\mu^\top w' = m$; write $d = w' - w$, so that $\mathbf{1}^\top d = 0$ and $\mu^\top d = 0$. Then $d^\top \Sigma w = \theta_1 \, \mathbf{1}^\top d + \theta_2 \, \mu^\top d = 0$, hence $w'^\top \Sigma w' = w^\top \Sigma w + d^\top \Sigma d \geq w^\top \Sigma w$, with equality if and only if $d = 0$ by positive definiteness. So w is the unique minimiser at target m, which is $w(m)$ by Theorem 11.14. ☐

Example 11.16 (The frontier of the running universe)**.** With Σ^{-1} from Example 11.8 and $\mu = (4\%, 7\%, 8\%)^\top$,

$$\Sigma^{-1}\mu = 25\,(0.06,\ 0.04,\ 0.14)^\top = (1.5,\ 1,\ 3.5)^\top,$$

so $A = 100$, $B = 1.5 + 1 + 3.5 = 6$, $C = 0.04(1.5) + 0.07(1) + 0.08(3.5) = 0.41$ and $D = 41 - 36 = 5$. The minimum-variance mean is $B/A = 6\%$. From (11.7),

$$g = \tfrac{1}{5}\big(0.41\,(50, 0, 50)^\top - 6\,(1.5, 1, 3.5)^\top\big) = (2.3,\ -1.2,\ -0.1)^\top,$$
$$h = \tfrac{1}{5}\big(100\,(1.5, 1, 3.5)^\top - 6\,(50, 0, 50)^\top\big) = (-30,\ 20,\ 10)^\top,$$

so $w(m) = (2.3 - 30m,\ -1.2 + 20m,\ -0.1 + 10m)^\top$ and $\sigma^2(m) = (100m^2 - 12m + 0.41)/5$.

- $m = 6\%$: $w = (0.5, 0, 0.5)^\top$, variance $(0.36 - 0.72 + 0.41)/5 = 0.01$, volatility 10% — the minimum-variance portfolio.
- $m = 7\%$: $w = (0.2, 0.2, 0.6)^\top$, variance $(0.49 - 0.84 + 0.41)/5 = 0.012$, volatility 10.95%. Direct check by (11.1): $0.01\,(0.08 + 0.08 + 0.72 + 0.08 + 0.24) = 0.012$.
- $m = 8\%$: $w = (-0.1, 0.4, 0.7)^\top$, variance $(0.64 - 0.96 + 0.41)/5 = 0.018$, volatility 13.42% — a short position appears.
- $m = 4\%$: $w = (1.1, -0.4, 0.3)^\top$, variance $(0.16 - 0.48 + 0.41)/5 = 0.018$, the same volatility as $m = 8\%$ by the symmetry of the quadratic (11.8) about $m = B/A$, but below the efficient threshold of Definition 11.12. Asset 1 alone (mean 4%, variance 0.02) is beaten even by this inefficient portfolio.

Corollary 11.17 (The frontier is a hyperbola with vertex at the minimum-variance portfolio)**.** *The frontier variance satisfies*

$$\sigma^2(m) = \frac{1}{A} + \frac{A}{D}\Big(m - \frac{B}{A}\Big)^2, \tag{11.10}$$

so in the (σ, m) plane the frontier is the right-hand branch of the hyperbola

$$A\sigma^2 - \frac{A^2}{D}\Big(m - \frac{B}{A}\Big)^2 = 1,$$

with vertex $(1/\sqrt{A},\ B/A)$ — the minimum-variance portfolio — and asymptotes $m = B/A \pm \sqrt{D/A}\,\sigma$. Against variance rather than volatility it is a parabola. The efficient frontier is the upper half $m \geq B/A$: each frontier portfolio with $m < B/A$ is dominated by its mirror image at the same volatility.

Proof. We complete the square in (11.8): $Am^2 - 2Bm + C = A(m - B/A)^2 + C - B^2/A = A(m - B/A)^2 + D/A$, and dividing by D gives (11.10). Multiplying through by A and rearranging gives the hyperbola equation; the vertex is where the squared term vanishes, $m = B/A$ and $\sigma^2 = 1/A$, which is w_g by the last sentence of Theorem 11.14. For $|m - B/A|$ large, $\sigma^2(m) \approx (A/D)(m - B/A)^2$, i.e. $m - B/A \approx \pm\sqrt{D/A}\,\sigma$, which are the asymptotes. Finally, the portfolios at $m = B/A + t$ and $m = B/A - t$ have the same variance $1/A + (A/D)t^2$ but means differing by $2t$; for $t > 0$ the upper one has strictly larger mean at equal volatility, so the lower one is dominated. ☐

In the running universe (11.10) reads $\sigma^2 = 0.01 + 20(m - 0.06)^2$, with asymptote slope $\sqrt{D/A} = \sqrt{0.05} = 0.2236$; Figure 19 draws this frontier, its asymptotes and the three asset points. The risk-free rate, the tangency portfolio and the capital market line drawn on the same figure are explained in §11.4.

Theorem 11.18 (Two-fund separation)**.** *Let Σ be positive definite and let $w(m)$ be the frontier map of (11.7). Then:*

(i) The frontier map is affine in the target mean: $w(m) = w_g + (m - B/A)\,h$, with the fixed direction h of (11.7), which satisfies $\mathbf{1}^\top h = 0$.

(ii) Let $w(m_1)$ and $w(m_2)$ be two distinct frontier portfolios ($m_1 \neq m_2$). Then for every m,

$$w(m) = \tau\, w(m_1) + (1 - \tau)\, w(m_2), \qquad \tau = \frac{m - m_2}{m_1 - m_2},$$

and conversely every combination $\tau w(m_1) + (1 - \tau) w(m_2)$, $\tau \in \mathbb{R}$, is the frontier portfolio at mean $\tau m_1 + (1 - \tau) m_2$. In particular the minimum-variance portfolio together with any one other frontier portfolio — for instance $\Sigma^{-1}\mu/B$ when $B \neq 0$, which has mean C/B — spans the entire frontier.

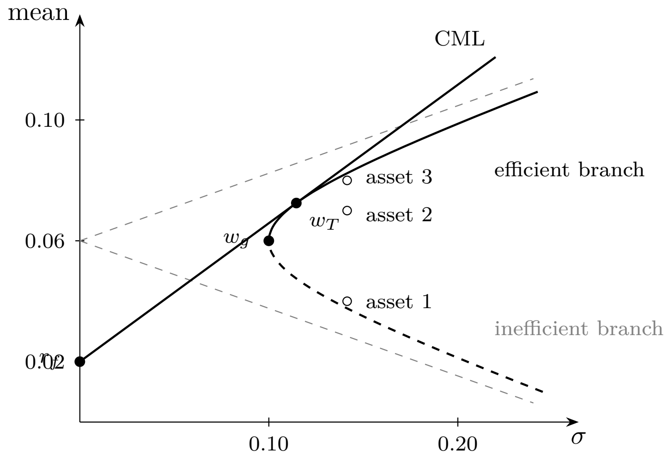


Figure 19: The mean–variance frontier of the running universe (Example 11.16) in the (σ, mean) plane. The vertex $w_g = (0.5, 0, 0.5)^\top$ at $(10\%, 6\%)$ is Theorem 11.6; the curve $\sigma^2 = 0.01 + 20(m - 0.06)^2$ is Theorem 11.14 and Corollary 11.17, with its asymptotes of slope ± 0.2236 dashed in gray; the upper branch is efficient, the lower (dashed) branch is dominated. The three assets, each at volatility 14.14%, lie strictly inside. The straight line from $r_f = 2\%$ on the vertical axis is the capital market line of Corollary 11.24 in §11.4; it lies weakly above the hyperbola everywhere and touches it only at the tangency portfolio w_T at $(11.46\%, 7.25\%)$.

Proof. We show that the frontier map $m \mapsto w(m)$ is affine and that an affine map sends the line through two points to the line through their images.

Part (i). By (11.7), $w(m) - w(B/A) = h\,(m - B/A)$, and $w(B/A) = w_g$ by Theorem 11.14; $\mathbf{1}^\top h = 0$ was shown there.

Part (ii). With $w(m) = g + hm$,

$$\tau\,(g + hm_1) + (1-\tau)\,(g + hm_2) = g + h\,(\tau m_1 + (1-\tau)m_2),$$

and the choice $\tau = (m - m_2)/(m_1 - m_2)$ makes $\tau m_1 + (1-\tau)m_2 = m$, so the right-hand side is $w(m)$. The combination is fully invested because $\tau \cdot 1 + (1-\tau)\cdot 1 = 1$. The converse reads the same identity from left to right: for any τ the combination equals $g + hm'$ with $m' = \tau m_1 + (1-\tau)m_2$, which is $w(m')$. For the last claim, $\Sigma^{-1}\mu/B$ is fully invested when $B \neq 0$ and satisfies $\Sigma(\Sigma^{-1}\mu/B) = \mu/B$, which is of the form $\theta_1\mathbf{1} + \theta_2\mu$ with $\theta_1 = 0$ and $\theta_2 = 1/B$; so by Corollary 11.15 it is the frontier portfolio at its own mean $\mu^\top\Sigma^{-1}\mu/B = C/B$, and it differs from w_g because μ is not a multiple of $\mathbf{1}$. □

The two multipliers of Theorem 11.14 are the two funds: a desk needs only two products — the minimum-variance portfolio and one other frontier portfolio — to span every efficient portfolio, and investors differ only in how they mix them (Black, 1972).

Example 11.19 (Two-fund checks in the running universe)**.** With the frontier map of Example 11.16, part (i) reads $w(8\%) = w_g + 0.02\,h = (0.5, 0, 0.5)^\top + (-0.6, 0.4, 0.2)^\top = (-0.1, 0.4, 0.7)^\top$, as computed there. For part (ii), the two funds $w(6\%) = w_g$ and $w(8\%)$ give $\frac{1}{2}w(6\%) + \frac{1}{2}w(8\%) = (0.2, 0.2, 0.6)^\top = w(7\%)$. The second spanning fund named in the theorem is $\Sigma^{-1}\mu/B = (1.5, 1, 3.5)^\top/6 = (0.25, 0.1667, 0.5833)^\top$, with mean $C/B = 0.41/6 = 6.833\%$; the frontier map confirms it, $w(0.06833) = (2.3 - 2.05,\ -1.2 + 1.3667,\ -0.1 + 0.6833)^\top = (0.25, 0.1667, 0.5833)^\top$. Reaching $w(7\%)$ from w_g and this fund requires $\tau = (0.07 - 0.06833)/(0.06 - 0.06833) = -0.2$: the mix is $-0.2\,w_g + 1.2\,(\Sigma^{-1}\mu/B) = (0.2, 0.2, 0.6)^\top$, which shorts the minimum-variance fund.

Intuition. Two constraints, two multipliers, two funds. The first-order condition $\Sigma w = \theta_1\mathbf{1} + \theta_2\mu$ says that on the frontier every asset's covariance with the portfolio is an affine function of its own mean — assets with higher expected return are allowed to co-move more with the book, in exact proportion. Everything else (the hyperbola, the vertex, the spanning by two funds) is bookkeeping on that one equation.

Interview trap. Sketching the frontier as a parabola in the (σ, mean) plane, or with its vertex on the vertical axis, is wrong: it is a hyperbola whose vertex is the minimum-variance portfolio at $\sigma = 1/\sqrt{A} > 0$, with straight-line asymptotes in σ, and it is a parabola only when plotted against *variance*. Calling the whole hyperbola "efficient" is the companion error: only the upper branch ($m \geq B/A$) is. In Example 11.16 the $m = 4\%$ and $m = 8\%$ portfolios share the volatility 13.42%, and one dominates the other.

Interview trap. Two-fund separation is a statement about the weights being affine in the target mean, not about any two portfolios: the spanning funds must both lie on the frontier, and the mix may short one of them ($\tau \notin [0, 1]$, as in Example 11.19). Adding an asset can never move the frontier inward — the old feasible set is a subset of the new one — but the new frontier can touch the old one at a single point: asset 2 carries zero weight only at $m = 6\%$ in Example 11.16, and since $w(m)$ is affine in m, zero weight on an interval of target means would mean zero weight everywhere.

11.4 The Risk-Free Asset, the Tangency Portfolio and the Capital Market Line

The single most-asked question in this area — "you run uncorrelated strategies with Sharpe ratios 0.5, 1 and 1; what is the best combined Sharpe ratio and how do you weight them?" — and its theory cousin "what does a risk-free asset do to the frontier?" are the same computation. Adding an asset with zero variance and return r_f changes the geometry from a hyperbola to a straight line: mixing cash with any risky portfolio traces a ray from $(0, r_f)$, and the best ray is tangent to the frontier. Its risky portfolio maximises the Sharpe ratio, its weights are proportional to $\Sigma^{-1}(\mu - r_f\mathbf{1})$, and its squared Sharpe ratio is $(\mu - r_f\mathbf{1})^\top\Sigma^{-1}(\mu - r_f\mathbf{1})$; for uncorrelated strategies this is the interview classic $\mathrm{SR}^2 = \sum_i \mathrm{SR}_i^2$, of which Theorem 10.11 and Corollary 10.12 are special cases. The proof is one application of Lemma 11.5(ii).

Setup. Fix a real number r_f. A portfolio is now a risky weight vector $w \in \mathbb{R}^n$ that need not sum to one; the remainder $1 - \mathbf{1}^\top w$ earns the deterministic rate r_f, negative values meaning borrowing at r_f. Its return is

$$R_p = (1 - \mathbf{1}^\top w)\, r_f + w^\top R = r_f + w^\top(R - r_f\mathbf{1}).$$

Write $\mu_e = \mu - r_f\mathbf{1}$, assumed nonzero. By Proposition 11.2, with the deterministic component contributing zero variance and zero covariance, $\mathbb{E}[R_p] = r_f + w^\top\mu_e$ and $\mathrm{Var}(R_p) = w^\top\Sigma w$. With the frontier constants (11.6), put

$$k = \mathbf{1}^\top\Sigma^{-1}\mu_e = B - r_f A.$$

Definition 11.20 (Risk-free asset, excess Sharpe ratio, tangency portfolio and capital market line)**.** The *risk-free asset* is the additional asset (cash) with deterministic return r_f and zero variance, and $\mu_e = \mu - r_f\mathbf{1}$ is the *excess-mean vector.* The *Sharpe ratio* of a portfolio with $w \neq 0$ is

$$\mathrm{SR}(w) = \frac{w^\top\mu_e}{\sqrt{w^\top\Sigma w}} = \frac{\mathbb{E}[R_p] - r_f}{\sqrt{\mathrm{Var}(R_p)}},$$

which reduces to Definition 10.1 at $r_f = 0$ — §10 took $\mathrm{SR} = \mu/\sigma$ with no risk-free adjustment, and that convention is the $r_f = 0$ case of this one — and is invariant under $w \mapsto tw$ for $t > 0$ (both numerator and denominator scale by t). A *tangency portfolio* is a fully invested maximiser of SR. The *capital market line* (CML) is the set of (σ, mean) pairs obtained by mixing the risk-free asset with the tangency portfolio.

Theorem 11.21 (Maximum Sharpe ratio and the tangency portfolio)**.** *Under the setup and Definition 11.20 above, with Σ positive definite and $\mu_e \neq 0$:*

(i) For every $w \neq 0$,

$$w^\top \mu_e \ \le\ \sqrt{w^\top \Sigma w}\,\sqrt{\mu_e^\top \Sigma^{-1}\mu_e},$$

with equality if and only if w is a positive multiple of $\Sigma^{-1}\mu_e$. Hence

$$\sup_{w\neq 0} \mathrm{SR}(w) = \mathrm{SR}^* := \sqrt{\mu_e^\top \Sigma^{-1}\mu_e} = \sqrt{C - 2r_f B + r_f^2 A},$$

attained exactly on the ray $\{t\,\Sigma^{-1}\mu_e : t > 0\}$.

(ii) If $k = B - r_f A > 0$ — equivalently $r_f < B/A$, the minimum-variance mean — the unique tangency portfolio is

$$w_T = \frac{\Sigma^{-1}\mu_e}{k} = \frac{\Sigma^{-1}(\mu - r_f\mathbf{1})}{\mathbf{1}^\top\Sigma^{-1}(\mu - r_f\mathbf{1})}, \tag{11.11}$$

with mean $\mu_T = (C - r_f B)/k$ and variance $\sigma_T^2 = (\mathrm{SR}^*)^2/k^2$. It lies on the efficient frontier, with $\mu_T - B/A = D/(Ak) > 0$.

(iii) The two portfolios coincide, $w_T = w_g$, if and only if μ_e is proportional to $\mathbf{1}$, i.e. if and only if every asset has the same expected excess return; under the standing assumption of §11.3 that μ is not a multiple of $\mathbf{1}$, this never happens and $w_T \neq w_g$.

Proof. We apply the Cauchy–Schwarz inequality of Lemma 11.5(ii) in the Σ-inner product, with the second vector chosen so that its Σ-product with w is the excess mean.

Part (i). Put $y = \Sigma^{-1}\mu_e$, so that $\Sigma y = \mu_e$ and $w^\top\mu_e = w^\top\Sigma y$. Lemma 11.5(ii) with $G = \Sigma$, $x = w$ gives

$$\begin{aligned}(w^\top\mu_e)^2 = (w^\top\Sigma y)^2 &\le (w^\top\Sigma w)(y^\top\Sigma y)\\ &= (w^\top\Sigma w)\,(\mu_e^\top\Sigma^{-1}\Sigma\Sigma^{-1}\mu_e) = (w^\top\Sigma w)\,(\mu_e^\top\Sigma^{-1}\mu_e),\end{aligned}$$

with equality if and only if w and y are linearly dependent. Both $w^\top\Sigma w$ and $\mu_e^\top\Sigma^{-1}\mu_e$ are strictly positive (positive definiteness of Σ and of Σ^{-1}, the latter by Lemma 11.5(i), with $w \neq 0$ and $\mu_e \neq 0$), so taking square roots gives $|w^\top\mu_e| \le \sqrt{w^\top\Sigma w}\,\mathrm{SR}^*$, and dividing by $\sqrt{w^\top\Sigma w}$ gives $\mathrm{SR}(w) \le \mathrm{SR}^*$. Equality in the inequality for $w^\top\mu_e$ itself (not its absolute value) requires $w = ty$ with $w^\top\mu_e = t\,y^\top\Sigma y > 0$, i.e. $t > 0$; negative multiples have Sharpe ratio $-\mathrm{SR}^*$. Every positive multiple attains SR^* by scale invariance. Finally, substituting $\mu_e = \mu - r_f\mathbf{1}$ and expanding with (11.6),

$$\begin{aligned}\mu_e^\top\Sigma^{-1}\mu_e &= \mu^\top\Sigma^{-1}\mu - 2r_f\,\mathbf{1}^\top\Sigma^{-1}\mu + r_f^2\,\mathbf{1}^\top\Sigma^{-1}\mathbf{1}\\ &= C - 2r_f B + r_f^2 A.\end{aligned}$$

Part (ii). The ray $\{ty : t > 0\}$ meets the hyperplane $\mathbf{1}^\top w = 1$ exactly when $t\,\mathbf{1}^\top y = tk = 1$, which has the unique solution $t = 1/k$ when $k > 0$; so $w_T = y/k$ is the unique fully invested point of the ray, hence the unique tangency portfolio, and $k = \mathbf{1}^\top\Sigma^{-1}\mu - r_f\,\mathbf{1}^\top\Sigma^{-1}\mathbf{1} = B - r_f A$, which is positive if and only if $r_f < B/A$ since $A > 0$. The mean is $w_T^\top\mu = \mu^\top\Sigma^{-1}(\mu - r_f\mathbf{1})/k = (C - r_f B)/k$ and the variance is $w_T^\top\Sigma w_T = y^\top\Sigma y/k^2 = \mu_e^\top\Sigma^{-1}\mu_e/k^2$. For frontier membership, $\Sigma w_T = \mu_e/k = -(r_f/k)\mathbf{1} + (1/k)\mu$ and w_T is fully invested, so by Corollary 11.15 (with $\theta_1 = -r_f/k$, $\theta_2 = 1/k$) it is the frontier portfolio at its own mean. For efficiency,

$$\mu_T - \frac{B}{A} = \frac{A(C - r_f B) - B(B - r_f A)}{Ak} = \frac{AC - B^2}{Ak} = \frac{D}{Ak} > 0$$

by Lemma 11.13, so w_T sits on the upper branch of Corollary 11.17.

Part (iii). By (11.4) and (11.11), $w_T = w_g$ if and only if $\Sigma^{-1}\mu_e/k = \Sigma^{-1}\mathbf{1}/A$; multiplying by the invertible matrix Σ shows this holds if and only if $\mu_e = (k/A)\mathbf{1}$. Conversely, if $\mu_e = u\mathbf{1}$ with $u > 0$ then $\Sigma^{-1}\mu_e = u\Sigma^{-1}\mathbf{1}$, $k = uA$, and $w_T = \Sigma^{-1}\mathbf{1}/A = w_g$. When μ is not a multiple of $\mathbf{1}$, neither is $\mu_e = \mu - r_f\mathbf{1}$, and the strict inequality $\mu_T - B/A = D/(Ak) > 0$ of part (ii) shows the two portfolios differ. □

Remark 11.22 (The case $r_f \geq B/A$). When $r_f \geq B/A$ the maximum-Sharpe ray of Theorem 11.21(i) still exists, but its fully invested normalisation is undefined ($k = 0$) or is a negative multiple of $\Sigma^{-1}\mu_e$ lying on the inefficient branch with Sharpe ratio $-\mathrm{SR}^*$ ($k < 0$). Every statement below that involves w_T assumes $k > 0$; the case receives no further treatment here.

Example 11.23 (The tangency portfolio of the running universe at $r_f = 2\%$)**.** With $\mu = (4\%, 7\%, 8\%)^\top$ and $r_f = 2\%$, $\mu_e = (2\%, 5\%, 6\%)^\top$ and, using Σ^{-1} from Example 11.8, $\Sigma^{-1}\mu_e = 25\,(0.02,\ 0.04,\ 0.10)^\top = (0.5,\ 1,\ 2.5)^\top$ (the three row sums being $0.06-0.10+0.06$, $-0.04+0.20-0.12$ and $0.02-0.10+0.18$), so $k = 4 = B - r_f A = 6-2$, and by (11.11) $w_T = (0.125, 0.25, 0.625)^\top$. The squared maximal Sharpe ratio is

$$\begin{aligned}(\mathrm{SR}^*)^2 = \mu_e^\top\Sigma^{-1}\mu_e &= 0.02(0.5) + 0.05(1) + 0.06(2.5) = 0.21\\ &= C - 2r_f B + r_f^2 A = 0.41 - 0.24 + 0.04,\end{aligned}$$

so $\mathrm{SR}^* = 0.458$. The tangency mean is $(C - r_f B)/k = (0.41 - 0.12)/4 = 7.25\%$ and the variance $0.21/16 = 0.013125$ (volatility 11.46%); direct check, $(7.25\% - 2\%)/11.46\% = 0.458$. It lies on the frontier: $\sigma^2(0.0725) = 0.01 + 20(0.0125)^2 = 0.013125$ by (11.10), and $\mu_T - B/A = 1.25\% = D/(Ak) = 5/400$. Minimum variance against maximum Sharpe: $w_g = (0.5, 0, 0.5)^\top$ has Sharpe ratio $(6\% - 2\%)/10\% = 0.40 < 0.458$, and the two portfolios differ because μ_e is not proportional to $\mathbf{1}$, as part (iii) says.

Corollary 11.24 (Capital market line)**.** *Assume $k > 0$.*

(i) Holding a fraction $a \geq 0$ of wealth in w_T and $1 - a$ in the risk-free asset gives mean $r_f + a(\mu_T - r_f)$ and volatility $a\sigma_T$, so the attainable pairs form the line

$$\textit{mean} = r_f + \mathrm{SR}^*\,\sigma, \qquad \sigma \geq 0. \tag{11.12}$$

(ii) Every portfolio (risky part w, remainder at r_f) satisfies $\mathbb{E}[R_p] - r_f \leq \mathrm{SR}^*\sqrt{\mathrm{Var}(R_p)}$, with equality if and only if w is a nonnegative multiple of $\Sigma^{-1}\mu_e$. Hence the CML lies weakly above the risky frontier of Corollary 11.17, touches it exactly at w_T, and is the efficient set once a risk-free asset exists (Figure 19).

Proof. We compute the moments of the mix and then rewrite Theorem 11.21(i).

Part (i). The risky weight vector of the mix is $a\,w_T$, so by the setup its mean is $r_f + a\,w_T^\top\mu_e = r_f + a(\mu_T - r_f)$ and its variance is $a^2 w_T^\top\Sigma w_T = a^2\sigma_T^2$, giving volatility $a\sigma_T$ for $a \geq 0$. Eliminating $a = \sigma/\sigma_T$ yields mean $= r_f + \big((\mu_T - r_f)/\sigma_T\big)\sigma$, and $(\mu_T - r_f)/\sigma_T = \mathrm{SR}(w_T) = \mathrm{SR}^*$ by Theorem 11.21.

Part (ii). For $w = 0$ both sides vanish. For $w \neq 0$, $\mathbb{E}[R_p] - r_f = w^\top\mu_e$ and $\sqrt{\mathrm{Var}(R_p)} = \sqrt{w^\top\Sigma w}$, so the inequality is Theorem 11.21(i), with equality exactly on the ray $t\,\Sigma^{-1}\mu_e$, $t > 0$. A fully invested risky portfolio therefore lies weakly below the line (11.12) in the (σ, mean) plane, and a frontier portfolio lies on it if and only if it is the fully invested point of the ray, which is w_T by part (ii) of the theorem. □

Example 11.25 (The capital market line of the running universe)**.** From Example 11.23, the CML (11.12) is mean $= 0.02 + 0.458\,\sigma$. At the 13.42% volatility of the frontier portfolio $w(8\%)$ it delivers $8.15\% > 8\%$, and at the 10.95% of $w(7\%)$ it gives $7.02\% > 7\%$: both frontier portfolios lie strictly below the line, as Corollary 11.24(ii) requires, and only w_T at $(11.46\%, 7.25\%)$ touches it. Reaching the volatility 13.42% along the line means holding $a = 0.1342/0.1146 = 1.17$ of wealth in w_T, financed by borrowing 17% at r_f.

Figure 19 shows the line: from $r_f = 2\%$ on the vertical axis, through the tangency point at $(11.46\%, 7.25\%)$, with slope $\mathrm{SR}^* = 0.458$; the hyperbola lies under it everywhere except at w_T.

Corollary 11.26 (Sharpe ratios of uncorrelated strategies add in quadrature)**.** *Under the setup and Definition 11.20 above:*

(i) If $\Sigma = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)$ *and* $\mathrm{SR}_i = (\mu_i - r_f)/\sigma_i$ *denotes the Sharpe ratio of asset* i *alone, then*

$$(\mathrm{SR}^*)^2 = \sum_{i=1}^{n} \mathrm{SR}_i^2,$$

attained at weights $w_i \propto (\mu_i - r_f)/\sigma_i^2 = \mathrm{SR}_i/\sigma_i$*, i.e. at* risk budgets $\sigma_i w_i \propto \mathrm{SR}_i$*.*

(ii) For every Σ, $\mathrm{SR}^* \geq \max_i \mathrm{SR}_i$*.*

Proof. We evaluate Theorem 11.21(i) for a diagonal matrix and at the basis vectors.

Part (i). With $\Sigma^{-1} = \operatorname{diag}(\sigma_1^{-2}, \ldots, \sigma_n^{-2})$, $\mu_e^\top \Sigma^{-1} \mu_e = \sum_i (\mu_i - r_f)^2/\sigma_i^2 = \sum_i \mathrm{SR}_i^2$, and $\Sigma^{-1}\mu_e$ has entries $(\mu_i - r_f)/\sigma_i^2 = \mathrm{SR}_i/\sigma_i$; multiplying entry i by σ_i gives the risk budget $\sigma_i w_i \propto \mathrm{SR}_i$.

Part (ii). Taking $w = e_i$ in the supremum of Theorem 11.21(i) gives $\mathrm{SR}(e_i) = (\mu_i - r_f)/\sigma_i = \mathrm{SR}_i \leq \mathrm{SR}^*$. ☐

Remark 11.27 (Theorem 10.11 and Corollary 10.12 as special cases)*.* Three earlier results are special cases of Corollary 11.26(i). With $n = 2$, $r_f = 0$, equal Sharpe ratios and equal volatilities, $\mathrm{SR}^* = \sqrt{2\,\mathrm{SR}_1^2} = \sqrt{2}\,\mathrm{SR}_1$ at equal weights, and the raw sum of the two legs is a positive multiple of the tangency direction: this is Theorem 10.11. With n legs of equal Sharpe ratio, $\mathrm{SR}^* = \sqrt{n}\,\mathrm{SR}_1$: Corollary 10.12. And the weights $w_i \propto \mu_i/\sigma_i^2$ are exactly the mean–variance weights promised in Remark 10.14; the equal-volatility hypothesis of Theorem 10.11 is unnecessary because the tangency weights rebalance risk automatically.

Example 11.28 (The interview classic: three uncorrelated strategies)**.** Three uncorrelated strategies have Sharpe ratios 0.5, 1.0, 1.0 and volatilities 10%, 20%, 5%; their returns are self-financing P&L streams, so $r_f = 0$ and the means are 5%, 20%, 5%. By Corollary 11.26, $\mathrm{SR}^* = \sqrt{0.25 + 1 + 1} = 1.5$, at weights proportional to $\mu_i/\sigma_i^2 = (5, 5, 20)$, i.e. $(1/6, 1/6, 2/3)$. The risk budgets $\sigma_i w_i = (1.67\%, 3.33\%, 3.33\%)$ are proportional to $(0.5, 1, 1) = \mathrm{SR}_i$. The combined mean is $5/6 + 20/6 + 10/3 = 7.5\%$ and the volatility $\sqrt{0.0167^2 + 0.0333^2 + 0.0333^2} = 5.0\%$, Sharpe ratio 1.5. Equal capital weights instead give mean 10%, volatility $\sqrt{0.0333^2 + 0.0667^2 + 0.0167^2} = 7.64\%$ and Sharpe ratio 1.31. The "average Sharpe" 0.83 and the "sum of Sharpes" 2.5 are both wrong. The two-strategy version — Sharpe 1 with Sharpe 0.5, uncorrelated — gives $\sqrt{1.25} = 1.118$: not 0.75, not 1.5.

Definition 11.29 (Mean–variance utility)**.** Fix a *risk aversion* $\gamma > 0$. The *mean–variance utility* of a portfolio with unconstrained risky weights $w \in \mathbb{R}^n$ is

$$U(w) = \mathbb{E}[R_p] - \tfrac{\gamma}{2}\operatorname{Var}(R_p) = r_f + w^\top \mu_e - \tfrac{\gamma}{2}\, w^\top \Sigma w.$$

Proposition 11.30 (Mean–variance utility and the tangency direction)**.** *The utility* U *of Definition 11.29 has the unique maximiser*

$$w^\star = \frac{1}{\gamma}\,\Sigma^{-1}\mu_e,$$

a multiple of $\Sigma^{-1}\mu_e$ *— hence of* w_T *when* $k > 0$ *— with* $\mathbf{1}^\top w^\star = k/\gamma$ *in risky assets, and maximal value* $U(w^\star) = r_f + (\mathrm{SR}^*)^2/(2\gamma)$*.*

Proof. We complete the square. Expanding the quadratic form in $w - w^\star$ and using $\Sigma w^\star = \mu_e/\gamma$,

$$\begin{aligned}\frac{\gamma}{2}(w-w^\star)^\top\Sigma(w-w^\star) &= \frac{\gamma}{2}\,w^\top\Sigma w - \gamma\,w^\top\Sigma w^\star + \frac{\gamma}{2}\,(w^\star)^\top\Sigma w^\star\\ &= \frac{\gamma}{2}\,w^\top\Sigma w - w^\top\mu_e + \frac{1}{2\gamma}\,\mu_e^\top\Sigma^{-1}\mu_e,\end{aligned}$$

so that

$$U(w) = r_f + \frac{1}{2\gamma}\,\mu_e^\top\Sigma^{-1}\mu_e - \frac{\gamma}{2}(w-w^\star)^\top\Sigma(w-w^\star).$$

The subtracted term is nonnegative and vanishes if and only if $w = w^\star$, by positive definiteness (strict concavity of U). The maximal value is the remaining constant, $r_f + (\mathrm{SR}^*)^2/(2\gamma)$, and $\mathbf{1}^\top w^\star = \mathbf{1}^\top\Sigma^{-1}\mu_e/\gamma = k/\gamma$. □

This is two-fund separation with a risk-free asset (Tobin, 1958): every mean–variance investor holds the same risky portfolio w_T and differs only in the scale $1/\gamma$ — the premise of the CAPM in §11.5. The case $\gamma = 1$ is revisited as the Kelly connection in Remark 11.52.

Example 11.31 (Utility scaling in the running universe)**.** With $\Sigma^{-1}\mu_e = (0.5, 1, 2.5)^\top$ and $k = 4$ from Example 11.23, Proposition 11.30 gives: at $\gamma = 4$, $w^\star = w_T$ exactly (100% risky); at $\gamma = 10$, $w^\star = (0.05, 0.10, 0.25)^\top$, 40% risky and 60% cash, with value $0.02 + 0.21/20 = 3.05\%$; at $\gamma = 1$, $w^\star = (0.5, 1, 2.5)^\top$, a 400% gross exposure financed by borrowing 300% at r_f, with volatility $\sqrt{0.21} = 45.8\%$. The risky mix is $(0.125, 0.25, 0.625)^\top$ in every case: the mean–variance weights fix a *direction*, and the leverage $1/\gamma$ is a separate decision (Remark 3.8).

Intuition. Uncorrelated strategies are orthogonal directions, and Sharpe ratios are the lengths of the edge along each: the best combination is the diagonal of the box, and its length is the root of the sum of squares. Correlation tilts the axes and can shorten or lengthen the diagonal depending on the Sharpe ratios; Σ^{-1} is exactly the change of basis that makes the axes orthogonal again, which is why the general answer is $\sqrt{\mu_e^\top\Sigma^{-1}\mu_e}$.

Interview trap. Sharpe ratios do not average and do not add. Uncorrelated strategies combine in quadrature, and only with risk budgets proportional to SR_i (weights SR_i/σ_i); equal capital weights across unequal volatilities give 1.31 instead of 1.5 in Example 11.28, and two uncorrelated Sharpe-1 strategies give 1.41, not 1 and not 2. Quoting the quadrature formula without saying "uncorrelated" is the companion error: for legs of equal Sharpe ratio, $\rho > 0$ shrinks the improvement (Example 10.13) and $\rho < 0$ enlarges it; with unequal Sharpe ratios a large enough positive ρ also raises SR^* above quadrature, because Σ^{-1} turns the weaker leg into a hedge — with unit volatilities and Sharpe ratios s_1, s_2, Theorem 11.21 gives $(\mathrm{SR}^*)^2 = (s_1^2 + s_2^2 - 2\rho s_1 s_2)/(1-\rho^2)$, which for $\rho > 0$ exceeds $s_1^2 + s_2^2$ exactly when $\rho > 2s_1s_2/(s_1^2+s_2^2)$, for instance 1.357 against 1.118 at $(s_1, s_2) = (1, 0.5)$ and $\rho = 0.9$. The general answer is $\sqrt{\mu_e^\top\Sigma^{-1}\mu_e}$, and the diagonal formula is its special case.

Interview trap. Minimum variance is not maximum Sharpe, and the tangency direction is $\Sigma^{-1}(\mu - r_f\mathbf{1})$, not $\Sigma^{-1}\mu$. The portfolios $w_g = \Sigma^{-1}\mathbf{1}/A$ and $w_T \propto \Sigma^{-1}\mu_e$ coincide only when every asset has the same expected excess return (Theorem 11.21(iii)); otherwise the tangency portfolio sits strictly higher up the efficient branch, $\mu_T - B/A = D/(Ak) > 0$, and in the running universe the two Sharpe ratios are 0.40 and 0.458. The minimum-variance portfolio does not depend on r_f at all, whereas raising r_f tilts the tangency portfolio toward higher-mean assets and, once r_f reaches B/A, off the efficient branch entirely (Remark 11.22). "The optimal portfolio" without naming r_f is incomplete.

11.5 Beta, the CAPM and the Security Market Line

The rapid-fire block: "what is beta?", "do you hedge with beta or with correlation?", "state the CAPM in one line", "what is alpha?", "what is the difference between the CML and the SML?".

Mathematically, beta is the regression slope of Corollary 5.3 read in population form, the variance decomposition is the algebra of Proposition 10.15, and the CAPM pricing identity is one first-order condition of Theorem 11.21; only the equilibrium claim that the market portfolio *is* the tangency portfolio is a Fact (Sharpe, 1964; Lintner, 1965).

Setup. A *reference portfolio* (a benchmark, or the market) has return R_M with mean μ_M and variance $\sigma_M^2 > 0$. An asset or portfolio has return R_i with mean μ_i, volatility $\sigma_i > 0$ and correlation $\rho_{iM} = \operatorname{Corr}(R_i, R_M)$ with the reference. The risk-free rate is r_f; excess returns are $R_i - r_f$ and $R_M - r_f$. A portfolio w over the n assets is fully invested, $\mathbf{1}^\top w = 1$. Beta here is a population quantity; its sample estimate is the $\hat\beta$ of §5, whose subsection on inverse regression, beta and hedging (Proposition 5.21, Remark 5.23) already contrasts beta with correlation. Residual variances are written $\sigma_{\varepsilon,i}^2$.

Definition 11.32 (Beta, alpha and the security market line relative to a reference portfolio)**.** The *beta* of asset i to the reference portfolio is

$$\beta_i = \frac{\operatorname{Cov}(R_i, R_M)}{\operatorname{Var}(R_M)} = \rho_{iM}\,\frac{\sigma_i}{\sigma_M},$$

the population slope of the regression of R_i on R_M: it is (5.3) of Corollary 5.3 with $x = R_M$ and $y = R_i$, and the coefficient β^* of Proposition 10.15 with the black-box output of that proposition taken as R_M and the new model's output as R_i. The beta of a portfolio is $\beta_w = \operatorname{Cov}(w^\top R, R_M)/\sigma_M^2 = \sum_i w_i \operatorname{Cov}(R_i, R_M)/\sigma_M^2 = w^\top\beta$, by bilinearity of covariance. The *security market line* (SML) is the map $\beta \mapsto r_f + \beta(\mu_M - r_f)$ in the $(\beta, \text{expected return})$ plane, the line through $(0, r_f)$ and $(1, \mu_M)$. *Jensen's alpha* of asset i is

$$\alpha_i = \mu_i - r_f - \beta_i(\mu_M - r_f),$$

its vertical distance from the line (Figure 20). The pair (α_i, β_i) is the population version of the intercept and slope $(\hat\alpha, \hat\beta)$ of (5.3) applied to *excess* returns. Contrast with the CML: the capital market line lives in the (σ, mean) plane and only efficient portfolios lie on it; the security market line lives in the (β, mean) plane and, under Fact 11.36 below, every asset lies on it.

Proposition 11.33 (Systematic and idiosyncratic variance)**.** *Define the residual $\varepsilon_i = (R_i - r_f) - \alpha_i - \beta_i(R_M - r_f)$. Then:*

(i) The residual has mean zero and is uncorrelated with the reference,

$$\mathbb{E}[\varepsilon_i] = 0, \qquad \operatorname{Cov}(\varepsilon_i, R_M) = 0,$$

and β_i is the unique coefficient with the latter property.

(ii) The variance splits as

$$\operatorname{Var}(R_i) = \beta_i^2\sigma_M^2 + \operatorname{Var}(\varepsilon_i), \tag{11.13}$$

with $\operatorname{Var}(\varepsilon_i) = (1 - \rho_{iM}^2)\,\sigma_i^2$*, so the* systematic *share $\beta_i^2\sigma_M^2/\sigma_i^2$ equals ρ_{iM}^2, the R^2 of Corollary 5.3.*

(iii) If the residuals $\varepsilon_1, \ldots, \varepsilon_n$ are pairwise uncorrelated, with residual variances $\sigma_{\varepsilon,i}^2 = \operatorname{Var}(\varepsilon_i)$, then

$$\operatorname{Var}(R_w) = \beta_w^2\sigma_M^2 + \sum_{i=1}^n w_i^2\,\sigma_{\varepsilon,i}^2;$$

for equal weights and a common residual variance σ_ε^2 the idiosyncratic part is σ_ε^2/n while the systematic part $\beta_w^2\sigma_M^2$ does not shrink with n.

Proof. We reduce parts (i) and (ii) to Proposition 10.15 and prove part (iii) by bilinearity of covariance.

Parts (i) and (ii). Rearranging the definition, $\varepsilon_i = (R_i - \beta_i R_M) - (r_f + \alpha_i - \beta_i r_f)$: it is the residual $B_\perp$ of Proposition 10.15 (the new model's output minus β^* times the black box), with the black box taken as R_M and the new model as R_i — so that $\beta^* = \mathrm{Cov}(R_i, R_M)/\operatorname{Var}(R_M) = \beta_i$ — shifted by a constant. Constants change neither covariances nor variances, so that proposition gives $\mathrm{Cov}(\varepsilon_i, R_M) = 0$, the uniqueness of β_i, and $\operatorname{Var}(\varepsilon_i) = (1 - \rho_{iM}^2)\sigma_i^2$. Taking expectations, $\mathbb{E}[\varepsilon_i] = (\mu_i - r_f) - \alpha_i - \beta_i(\mu_M - r_f) = 0$ by the definition of α_i. Since $R_i = r_f + \alpha_i + \beta_i(R_M - r_f) + \varepsilon_i$ has random part $\beta_i R_M + \varepsilon_i$, Proposition 10.10 gives $\operatorname{Var}(R_i) = \beta_i^2\sigma_M^2 + \operatorname{Var}(\varepsilon_i) + 2\beta_i \operatorname{Cov}(R_M, \varepsilon_i)$, whose cross term vanishes; this is (11.13), and the systematic share is $\beta_i^2\sigma_M^2/\sigma_i^2 = \rho_{iM}^2$.

Part (iii). Summing the decompositions of part (ii) with weights w_i and using $\sum_i w_i = 1$, $R_w = r_f + \sum_i w_i\alpha_i + \beta_w(R_M - r_f) + \sum_i w_i\varepsilon_i$ with $\beta_w = \sum_i w_i\beta_i$. The residual sum is uncorrelated with R_M by part (i), and its variance has no cross terms by the pairwise-uncorrelatedness hypothesis, so $\operatorname{Var}(R_w) = \beta_w^2\sigma_M^2 + \sum_i w_i^2\sigma_{\varepsilon,i}^2$. At $w = \frac{1}{n}\mathbf{1}$ with $\sigma_{\varepsilon,i}^2 = \sigma_\varepsilon^2$ the second term is $n \cdot \sigma_\varepsilon^2/n^2 = \sigma_\varepsilon^2/n$. □

Remark 11.34 (The one-factor model is the equicorrelated model). Part (iii) is the floor of Proposition 11.10(ii) with its source named. Suppose every asset has the same beta β and the same volatility σ to one reference, with pairwise-uncorrelated residuals of common variance σ_ε^2. Since constants do not affect covariances, for $i \neq j$ bilinearity and part (i) give

$$\begin{aligned}\mathrm{Cov}(R_i, R_j) &= \mathrm{Cov}(\beta R_M + \varepsilon_i,\ \beta R_M + \varepsilon_j)\\ &= \beta^2\sigma_M^2 + \beta\,\mathrm{Cov}(R_M, \varepsilon_j) + \beta\,\mathrm{Cov}(\varepsilon_i, R_M) + \mathrm{Cov}(\varepsilon_i, \varepsilon_j)\\ &= \beta^2\sigma_M^2,\end{aligned}$$

while $\operatorname{Var}(R_i) = \beta^2\sigma_M^2 + \sigma_\varepsilon^2 = \sigma^2$ by (11.13). So Σ is the equicorrelated model of Definition 11.9 with $v = \sigma^2$ and $\rho v = \beta^2\sigma_M^2$: the common covariance that averaging cannot remove is the shared factor exposure.

Setup (tangency reference). As in §11.4: Σ is positive definite, $r_f < B/A$, and the tangency portfolio w_T of (11.11) has return R_T, mean μ_T and variance σ_T^2; it is now taken as the reference portfolio, so that $\beta_{w,T} = \mathrm{Cov}(R_w, R_T)/\operatorname{Var}(R_T)$.

Proposition 11.35 (The tangency portfolio prices every asset (SML identity))**.** *For every fully invested portfolio w — in particular every single asset, $w = e_i$ —*

$$\mu_w - r_f = \beta_{w,T}\,(\mu_T - r_f), \qquad \mu_w = w^\top\mu :$$

every alpha measured against the tangency portfolio is zero. No equilibrium assumption is used.

Proof. We compute both covariances from (11.11), which gives $\Sigma w_T = \mu_e/k$. By bilinearity and $\mathbf{1}^\top w = 1$,

$$\mathrm{Cov}(R_w, R_T) = w^\top\Sigma w_T = \frac{w^\top\mu_e}{k} = \frac{w^\top\mu - r_f\,\mathbf{1}^\top w}{k} = \frac{\mu_w - r_f}{k}$$

and

$$\operatorname{Var}(R_T) = w_T^\top\Sigma w_T = \frac{w_T^\top\mu_e}{k} = \frac{\mu_T - r_f}{k}.$$

The second quantity is positive ($\mu_T - r_f = \mathrm{SR}^*\sigma_T > 0$), so dividing, $\beta_{w,T} = (\mu_w - r_f)/(\mu_T - r_f)$, which rearranges to the claim. □

Fact 11.36 (Capital asset pricing model (Sharpe, 1964; Lintner, 1965))**.** Suppose that all investors choose portfolios by mean–variance criteria over one common period from the same μ and Σ (homogeneous beliefs), can lend and borrow without limit at r_f, and trade in frictionless markets that clear. Then, by Proposition 11.30, every investor holds the risky assets in the proportions w_T; aggregating, the market-capitalisation portfolio M equals the tangency portfolio, and Proposition 11.35 becomes the pricing relation

$$\mathbb{E}[R_i] - r_f = \beta_i \left(\mathbb{E}[R_M] - r_f\right) \qquad \text{for every asset } i,$$

with β_i measured against the market. The CAPM's mathematical content is one first-order condition, proved above as Proposition 11.35; its economic content is the equilibrium step — market clearing, existence, and the aggregation of individual demands into the market portfolio — which amounts to the claim that the market portfolio is mean–variance efficient, and is what empirical tests of the CAPM actually test. A proof is beyond our scope.

Example 11.37 (One stock fully worked, then the identity checked)**.** (a) A stock has $\rho_{iM} = 0.6$, $\sigma_i = 30\%$, and the market has $\sigma_M = 15\%$. By Definition 11.32, $\beta_i = 0.6 \times 0.30/0.15 = 1.2$. By (11.13) the systematic variance is $1.44 \times 0.0225 = 0.0324$ (volatility 18%), the idiosyncratic variance is $0.09 - 0.0324 = 0.0576$ (volatility 24%), and the systematic share is $0.0324/0.09 = 0.36 = \rho_{iM}^2$. With $r_f = 2\%$ and $\mu_M = 8\%$ the CAPM-implied mean is $2\% + 1.2 \times 6\% = 9.2\%$; a forecast mean of 14% is an alpha of 4.8%. A second stock with $\rho_{iM} = 0.9$ and $\sigma_i = 10\%$ has $\beta_i = 0.9 \times 0.10/0.15 = 0.6$ — tighter co-movement, half the beta — and its hedge ratio is 0.6, not 0.9 (Remark 5.23; the direction of the regression matters, Proposition 5.21).

(b) The SML holds exactly against the tangency portfolio of the running universe ($r_f = 2\%$, $w_T = (0.125, 0.25, 0.625)^\top$, Example 11.23). The covariances are $\mathrm{Cov}(R_i, R_T) = (\Sigma w_T)_i$ with $\Sigma w_T = 0.01\,(0.5,\ 1.25,\ 1.5)^\top = (0.005, 0.0125, 0.015)^\top = \mu_e/4$ (the row sums being $0.25 + 0.25$, $0.125+0.5+0.625$ and $0.25+1.25$), and $\mathrm{Var}(R_T) = 0.013125$, so the betas are $(0.381, 0.952, 1.143)^\top$. Then $\beta_i \times (7.25\% - 2\%) = (2\%, 5\%, 6\%)^\top = \mu_e$ exactly: every alpha is zero, as Proposition 11.35 requires. Against the minimum-variance portfolio instead, every beta equals 1 (Theorem 11.6(ii)) while the excess means differ, so w_g does not price the assets.

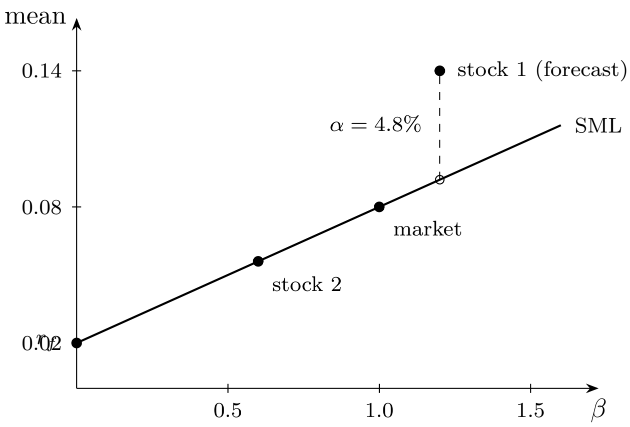


Figure 20: The security market line of Example 11.37(a): expected return against *beta*, not against σ (that is the CML of Figure 19). The line passes through $(0, r_f) = (0, 2\%)$ and $(1, \mu_M) = (1, 8\%)$. Stock 1 ($\beta = 1.2$) is forecast at 14% against a CAPM value of 9.2% (open circle); the dashed gap is its alpha of 4.8%. Stock 2 ($\beta = 0.6$) sits on the line. Under Fact 11.36 every asset sits on the line.

Trading application. The hedged position — long asset i, short β_i units of the market, the proceeds at r_f — has return $R_i - \beta_i(R_M - r_f) = r_f + \alpha_i + \varepsilon_i$, uncorrelated with R_M by Proposition 11.33(i). It is therefore a strategy with Sharpe ratio $\alpha_i/\,\mathrm{sd}(\varepsilon_i)$ (the *information ratio*) uncorrelated with the market, and by Corollary 11.26 the best combination with the market has $(\mathrm{SR}^*)^2 = \mathrm{SR}_M^2 + \left(\alpha_i/\,\mathrm{sd}(\varepsilon_i)\right)^2$: alpha adds to the market Sharpe ratio in quadrature through the information ratio. In Example 11.37(a), $\mathrm{SR}_M = 0.06/0.15 = 0.4$ and the information ratio is

$0.048/0.24 = 0.2$, giving $\sqrt{0.16 + 0.04} = 0.447$. One clause on a neighbouring construction: the beta-neutral projection of Corollary 6.26 strips the market mode (Fact 6.25) out of a signal vector *across* assets, whereas the time-series beta here is one asset's exposure through time; it is the same projection algebra in two different arenas.

Interview trap. Beta is not correlation: $\beta_i = \rho_{iM}\sigma_i/\sigma_M$ carries the volatility ratio, so a $\rho = 0.6$ stock can have $\beta = 1.2$ while a $\rho = 0.9$ stock has $\beta = 0.6$. Hedge with beta (it sets the magnitude); judge the hedge's quality with correlation (the residual share is $1 - \rho^2$) — Remark 5.23 restated. Nor does a high beta or a high volatility buy a higher Sharpe ratio: under the CAPM, dividing the pricing relation by σ_i and substituting $\beta_i = \rho_{iM}\sigma_i/\sigma_M$ gives $\mathrm{SR}_i = \rho_{iM}\,\mathrm{SR}_M \le \mathrm{SR}_M$, so no asset beats the market's Sharpe ratio; only the systematic part $\beta_i\sigma_M$ of an asset's volatility is priced, and Proposition 11.33(iii) shows why the idiosyncratic part carries no premium — it diversifies away.

Interview trap. CML versus SML: different horizontal axes (σ against β) and different membership (only efficient portfolios against every asset). An individual stock lies *below* the CML — its idiosyncratic variance is unrewarded — but *on* the SML under the CAPM; drawing the SML in the (σ, mean) plane is the most common whiteboard error in this block. Two bookkeeping slips travel with it: alpha is the intercept of the regression of *excess* returns on excess market returns, and with raw returns the intercept is $\alpha_i + r_f(1 - \beta_i)$, which is not α_i unless $\beta_i = 1$; and portfolio beta is a weighted average of betas, whereas portfolio volatility is not a weighted average of volatilities (§11.1).

11.6 Risk Contributions and Risk Parity

The desk questions: "decompose the book's volatility by position", "what are the risk-parity weights for a bond/equity portfolio, and do they depend on the correlation?", and the classic "60/40 is balanced, isn't it?". Capital weights say how money is allocated, not where the risk comes from. Because portfolio volatility is homogeneous of degree one in the weights, Euler's theorem splits it exactly into per-asset contributions; equalising those contributions defines risk parity; and the surprise is that the two-asset risk-parity weights $w_i \propto 1/\sigma_i$ hold for every correlation.

Setup. The covariance matrix Σ is positive definite. A position vector $w \in \mathbb{R}^n$, $w \neq 0$, need not sum to one (a book's exposures; the homogeneity argument does not use the budget constraint). The portfolio volatility is $\sigma_p(w) = \sqrt{w^\top \Sigma w} > 0$, so its partial derivatives exist. The beta of asset i to the portfolio is $\beta_{i,w} = \mathrm{Cov}(R_i, R_w)/\operatorname{Var}(R_w)$, Definition 11.32 with the portfolio itself as reference. In the equicorrelated case we write $\Delta = \operatorname{diag}(\sigma_1, \ldots, \sigma_n)$ and $\Sigma_\rho = (1-\rho)I + \rho\,\mathbf{1}\mathbf{1}^\top$ for the common correlation matrix (the equicorrelated model of Definition 11.9 with $v = 1$), so that $\Sigma = \Delta\Sigma_\rho\Delta$.

Definition 11.38 (Marginal and total risk contributions)**.** The *marginal risk contribution* of asset i is

$$\mathrm{MRC}_i = \frac{\partial \sigma_p}{\partial w_i} = \frac{(\Sigma w)_i}{\sigma_p},$$

the second equality being the chain rule on the square root, $\partial\sigma_p/\partial w_i = \frac{1}{2\sigma_p}\,\partial(w^\top\Sigma w)/\partial w_i$, with the gradient $2(\Sigma w)_i$ of (11.5). The *(total) risk contribution* is $\mathrm{RC}_i = w_i\,\mathrm{MRC}_i = w_i(\Sigma w)_i/\sigma_p$, and the *risk share* is RC_i/σ_p. Since $(\Sigma w)_i = \sum_j \Sigma_{ij} w_j = \mathrm{Cov}(R_i, R_w)$, one has $\mathrm{MRC}_i = \beta_{i,w}\,\sigma_p$ and the risk share equals $w_i\beta_{i,w}$.

Theorem 11.39 (Euler decomposition of portfolio volatility)**.** *For positive-definite Σ and the contributions of Definition 11.38:*

(i) For every $w \neq 0$,

$$\sigma_p(w) = \sum_{i=1}^{n} w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_{i=1}^{n} \mathrm{RC}_i; \tag{11.14}$$

equivalently the risk shares sum to one, $\sum_i w_i \beta_{i,w} = 1$.

(ii) At the minimum-variance portfolio w_g of Theorem 11.6, $\mathrm{MRC}_i = \sigma_g$ for every i: minimum variance equalises marginal *contributions, so the risk shares equal the capital weights, and the total contributions are equal only if the weights are.*

Proof. We proceed by homogeneity — Euler's theorem for functions homogeneous of degree one — and confirm the result by direct differentiation.

Part (i). For $t > 0$, $\sigma_p(tw) = \sqrt{t^2\, w^\top \Sigma w} = t\, \sigma_p(w)$. Differentiating both sides with respect to t, the chain rule on the left gives $\sum_i w_i\, (\partial \sigma_p/\partial w_i)(tw)$ and the right gives $\sigma_p(w)$; setting $t = 1$ yields (11.14). Direct check with Definition 11.38:

$$\sum_i w_i \frac{(\Sigma w)_i}{\sigma_p} = \frac{w^\top \Sigma w}{\sigma_p} = \sigma_p.$$

Dividing (11.14) by σ_p and using $\mathrm{RC}_i/\sigma_p = w_i \beta_{i,w}$ gives the statement about shares. The argument used only degree-one homogeneity, so it applies verbatim to any risk measure that is homogeneous of degree one in the positions; nothing further is claimed about other measures here.

Part (ii). By Theorem 11.6(ii), $(\Sigma w_g)_i = 1/A$ for every i, and $\sigma_g = 1/\sqrt{A}$, so $\mathrm{MRC}_i = (1/A)/(1/\sqrt{A}) = 1/\sqrt{A} = \sigma_g$. Multiplying by $w_{g,i}$, $\mathrm{RC}_i = w_{g,i}\sigma_g$ and the risk share is $w_{g,i}$. □

Definition 11.40 (Risk parity)**.** A long-only, fully invested portfolio $w > 0$, $\mathbf{1}^\top w = 1$, is *risk parity* (equal risk contribution) if $\mathrm{RC}_1 = \cdots = \mathrm{RC}_n$, equivalently if $w_i(\Sigma w)_i$ takes the same value for every i (σ_p is a common factor). The defining equations are invariant under $w \mapsto tw$, so the normalisation is imposed last.

Proposition 11.41 (Risk-parity weights: two assets for every correlation, and equicorrelated assets)**.** *Let Σ be positive definite. Then:*

(i) For $n = 2$ and any $\rho \in (-1, 1)$, $\mathrm{RC}_1 = \mathrm{RC}_2$ holds for positive weights if and only if $w_1\sigma_1 = w_2\sigma_2$, so the unique risk-parity portfolio is

$$w = \Big(\frac{\sigma_2}{\sigma_1 + \sigma_2}, \frac{\sigma_1}{\sigma_1 + \sigma_2}\Big)^\top, \qquad \sigma_p = \frac{\sigma_1\sigma_2\sqrt{2(1+\rho)}}{\sigma_1 + \sigma_2},$$

independent of ρ in its weights.

(ii) For $n \geq 2$ with a common pairwise correlation ρ ($1 + (n-1)\rho > 0$) and volatilities σ_i, the weights $w_i \propto 1/\sigma_i$ are risk parity.

(iii) If in addition every excess Sharpe ratio $(\mu_i - r_f)/\sigma_i$ equals a common $s_0 > 0$, the same portfolio is the tangency portfolio of Theorem 11.21, with $\mathrm{SR}^* = s_0\sqrt{n/(1 + (n-1)\rho)}$; at $\rho = 0$ this is $s_0\sqrt{n}$, the proof behind Remark 10.14 and Corollary 10.12: equal-Sharpe uncorrelated legs sized to equal volatility are simultaneously risk parity and maximum Sharpe.

Proof. We write out the equal-contribution equations and watch the correlation term cancel; for the equicorrelated case we factor Σ through its correlation matrix.

Part (i). With $(\Sigma w)_1 = w_1\sigma_1^2 + w_2\rho\sigma_1\sigma_2$ and $(\Sigma w)_2 = w_2\sigma_2^2 + w_1\rho\sigma_1\sigma_2$,

$$w_1(\Sigma w)_1 = w_1^2\sigma_1^2 + w_1 w_2\rho\sigma_1\sigma_2, \qquad w_2(\Sigma w)_2 = w_2^2\sigma_2^2 + w_1 w_2\rho\sigma_1\sigma_2.$$

The common term $w_1 w_2 \rho\sigma_1\sigma_2$ cancels identically, leaving $w_1^2\sigma_1^2 = w_2^2\sigma_2^2$, and for positive weights $w_1\sigma_1 = w_2\sigma_2$: equal volatility budgets. Normalising $w_1 + w_2 = 1$ gives the stated weights.

Substituting $w = \sigma_2/(\sigma_1+\sigma_2)$ into (11.2),

$$\sigma_p^2 = \frac{\sigma_2^2\sigma_1^2 + \sigma_1^2\sigma_2^2 + 2\rho\,\sigma_1\sigma_2\,\sigma_1\sigma_2}{(\sigma_1+\sigma_2)^2} = \frac{\sigma_1^2\sigma_2^2\,(2+2\rho)}{(\sigma_1+\sigma_2)^2},$$

and taking the square root gives the volatility.

Part (ii). Since the defining equations are scale-invariant, take $w_i = 1/\sigma_i$. Then $(\Sigma w)_i = \sum_j \rho_{ij}\sigma_i\sigma_j/\sigma_j = \sigma_i \sum_j \rho_{ij} = \sigma_i\,(1+(n-1)\rho)$, so $w_i(\Sigma w)_i = 1+(n-1)\rho$, the same for every i. (The computation shows more: inverse-volatility weights are risk parity whenever the correlation matrix has equal row sums, equicorrelation being the standard case.)

Part (iii). With $\Sigma = \Delta\Sigma_\rho\Delta$ as in the setup, $\Sigma^{-1} = \Delta^{-1}\Sigma_\rho^{-1}\Delta^{-1}$. The row-sum computation in the proof of Proposition 11.10(ii) (with $v=1$) gives $\Sigma_\rho\mathbf{1} = (1+(n-1)\rho)\mathbf{1}$, hence $\Sigma_\rho^{-1}\mathbf{1} = \mathbf{1}/(1+(n-1)\rho)$. The hypothesis says $\mu_e = s_0\,\Delta\mathbf{1}$, so

$$\Sigma^{-1}\mu_e = s_0\,\Delta^{-1}\Sigma_\rho^{-1}\Delta^{-1}\Delta\mathbf{1} = s_0\,\Delta^{-1}\Sigma_\rho^{-1}\mathbf{1} = \frac{s_0}{1+(n-1)\rho}\,\Delta^{-1}\mathbf{1},$$

whose entries are proportional to $1/\sigma_i$: by Theorem 11.21 the tangency portfolio is the inverse-volatility portfolio of part (ii). Its squared Sharpe ratio is $\mu_e^\top\Sigma^{-1}\mu_e = s_0^2\,\mathbf{1}^\top\Delta\,\Delta^{-1}\Sigma_\rho^{-1}\Delta^{-1}\Delta\,\mathbf{1} = s_0^2\,\mathbf{1}^\top\Sigma_\rho^{-1}\mathbf{1} = s_0^2\,n/(1+(n-1)\rho)$. □

Fact 11.42 (Existence and uniqueness of the risk-parity portfolio (Maillard, Roncalli and Teiletche, 2010))**.** For a general positive-definite Σ with $n \geq 3$ the defining equations $w_i(\Sigma w)_i = \text{const}$ of Definition 11.40 are nonlinear in w and are solved numerically; a long-only solution exists and is unique. A proof is beyond our scope.

Remark 11.43 (Risk parity versus minimum variance for two assets). The two-asset risk-parity weight $\sigma_2/(\sigma_1+\sigma_2)$ on asset 1 coincides with the minimum-variance weight $w^* = c/V$ of Proposition 11.3(ii) only at $\rho = -1$ (excluded by positive definiteness) or when $\sigma_1 = \sigma_2$: cross-multiplying $\sigma_2 V = c(\sigma_1+\sigma_2)$, with V and c as in (11.3), the reader should verify that the difference of the two sides is $\sigma_1\sigma_2(1+\rho)(\sigma_1-\sigma_2)$, which vanishes only in those two cases. In general, therefore, risk parity is neither the minimum-variance portfolio nor (by Theorem 11.21) the tangency portfolio, which depends on μ_e; Example 11.44 puts the three side by side.

Example 11.44 (Bonds and equities)**.** Take $\sigma_B = 5\%$, $\sigma_E = 15\%$ and $\rho = 0.2$, so

$$\Sigma = \begin{pmatrix} 0.0025 & 0.0015 \\ 0.0015 & 0.0225 \end{pmatrix}.$$

Risk parity by Proposition 11.41(i) is $w = (0.75, 0.25)^\top$: $\Sigma w = (0.001875 + 0.000375,\ 0.001125 + 0.005625)^\top = (0.00225, 0.00675)^\top$, so $w_i(\Sigma w)_i = 0.0016875$ for both assets; $\sigma_p = \sqrt{0.003375} = 5.81\%$, each side contributing 2.905%. The same weights are risk parity at $\rho = -0.2$ ($\sigma_p = 4.74\%$) and at $\rho = 0.6$ (6.71%): the split never moves, only the total. The classic 60/40 (0.4 bonds, 0.6 equities): $\Sigma w = (0.0019, 0.0141)^\top$, $\sigma_p = \sqrt{0.4(0.0019) + 0.6(0.0141)} = \sqrt{0.00922} = 9.60\%$, and by Definition 11.38 RC $= (0.79\%, 8.81\%)$, summing to 9.60% (the Euler check of Theorem 11.39); equities carry 91.8% of the risk (99.5% at $\rho = -0.2$). The marginal contributions for 60/40 are $(0.0198, 0.147)$: one extra point of equity weight adds 0.147 points of volatility, 7.4 times the bond marginal. The betas to the 60/40 portfolio are $(0.206, 1.529)$, and $0.4(0.206) + 0.6(1.529) = 1.000$, the shares summing to one. Levering the risk-parity mix up to the 9.60% volatility of 60/40 requires $9.60/5.81 = 1.65$ times the capital. Three portfolios side by side (Remark 11.43): the minimum-variance mix of Proposition 11.3(ii) is $w^* = 0.021/0.022 = 0.955$ in bonds, volatility 4.95%; risk parity is $(0.75, 0.25)$ at 5.81%; equal weight is $(0.50, 0.50)$ at $\sqrt{0.25(0.0025 + 0.0225 + 0.003)} = 8.37\%$. At $\rho = 0$ the minimum-variance mix moves to $(0.90, 0.10)$ while risk parity stays at $(0.75, 0.25)$.

Intuition. Volatility is a length, and Euler's theorem says a length is the sum of its shadows on the positions. Each shadow is the position size times how fast the length grows when that position is nudged. Minimum variance makes every nudge equally costly at the margin; risk parity makes every position's shadow equally long; the tangency portfolio makes every position's shadow proportional to its expected excess return. Three different symmetries, three different portfolios.

Interview trap. Capital weights are not risk weights: "60/40 is balanced" is 40/60 in capital and 8/92 in risk at $\rho = 0.2$. Standalone shares $w_i\sigma_i / \sum_j w_j\sigma_j$ are not risk contributions either — they ignore correlation and do not sum to σ_p; only the Euler contributions $w_i(\Sigma w)_i/\sigma_p$ do, and the sum is a one-line arithmetic check.

Interview trap. "Risk parity means inverse-volatility weights" is true for two assets at any correlation and whenever the correlation matrix has equal row sums — equicorrelation being the standard case (Proposition 11.41(i)–(ii)); with three or more assets and unequal correlations the contributions depend on the whole of Σ and the solution is numerical (Fact 11.42). Risk parity, minimum variance and maximum Sharpe are three different portfolios: the minimum-variance portfolio equalises marginal contributions (Theorem 11.39(ii)), risk parity equalises total contributions, and the tangency portfolio equalises the ratio of excess return to marginal risk ($\Sigma w_T \propto \mu_e$) — $(0.955, 0.045)$, $(0.75, 0.25)$ and a μ_e-dependent mix in Example 11.44. Risk parity and the tangency portfolio coincide under equal Sharpe ratios and equal correlations (Proposition 11.41(iii)); all three coincide when the volatilities are equal as well, in which case each is the equal-weight portfolio (Proposition 11.10(iii)). Risk parity, like the minimum-variance portfolio, uses no expected returns: it is a robustness choice, not an optimality claim, and the leverage needed to reach a target volatility is a separate decision.

11.7 Estimation Error: Why Naive Markowitz Fails in Practice

The open-ended follow-up at every fund, and the covariance-estimation question at research labs: "so why does nobody run raw Markowitz — what breaks, and what do you do instead?". Every formula in this section takes μ and Σ as known; in practice both are estimated, and the optimiser multiplies the estimation error by Σ^{-1}. The provable core is that Σ^{-1} amplifies expected-return error by up to $1/\lambda_{\min}$ along the smallest-eigenvalue direction, with a closed-form two-asset sensitivity factor $(1+\rho)/(1-\rho)$, the condition number of Σ when $\rho \geq 0$. The literature results — error maximisation, shrinkage, constraints as shrinkage, the $1/N$ horse race — are Facts with attribution, and a closing Remark ties the $\gamma = 1$ mean–variance objective to the Kelly criterion of §3.1, where estimation error becomes over-betting.

Setup. The true μ_e and Σ are unknown. The *plug-in* tangency portfolio is $\hat{w} \propto \hat{\Sigma}^{-1}\hat{\mu}_e$, with $\hat{\mu}_e$ and $\hat{\Sigma}$ estimated from T periods of returns. A *perturbation* $\delta \in \mathbb{R}^n$ is a change in the expected-return vector. By Fact 6.2, $\Sigma = Q\Lambda Q^\top$ with orthonormal eigenvectors $q_1, \dots, q_n$ and eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$; positive definiteness gives $\lambda_{\min} = \lambda_n > 0$ by Theorem 6.4. Here λ denotes an eigenvalue, as in the notation table, and $z_1, \dots, z_n$ denote the coordinates of a vector in the eigenbasis.

Proposition 11.45 (Σ^{-1} amplifies expected-return error)**.** *Under the setup above:*

(i) For the unnormalised tangency direction $w(\mu_e) = \Sigma^{-1}\mu_e$, a perturbation δ of the expected-return vector changes the weights by $\Sigma^{-1}\delta$, and

$$\|\Sigma^{-1}\delta\| \;\leq\; \frac{\|\delta\|}{\lambda_{\min}},$$

with equality if and only if δ lies in the eigenspace of $\lambda_{\min}$ (the direction q_n when λ_n is simple).

(ii) Take two assets with a common volatility σ, correlation $\rho \in (-1, 1)$, and excess means μ_1, μ_2 with $\mu_1 + \mu_2 > 0$. The normalised tangency weights satisfy

$$w_1 - w_2 = \frac{\mu_1 - \mu_2}{\mu_1 + \mu_2} \cdot \frac{1 + \rho}{1 - \rho} : \tag{11.15}$$

the relative gap between the two mean inputs is multiplied by $(1+\rho)/(1-\rho)$, which for $\rho \geq 0$ is the condition number of Σ (the ratio of its eigenvalues $\sigma^2(1+\rho)$ and $\sigma^2(1-\rho)$ along $(1, 1)^\top$ and $(1, -1)^\top$) and is unbounded as $\rho \to 1$; for $\rho < 0$ the factor is below one — the gap is damped — and equals the reciprocal of the condition number.

Proof. We expand the perturbation in the eigenbasis for part (i) and invert the 2×2 matrix explicitly for part (ii).

Part (i). By linearity, $w(\mu_e + \delta) - w(\mu_e) = \Sigma^{-1}\delta$. Write $\delta = \sum_i z_i q_i$ in the orthonormal eigenbasis of Fact 6.2, so that $\|\delta\|^2 = \sum_i z_i^2$. From $\Sigma q_i = \lambda_i q_i$ and $\lambda_i > 0$, multiplying by Σ^{-1} and dividing by λ_i gives $\Sigma^{-1} q_i = q_i/\lambda_i$, hence $\Sigma^{-1}\delta = \sum_i (z_i/\lambda_i) q_i$ and, by orthonormality,

$$\|\Sigma^{-1}\delta\|^2 = \sum_i \frac{z_i^2}{\lambda_i^2} \;\leq\; \frac{1}{\lambda_{\min}^2} \sum_i z_i^2 = \frac{\|\delta\|^2}{\lambda_{\min}^2},$$

since each $1/\lambda_i^2 \leq 1/\lambda_{\min}^2$. Equality holds if and only if $z_i = 0$ for every i with $\lambda_i > \lambda_{\min}$, i.e. δ lies in the eigenspace of $\lambda_{\min}$ — the argument of Theorem 6.12 applied to Σ^{-1}, whose largest eigenvalue is $1/\lambda_{\min}$.

Part (ii). With $\Sigma = \sigma^2 \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$, direct inversion gives

$$\Sigma^{-1} = \frac{1}{\sigma^2(1-\rho^2)} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}, \qquad \Sigma^{-1}\mu_e \propto (\mu_1 - \rho\mu_2,\ \mu_2 - \rho\mu_1)^\top.$$

The sum of the two entries is $(1-\rho)(\mu_1 + \mu_2) > 0$ and their difference is $(1+\rho)(\mu_1 - \mu_2)$. The normalised weights are the entries divided by their sum, so $w_1 - w_2$ is the difference divided by the sum, which is (11.15). The eigenvalues are the $d \pm c$ shortcut of Example 6.5 with $d = \sigma^2$, $c = \rho\sigma^2$: $\sigma^2(1+\rho)$ on $(1, 1)^\top$ and $\sigma^2(1-\rho)$ on $(1, -1)^\top$. For $\rho \geq 0$ the larger is $\sigma^2(1+\rho)$ and the ratio $\lambda_{\max}/\lambda_{\min}$ is $(1+\rho)/(1-\rho)$; for $\rho < 0$ the roles swap and the same expression is $\lambda_{\min}/\lambda_{\max} < 1$. $\square$

Remark 6.21 states the general principle: the small-eigenvalue directions are where inversion amplifies error. Here those directions are the near-arbitrage spreads between highly correlated assets.

Example 11.46 (One percentage point flips the portfolio)**.** Two assets, $\sigma = 20\%$ each, $\rho = 0.9$, $r_f = 0$, so

$$\Sigma = 0.04 \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}, \qquad \Sigma^{-1} = \begin{pmatrix} 131.6 & -118.4 \\ -118.4 & 131.6 \end{pmatrix},$$

and $\Sigma^{-1}\mu$ is proportional to $(\mu_1 - 0.9\mu_2,\ \mu_2 - 0.9\mu_1)$.

- Base case $\mu = (10\%, 10\%)$: proportional to $(0.01, 0.01)$, weights $(0.50, 0.50)$.
- Raise μ_1 to 11%: $(0.020, 0.001)$, weights $(0.952, 0.048)$ — one point of expected return moves 45 points of weight.
- Lower μ_1 to 9%: $(0, 0.019)$, weights $(0, 1)$.
- Lower μ_2 to 6% instead (keep $\mu_1 = 10\%$): proportional to $(0.10 - 0.054,\ 0.06 - 0.09) = (0.046, -0.030)$, sum 0.016, weights $(2.875, -1.875)$ — a 288% long position financed by a 188% short, from a four-point gap in means that is well inside one standard error (below). Check against (11.15): $(0.04/0.16)(19) = 4.75 = 2.875 - (-1.875)$.

Check against (11.15) at $\mu_1 = 11\%$: $(0.01/0.21)(1.9/0.1) = 0.905 = 0.952 - 0.048$. The amplification factor is 19, the condition number: eigenvalues 0.076 and 0.004, so $1/\lambda_{\min} = 250$ against $1/\lambda_{\max} = 13.2$, and the perturbation $(1,0)^\top = \frac{1}{2}((1,1)^\top + (1,-1)^\top)$ half-loads the spread direction. At $\rho = 0$ the same $+1$ point gives weights proportional to $(0.11, 0.10)$, i.e. $(0.524, 0.476)$, a 2.4-point move instead of 45: the amplification is entirely the factor $(1+\rho)/(1-\rho)$. The statistical scale of the shock: with $\sigma = 20\%$ and ten annual observations, the standard error of a sample mean is $0.20/\sqrt{10} = 6.3$ points (the variance of a T-period average is σ^2/T, the computation in the proof of Proposition 10.3) — six times the one-point change that already flipped the weights, and half again as large as the four-point gap that produced the levered long-short. The optimiser is responding to noise; expected returns are the least identifiable input, and fixing Σ alone is not enough.

Interview trap. The inverse Σ^{-1} amplifies estimation error — precisely along the small-eigenvalue directions, the near-arbitrage spreads between highly correlated assets where the sample covariance is least reliable (Remark 6.21) and where the optimiser bets hardest (a factor of 19 at $\rho = 0.9$). The three-hundred-percent position of Example 11.46 is the spread direction $(1,-1)^\top$ scaled by $1/\lambda_{\min}$: the optimiser is long the near-arbitrage spread between two almost identical assets, and a long-only constraint (Fact 11.49 below) or a shrunk $\hat{\mu}$ removes it. The mean vector, not the covariance, is the dangerous input: covariance estimates improve with more observations per unit of time, while the standard error of a mean shrinks only with calendar time, $\sigma/\sqrt{T}$ — a 6.3-point standard error against the 1-point shock that flipped the weights. A shrunk Σ with a raw $\hat{\mu}$ still produces error-maximised weights; minimum-variance and risk-parity portfolios avoid μ entirely, which is a large part of their practical appeal. The three-part answer to "why is Markowitz not used naively": the inputs are estimated, the inverse amplifies the error along low-variance directions, and the remedies are shrinkage, constraints, and the $1/N$ sanity benchmark.

Fact 11.47 (Error maximisation (Michaud, 1989), empirical)**.** A mean–variance optimiser fed sample estimates systematically overweights assets whose returns are overestimated and whose variances and covariances are underestimated, and underweights the reverse, because those are exactly the inputs that make an asset look attractive. The in-sample frontier therefore lies above the true frontier, and the realised out-of-sample Sharpe ratio of the "optimal" portfolio is typically worse than that of naive alternatives — Michaud's "estimation-error maximiser". The mechanism is selection: the optimiser picks the maximum of noisy estimates, and the maximum of noisy estimates is biased upward — the selection effect already flagged in item 4 of Remark 10.7 — operating along every low-variance direction of Proposition 11.45. A proof is beyond our scope.

Fact 11.48 (Shrinkage covariance estimation (Ledoit and Wolf, 2004))**.** Replace the sample covariance S by $(1-\zeta)S + \zeta F$, where F is a structured target (a scaled identity, or the constant-correlation matrix) and $\zeta \in [0,1]$ is the shrinkage intensity. The value of ζ minimising expected Frobenius loss has a consistent closed-form estimate. The result is well-conditioned and positive definite even when n exceeds the sample size — repair (1) of Remark 6.8 — and pulls the sample eigenvalues, which are biased outward (the largest too large, the smallest too small), toward their mean. It is the same medicine as ridge regression (Theorem 5.11, Remark 5.17) and the same bias–variance trade as Theorem 8.2 and Example 8.3. A proof is beyond our scope.

Fact 11.49 (Weight constraints as shrinkage (Jagannathan and Ma, 2003))**.** The no-short-sale minimum-variance portfolio — minimise $w^\top S w$ subject to $\mathbf{1}^\top w = 1$ and $w \geq 0$ — equals the unconstrained minimum-variance portfolio of the modified matrix $S - (\nu \mathbf{1}^\top + \mathbf{1}\nu^\top)$, where $\nu \geq 0$ is the vector of Lagrange multipliers of the sign constraints. Imposing the constraint is therefore equivalent to shrinking the covariances of the assets that would otherwise have been shorted, which are exactly the large sample covariances driving extreme positions — the interview one-liner is that constraints are implicit shrinkage. The argument runs through the Karush–Kuhn–Tucker conditions. A proof is beyond our scope.

Fact 11.50 (Out-of-sample performance of shrinkage and constraints (Ledoit and Wolf, 2004; Jagannathan and Ma, 2003), empirical)**.** On equity data, the shrinkage estimator of Fact 11.48 gives lower out-of-sample portfolio variance than the sample covariance S, and the constrained sample minimum-variance portfolio of Fact 11.49 performs as well out of sample as unconstrained portfolios built from shrunk or factor-based covariance matrices. These are measurements, not theorems. A proof is beyond our scope.

Fact 11.51 ($1/N$ versus optimised portfolios (DeMiguel, Garlappi and Uppal, 2009), empirical)**.** Across fourteen optimising models (sample-based mean–variance, Bayesian and shrinkage variants, minimum variance, constrained versions) and seven equity datasets, none consistently beat the equal-weight $1/N$ rule out of sample on Sharpe ratio, certainty-equivalent return or turnover. For 25 assets the sample-based mean–variance rule needs an estimation window of about 3,000 months (250 years) to outperform $1/N$, and about 6,000 months for 50 assets. This is evidence, not mathematics: at realistic sample sizes the gains from optimisation are smaller than the losses from estimation error. Proposition 11.10 explains the strength of the benchmark: $1/N$ already captures the bulk of diversification with zero estimation error, and it is itself the minimum-variance and the risk-parity portfolio under equal volatilities and equal correlations. A proof is beyond our scope.

Interview trap. The in-sample efficient frontier is not achievable: it is an upward-biased estimate of the true frontier (Fact 11.47), and a candidate who reports only an in-sample Sharpe ratio has not answered. "$1/N$ is naive" — it is the zero-estimation-error benchmark and wins the horse race at realistic sample sizes (Fact 11.51). "Constraints are ad hoc" — a no-short constraint is shrinkage in disguise (Fact 11.49).

Remark 11.52 (Mean–variance as second-order Kelly). For a portfolio return R_w small enough that $\ln(1+x) = x - x^2/2 + O(x^3)$ applies (Fact 7.30), the expected log growth of Definition 3.1 generalises to

$$\mathbb{E}[\log(1+R_w)] = w^\top\mu - \tfrac{1}{2}(w^\top\Sigma w + (w^\top\mu)^2) + \dots,$$

using $\mathbb{E}[R_w^2] = \mathrm{Var}(R_w) + (\mathbb{E}[R_w])^2$. Dropping the third-order terms and the $(w^\top\mu)^2$ term (second order in the small means) leaves $w^\top\mu - \frac{1}{2}w^\top\Sigma w$: the objective of Proposition 11.30 with $\gamma = 1$ and $r_f = 0$. With a risk-free asset, the same expansion of $\log(1 + r_f + w^\top(R - r_f\mathbf{1}))$ — treating r_f as small of the same order as the excess means, so that the corrections from the factors $(1+r_f)^{-1}$ are of the same second order as the dropped $(w^\top\mu_e)^2$ term — leads to $w^\top\mu_e - \frac{1}{2}w^\top\Sigma w$ up to a constant and terms of that order. Its maximiser $\Sigma^{-1}\mu_e$ is the multi-asset, continuous-return Kelly allocation. For one asset it reads $f = \mu_e/\sigma^2$, "edge over variance", the analogue of the edge-over-odds rule $f^* = (bp - q)/b$ of Theorem 3.3; Proposition 3.6 is precisely the one-asset case of this quadratic approximation: in that proposition's notation, where $\varepsilon = p - q$ is the edge of an even-money bet (not the residual of §11.5), $\tilde{g}(f) = \varepsilon f - f^2/2$, the bet having per-bet return of size f with mean εf and variance f^2 to leading order. Fractional Kelly is $\gamma > 1$, and log utility itself is item (1) of Remark 3.24. In the running universe, full Kelly is $w = \Sigma^{-1}\mu_e = (0.5, 1, 2.5)^\top$ (Example 11.31): 400% gross exposure, excess mean 21%, volatility 45.8%, and approximate growth rate $0.21 - 0.105 = 10.5\%$ per year; the fully invested tangency portfolio is that vector divided by 4. The estimation-error discussion of this subsection is therefore Remark 3.8(1) in matrix form: an overestimated edge in any direction of Σ^{-1} means over-betting in that direction, with the asymmetric penalty that remark describes — which is why practice sets γ well above 1.

Procedure 11.53 (Robust allocation pipeline)**.** Given T periods of returns on n assets:

1. Estimate Σ with shrinkage (Fact 11.48) or a factor structure (repair (3) of Remark 6.8).
2. Avoid or shrink μ: default to the minimum-variance, risk-parity or $1/N$ portfolio when views are weak, and shrink strategy Sharpe ratios toward a common prior, remembering their standard error from Fact 4.16.
3. Impose long-only, position and turnover constraints (Fact 11.49).
4. Size total leverage by fractional Kelly, i.e. γ well above 1 (Remark 11.52).

5. Benchmark out of sample against $1/N$ and report the gap between in-sample and out-of-sample Sharpe ratios.

Trading application. Procedure 11.53 is the answer to "what do you do instead of raw Markowitz": each step closes one channel through which Σ^{-1} turns estimation noise into positions, and the last step is the only honest measure of whether the optimiser earned its complexity.

12 Market Making I: the Economics of the Spread

A market maker earns the right to trade by quoting two prices at once: a bid at which it stands ready to buy and an ask at which it stands ready to sell. The gap between the two — the spread — is not arbitrary; it is the priced sum of identifiable costs and risks, plus a margin. This section names the components of the spread (§12.1), prices the subtlest component — adverse selection — exactly in a two-value Glosten–Milgrom miniature that is a standard interview derivation (§12.2), converts the theory into a measurement a trading desk can run on its own fills (§12.3), and closes with the decision rules for adjusting quotes when conditions change (§12.4).

12.1 Where the Spread Comes From

The most basic market-making interview question is also the most important: where does the bid–ask spread come from? The answer is a four-part decomposition, worth stating precisely because every quoting decision in this section and the next addresses one of its parts.

Setup. Consider a market maker quoting a single asset. All prices are in the asset's currency units.

Definition 12.1 (Bid, ask, and quoted spread)**.** A market maker posts a *bid* b, the price at which it stands ready to buy, and an *ask* $a \geq b$, the price at which it stands ready to sell (in any live market the inequality is strict). The *quoted spread* is $s = a - b$, and the *mid-price* (or *mid*) is $m = (a + b)/2$. A counterparty who sells to the maker at b is said to *hit the bid*; a counterparty who buys from the maker at a is said to *lift the ask*.

Definition 12.2 (Spread decomposition)**.** The economically sustainable spread decomposes into four components, each the compensation for a distinct cost borne by the maker:

$$s = \underbrace{\text{adverse selection}}_{\text{informed pickoffs}} + \underbrace{\text{inventory risk}}_{\text{price moves while held}} + \underbrace{\text{order processing and fees}}_{\text{infrastructure, exchange fees}} + \underbrace{\text{margin}}_{\text{profit}}.$$

Adverse selection is the expected loss to better-informed counterparties who choose to trade against the quote; inventory risk is the exposure to price moves while the maker holds the position a fill hands it; order processing covers infrastructure, exchange and clearing fees; the margin is the maker's profit.

Volatility widens the first two components: when prices move more, an informed counterparty's private information is worth more per fill (the pickoff profit per trade grows), and any inventory a fill hands the maker is riskier to carry. Competition between makers compresses the margin, and squeezes operational slack out of the processing component, but it cannot push the spread below the actuarial cost of the first two components: a maker quoting inside that cost loses money in expectation and exits. In §12.2 we compute the adverse-selection part of that floor exactly, in the smallest model that produces it (Theorem 12.4).

12.2 Adverse Selection: a Glosten–Milgrom Miniature

This subsection isolates the adverse-selection component of Definition 12.2 in a two-value miniature of the Glosten–Milgrom sequential-trade model. The derivation uses nothing beyond conditional probability, Bayes' rule, and conditional expectation, and it delivers the cleanest available answer to the question of why a spread must exist at all.

Setup. The following assumptions are in force for the rest of this subsection.

- An asset has an unknown value $V \in \{99, 101\}$ with $P(V = 101) = P(V = 99) = \frac{1}{2}$, so the prior mean is $\mathbb{E}[V] = 100$.
- A market maker posts a bid b and an ask a as in Definition 12.1, and a single trader arrives to trade one unit against these quotes.
- With probability $\pi \in [0, 1)$ the trader is *informed*: it knows V, buys (lifts the ask) when $V = 101$, and sells (hits the bid) when $V = 99$ — it trades only in the profitable direction.
- With probability $1 - \pi$ the trader is *noise*: it buys or sells with probability $\frac{1}{2}$ each, independently of V. The trader's type is independent of V.
- The maker is risk neutral and pays no fees and no inventory costs. The value V is revealed after the trade, so the maker's realized profit is $a - V$ on a lifted ask and $V - b$ on a hit bid.
- We write "buy" for the event that the arriving trader buys at the ask and "sell" for the event that it sells at the bid, and we restrict attention to quotes with $99 < b \le a < 101$, so that an informed trader strictly profits by trading on its information. The reader should verify that the quotes found in Theorem 12.4 satisfy this restriction for every $\pi \in [0, 1)$.

Definition 12.3 (Zero-profit quotes)**.** Quotes (b, a) are *zero-profit* if each side earns zero expected profit conditional on being traded against:

$$\mathbb{E}[\,a - V \mid \text{buy}\,] = 0 \qquad \text{and} \qquad \mathbb{E}[\,V - b \mid \text{sell}\,] = 0.$$

Since $\mathbb{E}[\,a - V \mid \text{buy}\,] = a - \mathbb{E}[V \mid \text{buy}]$ by linearity, the conditions say exactly that $a = \mathbb{E}[V \mid \text{buy}]$ and $b = \mathbb{E}[V \mid \text{sell}]$: each quote must equal the value of the asset conditional on the side that trades at it. Competition between makers drives quotes to this point — any maker quoting a strictly profitable side is undercut.

Theorem 12.4 (Glosten–Milgrom breakeven quotes)**.** *Under the setup above — values* 99 *and* 101 *equiprobable, informed fraction* $\pi \in [0, 1)$*, noise flow buying and selling with probability* $\frac{1}{2}$ *each — the unique zero-profit quotes are*

$$a = 100 + \pi, \qquad b = 100 - \pi,$$

and the breakeven spread is therefore $s = a - b = 2\pi$*.*

Proof. We proceed by conditioning: first on the trader's type to obtain the probability of a buy in each state of V, then by Bayes' rule to obtain the posterior on V given a buy, and finally by computing the conditional expectation that the zero-profit condition of Definition 12.3 equates to the ask.

Step 1 (buy probabilities in each state). Condition on $V = 101$: with probability π the trader is informed and buys with certainty; with probability $1 - \pi$ it is a noise trader and buys with probability $\frac{1}{2}$, independently of V. Conditioning on $V = 99$, an informed trader never buys. By the law of total probability,

$$P(\text{buy} \mid V = 101) = \pi + \frac{1 - \pi}{2} = \frac{1 + \pi}{2}, \qquad P(\text{buy} \mid V = 99) = \frac{1 - \pi}{2}. \tag{12.1}$$

Averaging over the equiprobable states gives the unconditional buy probability

$$P(\text{buy}) = \frac{1}{2} \cdot \frac{1+\pi}{2} + \frac{1}{2} \cdot \frac{1-\pi}{2} = \frac{1}{2}.$$

Step 2 (Bayes' rule). Using (12.1),

$$P(V = 101 \mid \text{buy}) = \frac{P(\text{buy} \mid V = 101)\, P(V = 101)}{P(\text{buy})} = \frac{\frac{1+\pi}{2} \cdot \frac{1}{2}}{\frac{1}{2}} = \frac{1+\pi}{2}.$$

Step 3 (conditional expectation). Consequently

$$\mathbb{E}[V \mid \text{buy}] = 101 \cdot \frac{1+\pi}{2} + 99 \cdot \frac{1-\pi}{2} = \frac{101+99}{2} + \frac{101-99}{2}\pi = 100 + \pi.$$

Step 4 (zero profit pins the ask). For any quotes in the restricted class $99 < b \le a < 101$, the behavior of both trader types — hence every conditional probability computed above — does not depend on the exact quote levels, so $\mathbb{E}[V \mid \text{buy}] = 100 + \pi$ regardless of a. The maker's expected profit conditional on a lift, $\mathbb{E}[\,a - V \mid \text{buy}\,] = a - (100 + \pi)$, is strictly increasing in a and vanishes if and only if $a = 100 + \pi$: the zero-profit ask exists and is unique.

Step 5 (the bid, by the mirrored computation). The model is invariant under simultaneously relabeling the states $99 \leftrightarrow 101$ and the sides buy $\leftrightarrow$ sell; carrying out the same three steps on the sell side gives

$$P(\text{sell} \mid V = 99) = \frac{1+\pi}{2}, \qquad P(\text{sell} \mid V = 101) = \frac{1-\pi}{2}, \qquad P(\text{sell}) = \frac{1}{2},$$

so that $P(V = 99 \mid \text{sell}) = \frac{1+\pi}{2}$ and

$$\mathbb{E}[V \mid \text{sell}] = 99 \cdot \frac{1+\pi}{2} + 101 \cdot \frac{1-\pi}{2} = 100 - \pi,$$

and the zero-profit bid is uniquely $b = 100 - \pi$. The breakeven spread is $s = a - b = (100 + \pi) - (100 - \pi) = 2\pi$. $\square$

(Problem P12.2 asks you to reproduce this proof from scratch, at $\pi = 0.3$.)

Figure 21 shows the geometry of the result on the value axis: the two possible values, the prior mean, the two quotes sitting symmetrically a distance π from it, and the flow that trades at each quote.

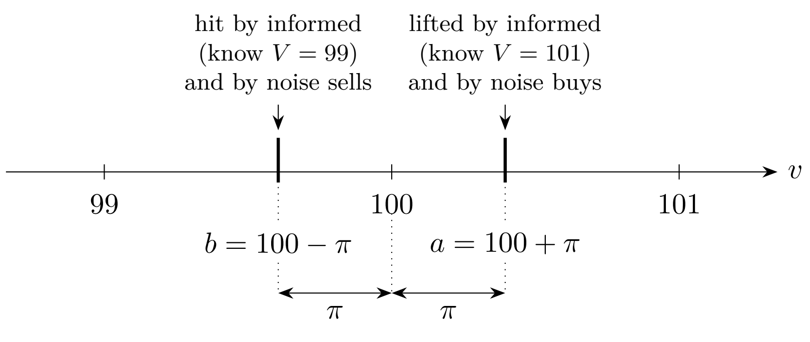


Figure 21: The Glosten–Milgrom miniature of Theorem 12.4 on the value axis, drawn for $\pi = 0.4$. The asset is worth 99 or 101 with equal probability, so the prior mean is 100. The zero-profit quotes sit symmetrically at $100 \pm \pi$: each quote equals the expected value of the asset conditional on the side that trades at it, so every fill breaks even in expectation and the quoted spread is $a - b = 2\pi$.

Example 12.5 (Breakeven quotes with 30 percent informed flow)**.** Take $\pi = 0.3$. By (12.1), $P(\text{buy} \mid V = 101) = 0.65$ and $P(\text{buy} \mid V = 99) = 0.35$, so $P(\text{buy}) = 0.5$ and Bayes' rule gives

$$P(V = 101 \mid \text{buy}) = \frac{0.65 \times 0.5}{0.5} = 0.65.$$

The zero-profit ask is therefore

$$a = \mathbb{E}[V \mid \text{buy}] = 101(0.65) + 99(0.35) = 65.65 + 34.65 = 100.30,$$

and by Theorem 12.4 the bid is $b = 99.70$, for a breakeven spread of $s = 0.60 = 2\pi$.

Direct profit check. Conditional on a lift, the asset the maker delivers is worth 100.30 on average, so an ask of 100.30 earns exactly zero per lifted offer; a maker who instead quoted the prior mean 100 would lose 0.30 in expectation on every lifted offer, with zero fees, zero inventory cost, and no profit margin in the model at all. As a small-case check, setting $\pi = 0$ collapses the quotes to $a = b = 100$ and the spread to zero.

Intuition. Even a maker with zero directional opinion must quote a spread: the flow that chooses to trade against a fixed quote is conditionally biased against it. The spread is an insurance premium, and Theorem 12.4 prices it by conditional expectation — each quote sits at the value of the asset given the bad news implicit in being traded with.

Interview trap. "Incoming buys and sells are equally likely, so a maker with no directional opinion can quote an arbitrarily tight spread around fair value and still break even." The premise is true — in the model, $P(\text{buy}) = \frac{1}{2}$ unconditionally — but the conclusion confuses unconditional flow with the flow that actually trades: a resting ask is lifted disproportionately in the state $V = 101$. Quotes must be set at the expected value conditional on being traded with, not at the unconditional mean, and any spread below 2π loses money in expectation even with zero fees, zero inventory cost, and no margin. This is also why competition cannot drive spreads to zero: it can compress only the margin component of Definition 12.2, never the actuarial floor.

Remark 12.6 (Limiting cases of the breakeven spread)*.* The breakeven spread 2π is increasing in the informed fraction, and its two limits are instructive. As $\pi \to 0$ the spread tends to zero: facing pure noise flow, a maker can quote at the mid and break even — the claim in the trap above is exactly this special case. As $\pi \to 1$ the quotes tend to the two possible values themselves, $b \to 99$ and $a \to 101$, and the spread tends to 2, the full width of the uncertainty: every fill breaks even exactly, and with no noise flow left there is no counterparty from whom a margin could be earned — the market is at the edge of shutting down. For every $\pi \in [0, 1)$ the quotes remain strictly inside $(99, 101)$, consistent with the restriction in the Setup.

12.3 Measuring Adverse Selection: Mark-Outs

Theorem 12.4 prices adverse selection in a model where the informed fraction π is known. A desk must instead estimate the toxicity of its actual flow from its own fills. The standard estimator — and the concrete, operational answer to the interview question "what is adverse selection?" — is the mark-out.

Setup. Fix a horizon $\tau > 0$, measured in seconds, and write m_u for the mid-price of Definition 12.1 at time u. Consider a fill executed at time t against the maker's quotes: either the maker's ask a is lifted (the maker sells at a) or its bid b is hit (the maker buys at b).

Definition 12.7 (Mark-out)**.** The *mark-out* of the fill at horizon τ is the mid-price change τ seconds after the fill, signed so that a negative value means the market moved against the maker's trade:

$$M_\tau = \begin{cases} a - m_{t+\tau} & \text{if the ask was lifted (the maker sold at } a\text{),} \\ m_{t+\tau} - b & \text{if the bid was hit (the maker bought at } b\text{).} \end{cases}$$

Desk practice averages M_τ over fills within buckets — by counterparty, by venue, and by size — and monitors the averages across several horizons τ.

The average mark-out is realized adverse selection, and it estimates the informed-flow premium of the model. To see the link, return to the miniature of §12.2 and suppose that after each trade the market impounds the trade's information, so that the post-trade mid equals the conditional expectation of V given the observed side. A maker who ignores Theorem 12.4 and quotes both sides at the prior mean 100 records $M_\tau = 100 - (100 + \pi) = -\pi$ on every lifted ask and $M_\tau = (100 - \pi) - 100 = -\pi$ on every hit bid: the average mark-out recovers exactly the premium π that the maker failed to charge. At the zero-profit quotes the same computation gives $M_\tau = 0$ on both sides. A persistently negative mark-out bucket is therefore under-priced toxicity in that bucket.

Example 12.8 (Mark-out computation)**.** A maker's ask is lifted at $a = 100.30$; five seconds later the mid stands at $m_{t+5} = 100.34$. The mark-out is

$$M_5 = 100.30 - 100.34 = -0.04,$$

an adverse move of four cents on a price near 100 — about 4 basis points against the maker. Separately, the maker's bid is hit at $b = 99.70$ and the mid five seconds later is 99.71, giving $M_5 = 99.71 - 99.70 = +0.01$: a mildly profitable fill. If fills of the first kind concentrate in a single counterparty bucket — say one counterparty's fills mark 4 basis points against the maker at five seconds while all other flow marks flat — that counterparty's flow is toxic: its trades systematically precede adverse moves.

Trading application. A market-maker dashboard's mark-outs-by-counterparty panel is exactly the estimator of Definition 12.7: average post-fill mid drift, bucketed by counterparty, venue, and size. In an interview, this panel is the concrete answer to "what is adverse selection and how would you measure it?" — realized adverse selection is the average mark-out on the desk's own fills, and the bucketing identifies which flow is informed.

12.4 Widen, Skew, or Pull

A maker that concludes its current quotes are wrong has exactly three controls: *widen* (increase the spread s around an unchanged mid), *skew* (shift both quotes in the same direction, moving the effective mid), and *pull* (withdraw quotes entirely). Which control answers which diagnosis follows directly from the spread decomposition of Definition 12.2 and the mark-out evidence of §12.3.

Remark 12.9 (Quote-adjustment decision rules)*.* The standard mapping from diagnosis to control is the following.

- *Volatility spikes: widen.* Both volatility-sensitive components of Definition 12.2 — adverse selection and inventory risk — grow with volatility. The flow itself has not become more informed, so the maker keeps quoting, at a spread that re-prices the risk.
- *Mark-outs deteriorate in an identifiable bucket: widen or pull for that bucket.* Flow that consistently marks against the maker is toxic; where the venue permits differentiation, widen against that bucket, cap the size shown to it, or stop quoting to it.
- *Toxic flow that cannot be distinguished: wider by default, for everyone.* Unidentifiable toxic flow raises the average informed fraction — the effective π of Theorem 12.4 — of all flow, and the breakeven spread rises with it.
- *News or a jump in uncertainty imminent: pull first.* Re-quote wide once the news is out, and tighten as the picture clears.
- *Inventory builds up: skew.* Skewing is driven by inventory risk rather than adverse selection; it is the subject of the next section.

Intuition. Pulling is not an admission that quoting has failed; it is the limiting zero-profit quote. Around an unscheduled announcement, the informed — or merely faster — fraction of flow spikes, so the breakeven spread 2π of Theorem 12.4 can momentarily exceed any spread that would still be filled: quoting nothing is then the only quote that does not lose money in expectation, and being on the wrong side of a single gap through a stale quote costs many spreads.

Practice: problems P12.1–P12.4 in §20, solutions in §21.

13 Market Making II: Inventory and Skewing

The preceding section priced the spread; this section decides where to place it. A maker who has been filled on one side is no longer indifferent to direction, and the center of the quote pair must move away from the mid to reflect the position being carried. We develop the reservation price of Avellaneda and Stoikov, write the make-me-a-market interview ritual as an explicit procedure, work through a calibration game exactly as played in a live assessment, and close with how a desk sheds inventory when skewing alone is too slow.

13.1 Inventory Risk and the Reservation Price

A market maker's quotes are supposed to earn the spread, but every fill also changes the book: inventory accumulates, and inventory converts price risk into profit-and-loss risk for as long as it is held. The central result of this subsection says that a maker carrying inventory should quote around a *shifted* center — the reservation price — rather than around the mid, and it makes the size of the shift explicit.

Setup. Fix a current time t and a risk horizon $T > t$. The mid price of the traded asset is m, and over the remaining horizon the mid is modeled as driftless with variance rate σ^2: writing m_T for the terminal mid, $\mathbb{E}[m_T] = m$ and $\mathrm{Var}(m_T) = \sigma^2(T-t)$, where σ^2 has units of price squared per unit time. The maker's signed inventory is q (in units of the asset; positive means long, negative means short) and the maker's cash is c. The risk-aversion parameter is $\gamma > 0$, with units $1/(\text{price} \times \text{units})$ so that $q\gamma\sigma^2(T-t)$ is a price. Quotes are a bid and an ask placed at half-width $\delta > 0$ around a chosen center. Within this section the symbols q and r denote inventory and the reservation price; they are unrelated to the Markov-chain notation $q = 1-p$ and $r = q/p$ used elsewhere in the document.

Fact 13.1 (Reservation price (Avellaneda–Stoikov, 2008))**.** Consider a market maker with exponential utility of terminal wealth (risk aversion γ) who quotes a bid and an ask around a driftless diffusion mid with variance rate σ^2, whose quotes are filled by market orders arriving with an intensity that decays with the quote's distance from the mid. The maker's indifference price for inventory — the *reservation price* — is

$$r = m - q\,\gamma\,\sigma^2\,(T-t), \tag{13.1}$$

and the optimal bid and ask straddle r, not the mid m. For $q > 0$ the correction is negative and both quotes shift *down*: the ask moves toward the mid (the long is offered out more eagerly) and the bid backs away (further buying is discouraged). For $q < 0$ the shift is upward, symmetrically. The full derivation, via the Hamilton–Jacobi–Bellman equation of the underlying stochastic control problem, is beyond our scope.

Although the control-theoretic proof is out of scope, every factor in (13.1) can be recovered from a one-line variance-penalty argument. The following proposition is that argument. It is a *heuristic* relative to the dynamic model — it freezes the book and ignores future quoting entirely — but it reproduces the formula exactly and explains why each of q, γ, σ^2, and $(T-t)$ belongs in the correction.

Proposition 13.2 (Inventory-penalty heuristic)**.** *Adopt the setup above and freeze the book: the maker holds cash c and inventory q from t to T with no further trading, and values random terminal wealth W by the mean–variance certainty equivalent*

$$\mathrm{CE}[W] = \mathbb{E}[W] - \frac{\gamma}{2}\,\mathrm{Var}(W).$$

Then the marginal indifference price for inventory — the price per unit at which selling an infinitesimal quantity leaves the certainty equivalent unchanged — equals

$$r(q) = m - q\,\gamma\,\sigma^2\,(T-t),$$

in agreement with (13.1).

Proof. We compute the certainty equivalent of the frozen book as a function of inventory and differentiate. Terminal wealth is $W(q) = c + q\,m_T$, so by the setup $\mathbb{E}[W(q)] = c + qm$ and $\mathrm{Var}(W(q)) = q^2\sigma^2(T-t)$, giving

$$\mathrm{CE}(q) = c + q\,m - \frac{\gamma}{2}\,q^2\sigma^2(T-t). \tag{13.2}$$

The mean grows linearly in q, but the variance penalty grows quadratically: each additional unit is worth less to the book than the one before. Now sell an infinitesimal quantity dq at a unit price r: inventory falls to $q - \mathrm{d}q$ and cash rises by $r\,\mathrm{d}q$. To first order the certainty equivalent changes by $(r - \partial\,\mathrm{CE}\,/\partial q)\,\mathrm{d}q$, so indifference requires the sale price to equal the marginal certainty-equivalent value of inventory:

$$r(q) = \frac{\partial\,\mathrm{CE}}{\partial q} = m - q\,\gamma\,\sigma^2(T-t).$$

This is (13.1). The reader should verify from (13.2) that the indifference price for selling one *whole* unit is $\mathrm{CE}(q) - \mathrm{CE}(q-1) = m - \gamma\sigma^2(T-t)\,(q - \frac{1}{2})$, which agrees with the marginal version up to the half-unit convention. □

(Problem P13.2 asks you to reproduce this argument from scratch and to stress-test each factor of the correction.)

Example 13.3 (Reservation-price quoting, numeric)**.** A maker quotes a coin whose mid is $m = 50.00$. The book is long $q = +40$ units, the risk aversion is $\gamma = 0.002$ per (price unit × unit), the variance rate is $\sigma^2 = 1.5$ squared price units per day, and $T - t = 0.5$ days remain in the risk horizon. The inventory shift in (13.1), computed stepwise, is

$$q\,\gamma\,\sigma^2(T-t) = 40 \times 0.002 \times 1.5 \times 0.5 = 0.08 \times 1.5 \times 0.5 = 0.12 \times 0.5 = 0.06,$$

so the reservation price is $r = 50.00 - 0.06 = 49.94$. With a fixed half-width $\delta = 0.10$ the quotes are 49.84 bid at 50.04 ask. Compared with mid-centered quotes (49.90 bid at 50.10 ask), *both* quotes sit 0.06 lower: the ask now undercuts, so the offer is more likely to be lifted and the long is sold down; the bid has backed off, so the book is less likely to be hit into an even larger long. The maker leans toward shedding inventory while remaining two-sided. If instead the book is short $q = -40$ units, the shift flips sign by linearity: $r = 50.06$, and the quotes become 49.96 bid at 50.16 ask, now leaning to buy the short back. Figure 22 sketches the geometry.

Intuition. The reservation price is where the maker is genuinely indifferent, and quotes should always straddle the point of indifference. To a long book the marginal unit is worth *less* than mid, because it adds variance the maker must carry to the horizon; the quoting center therefore drops below mid by exactly the marginal risk cost $q\gamma\sigma^2(T-t)$. Skewing is not a directional bet — it is the book repricing its own risk: sell more eagerly, buy less eagerly, and never go one-sided.

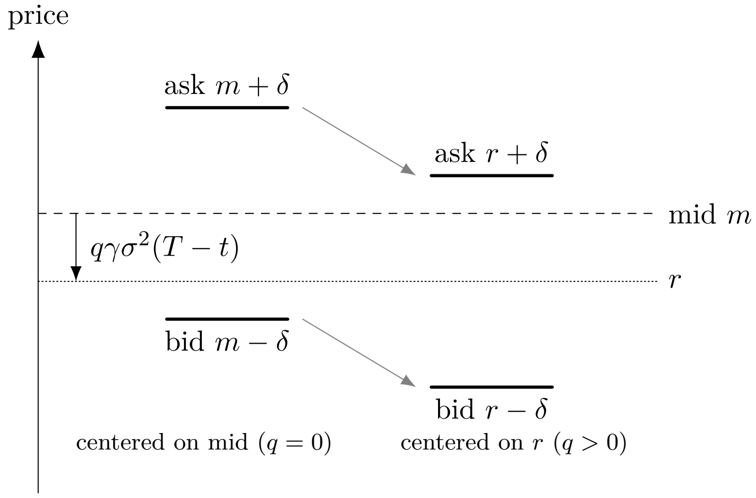


Figure 22: Mid price versus reservation price for positive inventory. A maker long $q > 0$ units centers the quote pair on the reservation price $r = m - q\gamma\sigma^2(T-t)$ of (13.1) rather than on the mid m: both quotes shift down by the same amount, so the ask moves toward the mid (the long is offered out more eagerly) while the bid backs away (further accumulation is discouraged). The half-width δ is unchanged — skewing relocates the quote pair; it does not widen it.

13.2 The Make-Me-a-Market Ritual

"Make me a market on X" is one of the most common live tests at trading firms, and underneath it sits exactly the machinery of §13.1: a fair-value estimate, a defensible width, Bayesian updating on the flow, and an inventory skew. The graded behavior can be written down as a procedure.

Procedure 13.4 (Make-me-a-market ritual)**.** Given a request to make a market on a quantity X:

1. **Compute a fair value silently.** Form the estimate before speaking; the market is quoted around it, not reasoned toward out loud.
2. **Quote two-sided, at a width that can be defended.** The width prices the uncertainty in the estimate (the spread economics of the preceding section); a width chosen at random is indefensible in the follow-up.
3. **After each fill, update the level and skew.** A fill is information: move the level in the direction of the flow (a Bayesian update on what the counterparty's willingness to trade reveals) *and* shift the center away from the accumulated inventory, as in (13.1).
4. **Track and narrate inventory aloud.** A maker long two units might say: "long two now, so 6.6 at 7.4." The narration shows the count is being kept and ties the skew to the position.
5. **Never freeze.** A committed, updatable price beats a perfect silent one.

Interview trap. The repeated-lift trap: the interviewer announces "I buy. I buy again. I buy again — same price?" Keeping the same offer through repeated lifts ignores both the information in the flow (an offer that keeps getting lifted is evidently cheap) and the maker's growing short position. Move the market — step 3 of Procedure 13.4 applies after *every* fill, not once.

13.3 The Calibration Game: Making a Market on an Estimate

A harder variant of the ritual fuses Fermi estimation with quoting: the player must first manufacture the fair value from scratch, state an honest confidence interval around it, and then trade on it. The game grades calibration and bookkeeping, not trivia. Throughout this subsection, k denotes thousands and M denotes millions, and cash is measured in the same price units as the quotes.

Definition 13.5 (Calibration game)**.** The interviewer names a quantity whose value the quoting player does not know precisely. The player must, within roughly 15 seconds, state a point estimate and a 95% confidence interval for the quantity, and then quote a two-sided market on it. The interviewer trades against the quotes at will, one unit per quote — a counterparty *buy* lifts the offer, meaning the player has *sold* one unit at the ask; a counterparty *sell* hits the bid, meaning the player has *bought* one unit at the bid — and may stop the game at any time. The game ends with the closing question: "are you long or short, and how much?" Grading rewards process — the four behaviors collected in Remark 13.7 below — not proximity of the estimate to the true value.

Example 13.6 (A calibration game as played: the complete ledger)**.** The quantity: the number of Bitcoin blocks mined since 2012, given that a block arrives every 10 minutes on average. Prices are quoted in blocks. The quoting player's rounds, with the counterparty trading one unit per quote, were as follows.

Round	Quote (bid–ask)	Counterparty	Player's fill	Cash flow	Position	Cash
1	100k–200k	buys	sold 1 at 200k	+0.2M	−1	+0.2M
2	400k–600k	buys	sold 1 at 600k	+0.6M	−2	+0.8M
3	900k–1M	buys	sold 1 at 1M	+1.0M	−3	+1.8M
4	4M–6M	sells	bought 1 at 4M	−4.0M	−2	−2.2M
5	2M–3M	no trade	—	0	−2	−2.2M

Closing accounting. Three units were sold and one was bought, so the position is **short 2 units**. Cash is $+0.2 + 0.6 + 1.0 - 4.0 = -2.2$M. Marked at the final mid $(2\text{M} + 3\text{M})/2 = 2.5$M, the mark-to-market equity is

$$\text{cash} + q \times \text{mid} = -2.2\text{M} + (-2)(2.5\text{M}) = -7.2\text{M} :$$

the book is down badly — three sales printed far below value before the single buy printed above it.

The flow read. Three consecutive buys said the offer was cheap, and the player correctly jumped the market in multiples rather than ticks (100k–200k to 400k–600k to 900k–1M to 4M–6M) until the flow flipped. The flip is the information payoff: the counterparty bought at the 1M offer and later sold at the 4M bid, so they were acting as if the value lay in the bracket $(1\text{M}, 4\text{M})$ — and the final quote 2M–3M correctly sits inside that bracket.

The underlying Fermi chain. At 10 minutes per block: 6 blocks per hour $\times\, 24 = 144$ per day; $\times\, 365 \approx 52.6$k per year; $\times\, 14$ years ≈ 735k blocks. The game grades process, not proximity to this number.

Remark 13.7 (Graded behaviors in the calibration game)*.* Four behaviors are graded, in order of importance.

1. **Width equals confidence.** A 95% interval on a number the player does not actually know must be *embarrassingly wide* — order-of-magnitude wide. A ±10% band is a memorized-fact interval; on a true unknown it will simply not contain the answer. The one losing move is refusing to give an interval at all (for instance, "the volatility is not known, so no interval can be given"): the interval *is* the statement of the player's uncertainty.
2. **Move with the flow.** Repeated buys mean the offer is cheap: jump the market up aggressively, in multiples rather than ticks, until the flow flips. The flip brackets the counterparty's view: a buy at offer a_1 followed later by a sell at bid b_2 means the counterparty acts as if the value lies in (a_1, b_2), and the next quote should sit inside that bracket.
3. **Track inventory out loud.** Every lift of the offer is a sale by the player; every hit of the bid is a purchase. Keep the running count: the closing question is the whole point of the game.

4. **Know the rough profit and loss.** Cash collected minus cash paid, with the open position marked at the final mid — exactly the closing accounting of Example 13.6.

Interview trap. The 15-second freeze: the components of the estimate are usually available — 10 minutes per block, 14 years — so the failure mode is not missing knowledge but an unpracticed chain. The drill is the rate-times-time chain said fast: per hour, then per day, then per year, then total. Training regimen: pick any quantity, produce a point estimate plus an honest 95% interval in 15 seconds, and score interval *coverage* over many repetitions — roughly 19 out of 20 intervals should contain the truth. Missing half the time means the intervals are far too narrow; never missing means they are too wide.

13.4 Hedging Inventory

Skewing sheds inventory only through fill probabilities, which is gradual; a maker can instead remove the risk directly by hedging the inventory with a closely related instrument, often on another venue. The crypto version of this is the cleanest illustration.

Spot inventory hedges with perpetual futures on the same or another venue: a maker absorbing spot buy flow on venue A shorts the perp on venue B, leaving the combined book approximately flat in the underlying. The position then carries the funding rate instead of directional risk (perpetual futures and their funding mechanics are treated in the section on crypto market structure). The hedge is not free of risk: the residual exposures are basis moves — the spot–perp spread can shift against the pair — and venue and counterparty risk on the leg now held at B.

Trading application. On a desk of any size, inventory management becomes a *routing and financing* problem, not just a quoting problem: which venue to hedge on, what funding carry to pay while the hedge is on, and where collateral sits. Quoting decides how inventory arrives; routing and financing decide where it goes and what it costs to hold in the meantime.

Practice: problems P13.1–P13.6 in §20, solutions in §21.

14 Market Microstructure

Market microstructure studies how trading itself moves prices and how the visible limit-order book informs the next price move. This section develops three tools that recur constantly in market-making and execution interviews: the empirical square-root law of market impact, Kyle's price-impact coefficient, and the pair of order-book signals — imbalance and the volume-adjusted mid — that summarize a book snapshot in one number each. All three connect a forecast or a quote screen to realizable profit, which is why the section closes each thread with the practical consequence for trading.

14.1 The Square-Root Law of Market Impact

Executing a large order moves the price against the trader, and the size of that move is the single most important number in execution. This subsection states the empirical law governing it, derives the law's exponents by a dimensional-analysis heuristic, and draws the practical consequence: at sufficient size, impact consumes the alpha that motivated the trade.

The setup is as follows. A *metaorder* is a large parent order of Q shares, executed over some horizon as a sequence of smaller child orders. The market in that asset trades an average of V shares per day, and the asset's daily volatility is σ. The *impact* I of the metaorder is the expected displacement of the price, in the direction of the trade, between the start and the end of the execution. Impact and volatility must be measured in the same price convention — both in currency units, or both in relative (return) units; the numerical examples below use relative units and quote results in basis points (bp), where $1\,\text{bp} = 0.01\%$. The constant Y appearing below is dimensionless.

Fact 14.1 (Square-root impact law, empirical)**.** Across asset classes, market epochs, and execution styles, the impact of a metaorder of Q shares satisfies

$$I \;=\; Y\,\sigma\,\sqrt{\frac{Q}{V}}, \qquad Y \approx 1, \tag{14.1}$$

where σ is daily volatility and V is average daily volume. The scaling and the near-universality of $Y \approx 1$ are documented in large metaorder studies (Almgren et al., 2005; Toth et al., 2011). The law is an empirical regularity: no derivation from first principles is settled, and a proof is beyond our scope.

Although the law itself is empirical, its exponents are not arbitrary. The following proposition — a *heuristic*, in the sense that its hypothesis is a modelling assumption rather than an established premise — shows that once one accepts a pure power law in the three variables Q, V, σ, the square root is forced by unit consistency alone.

Proposition 14.2 (Square-root exponents by dimensional analysis; heuristic)**.** *Suppose the impact of a metaorder depends on the trade size Q, the average daily volume V, and the daily volatility σ only, through a pure power law*

$$I \;=\; Y\,Q^{\alpha}\,V^{\beta}\,\sigma^{\gamma}$$

with Y a dimensionless constant. Then necessarily $\gamma = 1$, $\beta = -\frac{1}{2}$, and $\alpha = \frac{1}{2}$; that is, $I = Y\sigma\sqrt{Q/V}$ is the only dimensionally consistent power law in these three variables, recovering the form of (14.1).

Proof. We proceed by dimensional analysis, matching units on both sides of the ansatz. The units of the four quantities are

$$[\,I\,] = \text{price}, \qquad [\,Q\,] = \text{shares}, \qquad [\,V\,] = \frac{\text{shares}}{\text{day}}, \qquad [\,\sigma\,] = \frac{\text{price}}{\text{day}^{1/2}},$$

and Y carries no units. Substituting these into $I = YQ^{\alpha}V^{\beta}\sigma^{\gamma}$ gives the units equation

$$\text{price} \;=\; \text{shares}^{\alpha}\left(\frac{\text{shares}}{\text{day}}\right)^{\beta}\left(\frac{\text{price}}{\text{day}^{1/2}}\right)^{\gamma} \;=\; \text{shares}^{\,\alpha+\beta}\,\text{day}^{\,-\beta-\gamma/2}\,\text{price}^{\,\gamma}.$$

The three units are independent, so the exponents must match one unit at a time. Matching price exponents gives $\gamma = 1$. Matching shares exponents gives $\alpha + \beta = 0$. Matching day exponents gives $-\beta - \gamma/2 = 0$, hence $\beta = -\gamma/2 = -\frac{1}{2}$, and then $\alpha = -\beta = \frac{1}{2}$. Substituting back yields $I = Y\sigma\sqrt{Q/V}$. As a check, the units of $\sigma\sqrt{Q/V}$ are

$$\frac{\text{price}}{\text{day}^{1/2}} \times \left(\frac{\text{shares}}{\text{shares/day}}\right)^{1/2} = \frac{\text{price}}{\text{day}^{1/2}} \times \text{day}^{1/2} = \text{price},$$

as required. $\square$

(Problem P14.1 asks you to reproduce this proof from scratch and to name the hidden assumption.)

Remark 14.3 (What the ansatz assumes). The hypothesis of Proposition 14.2 does the heavy lifting. It declares that Q, V, and σ are the *only* relevant variables — excluding tick size, quoted spread, execution duration, and participation rate as independent quantities — and that the dependence is a pure power law with a universal dimensionless Y. Dimensional analysis then shows the square root is the only consistent power law in those three variables, but it cannot justify the variable selection itself. What justifies it is the empirical record: $Y \approx 1$ holds across markets and epochs, which is why the law is stated as Fact 14.1 and the derivation only as a heuristic.

Example 14.4 (Impact at two sizes)**.** Consider a stock with daily volatility $\sigma = 2\%$ and average daily volume $V = 16$ million shares, and take $Y = 1$ in (14.1).

- A metaorder of $Q = 160{,}000$ shares has $Q/V = 0.01$, so $\sqrt{Q/V} = 0.10$ and

$$I \approx 1 \times 2\% \times 0.10 = 0.20\% = 20 \text{ bp}.$$

- A metaorder of $Q = 1.6$ million shares has $Q/V = 0.1$, so $\sqrt{Q/V} \approx 0.316$ and $I \approx 2\% \times 0.316 \approx 63$ bp.

Ten times the size produces only about 3.16 times the impact — impact is concave in Q. The total dollar cost, however, scales like $Q \times I \propto Q^{3/2}$ (up to a size-independent factor reflecting that the average fill is displaced by a fraction of the final impact): the tenfold-larger order costs roughly 31.6 times as many dollars, and the *per-share* cost still grows like $\sqrt{Q}$.

Remark 14.5 (Slicing and the Almgren–Chriss trade-off). Because per-share cost grows like $\sqrt{Q}$, large parent orders are never sent at once: they are sliced over time on a schedule (TWAP spreads the order evenly in time; VWAP spreads it in proportion to expected volume). Scheduling opens a genuine trade-off, formalized by Almgren and Chriss (2000): trade too big and impact eats the alpha; trade too small and the position misses the forecast move; trade too slowly and the un-executed remainder bears the risk of adverse price moves while the forecast decays. Optimal execution balances impact cost against this timing risk.

Intuition. Two structural features of (14.1) are worth internalizing. First, doubling both Q and V leaves impact unchanged: what matters is the trade's size *relative to the market's turnover*, not its absolute size. Second, the square root means each additional share moves the price less than the one before — large orders are worked where and when liquidity is available, so the market absorbs them more gently than a linear extrapolation from small trades would predict.

Interview trap. A standard interview question runs: "The signal says the stock rises 1% over the next two hours, and the desired position is large — can the alpha be captured?" The tempting answer is yes, since the forecast is genuine. The correct answer is *no* at sufficient size: with impact growing as $\sqrt{Q}$ by (14.1), the trader's own impact — paid once entering and again exiting — can exceed the forecast move, so the net capture (alpha minus round-trip impact and costs) turns negative. Any claimed expected profit from a forecast must be quoted *after* subtracting the impact of getting in and out at the intended size.

14.2 Kyle's Lambda

The square-root law (Fact 14.1) describes the total impact of a worked metaorder. At the granularity of individual trades, impact is instead summarized by a single regression coefficient, universally called Kyle's lambda. Interviews probe three things: the definition, how to estimate it, and how a linear coefficient can coexist with the concave square-root law.

The setup is as follows. Trading is observed over short intervals indexed by $t = 1, 2, \dots$ (one-minute bars, say). Let S_t denote the price at the end of interval t, and let x_t denote the *signed volume* in interval t: the sum of trade sizes with buyer-initiated (aggressor-buy) trades counted positive and seller-initiated trades counted negative. Let ε_t be a noise term uncorrelated with x_t. In this section S denotes a price (the document-wide P is reserved for probability) and λ denotes Kyle's lambda, not an eigenvalue; no eigenvalues appear in this section, so the canonical symbol is retained.

Definition 14.6 (Kyle's lambda, via the price-impact regression)**.** Kyle's lambda is the coefficient λ in the linear price-impact model

$$S_t \;=\; S_{t-1} + \lambda\, x_t + \varepsilon_t, \tag{14.2}$$

that is, the expected price move per unit of signed order flow. It is estimated in practice as the OLS slope of a regression of price changes $S_t - S_{t-1}$ on signed volume x_t. A large λ means small flow moves the price a lot: λ is a direct measure of *illiquidity*, the inverse of market depth.

Example 14.7 (Estimating and using lambda)**.** A researcher regresses one-minute price changes on one-minute signed volume for two stocks and finds that 10,000 shares of net signed flow move stock A by 1.5 bp on average and stock B by 0.3 bp. Then

$$\lambda_A = 0.15 \text{ bp per 1,000 shares}, \qquad \lambda_B = 0.03 \text{ bp per 1,000 shares},$$

so stock A is five times as illiquid as stock B — equivalently, B is five times as deep. Under the linear model (14.2), a 20,000-share buy in stock A is expected to move its price by $\lambda_A \times 20 = 3.0$ bp.

Trading application. Three standard uses of an estimated λ: comparing liquidity across assets, venues, and time; feeding pre-trade cost models and position sizing (an impact budget: how much flow a given price concession buys); and monitoring for liquidity droughts — a spike in λ means the same flow suddenly moves the price much further, an early warning that depth has evaporated.

Fact 14.8 (Equilibrium origin of lambda (Kyle, 1985))**.** The regression coefficient of Definition 14.6 has a theoretical pedigree. In Kyle's single-period auction model — one informed trader who knows the terminal value $v \sim \mathcal{N}(\mu, \sigma_v^2)$, noise traders submitting exogenous flow $u \sim \mathcal{N}(0, \sigma_u^2)$, and a competitive risk-neutral market maker who observes only the aggregate flow — there is a unique linear equilibrium in which the market maker sets the price as a linear function of the flow with slope

$$\lambda \;=\; \frac{\sigma_v}{2\,\sigma_u}.$$

Impact is steep when there is much private information per unit of noise trading, and markets are deep when noise trading is plentiful: λ is the price of adverse selection, compensating the market maker for trading against informed flow. A proof is beyond our scope.

Remark 14.9 (Reconciling a linear lambda with concave impact). The model (14.2) is linear, yet Fact 14.1 says total impact is concave. Empirically the two are reconciled by letting the fitted slope depend on trade size: estimated within size buckets,

$$\lambda(Q) \;\propto\; Q^{-1/2}, \qquad \text{so total impact} \qquad \lambda(Q)\,Q \;\propto\; \sqrt{Q},$$

which is exactly the square-root law (14.1). A single OLS regression estimates an *average* slope over whatever size distribution was sampled; extrapolating that slope linearly to sizes far beyond the sample overstates the impact of large orders. For sizing a large trade, the concave prediction is the one to trust.

14.3 Order-Book Imbalance and the Volume-Adjusted Mid

The displayed limit-order book carries information about the next price move, and two one-number summaries of the top of the book are the workhorses of short-horizon prediction. This subsection defines both, computes them on a deliberately lopsided book, and closes with a naming distinction that matters in interviews at high-frequency firms.

The setup is as follows. At the top of the book, the best bid quotes a price S_b with displayed size q_b (shares resting at the bid), and the best ask quotes a price $S_a > S_b$ with displayed size q_a. The *mid price* is $m = (S_b + S_a)/2$. Here q_b and q_a denote displayed queue sizes — not the Markov-chain probability q of the document-wide notation — and S again denotes a price.

Definition 14.10 (Order-book imbalance)**.** The *order-book imbalance* of the top of the book is

$$\text{OBI} \;=\; \frac{q_b - q_a}{q_b + q_a} \;\in\; [-1, 1].$$

An imbalance of $+1$ means all displayed size is on the bid, -1 means all on the ask, and 0 means a balanced book.

Trading application. A positive imbalance signals buying pressure: heavy resting demand below and thin supply above make the next mid-price move more likely up than down. Order-book imbalance is the workhorse short-horizon (tick-scale) predictor on desks that trade against the book.

Definition 14.11 (Volume-adjusted mid price (VAMP))**.** The *volume-adjusted mid price* is

$$S_{\mathrm{VAMP}} \;=\; \frac{S_a\,q_b + S_b\,q_a}{q_b + q_a},$$

the convex combination of the two quoted prices in which each price is weighted by the *opposite* side's displayed size.

The reader should verify from Definition 14.11 that the two weights sum to one, so S_{VAMP} always lies between S_b and S_a; that a balanced book ($q_b = q_a$) gives $S_{\mathrm{VAMP}} = m$; and that $S_{\mathrm{VAMP}} \to S_a$ as $q_b/q_a \to \infty$. The next example shows why the opposite-side weighting is the correct one.

Example 14.12 (A one-sided book)**.** Consider the book with best bid $S_b = 99$ carrying $q_b = 1{,}000$ shares and best ask $S_a = 100$ carrying $q_a = 1$ share. Then

$$\mathrm{OBI} \;=\; \frac{1{,}000 - 1}{1{,}000 + 1} \;=\; \frac{999}{1{,}001} \;\approx\; +0.998,$$

$$S_{\mathrm{VAMP}} \;=\; \frac{100 \times 1{,}000 + 99 \times 1}{1{,}001} \;=\; \frac{100{,}099}{1{,}001} \;\approx\; 99.999,$$

while the plain mid is $m = 99.50$. The interpretation: to push the price down one tick, sellers must consume 1,000 resting shares, while a single share of buying lifts the entire ask — the next move is overwhelmingly likely to be up, so the "true" price sits essentially at the ask. Weighting the ask price by the *bid* size encodes exactly this: a heavy bid pins value toward the ask. The plain mid (99.50) ignores the pressure entirely, and the naive *same*-side weighting $(99\times 1{,}000 + 100\times 1)/1{,}001 \approx 99.001$ would drag the estimate toward the crowded side — backwards. Figure 23 shows this book together with two deeper levels on each side.

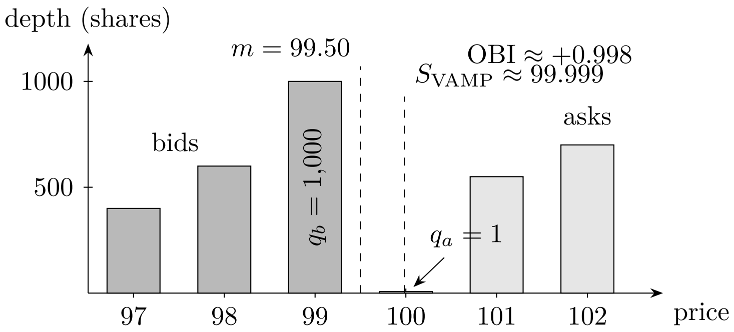


Figure 23: The limit-order book of Example 14.12: bid depth (dark bars, prices 97–99) against ask depth (light bars, prices 100–102), with a 1,000-share best bid facing a single displayed share at the best ask. The imbalance is extreme (OBI $\approx +0.998$), and the volume-adjusted mid sits essentially at the ask, far from the plain mid $m = 99.50$. As defined here, OBI and S_{VAMP} use top-of-book sizes only; the deeper levels are shown for context.

Remark 14.13 (Weighted mid versus the formal micro-price). Naming precision matters here. The quantity S_{VAMP} of Definition 14.11 is the volume-weighted (imbalance-weighted) *mid.* The formal *micro-price* of Stoikov (2018) is defined differently, as the limit of expected future mid prices conditional on the current state of the book; the weighted mid is a first-order proxy for it, not the object itself. In an interview at a high-frequency or market-making firm, stating the distinction — "this is the weighted mid, a first-order approximation to Stoikov's micro-price" — signals fluency with the literature rather than a memorized formula.

Practice: problems P14.1–P14.5 in §20, solutions in §21.

15 Crypto Market Structure

Crypto markets run continuously, fragment across hundreds of venues, and host two instrument classes with no direct equity-market analogue: perpetual futures on centralized exchanges and automated market makers on-chain. This section develops the small amount of arithmetic the subject genuinely requires — funding-rate annualization, the constant-product trade-cost formula, and the impermanent-loss formula — and organizes the surrounding market-structure facts (the venue landscape, liquidation cascades, maximal extractable value) that interviewers use to test whether a candidate has actually watched these markets trade.

15.1 The Centralized-Exchange Landscape

Centralized crypto exchanges (CeFi) differ from equity venues in ways that reshape what liquidity provision means in practice. The observations in this subsection are qualitative, but each carries a direct operational consequence for a trading desk.

Crypto trades 24 hours a day, 7 days a week, on hundreds of venues, and the same asset is listed on essentially all of them. There is no primary listing exchange, no opening or closing auction, and no market close at which risk can be handed off.

Remark 15.1 (Fair value is a cross-venue estimate)*.* Because the same asset trades on many venues simultaneously, no single venue's last price is authoritative. Fair value is an *estimate* aggregated across venues, typically a liquidity-weighted combination of prices wherever the asset trades, and every quote a market maker posts on one venue leans on that cross-venue estimate. The spot *index* against which derivatives venues settle and compute funding (Definition 15.3) is exactly such a composite.

Trading application. Liquidity provision on CeFi venues is inventory *logistics* as much as quoting. To quote on a venue, a desk must hold inventory on that venue; withdrawal and transfer times (blockchain confirmations plus exchange processing) are frictions that any cross-venue arbitrage must cross, so a price discrepancy that closes faster than assets can move is capturable only with inventory pre-positioned on both venues. Fee tiers shift where the edge lies: maker rebates and taker fees differ across venues and volume tiers, and the difference can flip a marginal quote from unprofitable to profitable.

Remark 15.2 (Custody is a first-class risk)*.* Since the failure of the FTX exchange (2022), *where inventory sits overnight* is a first-class risk decision, not an operational afterthought: assets held on a venue are exposed to the venue's own credit. Desks cap per-venue balances and treat the allocation of inventory across venues as part of the risk book.

15.2 Perpetual Futures and Funding

The perpetual future is the dominant crypto derivative. It has no expiry, so nothing contractual forces convergence to spot at a terminal date; a periodic cash flow between longs and shorts does the anchoring instead. Interviewers routinely ask a candidate to annualize a printed funding rate on the spot, which is the content of Proposition 15.4.

Setup. Let S denote the venue's spot index for the underlying (a cross-venue composite, Remark 15.1) and let F denote the perpetual's traded price. Write ϕ for the funding rate per interval; a position's *notional* is its size times the mark price. We assume three funding intervals per day and a 365-day year, hence $N = 3 \times 365 = 1095$ intervals per year.

Definition 15.3 (Perpetual future and funding rate)**.** A *perpetual future* (perp) is a linear futures contract with no expiry date. Once per *funding interval* — commonly 8 hours, hence three intervals per day — every open position pays or receives a *funding payment* equal to ϕ times its notional, where the venue sets the rate ϕ as a clamped (capped and floored) function of the perp's average premium $(F - S)/S$ over the interval. By convention, a positive rate means longs pay shorts, and a negative rate means shorts pay longs.

Intuition. The funding flow is what ties the perp to spot in the absence of an expiry. If the perp trades above the index, the premium is positive, so funding is positive: holding a long now *costs* ϕ per interval, which pushes longs to close and pays shorts to lean against the premium — pressure on the perp price is *downward*, back toward the index. Below the index the mechanism mirrors. The anchor is economic rather than contractual.

Proposition 15.4 (Funding annualization)**.** *Suppose the funding rate is a constant $\phi > -1$ per 8-hour interval. Then, as a fraction of a fixed notional:*

(i) the funding accrued per day is 3ϕ, and the simple *(non-compounded) annualized rate is $N\phi = 1095\,\phi$;*
(ii) if each payment is reinvested into the position, the compounded *annualized rate is $(1+\phi)^N - 1$;*
(iii) the compounded rate is never below the simple rate: $(1+\phi)^N - 1 \geq N\phi$, with equality if and only if $\phi = 0$.

Proof. Parts (i) and (ii) are direct computation; part (iii) is Bernoulli's inequality, which we prove by induction.

(i) Over one day the position pays or receives three payments of ϕ times notional; over a year, $N = 1095$ of them. Held against a fixed notional, the payments sum to $1095\,\phi$.

(ii) If each payment is reinvested, the account is multiplied by $(1 + \phi)$ each interval, so after N intervals it holds $(1 + \phi)^N$ times its starting value, a gain of $(1 + \phi)^N - 1$.

(iii) We claim $(1 + \phi)^n \geq 1 + n\phi$ for every integer $n \geq 1$ and every $\phi > -1$. The base case $n = 1$ holds with equality. For the inductive step, assume $(1+\phi)^n \geq 1+n\phi$. Since $1+\phi > 0$, multiplying both sides by $(1 + \phi)$ preserves the inequality:

$$(1 + \phi)^{n+1} \;\geq\; (1 + n\phi)(1 + \phi) \;=\; 1 + (n + 1)\phi + n\phi^2 \;\geq\; 1 + (n + 1)\phi.$$

The final inequality discards $n\phi^2 \geq 0$, which is strict unless $\phi = 0$; hence equality throughout forces $\phi = 0$, and conversely $\phi = 0$ gives equality. Applying the claim with $n = N$ and subtracting 1 proves (iii). ☐

(Problem P15.4 asks you to reproduce this arithmetic from scratch at a different rate.)

Example 15.5 (The 0.01% anchor)**.** Take $\phi = 0.01\% = 10^{-4}$ per 8-hour interval, the standard reference rate on major venues. Per day: $3 \times 0.01\% = 0.03\%$. Simple annualized, by Proposition 15.4(i):

$$1095 \times 0.0001 = 0.1095 \approx 11\% \text{ per year (simple).}$$

Compounded, by part (ii): $(1.0001)^{1095} = e^{1095 \ln(1.0001)} \approx e^{0.1095} \approx 1.1157$, i.e. about 11.6% per year. As a self-check on the gap, the second-order term of $e^x \approx 1+x+x^2/2$ at $x = 0.1095$ predicts an excess of $0.1095^2/2 \approx 0.6$ percentage points over the simple figure — which is exactly the difference 11.57% – 10.95%. The desk anchor worth memorizing: *0.01% per 8 hours is about 11% a year.*

Remark 15.6 (The compounding caveat). Two caveats attach to the 11% anchor. First, it is the *simple* figure; reinvestment lifts it to about 11.6%, and Proposition 15.4(iii) says the lift is always upward. Second, and more importantly, the rate is not constant: it resets every interval as a function of the premium, so annualizing one printed rate extrapolates a snapshot. A funding rate is a state variable, not a locked-in yield.

Trading application. *Cash-and-carry.* Buy spot, short an equal notional of the perp, and collect the funding while it is positive: a delta-neutral yield. The yield is capped by *basis risk* (no expiry ever forces convergence, so an unwind at an adverse basis can surrender accumulated carry) and by *venue risk* (both legs and the collateral sit on an exchange, Remark 15.2). The funding *sign* is also a positioning gauge: persistently positive funding means leveraged longs are paying a steady premium to stay long — crowded positioning — and at extremes it reads as a contrarian signal.

15.3 Liquidations and Cascades

Perpetual positions are levered: the trader posts margin far smaller than notional, and the venue force-closes (*liquidates*) the position when its equity falls to a maintenance threshold. For leverage L, an adverse move of roughly $1/L$ exhausts the initial margin, so every levered position carries a *liquidation price* that is known in advance — and clusters of similar positions carry clusters of nearby liquidation prices.

Remark 15.7 (Liquidation cascades). A liquidation is a *forced market order in the direction of the move*: a levered long liquidated in a falling market is closed with a forced sell. When price trades through a dense cluster of liquidation prices, the first tranche of forced sells pushes price lower, which triggers the next tranche, and so on — a mechanical positive-feedback loop called a *cascade*, producing relentless one-way flow and gapping prices.

Trading application. For a market maker a cascade is both danger and opportunity. Danger: the flow is toxic in the mark-out sense even though it is forced rather than informed — it is all same-side and it marks against every fill while the cascade runs, and prices gap through quote levels. Opportunity: liquidity premia spike, because the forced flow carries no information about fundamentals and must transact regardless of price; whoever still has risk capacity is paid unusually well for providing liquidity into it. In an interview, describing cascade dynamics accurately is a credible signal of having actually watched crypto trade.

15.4 Constant-Product Automated Market Makers

On-chain venues (DeFi) replace the limit order book with a deterministic pricing curve: a passive pool of two assets quotes every trade size algorithmically. The constant-product design is the canonical one, and its trade-cost formula is a one-line derivation that interviewers expect a candidate to produce on demand.

Setup. In the remainder of this section, p denotes a *price* and r a *gross price ratio*; these local meanings supersede the document-wide Markov-chain conventions for p and r. A pool holds reserves $x > 0$ of asset X (say ETH) and $y > 0$ of asset Y (say a dollar stablecoin), with invariant constant $k = xy$. We ignore trading fees throughout. Prices of X are quoted in units of Y.

Definition 15.8 (Constant-product AMM)**.** A *constant-product automated market maker* is a pool with reserves (x, y) that accepts any trade leaving the product of its reserves unchanged: a fee-free trade moving the reserves to (x', y') is admissible if and only if $x'y' = xy = k$. The pool's *marginal price* of X is $p = y/x$, the cost per unit of an infinitesimal purchase (justified in Proposition 15.9).

Proposition 15.9 (Cost of a buy from a constant-product pool)**.** *Consider a constant-product pool with reserves (x, y) and invariant $k = xy$. A trader who buys Δx units of X, where $0 < \Delta x < x$, must pay*

$$\Delta y \;=\; \frac{k}{x - \Delta x} - y \;=\; \frac{k\,\Delta x}{x\,(x - \Delta x)} \;>\; 0. \tag{15.1}$$

Moreover the cost per unit tends to y/x as $\Delta x \to 0$, which justifies calling $p = y/x$ the marginal price.

Proof. We compute directly from the invariant. After the trade the pool holds $x - \Delta x$ units of X and $y + \Delta y$ units of Y, and admissibility requires

$$(x - \Delta x)(y + \Delta y) = k.$$

Solving for the payment: $y + \Delta y = k/(x - \Delta x)$, hence $\Delta y = k/(x - \Delta x) - y$. For the second form, substitute $y = k/x$ and combine over a common denominator:

$$\Delta y = \frac{k}{x - \Delta x} - \frac{k}{x} = \frac{k\,x - k\,(x - \Delta x)}{x\,(x - \Delta x)} = \frac{k\,\Delta x}{x\,(x - \Delta x)},$$

which is positive since $0 < \Delta x < x$. Dividing by Δx gives the per-unit cost $k/\big(x(x - \Delta x)\big)$, which tends to $k/x^2 = y/x$ as $\Delta x \to 0$. $\square$

(Problem P15.7 asks you to reproduce this derivation and run the numbers on a live pool; Problem P15.8 applies it three times in a row.)

Corollary 15.10 (Price impact is built into the curve)**.** *In the setting of Proposition 15.9, write $f = \Delta x/x \in (0, 1)$ for the fraction of the reserve purchased. Then the average price paid and the post-trade marginal price are*

$$\frac{\Delta y}{\Delta x} = \frac{p}{1 - f}, \qquad \frac{y + \Delta y}{x - \Delta x} = \frac{p}{(1 - f)^2},$$

both strictly increasing and convex in f, and both diverging as $f \uparrow 1$. In particular the total cost $\Delta y = p\,x\,f/(1 - f)$ grows super-linearly in trade size: a constant-product pool has price impact built into its geometry, with no order book required.

Proof. We substitute $k = px^2$ (from $p = y/x$ and $k = xy$) into the formulas of Proposition 15.9. The average price is

$$\frac{\Delta y}{\Delta x} = \frac{k}{x(x - \Delta x)} = \frac{p\,x^2}{x^2(1 - f)} = \frac{p}{1 - f},$$

and the post-trade reserves are $x' = x(1 - f)$ and $y' = k/x'$, so the new marginal price is $y'/x' = k/x'^2 = p/(1 - f)^2$. Each expression is p times a positive integer power of $(1 - f)^{-1}$, hence strictly increasing and convex on $(0, 1)$ and unbounded as $f \uparrow 1$. Finally $\Delta y = \Delta x \cdot p/(1-f) = p\,x\,f/(1-f)$, whose second derivative in f is $2px/(1 - f)^3 > 0$: the cost is convex, i.e. super-linear. $\square$

Figure 24 shows the geometry: a buy slides the state point along the hyperbola $xy = k$ into its steepening region.

Example 15.11 (Buying 10 ETH from a 100-ETH pool)**.** A fee-free ETH/USDC pool holds $x = 100$ ETH and $y = 200{,}000$ USDC, so $k = 2 \times 10^7$ and the marginal price is $p = 200{,}000/100 =$ \$2,000 per ETH. A trader buys $\Delta x = 10$ ETH, a fraction $f = 0.1$ of the reserve. By (15.1),

$$\Delta y = \frac{2 \times 10^7}{90} - 200{,}000 = 222{,}222.22 - 200{,}000 = 22{,}222.22 \text{ USDC},$$

an average price of $22{,}222.22/10 =$ \$2,222.22 per ETH, matching Corollary 15.10: $p/(1 - f) = 2{,}000/0.9 = 2{,}222.22$. The pool's new marginal price is $p/(1 - f)^2 = 2{,}000/0.81 =$ \$2,469.14. Two self-checks: the invariant survives, $90 \times 222{,}222.22 = 2.0 \times 10^7$; and the reader should verify, from the two formulas of Corollary 15.10, that the average price is always the geometric mean of the pre- and post-trade marginal prices — here $\sqrt{2{,}000 \times 2{,}469.14} = 2{,}222.2$.

Remark 15.12 (Arbitrage is the AMM's price discovery). The pool has no view, no oracle, and no order book: after any trade its marginal price deviates from the price on centralized exchanges, and arbitrageurs trade against the pool until the two agree again, pocketing the difference. That external arbitrage is the entire price-discovery mechanism of an AMM, and it is what keeps AMM prices honest. Its cost to the passive side is quantified in Theorem 15.14 as impermanent loss.

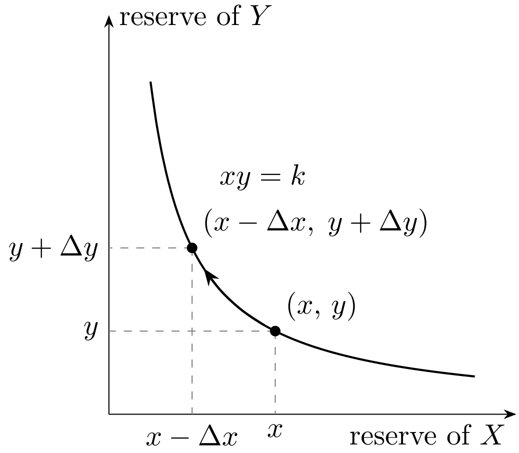


Figure 24: A buy of Δx units of X from a constant-product pool moves the state along the hyperbola $xy = k$ from (x, y) to $(x - \Delta x,\ y + \Delta y)$; the buyer pays $\Delta y = k/(x - \Delta x) - y$ by (15.1). The curve steepens as x shrinks, so successive units cost more: price impact is the geometry of the curve itself (Corollary 15.10).

15.5 Impermanent Loss

A liquidity provider (LP) deposits both assets into a constant-product pool and passively takes the other side of every trade, including the arbitrage flow of Remark 15.12. The classic interview derivation of this subsection prices what that passivity costs: relative to simply holding the deposited coins, the LP loses a deterministic function of the price move, negative in *both* directions.

Setup. Fix a fee-free constant-product pool with invariant k (Definition 15.8), and assume arbitrageurs keep the pool's marginal price equal to the external market price at all times. An LP deposits x_0 units of X and y_0 units of Y when the external price is p_0; the price then moves to $p_1 = r\,p_0$ for some gross ratio $r > 0$. Write V_{LP} for the value of the pool position at the new price, V_{HODL} for the value of simply holding the deposited coins (x_0 of X and y_0 of Y) to the same date, and define the *impermanent loss* $\mathrm{IL}(r) = V_{\mathrm{LP}}/V_{\mathrm{HODL}} - 1$. All values are in units of Y.

Lemma 15.13 (Arbitrage-aligned reserves). *A constant-product pool with invariant k and marginal price p has uniquely determined reserves*

$$x = \sqrt{k/p}, \qquad y = \sqrt{k\,p},$$

the value of the reserves at price p is $p\,x + y = 2\sqrt{k\,p}$, and the pool always holds equal values of the two assets ($p\,x = y$).

Proof. We solve the two defining equations simultaneously. The constraints are $xy = k$ and $y/x = p$ with $x, y > 0$. Substituting $y = p\,x$ into the invariant gives $p\,x^2 = k$, whose unique positive root is $x = \sqrt{k/p}$; then $y = p\,x = \sqrt{k\,p}$. The value at price p is $p\,x + y = \sqrt{k\,p} + \sqrt{k\,p} = 2\sqrt{k\,p}$, and the equal-split identity $p\,x = \sqrt{k\,p} = y$ is the middle step of that computation. ☐

Theorem 15.14 (Impermanent loss). *In the setup above, for every $r > 0$:*

(i) the pool position is worth $V_{\mathrm{LP}} = V_0\sqrt{r}$, where $V_0 = p_0 x_0 + y_0 = 2y_0$ is the deposit value;

(ii) the hold benchmark is worth $V_{\mathrm{HODL}} = V_0\,\dfrac{1+r}{2}$;

(iii) consequently

$$\mathrm{IL}(r) \;=\; \frac{2\sqrt{r}}{1+r} - 1 \;\le\; 0, \tag{15.2}$$

with equality if and only if $r = 1$.

Proof. We compute both terminal values in closed form via Lemma 15.13 and compare them with the AM–GM inequality.

Step 1: the deposit. At price p_0 the aligned pool holds $x_0 = \sqrt{k/p_0}$ and $y_0 = \sqrt{k\,p_0}$, and by the equal-split identity $p_0 x_0 = y_0$, so the deposit is automatically a 50/50 split with value $V_0 = 2y_0 = 2\sqrt{k\,p_0}$.

Step 2: the pool after the move. Arbitrageurs re-align the pool to the new price $p_1 = r\,p_0$, so by Lemma 15.13 the reserves rebalance to

$$x_1 = \sqrt{k/p_1} = \frac{x_0}{\sqrt{r}}, \qquad y_1 = \sqrt{k\,p_1} = y_0\sqrt{r}.$$

For $r > 1$ the pool now holds *fewer* units of X: it sold X into the rally. The position's value is

$$V_{\text{LP}} = p_1 x_1 + y_1 = 2\sqrt{k\,p_1} = 2\sqrt{k\,p_0}\,\sqrt{r} = V_0\sqrt{r},$$

proving (i): the LP's value grows like $\sqrt{r}$, not like r.

Step 3: the benchmark. Holding the deposited coins instead,

$$V_{\text{HODL}} = p_1 x_0 + y_0 = r\,p_0 x_0 + y_0 = (r+1)\,y_0 = V_0\,\frac{1+r}{2},$$

proving (ii).

Step 4: the comparison. Dividing,

$$\frac{V_{\text{LP}}}{V_{\text{HODL}}} = \frac{V_0\sqrt{r}}{V_0(1+r)/2} = \frac{2\sqrt{r}}{1+r},$$

which gives the formula in (15.2). By the AM–GM inequality applied to the pair $\{1, r\}$, we have $\sqrt{r} = \sqrt{1 \cdot r} \le (1+r)/2$, with equality if and only if $1 = r$; hence the ratio is at most 1 and $\mathrm{IL}(r) \le 0$, vanishing exactly at $r = 1$. $\square$

(Problem P15.9 asks you to reproduce this proof from scratch and extend it to the symmetry $\mathrm{IL}(r) = \mathrm{IL}(1/r)$ *and the small-move law; Problem P15.10 applies it to a benchmark trap.)*

Example 15.15 (Impermanent loss at $r = 2$ and $r = 4$)**.** First the pure ratios from (15.2). A doubling, $r = 2$: $\mathrm{IL}(2) = 2\sqrt{2}/3 - 1 = 0.9428 - 1 = -5.72\% \approx -5.7\%$. A quadrupling, $r = 4$: $\mathrm{IL}(4) = 2 \cdot 2/5 - 1 = -20\%$.

Now in dollars. Deposit \$5,000 of ETH and \$5,000 of USDC when ETH trades at \$2,000: $x_0 = 2.5$ ETH, $y_0 = 5{,}000$ USDC, $V_0 = \$10{,}000$, $k = 12{,}500$. ETH doubles to \$4,000 ($r = 2$): the pool rebalances to $x_1 = 2.5/\sqrt{2} = 1.7678$ ETH and $y_1 = 5{,}000\sqrt{2} = 7{,}071.07$ USDC, so

$$V_{\text{LP}} = 1.7678 \times 4{,}000 + 7{,}071.07 = \$14{,}142.14 = V_0\sqrt{2}, \qquad V_{\text{HODL}} = 2.5 \times 4{,}000 + 5{,}000 = \$15{,}000,$$

a shortfall of \$857.86 $\approx$ \$858, i.e. $858/15{,}000 = 5.72\%$ — matching IL(2). If instead ETH quadruples to \$8,000 ($r = 4$): $V_{\text{LP}} = 10{,}000\sqrt{4} = \$20{,}000$ against $V_{\text{HODL}} = 2.5 \times 8{,}000 + 5{,}000 = \$25{,}000$, a \$5,000 shortfall: exactly 20%. Figure 25 plots the curve through these points.

Intuition. The mechanism behind Theorem 15.14 is visible in Step 2 of the proof: as price rises the pool mechanically *sells* the appreciating asset (from x_0 down to $x_0/\sqrt{r}$), and as price falls it mechanically buys the depreciating one. A position that sheds winners and accumulates losers on every move, and whose value ($V_0\sqrt{r}$, concave in r) lags a linear benchmark in both directions, is *short gamma*: the LP has sold an option to the market, and the trading fees a real pool charges are the premium collected for it. Hence the one-line summary: *an LP profits precisely when fees exceed the option value given away.*

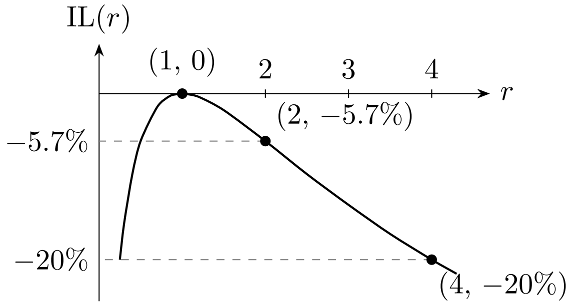


Figure 25: Impermanent loss $\mathrm{IL}(r) = 2\sqrt{r}/(1+r) - 1$ of (15.2) against the gross price ratio r. The curve touches zero only at $r = 1$ and is negative in both directions — the left branch shows that a sell-off hurts the LP too. Marked points: $(1, 0)$, $(2, -5.7\%)$, $(4, -20\%)$.

Interview trap. Two standard traps. First, the benchmark: an LP position can be up substantially in absolute terms and still have lost — in Example 15.15 the $r = 2$ position is up 41% on the deposit yet lags holding the same coins by \$858; quoting profit against the deposit hides the loss, and the honest benchmark is always *holding the deposited assets*. Second, the name: "impermanent" suggests the loss heals if one waits, but it is a property of the current price ratio — withdraw at any $r \neq 1$ and the loss is permanent.

15.6 Maximal Extractable Value and Decentralized Venues

Public-blockchain execution creates an adversarial ordering game with no CeFi analogue, and interviewers probe it with one-liners. Two facts suffice at this level.

Remark 15.16 (MEV and the sandwich attack). *Maximal extractable value* (MEV) is profit extracted by controlling the order of transactions within a block. The canonical instance is the *sandwich*: an attacker brackets a victim's pending AMM trade, front-running it with a buy (pushing the pool up the curve of Figure 24), letting the victim execute at the worsened price, then back-running with a sell that captures the victim's extra slippage. The attack is possible because pending transactions are visible in the *public mempool* before inclusion, and block producers — or searchers bidding to them — choose the ordering. Sending an order to a public mempool broadcasts intent to adversaries who can trade around it.

Trading application. Major market-making firms quote on decentralized venues alongside centralized ones; the two market structures share one trading business. In crypto-facing interviews, DeFi literacy — AMM mechanics, funding, liquidations, MEV — is table stakes, not a specialty.

Practice: problems P15.1–P15.10 in §20, solutions in §21.

16 Options and Volatility

This section develops the minimal option-pricing toolkit that trading interviews actually draw on: the price of a cash-or-nothing digital under the Black-76 model, the Breeden–Litzenberger identity that reads a full risk-neutral density off a listed strike curve, the square-root-of-time law for volatility, and the two empirical facts — the variance risk premium and the delta-hedged profit-and-loss identity — that explain what "trading volatility" means. The digital option is the connecting thread: its payoff is exactly that of a prediction-market binary contract, so pricing it from an options venue's quotes turns the theory into a live relative-value trade.

16.1 Digital Options under Black-76

A cash-or-nothing digital is the simplest nontrivial derivative: it pays one unit of currency if the underlying settles above a strike and nothing otherwise. Its price under the standard lognormal model is the single most quoted formula in options interviews — "the risk-neutral probability, discounted" — and this subsection derives it in full from the defining integral.

The setup is as follows. A futures contract on some underlying trades today at price $F > 0$ and settles at time $T > 0$ (measured in years) at the random price $F_T > 0$. Fix a strike $K > 0$ and an implied volatility $\sigma > 0$. Throughout this section r denotes the continuously compounded risk-free interest rate (not the gambler's-ruin ratio q/p used in the Markov-chain sections). Prices are computed under a probability measure $\mathbb{Q}$, the *risk-neutral measure*; we write $\mathbb{Q}(A)$ for the probability of an event A under this measure and $\mathbb{E}^{\mathbb{Q}}$ for the corresponding expectation, and the *pricing rule* values a payoff X paid at time T at $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[X]$ today. As everywhere in this document, Φ denotes the standard normal cumulative distribution function and $\varphi(z) = e^{-z^2/2}/\sqrt{2\pi}$ its density. A remark on notation: the options literature universally writes N for this same function, so the formula below — written here as "$\Phi(d_2)$" to keep one symbol to one meaning — is the one recognized on sight in that literature as "$N(d_2)$."

Definition 16.1 (Cash-or-nothing digital)**.** The *cash-or-nothing digital* (or *binary*) with strike K and expiry T is the contract paying $\mathbf{1}\{F_T > K\}$ at time T: one unit of currency if the futures settlement price exceeds the strike, zero otherwise.

Definition 16.2 (Black-76 lognormal model)**.** The *Black-76 model* (Black, 1976) posits that under $\mathbb{Q}$ the settlement price is lognormal with today's futures price as its mean:

$$F_T \;=\; F\exp\Bigl(-\tfrac{\sigma^2 T}{2} + \sigma\sqrt{T}\,Z\Bigr), \qquad Z \sim \mathcal{N}(0,1) \text{ under } \mathbb{Q}. \tag{16.1}$$

The correction $-\sigma^2 T/2$ in the exponent is exactly what makes the model drift-free: the reader should verify, using the normal moment generating function $\mathbb{E}\,e^{aZ} = e^{a^2/2}$, that (16.1) gives $\mathbb{E}^{\mathbb{Q}}[F_T] = F$, as befits a futures price, which costs nothing to enter and therefore earns no risk-neutral drift.

Theorem 16.3 (Digital price under Black-76)**.** *Under the Black-76 model of Definition 16.2, the cash-or-nothing digital of Definition 16.1 is worth*

$$\mathrm{Dig}(K) \;=\; e^{-rT}\,\Phi(d_2), \qquad \textit{where} \qquad d_2 \;=\; \frac{\ln(F/K) - \sigma^2 T/2}{\sigma\sqrt{T}}. \tag{16.2}$$

In words: the digital's price is the risk-neutral probability of settling above the strike, discounted.

Proof. We compute the price directly from the pricing rule, by reducing the event $\{F_T > K\}$ to an event about the standard normal variable Z and evaluating the resulting integral.

Step 1: the event. By (16.1) and the strict monotonicity of the exponential,

$$F_T > K \iff -\frac{\sigma^2 T}{2} + \sigma\sqrt{T}\,Z \;>\; \ln\frac{K}{F} \iff Z \;>\; \frac{\ln(K/F) + \sigma^2 T/2}{\sigma\sqrt{T}} \;=\; -d_2,$$

where the last equality is the definition of d_2 in (16.2) with both signs flipped.

Step 2: the integral. Since Z has density φ,

$$\mathbb{Q}(F_T > K) \;=\; \int_{-d_2}^{\infty} \varphi(z)\,dz \;=\; \int_{-\infty}^{d_2} \varphi(u)\,du \;=\; \Phi(d_2),$$

where the middle equality substitutes $u = -z$ (so $du = -dz$ and the limits swap) and uses the symmetry $\varphi(-u) = \varphi(u)$ of the standard normal density.

Step 3: the discount. The payoff is an indicator, and the expectation of an indicator is the probability of its event, so the pricing rule gives

$$\mathrm{Dig}(K) \;=\; e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\mathbf{1}\{F_T > K\}] \;=\; e^{-rT}\,\mathbb{Q}(F_T > K) \;=\; e^{-rT}\,\Phi(d_2). \qquad \square$$

Figure 26 shows the two objects of the proof side by side: the lognormal density of F_T, with the probability mass above the strike — the area $\Phi(d_2)$ — shaded, and the digital's step payoff drawn above it.

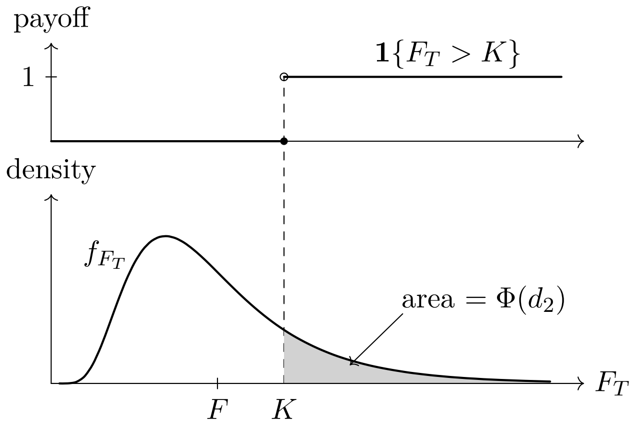


Figure 26: The cash-or-nothing digital under Black-76 (Theorem 16.3), drawn for $F = 1$, $\sigma\sqrt{T} = 0.5$, $K = 1.4$. Bottom: the lognormal density f_{F_T} of the settlement price, centred so that $\mathbb{E}^{\mathbb{Q}}[F_T] = F$; the shaded area above the strike is $\mathbb{Q}(F_T > K) = \Phi(d_2) \approx 0.18$. Top: the digital's step payoff $\mathbf{1}\{F_T > K\}$; the contract is worth the discounted shaded area, $e^{-rT}\Phi(d_2)$.

Example 16.4 (Pricing a digital on a futures contract)**.** A futures contract trades at $F = 100$; price the digital paying 1 if it settles above $K = 120$ in $T = 0.25$ years (three months), with implied volatility $\sigma = 40\%$ and rate $r = 2\%$. The pieces of (16.2): $\sigma\sqrt{T} = 0.40 \times 0.5 = 0.20$ (a 20% three-month volatility); $\ln(F/K) = -\ln 1.2 \approx -0.1823$; and $\sigma^2 T/2 = 0.16 \times 0.25/2 = 0.02$. Then

$$d_2 = \frac{-0.1823 - 0.02}{0.20} \approx -1.01, \qquad \Phi(-1.01) = 1 - \Phi(1.01) \approx 1 - 0.844 = 0.156,$$

and the price is $e^{-0.02\times 0.25} \times 0.156 = 0.995 \times 0.156 \approx 0.155$: about 15.5 cents per dollar of notional. As a sanity check, the required log-move 0.182 against a three-month volatility of 0.20 is about 0.91 standard deviations; the $-\sigma^2 T/2$ shift of the risk-neutral median adds another 0.10, and a 1.01-standard-deviation upper tail should indeed be in the mid-teens. The price also respects its structural bounds: it lies in $(0, e^{-rT})$ and well below the roughly 50 cents an at-the-money digital would command.

Intuition. The quantity d_2 is the log-distance from the strike to the risk-neutral *median* of F_T, measured in units of total volatility: by (16.1) the median of F_T is $Fe^{-\sigma^2 T/2}$ (the exponential of the median of the normal exponent), and $d_2 = \ln\big(Fe^{-\sigma^2 T/2}/K\big)/(\sigma\sqrt{T})$. A digital struck at the median is worth half the discount factor; every volatility-unit of distance moves the price along the normal CDF.

Interview trap. The phrase to say is "the *risk-neutral* probability, discounted." Calling $\Phi(d_2)$ "the probability the option finishes in the money" without the qualifier invites the follow-up "under which measure?" — under $\mathbb{Q}$ the futures price has no drift, while the real-world distribution generally does. The distinction is not pedantry: when the digital price is compared with a prediction-market quote (§16.2), part of any gap can be a legitimate risk premium separating the two measures rather than mispricing.

16.2 The Breeden–Litzenberger Density and the Prediction-Market Link

Theorem 16.3 prices a digital under one specific model. This subsection proves a far stronger, model-free statement: the entire risk-neutral density — and with it a digital at *any* strike — can be read off the curve of listed call prices, with no distributional assumption at all. The result is the engine behind extracting probabilities from an options venue's quotes, and it is a standard derivation request in options and crypto market-making interviews.

The setup is as follows. Fix an expiry T and let $S_T > 0$ denote the underlying's settlement price. Assume the risk-neutral distribution of S_T admits a continuous density $f_{\mathbb{Q}}$ with $\mathbb{E}^{\mathbb{Q}}[S_T] < \infty$, and that for every strike $K > 0$ the European call struck at K trades exactly at its discounted expected payoff,

$$C(K) \;=\; e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\Big[(S_T - K)^+\Big] \;=\; e^{-rT}\int_K^\infty (s-K)\,f_{\mathbb{Q}}(s)\,ds. \tag{16.3}$$

The digital of Definition 16.1 (with S_T in place of F_T) is again worth $e^{-rT}\,\mathbb{Q}(S_T > K)$ by the pricing rule.

Theorem 16.5 (Breeden–Litzenberger)**.** *Under the setup above, the call-price curve $K \mapsto C(K)$ is twice differentiable, and*

$$\frac{\partial C}{\partial K} \;=\; -\,e^{-rT}\,\mathbb{Q}(S_T > K), \tag{16.4}$$

$$\frac{\partial^2 C}{\partial K^2} \;=\; e^{-rT} f_{\mathbb{Q}}(K), \qquad \textit{equivalently} \qquad f_{\mathbb{Q}}(K) \;=\; e^{rT}\,\frac{\partial^2 C}{\partial K^2}. \tag{16.5}$$

In particular the cash-or-nothing digital struck at K is worth the negative slope of the call curve, $\mathrm{Dig}(K) = -\,\partial C/\partial K$*. The result is due to Breeden and Litzenberger (1978).*

Proof. We proceed by differentiating (16.3) under the integral sign, twice.

First derivative. Write $g(s,K) = (s-K)f_{\mathbb{Q}}(s)$ for the integrand of (16.3). By the Leibniz rule for an integral with a variable lower limit,

$$\frac{\partial}{\partial K}\int_K^\infty g(s,K)\,ds \;=\; -\,g(K,K) \;+\; \int_K^\infty \frac{\partial g}{\partial K}(s,K)\,ds.$$

The boundary term vanishes, because the integrand vanishes at its own lower limit: $g(K,K) = (K-K)f_{\mathbb{Q}}(K) = 0$. The partial derivative of the integrand is $\partial g/\partial K = -f_{\mathbb{Q}}(s)$, whose absolute value $f_{\mathbb{Q}}(s)$ is integrable and does not depend on K, so dominated convergence justifies the interchange of derivative and integral. Therefore

$$\frac{\partial C}{\partial K} \;=\; -\,e^{-rT}\int_K^\infty f_{\mathbb{Q}}(s)\,ds \;=\; -\,e^{-rT}\,\mathbb{Q}(S_T > K),$$

which is (16.4); since the digital is worth $e^{-rT}\,\mathbb{Q}(S_T > K)$, the slope identity $\mathrm{Dig}(K) = -\,\partial C/\partial K$ follows.

Second derivative. Differentiating the right-hand side of (16.4) in K once more, the fundamental theorem of calculus applied to $K \mapsto \int_K^\infty f_{\mathbb{Q}}(s)\,ds$ (valid because $f_{\mathbb{Q}}$ is continuous) gives

$$\frac{\partial^2 C}{\partial K^2} \;=\; -\,e^{-rT} \times \big(-\,f_{\mathbb{Q}}(K)\big) \;=\; e^{-rT} f_{\mathbb{Q}}(K),$$

which is (16.5). □

(Problem P16.2 asks you to reproduce this proof from scratch and apply it to listed quotes.)

Example 16.6 (Digital and density from three listed calls)**.** With $r = 0$, calls on some underlying trade at $C(90) = 11.20$, $C(100) = 5.00$, and $C(110) = 1.80$. Estimate the digital struck at 100 and the risk-neutral density there, replacing the derivatives in (16.4)–(16.5) by centred finite differences with spacing $h = 10$:

$$\mathrm{Dig}(100) \approx -\frac{C(110) - C(90)}{2h} = -\frac{1.80 - 11.20}{20} = 0.47,$$
$$f_{\mathbb{Q}}(100) \approx \frac{C(90) - 2\,C(100) + C(110)}{h^2} = \frac{11.20 - 10.00 + 1.80}{100} = \frac{3.00}{100} = 0.030$$

per unit of price. Three checks confirm the numbers are sensible. The two one-sided slopes, $(11.20 - 5.00)/10 = 0.62$ and $(5.00 - 1.80)/10 = 0.32$, bracket the centred estimate 0.47 and both lie in $[0, 1]$, the no-arbitrage band for a digital at $r = 0$. The second-difference numerator 3.00 is the price of the 90/100/110 butterfly, a non-negative payoff whose price must be non-negative — this is exactly convexity of C in K, and it is what guarantees the density estimate cannot come out negative. And a density of 0.030 over a 10-wide bucket puts roughly 30% of the mass near 100, consistent with a digital near one half.

Trading application. A cash-or-nothing digital has exactly the payoff of a prediction-market binary contract: "pays 1 if the price settles above K." Theorem 16.5 therefore prices such contracts model-free from any liquid strike curve: interpolate the call quotes of a listed crypto options venue (Deribit), differentiate at the contract's strike, and compare $-\partial C/\partial K$ with the prediction-market quote for the same event and settlement — the Deribit-to-prediction-market pipeline. A persistent gap between the two is a relative-value signal, tradeable by dealing the rich leg against a tight call spread on the options venue. Using the slope of the *actual* curve rather than $e^{-rT}\Phi(d_2)$ at a single implied volatility matters: the smile makes implied volatility vary with strike, and the curve's slope automatically carries the skew correction that the flat-volatility formula of Theorem 16.3 misses.

Remark 16.7 (The vol clock: intraday seasonality)*.* Crypto underliers exhibit strong intraday and day-of-week volatility seasonality: some hours are reliably far busier than others. For short-dated options — hours to a few days — measuring time to expiry in calendar units then misprices, because the constant-variance-rate assumption behind calendar scaling (made precise in Proposition 16.8 below) fails at that horizon. Practice replaces calendar time with a *deformed vol clock*: time is measured in expected accumulated variance, so that hours expected to be quiet contribute little clock and busy hours contribute much, and T in formulas such as (16.2) is read off that clock rather than the calendar.

16.3 How Volatility Scales with Horizon

Volatility quoted for one horizon must constantly be converted to another — annual to daily on a trading desk, daily to monthly in a risk report. The conversion follows a square-root law, and interviews test both the rule and the reason. This subsection proves the law under independent returns and records the conversion constants worth memorizing.

The setup is as follows. Trading periods are indexed $i = 1, \ldots, n$, and R_i denotes the *log-return* of period i: with P_i the price at the end of period i, $R_i = \ln(P_i/P_{i-1})$. Log-returns are the right object here because they add across periods: the cumulative n-period log-return is exactly $S_n = R_1 + \cdots + R_n = \ln(P_n/P_0)$, whereas simple returns compound rather than add. The per-period volatility is $\sigma_1 = \sqrt{\mathrm{Var}(R_i)}$, assumed finite, and the n-period volatility means $\sqrt{\mathrm{Var}(S_n)}$.

Proposition 16.8 (Square-root-of-time volatility scaling)**.** *Let $R_1, \ldots, R_n$ be iid log-returns with finite variance σ_1^2. Then the cumulative log-return $S_n = R_1 + \cdots + R_n$ satisfies*

$$\mathrm{Var}(S_n) = n\,\sigma_1^2, \qquad \sqrt{\mathrm{Var}(S_n)} = \sigma_1\sqrt{n}\,:$$

variance grows linearly in the horizon, and volatility grows as its square root.

Proof. We proceed by expanding the variance of the sum through bilinearity of covariance:

$$\mathrm{Var}(S_n) = \mathrm{Cov}\Big(\sum_{i=1}^{n} R_i, \sum_{j=1}^{n} R_j\Big) = \sum_{i=1}^{n}\sum_{j=1}^{n} \mathrm{Cov}(R_i, R_j) = \sum_{i=1}^{n} \mathrm{Var}(R_i) + \sum_{i\neq j} \mathrm{Cov}(R_i, R_j).$$

For $i \neq j$ the variables R_i and R_j are independent, and independent variables with finite variances are uncorrelated: $\mathbb{E}[R_i R_j] = \mathbb{E}[R_i]\,\mathbb{E}[R_j]$, so every cross term $\mathrm{Cov}(R_i, R_j)$ vanishes. What remains is $\mathrm{Var}(S_n) = \sum_{i=1}^{n} \sigma_1^2 = n\,\sigma_1^2$, and taking the square root (monotone on $[0,\infty)$) gives the volatility statement. Note that the proof used only that the returns are uncorrelated with a common finite variance; independence and identical distributions are sufficient, not necessary. □

(Problem P16.3 asks you to reproduce this scaling argument, and its failure modes, from scratch.)

Corollary 16.9 (Annualization convention)**.** *With $n = 252$ trading days per year, annual and daily volatility are related by*

$$\sigma_{\text{annual}} = \sqrt{252}\,\sigma_{\text{daily}} \approx 15.9\,\sigma_{\text{daily}}.$$

In particular a 16% annual volatility corresponds to about a 1% typical daily move — the desk anchor "16 vol is 1 percent a day."

Proof. Take one period to be a trading day and $n = 252$ in Proposition 16.8; numerically $\sqrt{252} \approx 15.87$, and $16\%/15.87 \approx 1.01\%$. □

Example 16.10 (Scaling a 16 percent annual volatility)**.** An asset carries $\sigma_{\text{annual}} = 16\%$. Its daily volatility is $16\%/\sqrt{252} = 16\%/15.87 \approx 1.01\%$, and its monthly volatility is $16\%/\sqrt{12} \approx 4.6\%$. The inverse check closes the loop: $1.01\% \times 15.87 \approx 16.0\%$. Note both conversions divide by the square root of the number of periods per year — never by the number of periods itself.

Interview trap. The classic annualization error is to scale volatility by the horizon instead of its square root: multiplying the 1.01% daily volatility of Example 16.10 by 252 "annualizes" it to 254%. What went wrong is that *variance* is the additive quantity in Proposition 16.8, not volatility: the factor 252 belongs to the variance, so applying it to the standard deviation double-counts time. A 254% figure would require every day to move the same direction with certainty. The correct multiplier for volatility is $\sqrt{252} \approx 15.9$, recovering 16%.

Remark 16.11 (What the scaling rule assumes). Proposition 16.8 needs uncorrelated increments with a stable variance, and markets violate both in measurable ways. Positive autocorrelation (momentum) makes the cross terms in the proof positive, so long-horizon volatility exceeds the square-root extrapolation; mean reversion makes them negative, so it falls short — the variance-ratio statistic $\mathrm{Var}(S_n)/(n\sigma_1^2)$ measures exactly this deviation from one. Volatility clustering breaks the stable-variance assumption, which is why a point-in-time daily estimate extrapolates poorly to long horizons, and the intraday seasonality of Remark 16.7 is the same failure at short horizons.

16.4 The Variance Risk Premium and Delta-Hedged Profit and Loss

What does it mean to "trade volatility"? The mechanical answer is a delta-hedged option position, and its profit-and-loss admits a one-line description: gamma-weighted realized-minus-implied variance. The empirical answer is that implied volatility systematically sits above realized, so the side that is short options usually collects. Both statements are recorded here as facts with honest labels — the first is a heuristic identity, the second an empirical regularity — because neither is a theorem of this document.

The setup is as follows. An option on an underlier with price S is bought at implied volatility σ_i and delta-hedged: the option's delta is offset with the underlier once per day. The position's gamma is Γ (the second derivative of option value in S) and its theta is Θ (the time decay per year); one day is $\Delta t = 1/252$ years; R_t denotes day t's simple return $\Delta S/S$, which agrees with the log-return of §16.3 to first order over one day. The volatility actually realized by the underlier over the option's life is σ_r. Rates are taken as approximately zero throughout.

Fact 16.12 (Delta-hedged profit and loss, heuristic)**.** The daily profit and loss of a delta-hedged long option position is approximately

$$\text{daily P\&L} \;\approx\; \tfrac{1}{2}\,\Gamma S^2\Big(R^2 - \sigma_i^2\,\Delta t\Big), \qquad \text{lifetime P\&L} \;\approx\; \sum_t \tfrac{1}{2}\,\Gamma_t S_t^2\Big(R_t^2 - \sigma_i^2\,\Delta t\Big): \tag{16.6}$$

a *gamma-weighted* sum of realized-minus-implied daily variance — the one-line explanation of "trading vol." The identity is a heuristic, not a theorem: it comes from a second-order Taylor expansion of the option value over one day (the delta term is hedged away, the gamma term contributes $\frac{1}{2}\Gamma S^2 R^2$, and theta contributes $\Theta\,\Delta t$) combined with the Black–Scholes relation $\Theta = -\frac{1}{2}\Gamma S^2\sigma_i^2$ for a delta-hedged book at zero rates, with the Greeks frozen over the day and the hedge rebalanced discretely; a rigorous continuous-time treatment is beyond our scope. The weights $\Gamma_t S_t^2$ make the sum path-dependent: variance realized while spot is near the strike, where gamma is largest, counts most.

Fact 16.13 (Variance risk premium, empirical)**.** Across major option markets — equity indices most prominently — implied volatility systematically exceeds the volatility subsequently realized over the option's life, by a few volatility points on average on equity indices. By (16.6), a systematically long, delta-hedged option book therefore loses money on average, and the short side earns a steady carry punctuated by occasional violent drawdowns. This is an empirical regularity, documented in delta-hedged-gains and variance-swap studies (Bakshi–Kapadia, 2003; Carr–Wu, 2009), not a theorem; there is nothing to prove, only data.

Intuition. The two facts fit together as an insurance market. A long option position is long realized variance and pays implied variance as rent: by (16.6), gamma gains accrue when the underlier moves and theta is the bill, priced at σ_i. Options function as insurance against large moves, and the seller's losses arrive precisely in high-volatility crisis states — so buyers pay a premium above actuarial value, implied sits above realized on average, and the gap persists because bearing it is genuinely risky rather than free money.

Interview trap. "You buy a one-month straddle at 25 implied and delta-hedge daily; realized volatility comes in at 20. Do you make or lose money?" The tempting answer is that a hedged position is flat; the correct answer is that you *lose*: by (16.6) the daily R_t^2 falls short of the theta rent $\sigma_i^2\Delta t$ on average, and to first order the loss is about $(1 - 20/25) = 20\%$ of the premium paid — more if the shortfall in realized volatility happened while spot sat near the strike, where the gamma weights concentrate. Problem P16.4 works the arithmetic in full.

Practice: problems P16.1–P16.4 in §20, solutions in §21.

Part VI

Interview Craft: Cost Models and Open-Ended Questions

17 Algorithmic Cost Models

This section develops the two cost-model calculations that live quant and AI assessments ask most often: when a low-rank factorisation of a matrix beats the dense product, and when pre-sorting or hashing a dataset beats scanning it. Both questions reduce to a single move — write down the two competing cost expressions and solve for the crossover — which we isolate as a named pattern in Remark 17.9.

17.1 Matrix Multiplication and the Decomposition Crossover

Every linear layer, factor exposure, and covariance update is ultimately a matrix product, so the first quantity to control is what a product costs. This subsection counts that cost exactly, then answers the question interviewers build on top of it: if a dense matrix is replaced by a low-rank factorisation, at what rank does the factored product become cheaper — and why does one of the two ways of associating the factored product *never* win?

Setup. Let N, P, R, and K denote positive integers. Here N, P, R, K are integer matrix dimensions; in particular P is not the probability operator of the notation table and R is not a return — no probabilities or returns appear in this section. All matrices in this subsection are real and dense: A is $N \times P$ and B is $P \times R$, so the product AB is $N \times R$. The cost model counts *scalar multiplications*: the cost of an algorithm is the number of scalar multiplications it performs, with scalar additions tracked alongside for completeness. This is the standard unit in numerical linear algebra, and the two counts agree to leading order in any case, as the next proposition shows.

Proposition 17.1 (Matrix multiplication count)**.** *Computing the product AB of an $N \times P$ matrix A and a $P \times R$ matrix B directly from the definition costs exactly NPR scalar multiplications and $NR(P-1)$ scalar additions. In total this is $2NPR - NR$ floating-point operations, which is $2NPR$ flops to leading order.*

Proof. We proceed by a counting argument. The product AB has NR entries, and entry (i, j) is the dot product of row i of A with column j of B:

$$(AB)_{ij} \;=\; \sum_{k=1}^{P} a_{ik}\, b_{kj}.$$

Evaluating one such length-P dot product takes P multiplications (one per term) and $P - 1$ additions (to sum P terms). Since the NR output entries are computed independently, the totals are $NR \cdot P = NPR$ multiplications and $NR(P-1)$ additions. Summing the two counts gives $NPR + NR(P-1) = 2NPR - NR$ operations; the deficit NR is lower order in P, so the flop count is $2NPR$ to leading order. ☐

(Problem P17.1 asks you to reproduce this proof from scratch.)

Example 17.2 (Counting a small product by hand)**.** Take $N = 2$, $P = 3$, $R = 2$, so A is 2×3 and B is 3×2. Proposition 17.1 predicts $NPR = 2 \cdot 3 \cdot 2 = 12$ multiplications and

$NR(P-1) = 2 \cdot 2 \cdot 2 = 8$ additions. Direct verification: the product has $NR = 4$ entries, each a length-3 dot product costing 3 multiplications and 2 additions, giving $4 \cdot 3 = 12$ multiplications and $4 \cdot 2 = 8$ additions, as predicted. The exact operation total is $12 + 8 = 20$, against the leading-order figure $2NPR = 24$; the gap is exactly $NR = 4$.

Intuition. The formula NPR is best remembered as a picture, not a symbol string: the output has $N \times R$ cells, and filling each cell means running a length-P dot product across the inner dimension. Output area times inner length. The flop convention $2NPR$ simply counts the multiply and the add of each multiply-accumulate step separately.

Setup (decomposition). Suppose now that a dense $N \times P$ matrix M has been replaced by a rank-K factorisation

$$M = UV, \qquad U \in \mathbb{R}^{N \times K}, \qquad V \in \mathbb{R}^{K \times P},$$

and that the object we actually need is the product MB with a data block $B \in \mathbb{R}^{P \times R}$ (for a single input vector, $R = 1$). Matrix multiplication is associative, so the factored product can be evaluated in two orders, $(UV)B$ or $U(VB)$, which give the same matrix at very different prices. The next theorem prices both orders and locates the rank below which the factorisation pays.

Theorem 17.3 (Decomposition crossover)**.** *Under the multiplication-count cost model of Proposition 17.1, with $M = UV$ as above:*

(i) the order $(UV)B$ costs $NKP + NPR$ multiplications, which exceeds the dense cost NPR of MB for every $K \geq 1$ — this association order never beats the dense product;

(ii) the order $U(VB)$ costs $KPR + NKR = KR(N+P)$ multiplications;

(iii) with the correct association $U(VB)$, the factored product beats the dense product if and only if

$$K \; < \; \frac{NP}{N+P}, \tag{17.1}$$

a threshold that does not depend on R.

Proof. We apply Proposition 17.1 to each subproduct and compare the resulting cost expressions.

Part (i). Forming UV multiplies an $N \times K$ matrix by a $K \times P$ matrix, costing NKP by Proposition 17.1, and the result is the full dense $N \times P$ matrix M. Multiplying it by B then costs NPR, for a total of

$$\operatorname{cost}\big[(UV)B\big] \; = \; NKP + NPR. \tag{17.2}$$

The second term alone already equals the dense cost of MB, so the total exceeds it by $NKP > 0$ whenever $K \geq 1$. The mechanism deserves stating in words: reassembling the factors recreates the dense matrix, hence the entire dense bill NPR, and the assembly cost NKP is paid on top. No choice of K rescues this order.

Part (ii). Forming VB multiplies a $K \times P$ matrix by a $P \times R$ matrix, costing KPR; the result is only $K \times R$. Multiplying U $(N \times K)$ by this small block costs NKR. The total is

$$\operatorname{cost}\big[U(VB)\big] \; = \; KPR + NKR \; = \; KR(N+P). \tag{17.3}$$

Part (iii). We compare (17.3) with the dense cost NPR and solve for K:

$$KR(N+P) \; < \; NPR \iff K(N+P) \; < \; NP \iff K \; < \; \frac{NP}{N+P},$$

where the first equivalence divides both sides by $R > 0$. The variable R cancels, which proves the threshold is independent of the width of the data block. $\square$

(Problem P17.3 asks you to reproduce this proof from scratch.)

Figure 27 shows all three cost expressions as functions of K at fixed N, P, R: the dense cost is flat, the correctly associated factored cost is a line through the origin that crosses it at $K^* = NP/(N+P)$, and the misassociated cost (17.2) starts *at* the dense cost when $K = 0$ and only climbs from there.

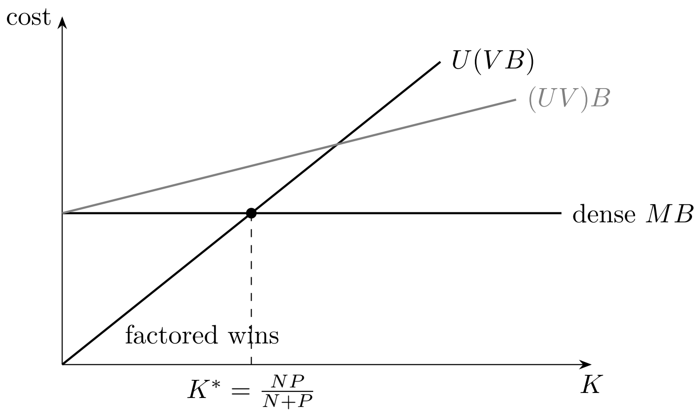


Figure 27: Multiplication cost as a function of the factor rank K, at fixed N, P, R. The dense product MB costs NPR regardless of K. The correctly associated factored product $U(VB)$ costs $KR(N+P)$, a straight line through the origin that crosses the dense cost at $K^* = NP/(N+P)$; for $K < K^*$ the factored path wins. The misassociated product $(UV)B$ costs $NKP + NPR$: it starts at the dense cost and rises, so it loses to the dense product for every $K \geq 1$.

Remark 17.4 (Harmonic-mean form of the threshold). The harmonic mean of N and P is $H = 2NP/(N+P)$, so the threshold (17.1) reads $K < H/2$: the break-even rank is half the harmonic mean of the two dense dimensions. For a square matrix, $N = P$ gives $K < N/2$ — any factorisation with rank below half the side length is a win. Because R cancels, the comparison is effectively per column of data: pushing more data through the layer changes both bills proportionally and moves the crossover not at all.

Example 17.5 (Low-rank arithmetic at LoRA scale)**.** A dense 4096×4096 layer is replaced by a rank-64 factorisation, so $N = P = 4096$ and $K = 64$; take a single input vector, $R = 1$.

- *Dense cost:* $NPR = 4096 \cdot 4096 = 16{,}777{,}216$ multiplications.
- *Factored cost, correct order* $U(Vx)$*:* $KR(N+P) = 64 \cdot 8192 = 524{,}288$ multiplications.
- *Speedup:* $16{,}777{,}216/524{,}288 = 32$, an exact 32-fold reduction.
- *Break-even rank:* $K^* = NP/(N+P) = 4096^2/8192 = 2048$, which is $N/2$ as Remark 17.4 predicts for a square layer; the speedup can be read off as $K^*/K = 2048/64 = 32$, consistent with the direct division.
- *Misassociated order* $(UV)x$*:* $NKP + NPR = 4096 \cdot 64 \cdot 4096 + 16{,}777{,}216 = 1{,}073{,}741{,}824 + 16{,}777{,}216 = 1{,}090{,}519{,}040$ multiplications — 65 times the dense cost, instead of 32 times cheaper.

Trading application. This crossover is the economics behind LoRA-style adapters: a rank-K update with $K \ll \min(N, P)$ sits far below the threshold (17.1), so the factored path is cheap — rank 64 against a break-even of 2048 in Example 17.5. The operational rule that comes with it: keep the factors separate and always multiply the data into V first. Re-multiplying the factors back together before touching the data forfeits the entire saving and adds an assembly bill on top, by Theorem 17.3(i).

Interview trap. Association order is the hidden half of the question. An interviewer who asks for the cost of the factored product expects both orders priced: quoting only $(UV)B$ at $NKP+NPR$, or asserting that "the factorisation makes it cheaper" without naming the order, misses that one order *always* loses — for every rank, by Theorem 17.3(i), and by a factor of 65 at the scale of Example 17.5.

17.2 Membership Queries: Scan vs Sort vs Hash

The second house calculation prices a data-structure decision instead of a linear-algebra one: given a batch of membership queries against a static dataset, when does the up-front cost of sorting (or hashing) pay for itself? The move is the same as in Theorem 17.3(iii): write both total costs and solve for the crossover.

Setup. A static dataset holds N elements, with $N \geq 2$, and must answer q membership queries ("is x in the dataset?"). In this subsection q denotes the number of queries — not the complement probability $1-p$ of the notation table; the symbol is re-purposed locally. Logarithms in cost expressions are written $\log_2$ because the counts arise from repeated halving; inside $O(\cdot)$ the base only shifts a constant factor. Three designs compete, with the following idealised comparison counts:

$$\begin{aligned}
&\text{scan: } qN,\\
&\text{sort + binary search: } N\log_2 N + q\log_2 N = (N+q)\log_2 N,\\
&\text{hash set: } O(N) \text{ expected build} + O(q) \text{ expected probes.}
\end{aligned} \tag{17.4}$$

Here a linear scan costs N comparisons per query with no preprocessing; a comparison sort costs $N\log_2 N$ comparisons once, after which each binary search costs $\log_2 N$ comparisons; and a hash set costs $O(N)$ expected time to build, then $O(1)$ expected time per probe, for $O(N+q)$ expected in total.

Proposition 17.6 (Scan-versus-sort crossover)**.** *Under the cost model (17.4), sorting once and binary-searching each query beats scanning per query if and only if*

$$q \;>\; \frac{N\log_2 N}{N-\log_2 N}, \tag{17.5}$$

and the threshold satisfies $N\log_2 N/(N-\log_2 N) \to \log_2 N$ in ratio as $N\to\infty$: for large N, sorting pays for itself after roughly $\log_2 N$ queries.

Proof. We compare the two total costs in (17.4) directly and solve the resulting inequality for q. Sorting wins exactly when

$$(N+q)\log_2 N \;<\; qN \quad\Longleftrightarrow\quad N\log_2 N \;<\; q\,(N-\log_2 N).$$

Since $2^N > N$ for every $N \geq 1$, we have $N > \log_2 N$, so the factor $N-\log_2 N$ is positive and dividing by it preserves the inequality, giving (17.5). For the asymptotic claim, divide the threshold by $\log_2 N$:

$$\frac{1}{\log_2 N}\cdot\frac{N\log_2 N}{N-\log_2 N} \;=\; \frac{N}{N-\log_2 N} \;=\; \left(1-\tfrac{\log_2 N}{N}\right)^{-1} \;\longrightarrow\; 1 \qquad (N\to\infty),$$

because $\log_2 N/N \to 0$. The threshold is therefore $\log_2 N$ up to a factor tending to one. ☐

(Problem P17.5 asks you to reproduce this derivation from scratch.)

Example 17.7 (Crossover at $N = 1024$)**.** Take $N = 1024$, so $\log_2 N = 10$. The exact threshold (17.5) is

$$q \;>\; \frac{1024\cdot 10}{1024-10} \;=\; \frac{10{,}240}{1014} \;\approx\; 10.1,$$

against the approximation $\log_2 N = 10$ — the approximation is tight already at this size. Checking the two integers either side of the threshold: at $q = 10$, scanning costs $10 \cdot 1024 = 10{,}240$ while sort-plus-search costs $(1024 + 10) \cdot 10 = 10{,}340$, so scanning still wins; at $q = 11$, scanning costs $11 \cdot 1024 = 11{,}264$ while sort-plus-search costs $(1024 + 11) \cdot 10 = 10{,}350$, and sorting has already paid for itself. Ten queries against a thousand elements — a strikingly small number.

Fact 17.8 (Expected cost of hashing)**.** Under the standard uniform-hashing assumptions, building a hash set over N elements takes $O(N)$ expected time (amortised $O(1)$ per insertion with table doubling) and each membership probe takes $O(1)$ expected time, so answering q queries costs $O(N + q)$ expected in total — asymptotically better than both alternatives in (17.4). The guarantees are expected-case, not worst-case: adversarial or heavily colliding keys degrade probes toward linear time. The analysis is classical (Knuth, 1998); a proof is beyond our scope.

Trading application. The hash set beats both scanning and sorting asymptotically, but the win is bought with three concessions: extra memory for the table, the loss of all ordering information, and worst-case degradation under collisions. In an interview, frame it as "expected $O(1)$ lookups, amortised build" — the qualifiers are the point, because they show the caveats are understood rather than papered over.

Interview trap. Quoting the hash table as flatly "$O(1)$" drops both qualifiers that make the claim true. The lookup bound is *expected*, not guaranteed — collisions, and in the worst case adversarial keys, push probes toward linear time — and the build is *amortised* over insertions. An answer that prices hashing without saying "expected" and "amortised" invites exactly the follow-up it fails to anticipate.

Remark 17.9 (The house pattern: equate the two costs). Every crossover question in this section is the same calculation wearing different clothes: write down the total cost of each competing design as an expression in the problem's parameters, set the two expressions equal, and solve for the variable the question asks about. Theorem 17.3(iii) is this pattern solved for the rank K; Proposition 17.6 is the same pattern solved for the query count q. Live assessments have asked exactly this shape ★, and the pattern generalises to any "at what point does design A beat design B" question: two cost lines, one intersection, solve.

Practice: problems P17.1–P17.6 in §20, solutions in §21.

18 Forecasting Returns: the Open-Ended Question

"How would you predict the returns of an index?" is a standard closing question in quantitative interviews, and it is graded differently from every puzzle that precedes it: the expected answer is a *process*, not a model name. This section fixes that process as a four-step numbered procedure (Procedure 18.1), supplies the rationale behind each step, examines the one genuine architectural fork inside the pipeline — modelling the index directly versus modelling its constituents and aggregating (§18.2) — and makes the validation protocol and its metrics precise (§18.3). Figure 28 shows the whole pipeline at a glance.

18.1 The Pipeline as a Procedure

This subsection states the pipeline formally and gives the rationale for its first two steps; the modelling fork (Step 3) and the validation protocol (Step 4) carry enough content to receive their own subsections below.

The setup is as follows. Fix an equity index and a forecast horizon, and let time run in discrete periods of that horizon. The quantity to be forecast is the index return R_{t+1} over the period from t to $t + 1$. The forecaster observes, at time t, a feature vector $X_t \in \mathbb{R}^d$ and issues a forecast

$\hat{R}_{t+1}$ of R_{t+1}. The standing assumption on all features is *measurability at forecast time*: every component of X_t must be computable from information actually available when the forecast is issued. Violations of this assumption are the look-ahead bias that the interview-trap block of §18.3 names first.

Procedure 18.1 (Index-return forecasting pipeline)**.** Given the task of forecasting the return of an equity index, proceed in four steps.

1. **Define the target.** Fix the return convention — log or simple — and the horizon (one minute, one day, one week), and state *why*: every downstream choice depends on the target.
2. **Explore the data, then build features in layers.** Exploratory analysis first: stationarity, outliers, and regime breaks; the correlation matrix; PCA for the factor structure. Then assemble X_t in four layers: (i) *endogenous* — past returns, realized volatility, volume, spread, plus order-book features when the horizon is intraday; (ii) *exogenous lead–lag* — markets that move earlier (the US close leads the European open; overnight index futures; rates, FX, commodities); (iii) *composition-specific* — features implied by what the index actually holds (the CAC 40's heavy luxury-sector weight makes China sentiment a feature; TotalEnergies makes Brent one); (iv) *sentiment* — news NLP and positioning data.
3. **Choose the modelling route: direct or factorial** (Definition 18.4). In either route, fit the simplest credible model first — ridge regression on standardized features — and escalate to gradient-boosted trees (LightGBM) or sequence models only if the added capacity is justified out of sample.
4. **Validate against a baseline, walking forward.** Establish a trivial baseline and ask first whether the information coefficient (Definition 18.6) is positive at all; split the data chronologically (`TimeSeriesSplit` / walk-forward), never shuffled; score the information coefficient, the hit rate, and the Sharpe ratio of the simulated strategy; and name the deployment risks collected in the interview-trap block of §18.3 without being prompted.

(Problem P18.1 asks for exactly this procedure, narrated end to end under interview conditions.)

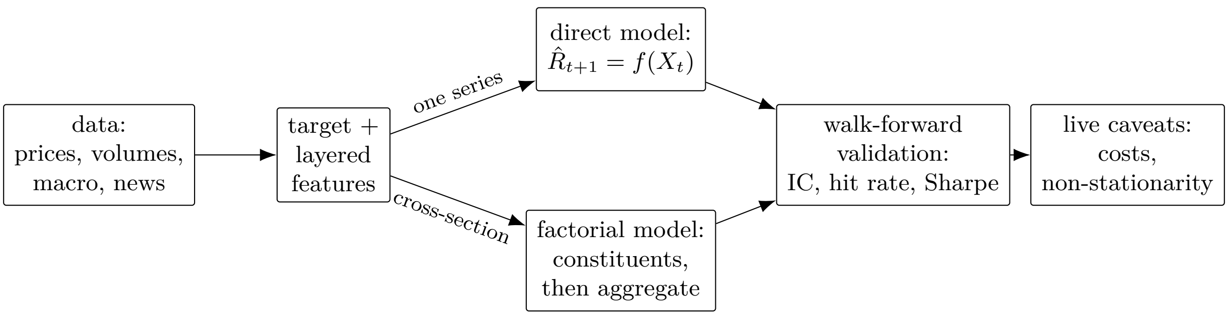


Figure 28: The forecasting pipeline of Procedure 18.1. Data feed layered features; the modelling stage forks into the direct route (one model fitted to the index series) and the factorial route (constituent-level models aggregated by index weights, Definition 18.4); both routes are scored by the same walk-forward validation, and whatever survives still faces the live caveats collected in the interview-trap block of §18.3.

Interview trap. The model-name answer. Replying "an LSTM" or "gradient boosting" to this question fails regardless of which model is named, because the question grades process, not taste in architectures. The strong answer walks Procedure 18.1 in order — target first, model third — and volunteers the risks of §18.3 unprompted.

Remark 18.2 (Rationale for Step 1: the target determines everything downstream). Two choices are made in Step 1, and neither is cosmetic. First, the return convention: log returns add across time, which suits multi-period and volatility work, while simple returns add across assets, which is what a weighted aggregation over index constituents requires — so the convention chosen here interacts directly with the fork of Step 3 (see (18.1)). Second, the horizon: it sets the noise level of the target, determines which features exist at all (order-book features are meaningful intraday and not at a weekly horizon), governs how quickly any signal decays, and fixes how often the resulting strategy must trade — hence its exposure to transaction costs. Stating the "why" aloud is part of what is graded: a forecaster who cannot say why the horizon is one day has not yet defined the problem.

Remark 18.3 (Rationale for Step 2: explore before modelling, layer the features). The exploratory pass earns its place ahead of any model. Stationarity checks decide whether tools that assume a stable distribution are usable at all; outliers decide the loss function and winsorization policy; detected regime breaks warn that validation must span regimes rather than sit inside one; and the correlation matrix with a PCA on top exposes the factor structure that Step 3's factorial route will reuse. The layers themselves are an audit device: each layer draws on a different source of information — the index's own history, other markets that trade earlier (an empirical lead–lag regularity, as with the US close leading the European open), the macro exposures implied by the index's actual holdings, and text or positioning data. Walking the layers in order shows the coverage is systematic rather than anecdotal, and the composition-specific layer is what distinguishes forecasting *this* index from forecasting a generic one.

18.2 The Modelling Fork: Direct versus Factorial

Step 3 of Procedure 18.1 contains the pipeline's one genuine architectural decision: fit one model to the index series itself, or model the constituents and aggregate. This subsection defines both routes precisely and gives the rationale for preferring the second.

The setup is as follows. The index has n constituents with weights $w_1, \dots, w_n$, positive and summing to one, held fixed over the forecast horizon for simplicity. Constituent i has simple return $R_{i,t+1}$ over the period from t to $t+1$, and the index simple return is the weighted aggregate

$$R_{t+1} \;=\; \sum_{i=1}^{n} w_i \, R_{i,t+1}, \tag{18.1}$$

which is an identity for simple returns (and only approximate for log returns — the convention choice of Step 1 at work). The symbol $R_{m,t+1}$ denotes the market return over the same period, and β_i the market beta of constituent i.

Definition 18.4 (Direct and factorial forecasts)**.** Fix the target R_{t+1} and the feature vector X_t of §18.1.

1. The *direct route* fits a single model f to the index series itself and forecasts $\hat{R}_{t+1} = f(X_t)$.
2. The *factorial route* decomposes each constituent by the market model

$$R_{i,t+1} \;=\; \beta_i \, R_{m,t+1} + \alpha_{i,t+1}, \tag{18.2}$$

   where $\alpha_{i,t+1}$ is the beta-neutralized residual return of constituent i; builds forecasts at the constituent level — for the market component and for the residuals — and aggregates them into an index forecast through (18.1): $\hat{R}_{t+1} = \sum_{i=1}^{n} w_i \, \hat{R}_{i,t+1}$.

Remark 18.5 (Rationale for Step 3: why the factorial route is often better). Three observations favor the factorial route.

1. *PCA reveals the structure the route exploits.* An eigendecomposition of the constituent correlation matrix typically shows the first principal component approximating the market mode and the next few components (roughly PC2 through PC5) approximating sector modes. This decomposition supplies the market factor used for the residualization in (18.2) and compresses the n correlated constituent series into a small number of drivers.

2. *The beta-neutralized residual is the cleaner target.* At short horizons the raw constituent series is dominated by the common market mode, which is the noisiest and hardest component to predict; stripping it out by (18.2) and forecasting the residual $\alpha_{i,t+1}$ gives a prediction target with cleaner structure than the raw series.
3. *The cross-section multiplies the data.* Modelling n related series instead of one provides many more observations of the shared structure per unit of calendar time than the single index series does.

The aggregation step itself relies on (18.1) holding exactly, which is one more reason the return convention is fixed in Step 1 before any model is chosen.

(Problem P18.3 asks for this case to be argued in full, including the ways the weighted aggregation can fail.)

Intuition. The direct route asks one model to predict the hardest object in the market — the market mode itself — from a single noisy history. The factorial route replaces that one hard problem with many easier, related ones plus a known aggregation rule: model what has structure (the residuals, the factor loadings), and let the arithmetic of (18.1) do the rest.

18.3 Validation: Protocol, Metrics, and the Risks to Name

Step 4 is where forecasting pipelines are separated from curve-fitting exercises. This subsection defines the three evaluation metrics precisely, gives the rationale for the walk-forward protocol and for rejecting mean squared error as the headline metric, works a fully numeric example, and closes with the structured list of risks that a complete answer names unprompted.

The setup is as follows. The model is evaluated on a window of T periods, producing pairs $(\hat{R}_t, R_t)$ for $t = 1, \dots, T$, where $\hat{R}_t$ denotes the forecast of R_t issued one period earlier; every forecast obeys the measurability assumption of §18.1. Sample means over the window are written $\bar{\hat{R}}$ and $\bar{R}$. The sign function takes values in $\{-1, 0, +1\}$. The position held in period t is written w_t, with the convention $w_0 = 0$ — a traded position, unrelated to the constituent weights w_i of §18.2 — and $c \geq 0$ denotes the per-unit transaction cost charged on position changes.

Definition 18.6 (Forecast evaluation metrics: IC, hit rate, strategy Sharpe)**.** Over the evaluation window:

1. The *information coefficient* (IC) is the Pearson sample correlation between forecasts and realizations,
$$\mathrm{IC} \;=\; \frac{\sum_{t=1}^{T}(\hat{R}_t - \bar{\hat{R}})(R_t - \bar{R})}{\sqrt{\sum_{t=1}^{T}(\hat{R}_t - \bar{\hat{R}})^2}\,\sqrt{\sum_{t=1}^{T}(R_t - \bar{R})^2}}.$$
2. The *hit rate* is the fraction of periods on which the forecast calls the direction correctly, $H = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\{\mathrm{sign}(\hat{R}_t) = \mathrm{sign}(R_t)\}$.
3. The *strategy Sharpe* is computed by trading the forecast: take the position $w_t = \mathrm{sign}(\hat{R}_t)$ each period, record the per-period profit and loss net of transaction costs, $\Pi_t = w_t R_t - c\,|w_t - w_{t-1}|$ (the per-unit cost c charged on position changes), and report the Sharpe ratio of the series $\Pi_1, \dots, \Pi_T$ — its sample mean over its sample standard deviation, annualized by the square root of the number of periods per year (the Sharpe conventions of §10).

Remark 18.7 (Rationale for Step 4, part one: why these metrics and not MSE). Raw mean squared error is direction-blind: it measures distance between forecast and realization, while trading profit depends on the *sign* of the forecast through the position w_t. A squared-error criterion can therefore strictly prefer a money-losing forecast to a money-making one — Example 18.9 exhibits the failure numerically. The three metrics of Definition 18.6 are chosen to respect the use case: the IC measures monotone co-movement, the hit rate scores direction itself, and the strategy Sharpe is the economic bottom line after costs.

Remark 18.8 (Rationale for Step 4, part two: baseline first, walk forward, never shuffle)*.* The baseline comes first because it defines the null: trivial forecasts (identically zero, or the historical mean) cost nothing and encode no skill, so the first question for any candidate model is whether it achieves an out-of-sample IC greater than zero at all. The evaluation must be chronological: train on a window of past data, test on strictly later data, then roll the window forward — the scheme implemented by `TimeSeriesSplit` and generically called walk-forward validation. Shuffled cross-validation is never acceptable for this problem: shuffling places observations from the test period's future inside the training set, which is look-ahead bias built directly into the protocol, and it produces validation scores that cannot be earned in live trading, where the future is genuinely unavailable.

Example 18.9 (Scoring a five-day toy forecast)**.** A daily index forecast is evaluated over $T = 5$ days. Returns are in percent.

Day t	1	2	3	4	5
Realized R_t	$+1.0$	-0.5	$+0.2$	-0.8	$+0.6$
Forecast $\hat{R}_t$	$+0.4$	-0.1	-0.3	-0.2	$+0.5$
Direction hit	yes	yes	no	yes	yes
P&L $\Pi_t = w_t R_t$ (at $c = 0$)	$+1.0$	$+0.5$	-0.2	$+0.8$	$+0.6$

Hit rate. Four of the five signs agree, so $H = 4/5 = 80\%$.

Information coefficient. The sample means are $\bar{\hat{R}} = 0.3/5 = 0.06$ and $\bar{R} = 0.5/5 = 0.10$. The centered cross-products sum to

$$(0.34)(0.90) + (-0.16)(-0.60) + (-0.36)(0.10) + (-0.26)(-0.90) + (0.44)(0.50) = 0.820,$$

while the centered sums of squares are 0.5320 for the forecasts and 2.24 for the realizations, so

$$\mathrm{IC} = \frac{0.820}{\sqrt{0.5320 \times 2.24}} = \frac{0.820}{1.0916} \approx 0.75.$$

Strategy Sharpe (before costs). The P&L series has sample mean $2.7/5 = 0.54\%$ per day and sample standard deviation $\sqrt{0.8320/4} \approx 0.456\%$, giving a per-day Sharpe of $0.54/0.456 \approx 1.18$ and an annualized Sharpe of $1.18 \times \sqrt{252} \approx 18.8$.

Why MSE is the wrong headline metric. Consider day 1, on which the realized return is $+1.0$. A forecast of -0.2 has squared error $(-1.2)^2 = 1.44$ and puts the strategy short, losing 1.0; a forecast of $+2.4$ has the *larger* squared error $(1.4)^2 = 1.96$ yet puts the strategy long, making $+1.0$. Squared error prefers the money-losing forecast, which is the direction-blindness of Remark 18.7 in one line.

Calibration of the numbers. The toy window is five days, so an IC of 0.75 and an annualized Sharpe of 18.8 carry no evidential weight — and in a real pipeline a number like 18.8 is precisely the "too-beautiful" Sharpe that the trap block below says to treat as a bug. Good daily-horizon index pipelines sit only a few points of hit rate above 50%; a backtest printing far above that level is evidence of leakage, not of edge.

Interview trap. The risks to name unprompted. Volunteering the deployment risks — rather than waiting to be asked — is what separates a process answer from a model-name answer. Four must be named.

1. *Look-ahead bias.* Any feature, label, or preprocessing step that touches information from after the forecast time violates the measurability assumption of §18.1; the walk-forward protocol of Remark 18.8 exists to prevent the protocol-level version of it.
2. *Overfitting.* A too-beautiful Sharpe is a bug: an implausibly good backtest is evidence of leakage or overfitting, not of edge, and the correct response is to hunt for the error (Example 18.9 calibrates what "implausible" means at a daily horizon).

3. *Transaction costs.* A gross edge at high turnover can be a net loss; the strategy Sharpe of Definition 18.6 is scored after costs for exactly this reason.
4. *Non-stationarity.* The relationships the model learns drift, and regimes break; a model validated inside one regime may fail in the next, which is why regime breaks are hunted in Step 2's exploratory pass and why walk-forward performance is examined for stability across sub-periods rather than only in aggregate.

(Problem P18.2 applies exactly this checklist to a strategy that "backtests very well.")

Practice: problems P18.1–P18.3 in §20, solutions in §21.

Part VII

Computer Science for Research Engineering

19 Computer Science for Research Engineers: Python Data Structures, Parallelism, and Time

Research-lab on-site rounds for research-engineering roles contain a distinctive interview: one long-form "algorithmic reasoning" question in escalating parts, answered in pseudocode on paper, with no live coding and a grade for the quality of the reasoning rather than for syntax. The early parts ask for a container and its cost; the later parts move the same task onto many cores, then onto many machines, and ask what breaks. Two topics decide those later parts more than any other — the difference between wall-clock time and CPU time, and race conditions — and this section is built around them. It develops, in order, the cost of Python's containers as CPython implements them (§19.1), the handful of algorithmic patterns that recur (§19.2), the two clocks and how to read their ratio (§19.3), the speedup laws of Amdahl and of the work–span model (§19.4), what Python's threads, processes and coroutines can and cannot parallelise (§19.5), race conditions, the lost update and deadlock (§19.6), associativity as the licence to parallelise and what floating point does to it (§19.7), and distributed aggregation with the top-k result that candidates most often get wrong (§19.8). It builds directly on the cost models of Section 17: the crossover pattern of Remark 17.9 reappears solved for the number of workers, and the hashing caveat of Fact 17.8 receives its defence.

19.1 Cost Qualifiers and Python's Containers as CPython Implements Them

Every algorithmic-reasoning answer starts by naming a container and quoting what its operations cost, and the correct quotation carries qualifiers — amortised, expected, worst-case — that this subsection defines and then proves for the containers a research engineer uses in Python. It builds on the membership cost model of Section 17.2: Fact 17.8 priced hashing with a collision caveat, and the hash-randomisation material below is the defence to that caveat.

Setup. Costs are counted in the RAM model: one unit per pointer read or write, per comparison, or per hash evaluation, with memory access at unit cost. The symbol n denotes the number of elements currently stored in a container, m the length of a sequence of operations, and k the length of a slice or the size of a sub-collection; none of these is a probability or a random-walk quantity of the notation table. A CPython container stores *pointers* to boxed objects, never the values themselves (Remark 9.13); on a 64-bit build a pointer occupies 8 bytes. Every implementation claim below refers to CPython 3.12 unless a version is named, and is an implementation choice rather than a language guarantee unless it is stated as one. The symbol $g > 1$ denotes a growth factor and $c_0 \geq 1$ an initial capacity.

Definition 19.1 (Worst-case, amortised and expected cost)**.** Let an operation on a container be applied repeatedly.

(i) Its *worst-case cost* is a bound that holds for every single application, whatever the state of the container.
(ii) Its *amortised cost* is a bound a such that any sequence of m applications, starting from an empty container, costs at most am in total (the *aggregate method*): one expensive application may be paid for by many cheap ones, and the bound says nothing about any single application.
(iii) Its *expected cost* is the average over the algorithm's own randomness — for hash tables, the choice of hash function — for a *fixed* input; it says nothing about the cost on a single adversarial input.

The three qualifiers are independent: an operation can be amortised $O(1)$ and worst-case $O(n)$ (a resize), or expected $O(1)$ and worst-case $O(n)$ (a hash probe under collisions). Fact 17.8 attached "expected" and "amortised" to hashing; every row of Table 3 carries one of the three.

Fact 19.2 (List and tuple layout (CPython 3.12, Objects/listobject.c and Objects/tupleobject.c))**.** A `list` is a header holding a pointer to one contiguous array of object pointers, the current length, and an allocated capacity. When an append finds the array full, `list_resize` allocates a new capacity of newsize + $\lfloor$newsize/8$\rfloor + 6$ rounded down to a multiple of 4 and copies every pointer across. Read through `sys.getsizeof` on the installed 3.11, the capacities of a list grown by appends are $4, 8, 16, 24, 32, 40, 52, 64, 76, 92, \ldots,$ so the growth ratio tends to $9/8$; the 3.12 formula is the same. Consequently indexing and item assignment cost $O(1)$, `pop()` at the end $O(1)$, `append` amortised $O(1)$ (Theorem 19.3), `insert(0, x)` and `pop(0)` $\Theta(n)$ (every remaining pointer moves), membership `x in L` $O(n)$, and a slice of length k costs $O(k)$. A `tuple` is the same pointer array with no spare capacity; it is immutable, and it is hashable exactly when every element is hashable.

Theorem 19.3 (Dynamic-array costs: geometric growth gives amortised $O(1)$ append; front operations are linear)**.** *Consider a dynamic array with initial capacity $c_0 \geq 1$ that, whenever an append finds it full, allocates a new array and copies its contents. Let n appends be performed on an initially empty array.*

(i) If the capacity is multiplied by a fixed factor $g > 1$ at each overflow, the total number of pointer copies is less than $gn/(g-1)$, so the amortised cost of an append — copies plus the one direct write — is at most $1 + g/(g-1)$, a constant independent of n.
(ii) If instead the capacity grows by a fixed constant $c \geq 1$ at each overflow (take $c_0 = c$), the total number of copies is $n^2/(2c) + O(n)$: quadratic in n.
(iii) Inserting at index i of a contiguous array of length n moves the $n - i$ pointers behind it, and deleting index i moves $n - i - 1$; hence `insert(0, x)` and `pop(0)` cost $\Theta(n)$, and draining a list of n items from the front with `pop(0)` costs $n(n-1)/2 = \Theta(n^2)$ pointer moves.

Proof. We use the aggregate method of Definition 19.1(ii) for (i) and (ii), and a contiguity argument for (iii).

Part (i). Resizes occur when the array is full, at sizes $c_0, c_0 g, c_0 g^2, \ldots, c_0 g^m$, and the j-th resize copies the $c_0 g^j$ pointers then stored. The last resize is triggered by an append that found the array full at size $c_0 g^m$, so $c_0 g^m \leq n - 1 < n$. The copies form a geometric series:

$$\sum_{j=0}^{m} c_0 g^j \;=\; c_0 \frac{g^{m+1} - 1}{g - 1} \;<\; \frac{c_0 g^{m+1}}{g - 1} \;=\; \frac{g}{g-1}\, c_0 g^m \;<\; \frac{g\, n}{g - 1}.$$

Adding the n direct writes (one per append) and dividing by n gives an amortised cost below $1 + g/(g-1)$ per append.

Part (ii). With $c_0 = c$ and additive growth, the array is full at sizes $c, 2c, 3c, \dots$, so the j-th resize copies jc pointers, and resizes happen for $j = 1, \dots, m$ with $m = \lfloor (n-1)/c \rfloor$. By the arithmetic series,

$$\sum_{j=1}^{m} jc \;=\; c\,\frac{m(m+1)}{2} \;=\; \frac{n^2}{2c} + O(n),$$

because $m = n/c + O(1)$.

Part (iii). The list invariant is contiguity: before and after every operation, element j lives at address base $+ 8j$. Inserting a new element at index i therefore requires elements $i, \dots, n-1$ to occupy slots $i+1, \dots, n$ afterwards: each of the $n-i$ of them moves once (a single `memmove`, but $n-i$ pointer copies). Deleting index i symmetrically moves elements $i+1, \dots, n-1$ down by one slot, $n-i-1$ moves. With $i = 0$ both counts are $\Theta(n)$. Draining by `pop(0)` from length n moves $(n-1) + (n-2) + \cdots + 0 = n(n-1)/2$ pointers in total, again by the arithmetic series. □

Remark 19.4 (The potential-function view and CPython's trade). The same bound follows from the potential method: with $g = 2$, define $\Phi = 2 \cdot \text{size} - \text{capacity}$, which is 0 just after a resize and equals the size just before one. An append without resize costs 1 and raises Φ by 2, amortised 3; an append with resize at size s costs $s+1$ and changes Φ from s to 2, amortised $s+1+(2-s) = 3$. CPython's growth ratio tends to 9/8 rather than 2, so the bound of Theorem 19.3(i) is $1 + (9/8)/(1/8) = 10$ writes per append instead of 3; in exchange the slack memory after a resize is about one eighth of the array instead of up to one half. The append stays amortised $O(1)$ for *any* $g > 1$; only the constant moves.

Example 19.5 (Appending a million pointers)**.** Take $n = 10^6$ appends from $c_0 = 4$ with doubling, $g = 2$. Resizes occur at sizes $4, 8, 16, \dots, 524{,}288 = 2^{19}$, and the copies sum to $4 + 8 + \cdots + 2^{19} = 2^{20} - 4 = 1{,}048{,}572$. With the 10^6 direct writes the total is 2,048,572 writes, about 2.05 per append, inside the bound $1 + g/(g-1) = 3$ of Theorem 19.3(i). With CPython's ratio tending to 9/8 the bound is 10 per append; the constant is larger, the order the same. Additive growth by $c = 100$ slots instead costs about $n^2/(2c) = 10^{12}/200 = 5 \times 10^9$ copies by part (ii): three and a half orders of magnitude worse for the same million appends. For part (iii), draining a queue of $n = 10^6$ items with `pop(0)` moves $n(n-1)/2 \approx 5 \times 10^{11}$ pointers; `deque.popleft` (Fact 19.9) moves none.

Interview trap. "A list is a fine queue." Dequeuing with `pop(0)` costs $\Theta(n)$ by Theorem 19.3(iii), so a breadth-first search written with a list as its queue costs $O(V^2)$ instead of $O(V+E)$ (Proposition 19.20). The queue is a `collections.deque`; a list is a stack.

Fact 19.6 (dict and set layout (CPython 3.6 and later, Objects/dictobject.c and Objects/setobject.c))**.** A `dict` is an open-addressing hash table in the *compact* layout adopted in 3.6 (following Hettinger's 2012 proposal): a sparse *index* array of small integers (1, 2, 4 or 8 bytes per slot, whichever fits) points into a dense *entries* array of (hash, key, value) triples stored in insertion order; a lookup hashes the key, probes the index array along a hash-perturbed sequence, and compares the stored hash and then the key. The usable slots are two thirds of the index size, and when they are exhausted the table grows geometrically: on the installed 3.11, `sys.getsizeof` shows resizes after $5, 10, 21, 42, 85$ and 170 insertions, the index doubling each time. Insertion order was an implementation detail in 3.6 and is a *language guarantee* since 3.7. Keys must be *hashable*: the object must return a hash that never changes over its lifetime and that agrees on objects comparing equal (the `__hash__`/`__eq__` contract of the data model); `int`, `str`, `bytes`, tuples of hashables and `frozenset` qualify, while `list`, `dict` and `set` do not. Get, set and delete cost expected $O(1)$ and worst-case $O(n)$ under collisions (Fact 17.8); iteration costs $O(n)$ in insertion order. A `set` is its *own* open-addressing table of (hash, key) slots in `Objects/setobject.c`, with no values and no dense ordered entries array: it offers the same expected costs (add, membership and discard expected $O(1)$; union and intersection expected $O(|A| + |B|)$) but makes no ordering promise at all; `frozenset` is the hashable version. Theorem 19.3(i) covers the geometric growth of both tables: insertions are amortised $O(1)$ on top of being expected $O(1)$.

Fact 19.7 (Hash randomisation for str and bytes (PEP 456, 2013; Python/pyhash.c))**.** The hash of a `str` or `bytes` object is computed by a keyed function of the SipHash family with a per-process random key. Randomisation has been on by default since 3.3, as the response to the 2011 hash-collision denial-of-service attack on web frameworks (CVE-2012-1150); SipHash-2-4 was adopted as the default function in 3.4 and a faster variant later (the installed 3.11 reports `siphash13` through `sys.hash_info`); the environment variable `PYTHONHASHSEED` fixes the key. Integers are not randomised: `hash(n)` equals n for small non-negative n, and numeric hashes are reduced modulo $2^{61} - 1$ on 64-bit builds. This is the defence to the adversarial caveat of Fact 17.8: an attacker who cannot predict the key cannot precompute strings that share a probe chain and so cannot drive `dict` operations to $O(n)$. Two consequences are carried forward: a set of strings iterates in a run-dependent order (the smallest form of the determinism trap of §19.6), and `hash(word) % R` takes different values in different worker processes unless the seed is fixed or a stable hash such as `hashlib` is used (a shuffle trap, §19.8).

Example 19.8 (Run-dependent order, and the flood that randomisation prevents)**.** The one-line program

```
print(list({"apple", "banana", "cherry"}))
```

run three times on the installed 3.11 with `PYTHONHASHSEED` unset, printed `apple, banana, cherry`, then `banana, cherry, apple`, then `apple, banana, cherry`; with `PYTHONHASHSEED=0` it printed `banana, cherry, apple` on every run. A list or a dict built from the same three strings preserves insertion order on every run. For the flood: a dict of $n = 10^5$ string keys engineered to share one probe chain — possible only if the hash function is known, which randomisation prevents — costs about n probes per lookup, so 10^5 lookups cost about 10^{10} probes instead of the expected 10^5, a factor of 10^5; this is the 2011 attack that PEP 456 answers.

Interview trap. Three wrong sentences about hash tables. "A hash map is $O(1)$" drops both qualifiers; the answer is *expected* $O(1)$ under a randomised hash and *amortised* over geometric resizes, plus "keys must be hashable" — the same trap as the one following Fact 17.8. "Sets keep insertion order like dicts" is false: a set is a separate table with no ordering promise (Fact 19.6), and a set of strings iterates in a run-dependent order. "A dict keeps its keys sorted" confuses insertion order (guaranteed since 3.7) with sorted order, which costs $O(n \log n)$ to produce or a bisect-maintained list to keep (Fact 19.13).

Fact 19.9 (collections.deque and heapq (Modules/_collectionsmodule.c; Lib/heapq.py with the _heapq accelerator))**.** A `deque` is a doubly linked list of fixed-size blocks of 64 object pointers: `append`, `appendleft`, `pop` and `popleft` cost $O(1)$ worst case, indexing near the middle costs $O(n)$ (documented), and `maxlen` turns it into a ring buffer that discards from the far end. It is the standard-library queue for breadth-first search and for producer–consumer pipelines; `queue.Queue` is a deque plus a lock and condition variables. The `heapq` module stores a binary min-heap in a plain list: the children of index i sit at $2i+1$ and $2i+2$, the parent at $\lfloor (i-1)/2 \rfloor$, and the invariant is $h[i] \le h[2i+1]$ and $h[i] \le h[2i+2]$ wherever those indices exist. Because the list is a complete binary tree of height $\lfloor \log_2 n \rfloor$, `heappush` (append as the last leaf, then sift up) and `heappop` (move the last leaf to the root, then sift down) each walk one root-to-leaf path and cost $O(\log n)$; the minimum is $h[0]$ in $O(1)$. The invariant is local, so there is no search, no sorted iteration and no decrease-key. The documented lazy-deletion idiom ("Priority Queue Implementation Notes" in the library reference) replaces decrease-key: push a fresh (priority, item) entry and skip the stale one when it surfaces; the heap may then hold one entry per update rather than per item, which is what the bound of Proposition 19.22 below charges for.

Theorem 19.10 (heapify runs in linear time)**.** *Building a heap on n items in place by sifting down the internal nodes from index $\lfloor n/2 \rfloor - 1$ down to 0 performs at most $2n$ swaps. By contrast, n successive* `heappush` *calls perform $\Theta(n \log n)$ swaps in the worst case.*

Proof. We count swaps by the height of the node they start from. Define the *height* of a node as the number of edges on the longest downward path from it to a leaf, so leaves have height 0 and the root has height $H = \lfloor \log_2 n \rfloor$. A sift-down from a node of height h moves it down one level per swap, so it makes at most h swaps.

Step 1: at most $\lceil n/2^{h+1} \rceil$ nodes have height h. By induction on h. The internal nodes are the indices $0, \dots, \lfloor n/2 \rfloor - 1$, so there are $\lceil n/2 \rceil$ leaves, which is the case $h = 0$. For $h \geq 1$, delete all leaves: what remains is the array prefix of length $n' = \lfloor n/2 \rfloor$, itself a complete binary tree, in which every surviving node has height exactly one less than before (its longest downward path lost its final edge, and every new leaf was an old internal node). The nodes of height h in the original tree are the nodes of height $h - 1$ in the new one, so by the induction hypothesis there are at most $\lceil n'/2^h \rceil \leq \lceil n/2^{h+1} \rceil$ of them.

Step 2: summation. For $1 \leq h \leq H$ we have $n/2^{h+1} \geq 1/2$ because $n \geq 2^h$, and $\lceil x \rceil \leq 2x$ whenever $x \geq 1/2$; hence $\lceil n/2^{h+1} \rceil \leq n/2^h$. Nodes of height 0 contribute no swaps. The total is therefore at most

$$\sum_{h=1}^{H} h \cdot \frac{n}{2^h} \;\leq\; n \sum_{h \geq 1} \frac{h}{2^h} \;=\; 2n,$$

where the series is evaluated by differentiating the geometric series: $\sum_{h \geq 0} x^h = 1/(1-x)$ for $|x| < 1$ gives $\sum_{h \geq 1} h x^{h-1} = 1/(1-x)^2$, so $\sum_{h \geq 1} h x^h = x/(1-x)^2$, which equals $(1/2)/(1/4) = 2$ at $x = 1/2$.

The contrast. Pushing a strictly decreasing sequence makes every new element climb to the root: the i-th push sifts up through $\lfloor \log_2 i \rfloor$ levels, and the last $n/2$ pushes each climb at least $\lfloor \log_2 n \rfloor - 1$ levels, for $\Theta(n \log n)$ swaps in total. □

Example 19.11 (heapify against repeated pushes)**.** For $n = 15$, a full tree of height 3, the heights $0, 1, 2, 3$ hold $8, 4, 2, 1$ nodes. Sifting down the internal nodes costs at most $4 \times 1 + 2 \times 2 + 1 \times 3 = 11$ swaps, and $11 \leq 2n = 30$; running the procedure on the reversed sequence $15, 14, \dots, 1$ makes exactly 11 swaps, while 15 successive pushes of that sequence make 34. For $n = 2^{20}$ the bound is $2n \approx 2.1 \times 10^6$ swaps, against about $n \log_2 n \approx 2.1 \times 10^7$ for pushes.

Example 19.12 (Top-k with a size-k min-heap)**.** To find the k largest of n items, keep a min-heap of size k: for each item, compare it with the root; if it is larger, replace the root and sift down (a `heapreplace`). The invariant "the heap holds the k largest items of the prefix seen so far" holds after the first k items and is preserved by each step, because an item that fails to beat the heap's minimum cannot be among the k largest, and one that does beat it displaces exactly the item that is no longer among them. Cost: $O(n \log k)$ time and $O(k)$ memory, against $O(n \log n)$ and $O(n)$ for sorting. For $k = 100$ of $n = 10^7$ items: $n \log_2 k \approx 10^7 \times 6.64 = 6.6 \times 10^7$ heap steps and 100 slots, against $n \log_2 n \approx 2.3 \times 10^8$ and 10^7 slots. The same heap merges M sorted shard lists in $O(Mk \log M)$ (Proposition 19.103).

Interview trap. "`heapq` gives sorted iteration." The invariant is local: only the minimum is $O(1)$; finding any other element, the maximum included, costs $O(n)$, and the list underlying the heap is not sorted.

Fact 19.13 (Timsort and bisect (Peters, 2002; Objects/listsort.txt; Lib/bisect.py))**.** `list.sort` and `sorted` run Timsort, a merge sort over detected monotone runs: it is stable, uses $O(n \log n)$ comparisons in the worst case and $O(n)$ on input that is already sorted or reverse-sorted, needs $O(n)$ auxiliary memory in the worst case, and evaluates `key=` once per element. Version 3.11 changed the run-merging policy to powersort (Munro and Wild, 2018) without changing these bounds. `bisect_left` and `bisect_right` give $O(\log n)$ search on a sorted list (the loop of Proposition 19.18), but `insort` pays the $O(n)$ shift of Theorem 19.3(iii). The standard library has no balanced binary search tree, so the "ordered map" row of the classical cost table — insert, delete, search, predecessor, successor and range queries all in $O(\log n)$, the last three being what a dict cannot answer — has no stdlib implementation; a sorted list with `bisect` is the read-mostly substitute, and the third-party `sortedcontainers` package the general one. A proof of the Timsort bounds is beyond our scope.

Proposition 19.14 (Repeated string concatenation is quadratic; join is linear). *Strings are immutable, so* `s = s + t` *allocates a fresh object and copies both operands. Building a string from n pieces of length L by repeated concatenation copies $L\,n(n+1)/2 = \Theta(n^2L)$ characters;* `"".join(pieces)` *copies $\Theta(nL)$ characters.*

Proof. We count characters copied and sum an arithmetic series. After $i-1$ concatenations the accumulated string has length $(i-1)L$, and the i-th concatenation copies it together with the new piece: iL characters. Summing over $i = 1, \ldots, n$ gives $L(1+2+\cdots+n) = L\,n(n+1)/2$. The `join` method first sums the lengths of the pieces (n reads), allocates once, and copies each of the nL characters exactly once. □

Remark 19.15 (The refcount-one special case). CPython may extend the left operand in place when no other reference to it exists (a specialisation of `BINARY_OP` for in-place unicode addition in 3.11 and 3.12, in `Python/ceval.c` and its specialisation file), which can hide the quadratic cost in a micro-benchmark of `s += piece`. The optimisation disappears as soon as another reference to the string exists and is not a language guarantee, so the rule stands: build with `join`.

Example 19.16 (Joining a hundred thousand pieces). With $n = 10^5$ pieces of $L = 10$ characters, repeated concatenation copies $10 \times 10^5 \times (10^5+1)/2 \approx 5.0 \times 10^{10}$ characters by Proposition 19.14; `join` copies 10^6. The ratio is 50,000.

Table 3 collects the costs proved or recorded above; it is the table to reproduce, qualifiers included, when a question opens with "which container, and what does it cost?".

Reading the table by qualifier: the dict and set rows are the only *expected* costs, and they carry the two provisos that keys be hashable and that the hash be randomised (Fact 19.7); the `append` row and the insertion half of the dict row are *amortised*, with the geometric-resize argument of Theorem 19.3 behind both; every other row is a *worst-case* bound. A candidate who says "dict lookup is expected $O(1)$, list append is amortised $O(1)$, deque popleft is $O(1)$ worst case" has answered the qualifier question before it is asked.

Example 19.17 (Membership queries, revisited with a set). Example 17.7 priced $N = 1024$ elements and q queries for scanning against sorting. At $q = 100$ queries, scanning costs $100 \times 1024 =$ 102,400 comparisons and sort-plus-search costs $(1024 + 100) \times 10 = 11{,}240$ by (17.4). A `set` costs 1024 expected insertions plus 100 expected probes, about 1,124 operations — a further factor of ten — but loses all ordering information, requires hashable elements, and is expected rather than worst-case. This is the three-way decision of Section 17.2 with its qualifiers now attached.

Trading application. A tick-data pipeline is the same decision made several times over. Per-symbol state (last price, running volume) lives in a dict keyed by symbol string: expected $O(1)$ per update, insertion order irrelevant. A rolling window of the last w trades is a `deque` with `maxlen = w`: $O(1)$ per tick, and the oldest trade falls off the far end without a shift. A "top ten movers" panel is a size-10 min-heap over the symbol universe (Example 19.12), not a sort. A limit order book, which needs the best bid and best ask *and* the next levels in price order, is the one structure on the list for which the standard library has no direct answer: a sorted list with `bisect` works while the book is small, and a balanced tree or a price-indexed array is the production answer (Fact 19.13).

Interview trap. Two further wrong sentences. "My `s += piece` loop benchmarked fine": the refcount-one special case of Remark 19.15 hid the quadratic cost, and it is not guaranteed. "I will just use a list and `sort()` each time I need order": re-sorting after every insertion costs $O(n \log n)$ per insertion where `insort` costs $O(n)$ and a balanced tree $O(\log n)$; the sorted-list substitute is for read-mostly workloads.

Operation	Cost	Qualifier and reason
`L[i]`, `L[i] = x`	$O(1)$	worst case; contiguous pointer array (Fact 19.2)
`L.append(x)`	$O(1)$	amortised; geometric resize (Theorem 19.3(i))
`L.pop()`	$O(1)$	worst case; no shift
`L.insert(0, x)`, `L.pop(0)`	$\Theta(n)$	worst case; every pointer moves (Theorem 19.3(iii))
`x in L`	$O(n)$	worst case; linear scan
`L[a:b]` of length k	$O(k)$	worst case; pointers copied
`T[i]` (tuple)	$O(1)$	worst case; same array, no capacity
`D[k]`, `D[k] = v`, `del D[k]`	$O(1)$ / $O(n)$	expected / worst under collisions (Fact 19.6); insert also amortised over resizes
iterate `D`	$O(n)$	worst case; insertion order (3.7 guarantee)
`S.add(x)`, `x in S`	$O(1)$ / $O(n)$	expected / worst; no order (Fact 19.6)
`A | B`, `A & B` (sets)	$O(\lvert A\rvert + \lvert B\rvert)$	expected
`dq.append`, `dq.popleft` (both ends)	$O(1)$	worst case; linked blocks (Fact 19.9)
`dq[i]` (middle)	$O(n)$	worst case; block walk
`heappush`, `heappop`	$O(\log n)$	worst case; one root-to-leaf path
`heapify`	$O(n)$	worst case (Theorem 19.10)
`h[0]` (peek)	$O(1)$	worst case; invariant puts the minimum at the root
`sorted`, `L.sort()`	$O(n \log n)$	worst case; $O(n)$ on presorted input; stable (Fact 19.13)
`bisect_left`	$O(\log n)$	worst case (Proposition 19.18)
`insort`	$O(n)$	worst case; the shift dominates the search
`s += t` in a loop	$\Theta(n^2 L)$	total, not guaranteed away (Proposition 19.14)
`"".join(pieces)`	$\Theta(nL)$	total length copied once

Table 3: Costs of Python's containers as CPython implements them. Every row names its qualifier from Definition 19.1: "expected" rows average over the randomised hash and can degrade to $O(n)$ on colliding keys; "amortised" rows contain an occasional $O(n)$ resize paid for by the cheap operations around it; "worst case" rows hold for every call.

19.2 A Little Algorithms: Invariants, Costs, and the Recursion Limit

Long-form reasoning questions rarely ask for a named algorithm, but a strong answer recognises one of half a dozen patterns and states its cost unprompted, and the parallel questions that follow reuse them as bricks: breadth-first layers become schedule levels, prefix sums become scans, hash sets replace sorts. This subsection proves the invariant behind each pattern in the containers of §19.1 and records Python's recursion limit as the practical reason to write searches iteratively.

Setup. For binary search, $a[0], \dots, a[n-1]$ is a list sorted in non-decreasing order and t is a target value; indices are half-open, $[lo, hi)$ denoting $lo \le i < hi$. A graph $G = (V, E)$ is stored as adjacency lists (a dict mapping each vertex to a list of its out-neighbours); the letters V and E also denote the numbers of vertices and edges, and $\deg v$ is the out-degree of v. For Dijkstra's algorithm every edge (u, v) carries a weight $w(u, v) \ge 0$, and $\delta(v)$ denotes the true shortest-path distance from a fixed source s. Logarithms in operation counts are base 2. The cost model is that of §19.1.

Proposition 19.18 (Binary search: invariant, correctness and iteration count)**.** *Initialise $lo = 0$ and $hi = n$, and while $lo < hi$ set $mid = \lfloor (lo + hi)/2 \rfloor$ and move lo to $mid + 1$ if $a[mid] < t$, or hi*

to mid otherwise. Then:

(i) the invariant "every $a[i]$ with $i < lo$ is $< t$, and every $a[i]$ with $i \geq hi$ is $\geq t$" holds before every iteration;

(ii) the loop terminates after at most $\lfloor \log_2 n \rfloor + 1$ iterations with $lo = hi$ equal to the number of elements strictly below t (the `bisect_left` contract);

(iii) t is present if and only if $lo < n$ and $a[lo] = t$.

Proof. We proceed by induction on the number of iterations for (i) and by bounding the interval length for (ii). The invariant holds initially because no index satisfies $i < 0$ or $i \geq n$. Suppose it holds before an iteration. If $a[mid] < t$ then, by sortedness, every $a[i]$ with $i \leq mid$ is at most $a[mid] < t$, so setting $lo = mid + 1$ preserves the first half of the invariant and leaves the second untouched. Otherwise $a[mid] \geq t$, so every $a[i]$ with $i \geq mid$ is $\geq t$ and setting $hi = mid$ preserves the second half. This proves (i).

For (ii), write $\ell = hi - lo$ for the interval length. Since $lo \leq mid < hi$ whenever $\ell \geq 1$, the new length is either $hi - mid - 1 = \lceil \ell/2 \rceil - 1 \leq \lfloor \ell/2 \rfloor$ or $mid - lo = \lfloor \ell/2 \rfloor$; in both cases ℓ at least halves, so after k iterations $\ell \leq \lfloor n/2^k \rfloor$, which is 0 once $k = \lfloor \log_2 n \rfloor + 1$. On exit $lo = hi$, and the invariant then says that the elements below t are exactly $a[0], \ldots, a[lo-1]$, so lo counts them.

For (iii), if t is present, sortedness places all its copies immediately after the elements below it, so $a[lo] = t$; conversely $a[lo] = t$ exhibits it. $\square$

Python's arbitrary-precision integers make the overflow hazard of $lo + hi$ in C irrelevant. What the proof does settle is the loop test: with the invariant stated on half-open intervals the test is $lo < hi$, and the ± 1 decisions follow from the invariant, not from memory.

Example 19.19 (A binary-search trace)**.** Take $a = [2, 3, 5, 7, 11, 13, 17, 19]$ and $t = 13$. Iteration 1: $(lo, hi) = (0, 8)$, $mid = 4$, $a[4] = 11 < 13$, so $lo = 5$. Iteration 2: $(5, 8)$, $mid = 6$, $a[6] = 17 \geq 13$, so $hi = 6$. Iteration 3: $(5, 6)$, $mid = 5$, $a[5] = 13 \geq 13$, so $hi = 5$. The loop exits with $lo = 5$ and $a[5] = 13$: found in 3 iterations, within the bound $\lfloor \log_2 8 \rfloor + 1 = 4$. For $t = 4$ the trace is $(0, 8) \to (0, 4) \to (0, 2) \to (2, 2)$, exiting with $lo = 2$, the insertion point between 3 and 5; `bisect_left` returns 5 and 2 respectively. At $n = 10^6$ each query costs at most 20 iterations, so 10^6 queries cost about 2×10^7 comparisons against 10^{12} for repeated scanning.

Proposition 19.20 (BFS, DFS and Kahn's order cost $O(V + E)$; BFS layers are shortest unweighted distances)**.** *Let a search mark each vertex when it is first* discovered *(on insertion into the container, not on removal) and scan each vertex's adjacency list once, when the vertex is removed from the container.*

(i) Breadth-first search with a deque as a FIFO queue, depth-first search with a list as a stack, and Kahn's topological ordering with an in-degree counter and a container of in-degree-zero vertices all run in $O(V + E)$.

(ii) In breadth-first search from s, every vertex v is discovered from a vertex at distance $\delta(v) - 1$, where δ is the unweighted shortest-path distance; hence the layer in which v is discovered is $\delta(v)$ and the discovery pointers form a shortest-path tree.

Proof. We count container operations for (i) and use induction on the distance for (ii).

Part (i). Marking on discovery ensures that each vertex enters the container at most once, hence is removed at most once, hence has its adjacency list scanned at most once. The total work is $\sum_v (1 + \deg v) = V + E$ container operations and edge inspections, each $O(1)$ for a deque, a list used as a stack, or a decrement of an in-degree counter (Table 3). Kahn's algorithm additionally initialises the in-degrees in one pass over the edges, $O(E)$.

Part (ii). Call a vertex *pending* if it has been discovered but not yet dequeued. The claim, proved by induction on d, is: at the moment the first vertex at distance d is dequeued, every vertex at distance $\leq d$ has been discovered in the layer equal to its distance, every pending vertex has distance d or $d+1$, and the pending vertices sit in the queue in non-decreasing order of distance. For $d = 0$ the queue holds only s. Assume the claim for d and follow the phase during which the distance-d vertices are dequeued. Each of them scans its out-neighbours: a neighbour that is already marked is ignored; an unmarked neighbour has distance exactly $d+1$ (it cannot be closer, since everything closer is already marked, and it cannot be farther, since it is adjacent to a vertex at distance d), and it is discovered now and appended behind the pending distance-$(d+1)$ vertices. Every vertex at distance $d+1$ is adjacent to some vertex at distance d, which is dequeued during this phase, so it is discovered by the end of the phase. When the last distance-d vertex has been dequeued, the pending vertices are therefore exactly the distance-$(d+1)$ vertices, all discovered in layer $d+1$, which is the claim for $d+1$. Since v is discovered while a vertex at distance $\delta(v) - 1$ is being scanned, its discovery pointer leads to such a vertex, and following the pointers back to s traces a path of length $\delta(v)$. □

The only difference among the three searches is the container: FIFO order gives layers, LIFO order gives depth-first branches, and "in-degree just reached zero" gives a topological order. Kahn's in-degree-zero waves are exactly the schedule levels of Definition 19.44 in §19.4: the computational graph of Figure 15 is such a DAG, and its topological order is the order of the forward pass.

Example 19.21 (Kahn's waves on a build graph)**.** Consider the build DAG with edges `fetch` $\to$ `compile`, `fetch` $\to$ `lint`, `compile` $\to$ `test`, `compile` $\to$ `package`, `lint` $\to$ `package`, `test` $\to$ `deploy`, `package` $\to$ `deploy`, and an isolated task `docs`: $V = 7$ and $E = 7$. The in-degrees are $0, 1, 1, 1, 2, 2, 0$ for `fetch`, `compile`, `lint`, `test`, `package`, `deploy`, `docs`. Kahn's waves are $\{$`fetch`, `docs`$\}$, then $\{$`compile`, `lint`$\}$, then $\{$`test`, `package`$\}$ (`package`'s in-degree reaches zero only after both `compile` and `lint`), then $\{$`deploy`$\}$: 7 unit tasks in 4 waves. Each vertex is enqueued once (7 enqueues) and each edge decrements one counter (7 decrements), confirming $V + E = 14$ operations. In the language of Definition 19.44, the work is 7, the span 4 and the parallelism $7/4 = 1.75$.

Proposition 19.22 (Dijkstra with heapq and lazy deletion runs in $O((V + E) \log V)$ and is correct for non-negative weights)**.** *Maintain $dist[v]$, initially ∞ except $dist[s] = 0$, and a min-heap of pairs (d, v) initially holding $(0, s)$. Repeatedly pop the minimum (d, u); if $d > dist[u]$ the entry is* stale *and is skipped; otherwise u is* settled*: for each edge (u, v) with $dist[u] + w(u, v) < dist[v]$, set $dist[v]$ to that value and push $(dist[v], v)$. Then every vertex is settled at most once with $dist[u] = \delta(u)$, and the algorithm performs at most $E + 1$ pushes and pops on a heap of size at most $E + 1$, for a total cost of $O((V + E) \log V)$.*

Proof. We bound the cost by counting relaxations, and prove correctness by induction on the order in which vertices are settled.

Cost. Entries for the same vertex are pushed with strictly decreasing keys, because a push happens only when $dist$ strictly decreases; so at any time at most one entry per vertex — the last one pushed — is non-stale. Once u is settled, $dist[u] = \delta(u)$ by the correctness argument below, and since $dist$ never drops below δ it never decreases again, so no further entry for u is ever pushed. Hence each vertex is settled at most once, each edge is relaxed at most once (when its tail is settled), and each successful relaxation pushes one entry: at most E pushes beyond the initial one. Pops never exceed pushes, and the heap never exceeds $E + 1$ entries, so each heap operation costs $O(\log(E + 1)) = O(\log V)$ because $E \leq V^2$. Adding the $O(V + E)$ adjacency scans gives $O((V + E) \log V)$.

Correctness. Throughout, $dist[v]$ is either ∞ or the length of some actual path to v, so $dist[v] \geq \delta(v)$. The claim, by induction on settling order, is that a vertex is settled only with $dist[u] = \delta(u)$. The source is settled first with $0 = \delta(s)$. Suppose every vertex settled so far has the correct distance, and let (d, u) be popped and not stale, so $d = dist[u]$; suppose for contradiction that $\delta(u) < d$. Fix a shortest path $s = v_0, v_1, \ldots, v_m = u$ and let $y = v_j$ be its first unsettled vertex ($j \geq 1$, since

s is settled). Its predecessor v_{j-1} is settled with $dist[v_{j-1}] = \delta(v_{j-1})$, and when it was settled the edge (v_{j-1}, y) was relaxed, so at that moment $dist[y] \le \delta(v_{j-1}) + w(v_{j-1}, y) = \delta(y)$, the last equality because a prefix of a shortest path is a shortest path; hence $dist[y] = \delta(y)$ from then on, and an entry $(\delta(y), y)$ was pushed. Now

$$\delta(y) \;\le\; \delta(u) \;<\; d,$$

the first inequality because the path from y to u has non-negative weight — *this is the one place the hypothesis* $w \ge 0$ *is used.* If $y \ne u$, the entry $(\delta(y), y)$ is still in the heap (y is unsettled and its entry is not stale), contradicting the choice of (d, u) as the minimum. If $y = u$, then $dist[u] = \delta(u) < d$, so the popped entry was stale, contradicting the assumption. Either way $\delta(u) = d$. □

Example 19.23 (Heap Dijkstra against the array version)**.** On a sparse graph with $V = 10^4$ and $E = 10^5$, the heap version performs at most 10^5 pushes and pops at $\lceil \log_2 10^5 \rceil = 17$ levels each, about 1.7×10^6 heap steps; the array version that scans all unsettled vertices for the minimum costs $V^2 = 10^8$. On a dense graph with $E = V^2/2 = 5 \times 10^7$ the heap version costs about $5 \times 10^7 \times 26 = 1.3 \times 10^9$ and the array scan wins at 10^8. The crossover is the house pattern of Remark 17.9 once more: $E \log V$ against V^2.

Proposition 19.24 (Linear passes: two pointers, sliding windows, and merging)**.** *Let a scan maintain two indices* $lo \le hi$ *in* $\{0, \ldots, n\}$ *such that neither index ever decreases, each iteration advances at least one of them by at least one, and each iteration does* $O(1)$ *work. Then the scan performs at most* $2n$ *iterations and costs* $O(n)$*.*

Proof. We use the potential $lo + hi$. It is a non-negative integer, at most $2n$, and each iteration increases it by at least one; hence there are at most $2n$ iterations, each $O(1)$. □

Instances: the longest window with sum at most S in an array of non-negative numbers (advance hi, and advance lo while the window sum exceeds S); pair-sum on a sorted array (pointers at both ends, moving inward); and merging two sorted lists of lengths n and m in $O(n + m)$, which is the instance the k-way shard merge of §19.8 builds on. The companion pattern "sort first, then one linear pass" (duplicates, interval merging, grouping) costs $O(n \log n)$ and competes with hash-set membership at $O(n)$ expected; Proposition 17.6 gives the crossover shape and Section 17.2 the cases in which order is worth paying for.

Example 19.25 (A sliding-window trace)**.** Take $[2, 1, 5, 1, 3, 2]$ and $S = 7$. Advancing hi from 0: sums 2, 3, 8 (exceeds 7: drop 2, $lo = 1$, sum 6), 7, 10 (drop 1 then 5, $lo = 3$, sum 4), 6. The right pointer visits 6 positions and the left pointer moves 3 times: 9 iterations, within $2n = 12$. The longest windows have length 3: $[1, 5, 1]$ with sum 7 and $[1, 3, 2]$ with sum 6.

Fact 19.26 (CPython's recursion limit (sys.getrecursionlimit, default 1000; library reference))**.** CPython raises `RecursionError` when the depth of Python frames exceeds the recursion limit, whose default is 1000; the limit is adjustable, and 3.12 bounds C-stack recursion separately. A recursive depth-first search on a path of a few thousand vertices therefore fails, and raising the limit is fragile. The fix is the iterative version with an explicit list as the stack — the call stack made visible — pushing children in reverse order to reproduce the recursive visiting order, and keeping (node, state) pairs when a post-order action is needed. The same rewrite applies to tree traversals and to divide-and-conquer on long inputs.

Fact 19.27 (Master theorem (Bentley, Haken and Saxe, 1980; CLRS form))**.** Let $T(n) = a\,T(n/b) + \Theta(n^d)$ with $a \ge 1$, $b > 1$ and $d \ge 0$. Then $T(n) = \Theta(n^d)$ if $d > \log_b a$; $T(n) = \Theta(n^d \log n)$ if $d = \log_b a$; and $T(n) = \Theta(n^{\log_b a})$ if $d < \log_b a$. Instances: binary search, $(a, b, d) = (1, 2, 0)$, gives $\Theta(\log n)$; merge sort, $(2, 2, 1)$, gives $\Theta(n \log n)$; Karatsuba multiplication, $(3, 2, 1)$, gives $\Theta(n^{\log_2 3}) = \Theta(n^{1.585})$; Strassen's algorithm, $(7, 2, 2)$, gives $\Theta(n^{2.807})$ against the n^3 of Proposition 17.1. The hypotheses — equal-size subproblems and a polynomial driving term — are part of any quotation of the theorem. A proof is beyond our scope.

Example 19.28 (Depth, and two recurrences)**.** A recursive depth-first search on a chain of 5,000 vertices raises `RecursionError` at depth 1000; the explicit-stack version pushes and pops 5,000 entries and scans 4,999 edges, $V + E = 9{,}999$ units of work, and finishes. Karatsuba on $n = 2^{10}$ digits performs $3^{10} = 59{,}049$ single-digit multiplications against $2^{20} = 1{,}048{,}576$ for the schoolbook method. Merge sort on $n = 2^{20}$ items costs 20 levels of 2^{20} work each, about 2.1×10^7 comparisons, consistent with the worst case of Fact 19.13.

Interview trap. Five wrong moves. An off-by-one in binary search: write the invariant first (Proposition 19.18(i)); it decides the loop test and the ± 1. "BFS with `list.pop(0)`": $O(n)$ per dequeue makes the search $O(V^2)$; the queue is a deque (Fact 19.9). "Dijkstra needs decrease-key": lazy deletion with stale-entry skipping suffices; and "Dijkstra works with negative edges" fails at the one inequality in the proof of Proposition 19.22 that uses $w \geq 0$. "Recursion is fine in Python": the default depth is 1000 (Fact 19.26); say "iterative with an explicit stack" unprompted. Quoting the master theorem for unequal splits or a non-polynomial driving term: the hypotheses are part of the answer.

19.3 Wall-Clock Time versus CPU Time

"This job took 100 seconds" is ambiguous until the clock is named, and the ratio of the two clocks is the fastest single diagnostic of where a program spends its life; research-lab on-site rounds ask exactly this ("100 seconds of wall time, 40 seconds of CPU: what happened?"). This subsection defines the clocks, proves the accounting inequality that relates them and turns it into a diagnosis table, separates latency from throughput, and ends with how to measure without fooling oneself, the asynchronous GPU included. It is the first of the two subsections around which the section is built.

Setup. A process runs on a machine with $C \geq 1$ cores over a wall-clock interval of length $W > 0$ seconds; the operating system schedules the process's threads onto cores. For each instant t in the interval, $k(t) \in \{0, 1, \ldots, C\}$ denotes the number of cores executing a thread of this process at t; the function k is piecewise constant. Let U and S be the user and system CPU time the process accumulates over the interval (defined below), $r = (U + S)/W$ the *CPU-to-wall ratio*, and $\rho = (U + S)/(CW)$ the *utilisation*; in this subsection ρ is a utilisation, not a correlation, and S is system time, not a P&L. Let m denote the largest number of the process's threads that are ever simultaneously on cores. Time is in seconds.

Definition 19.29 (Wall-clock time, CPU time, latency and throughput)**.** *Wall-clock time* (elapsed time, real time) is the length W of the interval between two readings of a clock that advances with physical time whatever the process does. *CPU time* is the integral

$$U + S \;=\; \int_0^W k(t)\,dt,$$

the number of core-seconds consumed by all the process's threads; it splits into *user time* U, spent executing the program's own code, and *system time* S, spent in the kernel on the program's behalf (system calls, page faults, I/O setup). Per-thread CPU time is the same integral restricted to one thread. *Waiting* — blocked on I/O, on a lock, on a network reply, on a GPU kernel, or in `time.sleep` — accrues no CPU time at all. *Latency* is the wall time from the submission of one item of work to its completion; *throughput* is the number of items completed per unit of wall time in steady state. A batch of size B processed with a fixed overhead c and a per-item cost τ has latency $c + B\tau$ and throughput $B/(c + B\tau)$, which tends to $1/\tau$ as B grows while the latency grows linearly.

Fact 19.30 (Python's clocks (PEP 418, 2012; the time module, 3.3 and later) and the shell time command)**.** `time.time()` returns seconds since the epoch from the system clock, which the operating system may step forwards or backwards (NTP corrections, manual changes), so its differences are not reliable intervals. `time.perf_counter()` is monotonic with the highest available resolution and is the clock for measuring intervals (the `timeit` module uses it); `time.monotonic()` is monotonic with lower resolution. `time.process_time()` returns user plus system CPU time of the current process summed over all its threads; `time.thread_time()` (3.7) returns the calling thread's CPU time. The POSIX `time` command prints `real` (wall), `user` and `sys`, the last two summed over the process and the children it waited for.

Theorem 19.31 (The accounting inequality and the three regimes)**.** *Under the setup above:*

(i) $0 \le U + S \le \min(C, m)\, W$*; hence* $\rho \in [0, 1]$ *and* $W \ge (U + S)/C$*.*

(ii) If $r < 1$*, the process was on no core at all for at least a fraction* $1 - r$ *of the interval: it waited. If the process has a single thread, then* $r \le 1$*, and* $r = 1$ *means that thread occupied a core throughout. If* $r = k > 1$*, then an average of* k *cores were busy on this process over the interval, and necessarily* $k \le \min(C, m)$*.*

Proof. We integrate the busy-core count. At every instant a core executes at most one thread, and a thread runs on at most one core, so $0 \le k(t) \le \min(C, m)$ pointwise; integrating over $[0, W]$ gives (i), and dividing by CW and by C gives the two consequences.

For (ii), let $I = \{t : k(t) \ge 1\}$ be the set of instants at which the process is on some core. Since $k \ge 1$ on I and $k \ge 0$ elsewhere, the measure of I is at most $\int_0^W k(t)\, dt = rW$, so the process is on no core for a set of instants of measure at least $W - rW = (1 - r)W$. For a single thread, $k(t) \le 1$ pointwise gives $U + S \le W$, and $U + S = W$ forces $k = 1$ almost everywhere. The last statement restates $r = (1/W) \int_0^W k(t)\, dt$ as the time average of k, bounded by (i). □

This is the inequality candidates get backwards: CPU time exceeds wall time exactly when more than one core is busy, and the bound is $U + S \le C\, W$, not $U + S \le W$.

Example 19.32 (Diagnosing 100 seconds of wall time and 40 of CPU)**.** With $W = 100$ s and $U + S = 40$ s, $r = 0.4$, so by Theorem 19.31(ii) the process was on no core for at least 60 s: it waited. It cannot be a single-threaded CPU-bound program (that gives r near 1), and more CPU cores cannot help the 60 s. Parallelising the 40 s of compute perfectly over 4 cores leaves $10 + 60 = 70$ s, a speedup of $100/70 = 1.43$, which is Amdahl's law with $p = 0.4$ and $P = 4$ (Theorem 19.41: $1/(0.6 + 0.4/4) = 1.43$, ceiling $1/0.6 = 1.67$). The two follow-up diagnoses are: if the wait is I/O, overlap it (threads, `asyncio`, prefetching, §19.5) and the 60 s shrinks toward the compute; if the wait is a GPU, the CPU is not the bottleneck and the number to measure is device time (Fact 19.39).

Example 19.33 (Utilisation of eight workers)**.** Eight worker processes report a total of $U + S = 700$ core-seconds over $W = 100$ s: $r = 7$, $\rho = 700/800 = 87.5\%$, and 100 core-seconds were idle; the accounting inequality holds as $700 \le 8 \times 100$. Two shapes explain the idle 12.5%: a straggler (seven workers finish at 87.5 s and one at 100 s) or every worker idling 12.5% of the time at a per-step barrier. Both are the span term of Theorem 19.45, and the fixes differ — rebalance the tasks in the first case, shorten the synchronised section in the second.

Example 19.34 (CPU time exceeding wall time)**.** A numpy product of two 8000×8000 double-precision matrices on 8 BLAS threads reports `real` 1.9 s, `user` 14.6 s, `sys` 0.2 s: $r = 14.8/1.9 = 7.8$ cores on average. By Proposition 17.1 the work is $2 \times 8000^3 = 1.02 \times 10^{12}$ flops, so the aggregate rate is $1.02 \times 10^{12}/1.9 = 540$ GFLOP/s, about 69 GFLOP/s per busy core — plausible for vectorised double precision. CPU time above wall time is the signature of parallelism, not an accounting error.

Remark 19.35 (Reading r: the diagnosis table). The ratio r of Theorem 19.31 is read as follows.

r	Reading
near 0	the process slept or waited nearly the whole interval: I/O, a lock, a network reply, a synchronising GPU wait, `sleep`
$0 < r < 1$	one thread that waits part of the time; the waiting fraction is at least $1 - r$
near 1	one core saturated: the signature of CPU-bound pure Python under the GIL (Fact 19.53)
$k > 1$	k cores busy on average: BLAS threads, multiprocessing, or GIL-releasing extensions
large system share	system-call or page-fault churn: many small reads, allocation, pickling
near C	the machine is saturated; adding workers cannot help (Theorem 19.41)

GPU busy time is not CPU time: a GPU-bound training loop shows r well below 1 and looks like waiting from the CPU's viewpoint, which is correct, and the device must be profiled separately (Fact 19.39).

Remark 19.36 (Pipelines, bounded queues and backpressure). For a chain of stages with steady-state rates $r_1, \ldots, r_m$ items per second, every item passes through every stage, so the pipeline's throughput cannot exceed $\min_i r_i$; and the slowest stage runs continuously once its input queue is non-empty, so the bound is attained. A *bounded* queue in front of the slowest stage fills, and the faster stages block when they find it full — *backpressure* — so they run at the slow stage's rate and memory stays bounded; an unbounded queue grows without limit. This is the shape of a CPU data loader feeding a GPU and of the `queue.Queue` and `asyncio.Queue` designs of §19.5 and §19.6.

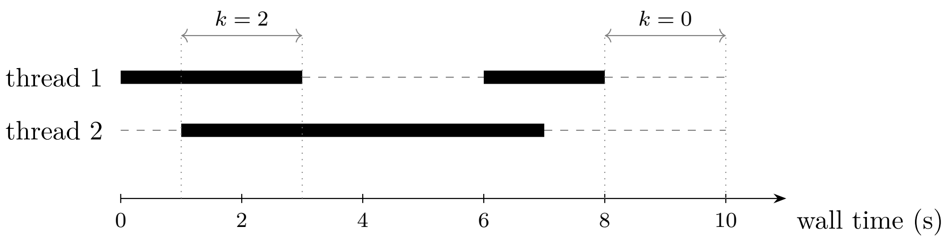
Example 19.37 (Latency against throughput under batching). A batched model call has launch overhead $c = 5$ ms and per-item cost $\tau = 100$ ns. A batch of $B = 10^5$ items has latency $5 + 10 = 15$ ms and throughput $10^5/0.015 = 6.7 \times 10^6$ items per second; a batch of $B = 10^3$ has latency 5.1 ms and throughput 2.0×10^5 per second. Batching buys $34\times$ the throughput for $2.9\times$ the latency. The point at which overhead equals compute, $B^* = c/\tau = 50{,}000$ items, is the house crossover of Remark 17.9 solved for the batch size.

Procedure 19.38 (Measuring time correctly).
1. Read `time.perf_counter()` for wall time and `time.process_time()` for CPU time immediately around the region of interest, never `time.time()`; record both and their ratio r.
2. Warm up: discard the first call, whose cost includes imports, allocator growth, cache misses, kernel autotuning and CUDA context creation.
3. Repeat, and report the minimum or the median together with the spread, never a single run.
4. Synchronise the accelerator before reading the clock (Fact 19.39).
5. Pin the environment and state it: BLAS thread counts (`OMP_NUM_THREADS`, `MKL_NUM_THREADS`), other tenants on the machine, CPU frequency scaling, and the core count, because $k > 1$ changes the reading.
6. Measure the quantity the question asks for — latency per item or throughput — and say which; convert a measured time into a rate with a flop count (Proposition 17.1) as a plausibility check.

Fact 19.39 (Asynchronous GPU execution requires synchronisation before timing (CUDA C Programming Guide; PyTorch CUDA semantics documentation)). A kernel launch returns control to the host before the kernel has executed — the asynchrony named in Remark 9.14 — so a `perf_counter` difference taken around an unsynchronised launch measures the launch (microseconds) and the device time appears later, at the next synchronising operation: `.item()`, a copy to host, `torch.cuda.synchronize()`, or CUDA events recorded on the stream. Unsynchronised timings therefore underestimate GPU work by orders of magnitude. CPU time is unaffected either way, because the CPU idled while the device worked, and $r < 1$ in the sense of Theorem 19.31(ii) is the signature of a GPU-bound program. The 10–100× of Fact 9.15 is measurable only with the synchronisation in place.

Example 19.40 (The same kernel timed three ways)**.** A GPU kernel is timed by `perf_counter` around the bare launch: 0.05 ms. Timed by `perf_counter` after `torch.cuda.synchronize()`: 12.3 ms. Timed by `process_time` in either arrangement: about 0.05 ms, because the CPU idled while the device worked. The first number is the trap; the third explains why a GPU-bound loop reads as "waiting" in Remark 19.35, and the honest wall-clock figure is the second.

Figure 29 draws the accounting inequality for a two-thread process: the bars are the busy intervals, and CPU time is their total length, which exceeds the wall time wherever the bars overlap.



CPU time $= 5 + 6 = 11$ s $>$ wall time $= 10$ s; $r = 1.1$, $\rho = 11/(2 \times 10) = 0.55$ on $C = 2$ cores

Figure 29: The accounting inequality on a timeline. Black bars are intervals during which a thread is on a core; dashed gray segments are waiting. Thread 1 is busy on $[0, 3]$ and $[6, 8]$, thread 2 on $[1, 7]$, so the process's CPU time is $5 + 6 = 11$ s over a wall-clock interval of 10 s: $r = 1.1 > 1$ because two cores are busy on $[1, 3]$ and $[6, 7]$ ($k = 2$), while on $[8, 10]$ the process is entirely waiting ($k = 0$). By Theorem 19.31(i), $11 \le \min(C, m) \times 10 = 20$.

Trading application. A daily research job that reloads a year of tick data, computes features and fits a model reads `real` 100 s, `user` 38 s, `sys` 2 s. By Example 19.32 the job is waiting for most of its life — almost certainly on the data reads — and the lever is to overlap or cache the loading (columnar files, a prefetching reader, a process that keeps the frame in memory), not to optimise the model fit. The opposite reading, `user` far above `real`, means the BLAS underneath the fit is already using every core, and the next lever is algorithmic (a smaller cross-validation grid, a cheaper solver), not more threads.

Interview trap. Five wrong sentences about time. "CPU time cannot exceed wall time": it does whenever more than one core is busy; the inequality is $U + S \le C\,W$, and $U + S > W$ is the signature of parallelism (Theorem 19.31). "Wall time measures my algorithm": it includes waiting on I/O, locks, other tenants and unsynchronised device work; the ratio r says which (Example 19.32). "`time.time()` is fine for benchmarks": the epoch clock can step; `perf_counter` is the interval clock (Fact 19.30). "The GPU op took 50 microseconds": an unsynchronised launch timing (Fact 19.39). Reading `real` $\gg$ `user` + `sys` as "slow code" (it is waiting) or `real` $\ll$ `user` + `sys` as "an error" (it is parallelism): only `user` + `sys` = `real` with one thread means CPU-bound and single-threaded.

19.4 Speedup Laws: Amdahl and the Work–Span Model

"How much faster with P machines?" has two textbook answers, both proved here: Amdahl's law for a program with a serial fraction, and the greedy-scheduler bound of the work–span model for a computation with a dependency graph. The diagnostics of §19.3 supply their inputs — the serial fraction is the part of the run where $r = 1$, and the span is what a straggler or a barrier exposes.

Setup. Let $T_1 > 0$ be the running time of a program on one processor. A fraction $p \in [0, 1)$ of T_1 parallelises perfectly (it divides exactly by the number of processors) and the remaining fraction $1 - p$ does not parallelise at all; $P \geq 1$ is the number of processors and T_P the running time on P of them; the speedup is $S(P) = T_1/T_P$. In this subsection p is the parallel fraction and P the processor count, not the step probability and probability operator of the notation table, and S is a speedup, not a P&L or system time. For the work–span model, a computation is a directed acyclic graph of unit-time *tasks* (a task of duration d is a chain of d unit tasks), with an edge $u \to v$ meaning that v may start only after u has finished. A *schedule* assigns each task to a step, respecting the edges, with at most P tasks per step; T_P is its number of steps. A task is *ready* at a step if all its predecessors have executed in earlier steps. A *greedy* schedule executes P ready tasks whenever at least P are ready, and all ready tasks otherwise; a step is *complete* if P tasks execute in it and *incomplete* otherwise.

Theorem 19.41 (Amdahl's law (Amdahl, 1967) and its inversions)**.** *Under the setup above:*

(i) $T_P = (1 - p)\, T_1 + p\, T_1/P$, *so that*

$$S(P) \;=\; \frac{1}{(1-p) + p/P}. \tag{19.1}$$

(ii) S *is increasing in* P, $S(P) < 1/(1-p)$ *for every* P, *and* $S(P) \to 1/(1-p)$ *as* $P \to \infty$.

(iii) Inversions: a target speedup s *with* $1 \leq s < 1/(1-p)$ *requires* $P = p/(1/s - (1-p))$ *processors;* $S(P)$ *equals half its ceiling exactly at* $P = p/(1-p)$*; and a measured speedup* S *on* $P > 1$ *processors implies the serial fraction*

$$1 - p \;=\; \frac{P/S - 1}{P - 1} \qquad \text{(Karp and Flatt, 1990)},$$

so a scaling measurement becomes a diagnosis, and a serial fraction that grows with P *signals overhead rather than a fixed serial part.*

Proof. We compute directly from the hypotheses. Split the serial run into its serial part $(1-p)T_1$ and its parallel part pT_1; by hypothesis the parallel part takes pT_1/P on P processors and the serial part is unchanged, which gives (i), and $S(P) = T_1/T_P$ is (19.1). For (ii), the term p/P is positive, decreases in P, and tends to 0, so the denominator of (19.1) decreases toward $1-p$ from above; hence S increases, stays below $1/(1-p)$, and converges to it. For (iii), solve $1/s = (1-p) + p/P$ for P, which is possible exactly when $1/s > 1-p$; set $S(P) = 1/(2(1-p))$, that is $(1-p)+p/P = 2(1-p)$, which gives $P = p/(1-p)$; and rewrite $1/S = (1-p) + p/P = (1-p)(1 - 1/P) + 1/P$, so that $1 - p = (1/S - 1/P)/(1 - 1/P) = (P/S - 1)/(P - 1)$. □

Example 19.42 (Amdahl arithmetic)**.** With $p = 0.95$: $S(8) = 1/(0.05 + 0.11875) = 5.93$; $S(32) = 1/(0.05 + 0.0297) = 1/0.0797 = 12.5$; $S(64) = 15.4$; $S(128) = 17.4$; the ceiling is 20. Reaching $15\times$ needs $P = 0.95/(1/15 - 0.05) = 0.95/0.01667 = 57$ processors, and half the ceiling ($10\times$) is reached at $P = 0.95/0.05 = 19$. With $p = 0.9$ and $P = 100$, $S = 1/(0.1 + 0.009) = 9.17$ against a ceiling of 10. The inversion: a measured $12.5\times$ at $P = 32$ implies $1-p = (32/12.5-1)/31 = 1.56/31 = 0.050$, and then $S(64) = 1/(0.050 + 0.950/64) = 15.4$, not 25. The $p = 0.4$ of Example 19.32 has ceiling $1/0.6 = 1.67$.

Figure 30 plots (19.1) for three values of p: the curves bend away from the ideal line almost immediately and flatten under their ceilings.

Remark 19.43 (Gustafson's scaled speedup (Gustafson, 1988))*.* Hold the *parallel* run at unit time with serial part s' and parallel part $1 - s'$. The same work on one processor would take $s' + (1-s')P$, so the *scaled speedup* is $s' + (1-s')P$, linear in P. Amdahl's law is fixed-size scaling and Gustafson's is fixed-time scaling: the same identity read from different baselines. The practical content is that the serial fraction of a training run — data loading, logging, a single-device optimiser step — is an engineering choice, not physics, and growing the problem with the machine keeps it small.

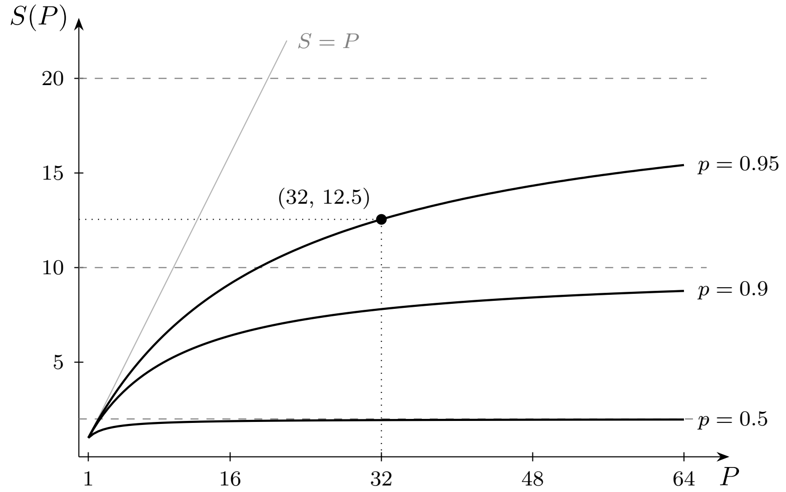


Figure 30: Amdahl speedup (19.1) against the processor count for $p = 0.5$, 0.9 and 0.95, with the ceilings $1/(1-p) = 2$, 10 and 20 as dashed lines and the ideal $S = P$ in light gray. The marked point is $S(32) = 12.5$ at $p = 0.95$ (Example 19.42); doubling to 64 processors reaches only 15.4.

Definition 19.44 (Computation DAG, work, span, parallelism, greedy schedule)**.** For a computation DAG of unit-time tasks, the *work* T_1 is the number of tasks, the *span* T_∞ is the number of tasks on a longest directed path (the *critical path*), and the *parallelism* is T_1/T_∞, the largest number of processors that can be kept fully busy on average. A schedule, its length T_P, ready tasks, greedy schedules and complete and incomplete steps are as in the Setup. Figure 31 shows a DAG with $T_1 = 8$ and $T_\infty = 5$. The computational graph of Figure 15 is a DAG of this kind, and the in-degree-zero waves of Proposition 19.20 are its levels.

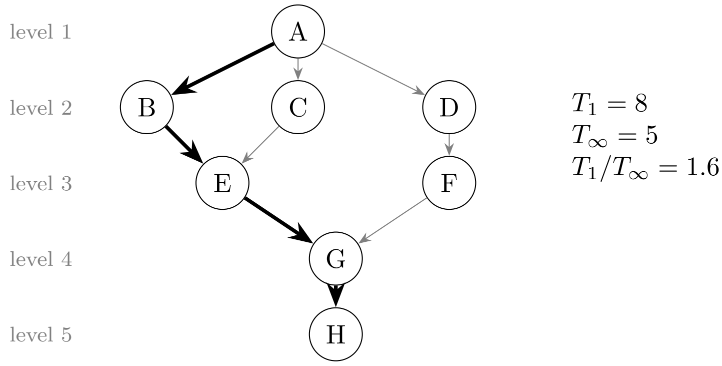


Figure 31: The computation DAG of Example 19.46: eight unit tasks, with the critical path A–B–E–G–H (thick) of length $T_\infty = 5$; the remaining edges are gray. Levels are the earliest steps at which tasks can run, which are Kahn's waves of Proposition 19.20. With three processors the greedy schedule finishes in 5 steps, so the span law of Theorem 19.45(i) is tight and a fourth processor buys nothing.

Theorem 19.45 (Work and span laws and the greedy-scheduler bound (Graham, 1969; Brent, 1974))**.** *For a computation DAG with work T_1 and span T_∞ scheduled on P processors:*

(i) every schedule satisfies $T_P \geq T_1/P$ (the work law) and $T_P \geq T_\infty$ (the span law);

(ii) every greedy schedule satisfies $T_P \leq T_1/P + T_\infty$;

(iii) consequently a greedy schedule is within a factor 2 of optimal, $T_P \leq 2\max(T_1/P, T_\infty)$, and if $P \leq \varepsilon T_1/T_\infty$ for some $\varepsilon > 0$ then its speedup satisfies $T_1/T_P \geq P/(1+\varepsilon)$: near-linear speedup whenever P is well below the parallelism.

Proof. We prove (i) by counting, and (ii) by classifying the steps of a greedy schedule as complete or incomplete.

Part (i). Each step executes at most P tasks, so $T_1 \le P\,T_P$. The tasks of a longest path must execute in distinct steps in path order, because each is a predecessor of the next, so $T_P \ge T_\infty$.

Part (ii). A complete step executes P tasks, so there are at most $\lfloor T_1/P \rfloor$ complete steps. Consider an incomplete step and the DAG R of tasks not yet executed before it. Fewer than P tasks are ready, so the greedy rule executes *every* ready task. Take any longest path in R; its first task has no unexecuted predecessor (otherwise prepending that predecessor would give a longer path in R), so it is ready and executes in this step. A step executes only tasks with no unexecuted predecessors, and any path in R contains at most one such task, namely its first; so every path in R shrinks by at most one during the step, every longest path shrinks by exactly one, and the span of the remaining DAG decreases by exactly one at every incomplete step. The span starts at T_∞ and cannot go below 0, so there are at most T_∞ incomplete steps. Adding the two counts gives $T_P \le \lfloor T_1/P \rfloor + T_\infty$.

Part (iii). Each of the two terms in (ii) is at most $\max(T_1/P, T_\infty)$, which gives the factor 2 against the lower bounds of (i). If $P \le \varepsilon T_1/T_\infty$, then $T_\infty \le \varepsilon T_1/P$, so (ii) gives $T_P \le (1+\varepsilon)\,T_1/P$ and hence $T_1/T_P \ge P/(1+\varepsilon)$. □

Example 19.46 (Scheduling the DAG of Figure 31)**.** The tasks A to H have edges A→B, A→C, A→D, B→E, C→E, D→F, E→G, F→G, G→H. The work is $T_1 = 8$, the span is $T_\infty = 5$ along A–B–E–G–H, and the parallelism is 1.6. With $P = 2$, the lower bounds of Theorem 19.45(i) are $\max(8/2, 5) = 5$, the greedy bound of (ii) is $8/2 + 5 = 9$, and the greedy schedule $\{A\}, \{B, C\}, \{D, E\}, \{F\}, \{G\}, \{H\}$ takes 6 steps: the steps $\{D, E\}$ and $\{B, C\}$ are complete, the other four incomplete, each of the latter reducing the remaining span by one. With $P = 3$, the bound is $8/3 + 5 = 7.67$ and the greedy schedule $\{A\}, \{B, C, D\}, \{E, F\}, \{G\}, \{H\}$ takes $5 = T_\infty$ steps, so the span law is tight and a fourth processor buys nothing.

Example 19.47 (Reduction trees and the chunked reduction)**.** Summing $n = 8$ values as a balanced binary tree has work 7 and span $3 = \log_2 8$. For $n = 2^{20}$ the work is about 10^6 and the span 20, a parallelism of about 5×10^4; on $P = 64$ cores, Theorem 19.45(iii) with $\varepsilon = 64 \times 20/(2^{20} - 1) = 0.0012$ guarantees a greedy speedup of at least $64/1.0012 = 63.9$. The P-worker *chunked* reduction — each worker folds a contiguous block of $\lceil n/P \rceil$ values, then the P partial results merge in a tree — has span $\lceil n/P \rceil + \lceil \log_2 P \rceil$, which is $16{,}384 + 6 = 16{,}390$ at $n = 2^{20}$ and $P = 64$; this is the shape reused by Proposition 19.93 and by the all-reduce of §19.5.

Remark 19.48 (Unequal tasks: the finish time is the maximum, not the mean)*.* For independent tasks of unequal durations handed, in some order, to whichever worker is free, the argument of Theorem 19.45(ii) gives Graham's list-scheduling bound $T_P \le T_1/P + t_{\max}$, where $t_{\max}$ is the largest task: the task that finishes last started at a moment when every worker was busy, hence no later than the average load T_1/P, and it then ran for at most $t_{\max}$. The $t_{\max}$ term is the price of a large task handed out late; sorting the tasks in decreasing size ("longest processing time first") tightens it. The whiteboard trap is to compute the mean load and call it the finish time: the finish time is the *maximum*, which is why a straggler sets the wall time of Example 19.33 and why many more tasks than workers are used in §19.8.

Example 19.49 (The utilisation example as a greedy budget)**.** Read the eight workers of Example 19.33 through Theorem 19.45(ii): $T_1 = 700$ s, $T_1/P = 87.5$ s, observed 100 s, so a span-like term of at least 12.5 s — a straggler task, in the sense of Remark 19.48, or a per-step barrier — is present. Halving the largest task or shortening the barrier is the lever; adding workers shrinks only the 87.5 s term.

Proposition 19.50 (The overhead crossover: how many workers)**.** *Suppose a total amount of work $W > 0$ divides perfectly among P workers, but each worker adds a fixed cost $c > 0$ (spawning, importing, loading a model, gathering its result) that is paid serially, so that $T(P) = cP + W/P$.*

Then T is minimised at $P^* = \sqrt{W/c}$, where $T^* = 2\sqrt{Wc}$ and the two terms are equal; past P^* every added worker costs more in coordination than it removes in compute.

Proof. We apply the arithmetic–geometric mean inequality: for positive a, b, $a + b \geq 2\sqrt{ab}$ with equality if and only if $a = b$. With $a = cP$ and $b = W/P$, $T(P) \geq 2\sqrt{cP \cdot W/P} = 2\sqrt{Wc}$, with equality if and only if $cP = W/P$, that is $P = \sqrt{W/c}$. Equivalently, $T'(P) = c - W/P^2$ vanishes at P^* and $T''(P) = 2W/P^3 > 0$ there. □

This is the "equate the two costs" pattern of Remark 17.9 solved for the worker count; it is a small extension of the settled material, included because it is the crossover shape live assessments have asked.

Example 19.51 (Six workers, not sixteen)**.** With $W = 150$ s of work and $c = 4$ s per worker, $P^* = \sqrt{37.5} = 6.1$, so six workers, with $T^* = 2\sqrt{600} = 49$ s. Checking: $T(1) = 154$ s, $T(4) = 16 + 37.5 = 53.5$ s, $T(6) = 24 + 25 = 49$ s, $T(8) = 32 + 18.75 = 50.75$ s, $T(16) = 64 + 9.4 = 73.4$ s — sixteen workers are slower than six. A second instance: $W = 10$ s and $c = 0.01$ s give $P^* = 31.6$ and $T^* = 0.63$ s.

Remark 19.52 (Amdahl's law as a span statement)*.* A program with serial fraction $1 - p$ is the DAG "a chain of $(1 - p)T_1$ tasks followed by pT_1 independent tasks": its span is $(1 - p)T_1 + 1$, so the span law of Theorem 19.45(i) reproduces the ceiling $1/(1 - p)$, and the greedy bound reproduces Amdahl's T_P up to one step. The span is the general notion and the serial fraction its simplest instance; a per-step barrier — an all-reduce, a simulation time step — contributes to the span in the same way.

Interview trap. Five wrong sentences about speedup. "Just add machines": the ceiling $1/(1 - p)$, the span law and the overhead crossover P^* all cap it, and past P^* the job gets slower (Example 19.51). "The work divides, so the speedup is P": the span law forbids beating T_∞, and a chain of dependencies is unaffected by P. "The mean load is the finish time": the finish time is the maximum, and Graham's $t_{\max}$ term is the price of a large task handed out late (Remark 19.48). "95% parallel is basically all parallel": at most 20× ever, and 12.5× on 32 processors. "12.5× on 32 cores, so 25× on 64": the inversion of Theorem 19.41(iii) gives $1 - p = 0.05$ and $S(64) = 15.4$.

19.5 Parallelism in Python: Threads, Processes, the GIL, and Vectorisation

Python offers three ways to do two things at once — threads, processes and coroutines — and the interpreter's global lock decides which of them buys wall-clock time. This subsection states the global interpreter lock and its planned removal as Facts with version qualifiers, proves the one consequence that follows from it (CPU-bound pure-Python threads cannot speed up), prices multiprocessing, places vectorisation as the first form of parallelism, and ends with data-parallel training as the application, using the diagnostics of §19.3 to classify a workload before any tool is chosen.

Setup. A workload is *CPU-bound* if a single-threaded run shows r near 1 in the sense of Theorem 19.31, and *I/O-bound* if r is well below 1. *Concurrency* means that several tasks are in progress during overlapping intervals; *parallelism* means $k(t) > 1$, more than one core busy at once. A *thread* (the `threading` module) shares its process's interpreter and heap; a *process* (`multiprocessing`, or the `ProcessPoolExecutor` of `concurrent.futures`) has its own interpreter and memory; a *coroutine* (`asyncio`) is cooperatively scheduled on one thread and yields control only at an `await`. As in §19.4, P denotes a count of workers and p a parallel fraction. For the training application, θ denotes the model parameters, $\ell_i(\theta)$ the loss on example i, and ∇ the gradient with respect to θ.

Fact 19.53 (The global interpreter lock and the free-threaded build (Python/ceval_gil.c, the "new GIL" of C
In the default CPython build a thread must hold the process-wide global interpreter lock (GIL) to execute Python bytecode, so at most one thread executes bytecode at any instant, however many cores exist. A thread holding the lock is asked to release it after the *switch interval* (`sys.getswitchinterval()`, default 0.005 s) whenever another thread is waiting for it. The lock is released voluntarily around blocking system calls — the `Py_BEGIN_ALLOW_THREADS` macros around file and socket I/O, `sleep` and lock waits — and by extension modules that opt out during C-level work: numpy for many array and BLAS operations (its `NPY_BEGIN_THREADS` macros), and the `hashlib`, `zlib` and `socket` modules among others. The GIL protects the interpreter's own data structures, not the program's invariants.

(ii) CPython 3.13 ships an experimental build without the GIL (the `python3.13t` executable; per-object locks and biased reference counting, PEP 703) alongside the default build. Extensions must be rebuilt for it; its single-threaded overhead was reported in the release notes as tens of percent in 3.13 and reduced to several percent in 3.14; and PEP 779 sets the criteria under which later releases move it toward supported status.

Consequence: threads overlap *waits* and GIL-releasing C work; they do not parallelise pure-Python CPU work; and no correctness argument may depend on the lock in either direction (§19.6).

Proposition 19.54 (No CPU-bound speedup from threads under the GIL)**.** *In the default build, if m threads execute only Python bytecode, never block, and their bytecode work totals T_1 seconds when run alone, then the wall time of the multi-threaded run satisfies $W \geq T_1$ whatever the number of cores; the run has $r \leq 1$, and Amdahl's law (19.1) applies with $p = 0$. In practice the lock hand-offs make the multi-threaded run slightly slower than the single-threaded one.*

Proof. We integrate the busy-core count as in Theorem 19.31. By Fact 19.53(i) the intervals during which different threads execute bytecode are pairwise disjoint, so the number of cores executing bytecode for this process is at most 1 at every instant. Integrating over the run gives $U + S \leq W$, that is $r \leq 1$. All T_1 seconds of bytecode work must be performed during the run, so $T_1 \leq U + S \leq W$. □

A multi-threaded pure-Python program showing r near 1 on an eight-core machine is GIL-bound; the fix is processes or C-level work, not more threads.

Fact 19.55 (multiprocessing: start methods, pickling, memory (library reference; Lib/multiprocessing))**.** Worker processes are separate interpreters, each with its own GIL. Three start methods exist. `fork` (Unix only) duplicates the parent through copy-on-write, so large read-only data is inherited free; it is unsafe when the parent has running threads, and 3.12 emits a warning when it is used in that situation. `spawn` (always on Windows; the macOS default since 3.8) starts a fresh interpreter that re-imports the main module, so everything sent to the worker must be picklable. `forkserver` forks workers from a clean helper process. The Linux default was `fork` through 3.13, and the 3.14 changelog documents a change of the default to `forkserver` on platforms other than macOS and Windows. Arguments and results cross process boundaries by pickling through pipes, at a cost proportional to their size; lambdas and open handles cannot cross; no memory is shared by default, the `shared_memory` submodule (3.8) being the escape hatch; and starting a process costs tens to hundreds of milliseconds, plus module import time under `spawn`.

Example 19.56 (Threads against processes on two workloads)**.** A CPU-bound pure-Python job of 40 s: four threads take about 40 s of wall time (r near 1, often slightly more wall from hand-offs, by Proposition 19.54); four processes take about 10 s plus roughly 0.3 s of start-up and pickling. An I/O-bound job of 100 network requests, each with 0.5 s of latency and 5 ms of CPU work: serially 50 s; with 20 threads, five rounds of 0.5 s, about 2.5 s of wall time with 0.5 s of CPU time in total, $r = 0.2$. The threads win despite the GIL, because the waits overlap and the lock is released during each of them.

Remark 19.57 (asyncio is concurrency, not parallelism). One thread, cooperative scheduling: a coroutine yields only at `await`, so thousands of pending I/O operations coexist cheaply, but one CPU-bound coroutine stalls the whole event loop; `run_in_executor` hands such work to a thread or process pool. A bounded `asyncio.Queue` between producer and consumer coroutines is the backpressure pattern of Remark 19.36.

Example 19.58 (The price of pickling and of tiny tasks)**.** Shipping an 800 MB numpy array to 8 workers by pickling copies 6.4 GB, about 3.2 s at 2 GB/s before any work is done; a fork-inherited read-only array or a `shared_memory` block costs nothing per worker. Splitting a 10^6-item job into 8 chunks pays 8 round trips; splitting it into 10^6 tasks pays 10^6 round trips at 50–100 microseconds each, 50–100 s of pure overhead. This is the cP term of Proposition 19.50 with the per-task cost in place of the per-worker one.

Procedure 19.59 (Choosing the concurrency tool)**.**

1. Measure r on a single-threaded run (Remark 19.35).
2. If waiting dominates, use threads (a few tasks) or `asyncio` (thousands of waits) so that the waits overlap; the wall time becomes the maximum of the waits rather than their sum.
3. If CPU-bound and the hot loop is numpy, BLAS or another GIL-releasing extension, check whether r already exceeds 1 (BLAS threads) before adding anything.
4. If CPU-bound pure Python, first vectorise (Remark 19.60); then use multiprocessing with chunked tasks — many more tasks than workers, each chunk large enough to amortise its pickling — priced with Proposition 19.50.
5. Compare the result with the Amdahl ceiling of the measured serial fraction (Theorem 19.41(iii)) before promising a speedup.
6. For training, run one process per GPU (Fact 19.63).

Remark 19.60 (Vectorisation is the first parallelism). Remark 9.13 explains the compiled-loop win as a memory-layout and dispatch story. On top of it, SIMD instructions process several lanes per instruction (4 doubles per AVX2 instruction, 8 per AVX-512), and BLAS libraries (OpenBLAS, MKL) run large products on several threads by default, controlled by `OMP_NUM_THREADS` and `MKL_NUM_THREADS`. This is why a "single-threaded" numpy script shows $r = 8$ (Example 19.34), and why *oversubscription* — eight processes each starting eight BLAS threads — collapses throughput while r stays high. A GPU kernel is the same idea with thousands of lanes (Fact 9.15, with its caveats). The order of escalation for a slow Python loop is: vectorise, then BLAS or GPU, then processes.

Example 19.61 (BLAS threads make $r = 8$ without a single Python thread)**.** A 4096×4096 double-precision product costs $2 \times 4096^3 = 1.37 \times 10^{11}$ flops by Proposition 17.1; at 50 GFLOP/s per core it takes 2.7 s single-threaded. With 8 BLAS threads the same call shows $W \approx 0.35$ s and $U + S \approx 2.7$ s, so $r \approx 8$: parallel, with no thread created by the Python program.

Proposition 19.62 (Data-parallel gradients: size-weighted shard gradients merge to the full-batch gradient)**.** *Let the batch loss be $L(\theta) = \frac{1}{B}\sum_{i=1}^{B} \ell_i(\theta)$ over B examples split into P shards $I_1, \dots, I_P$ of sizes $B_1, \dots, B_P$ with $\sum_j B_j = B$, and let worker j compute its shard-mean gradient $g_j = \frac{1}{B_j}\sum_{i \in I_j} \nabla \ell_i(\theta)$. Then*

$$\sum_{j=1}^{P} \frac{B_j}{B}\, g_j \;=\; \nabla L(\theta);$$

with equal shards this is the plain mean of the g_j, that is, an all-reduce sum divided by P.

Proof. We use linearity of the gradient and regroup the sum over the batch as a sum over shards:

$$\nabla L(\theta) = \frac{1}{B}\sum_{i=1}^{B}\nabla \ell_i(\theta) = \frac{1}{B}\sum_{j=1}^{P}\sum_{i\in I_j}\nabla \ell_i(\theta)$$
$$= \sum_{j=1}^{P}\frac{B_j}{B}\cdot\frac{1}{B_j}\sum_{i\in I_j}\nabla \ell_i(\theta) = \sum_{j=1}^{P}\frac{B_j}{B}\,g_j.$$

With $B_j = B/P$ for all j the weights are $1/P$. □

The merge is addition, which is associative and commutative, so any reduction tree is legal in exact arithmetic — proved as Theorem 19.81 in §19.7 — while in floating point the tree changes the low bits (Fact 19.84). The training loop this instantiates is the one of Section 9.

Fact 19.63 (Ring all-reduce bandwidth (Patarasuk and Yuan, 2009))**.** An all-reduce of a vector of D bytes over P workers arranged in a ring — a reduce-scatter phase followed by an all-gather phase — has each worker send and receive $2\,\frac{P-1}{P}\,D$ bytes in $2(P-1)$ steps; the per-worker traffic is independent of P to leading order, and the scheme is bandwidth-optimal. It requires the reduction to be associative and commutative. A proof is beyond our scope. In data-parallel training, each of P workers computes the gradient on its shard (Proposition 19.62), an all-reduce forms the sum, and every replica applies the identical update; the per-step wall time is $T_1/P + T_{\mathrm{sync}}$, a greedy-bound-shaped budget whose synchronisation term is the span (Remark 19.52). One process per GPU is the rule because of Fact 19.53.

Example 19.64 (One data-parallel step)**.** Take $P = 8$ GPUs, 120 ms of compute per step on one shard, and a model of 350 million float32 parameters, 1.4 GB of gradient. The ring all-reduce moves $2 \times \frac{7}{8} \times 1.4 = 2.45$ GB per GPU, about 12 ms at 200 GB/s, so a step takes 132 ms and the efficiency is $120/132 = 91\%$. Over 10 GB/s Ethernet the same all-reduce takes 245 ms and the efficiency falls to $120/365 = 33\%$: the synchronisation term dominates. For the merge itself, shards of sizes 3 and 1 with shard-mean gradients 1 and 5 (one coordinate shown) merge to $(3 \times 1 + 1 \times 5)/4 = 2$, the full-batch gradient by Proposition 19.62; the unweighted mean 3 is wrong.

Trading application. A parameter sweep over 10^4 backtest configurations is CPU-bound pure Python around a numpy core. Procedure 19.59 reads: measure r first (if the numpy core already drives r to 8, more processes will oversubscribe); then hand the configurations to a process pool in chunks of, say, 50 (many more tasks than workers, each chunk large enough to amortise the pickling of the shared price panel, which should be fork-inherited or placed in shared memory rather than pickled per task); and expect at most the Amdahl ceiling of the serial part — loading the panel, writing the results table — however many cores the machine has.

Interview trap. Six wrong sentences about Python parallelism. "The GIL makes Python thread-safe": it serialises bytecodes, not statements; `+=` is several bytecodes (§19.6), and the free-threaded build has no GIL at all. "More threads means faster": for CPU-bound Python, no (Proposition 19.54); for I/O-bound work, yes, up to the waiting fraction. "Threads never help in Python": they do for I/O waits and for GIL-releasing C code; numpy-heavy threads scale. "`multiprocessing` shares my big array for free": every task pickles its arguments and `spawn` re-imports the module (Fact 19.55); price it with Proposition 19.50. "My script is single-threaded, so r must be 1": BLAS threads make $r = 8$, and oversubscription keeps r high while throughput falls. "`asyncio` is parallel": it is single-threaded concurrency; one CPU-bound coroutine blocks everything (Remark 19.57).

19.6 Race Conditions, the Lost Update, and Deadlock

A race condition is the failure no test reliably reproduces and no interviewer lets pass: two workers touch the same state without an agreed order, and the result depends on the scheduler. The standard on-site form is “two threads increment a shared counter: what can go wrong, how wrong can it get, and how do you fix it?”. This subsection defines data races precisely, shows on Python bytecode that the GIL of Fact 19.53 does not make an increment atomic, works the lost-update interleaving on paper (Figure 32), proves exactly how low the counter can go, builds the fix ladder whose last rung — private state plus an associative merge — §19.7 then justifies, proves that lock ordering prevents deadlock, and ends with the two properties that make retries safe. It is the second of the two subsections around which the section is built.

Setup (the interleaving model). Threads $t_1, \ldots, t_m$ share memory. Each thread executes a fixed sequence of *atomic steps* — one memory read, one memory write, or one interpreter bytecode — and an *execution* is any interleaving of the threads' sequences that preserves each thread's own order (sequential consistency; weaker hardware memory models are out of scope). A shared integer c starts at 0. Threads A and B (more generally $K \geq 2$ threads) each perform $N \geq 1$ increments; one increment is the three steps *load* (read c into a private temporary), *add* (add one to the temporary, a private step), and *store* (write the temporary back to c). The final value of c is the value written by the last store of the execution. In this subsection N is a number of increments and K a number of threads, not the random-walk target and the number of Kahan terms.

Definition 19.65 (Data race, race condition, critical section, atomic operation)**.** Two accesses to the same memory location *conflict* if they are made by different threads and at least one is a write. A *data race* is a pair of conflicting accesses that no synchronisation orders, so that some execution places them adjacent in either order. A program has a *race condition* if its outcome depends on the interleaving; a data race is the usual cause but not the only one — a check-then-act on a file name, or a size check followed by an append on a structure that is locked during each operation but not across the pair, is a race condition without a data race. A *critical section* is a region of code that must appear atomic to other threads; an operation is *atomic* if no execution observes an intermediate state of it; *mutual exclusion* is the property that at most one thread is inside a given critical section at any instant.

Fact 19.66 (counter += 1 is four bytecodes, and the GIL may change hands between them (the dis module; Python/ceval.c, CPython 3.11 and 3.12))**.** Inside a function that declares `global counter`, the statement `counter += 1` compiles to `LOAD_GLOBAL counter`, `LOAD_CONST 1`, `BINARY_OP +=`, `STORE_GLOBAL counter` (verified with `dis` on the installed 3.11; identical on 3.12; an attribute target compiles to `LOAD_ATTR` and `STORE_ATTR` in the same shape) — logically load, add, store — and the language guarantees no atomicity for the statement. CPython transfers the GIL only at its *eval-breaker* check points, which in 3.11 and 3.12 sit at function entry and resumption, at backward jumps, and around calls, and whose placement has changed across versions. Whether a particular straight-line statement in a particular version can be split between its load and its store is therefore an implementation accident; any statement containing a call or a GIL-releasing operation between its load and its store can be split; the Python FAQ's list of operations that are atomic in CPython because they compile to a single bytecode (`list.append`, dict item assignment) is an implementation property, not a guarantee; and the free-threaded build of Fact 19.53(ii) removes the lock entirely. The precise statement is the weak one: the interleaving model permits the split, a correct program never depends on its absence, and a test that fails to reproduce the lost update is not evidence of atomicity.

Example 19.67 (The lost update)**.** Start from $c = 0$ with one increment per thread. Thread A loads 0 and adds, holding 1 in its temporary; the scheduler switches; thread B loads 0, adds, and stores 1; the scheduler switches back; thread A stores its stale 1. Two increments were executed and $c = 1$: B's update is lost. Figure 32 draws this interleaving; the fully alternating one (load A, load B, add A, add B, store A, store B) also ends at 1. Scaled up by the schedule of Proposition 19.68, $N = 1000$ increments per thread can end at $c = 2$ instead of 2000.

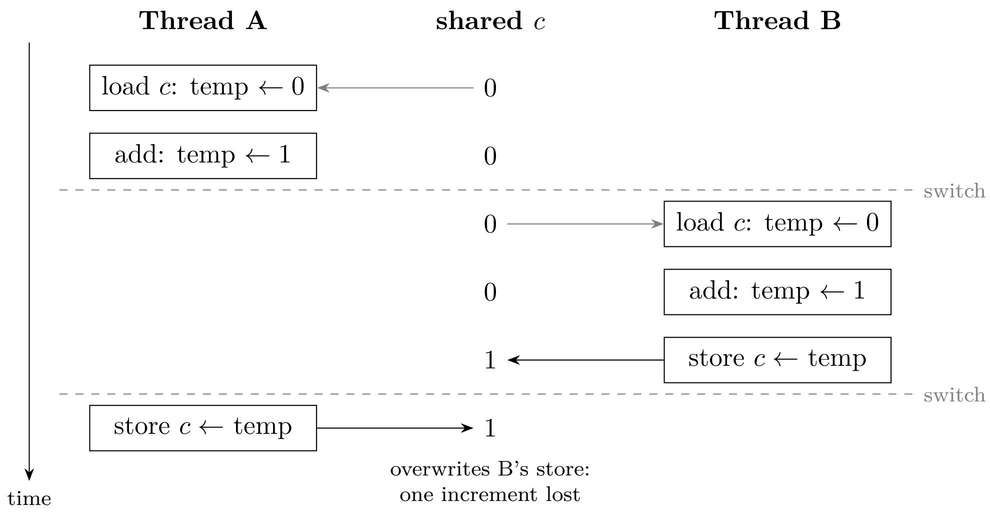


Figure 32: The lost update of Example 19.67. Time runs downward; the middle column shows the shared value after each step. Thread A loads 0 and computes 1, the scheduler switches (dashed line) at the first hand-off, thread B loads the same 0, computes 1 and stores it, and after the second hand-off thread A stores its stale 1 over B's. Two increments, final value 1. Nothing in the picture depends on the GIL: the two switches are the hand-offs Fact 19.66 permits.

Proposition 19.68 (How low a racy counter can go)**.** *Let $K \geq 2$ threads each perform $N \geq 2$ load–add–store increments of c from* 0 *in the interleaving model. Then every reachable final value v satisfies $2 \leq v \leq KN$, and both bounds are attained. In particular two threads performing N increments each can finish with $c = 2$.*

Proof. We prove the upper bound by an invariant, the lower bound by examining the last store, and attainment by exhibiting schedules.

Upper bound. The invariant is: at every moment, c is at most the number of stores completed so far. It holds initially ($0 \leq 0$). A store writes temp + 1, where temp was loaded at an earlier moment, when c was at most the number of stores completed by then, which is at most the number completed just before this store; so the value written is at most that number plus one, which is the number of stores completed after this store. The run has KN stores, so $v \leq KN$.

Lower bound. Every store writes temp + 1 ≥ 1, since every loaded value is at least 0; hence once any store has completed, $c \geq 1$ forever. The last store of the run is some thread's N-th store, and in that thread's own order its N-th load comes after its $(N-1)$-th store, which exists because $N \geq 2$. At the moment of that load, therefore, at least one store had completed and $c \geq 1$; the thread loaded at least 1 and its final store wrote at least 2. So $v \geq 2$.

Attaining 2. Thread A loads 0. All other threads then complete all their increments except that thread B withholds its last one; c is now at least $N - 1 \geq 1$. Thread A stores 1. Thread B loads 1. Thread A completes its remaining $N - 1$ increments in isolation, bringing c to N. Thread B adds and stores 2, the last store of the run. Every step respects each thread's own order, so this is an execution, and $v = 2$.

Attaining KN. Run the threads one after another with no interleaving; each increment then reads the previous store, and $v = KN$. ☐

Brute-force enumeration of all interleavings confirms the reachable sets $\{2, 3, 4\}$ for $K = 2$, $N = 2$ and $\{2, \ldots, 6\}$ for $K = 2$, $N = 3$ and for $K = 3$, $N = 2$. The damage in the worst case is near-total, not a small fraction.

Remark 19.69 (Why the bug is rare and why that is no comfort). A lost update needs a hand-off to land between a load and its store, a window of the order of 100 ns, while the default switch interval of Fact 19.53(i) is 5 ms; so the per-increment loss probability is of the order of 10^{-5} to 10^{-4} under the default interval, and a run may lose a handful of increments or — depending on where that version's check points sit relative to the statement — none at all. Setting `sys.setswitchinterval(1e-6)` makes the race frequent enough to demonstrate. Rarity is a property of one scheduler on one version; the proposition is a property of the program.

Example 19.70 (Hand-offs against increments)**.** A tight Python loop performs of the order of 10^6 to 10^7 increments per second, so a 5 ms switch interval means a hand-off every few thousand increments, and only a hand-off inside a load–store window loses an update. A test that runs two threads of 10^6 increments each and observes 2,000,000 has observed one schedule; Proposition 19.68 says another schedule ends at 2. This is why the bug is invisible in tests and catastrophic in the worst case.

Procedure 19.71 (The fix ladder, best last)**.**

1. *A lock around the critical section* (`threading.Lock` as a context manager). Correct, but the section runs serially: if it is a fraction s of the work, Theorem 19.41 with $1 - p = s$ caps the speedup at $1/s$ regardless of the worker count; each acquisition costs tens of nanoseconds plus contention. Hold locks for the shortest possible span.
2. *An atomic read-modify-write*: a fetch-and-add, or a compare-and-swap loop (read old, compute new, `CAS(old, new)`, retry on failure), available in C and in CUDA's `atomicAdd`, not for Python integers in the standard library. No blocking, but every update still funnels through one contended memory location (and one cache line), and floating-point atomics reduce in a nondeterministic order (§19.7).
3. *Private per-thread state plus an associative merge at the end.* Each worker accumulates into its own variable, and a single merge follows the join; no synchronisation during the work. Correct for any chunking by Theorem 19.81, proved in §19.7; this is the pattern behind combiners (§19.8) and the all-reduce of §19.5, and it is the answer interviewers want.
4. *Immutability and message passing.* Transfer ownership through a `queue.Queue` — a deque plus a lock and condition variables (Fact 19.9), bounded for backpressure (Remark 19.36) — instead of sharing; a value that is never mutated cannot race.

Example 19.72 (A lock against private counters)**.** Eight workers — processes, or threads on a free-threaded build — each perform 10^7 increments. With one shared lock, 8×10^7 serialised acquisitions at about 25 ns each cost 2 s however many workers there are, and the speedup is below 1. If the locked section is instead 10% of each task, Amdahl's law with $1 - p = 0.1$ caps the speedup at $10\times$ regardless of P. With private counters, each worker performs 10^7 local increments in parallel and the merge is one eight-term sum, for a speedup close to 8.

Definition 19.73 (Deadlock and the wait-for graph)**.** A set of threads is *deadlocked* if each of them is waiting for a resource held by another member of the set, so that none can ever proceed. The *wait-for graph* has an edge from each waiting thread to the holder of the lock it awaits; a deadlock is a cycle in it.

Fact 19.74 (Coffman conditions (Coffman, Elphick and Shoshani, 1971))**.** Four conditions are jointly necessary for deadlock: *mutual exclusion* (a resource is held by at most one thread), *hold-and-wait* (a thread holds resources while waiting for more), *no preemption* (a held resource is released only by its holder), and *circular wait* (a cycle in the wait-for graph). Breaking any one of them prevents deadlock: try-locks with release on failure break hold-and-wait, timeouts break no-preemption, and a global lock order breaks circular wait, which is the standard practical choice (Proposition 19.75). A proof is beyond our scope.

Proposition 19.75 (A global lock order prevents deadlock)**.** *Fix a total ranking of all locks. If every thread acquires locks in strictly increasing rank — it never waits for a lock of lower rank than one it already holds — then no deadlock can occur.*

Proof. We argue by contradiction through the wait-for cycle. Suppose threads $t_1, \ldots, t_k$ are deadlocked, with t_i holding lock l_i and waiting for l_{i+1} held by t_{i+1}, indices taken modulo k. The discipline gives $\mathrm{rank}(l_i) < \mathrm{rank}(l_{i+1})$ for every i, since t_i waits for l_{i+1} while holding l_i. Composing these strict inequalities around the cycle yields $\mathrm{rank}(l_1) < \mathrm{rank}(l_1)$, which is impossible. ☐

Example 19.76 (Two locks suffice)**.** Thread A acquires L_1 then L_2; thread B acquires L_2 then L_1. The interleaving "A takes L_1; B takes L_2; A waits for L_2; B waits for L_1" satisfies all four Coffman conditions — each lock has one holder, each thread holds one lock while waiting, neither lock is taken away, and the wait-for graph is the cycle $A \to B \to A$ — and both threads wait forever. Impose the order $L_1 < L_2$: B must now take L_1 first, so it blocks on L_1 while holding nothing, A completes and releases both locks, and B proceeds. This is Proposition 19.75 in action.

Definition 19.77 (Determinism and idempotence)**.** A task is *deterministic* if its output is a function of its input alone — no dependence on scheduling, on the clock, or on unseeded randomness. A task is *idempotent* if executing it twice leaves the same state as executing it once.

Remark 19.78 (Retries are safe only for idempotent tasks). Fault tolerance by re-execution — the task restarts of Fact 19.99 in §19.8, a data-loader worker respawn, a job-array retry — is correct only for idempotent tasks. Writing `results[task_id] = y`, or writing to a per-task temporary file and then renaming it atomically into place, is idempotent; `results.append(y)`, `counter += 1` and "append a line to a shared log" are not. At-least-once delivery combined with an idempotent handler behaves as exactly-once. Random numbers must be keyed by (seed, task id) so that a re-execution reproduces the original, which also makes the task deterministic in the sense of Definition 19.77.

Example 19.79 (A duplicated shard)**.** A shard task that appends its 100 partial word counts to a shared list is re-executed after a timeout, and the original also completes; the list now holds 200 records, 100 of them duplicates, and the word "the" is counted twice. With `results[shard_id] = partial` — or write-to-temporary-then-rename — the re-execution overwrites the entry with an identical value and the total is correct.

Trading application. A backtest sweep that writes each configuration's Sharpe ratio into one shared results dict from several threads is the counter of Example 19.67 in disguise if any entry is ever read-modified-written (a running best, a count of configurations tried). Rung 3 of Procedure 19.71 is the design: each worker keeps its own results dict keyed by configuration id, the merge after the join is a dict union whose keys are disjoint, and a retried configuration overwrites its own key — idempotent by Remark 19.78. The rolling best is then computed once, at the end, from the merged dict.

Interview trap. Six wrong sentences about races. "The GIL makes Python thread-safe": it serialises bytecodes, not statements; `+=` is four bytecodes and the GIL may change hands between them (Fact 19.66, Example 19.67). "I ran it a thousand times and the count was right, so it is atomic": absence of the bug is not absence of the race (Remark 19.69); check-point placement is an implementation accident, and the free-threaded build removes the lock. "Just add a lock": correct, but it serialises the critical section (Amdahl), and two locks invite deadlock; the house answer is private state plus an associative merge. "Deadlock needs many locks": two suffice (Example 19.76). "Retry the failed task": only if the task is idempotent; appends and increments are not (Remark 19.78). "A race needs two writers": one writer and one reader with no ordering between them is already a data race (Definition 19.65).

19.7 Associativity: the Licence to Parallelise, Floating Point, Merged Statistics, and Scans

Every parallel aggregation — a thread's private sums merged at the end, a gradient all-reduce, a MapReduce combiner, a prefix scan — rests on one algebraic property, and this subsection proves it: an associative operation may be evaluated by any chunking or tree, and a commutative one in any order. It then shows that floating-point addition lacks the property and what that does to reproducibility, proves that the mean and the variance become associative merges once the state is lifted, and shows how a dependent loop becomes parallel when its operation is associative.

Setup. Let $\mathcal{S}$ be a set with a binary operation written $\oplus$ and, where needed, an identity element e with $e \oplus x = x \oplus e = x$. For a sequence $x_1, \dots, x_n$ in $\mathcal{S}$ the *left fold* is

$$\mathrm{fold}(x_1, \dots, x_n) \;=\; (\cdots((x_1 \oplus x_2) \oplus x_3) \oplus \cdots) \oplus x_n,$$

with $\mathrm{fold}(x_1) = x_1$. A *bracketing* of $x_1, \dots, x_n$ is any full parenthesisation of the expression $x_1 \oplus x_2 \oplus \cdots \oplus x_n$ with the operands in their given order; a *chunking* partitions the sequence into contiguous blocks, folds each block separately, and combines the block results by some bracketing; a permutation σ reorders the operands. IEEE 754 binary64 has unit roundoff $u = 2^{-53} \approx 1.1 \times 10^{-16}$. For a finite multiset D of real numbers with $n \geq 1$ elements, the *summary state* is $\mathrm{st}(D) = (n, m, M_2)$, where m is the mean of D and $M_2 = \sum_{x \in D}(x - m)^2$ is the sum of squared deviations from the mean; the population variance is M_2/n and the sample variance $M_2/(n-1)$. In this subsection n is a count of elements and m a mean, not a thread count.

Definition 19.80 (Associative, commutative, monoid)**.** The operation $\oplus$ is *associative* if $(a \oplus b) \oplus c = a \oplus (b \oplus c)$ for all $a, b, c \in \mathcal{S}$, and *commutative* if $a \oplus b = b \oplus a$ for all a, b. A *monoid* is a set with an associative operation and an identity e; the identity is what lets empty chunks merge harmlessly. Examples: integer addition and multiplication, max, min, logical and, logical or, set union and gcd are associative and commutative; string concatenation, “last non-null”, “first value” and the matrix product are associative but not commutative (the associativity of the matrix product is what Theorem 17.3 relies on); subtraction and division are neither; the mean and the variance as written are neither, though Proposition 19.89 repairs them; floating-point addition is commutative but not associative (Fact 19.84).

Theorem 19.81 (Fold invariance: associativity licenses any bracketing, commutativity any reordering)**.** *(i) If $\oplus$ is associative, every bracketing of $x_1 \oplus \cdots \oplus x_n$ equals* $\mathrm{fold}(x_1, \dots, x_n)$*; hence every contiguous chunking and every reduction tree computes the same value.*

(ii) If $\oplus$ is associative and commutative, every bracketing of every permutation $x_{\sigma(1)} \oplus \cdots \oplus x_{\sigma(n)}$ equals $\mathrm{fold}(x_1, \dots, x_n)$*; hence partial results may be merged in any arrival order.*

Proof. We proceed in three named steps: a concatenation lemma for folds, a strong induction on n for bracketings, and a decomposition of permutations into adjacent transpositions.

Step 1 (folds concatenate). For associative $\oplus$ and $1 \leq k < n$,

$$\mathrm{fold}(x_1, \dots, x_k) \oplus \mathrm{fold}(x_{k+1}, \dots, x_n) \;=\; \mathrm{fold}(x_1, \dots, x_n). \tag{19.2}$$

By induction on $n - k$. If $n - k = 1$ the right-hand fold is $\mathrm{fold}(x_{k+1}) = x_{k+1}$ and (19.2) is the definition of the left fold. For $n - k \geq 2$, write $\mathrm{fold}(x_{k+1}, \dots, x_n) = \mathrm{fold}(x_{k+1}, \dots, x_{n-1}) \oplus x_n$ by definition; then

$$\begin{aligned}\mathrm{fold}(x_1, \dots, x_k) \oplus \big(\mathrm{fold}(x_{k+1}, \dots, x_{n-1}) \oplus x_n\big) &= \big(\mathrm{fold}(x_1, \dots, x_k) \oplus \mathrm{fold}(x_{k+1}, \dots, x_{n-1})\big) \oplus x_n \\ &= \mathrm{fold}(x_1, \dots, x_{n-1}) \oplus x_n,\end{aligned}$$

regrouping once by associativity and then applying the induction hypothesis to the shorter gap $(n-1) - k$; the last expression is $\mathrm{fold}(x_1, \dots, x_n)$ by definition.

Step 2 (part (i)). By strong induction on n. For $n = 1$ there is nothing to prove. For $n \geq 2$, any bracketing has an outermost $\oplus$ that splits the operands as

$$(\text{a bracketing of } x_1, \dots, x_k) \;\oplus\; (\text{a bracketing of } x_{k+1}, \dots, x_n)$$

for some $1 \le k < n$; by the induction hypothesis the two sides equal $\mathrm{fold}(x_1, \ldots, x_k)$ and $\mathrm{fold}(x_{k+1}, \ldots, x_n)$, and Step 1 joins them into $\mathrm{fold}(x_1, \ldots, x_n)$. A chunking followed by a tree of merges is itself a bracketing, so the consequence follows.

Step 3 (part (ii)). Every permutation is a product of adjacent transpositions (bubble sort exhibits the factorisation), so it suffices to show that swapping x_j and x_{j+1} does not change the fold. By part (i), $\mathrm{fold}(x_1, \ldots, x_n)$ equals the bracketing in which $x_j \oplus x_{j+1}$ is formed first as a pair; commutativity replaces that pair by $x_{j+1} \oplus x_j$ without changing its value; the result is a bracketing of the transposed sequence, which equals the fold of the transposed sequence by part (i). Induction on the number of transpositions finishes the argument. □

Rung 3 of Procedure 19.71 is therefore correct for any associative merge and any assignment of contiguous chunks to workers, and needs commutativity as well whenever workers finish in an unscheduled order — which they do.

Example 19.82 (Three ways to fold eight integers)**.** Take $x = [3, 1, 7, 0, 4, 1, 6, 3]$ under integer addition. The left fold gives 25; the balanced tree $((3 + 1) + (7 + 0)) + ((4 + 1) + (6 + 3)) = 11 + 14 = 25$; two chunks of four folded separately and then merged give $11 + 14 = 25$, as Theorem 19.81(i) promises. Under subtraction the left fold gives -19 and the balanced tree $((3 - 1) - (7 - 0)) - ((4 - 1) - (6 - 3)) = -5 - 0 = -5$: subtraction is not associative, and no parallel version of the loop exists as written.

Remark 19.83 (Which operations qualify, and the last-non-null trap). Read Definition 19.80 as a licence. Associative and commutative operations admit any tree and any arrival order. Associative but non-commutative operations — concatenation, last non-null, the matrix product — admit any tree but not an unordered merge, so partial results must be tagged with their chunk position. The mean and the variance are repaired by lifting the state (Proposition 19.89). The median is neither and cannot be merged without shipping data or approximating. Floating-point addition is treated next.

Fact 19.84 (Floating-point addition is not associative (IEEE 754, 1985 and 2008; Higham, 2002))**.** Each floating-point addition rounds its exact result to the nearest representable number, so the sequence of roundings, and hence the computed sum, depends on the bracketing. Verified binary64 identities: $(0.1 + 0.2) + 0.3 = 0.6000000000000001$ while $0.1 + (0.2 + 0.3) = 0.6$; $(10^{16} + 1.0) + 1.0 = 10^{16}$ while $10^{16} + (1.0 + 1.0) = 1.0000000000000002 \times 10^{16}$ (the spacing of doubles near 10^{16} is 2); and $(10^{20} + (-10^{20})) + 1.0 = 1.0$ while $10^{20} + ((-10^{20}) + 1.0) = 0.0$. In binary32, adding 0.1 one million times sequentially gives 100958.34, while pairwise summation of the same array gives 100000.09, against the exact 100000. The forward error of the recursive (left-to-right) sum is bounded by $(n - 1)\, u \sum_i |x_i|$ to first order in u. Consequently Theorem 19.81 does not apply, and a parallel sum whose bracketing depends on the thread count, the chunking or the arrival order is run-to-run nondeterministic in its low bits, which chaotic downstream computations — an optimiser trajectory, a simulation — amplify.

Example 19.85 (The identities, digit by digit)**.** In binary64, $0.1 + 0.2 = 0.30000000000000004$, and adding 0.3 gives 0.6000000000000001; but $0.2 + 0.3 = 0.5$ exactly, and adding 0.1 gives 0.6. Near 10^{16} consecutive doubles differ by 2, so $10^{16} + 1.0$ rounds back to 10^{16} (ties to even), whereas $1.0 + 1.0 = 2.0$ is exactly representable on top of 10^{16}. The binary32 accumulation of Fact 19.84 gives 100958.34 sequentially against 100000.09 pairwise against the exact 100000: a relative error of 1% for the sequential sum. A test asserting equality between an 8-thread and a 16-thread sum fails on such inputs; a relative tolerance of 10^{-6} passes (Remark 19.87).

Fact 19.86 (Compensated and pairwise summation (Kahan, 1965; Higham, 2002))**.** Kahan summation carries the rounding error of each addition in a compensation term and achieves a forward error of $2u \sum_i |x_i| + O(nu^2)$, independent of n to first order; pairwise (tree) summation achieves a relative error of $O(u \log_2 n)$ at the same work, and is what `numpy.sum` uses on contiguous floating-point arrays; `math.fsum` returns the correctly rounded sum (Shewchuk's algorithm). A proof is beyond our scope.

Remark 19.87 (The fixes for floating-point nondeterminism)*.* Four fixes, in decreasing order of strength. A *fixed reduction tree*: a bracketing that does not depend on the worker count or the arrival order (or the framework's deterministic-algorithms flag), at a cost in speed. *Compensated or pairwise summation* (Fact 19.86), which shrinks the discrepancy without removing it. *Integer or fixed-point accumulation* when exactness is required, since integer addition is associative. *Tolerance-based tests*: $|a-b| \le \mathrm{atol} + \mathrm{rtol}\,|b|$ (`math.isclose`, `numpy.allclose`), never equality. Remark 6.21 is the sibling numerical caution for linear algebra.

Example 19.88 (Summation error at a billion terms)**.** Summing 10^9 doubles of magnitude about 1: the recursive bound $(n-1)u\sum|x_i| \approx 10^9 \times 1.1 \times 10^{-16} \times 10^9 = 1.1 \times 10^2$ absolute on a sum of about 10^9, a relative error of order 10^{-7}; pairwise summation about $\log_2(10^9)\,u = 30 \times 1.1 \times 10^{-16} = 3.3 \times 10^{-15}$ relative; Kahan about $2u = 2.2 \times 10^{-16}$. The bounds are worst cases; typical errors are smaller, but the *difference* between two bracketings is what a parallel run exposes.

Proposition 19.89 (Mean and variance are associative merges of summary states (Chan, Golub and LeVeque, 1979))**.** *For the mean, the state (n,s) with s the sum merges as $(n_A,s_A)\oplus(n_B,s_B) = (n_A+n_B, s_A+s_B)$, and the mean of the merged multiset is s/n. For the variance, define on summary states*

$$(n_A, m_A, M_2^A) \oplus (n_B, m_B, M_2^B) \;=\; \Big(n,\; m_A + \delta\,\frac{n_B}{n},\; M_2^A + M_2^B + \delta^2\,\frac{n_A n_B}{n}\Big),$$

where $\delta = m_B - m_A$ and $n = n_A + n_B$. Then $\mathrm{st}(A)\oplus\mathrm{st}(B) = \mathrm{st}(A \uplus B)$ for all non-empty multisets A and B, where $\uplus$ is multiset union. Consequently $\oplus$ is associative and commutative on summary states of non-empty multisets, and the variance is read off the final state as M_2/n or $M_2/(n-1)$.

Proof. We verify the merge identity by expanding a square, and then obtain associativity and commutativity from the fact that st is a homomorphism.

Mean. The mean of $A \uplus B$ is $(n_A m_A + n_B m_B)/n = m_A + n_B(m_B - m_A)/n = m_A + \delta\, n_B/n$, as the merge states; for the sum-based state the claim is the additivity of sums.

Squared deviations. Write m for the merged mean. Split $M_2(A \uplus B) = \sum_{x\in A}(x-m)^2 + \sum_{x\in B}(x-m)^2$ and expand the first sum around m_A:

$$\sum_{x\in A}(x-m)^2 = \sum_{x\in A}\big[(x-m_A)+(m_A-m)\big]^2 = M_2^A + 2(m_A-m)\sum_{x\in A}(x-m_A) + n_A(m_A-m)^2 = M_2^A + n_A(m_A-m)^2,$$

because the deviations from m_A sum to zero (the same cancellation as in Proposition 10.10). Likewise $\sum_{x\in B}(x-m)^2 = M_2^B + n_B(m_B-m)^2$. Now $m_A - m = -\delta\, n_B/n$ and $m_B - m = \delta - \delta\, n_B/n = \delta\, n_A/n$, so the two constant terms add to

$$n_A\,\frac{\delta^2 n_B^2}{n^2} + n_B\,\frac{\delta^2 n_A^2}{n^2} = \delta^2\,\frac{n_A n_B(n_B+n_A)}{n^2} = \delta^2\,\frac{n_A n_B}{n},$$

which is the merge's third component.

Associativity and commutativity. Every summary state that a fold produces is $\mathrm{st}(D)$ for some non-empty multiset D, and the merge identity says $\mathrm{st}(A)\oplus\mathrm{st}(B) = \mathrm{st}(A \uplus B)$. Hence for any three multisets,

$$(\mathrm{st}(A)\oplus\mathrm{st}(B))\oplus\mathrm{st}(C) = \mathrm{st}(A\uplus B\uplus C) = \mathrm{st}(A)\oplus(\mathrm{st}(B)\oplus\mathrm{st}(C)),$$

and $\mathrm{st}(A)\oplus\mathrm{st}(B) = \mathrm{st}(A\uplus B) = \mathrm{st}(B\uplus A) = \mathrm{st}(B)\oplus\mathrm{st}(A)$, because multiset union is associative and commutative. No separate algebraic check of the formula is needed. ☐

Remark 19.90 (The naive state and catastrophic cancellation). The state $(n, \sum x, \sum x^2)$ is also a homomorphism in exact arithmetic — all three components are plain sums — but it computes the variance as $\mathbb{E}[x^2]-m^2$, a difference of two nearly equal large numbers. For $x = 10^8+1$, 10^8+2, 10^8+3 it returns 0.0 in binary64 against the true population variance 2/3 (verified). The merged-deviation form of Proposition 19.89, which never forms $\sum x^2$, is the house answer.

Example 19.91 (Merging two shards' statistics)**.** Shard $A = \{1, 2, 3\}$ has state $(3, 2, 2)$ and shard $B = \{10, 14\}$ has $(2, 12, 8)$; $\delta = 10$ and $n = 5$. The merged mean is $2 + 10 \times 2/5 = 6$ and $M_2 = 2 + 8 + 100 \times 3 \times 2/5 = 130$. Direct check on $\{1, 2, 3, 10, 14\}$: mean 6, squared deviations $25 + 16 + 9 + 16 + 64 = 130$. Merging in the other order, $\delta = -10$: mean $12 - 10 \times 3/5 = 6$ and $M_2 = 8 + 2 + 100 \times 6/5 = 130$, confirming commutativity. For unequal shards $\{10\}$ and $\{2, 2, 2\}$, the size-weighted mean is $16/4 = 4$; the unweighted mean of the shard means, $(10 + 2)/2 = 6$, is wrong.

Definition 19.92 (Prefix sums (scan))**.** For a monoid $(\mathcal{S}, \oplus, e)$, the *inclusive scan* of $x_1, \ldots, x_n$ is the sequence $y_i = x_1 \oplus \cdots \oplus x_i$, and the *exclusive scan* is $y_i = e \oplus x_1 \oplus \cdots \oplus x_{i-1}$ (so $y_1 = e$). The sequential loop $y_i = y_{i-1} \oplus x_i$ has work $\Theta(n)$ and, as written, span $\Theta(n)$, because each output depends on the previous one. The question is whether that dependency is fundamental or an artifact of the left-to-right bracketing.

Proposition 19.93 (Parallel prefix sum by blocks)**.** *Let $\oplus$ be associative and let P workers hold contiguous blocks of at most $\lceil n/P \rceil$ elements each. The three-phase scheme — (1) each worker folds its block; (2) an exclusive scan of the P block totals is computed, sequentially in $O(P)$ or by a tree in $O(\log P)$ span; (3) each worker rescans its block, seeded with its block's offset from phase (2) — computes all n inclusive prefixes with work $2n + O(P)$ and span $2\lceil n/P \rceil + O(\log P)$.*

Proof. We use the concatenation identity (19.2) from the proof of Theorem 19.81. Let position i lie in block j, whose first element is x_a, and let o_j be the block's offset, the exclusive scan value $t_1 \oplus \cdots \oplus t_{j-1}$ of the block totals $t_1, \ldots, t_{j-1}$ (with $o_1 = e$). Repeated application of (19.2) shows that $o_j = \mathrm{fold}(x_1, \ldots, x_{a-1})$, the fold of everything before block j. The seeded rescan computes $\mathrm{fold}(o_j, x_a, \ldots, x_i)$, which by (19.2) applied to the sequence $(o_j, x_a, \ldots, x_i)$ with $k = 1$ equals

$$o_j \oplus \mathrm{fold}(x_a, \ldots, x_i) = \mathrm{fold}(x_1, \ldots, x_{a-1}) \oplus \mathrm{fold}(x_a, \ldots, x_i) = \mathrm{fold}(x_1, \ldots, x_i) = y_i,$$

using (19.2) once more for the last step. For the costs, phase (1) performs at most n operations across the workers with span $\lceil n/P \rceil$, phase (2) performs $O(P)$ operations, and phase (3) performs n operations with span $\lceil n/P \rceil$; adding gives the claim. The dependency of Definition 19.92 was an artifact of the bracketing. □

Example 19.94 (A block-scan trace and a Blelloch trace)**.** Scan $[3, 1, 4, 1, 5, 9, 2, 6]$ with $P = 2$: the block totals are 9 and 22; the exclusive offsets are 0 and 9; the seeded rescans give $[3, 4, 8, 9]$ and $[14, 23, 25, 31]$, the correct inclusive prefixes. Work: $3 + 3$ additions in phase (1), 1 in phase (2), $4 + 4$ in phase (3), 15 in all against 7 sequentially; span $3 + 1 + 4 = 8$ against 7 here, but about $2n/P + \log_2 P$ in general — for $n = 2^{20}$ and $P = 64$ about 32,800 against 10^6. The Blelloch scheme of Fact 19.95 on $[3, 1, 7, 0, 4, 1, 6, 3]$: the up-sweep forms the pair sums $[4, 7, 5, 9]$, then $[11, 14]$, then the root 25; the down-sweep, with the root reset to 0, yields the exclusive scan $[0, 3, 4, 11, 11, 15, 16, 22]$; 14 additions at depth 6 against 7 additions at depth 7 sequentially.

Fact 19.95 (Work-efficient scan (Blelloch, 1990; Hillis and Steele, 1986))**.** Blelloch's *up-sweep* (a reduction tree that stores each subtree's fold) followed by a *down-sweep* on the same tree (the root is set to e; each node passes its own value to its left child and its value $\oplus$ the left child's stored fold to its right child) computes the exclusive scan for any monoid with work $O(n)$ and span $O(\log n)$. Hillis and Steele's scheme — in round j, every position i absorbs position $i - 2^j$ — achieves span $O(\log n)$ with work $O(n \log n)$: span-optimal but not work-efficient, and slower than the sequential loop when $P \ll n$. A proof is beyond our scope.

Remark 19.96 (What scans are for, and what they cannot do). Scan is the escape hatch for loops whose dependency is an associative operator in disguise: running totals; output offsets in count-then-fill layouts; stream compaction; and linear recurrences $s_{t+1} = A_t s_t + b_t$, whose (A, b) pairs compose associatively as affine maps (the state-space models of sequence modelling). A nonlinear recurrence such as a physics time step has no associative composition, keeps span $\Theta(T)$ over T steps, and the honest answer says so; the parallelism then lives *within* each step.

Example 19.97 (Last non-null: a tree is fine, an unordered merge is not)**.** Chunks $[\text{None}, 5]$, $[\text{None}, \text{None}]$ and $[7, \text{None}]$ under the merge "last non-null", which is associative but not commutative. Any tree that respects chunk order gives 7; merging in the arrival order chunk 3, chunk 1, chunk 2 gives 5, the wrong answer. Commutativity is exactly what an unordered merge needs (Theorem 19.81(ii)).

Interview trap. Five wrong sentences about associativity. "A parallel floating-point sum is deterministic": only with a fixed bracketing; thread count, chunking and arrival order all change the rounding sequence (Fact 19.84), and the fix should be named in the same breath (Remark 19.87). "Mean is associative, average the averages": the merge needs the counts, either (sum, count) or the state of Proposition 19.89; the unweighted average is wrong for unequal shards (Example 19.91). "Variance from the sum and the sum of squares": exact in algebra, catastrophic cancellation in floating point (Remark 19.90). "Prefix sums are inherently sequential": the dependency is the bracketing, and associativity gives a scan with span $O(\log n)$ (Proposition 19.93, Fact 19.95); the honest answer for a nonlinear recurrence is span $\Theta(n)$. "`assert a == b` after a parallel run": use a tolerance.

19.8 Distributed Aggregation: MapReduce, Combiners, and Top-k Across Shards

The long-form on-site question escalates from one machine to many. This subsection assembles §19.1 (a dict per machine), §19.6 (idempotent retries) and §19.7 (associative merges) into the MapReduce pattern, identifies the shuffle as the bottleneck, and proves the result candidates most often get wrong: top-k must aggregate before it selects.

Setup. There are n records whose keys are drawn from d distinct keys, spread over M machines with about n/M records each; h is a hash partition from keys to $\{1, \ldots, M\}$; each machine holds a local dict; communication is measured in records crossing the network; k is the number of top items requested; d_j denotes the number of distinct keys present on machine j, so that $d_j \le \min(n/M, d)$. In this subsection M is a machine count and k a rank, not a heap height.

Procedure 19.98 (MapReduce word count with combiners)**.**

1. *Map.* Each machine emits one (word, 1) pair per occurrence in its own data: embarrassingly parallel, $O(n/M)$ expected dict updates per machine (Fact 19.6).
2. *Combine.* Each machine pre-reduces its own emissions into a local dict of (word, local count). This is legal exactly because integer addition is associative and commutative (Theorem 19.81(ii)): the partial counts will later be merged in an arbitrary order.
3. *Shuffle.* Each (word, partial count) record travels to machine $h(\text{word})$: an all-to-all network exchange, after which every occurrence of a word has its partial counts on exactly one machine.
4. *Reduce.* Each owner folds the partials of its keys and emits (word, total).

Communication. Without the combiner, n records cross the network. With it, machine j sends at most $d_j \le \min(n/M, d)$ records — one per distinct key it holds — so the total is at most Md: the combiner collapses shuffle traffic from the number of records to the number of distinct keys per machine. The shuffle's all-to-all phase is the bottleneck and must be named as such.

Fact 19.99 (MapReduce fault tolerance and combiners (Dean and Ghemawat, 2004))**.** The framework re-executes the tasks of failed machines and speculatively re-executes straggler tasks near the end of a job, which requires tasks to be deterministic and idempotent (Definition 19.77, Remark 19.78); combiner functions are the optional local reducer run on each mapper's output before the shuffle. These are properties of the published system; a proof is beyond our scope.

Example 19.100 (Word-count costs at scale)**.** Take $n = 10^{10}$ words over $M = 1000$ machines with $d = 10^6$ distinct words: 10^7 words are mapped per machine, about 1 s at 100 ns per expected dict update. Without a combiner, 10^{10} records cross the network; with one, at most 10^6 per machine and at most 10^9 in total. At 8 bytes per record the cluster-wide shuffle volume is 80 GB against at most 8 GB, and each machine's share is 80 MB against at most 8 MB — 0.8 s against 0.08 s at a 100 MB/s share of the network, so the combiner turns a shuffle that costs as much as the map into one that costs a tenth of it.

Remark 19.101 (Skew, hot keys and stragglers)*.* Hash partitioning balances *keys*, not *occurrences*: a hot key floods its owner. The combiner caps that key's partials at M, one per mapper; for aggregates that stay hot, *salting* splits the key into r sub-keys ("the#1" to "the#r") aggregated in two rounds. The finish time is the maximum over machines (Remark 19.48), so the standard levers are many more tasks than workers drawn from a work queue, speculative re-execution of stragglers, and, in one sentence, power-of-two-choices assignment (send each task to the less loaded of two random workers).

Example 19.102 (A hot key)**.** The word "the" occurs 10^9 times in the corpus of Example 19.100. Without a combiner, 10^9 records (8 GB) converge on one reducer, which at 100 MB/s receives for 80 s while every other machine finishes in under a second: a straggler by construction. With the combiner, at most 1000 partials arrive. With salt $r = 10$, the ten sub-keys land on ten machines and a second aggregation of ten partials finishes the job.

Proposition 19.103 (Aggregate, then select: correctness of partitioned top-k)**.** *(i) Selecting each shard's local top-k of* raw occurrences *and merging the lists can miss the global winner: Example 19.104 is a witness, and (i) needs no proof beyond it.*

(ii) After hash-partitioning by key, so that every key's complete *count lives on exactly one machine, the union of the machines' local top-k lists contains the global top-k. Hence "aggregate, select locally, merge M lists of k" is exact, with communication $O(d)$ for the shuffle plus $O(Mk)$ for the merge, and a merge cost of $O(Mk \log M)$ by a k-way heap merge (Example 19.12, Proposition 19.24).*

Proof. We prove (ii) by a ranking argument. Let x be any key in the global top-k by complete count, so at most $k-1$ keys have a strictly larger global count. The owner of x ranks its keys by these same complete counts, since after partitioning it holds every occurrence of each of its keys; so at most $k-1$ keys outrank x locally, and x is in the owner's local top-k list. Thus every global top-k key appears in the union of the local lists, with its exact count, and the coordinator that merges the M lists selects the global top-k correctly. Merging M sorted lists of length k with a heap of M heads costs $O(Mk \log M)$, each of the Mk outputs being one pop and one push on a heap of size M. □

The invariant to state aloud: *after key-partitioning, local top-k lists are globally exhaustive candidates.*

Example 19.104 (The select-then-aggregate counterexample)**.** Let $k = 1$ with three shards. The word a appears $2, 2, 2$ times on the three shards (total 6); words b, c and d appear 3 times each, each on one shard only. The local top-1 lists are b, c and d, and the true global top-1, a, is never reported, whatever the tie-breaking rule. Select-then-aggregate is wrong.

Example 19.105 (The same shards, aggregated first)**.** Repartition the three shards by key: a's total 6 now lives on one machine, which reports it as its local top-1; the coordinator merges three lists of one and returns a. The costs are $O(d)$ records for the shuffle and $O(3)$ for the merge, as Proposition 19.103(ii) states.

Remark 19.106 (Refinements: threshold algorithms, sketches, and the randomised-hash shuffle trap)*.* Threshold algorithms (Fagin, Lotem and Naor, 2001) fetch the local lists incrementally and stop as soon as no unseen key can enter the top-k, reducing communication below $O(Mk)$ when the lists are skewed. For streaming data a count-min sketch plus a heap per shard, with the sketches merged at the coordinator, is exact for the merge — cell-wise addition is associative and commutative (Theorem 19.81) — and approximate only in the counts. The coordinator's $O(Mk)$ load is never the bottleneck; the shuffle is. Finally, a home-made shuffle that partitions by `hash(word) % M` in separate Python processes is wrong unless `PYTHONHASHSEED` is fixed or a stable hash from `hashlib` is used, because string hashes are randomised per process (Fact 19.7): the same word would be sent to different owners by different mappers, and no machine would hold its complete count.

Trading application. "Which ten symbols traded the most notional today, across a feed sharded by exchange?" is Proposition 19.103 verbatim: a symbol that trades moderately on every venue can out-total one that is large on a single venue, so per-shard top tens are not exhaustive candidates. Aggregate per symbol first — a shuffle by symbol with local combiners summing notional, an associative and commutative merge — then select. The same argument applies to "top signals by information coefficient across date shards" when the statistic is a size-weighted mean (Proposition 19.89).

Interview trap. Five wrong sentences about distributed aggregation. "Top-k of each shard, then merge": wrong (Example 19.104); aggregate first. "Hash partitioning balances the load": it balances keys, not occurrences; skew lives in hot keys (Remark 19.101). "Combiners always apply": only for associative and commutative reducers; a median cannot be combined without shipping values or approximating (Remark 19.83). "Re-run the failed task": only if the task is idempotent (Remark 19.78). "`hash(word) % M` partitions consistently across my worker processes": not for string keys under per-process hash randomisation (Fact 19.7).

Part VIII

Problems and Solutions

20 Problem Bank

Two-plus problems per subsection of the body, mixed difficulty, no coding. A star (★) after a problem means it reproduces, from memory, a question asked in a live 2026 interview; a diamond (⧫) means the same for a timed online assessment; the firms are deliberately not named. Work them closed-book, under a rough 5-minute clock each; full worked solutions are in §21. Every answer was re-derived by two independent blind solvers before inclusion.

20.1 Problems — §1 Expected Value for Trading Games

P1.1 *Independence and the product rule* ⧫

The probability I order a burger from the office Deliveroo on a given day is 0.4. The probability my stomach hurts on a given day is 0.1. Assuming the two events are independent, what is the probability that I order a burger *and* my stomach hurts that day?

P1.2 *Binomial counting* ♦

A fair coin is flipped 5 times. What is the probability of exactly 3 heads?

P1.3 *Two aces (hypergeometric draw)* ♦

Cards are dealt from a well-shuffled standard 52-card deck, and a specific player receives two of them. What is the probability that this player is dealt two aces?

P1.4 *Dice outcome counting* ♦

Two fair eight-sided dice (faces numbered 1–8) are thrown. What is the probability that the sum of the two faces is greater than or equal to 14?

P1.5 *Binomial at-least* ♦

In foosball I have a 25% chance of scoring each time I shoot, independently of all other shots. I shoot 3 times. What is the probability that I score 2 or more goals?

P1.6 *Distinct faces in ten rolls*

A fair die is rolled 10 times. What is the expected number of *distinct* faces that appear? Give an exact expression and a numeric value — and explain why your calculation is legitimate even though the quantities you sum are not independent.

P1.7 *Gold and silver balls* ♦

A bag contains 100,000 gold balls and 100,000 silver balls. Employees take turns drawing balls from the bag at random, one at a time, without replacement; each employee stops their turn as soon as they draw a gold ball, and then the next employee begins. This continues until all the gold balls have been drawn. What is the expected proportion of gold balls among all the balls pulled out of the bag? Give the closest simple value and justify it.

P1.8 *Geometric waiting time* ♦

What is the expected number of times I need to roll a fair die in order to get a 6?

P1.9 *Geometric waiting time: memorylessness*

You are working a resting limit order. Each second, independently of everything that has happened so far, the order is filled with probability $p = 0.05$.

(a) What is your expected wait, in seconds, until the fill?

(b) You have already waited 30 seconds without a fill. What is your expected *additional* wait from this moment?

(c) A different game: a fair coin is flipped repeatedly. What is the expected number of flips until you have seen *both* a head and a tail at least once?

P1.10 *Pattern waiting times: HH versus HT*

A fair coin is flipped until the pattern HH first appears; in a separate game, until HT first appears. Compute both expected waiting times by first-step analysis. Then explain, in one sentence, why the answers differ even though HH and HT are equally likely to occupy any two fixed flips.

P1.11 *Pattern waiting times: the mental closed-form arsenal* ♦

A timed online numerical test — no pen and paper allowed — hits you with a rapid block of expected-number-of-rolls questions: expected rolls or flips to reach a given outcome or pattern. Under those conditions the hard part is not the concepts but tracking states and transitions purely in your head. State *from memory*, each with a one-line justification you could reconstruct under pressure:

(a) expected trials until a first success of probability p;

(b) expected flips of a fair coin until a run of k consecutive heads;
(c) expected rolls of a fair die until a run of k consecutive sixes;
(d) expected flips until the first appearance of HH, of HT, and of HTH;
(e) expected rolls of a die until every face has been seen.

Briefly: why does memorising these dominate deriving them live in this test format?

P1.12 *Pattern races: which comes first*

A single fair coin is flipped repeatedly.

(a) Your colleague notes that $\mathbb{E}[\text{flips to HT}] = 4$ while $\mathbb{E}[\text{flips to HH}] = 6$, and concludes that in a head-to-head race on the same coin, HT beats HH more often than not. What is the actual probability that HT appears before HH, and what exactly is wrong with the reasoning?
(b) Now race TH against HH on one coin. A counterparty offers you an even-money bet on HH appearing first. Do you take it, and what would the fair odds be?

P1.13 *Coupon collector*

You roll a fair die repeatedly. How many rolls do you expect to need until every face 1–6 has appeared at least once? Give the exact value, then the rough growth rate of the answer for an n-faced die.

P1.14 *Coupon collector variant: subset of coupons amid noise*

Same fair die, but now you only care about the faces 1, 2 and 3. How many rolls do you expect to need until each of 1, 2 and 3 has appeared at least once? (Faces 4–6 still come up as usual — they are simply useless to you.) Before computing: should the answer be the 3-face coupon-collector number $3(1 + \frac{1}{2} + \frac{1}{3}) = 5.5$?

P1.15 *Order statistics of dice: make-a-market ladder*

An interviewer rolls two fair dice X and Y behind a screen and asks you to make a market — state your fair value and how you would quote around it — on each of:

(a) $\max(X, Y)$;
(b) $\min(X, Y)$;
(c) $|X - Y|$.

P1.16 *Option value of a re-roll*

You roll a fair die and are paid its face value in dollars.

(a) Before rolling, you are granted the right to discard the first roll and re-roll once. What is the optimal policy and the value of the game?
(b) With the right to re-roll up to two times?
(c) What is the single re-roll option itself worth? And if using the re-roll costs a fee of \$0.50 payable at the moment you use it, what is the game worth — is it simply the answer to (a) minus \$0.50?

P1.17 *First 6 given all rolls even*

A fair die is rolled repeatedly until the first 6 appears. You are told that every roll in the sequence — including the final 6 — came up even. Given this, what is the expected number of rolls? Most candidates instantly answer 3, reasoning that the conditioning "reduces the die to the three even faces $\{2, 4, 6\}$", making the wait geometric with $p = 1/3$. Give the correct answer and explain precisely why the instant answer is wrong.

P1.18 *Hat-check fixed points*

At the end of a conference dinner, the coat-check tickets of n traders are shuffled and handed back uniformly at random, one ticket each. Let X be the number of traders who receive their own ticket.

(a) Compute $\mathbb{E}[X]$.
(b) Compute $\mathrm{Var}(X)$ for $n \geq 2$.
(c) For large n, a colleague offers you an even-money bet that *nobody* gets their own ticket back. Which side do you take, and what is your edge?

P1.19 *Random chord longer than the radius* ♦

Two points are chosen independently and uniformly at random on a circle. What is the probability that the length of the chord joining them is larger than the circle's radius?

(A) $1/\pi$
(B) $2/3$
(C) $1/2$
(D) $3/4$
(E) $\pi/4$

P1.20 *Max of four dice, exactly* ♦

You roll 4 fair six-sided dice and keep the highest value. What is the probability that the highest value is exactly 3?

(A) $80/1296$
(B) $1/81$
(C) $65/1296$
(D) $1/16$

20.2 Problems — §2 Bayesian Updating and Reading Flow

P2.1 *Bayes odds form, base rates* ♦

In a large population, 0.1% of people are infected with a disease. A test is positive for 100% of infected people and for 10% of uninfected people. A randomly chosen person tests positive. What is the probability that they are infected? Give the exact fraction, then an approximation you could say out loud.

P2.2 *Conditioning on the decisive event* ♦

The probability I beat my arch-nemesis at chess is 0.4, the probability they beat me is 0.2, and the probability we draw is 0.4. If we draw, we replay, and keep replaying until somebody wins. What is the probability that I win overall?

P2.3 *Odds form, prior vs evidence* ♦

A gag coin with heads on both sides is dropped into a bag containing 99 fair coins. One coin is chosen at random from the hundred and flipped 5 times, coming up heads all 5 times. What is the probability that the gag coin was chosen?

P2.4 *Sequential likelihood ratios*

A bag holds two dice. One is fair; the other is loaded so that a six appears with probability 1/2 (the other five faces equally likely). You pick one die at random and start rolling it.

(a) The first roll is a six. What is the probability you hold the loaded die?

(b) The first three rolls are all sixes. Now what is it?
(c) How many consecutive sixes would you need to be at least 99% sure?
(d) After three sixes your fourth roll is a two. Does your probability stay where it was, and if not, where does it go?

P2.5 *Correlated evidence*

You make markets in a mid-cap altcoin perp. Historically 10% of the accounts that trade aggressively against you are informed. Each aggression from an informed account lifts your offer (rather than hitting your bid) with probability 0.8; from a noise account, with probability 0.2. Your junior sees the offer lifted three times within one second and computes: prior odds $1 : 9$, likelihood ratio $(0.8/0.2)^3 = 64$, posterior odds $64 : 9$, hence $P(\text{informed}) \approx 88\%$ — and pulls every quote. The execution log then shows that all three fills were child slices of a single iceberg order worked by one account. What is wrong with the 88%, and what is a defensible update?

P2.6 *Flow reading, sequential update to a threshold*

You quote both sides of a crypto perp. Base rate: 5% of aggressive counterparties are informed buyers; the rest are noise. Each aggression from an informed buyer lifts your offer (rather than hitting your bid) with probability 0.9; a noise account lifts or hits with probability 0.5 each. Aggressions are conditionally independent given the type.

(a) One account lifts your offer twice in a row. What is the probability you are facing an informed buyer?
(b) After how many consecutive lifts does the posterior first exceed 90%?
(c) As the posterior climbs, what is the correct quoting response — and what exactly is wrong with "keep filling them at the same offer"?

20.3 Problems — §3 Betting, Bankroll, Ruin and Markov Chains

P3.1 *Kelly criterion*

You can bet any fraction of your bankroll, repeatedly, on independent even-money flips of a coin that lands your way with probability $p = 0.55$.

(a) For the general bet paying b-to-1 with win probability p, derive the fraction f^* that maximises the long-run growth rate, and check it is a maximum.
(b) Evaluate f^* for the coin above, and for a bet paying 2-to-1 that wins 40% of the time.
(c) For the coin, compute the growth rate at f^* and at $2f^*$. What is the moral?

P3.2 *Kelly under estimation error*

A junior trader backtests a signal over 25 independent trades and wins 15 of them ($\hat{p} = 0.60$, even-money payoffs). He computes full Kelly $f^* = 2\hat{p} - 1 = 20\%$ and starts betting a fifth of the book on every trade.

(a) What is wrong, before any further calculation?
(b) Suppose the true win probability is 0.55. Compute the growth rate of his 20% sizing and compare it with correct Kelly sizing and with betting half of his *estimated* Kelly.
(c) Quantify the noise in $\hat{p}$ itself. What sizing discipline does this justify?

P3.3 *Gambler's ruin, fair case*

You and a rival flip a fair coin for \$1 a flip; you start with \$30, they start with \$70, and you play until one of you is broke.

(a) What is the probability you end up with all \$100?

(b) What is the expected number of flips?
(c) You switch to \$5 per flip. What changes?

P3.4 *Gambler's ruin, biased*

(a) At a casino you win each \$1 round with probability 0.49 and lose it with probability 0.51. You start with \$50 and commit to playing until you either reach \$100 or go broke. Estimate your probability of reaching \$100, cleanly enough to do at a whiteboard.
(b) A market-making strategy earns +1 tick with probability 0.51 and −1 tick with probability 0.49 on each trade, and trades forever — there is no upper target. Starting with a 20-tick capital buffer, what is the probability it ever goes bust? What buffer keeps lifetime ruin below 1%?

P3.5 *First-step analysis / ruin reduction*

You win each point against an opponent independently with probability 0.6. Starting level, the match continues until one player leads by two points; that player wins (deuce rules). What is your probability of winning the match? Give two independent derivations, and comment on why the answer is not 0.6.

P3.6 *First-step analysis, pattern race*

A fair coin is flipped repeatedly. What is the probability that the pattern HH appears before the pattern TH? Answer with a structural argument, verify by first-step analysis, and explain why comparing the two patterns' expected waiting times (6 flips for HH, 4 for TH) does not answer the question.

P3.7 *Stationary distribution, two-state chain*

A market is either trending or choppy day to day. From trending it switches to choppy with probability 0.1; from choppy it switches to trending with probability 0.3; otherwise it stays where it is.

(a) In the long run, what fraction of days are trending?
(b) Today is trending. What is the expected number of days until the *next* trending day?
(c) Verify your stationary vector is actually stationary.

P3.8 *Stationary distribution on a graph, return time*

A knight sits on a corner square of an otherwise empty chessboard and makes a uniformly random legal knight move at each step, forever. What is the expected number of moves until it first returns to its starting corner? (No 64-state linear system allowed — find the structure.)

P3.9 *St. Petersburg, capped EV*

A casino flips a fair coin until the first head; if the first head arrives on flip n, you are paid 2^n dollars.

(a) Show the expected payoff is infinite.
(b) The casino's bankroll actually caps the payout at 2^{20} dollars (about a million). What is the fair price now?
(c) The interviewer pushes: "so how much would you personally pay to play once?" Name the considerations that justify an answer far below even the capped expectation.

P3.10 *Utility shape, risk aversion* ♦

What shape should the utility function of a rational, risk-averse investor take? State the two defining properties, justify each, and give one consequence of each: why does such an investor decline a fair \$100 coin flip, and what does this shape have to do with Kelly betting and the St. Petersburg paradox?

20.4 Problems — §4 Statistics I: Estimators and Inference

P4.1 *Second moment of the standard normal* ♦

Let X be a standard normal random variable. What is $E[X^2]$?

P4.2 *Bias-variance-MSE trade-off in variance estimation*

Let $X_1, \ldots, X_n$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$ with both parameters unknown, and let $S = \sum_{i=1}^{n}(X_i - \bar{X})^2$. Consider the family of variance estimators $\hat{\sigma}_c^2 = S/c$. Two standard choices are $c = n-1$ (unbiased) and $c = n$ (the maximum-likelihood estimator). Which value of c minimises the mean squared error $E\big[(\hat{\sigma}_c^2 - \sigma^2)^2\big]$? Rank the three candidate divisors by MSE, and state the moral about unbiasedness.

P4.3 *Bessel's correction and the bias of s*

You estimate the variance of a strategy's daily returns from an i.i.d. sample $X_1, \ldots, X_n$ with unknown mean μ and variance σ^2.

(a) Show that $E\big[\sum_{i=1}^{n}(X_i - \bar{X})^2\big] = (n-1)\sigma^2$, so the unbiased variance estimator divides by $n-1$.

(b) If instead the true mean μ were known and you centred the deviations on μ, what divisor makes the estimator unbiased?

(c) Your colleague concludes that $s = \sqrt{s^2}$, with s^2 the Bessel-corrected estimator from (a), is therefore an unbiased estimator of the volatility σ. Is that right?

P4.4 *Sum of i.i.d. normals* ♦

For independent random variables $X_1, \ldots, X_k$, each distributed $\mathcal{N}(m, s^2)$, what is the distribution of the sum $X_1 + \cdots + X_k$?

P4.5 *Sum of normals without independence* ♦

One random variable is distributed $\mathcal{N}(5, 4)$ and another is distributed $\mathcal{N}(3, 9)$ (means 5 and 3, variances 4 and 9). What is the distribution of their sum?

P4.6 *Distribution recognition (binomial)* ♦

Every IOC (immediate-or-cancel) buy order placed at the best offer on the Coinbase SOL/EUR order book fills with probability 0.7, independently of all other orders. If 50 such IOC orders are sent, what is the distribution of the number of orders filled?

P4.7 *Normal approximation to the binomial* ♦

What Normal distribution should you use to approximate a Binomial($n = 100,\ p = 0.2$) distribution?

P4.8 *CLT - what it does not say*

A junior quant on your desk argues: "Our daily P&L is fat-tailed, but annual P&L is the sum of 252 daily P&Ls, so by the central limit theorem it is normal, and I have computed our 4-sigma annual loss probability from the normal table." Give at least three distinct reasons this argument fails, and say in which direction the normal-table answer errs.

P4.9 *Confidence interval: variance vs sd* ♦

A variable is normally distributed with mean 1000 and variance 100. What is the approximate 95% confidence interval?

P4.10 *Hypothesis testing recipe*

A strategy was up on 140 of the last 252 trading days, and your PM declares the hit rate "obviously better than a coin flip." Run the test properly: state the null hypothesis, the test statistic and its distribution under the null, compute the two-sided p-value, and give your conclusion at the 5% level. What changes with a one-sided test — and what is the danger of switching to one after seeing the data?

P4.11 *Sharpe ratio standard error (the humility formula)*

A backtest over 3 years of daily data shows an annualised Sharpe ratio of 1.2. Using the i.i.d. standard-error formula $\mathrm{SE}(\widehat{\mathrm{SR}}) \approx \sqrt{(1+\mathrm{SR}^2/2)/n}$ (with SR and n at the observation frequency), estimate the standard error of the *annualised* Sharpe estimate. Is the Sharpe statistically distinguishable from 0? From 1? Roughly how many years of daily data would you need to pin the annualised Sharpe down to ± 0.25 (one standard error)?

P4.12 *Multiple testing / deflated-Sharpe logic*

Your team backtested 100 signal variants this quarter. The best one shows a t-statistic of 2.5 on 1000 days of returns, and a colleague notes "$p < 1.3\%$, let's allocate." Suppose, for simplicity, that all 100 backtests are independent and the signals are pure noise. (a) What is the probability that the *best* of the 100 shows $|t| \geq 2.5$? (b) Roughly what t-statistic should the best-of-100 need to clear for 5% family-wise significance? (c) What is the practical recipe when someone shows you the best result of many trials?

20.5 Problems — §5 Regression and Regularisation

P5.1 *OLS univariate slope and R-squared*

Desk quickie, no calculator. You regress daily returns of stock Y on daily returns of stock X (with an intercept). Sample statistics: correlation $\rho = 0.4$, $\sigma_X = 1\%$ daily, $\sigma_Y = 3\%$ daily. (a) What slope does the regression give? (b) What is the R^2? (c) X is up 2% today; ignoring the (tiny) intercept, what move does the regression predict for Y? (d) One sentence: how can the slope exceed 1 while the fit explains so little of Y's variance?

P5.2 *OLS derivation and projection*

From first principles, for $y = X\beta + \varepsilon$ with $X \in \mathbb{R}^{n\times p}$ of full column rank: (a) derive the OLS estimator and confirm you have found a minimum; (b) show the residual vector is orthogonal to every column of X; (c) show the hat matrix $H = X(X^\top X)^{-1}X^\top$ satisfies $H^\top = H$, $H^2 = H$ and $\mathrm{tr}(H) = p$, and interpret the trace; (d) check that your matrix formula reduces to $\widehat{\mathrm{Cov}}(x,y)/\widehat{\mathrm{Var}}(x)$ in the univariate centred case.

P5.3 *Gauss–Markov assumptions and violations*

(a) State the four Gauss–Markov assumptions and the theorem's exact conclusion. Is normality of the errors assumed? (b) For each desk failure below, name the assumption broken (if any), the consequence for OLS, and the standard fix: (i) the "alpha" signal you regress on was built with data that leaks future information; (ii) residual variance triples in high-volatility regimes; (iii) two of your factors have sample correlation 0.98; (iv) residuals are fat-tailed but independent and homoskedastic.

P5.4 *Omitted-variable bias*

The true daily return process of a stock is $y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$, where x_1 is your new signal and x_2 is the market return, with $\mathbb{E}[\varepsilon \mid x_1, x_2] = 0$. In truth $\beta_1 = 0$ (the signal has no alpha) and $\beta_2 = 1$. Your backtest regresses y on x_1 alone, and in the sample $\widehat{\mathrm{Cov}}(x_1,x_2)/\widehat{\mathrm{Var}}(x_1) = 0.3$. (a)

Derive the expectation of the estimated coefficient on x_1 in general. (b) What number does your backtest report as "alpha"? (c) Name the finance mistake and the standard fix.

P5.5 *Gauss–Markov BLUE proof + ridge follow-up*

(a) Prove the Gauss–Markov theorem: under linearity, full rank, strict exogeneity and spherical errors, any estimator $\tilde{\beta} = Cy$ that is linear in y and unbiased for every β satisfies $\mathrm{Var}(\tilde{\beta}) - \mathrm{Var}(\hat{\beta}_{\mathrm{OLS}}) \succeq 0$. (b) The interviewer follows up: "Ridge regularly beats OLS out of sample. Doesn't that contradict the theorem you just proved?" (c) Verify the theorem's content in the simplest case: estimating a mean μ from n i.i.d. observations with variance σ^2, which linear unbiased weighting $\sum_i w_i y_i$ is best, and why?

P5.6 *Ridge derivation, invertibility, shrinkage*

(a) Derive the ridge estimator minimising $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$ for $\lambda > 0$. (b) Explain why $(X^\top X + \lambda I)$ is always invertible, even when $X^\top X$ is singular (duplicated features, or $p > n$). (c) In the orthonormal-design case $X^\top X = I$, show ridge is pure shrinkage of OLS; if an OLS coefficient is 1.2 and $\lambda = 0.5$, what is the ridge coefficient? (d) What is the Bayesian interpretation of ridge, and how does λ relate to the prior?

P5.7 *Regularisation menu* ★

"Give five examples of regularization in machine learning." Give the five with one line each on the mechanism — and have a second batch ready in case the interviewer pushes with "give me more," including tree-model regularisers and at least one finance-specific example.

P5.8 *Lasso soft-thresholding*

In the orthonormal-design case the lasso objective separates coordinate by coordinate into $\min_b (b - \hat{\beta}_j)^2 + \lambda|b|$, where $\hat{\beta}_j$ is the OLS coefficient. (a) Solve this one-dimensional problem exactly, handling the kink at zero carefully. (b) Apply it with $\lambda = 1$ to OLS coefficients $(2.0,\ 0.6,\ -0.3)$. (c) Explain why no positive λ ever produces an exact zero in ridge, yet lasso does.

P5.9 *Lasso vs ridge in alpha research*

You have 500 candidate alpha signals, many of them highly correlated with one another, and about three years of daily observations ($n \approx 750$) for a cross-sectional return model. (a) Give the geometric picture for why an L1 penalty produces exact zeros while L2 does not. (b) Which penalty do you reach for here, and what is the known failure mode of lasso with groups of correlated features? (c) What does elastic net do, and why is it the pragmatic default in exactly this setting?

P5.10 *Inverse regression*

A colleague regresses Y on X and gets slope 0.5. For a different report he needs the slope of X regressed on Y, and — short on time — quotes $1/0.5 = 2$. The sample correlation between X and Y is 0.5. (a) What exactly is wrong with the shortcut? (b) Given $\sigma_X = \sigma_Y$, compute the actual X-on-Y slope. (c) Prove the general identity linking the two slopes, and state the one case where naive inversion is legitimate.

P5.11 *Hedge-ratio inversion, market application*

Stock S has daily volatility 2%, ETF A has daily volatility 1%, and their daily-return correlation is 0.6. You are long \$1M of S and hedge by shorting A. (a) What notional of A minimises the variance of the hedged position? (b) An intern instead regresses A's returns on S's returns, takes that slope, and inverts it to get the hedge ratio. What notional does he short, and by what factor is he over-hedged? (c) Compute the daily P&L volatility in dollars of the unhedged position, the correctly hedged position, and the intern's position. What is the punchline?

P5.12 *Beta vs correlation*

(a) True or false: "a stock with beta 2 on the index must be highly correlated with the index." If false, construct an explicit counterexample. (b) In hedging, which quantity sizes the hedge and which tells you whether the hedge is worth doing? (c) Quick numbers: a stock has daily vol 10%, the index 1%, correlation 0.2. Compute the stock's beta and the fraction of its variance the index explains — and reconcile the two numbers in one sentence.

P5.13 *Choosing a hedge instrument*

You must hedge a $1M long position in stock Y (daily vol 4%) with exactly one of two futures: F_1 has correlation 0.9 with Y and daily vol 8%; F_2 has correlation 0.6 with Y and daily vol 1%. (a) For each instrument, compute the minimum-variance hedge ratio (in $ notional) and the residual daily vol of the hedged position. (b) Which instrument do you choose, and what single quantity decides it? (c) A desk rule of thumb says "hedge with the higher-beta instrument." Critique it using your numbers.

P5.14 *WLS precision weighting*

You observe the same unknown quantity μ twice with independent noise: $x_1 = \mu + \varepsilon_1$ with $\mathrm{sd}(\varepsilon_1) = 1$, and $x_2 = \mu + \varepsilon_2$ with $\mathrm{sd}(\varepsilon_2) = 2$. (a) Find the best (minimum-variance) linear unbiased combination and its variance — is it better than using the good measurement alone? (b) Show this is exactly weighted least squares, and state the general weighting rule. (c) In a returns regression whose residual variance is four times higher in crisis months, what does unweighted OLS do wrong, and what are the fixes?

P5.15 *Residual weighting vs GLS*

Teammate 1 fits model A to daily returns. Teammate 2 then trains a second model B on the same data using sample weights $w_i = |y_i - \hat{y}_{A,i}|$ and averages the two models' predictions. Teammate 3 objects: "that is just WLS, and WLS weights are supposed to be *inverse*-variance — your weights are upside-down." Adjudicate: what is B actually doing, which classical technique does the scheme resemble, what is the right way to combine the two models, and in what market regime does the scheme backfire?

P5.16 *Dart problem, Gaussian, pitchfork*

You throw a single dart. Aiming at point a on the real line, the dart lands at $a + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. Two targets sit at x_1 and $x_2 = x_1 + d$, and your expected score is proportional to the landing density summed over the targets:

$$S(a) = e^{-(a-x_1)^2/(2\sigma^2)} + e^{-(a-x_2)^2/(2\sigma^2)}.$$

(a) Show the midpoint $m = (x_1 + x_2)/2$ is a critical point for every d. (b) For which d is the midpoint optimal, and what happens to the optimal aim as d grows past the threshold? (c) Follow-up: "what does the square in the exponent have to do with regression?"

P5.17 *Dart problem, L1, median vs endpoint*

(a) For a random variable X with continuous CDF F, show that c minimises $g(c) = \mathbb{E}|X - c|$ exactly when $F(c) = 1/2$, i.e. c is a median. With an empirical distribution on just two points $x_1 < x_2$, which aims are optimal under absolute loss? (b) Now a dart game with Laplace tails: aiming at a, the expected score contributed by a target at distance r is e^{-r} (distances in units of the Laplace scale). With targets at x_1 and $x_2 = x_1 + d$, so $S(a) = e^{-|a-x_1|} + e^{-|a-x_2|}$, prove the optimal aim is at one of the targets, and compute how much an endpoint beats the midpoint by. (c) Follow-up: "does this remind you of anything in regression?" Give the mapping and the classic conflation to avoid.

20.6 Problems — §6 Linear Algebra, PCA and Covariance

P6.1 *Vector norms / reverse triangle inequality* ♦

Vector A has length 5 and vector B has length 7. What is the lowest possible value of $|A - B|$? Justify your answer and describe the configuration that attains it.

P6.2 *Matrix transformations rapid-fire* ♦

Rapid-fire, multiple-choice pace — one line of justification each.

- **(a)** You want to apply a rotation R to a column vector x, then a shear S to the result. Which expression represents the combined transformation: SR or RS?
- **(b)** The matrix $\begin{pmatrix} \cos t & 0 & \sin t \\ 0 & 1 & 0 \\ -\sin t & 0 & \cos t \end{pmatrix}$ represents a rotation about which axis of $\mathbb{R}^3$?
- **(c)** If the columns of a square matrix A are linearly dependent, what must be true of $\det(A)$?
- **(d)** If A and B are invertible matrices, what is $(AB)^{-1}$?
- **(e)** A 2×2 matrix has determinant -2. Describe accurately the geometric effect of the transformation on the 2D plane.

P6.3 *Eigenvectors of a 2D rotation* ♦

Consider the rotation by 90 degrees counter-clockwise in the standard real plane $\mathbb{R}^2$. What is true of its eigenvectors? (One tempting multiple-choice option read: "exactly one eigenvector, along the axis of rotation.")

P6.4 *Correlation-matrix feasibility*

A risk report shows the following correlations for three strategies A, B, C, each entry estimated pairwise on a *different* (pairwise-complete) sample of dates:

$$\rho_{AB} = 0.9, \qquad \rho_{AC} = -0.9, \qquad \rho_{BC} = 0.8.$$

- **(a)** Show that no joint distribution of (A, B, C) can have this correlation matrix.
- **(b)** Explain how the pairwise estimation procedure can produce such a matrix anyway.
- **(c)** Derive the feasible range of ρ_{BC} given the other two entries, and name two standard repairs used in practice.

P6.5 *PCA derivation and explained variance*

(a) A dataset is centred and has sample covariance matrix Σ. Derive the direction w with $\|w\| = 1$ that maximises the variance of the projected data, and identify the maximised value. **(b)** For $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, find both principal directions and the fraction of total variance explained by the first principal component. **(c)** If you skip the centring step and the data mean is far from the origin, what does the leading "principal component" of $X^\top X$ actually capture?

P6.6 *Equicorrelated covariance / PC1 as market mode*

A universe of $n = 100$ stocks has, for simplicity, identical variance σ^2 and identical pairwise correlation $\rho = 0.3$, so the covariance matrix is

$$\Sigma = \sigma^2\left[(1 - \rho)I + \rho\,\mathbf{1}\mathbf{1}^\top\right],$$

where $\mathbf{1}$ is the all-ones vector. **(a)** Find all eigenvalues of Σ with their multiplicities (hint: $\mathbf{1}\mathbf{1}^\top$ has rank one). **(b)** Identify PC1 and compute the fraction of total variance it explains for the given numbers. **(c)** Use the result to justify the claim "PC1 is the market mode," and state the consequence for a signal whose weights do not sum to zero.

P6.7 *PCA via the SVD / conditioning*

Let X be an $n \times p$ centred data matrix with thin SVD $X = USV^\top$. **(a)** Show that the principal directions of the data are the columns of V, and express the sample variance along the i-th direction in terms of the singular value s_i (use the $1/(n-1)$ convention). **(b)** With $n = 5$ observations and singular values $s_1 = 4$, $s_2 = 2$, give the variance along each principal direction and the share of variance explained by PC1. **(c)** Your library offers both an SVD of X and an eigendecomposition of $X^\top X$. Which do you call for PCA, and why?

P6.8 *Gram-Schmidt / orthogonalising correlated signals*

Your desk runs a production signal A that you cannot modify. Your candidate signal B has correlation $\rho = 0.8$ with A; both are demeaned and scaled to unit variance. **(a)** Write the orthogonalised signal $B_\perp$ via the projection formula and verify $\mathrm{Corr}(A, B_\perp) = 0$. **(b)** Compute $\mathrm{Var}(B_\perp)$. What fraction of B's variance was already contained in A? **(c)** Why must the signals be demeaned for "orthogonal" to mean "uncorrelated"? **(d)** What does $B_\perp$ represent economically, and when is trading $B_\perp$ rather than raw B the right call?

P6.9 *Beta neutralisation via PC1*

You run PCA on the returns of three assets and find the first principal component is proportional to $M = (1,1,1)^\top$ — the market mode. Your raw signal is $S = (2,-1,1)^\top$. **(a)** Beta-neutralise S by projecting off M: compute $S' = S - \frac{\langle S,M\rangle}{\langle M,M\rangle} M$. **(b)** Verify neutrality and interpret the property the new weights satisfy. **(c)** Show that when the market mode has exactly equal weights this projection coincides with a familiar simpler operation, and explain why the two differ for a real, estimated PC1.

P6.10 *A-orthogonal matrices* ♦

Define $\langle x, y\rangle_A = x^\top A y$, where A is symmetric positive definite. A matrix Q is *A-orthogonal* if $\langle Qx, Qy\rangle_A = \langle x, y\rangle_A$ for all x, y. Take $A = \mathrm{diag}(2,3,5)$. Which of the following is A-orthogonal?

(A) the permutation swapping $e_1 \leftrightarrow e_3$
(B) $Q = \mathrm{diag}(1,-1,1)$
(C) the permutation swapping $e_1 \leftrightarrow e_2$
(D) the rotation by angle t in the x–y plane

20.7 Problems — §7 Calculus for Quant Interviews

P7.1 *Derivatives toolkit: power rule* ♦

What is the derivative of $\sqrt{x}$ with respect to x?

P7.2 *Derivatives toolkit: chain-rule cycle* ♦

What is the fourth derivative of $\sin(4x)$?

P7.3 *Derivatives toolkit: logistic identity*

A prediction-market contract's price is modelled as $p(x) = \dfrac{1}{1+e^{-x}}$, where x is the log-odds implied by the news flow. (a) Show that $\dfrac{dp}{dx} = p(1-p)$. (b) The contract currently trades at 80 cents; by how much does the price move per small unit of log-odds? (c) At what price is the contract most sensitive to news, and why does a trader care?

P7.4 *Derivatives toolkit: optimisation recipe*

You are quoting a two-sided market and must choose your half-spread $s > 0$ (in ticks). Wider quotes earn more per fill but are filled less often: model the fill probability as $e^{-s/2}$, so the expected profit per quote is $f(s) = s\,e^{-s/2}$. Find the half-spread that maximises expected profit, justify that it is a maximum, and report the maximum value.

P7.5 *Integration by parts* ♦

Compute the indefinite integral $\displaystyle\int \ln x\,dx$.

P7.6 *Integration: substitution to Gamma + symmetry caveat*

(a) Evaluate $\displaystyle\int_0^\infty x^3 e^{-x^2}\,dx$. (b) Evaluate $\displaystyle\int_{-\infty}^\infty x^3 e^{-x^2}\,dx$, and explain why the symmetry argument you use is legitimate here but would *not* justify writing $\int_{-\infty}^\infty x\,dx = 0$.

P7.7 *Improper integrals: p-tests, comparison, log refinement*

For each integral, state *converge* or *diverge* with a one-line justification (verdict plus one line is full credit):

- (a) $\displaystyle\int_1^\infty \frac{dx}{x^{3/2}}$
- (b) $\displaystyle\int_0^1 \frac{dx}{\sqrt{1-x}}$
- (c) $\displaystyle\int_2^\infty \frac{dx}{x\ln x}$
- (d) $\displaystyle\int_1^\infty \frac{2+\sin x}{x}\,dx$
- (e) $\displaystyle\int_0^\infty x^7 e^{-x/10}\,dx$

P7.8 *Improper integrals: interior singularity*

A candidate writes

$$\int_{-1}^1 \frac{dx}{x^2} = \Big[-\frac{1}{x}\Big]_{-1}^1 = (-1) - (1) = -2.$$

The interviewer raises an eyebrow. What is wrong, and what is the correct conclusion?

P7.9 *Improper integrals: combined p-tests via limit comparison*

Find all real values of p for which

$$I(p) = \int_0^\infty \frac{dx}{x^p(1+x)}$$

converges.

P7.10 *Moments as integrals: normalisation via Gamma*

A nonnegative random variable X has density $f(x) = c\,xe^{-x/2}$ for $x \ge 0$ (and 0 otherwise). Find c, $\mathbb{E}[X]$, and $\mathrm{Var}(X)$.

P7.11 *Moments as integrals: Gaussian fourth moment*

Daily returns on a book are modelled as $X \sim \mathcal{N}(0, \sigma^2)$. (a) Derive $\mathbb{E}[X^4]$ from the integral definition. (b) On real daily returns your desk measures $\widehat{\mathbb{E}[X^4]}\,/\,\hat{\sigma}^4 \approx 9$. What does the mismatch tell you, and give one practical consequence for risk management.

P7.12 *Taylor micro-kit: limit*

Compute $\displaystyle\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$.

P7.13 *Taylor micro-kit: compounding estimate*

Without a calculator, estimate $(1.02)^{30}$ — e.g. the growth factor of 30 periods of a steady 2% return. State the approximation you use and how you would sanity-check the result.

P7.14 *Change of variables in an ODE* ⧫

Suppose $dx/dt = x$. Define $\tau = 1/t$ (for $t > 0$). Which equation does $x(\tau)$ satisfy?

(A) $dx/d\tau = -x$
(B) $dx/d\tau = -x/\tau$
(C) $dx/d\tau = -x/\tau^2$
(D) $dx/d\tau = \tau x$

P7.15 *Differentiate a parameter-dependent integral* ⧫

Let $F(x) = \displaystyle\int_x^{2x} s\,x\,ds$ (the integration variable is s; x appears both in the limits and in the integrand). Compute $F'(x)$.

(A) $2x\int_0^x s\,ds$
(B) $\frac{9}{2}x^2$
(C) x^2
(D) $x^2/2$

P7.16 *Probability by integration* ⧫

Let $X, Y \sim \text{Uniform}(0,1)$ be i.i.d. What is $P(X^2 > Y)$?

(A) $1/2$
(B) $1/\pi$
(C) $1/3$
(D) $1/4$
(E) $\pi/4$

P7.17 *Newton's method: the slowest root* ⧫

We use Newton iteration to find the roots of $f(x) = \tanh(x^4 - x^2)$. Assuming correct initialization near each root, which root has the slowest convergence?

(A) The largest one.
(B) Newton's method always converges quadratically.
(C) The smallest one.
(D) The one with the smallest absolute value.

20.8 Problems — §8 ML Theory: Trees, Ensembles, Bias–Variance

P8.1 *Bias-variance decomposition*

Let $y = f(x) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$, $\text{Var}(\varepsilon) = \sigma^2$, and ε independent of the training data. A learning algorithm trained on a random training set D produces a predictor $\hat{f}_D$. At a fixed test point x_0, derive the decomposition of the expected squared prediction error $\mathbb{E}_{D,\varepsilon}\left[(y - \hat{f}_D(x_0))^2\right]$ into bias squared, variance, and irreducible noise, justifying precisely why each cross term vanishes. Then: if you collect more training data (same model class), which of the three terms shrink?

P8.2 *Decision trees: Gini and information gain*

A classification node holds 16 samples: 8 positive, 8 negative. A candidate split sends 8 samples left (6 positive, 2 negative) and 8 right (2 positive, 6 negative). (a) Compute the Gini impurity of the parent, the weighted Gini after the split, and the impurity decrease. (b) Compute the information gain of the split in bits. (c) Which of these numbers change if you first apply the monotone transform $x \mapsto \log x$ to the split feature?

P8.3 *Trees cannot extrapolate*

You train a gradient-boosted tree model on 2015–2019 data to predict next-day realised volatility of an equity index. One feature is the VIX level, which ranged between 10 and 30 in the training window. In March 2020 the VIX prints 80. Roughly what will the model output, and why? What would a linear regression trained on the same data do differently? Name two feature-engineering changes that would have made the tree model behave better out of range.

P8.4 *Ensemble variance formula and the correlation floor*

N models produce predictions $X_1, \dots, X_N$, each with variance σ^2 and common pairwise correlation ρ. (a) Derive $\mathrm{Var}\left(\frac{1}{N}\sum_i X_i\right)$. (b) Take $N \to \infty$ and explain what the limit means for random forests — in particular, why do they subsample *features at every split* rather than only bootstrapping rows? (c) How negative can ρ be for the covariance structure to remain valid, and what is the ensemble variance at that boundary? (d) With $\sigma^2 = 1$ and $\rho = 0.2$, how many models are needed for the ensemble variance to be within 0.05 of its large-N floor?

P8.5 *Bootstrap and out-of-bag*

A bootstrap sample draws n times with replacement from n training points. (a) What is the probability that a given training point never appears in the sample, and what is its limit for large n? (b) On average, what fraction of *distinct* training points does each tree in a bagged ensemble see? (c) What practical free lunch does this give random forests?

P8.6 *Boosting as functional gradient descent*

Gradient boosting builds $F_t = F_{t-1} + \eta\, h_t$. (a) For squared loss $L(y, F) = \frac{1}{2}(y - F)^2$, show that the negative functional gradient evaluated at the current model is exactly the residual — so "fit the next tree to the residuals" is literally gradient descent in function space. (b) What does the fitting target become under absolute loss $L = |y - F|$? (c) Boosting and bagging both combine many trees: contrast which error component each attacks, and list the knobs that control overfitting in boosting. Does adding more trees eventually overfit in each case?

P8.7 *Good features for tree models* ★

What makes a good feature for LightGBM — or for tree-based models in general?

20.9 Problems — §9 Deep Learning

P9.1 *Backprop mechanics and vanishing gradients*

Consider a deep scalar chain $a_\ell = \sigma(w_\ell\, a_{\ell-1})$ for $\ell = 1, \dots, L$ with input a_0, $|a_0| \le 1$, and sigmoid activation σ. (a) Backpropagation and finite differences both produce gradients: what exactly does backprop reuse that makes it cheap, and roughly what would finite differences cost per gradient step for a network with 10^7 parameters and a 10 ms forward pass? (b) Write $\partial a_L / \partial w_1$ as a product and show why it decays exponentially in depth for sigmoid activations; give a numeric bound for $L = 20$ when every $|w_\ell| \le 1$. (c) For each of ReLU, residual connections, LSTM gating, and batch normalisation, state in one line which factor of that product it repairs.

P9.2 *Self-attention and the sqrt(dk) scaling*

Scaled dot-product attention computes $\mathrm{softmax}(QK^\top/\sqrt{d_k})V$. (a) Assuming query and key components are independent with zero mean and unit variance, compute $\mathrm{Var}(q \cdot k)$. (b) For

$d_k = 64$, what is the standard deviation of an *unscaled* score, and what does softmax do to logits at that scale — give a concrete two-logit example? (c) Why does that kill training — what happens to the gradient through the softmax? (d) What property does dividing by $\sqrt{d_k}$ restore?

P9.3 *Transformers on tick data*

A desk proposes running a transformer directly on raw ticks: a liquid instrument prints about 500,000 ticks per day and they want a full-day context window. (a) How many entries does the attention score matrix have per head per layer, and how much memory is that at 2 bytes per entry? (b) Someone suggests FlashAttention as the fix — what does it actually fix, and what does it not fix? (c) In live trading a new tick arrives and you want an updated prediction: compare the per-tick inference cost shape of the transformer (with a KV cache) against a state-space model such as S4/Mamba. (d) What would a pragmatic first system do instead?

P9.4 *The model stopped working*

A systematic strategy performed well from 2005 to 2018, then decayed to zero and stayed there. Give five distinct hypotheses for what happened, and for each one a concrete diagnostic you would run. Which two hypotheses do you check first, and why?

P9.5 *Numpy vs lists, and why PyTorch exists* ★

Why are numpy arrays faster than Python lists? And given that, why doesn't PyTorch just use numpy arrays by default?

P9.6 *GPU vs CPU speedup* ★

What order of magnitude of speedup do you get from GPU versus CPU parallelisation? When does that answer change?

P9.7 *Python class variables* ⧫

In Python, what is true of a class (static) variable?

P9.8 *Training bottleneck diagnosis* ⧫

You are training a model and observe low GPU and CPU utilisation while the hard drive runs at 100% capacity reading data files. What standard term describes this system state?

P9.9 *Fail fast vs fail silent* ⧫

A data pipeline takes 3 days to run. If it detects an unexpected null/NaN in a critical feature column during the first hour, why is it generally better to crash immediately ("fail loudly") rather than skip the bad data and continue ("fail silently")?

P9.10 *Immutability and concurrency* ⧫

Functional programming encourages immutability — data cannot be changed once created. Why is this property highly valuable for multi-threaded systems?

P9.11 *Shared-node OOM forensics* ⧫

A job on a shared cluster node runs without issue, but neighbouring jobs on the same node crash with out-of-memory (OOM) errors. What is the most likely cause?

20.10 Problems — §10 Sharpe, Loss Probabilities and Model Combination

P10.1 *P(loss) = Phi(-SR) and time scaling*

A strategy's annual returns are normally distributed with annualised Sharpe ratio 1.5 (take the mean/vol ratio directly, no risk-free adjustment). (a) What is the probability of a losing year? (b)

Assuming returns are i.i.d. across time, what is the probability of a losing month? (c) Why does the monthly number look so much worse when the strategy has not changed?

P10.2 *Sharpe to loss probability with assumption demolition* ★

A strategy has an annual Sharpe ratio of 2 and annual P&L volatility of $50M. What is the probability of losing $100M? State your assumptions and justify them. Is the true probability higher or lower in reality?

P10.3 *Cantelli's inequality: proof, application, tightness*

(a) Prove Cantelli's inequality (the one-sided Chebyshev bound): for any random variable X with mean μ and variance σ^2, and any $k > 0$,

$$P(X - \mu \le -k\sigma) \;\le\; \frac{1}{1 + k^2}.$$

(b) A strategy has annual mean P&L $100M and volatility $50M, distribution otherwise unknown. Give the best distribution-free bound on the probability of losing at least $100M in a year, and compare it with the Gaussian answer. (c) Show the bound is tight by exhibiting a two-point distribution that achieves it.

P10.4 *Sharpe vs EV at maximum capacity* ♦

Assuming all strategies are at maximum capacity (none can be scaled up), which strategy would a market-making firm most prefer?

(A) Expected value $5,000/day with standard deviation $1/day.
(B) Expected value $20,000/day with standard deviation $30,000/day.
(C) Expected value $50,000/day with standard deviation $100,000/day.
(D) Expected value $100,000/day with standard deviation $400,000/day.

P10.5 *Var(A+B) and the Sharpe of combined strategies*

Strategies A and B each have annualised expected P&L $10M and annualised volatility $10M (Sharpe 1 each). You run both at full size, so the book's P&L is $A + B$.

(a) If $\mathrm{Corr}(A, B) = 0.5$, what are the book's volatility and Sharpe?
(b) If $\mathrm{Corr}(A, B) = 0$?
(c) If $\mathrm{Corr}(A, B) = -0.3$?
(d) Derive the rule for N uncorrelated strategies with equal Sharpe and equal vol, and explain why the general claim $\mathrm{SR}_{\text{total}} = \sqrt{\sum_i \mathrm{SR}_i^2}$ requires risk-balanced weights rather than a raw sum.

P10.6 *Decorrelating a new model from a black box*

Your desk already runs a profitable black-box model A — you can observe its outputs and P&L but not its internals. You have built a new model B whose output correlates 0.7 with A's. (a) A colleague proposes: "drop every feature from B's training set that is correlated with A's output, then retrain." Give two reasons this is a poor primary approach. (b) What would you do instead? Give the one-line exact method and one training-time alternative, and state the trade-off in each case.

20.11 Problems — §12 Market Making I: the Economics of the Spread

P12.1 *Spread decomposition*

An interviewer asks: "Where does the bid-ask spread come from? Name its components, say which of them widen when volatility rises, and which get compressed by competition between market makers."

P12.2 *Glosten-Milgrom zero-profit quotes*

A friend claims: "A market maker with no directional opinion can quote an arbitrarily tight spread around fair value and still break even, because incoming buys and sells are equally likely." Test the claim in a miniature model: the asset's value is $V \in \{99, 101\}$, equally likely. A fraction $\pi = 0.3$ of order flow is informed (knows V and trades in the profitable direction); the rest is noise, buying or selling with probability $\frac{1}{2}$ each. (a) Where must a zero-profit maker set the ask and the bid? (b) What is the breakeven spread, and what exactly is wrong with the friend's claim?

P12.3 *Glosten-Milgrom with asymmetric prior*

Take the standard Glosten–Milgrom miniature — $V \in \{99, 101\}$, informed fraction $\pi = 0.2$, noise traders buy/sell with probability $\frac{1}{2}$ — but make the prior asymmetric: $P(V = 101) = p = 0.6$. (a) Compute the zero-profit ask and bid. (b) Is the spread still 2π? Is the quote midpoint still the prior mean? Which quote updates further from the prior mean, and why? (c) Sanity-check your quotes in the limits $p \to 1$ and $\pi \to 0$.

P12.4 *Mark-outs*

You run a market-making book. (a) An interviewer says: "Adverse selection is a nice theory word — how would you actually *measure* it in your fills?" (b) Your measurement then shows that fills against counterparty X mark 4 bp against you 5 seconds after trading, while all other flow marks flat. (c) Separately, volatility doubles on a macro print. (d) Separately, the exchange announces it will publish unscheduled news in 30 seconds. What do you do to your quotes in each of (b), (c), (d), and why?

20.12 Problems — §13 Market Making II: Inventory and Skewing

P13.1 *Reservation price computation*

You are quoting a coin whose mid is $s = 100$. Your inventory is $q = +60$ units (long), your risk-aversion parameter is $\gamma = 0.001$, the variance rate is $\sigma^2 = 2.5$ per day (in squared price units), and $T - t = 0.4$ days remain in your risk horizon. (a) Compute the Avellaneda–Stoikov reservation price $r = s - q\gamma\sigma^2(T - t)$. (b) If you keep a fixed half-spread of 0.10 around r, what are your quotes, and what are they doing in trading terms? (c) What are the quotes if instead you are short 60 units?

P13.2 *Reservation-price anatomy, skew vs widen*

An interviewer pushes on the reservation-price formula $r = s - q\gamma\sigma^2(T - t)$. (a) Justify in one line why each factor q, γ, σ^2, $(T - t)$ belongs in the correction. (b) What happens to the skew as $T - t \to 0$, and does that make trading sense? (c) Why does the maker shift *both* quotes rather than only improving the ask when long? (d) Skewing and widening are both defensive moves — what different risk does each address?

P13.3 *Fermi estimation + interval calibration* ★

You have 15 seconds. Bitcoin blocks arrive on average every 10 minutes. Estimate how many blocks have been mined since 2012, and give a 95% confidence interval on your estimate. (After you answer, the interviewer will make you trade a market on this number — both the point estimate and, especially, the interval are graded.)

P13.4 *Make-a-market game: position and P&L accounting* ★

You have just estimated the number of Bitcoin blocks mined since 2012 (a quantity you do not know precisely). The interviewer says: “Now make a market on your estimate.” One unit trades per quote, prices in blocks. The sequence, as played:

- You quote 100k–200k — interviewer **BUYS**.
- You quote 400k–600k — **BUYS**.
- You quote 900k–1M; asked “how confident are you?” you say 10% — **BUYS**.
- You quote 4M–6M — **SELLS**.
- You quote 2M–3M — no trade; game stops.

Closing question (the one the game builds to): are you long or short, and how much? Also give your cash, your mark-to-market P&L at your final mid, and what the buy-buy-buy-then-sell pattern told you about the interviewer’s view while it was happening.

P13.5 *Pre-trade risk controls / position limits* ♦

What is the best way to stop trading algorithms from accidentally entering very large positions which may cause extreme losses during price movements?

(A) Have traders monitor the algorithms’ positions on a dashboard and intervene manually.
(B) Send an automated alert to the team whenever a position becomes large.
(C) Enforce hard position limits that automatically block the algorithm from trading further on the side that would increase the breach.
(D) Backtest the algorithm thoroughly before deploying it.

P13.6 *Inventory hedging with perps*

You make markets in spot BTC on venue A, and persistent one-way retail flow has left you long a large spot inventory. Skewing your quotes has not shed it fast enough, and dumping the inventory through your own market would move the price against you and against your own resting quotes. A colleague says: “just pull your quotes until it clears.” Propose a better solution using instruments or venues beyond your own book, and name the residual risks you now carry.

20.13 Problems — §14 Market Microstructure

P14.1 *Square-root impact law by dimensional analysis*

Empirically, executing a metaorder of Q shares in a market with daily volume V and daily volatility σ moves the price by $\Delta P \approx Y\sigma\sqrt{Q/V}$ with $Y \approx 1$. Derive the exponents by dimensional analysis: posit $I = Y\,Q^{\alpha}V^{\beta}\sigma^{\gamma}$ with units $[I]$ = price, $[Q]$ = shares, $[V]$ = shares/day, $[\sigma]$ = price/$\sqrt{\text{day}}$, and show that α, β, γ are forced. Then say what extra modelling assumption the ansatz itself sneaked in.

P14.2 *Impact vs alpha capture*

A stock has daily volatility 2% and average daily volume 10 million shares. Your signal says the stock will rise 25 bp over the next two hours, and your desired position is 250,000 shares. (a) Using the square-root law with $Y = 1$, estimate your market impact. (b) Can you capture the alpha? (c) A colleague suggests slicing the order over five days to shrink the impact — what is the catch?

P14.3 *Kyle’s lambda and reconciliation with concave impact*

You regress one-minute price changes on signed volume, $\Delta P_t = \lambda Q_t + \varepsilon_t$, and find that trades of 10,000 shares move the price 2 bp on average. (a) What does λ measure, and what would you use it for? (b) Predict the average move for a 40,000-share trade under (i) the linear Kyle model and (ii) the empirical square-root law. (c) The two predictions disagree — how does the empirical literature reconcile a linear λ with concave total impact, and which prediction should you trust?

P14.4 *OBI and VAMP computation*

Top of book: bid 99 with 1,000 shares, ask 100 with 100 shares. (a) Compute the order-book imbalance $\text{OBI} = \frac{q_b - q_a}{q_b + q_a}$ and the volume-adjusted mid $P_{\text{VAMP}} = \frac{P_a q_b + P_b q_a}{q_b + q_a}$. (b) The VAMP formula weights each price by the *opposite* side's volume — explain the intuition using this book. (c) An interviewer at an HFT-adjacent shop asks: "so is that the micro-price?" Give the precise answer.

P14.5 *OBI signal monetisation*

A junior researcher backtests the rule "when OBI > 0.8, cross the spread and buy; exit after 10 seconds" on top-of-book data for a liquid crypto pair, ignoring costs, and reports a 58% hit rate on next-tick direction. He proposes going live. Give at least three distinct reasons this strategy loses money live *even if the 58% is real*, and describe how a market maker would actually monetise the same signal.

20.14 Problems — §15 Crypto Market Structure

P15.1 *Bitcoin mechanics* ★

Walk me through the basics of how Bitcoin works: how are new blocks mined, how are blocks linked into a chain, how does a user get a transaction included, and how do miners get paid?

P15.2 *Cross-exchange arbitrage, leg risk* ♦

You run a strategy that fires simultaneous orders — buy on centralized exchange A, sell on centralized exchange B — whenever the same coin shows a small, fleeting price discrepancy between the two venues. What is the biggest risk in executing this arbitrage for very fast, small opportunities?

- (a) Trading fees eat the margin.
- (b) Network latency makes you miss many of the opportunities.
- (c) Only one of the two legs fills, leaving an unhedged position.
- (d) Withdrawal and transfer times lock up your capital between venues.

P15.3 *The market-maker mandate* ♦

A token project engages a market-making firm as a designated liquidity provider for its token. Which of the following does the market maker promise the project: higher prices, or higher volumes?

P15.4 *Perpetual funding arithmetic*

A perpetual future on ETH trades above its spot index, and the venue prints a funding rate of 0.015% per 8-hour interval. (a) Who pays whom, and which way does the mechanism push the perp price? (b) Annualise the rate, simple and (approximately) compounded. (c) You are long $2M of the perp; roughly what does funding cost you per week if the rate persists?

P15.5 *Cash-and-carry*

Funding on a major venue's BTC perpetual has run at an annualised +30% for two weeks. A colleague proposes: buy $10M of spot BTC, short $10M of the perp, collect 30% a year delta-neutral — "free money." Present the strongest version of the trade, then explain what is wrong with the free-money claim (give at least four distinct risks), and say what the persistently positive funding is telling you about market positioning.

P15.6 *Liquidation price and cascades*

A trader opens a 10x leveraged long on a BTC perpetual at $50,000, posting initial margin equal to 10% of entry notional. The venue liquidates the account when equity (initial margin plus mark-to-market P&L) falls to the maintenance requirement of 0.5% of position notional marked at the current price. (a) At approximately what price is the position liquidated? Give the general rule of thumb for leverage L. (b) Venue data shows a dense cluster of similar longs with liquidation prices just below \$45,500. Describe what happens if price trades down into the cluster, and how you would react as a market maker quoting this book.

P15.7 *Constant-product AMM pricing and impact*

An ETH/USDC constant-product pool ($xy = k$, no fees) holds $x = 100$ ETH and $y = 200{,}000$ USDC. (a) What price does the pool currently quote? (b) A trader buys 10 ETH from the pool: how much USDC do they pay, what average price do they get, and what is the pool's new marginal price? (c) Suppose ETH trades at \$2,000 on a centralized exchange throughout. What happens next, and why is that mechanism essential to how an AMM tracks fair value?

P15.8 *MEV sandwich attack*

Same pool as the previous problem: $x = 100$ ETH, $y = 200{,}000$ USDC, $xy = k$, no fees. A victim's order to buy 10 ETH sits in the public mempool with a loose slippage tolerance. An attacker sandwiches it: front-runs with their own 10 ETH buy, lets the victim's buy execute, then back-runs by selling 10 ETH. (a) Compute the attacker's profit, ignoring gas and pool fees. (b) Show the attacker ends flat in ETH and identify exactly whose money the profit is. (c) What limits this attack in practice, and what does its existence imply about sending orders to a public mempool?

P15.9 *Impermanent loss*

You deposit equal dollar values of ETH and USDC into a constant-product AMM ($xy = k$, no fees); arbitrageurs always keep the pool price equal to the external market price. The ETH price then moves from p_0 to $r\,p_0$. (a) Derive the pool's holdings and the value of the LP position as a function of r. (b) Show that $V_{\mathrm{LP}}/V_{\mathrm{HODL}} = 2\sqrt{r}/(1+r)$, conclude $\mathrm{IL}(r) = 2\sqrt{r}/(1+r) - 1 \leq 0$, and evaluate at $r = 2$ and $r = 4$. (c) Show $\mathrm{IL}(r) = \mathrm{IL}(1/r)$ and that for a small move $r = 1 + \varepsilon$, $\mathrm{IL} \approx -\varepsilon^2/8$. What option-like position does this make the LP, and when is providing liquidity rational?

P15.10 *Impermanent-loss benchmark*

You deposited \$5,000 of ETH and \$5,000 of USDC into a constant-product pool when ETH traded at \$2,000. ETH now trades at \$4,000 and your LP position is worth \$14,142 — up 41% — so you conclude that providing liquidity beat holding. What is wrong with that conclusion, exactly how much did LPing cost you, and what would have to be true for the LP position to have genuinely won?

20.15 Problems — §16 Options and Volatility

P16.1 *Black-76 digital, prediction-market link*

BTC futures for delivery in 30 days trade at $F = 60{,}000$; the 70,000-strike implied volatility on the options venue is 60% and rates are approximately zero. (a) Under Black-76, price the cash-or-nothing digital paying \$1 if the future settles above 70,000 (take $T = 30/365$). You may use $\ln(7/6) \approx 0.154$ and $\Phi(0.98) \approx 0.837$. (b) A prediction market quotes "BTC above 70k in 30 days" at 25 cents. Interpret the gap and name the main caveats you would check before trading it.

P16.2 *Breeden-Litzenberger*

(a) Starting from $C(K) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[(S_T - K)^+\big]$, derive the price of a cash-or-nothing digital (paying 1 if $S_T > K$) and the Breeden–Litzenberger identity for the risk-neutral density $f_{\mathbb{Q}}(K)$. (b) With $r = 0$, listed calls trade at $C(95) = 7.00$, $C(100) = 4.00$, $C(105) = 2.00$. Estimate the price of the digital struck at 100 and the risk-neutral density at 100. (c) Which no-arbitrage condition guarantees your density estimate is non-negative, and what is this construction used for in practice?

P16.3 *Sqrt-t vol scaling*

An index has an annualised volatility of 20%. (a) What is its typical daily move? Its typical weekly move? (b) A colleague annualises a 1.26% daily vol by multiplying by 252 and reports 318% — what exactly went wrong? (c) State the assumption behind the square-root-of-time rule and give one realistic way markets violate it.

P16.4 *Variance risk premium, delta-hedged P&L*

You buy a 1-month at-the-money straddle on an index at 25% implied volatility and delta-hedge daily; realised volatility over the month comes in at 20%. (a) Do you make or lose money? Derive the standard expression for the daily P&L of a delta-hedged option that justifies your answer. (b) Roughly how much do you lose as a fraction of the premium paid? (You may use the approximation that an ATM straddle costs about $0.8\, S\, \sigma \sqrt{T}$.) (c) This outcome is the norm rather than the exception — name the phenomenon, explain why it persists, and describe the risk profile of the trader on the other side of your trade.

20.16 Problems — §17 Algorithmic Cost Models

P17.1 *Matmul operation count* ★

What is the complexity — the number of operations — of matrix multiplication? Cover the count for a general $(m \times n)(n \times p)$ product and for the square $n \times n$ case, say which operation is conventionally counted and why, and mention what is known below the naive exponent.

P17.2 *Why count multiplications* ⧫

When estimating the speed of matrix operations in AI, why is it common practice to count the number of multiplications rather than the number of additions?

P17.3 *Low-rank decomposition crossover* ★

Multi-step, pen and paper. **(a)** What is the cost, in multiplications, of multiplying an $N \times P$ matrix by a $P \times R$ matrix? **(b)** A dense $N \times P$ matrix M is replaced by a low-rank factorisation $M = UV$, with U of size $N \times K$ and V of size $K \times P$. To apply M to a $P \times R$ block of data B you may compute $(UV)B$ or $U(VB)$: cost both association orders, state which one you must use, and derive the condition on K under which the factored form beats the dense product MB.

P17.4 *Low-rank speedup, numeric*

An inference engineer replaces a dense 4096×4096 layer with a rank-64 factorisation $(4096 \times 64)(64 \times 4096)$ and claims each forward pass through the layer becomes about 32 times cheaper. **(a)** Verify or refute the claim with a multiplication count for a single input vector. **(b)** What is the break-even rank? **(c)** Name the first-order cost of this trade that the multiplication count does not capture.

P17.5 *Membership: scan vs sort vs hash* ★

Multi-step. You must answer q membership queries ("is x in the dataset?") against a static dataset of N items. Compare three designs: (i) a linear scan per query; (ii) sort once, then binary-search each query; (iii) build a hash set once, then probe each query. Give the total cost of each, derive roughly how many queries make sorting pay for itself relative to scanning, and state when each design is the right choice.

P17.6 *CS fundamentals rapid-fire* ⧫

Rapid-fire, multiple-choice pace — one line of justification each.

- **(a)** A slow function always returns the same output for the same input; you store results in a dictionary so repeat calls return instantly. What is this optimisation technique called? (One tempting option read: "dynamic programming.")
- **(b)** A brute-force nearest-neighbour search compares the query against every record. If the dataset size doubles, roughly what happens to the search time?
- **(c)** A stack (LIFO) undergoes, in order: Push(10), Push(20), val = Pop(), Push(40), Push(30), Push(val), Pop(), Pop(). What is the value of the last item popped?
- **(d)** A recursive function calls itself to depth N without tail-call optimisation. What is the impact on space complexity?
- **(e)** Simplify the Boolean expression $[(A \vee B) \wedge (A \vee \neg B)] \vee [(C \wedge D) \wedge (C \wedge \neg D)]$ to its most minimal form.

20.17 Problems — §18 Forecasting Returns: the Open-Ended Question

P18.1 *Open-ended process question*

An interviewer closes your first round with: "Suppose you wanted to predict tomorrow's return of a major equity index — say the Euro Stoxx 50. Walk me through, end to end, how you would approach it." Give the full process you would narrate, including the design choices you would flag explicitly and the risks you would name unprompted.

P18.2 *Backtest critique* ⧫

A strategy of buying Ethereum at 9am every day and closing the position at 7pm backtests very well over the past seven years. What would be the main issue in executing this strategy?

P18.3 *Open-ended design choice: direct vs factorial*

You are building a weekly return forecast for the CAC 40. A colleague fits models directly to the index return series. You propose instead modelling the 40 constituents and aggregating the forecasts with index weights. Make the case for the constituent (factorial) route, explain where PCA enters, and name at least two ways the aggregation can fail.

20.18 Problems — Puzzle Classics

PZ.1 *Monty Hall*

A game show has three doors: one hides a car, two hide goats. You pick a door. The host — who knows where the car is and always opens a goat door from the two you did not pick — opens one, revealing a goat, and offers you the chance to switch to the remaining closed door. Should you switch, and what is your probability of winning the car if you do? Would your answer change if the host had opened one of the other doors uniformly at random and it merely *happened* to show a goat?

PZ.2 *Two ropes, 45 minutes*

You have two ropes and a lighter. Each rope takes exactly 60 minutes to burn from one end to the other, but they burn *non-uniformly*: half the rope may take 5 minutes, the rest 55. How do you measure exactly 45 minutes?

PZ.3 *Clock-hands angle* ⧫

Mental only, 30 seconds. What is the angle between the hands of a clock at 09:30?

PZ.4 *100 lockers*

100 lockers are all closed. Person 1 toggles every locker; person 2 toggles lockers $2, 4, 6, \ldots$; person k toggles every k-th locker; and so on through person 100. After all 100 passes, how many lockers are open, and which ones?

PZ.5 *25 horses, 5 lanes*

You have 25 horses and a five-lane track with no stopwatch: each race ranks its five runners, nothing more. Each horse always runs at its own constant speed. What is the minimum number of races needed to find the fastest three horses, in order?

PZ.6 *Sixteen gamers, four per game* ♦

In a gaming competition, 4 players play each game and each game produces exactly one winner. Assuming the best player always wins any game they play, how many games are needed to find the best gamer in a group of 16 employees?

PZ.7 *Prisoners' hats in a line*

n prisoners stand in a line, each wearing a black or white hat, each seeing only the hats *in front* of them. Starting from the back (the prisoner who sees all $n-1$ others), each must call out "black" or "white"; a prisoner is freed iff they call their own hat's color. Everyone hears every call. The prisoners may agree on a strategy beforehand. How many can be guaranteed freedom, and by what strategy?

PZ.8 *Hats: guess or pass*

Three players are each given a hat, independently red or blue with probability $1/2$. Each player sees the other two hats but not their own. Simultaneously, and with no communication once the hats are on, each player must either name a colour or pass. The team wins if at least one player correctly names their own hat colour and nobody names wrongly; passing is always safe. The team agrees on a strategy in advance. A single designated guesser wins with probability $1/2$ — find a strategy that does strictly better, compute its winning probability, and prove no strategy can beat it.

PZ.9 *Broken stick triangle*

A stick is broken at two points chosen independently and uniformly at random along its length. What is the probability that the three pieces can form a triangle?

PZ.10 *Four points on a quarter arc*

Four points are chosen independently and uniformly at random on a circle. What is the probability that all four lie on some arc spanning a quarter of the circumference? Give also the general formula for n points and an arc that is a fraction $f \leq 1/2$ of the circumference.

PZ.11 *Random walk on a solid*

An ant stands on one vertex of a regular octahedron and walks along edges, choosing uniformly at random among the available edges at every vertex. What is the expected number of steps for it to reach the opposite (antipodal) vertex? (The octahedron has 6 vertices, each of degree 4; each vertex is adjacent to every vertex except its antipode.)

PZ.12 *Bold vs timid play*

You walk into a casino with \$100 and must walk out with \$200 by midnight — anything less is worthless to you. You may stake any amounts on even-money bets that win with probability $p = 18/38$ (roulette red). Compare two plans: (a) one bold bet of \$100; (b) timid play, betting \$1 at a time until you reach \$200 or go bust. Compute both success probabilities, state the general lesson, and give the trading analogue.

PZ.13 *Fair coin to probability 1/3* ★

Using only a fair coin, construct an event that occurs with probability exactly 1/3. Follow-ups to be ready for: what is the expected number of flips your construction uses, and can exact probability 1/3 be achieved with a fixed, predetermined number of flips?

21 Solutions

21.1 Solutions — §1 Expected Value for Trading Games

P1.1 *Independence and the product rule* ♦

Independence is given, so the joint probability is the product:

$$P(\text{burger} \cap \text{stomach}) = P(\text{burger})\,P(\text{stomach}) = 0.4 \times 0.1 = 0.04.$$

The entire content of the question is knowing that independence licenses multiplication. Without the independence assumption, all that could be said is $0 \le P(A \cap B) \le \min(0.4,\, 0.1) = 0.1$ — the word "assuming" is doing real work.

Self-check. Consistency bounds: $P(A \cap B) = 0.04 \le \min(0.4, 0.1)$, and inclusion-exclusion gives $P(A \cup B) = 0.4 + 0.1 - 0.04 = 0.46 \le 1$. Both hold, so 0.04 is a legal joint probability for these marginals.

Answer: 0.04.

P1.2 *Binomial counting* ♦

Each of the $2^5 = 32$ sequences is equally likely; the favourable ones are the $\binom{5}{3} = 10$ sequences with heads in exactly 3 of the 5 positions:

$$P(\text{exactly 3 H}) = \frac{\binom{5}{3}}{2^5} = \frac{10}{32} = \frac{5}{16} = 0.3125.$$

Self-check. The full binomial row is $1, 5, 10, 10, 5, 1$, which sums to 32 as it must; and by the symmetry $H \leftrightarrow T$, $P(3\text{ heads}) = P(2\text{ heads}) = 10/32$, consistent with the row.

Answer: $5/16 = 0.3125$.

P1.3 *Two aces (hypergeometric draw)* ♦

The player's two cards are a uniformly random 2-subset of the deck (which other players also get cards does not matter — a specific player's hand is exchangeable with any other 2-subset). Favourable hands: $\binom{4}{2} = 6$ pairs of aces out of $\binom{52}{2} = 1326$ possible hands:

$$P = \frac{\binom{4}{2}}{\binom{52}{2}} = \frac{6}{1326} = \frac{1}{221} \approx 0.45\%.$$

Self-check. Sequentially: first card an ace with probability $4/52 = 1/13$, then second ace $3/51 = 1/17$, product $\frac{1}{13} \cdot \frac{1}{17} = \frac{1}{221}$ — same number by a different route.

Answer: $\binom{4}{2}/\binom{52}{2} = 1/221 \approx 0.45\%$.

P1.4 *Dice outcome counting* ♦

There are $8 \times 8 = 64$ equally likely ordered outcomes. Count the favourable sums:

- sum $= 14$: $(6, 8), (7, 7), (8, 6)$ — 3 outcomes;
- sum $= 15$: $(7, 8), (8, 7)$ — 2 outcomes;
- sum $= 16$: $(8, 8)$ — 1 outcome.

Total 6 of 64:
$$P(\text{sum} \geq 14) = \frac{6}{64} = \frac{3}{32} \approx 9.4\%.$$

Self-check. Reflect each die by $x \mapsto 9 - x$: this bijection maps $\{\text{sum} \geq 14\}$ onto $\{\text{sum} \leq 4\}$, which has $1 + 2 + 3 = 6$ outcomes (sums $2, 3, 4$) — the same count from the other tail.

Answer: $3/32 \approx 9.4\%$.

P1.5 *Binomial at-least* ♦

"2 or more" out of 3 means *both* the exactly-2 case and the all-3 case — under time pressure the second term is exactly what gets dropped:

$$P(\geq 2) = \binom{3}{2}\left(\tfrac{1}{4}\right)^2\left(\tfrac{3}{4}\right) + \left(\tfrac{1}{4}\right)^3 = \frac{9}{64} + \frac{1}{64} = \frac{10}{64} = \frac{5}{32} \approx 15.6\%.$$

On the live test this question was the one confirmed miss: any answer built from a single term (or from mangled terms) is wrong because the event is a union of two disjoint outcomes.

Self-check. Complement: $P(0) = (3/4)^3 = 27/64$ and $P(1) = 3 \cdot \frac{1}{4} \cdot (3/4)^2 = 27/64$, so $P(\geq 2) = 1 - \frac{27+27}{64} = \frac{10}{64}$ — the same value computed from the other side of the distribution.

Answer: $5/32 = 10/64 \approx 15.6\%$.

P1.6 *Distinct faces in ten rolls*

Let I_k be the indicator that face k appears at least once in the 10 rolls. The number of distinct faces is $D = \sum_{k=1}^{6} I_k$, and

$$\mathbb{E}[I_k] = P(\text{face } k \text{ appears}) = 1 - \left(\tfrac{5}{6}\right)^{10},$$

so by linearity
$$\mathbb{E}[D] = 6\Big(1 - \left(\tfrac{5}{6}\right)^{10}\Big) \approx 6\,(1 - 0.1615) \approx 5.03.$$

Legitimacy: linearity of expectation holds for *any* random variables, dependent or not. The I_k here are negatively dependent (one face appearing crowds out others), but that only matters for the variance of D, never for its mean. This is the master move of the whole section: write the count as a sum of indicators and sum the probabilities.

Self-check. Small case $n = 2$ rolls: the formula gives $6(1 - (5/6)^2) = 6 \cdot \frac{11}{36} = \frac{11}{6}$; directly, $\mathbb{E}[D] = 2 - P(\text{second roll repeats the first}) = 2 - \frac{1}{6} = \frac{11}{6}$. Match. Bounds: $5.03 < 6$ (cannot exceed the number of faces) and it is close to 6 minus about one face — consistent with the coupon-collector fact that 10 rolls usually leaves roughly one face unseen.

Answer: $6(1 - (5/6)^{10}) \approx 5.03$ distinct faces. □

P1.7 *Gold and silver balls* ♦

The turn structure is a red herring: who is drawing changes nothing about *which* balls leave the bag. Drawing without replacement just reveals a uniformly random permutation of all 200,000 balls one position at a time, and the process stops exactly when the 100,000th — i.e. the *last* — gold ball is revealed. So the balls pulled out are precisely every ball up to and including the last gold in a random permutation.

All 100,000 golds are pulled by construction. The only randomness is how many silvers are left behind, namely those lying *after* the last gold. For a fixed silver ball, consider just its position relative to the 100,000 golds: all 100,001 relative orderings are equally likely, so it lies after every gold with probability $\frac{1}{100{,}001}$. By linearity over the silvers,

$$\mathbb{E}[\#\text{silver left in bag}] = \frac{100{,}000}{100{,}001} \approx 1.$$

So on average $\approx$ 199,999 balls are pulled, of which 100,000 are gold: the proportion is

$$\approx \frac{100{,}000}{199{,}999} \approx 0.5000025,$$

a hair *above* one half. (Strictly, the expected ratio is not the ratio of expectations, but with 2×10^5 balls the pulled count concentrates completely relative to any plausible answer spacing.) The closest simple value is 0.5.

Self-check. Tiny case, 1 gold + 1 silver: the two orders are GS (pull only the gold; proportion 1) and SG (pull both; proportion $\frac{1}{2}$), so the expected proportion is $\frac{3}{4}$ — above $\frac{1}{2}$, as our analysis predicts, and the excess shrinks like $1/N$: at $N = 10^5$ it is invisible. Also, the expected-silvers-left formula gives $\frac{1}{2}$ for this tiny case ($S/(G+1) = 1/2$), matching the direct average of $\{1, 0\}$ silvers left.

Answer: 0.5 (exactly: a shade above — about 0.5000025, since on average only ≈ 1 silver ball is never pulled).

P1.8 *Geometric waiting time* ♦

The number of rolls is geometric with success probability $p = \frac{1}{6}$. First-step analysis: one roll is spent; with probability $\frac{5}{6}$ nothing has changed (memorylessness) and the expected remaining count is again E:

$$E = 1 + \tfrac{5}{6}E \quad \Longrightarrow \quad \tfrac{1}{6}E = 1 \quad \Longrightarrow \quad E = 6.$$

In general $E = 1/p$: the single most-used closed form in this family.

Self-check. Direct series: $E = \sum_{k \geq 1} k \left(\frac{5}{6}\right)^{k-1} \frac{1}{6} = \frac{1}{6} \cdot \frac{1}{(1-5/6)^2} = \frac{1}{6} \cdot 36 = 6$. Same value without the recursion.

Answer: 6 rolls.

P1.9 *Geometric waiting time: memorylessness*

(a) Geometric waiting time: $\mathbb{E}[T] = 1/p = 1/0.05 = 20$ seconds.

(b) Still 20 seconds. The geometric distribution is memoryless: $P(T > s + t \mid T > s) = P(T > t)$, because the per-second fill events are independent with constant probability — the 30 fruitless seconds carry no information about the future. If waiting *did* change your forecast, the per-second fill probability would have to depend on the past, contradicting the model. (The trading caveat worth saying out loud: real fill processes are *not* memoryless — a long wait often does carry adverse information, e.g. your quote has gone stale or you are deep in the queue. The clean answer holds only under the stated independence.)

(c) The first flip shows one of the two faces for free. From then on you wait for the *opposite* face, a geometric wait with $p = \frac{1}{2}$, so

$$\mathbb{E} = 1 + \frac{1}{1/2} = 3.$$

Self-check. For (c), compute from the distribution: both faces first complete at flip $k \geq 2$ with probability $P(T = k) = 2 \cdot (1/2)^k = (1/2)^{k-1}$ (the first $k - 1$ flips all equal, times 2 choices of which face repeats). Then $\mathbb{E}[T] = \sum_{k \geq 2} k(1/2)^{k-1} = \left(\sum_{k \geq 1} k(1/2)^{k-1}\right) - 1 = 4 - 1 = 3$. For (b), a one-line sanity: the process restarts identically each second, so conditioning on any fill-free past leaves the same law — consistent with (a).

Answer: (a) 20 s; (b) 20 s (memorylessness); (c) 3 flips.

P1.10 *Pattern waiting times: HH versus HT*

HT. States: S_0 (no H held yet), S_1 (currently hold an H). From S_1, a T finishes; an H *stays in* S_1 — failure does not destroy progress:

$$E_1 = 1 + \tfrac{1}{2} \cdot 0 + \tfrac{1}{2}E_1 \;\Rightarrow\; E_1 = 2, \qquad E_0 = 1 + \tfrac{1}{2}E_1 + \tfrac{1}{2}E_0 \;\Rightarrow\; E_0 = 2 + E_1 = 4.$$

HH. Same states, but now from S_1 a T sends you all the way back to S_0:

$$E_1 = 1 + \tfrac{1}{2} \cdot 0 + \tfrac{1}{2}E_0, \qquad E_0 = 1 + \tfrac{1}{2}E_1 + \tfrac{1}{2}E_0.$$

The second equation gives $E_0 = 2 + E_1$; substituting into the first, $E_1 = 1 + \frac{1}{2}(2 + E_1)$, so $E_1 = 4$ and $E_0 = 6$.

The sentence. The per-window probabilities are identical; the waiting times differ because of what a *failure* does to your progress: after H, a bad flip for HT (another H) leaves you still holding an H, while a bad flip for HH (a T) resets you to zero — self-overlapping patterns are slower to arrive. (Same mathematics as losing streaks that reset a strategy entirely: full-reset events cost more than their probability suggests.)

Self-check. Conway's overlap rule: $\mathbb{E}[\text{wait}] = \sum 2^k$ over every length k at which the pattern's prefix equals its suffix. HH overlaps with itself at $k = 2$ and $k = 1$: $4 + 2 = 6$. HT only at $k = 2$: 4. Both match the state solutions.

Answer: $\mathbb{E}[\text{flips to HH}] = 6$; $\mathbb{E}[\text{flips to HT}] = 4$. □

P1.11 *Pattern waiting times: the mental closed-form arsenal* ◆

The arsenal, with one-line reconstructions:

(a) $E = 1/p$. Justification: $E = 1+(1-p)E$ (one trial spent; failure restarts by memorylessness).

(b) $E_k = 2^{k+1} - 2$. Justification: to extend a run of $k-1$ heads, flip once more; tails restarts from scratch: $E_k = E_{k-1} + 1 + \frac{1}{2}E_k$, i.e. $E_k = 2E_{k-1} + 2$ with $E_1 = 2$. Values: $2, 6, 14, 30, \ldots$

(c) $E_k = \dfrac{6^{k+1} - 6}{5}$. Same recursion with the die: $E_k = E_{k-1} + 1 + \frac{5}{6}E_k$, i.e. $E_k = 6E_{k-1} + 6$ with $E_1 = 6$. Values: $6, 42, 258, \ldots$

(d) HH $= 6$, HT $= 4$, HTH $= 10$. Justification: state equations (see the HH/HT problem), or Conway's overlap rule $\sum 2^k$ over self-overlaps — HTH overlaps at $k = 3$ and $k = 1$: $8 + 2 = 10$.

(e) Coupon collector: $6(1 + \frac{1}{2} + \cdots + \frac{1}{6}) = 14.7$. Justification: with j faces unseen the wait is geometric with $p = j/6$; sum $6/j$.

Two neighbours from the same first-step-analysis family are worth keeping equally warm for such tests (they live with the Markov-chain material): fair-game ruin duration $i(N-i)$ and the ant-on-a-cube answer 10.

Why memorise. Each of these is a 60–120 second, multi-equation derivation if done honestly — and doing it *mentally* means holding two or three unknowns and their coefficients in working memory while a clock runs. A stored closed form turns the question into arithmetic on the given parameters, and the one-line recursions above are the fallback if the test throws a variant (biased coin, different alphabet). Under a no-paper format, the binding constraint is working memory, not understanding; the closed forms are how you buy it back.

Self-check. Each run formula must reduce to (a) at $k = 1$: $2^2 - 2 = 2 = 1/(1/2)$ and $(6^2 - 6)/5 = 6 = 1/(1/6)$ — both pass. And HTH by direct states (no progress / H / HT): $e_2 = 1 + \frac{1}{2} \cdot 0 + \frac{1}{2}e_0$, $e_1 = 1 + \frac{1}{2}e_1 + \frac{1}{2}e_2$, $e_0 = 1 + \frac{1}{2}e_1 + \frac{1}{2}e_0$ gives $e_1 = 2 + e_2$, $e_0 = 2 + e_1 = 4 + e_2$, then $e_2 = 1 + \frac{1}{2}(4 + e_2)$ so $e_2 = 6$ and $e_0 = 10$, agreeing with the overlap rule.

Answer: (a) $1/p$; (b) $2^{k+1} - 2$; (c) $(6^{k+1} - 6)/5$; (d) HH 6, HT 4, HTH 10; (e) 14.7.

P1.12 *Pattern races: which comes first*

(a) Neither pattern can complete before the first H appears (each needs an H strictly before its final flip). At the first H, nothing has completed, and the *very next flip decides the race*: H completes HH, T completes HT. So

$$P(\text{HT before HH}) = \tfrac{1}{2},$$

despite the waiting times being 4 versus 6. The flaw: expected waiting times are *marginal* statistics of two separate games, while a race depends on the joint structure on one coin. The gap between 6 and 4 comes entirely from what happens *after a failure* (HH resets fully, HT keeps its H) — and in a race, the moment one pattern completes the game is over, so the post-failure behaviour that inflates $\mathbb{E}[\text{HH}]$ never gets a chance to matter.

(b) Claim: HH beats TH only when the game's first two flips are HH. If they are, HH has won at flip 2 (TH cannot have appeared inside one flip). Otherwise a T occurs before any HH has completed; consider the *first H after that T*. All flips between that T and this H are tails, so this H is immediately preceded by a T — it completes TH. And no HH can have completed earlier, since there were no heads between the T and this H. Hence

$$P(\text{HH first}) = P(\text{first two flips are HH}) = \tfrac{1}{4},$$

so TH wins with probability $\frac{3}{4}$. Decline the even-money bet on HH emphatically — fair odds are 3:1 against HH. (This asymmetry — every pattern is beaten by some other pattern chosen in response — is the Penney-game structure; the practical lesson is that "my signal fires eventually" and "my signal fires *first*" are different claims.)

Self-check. (a): enumerate from the first H — there is no path on which both patterns stay alive past the next flip, so $\frac{1}{2}$ is airtight. (b): condition on the first two flips: HH ($\frac{1}{4}$) gives HH the win; HT, TH, TT ($\frac{3}{4}$ total) each contain a T before HH can have completed, and the argument above then hands the win to TH; in the TH case it has already won at flip 2. The two computations agree, and (b) is consistent with the known Penney result that TH beats HH exactly 3:1.

Answer: (a) 1/2 — expected waiting times do not determine race odds; (b) do not take it: $P(\text{HH first}) = 1/4$, fair odds 3:1 against HH.

P1.13 *Coupon collector*

Decompose the collection into stages by how many faces remain unseen. While j faces are unseen, each roll reveals a new face with probability $j/6$, so that stage lasts a geometric time with mean $6/j$. By linearity, summing over $j = 6, 5, \dots, 1$ (equivalently $j = 1, \dots, 6$):

$$\mathbb{E}[T] = \sum_{j=1}^{6} \frac{6}{j} = 6\Big(1 + \tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{4} + \tfrac{1}{5} + \tfrac{1}{6}\Big) = 6 \cdot \frac{49}{20} = \frac{147}{10} = 14.7 \text{ exactly.}$$

The stages are independent geometrics, but linearity would not care if they weren't. For an n-faced die, $\mathbb{E}[T] = nH_n \approx n \ln n + \gamma n + \frac{1}{2}$ with $\gamma \approx 0.577$: slightly super-linear, and dominated by the long hunt for the last few faces ($6/1 + 6/2 = 9$ of the 14.7 rolls are spent on the last two).

Self-check. Two checks. (i) $n = 2$ "die" (a coin, collect both faces): formula gives $2H_2 = 3$, matching the direct answer $1 + 2 = 3$ from the geometric subsection. (ii) Asymptotic consistency at $n = 6$: $6(\ln 6 + 0.577) + 0.5 \approx 6 \times 2.369 + 0.5 \approx 14.7$ — the approximation already reproduces the exact value to one decimal.

Answer: $6H_6 = 147/10 = 14.7$ rolls; for n faces, $nH_n \approx n(\ln n + 0.577)$. □

P1.14 *Coupon collector variant: subset of coupons amid noise*

No — the noise faces do not help you, but they do consume rolls. While j of the three target faces remain unseen, each roll hits a new target with probability $j/6$ (not $j/3$), so the stage means are $6/3, 6/2, 6/1$:

$$\mathbb{E}[T] = \frac{6}{3} + \frac{6}{2} + \frac{6}{1} = 2 + 3 + 6 = 11.$$

Note the clean structure: $\sum_j 6/j = 2 \sum_j 3/j$, i.e. exactly double the 3-face collector 5.5. That is thinning: only half of all rolls land in $\{1, 2, 3\}$ at all, and conditional on landing there the process looks like the 3-face collector, so every stage — and hence the total — is stretched by the factor 2.

Recognising "coupon collector run at acceptance rate q" as "divide by q" is the fast route in an interview.

Self-check. Inclusion-exclusion on the maximum. $T = \max(T_1, T_2, T_3)$ where T_i is the first appearance of face i, each geometric with mean 6. Using $\max = \sum T_i - \sum_{i<j} \min(T_i, T_j) + \min(T_1, T_2, T_3)$ and the fact that the min over a set of faces is the wait for *any* of them (geometric: mean 3 for two faces, 2 for three faces):

$$\mathbb{E}[T] = 3(6) - 3(3) + 2 = 18 - 9 + 2 = 11.$$

Same answer by an entirely different decomposition. Bounds also pass: 11 sits between the 3-face collector 5.5 (too optimistic) and the full 6-face collector 14.7 (which demands more than we need).

Answer: 11 rolls. □

P1.15 *Order statistics of dice: make-a-market ladder*

(a) CDF first: $P(\max \le k) = (k/6)^2$, so $P(\max = k) = \frac{k^2-(k-1)^2}{36} = \frac{2k-1}{36}$ and

$$\mathbb{E}[\max] = \sum_{k=1}^{6} k\,\frac{2k-1}{36} = \frac{161}{36} \approx 4.47.$$

The trap to dodge out loud: quoting 3.5 "because dice average 3.5" ignores that an extreme is biased away from the mean — always ask whether the payoff is a mean, an extreme, or a median of the randomness.

(b) No new sum needed: $\min + \max = X + Y$ pointwise, so by linearity

$$\mathbb{E}[\min] = 7 - \frac{161}{36} = \frac{91}{36} \approx 2.53.$$

(c) Also no new sum: $|X - Y| = \max - \min$ pointwise, so

$$\mathbb{E}\,|X - Y| = \frac{161}{36} - \frac{91}{36} = \frac{70}{36} = \frac{35}{18} \approx 1.94.$$

Quoting: centre a two-sided market on the fair value with a spread you can defend — e.g. 4.3 bid at 4.6 offer on the max — wide enough to cover the payoff's dispersion and the risk that the counterparty (the interviewer) knows something; tighten only under pressure. The structural lesson is that (b) and (c) cost nothing once (a) is done: pointwise identities plus linearity replace two fresh distributions.

Self-check. (c) by brute count: over the 36 ordered pairs, $|i-j| = 0, 1, 2, 3, 4, 5$ occurs $6, 10, 8, 6, 4, 2$ times, so $\sum |i - j| = 0 + 10 + 16 + 18 + 16 + 10 = 70$ and $\mathbb{E} = 70/36 = 35/18$ — matches. Consistency: $\mathbb{E}[\max] + \mathbb{E}[\min] = 252/36 = 7 = \mathbb{E}[X + Y]$, as the identity demands.

Answer: (a) $161/36 \approx 4.47$; (b) $91/36 \approx 2.53$; (c) $35/18 \approx 1.94$.

P1.16 *Option value of a re-roll*

(a) A fresh roll is worth 3.5, so keep any face beating that: keep $\{4, 5, 6\}$ (conditional mean 5), re-roll $\{1, 2, 3\}$:

$$V_1 = \tfrac{1}{2} \cdot 5 + \tfrac{1}{2} \cdot 3.5 = 4.25.$$

(b) Now the continuation after discarding the first roll is the one-re-roll game, worth 4.25 — so at the first stage keep only $\{5, 6\}$ (mean 5.5):

$$V_2 = \tfrac{1}{3} \cdot 5.5 + \tfrac{2}{3} \cdot 4.25 = \tfrac{11}{6} + \tfrac{17}{6} = \tfrac{28}{6} = \tfrac{14}{3} \approx 4.67.$$

The keep-threshold rises with remaining optionality: more chances make you pickier.

(c) The option value is $V_1 - 3.5 = 0.75$. With a \$0.50 usage fee, re-rolling nets $3.5 - 0.5 = 3.0$, so keep any face ≥ 3: keep $\{3, 4, 5, 6\}$ (mean 4.5) with probability $\frac{2}{3}$, re-roll $\{1, 2\}$:

$$V_1^{\text{fee}} = \tfrac{2}{3} \cdot 4.5 + \tfrac{1}{3} \cdot 3.0 = 3 + 1 = 4.00.$$

This is *not* $4.25 - 0.50 = 3.75$: the fee is contingent — you only pay it when you exercise, and the fee itself changes the exercise policy (you now keep a 3), so the option still adds $4.00 - 3.5 = 0.50$ of value, not 0.25. Never net a contingent cost against an unconditional value. Market framing: the re-roll is a call-like claim on a volatile underlying; its value grows with dispersion and with the number of exercise dates, and shrinks — but less than one-for-one — with exercise costs.

Self-check. Ladder monotonicity and diminishing returns: $3.5 < 4.25 < 4.67 < 6$ (must be bounded by the max face) with increments 0.75, then $\frac{5}{12} \approx 0.42$ — optionality has diminishing marginal value, as it should. Fee case tie-check: at face 3 you are indifferent (keep 3.0 vs re-roll net 3.0); computing the value with the other tie-break, re-rolling on 3 as well, gives $\frac{1}{2} \cdot 5 + \frac{1}{2} \cdot 3.0 = 4.00$ — identical, confirming the indifference and the value 4.00.

Answer: (a) keep $\{4, 5, 6\}$: value 4.25; (b) keep $\{5, 6\}$ first stage: value $14/3 \approx 4.67$; (c) option worth 0.75; with the \$0.50 usage fee the game is worth 4.00 (keep $\{3, 4, 5, 6\}$) — the option still adds 0.50, not 0.25. ☐

P1.17 *First 6 given all rolls even*

Work from the joint law. The sequence has $N = n$ with all rolls even iff the first $n - 1$ rolls land in $\{2, 4\}$ and roll n is a 6:

$$P(N = n, \text{ all even}) = \left(\tfrac{2}{6}\right)^{n-1} \tfrac{1}{6} = \left(\tfrac{1}{3}\right)^{n-1} \tfrac{1}{6}, \qquad P(\text{all even}) = \frac{1/6}{1 - 1/3} = \frac{1}{4}.$$

Hence the conditional distribution is

$$P(N = n \mid \text{all even}) = 4 \cdot \left(\tfrac{1}{3}\right)^{n-1} \tfrac{1}{6} = \tfrac{2}{3} \left(\tfrac{1}{3}\right)^{n-1},$$

which is geometric with success probability $\frac{2}{3}$, so

$$\mathbb{E}[N \mid \text{all even}] = \frac{1}{2/3} = \frac{3}{2}.$$

Why 3 is wrong. “Roll a 3-face die $\{2, 4, 6\}$ until a 6” is a *different experiment*: there, each roll is guaranteed even by construction. Here, the conditioning is a statement about the *whole realised sequence*: a length-n sequence must dodge the odd faces n times, so long sequences are exponentially down-weighted — $P(\text{all even} \mid N = n) = (2/5)^{n-1}$ decays in n (the $n - 1$ non-6 rolls are uniform on five faces). Conditioning reweights sequences, not the die’s faces, and it shifts mass towards short sequences: the conditional mean drops below even the unconditional geometric structure the naive answer assumes. The general trigger: whenever you condition on an event whose probability depends on the length (or path) of the realisation, you may not just “restrict the alphabet”.

Self-check. The conditional distribution sums to 1: $\sum_{n \geq 1} \frac{2}{3}(1/3)^{n-1} = \frac{2/3}{2/3} = 1$. And $P(N = 1 \mid \text{all even}) = \frac{2}{3}$, versus $\frac{1}{3}$ under the naive 3-face model — the conditioning visibly favours stopping immediately, consistent with a mean of $1.5 < 3$. As a bound, the answer must be ≥ 1 and ≤ 3 (the naive model maximally favours long survival); 3/2 sits inside.

Answer: 3/2.

P1.18 *Hat-check fixed points*

(a) Let I_i indicate that trader i gets their own ticket; $P(I_i = 1) = \frac{1}{n}$ (their ticket is a uniform draw). By linearity,

$$\mathbb{E}[X] = n \cdot \tfrac{1}{n} = 1 \quad \text{for every } n,$$

independence nowhere required.

(b) Second moment via pairs: $X^2 = \sum_i I_i + \sum_{i \neq j} I_i I_j$, and for $i \neq j$,

$$P(I_i = I_j = 1) = \frac{1}{n} \cdot \frac{1}{n-1} = \frac{(n-2)!}{n!},$$

(the permutation must fix both positions). So

$$\mathbb{E}[X^2] = 1 + n(n-1) \cdot \frac{1}{n(n-1)} = 2 \qquad \Longrightarrow \qquad \mathrm{Var}(X) = 2 - 1^2 = 1 \text{ exactly, for all } n \geq 2.$$

The positive correlation between fixed points exactly offsets the small-p Bernoulli variance shortfall — mean 1, variance 1, for every $n \geq 2$.

(c) Mean 1 and variance 1, with dependence vanishing as n grows: X converges to Poisson(1), so

$$P(X = 0) \to e^{-1} \approx 0.368, \qquad P(X \geq 1) \to 1 - e^{-1} \approx 0.632.$$

Take the side "at least one match": at even money your edge is $0.632 - 0.368 \approx 26$ cents per dollar staked. (The general lesson for making markets on "no coincidence" events: rare-per-trial matches with many trials are Poisson, and $P(\text{none}) = e^{-\lambda}$ with λ the expected count — here $\lambda = 1$ regardless of n.)

Self-check. Exact tiny case $n = 2$: the two permutations give $X = 2$ (identity) or $X = 0$ (swap), each with probability $\frac{1}{2}$: $\mathbb{E}[X] = 1$, $\mathbb{E}[X^2] = \frac{1}{2}(4 + 0) = 2$, Var $= 1$ — all three formulas verified exactly. And the derangement probability converges fast: at $n = 4$ it is $9/24 = 0.375$, already within 0.007 of e^{-1}, so the "large n" claim in (c) is safe for any realistic dinner.

Answer: (a) 1; (b) 1; (c) bet on at least one match: it wins with probability $\approx 1 - 1/e \approx 0.632$, an edge of about 26 cents per dollar at even money. □

P1.19 *Random chord longer than the radius* ♦

By rotational symmetry, fix the first point and let the second sit at central angle $\theta \sim \mathrm{U}[0, 2\pi)$. The chord subtending central angle θ has length

$$L = 2R\sin(\theta/2),$$

(the isoceles triangle with two radii and the chord, split down the middle). The condition $L > R$ reads

$$2R\sin(\theta/2) > R \iff \sin(\theta/2) > \tfrac{1}{2} \iff \theta/2 \in \left(\tfrac{\pi}{6}, \tfrac{5\pi}{6}\right) \iff \theta \in \left(\tfrac{\pi}{3}, \tfrac{5\pi}{3}\right),$$

an arc of measure $\frac{5\pi}{3} - \frac{\pi}{3} = \frac{4\pi}{3}$ out of 2π: probability $\frac{4\pi/3}{2\pi} = \frac{2}{3}$.

Sanity check: a chord equal to the radius subtends 60° (equilateral triangle with the two radii), so the chord exceeds the radius unless the second point falls within 60° on either side of the first — 120° of 360° fails, 240° succeeds: 2/3 again.

Answer: (B) 2/3.

P1.20 *Max of four dice, exactly* ♦

The max-of-dice pipeline: go through the CDF, then difference.

$$P(\max \leq k) = \left(\tfrac{k}{6}\right)^4.$$

$$P(\max = 3) = P(\max \leq 3) - P(\max \leq 2) = \left(\tfrac{3}{6}\right)^4 - \left(\tfrac{2}{6}\right)^4 = \frac{81 - 16}{1296} = \frac{65}{1296}.$$

The trap is option (D): $1/16 = 81/1296 = P(\max \leq 3)$ — the *at most* probability, which forgets to remove the outcomes where no die shows a 3 at all. The option set is built for exactly this slip: under a clock, anyone who computes $(1/2)^4$ and stops sees a clean 1/16 waiting for them.

Sanity check: $65/1296 \approx 5\%$, sensibly small — with four dice the max is usually 5 or 6 ($P(\max \geq 5) = 1 - (4/6)^4 \approx 80\%$), so "exactly 3" should be rare.

Answer: (C) 65/1296.

21.2 Solutions — §2 Bayesian Updating and Reading Flow

P2.1 *Bayes odds form, base rates* ♦

Odds form. Prior odds of infected against not: $0.001 : 0.999 = 1 : 999$. Likelihood ratio of a positive test: $P(+ \mid \text{inf})/P(+ \mid \text{not}) = 1/0.1 = 10$. Posterior odds $= 10 : 999$, so

$$P(\text{infected} \mid +) \;=\; \frac{10}{10+999} \;=\; \frac{10}{1009} \;\approx\; 0.99\%.$$

The headline: a positive result from a test that *never misses an infection* still leaves you around 1%, because the prior was $999 : 1$ against and the false-positive channel (10% of everyone healthy) is a hundred times wider than the prevalence.

Self-check (counting version). Take 10,000 people: 10 are infected and all test positive; of the 9,990 uninfected, 10%, i.e. 999, also test positive. Among the 1,009 positives, 10 are infected: 10/1009. Sanity bound: when prevalence $\ll$ false-positive rate, $P \approx$ prevalence/FP rate $= 0.001/0.1 = 1\%$ — consistent.

Answer: $10/1009 \approx 0.99\%$.

P2.2 *Conditioning on the decisive event* ♦

Condition on the game that ends the match. A draw carries no information and simply restarts the process, so the match is decided by the first decisive game:

$$P(\text{I win}) \;=\; \frac{0.4}{0.4+0.2} \;=\; \frac{2}{3}.$$

In odds language: my win-to-loss odds are $0.4 : 0.2 = 2 : 1$ in every game, and replaying draws cannot change those odds — only renormalise them.

Self-check 1 (first-step analysis). Let W be my overall win probability. Conditioning on game one: $W = 0.4 + 0.4\,W$, so $0.6\,W = 0.4$ and $W = 2/3$.

Self-check 2 (series). $W = \sum_{n\geq 0}(0.4)^n \cdot 0.4 = 0.4/(1-0.4) = 2/3$; the opponent gets $0.2/0.6 = 1/3$, and $2/3 + 1/3 = 1$ with zero mass left on drawing forever ($0.4^n \to 0$).

Answer: 2/3.

P2.3 *Odds form, prior vs evidence* ♦

Odds form. Prior odds gag against fair: $1 : 99$. Likelihood ratio of five heads: $1/(1/2)^5 = 32$. Posterior odds $= 32 : 99$, so

$$P(\text{gag} \mid 5 \text{ heads}) \;=\; \frac{32}{32+99} \;=\; \frac{32}{131} \;\approx\; 24.4\%.$$

Five straight heads and the fair coins are *still* the favourite: a $99 : 1$ prior beats a $32 : 1$ likelihood ratio.

Self-check (direct Bayes).

$$\frac{\frac{1}{100}\cdot 1}{\frac{1}{100}\cdot 1 + \frac{99}{100}\cdot\frac{1}{32}} = \frac{1}{1+99/32} = \frac{32}{131}.$$

Sanity: a sixth head doubles the odds to $64 : 99 \approx 39\%$, and only at seven heads ($128 : 99$) does the gag coin finally become the favourite — the same shape as the notes' 1000-coin anchor, where even ten heads ($2^{10} = 1024$ vs 999) lands at barely over 50%.

Answer: $32/131 \approx 24.4\%$.

P2.4 *Sequential likelihood ratios*

The likelihood ratio of a single six is $\text{LR} = \frac{1/2}{1/6} = 3$, and rolls are conditionally independent given the die, so sequential LRs multiply.

(a) Prior odds $1:1$; posterior odds $3:1$, so $P = 3/4$.

(b) Odds $3^3 : 1 = 27 : 1$, so $P = 27/28 \approx 96.4\%$.

(c) We need posterior odds $\geq 99 : 1$, i.e. $3^n \geq 99$. $3^4 = 81$ gives $81/82 \approx 98.8\%$ — not enough; $3^5 = 243$ gives $243/244 \approx 99.6\%$. So $n = 5$.

(d) It drops. A non-six also updates: $P(\text{non-six} \mid \text{loaded}) = 1/2$ versus $5/6$ for the fair die, so its LR is $\frac{1/2}{5/6} = 3/5 < 1$. After three sixes and a two the odds are $27 \cdot \frac{3}{5} = 16.2$, i.e. $P = 16.2/17.2 \approx 94.2\%$. The absence of the signal is itself evidence — the flow-reading version: the aggressive trades that do *not* arrive also move your posterior.

Self-check. Recompute (b) directly: $\dfrac{(1/2)^3}{(1/2)^3 + (1/6)^3} = \dfrac{1/8}{1/8 + 1/216} = \dfrac{27}{28}$. General consistency check on the LRs: under the fair-die hypothesis the expected likelihood ratio of one roll must equal 1, and indeed $\frac{1}{6} \cdot 3 + \frac{5}{6} \cdot \frac{3}{5} = \frac{1}{2} + \frac{1}{2} = 1$.

Answer: (a) $3/4$; (b) $27/28 \approx 96.4\%$; (c) 5; (d) no — odds $16.2 : 1$, $P \approx 94.2\%$. □

P2.5 *Correlated evidence*

Multiplying likelihood ratios is valid only when the pieces of evidence are *conditionally independent given the hypothesis*. Three child fills of one parent order are one trading decision executed in slices, not three independent draws of the account's behaviour: given that this account decided to lift, the second and third slices were near-certain *whether it is informed or noise*, so their conditional likelihood ratios are ≈ 1. The junior counted one bit of evidence three times.

A defensible update treats the burst as a single lift event: odds $= \frac{1}{9} \times 4 = \frac{4}{9}$, so $P = 4/13 \approx 31\%$. (If total order *size* itself discriminates informed from noise, that is a separate, measured likelihood term — it is not recovered by re-multiplying the same fill.)

The general failure mode is correlated evidence: three quotes copied from one pricing source, three headlines rewriting one press release, three fills from one parent order. Independence is an assumption about the *generating process*, not about the number of lines in the log.

Self-check. Bounds: the truth must lie between the no-update prior (10%) and the fully-independent computation ($64 : 9 \Rightarrow 64/73 \approx 87.7\%$); the one-event posterior $4/13 \approx 30.8\%$ does. Conversion check: odds $4 : 9 \Rightarrow 4/(4+9) = 4/13$. Note the junior's *arithmetic* was right ($64/73 \approx 88\%$) — the independence assumption was the error.

Answer: LRs multiply only for conditionally independent evidence; three slices of one iceberg are one event, so the posterior is $\approx 4/13 \approx 31\%$, not 88%.

P2.6 *Flow reading, sequential update to a threshold*

The likelihood ratio of one lift is $\text{LR} = 0.9/0.5 = 1.8$; prior odds are $5 : 95 = 1 : 19$.

(a) Posterior odds $= 1.8^2/19 = 3.24/19 \approx 0.1705$, so

$$P = \frac{0.1705}{1.1705} \approx 14.6\%.$$

Two straight lifts and it is still only $\sim 15\%$: with mostly-noise flow the base rate dominates early evidence — exactly the notes' base-rate trap.

(b) Need odds $> 9 : 1$, i.e. $1.8^n > 9 \times 19 = 171$. Powers of 1.8: $1.8^2 = 3.24$, $1.8^4 = 10.50$, $1.8^8 = 110.2$ (odds 5.80, $P \approx 85.3\%$ — not yet), $1.8^9 = 198.4$ (odds 10.44, $P \approx 91.3\%$). So $n = 9$.

(c) Raise both quotes (skew the whole market up), and widen if uncertainty about the size of the move is growing. Each fill at the stale offer is a sale to a counterparty increasingly likely to know the price is going up; the posterior is precisely the probability weight on the adverse scenario, so the expected adverse-selection cost per fill grows with it. "Keep selling at the same price" treats the 9th lift like the 1st — it ignores that the flow has already told you which side the informed money is on. Moving the market both cuts your expected loss and makes the informed trader pay through the move.

Self-check. (a) directly: $0.05 \times 0.9^2 = 0.0405$ and $0.95 \times 0.5^2 = 0.2375$; $0.0405/(0.0405 + 0.2375) = 0.0405/0.2780 \approx 14.6\%$. (b) by logs: $n > \ln 171 / \ln 1.8 = 5.142/0.588 \approx 8.75 \Rightarrow n = 9$, agreeing with the power table.

Answer: (a) $\approx 14.6\%$; (b) 9 consecutive lifts; (c) skew both quotes up (and widen) — filling at a stale offer sells cheap insurance to informed flow.

21.3 Solutions — §3 Betting, Bankroll, Ruin and Markov Chains

P3.1 *Kelly criterion*

(a) Betting fraction f: wealth multiplies by $1 + bf$ (probability p) or $1 - f$ (probability $q = 1 - p$). Maximise the expected log growth

$$g(f) = p \ln(1 + bf) + q \ln(1 - f), \qquad g'(f) = \frac{pb}{1 + bf} - \frac{q}{1 - f} = 0 \;\Rightarrow\; f^* = \frac{bp - q}{b}.$$

Second derivative $g''(f) = -\dfrac{pb^2}{(1 + bf)^2} - \dfrac{q}{(1 - f)^2} < 0$ everywhere, so g is strictly concave and f^* is the unique maximum.

(b) Even money ($b = 1$): $f^* = p - q = 0.10$. For $b = 2$, $p = 0.4$: $f^* = (0.8 - 0.6)/2 = 0.10$. Both bets: 10% of bankroll.

(c) $g(0.10) = 0.55 \ln 1.1 + 0.45 \ln 0.9 \approx 0.0524 - 0.0474 = 0.0050$, about 0.50% compounded per bet. At $2f^* = 0.20$: $g(0.20) = 0.55 \ln 1.2 + 0.45 \ln 0.8 \approx 0.10028 - 0.10041 = -0.00014 < 0$. Doubling Kelly annihilates the growth. Small-edge picture: $g(f) \approx (p - q)f - f^2/2$, a parabola with roots at 0 and $2(p - q) = 2f^*$ — the growth curve crosses zero at exactly twice Kelly. Moral: the arithmetic EV per bet, $f(p - q)$, is positive at *any* f, yet compounding is negative beyond $2f^*$: positive edge per trade and a shrinking bankroll are perfectly compatible.

Self-check. Differentiate back: $g'(0.10) = \frac{0.55}{1.1} - \frac{0.45}{0.9} = 0.5 - 0.5 = 0$. The parabola approximation at f^* predicts $g \approx (0.1)(0.1) - (0.1)^2/2 = 0.005$, matching the exact 0.0050.

Answer: (a) $f^* = (bp - q)/b$; (b) 10% in both cases; (c) $g(f^*) \approx +0.50\%$ per bet, $g(2f^*) \approx -0.014\%$ — past twice Kelly you compound downward despite positive EV per bet. □

P3.2 *Kelly under estimation error*

(a) Kelly requires the *true* p; $\hat{p}$ from 25 trades is extremely noisy, and the Kelly penalty is asymmetric: under-betting gives up a little growth, while betting beyond twice the true Kelly produces *negative* growth. Full Kelly on an estimated edge is structural over-betting.

(b) True $p = 0.55$ makes the true Kelly $f^* = 0.10$ with $g(0.10) = 0.55 \ln 1.1 + 0.45 \ln 0.9 \approx +0.0050$ per trade. His $f = 0.20$ is exactly twice the true Kelly:

$$g(0.20) = 0.55 \ln 1.2 + 0.45 \ln 0.8 \approx -0.00014 < 0,$$

so he slowly grinds the book to zero while holding a genuine 5-point edge. Half of his estimated Kelly is 0.10 — exactly the true optimum here. That is the fractional-Kelly defence in one example: for small edges $g(cf^*) \approx c(2 - c)\, g(f^*)$, so half-Kelly keeps $\approx 75\%$ of the growth at half the volatility (a quarter of the variance), and it is robust to precisely this estimation error.

(c) $\mathrm{SE}(\hat{p}) = \sqrt{\hat{p}\hat{q}/n} = \sqrt{0.24/25} \approx 0.098$ — ten percentage points. A 95% interval is roughly 0.60 ± 0.19, i.e. $(0.41,\ 0.79)$: it contains 0.5, so the data cannot even rule out *no edge*. Discipline: half-Kelly or less, hard caps per bet, and scale up only as the sample grows.

Self-check. The rule $g(cf^*) \approx c(2 - c)g(f^*)$ gives 0 at $c = 2$, matching the exact $g(0.20) \approx 0$; and gives 1 at $c = 1$. Binomial check of (c): the win count has SD $\sqrt{25 \times 0.24} \approx 2.45$ wins, so 15 ± 4.8 wins spans $p \in (0.41, 0.79)$ — same interval.

Answer: (a) full Kelly on a 25-trade estimate; (b) $g(0.20 \mid p{=}0.55) \approx -0.014\%$ per trade vs $g(0.10) \approx +0.50\%$; half of estimated Kelly $= 0.10 =$ the true optimum; (c) SE ≈ 0.10, CI includes 0.5 — bet fractional Kelly with caps.

P3.3 *Gambler's ruin, fair case*

(a) Fair gambler's ruin from i with absorbing barriers 0 and N: the harmonic relation $p_i = \frac{1}{2}p_{i-1} + \frac{1}{2}p_{i+1}$ forces p_i linear in i, so $P = i/N = 30/100 = 0.30$.

(b) Expected duration $= i(N - i) = 30 \times 70 = 2100$ flips.

(c) Work in units of \$5: $i = 6$, $N = 20$. Win probability $6/20 = 0.30$ — unchanged: in a fair game it depends only on your *fraction* of the total capital. Duration $= 6 \times 14 = 84$ flips: 25 times fewer. Duration scales as $1/(\text{stake})^2$.

Self-check (martingale/EV argument, independent of the recursions). The game is fair, so your expected final wealth equals your initial \$30; the final wealth is \$100 with probability P and \$0 otherwise, hence $100P = 30$, $P = 0.30$ — for *any* stake size. Duration scaling: each \$5 flip moves wealth with 25 times the per-step variance of a \$1 flip while the distances to the barriers are fixed, so the expected time shrinks by 25: $2100/25 = 84$.

Answer: (a) 0.30; (b) 2100 flips; (c) win probability still 0.30; expected duration drops to 84 flips.

P3.4 *Gambler's ruin, biased*

(a) With $r = q/p = 51/49$, $P(\text{reach } N) = \dfrac{1 - r^i}{1 - r^N}$, here $i = 50$, $N = 100$. Since $N = 2i$,

$$\frac{1 - r^i}{1 - r^{2i}} = \frac{1}{1 + r^i}.$$

Mental math: $\ln \frac{51}{49} = \ln \frac{1+0.02}{1-0.02} \approx 2(0.02) = 0.04$, so $r^{50} \approx e^2 \approx 7.39$ and $P \approx 1/8.39 \approx 12\%$. A 1% per-round edge against you turns a coin-flip doubling shot into roughly 1-in-8.

(b) Now the *up* probability is 0.51, so $r = 49/51 < 1$. Let $N \to \infty$ in $\frac{1-r^i}{1-r^N}$: the survival probability is $1 - r^i$, hence

$$P(\text{ruin}) = r^i = \left(\tfrac{49}{51}\right)^{20} \approx e^{-20\times 0.04} = e^{-0.8} \approx 0.45.$$

Despite a genuine positive edge, the under-capitalised desk dies before compounding about 45% of the time. For ruin $\le 1\%$: $i \times 0.04 \ge \ln 100 \approx 4.61$, so $i \ge 115.2$ — a buffer of about 116 ticks, nearly six times larger.

Self-check. Exact $\ln(51/49) = 3.9318 - 3.8918 = 0.0400$, so the $2x$ approximation is essentially exact. Recompute (a) by the full formula: $(1 - 7.39)/(1 - 54.6) = 6.39/53.6 = 0.119 = 1/8.39$. Check the 1% buffer: $e^{-116\times 0.04} = e^{-4.64} \approx 0.0097 < 1\%$. Symmetry sanity: with the edge *for* you in setup (a), success would be $1/(1 + e^{-2}) \approx 88\%$ — the mirror image of 12%, as it must be.

Answer: (a) $1/(1 + (51/49)^{50}) \approx 12\%$; (b) ruin $= (49/51)^{20} \approx e^{-0.8} \approx 45\%$; need ≈ 116 ticks for ruin below 1%.

P3.5 *First-step analysis / ruin reduction*

Derivation 1 (blocks of two points). Group points in pairs. A pair goes win-win with probability p^2 (you win the match from level), lose-lose with q^2 (you lose), and split with $2pq$ (back to level, no memory). Condition on the first decisive pair:

$$P = \frac{p^2}{p^2 + q^2} = \frac{0.36}{0.36 + 0.16} = \frac{9}{13} \approx 69.2\%.$$

Derivation 2 (gambler's-ruin reduction). Your lead performs a random walk from 0 with absorption at ± 2 — i.e. ruin with $i = 2$, $N = 4$, $r = q/p = 2/3$:

$$P = \frac{1 - r^2}{1 - r^4} = \frac{1 - 4/9}{1 - 16/81} = \frac{5/9}{65/81} = \frac{9}{13}.$$

Self-check (first-step analysis, a third route). States = lead $\in \{-1, 0, +1\}$: $W_{+1} = p + qW_0$, $W_{-1} = pW_0$, $W_0 = pW_{+1} + qW_{-1}$. Substituting: $W_0 = p^2 + 2pq\,W_0$, so $W_0 = \dfrac{p^2}{1-2pq} = \dfrac{p^2}{p^2+q^2}$ (since $(p+q)^2 = 1$). Limits: $p = \frac{1}{2} \Rightarrow \frac{1}{2}$; $p = 1 \Rightarrow 1$.

Why more than 0.6: requiring two clear points forces a longer contest, and longer formats favour the better player — each extra independent point is another draw from a distribution tilted your way. Trading version: a desk with a small per-trade edge wants *many independent* expressions of it; the format (or the trade count) amplifies edge exactly as deuce amplifies 0.6 into 0.69.

Answer: $9/13 \approx 69.2\%$. □

P3.6 *First-step analysis, pattern race*

Structural argument. HH appears first if and only if the first two flips are both heads. Indeed, suppose a tail appears at any point before HH has been completed. Every later occurrence of HH begins with a head, and the *first* head to arrive after that tail completes TH immediately — so TH beats any subsequent HH. Hence HH can only win by arriving before any tail: flips 1 and 2 are HH, probability 1/4.

First-step verification. States: $\varnothing$ (start), H (last flip heads, no tail seen yet), T (a tail is pending). From T, the next head finishes TH and more tails stay in T: $P_T = 0$. From H: heads finishes HH, tails moves to T: $P_H = \frac{1}{2}(1) + \frac{1}{2}(0) = \frac{1}{2}$. From $\varnothing$: $P_\varnothing = \frac{1}{2}P_H + \frac{1}{2}P_T = \frac{1}{4}$.

Why waiting times mislead. $E[\text{wait for HH}] = 6$ and $E[\text{wait for TH}] = 4$, but the race odds are $3:1$, not $6:4$ — and the comparison is not even directionally reliable: HH vs HT also have waits 6 vs 4, yet that race is exactly 50 : 50 (once the first head arrives, the very next flip decides it). Races are governed by the overlap structure of the patterns (what happens to your progress when you "fail"), not by marginal waiting times. This is the engine of Penney's game: the second player can always pick a pattern that beats the first player's.

Self-check (enumeration of openings). The four equally likely opening pairs: HH $\Rightarrow$ HH wins; HT $\Rightarrow$ tail pending $\Rightarrow$ TH wins eventually; TH $\Rightarrow$ TH has already won; TT $\Rightarrow$ tail pending $\Rightarrow$ TH wins. HH wins in exactly one case out of four.

Answer: 1/4.

P3.7 *Stationary distribution, two-state chain*

(a) Two-state chain with exit probabilities $a = 0.1$ (leave trending) and $b = 0.3$ (leave choppy): $\pi = \left(\frac{b}{a+b}, \frac{a}{a+b}\right) = \left(\frac{0.3}{0.4}, \frac{0.1}{0.4}\right) = \left(\frac{3}{4}, \frac{1}{4}\right)$. Time is shared inversely to how easily each state is left: choppy is exited three times more readily than trending, so it carries a third as much weight — a 1:3 split, i.e. 1/4 choppy versus 3/4 trending.

(b) Expected return time to a state is $1/\pi_{\text{state}}$ (Kac's formula — in the long run a fraction π of days are trending, so trending days are on average $1/\pi$ apart): $1/(3/4) = 4/3$ days.

(c) Next-day trending probability under π: $0.75 \times 0.9 + 0.25 \times 0.3 = 0.675 + 0.075 = 0.75$. Stationary.

Self-check (independent derivation of (b) by first-step). From a trending day: with probability 0.9 tomorrow is trending — return time 1; with probability 0.1 we fall into choppy, from which the wait to switch back is geometric with success 0.3, mean 10/3. So $E = 0.9(1) + 0.1(1 + \frac{10}{3}) = 0.9 + 0.1 \times \frac{13}{3} = \frac{4}{3}$ — Kac's formula confirmed by hand.

Answer: (a) 3/4; (b) 4/3 days; (c) $0.75(0.9) + 0.25(0.3) = 0.75$.

P3.8 *Stationary distribution on a graph, return time*

Step 1: stationary distribution of a random walk on a graph. For uniform random moves on an undirected graph, $\pi_v = \dfrac{d_v}{2E}$ (d_v = degree, E = number of edges) is stationary: each neighbour u of v contributes $\pi_u \cdot \frac{1}{d_u} = \frac{1}{2E}$ of inbound probability, and v has d_v neighbours, so $(\pi P)_v = \frac{d_v}{2E} = \pi_v$.

Step 2: Kac's return-time formula. In the long run the walk spends a fraction π_v of its steps at v, so visits to v are on average $1/\pi_v$ steps apart:

$$E[\text{return time to } v] = \frac{1}{\pi_v} = \frac{2E}{d_v}.$$

Step 3: count. Knight moves come in 8 vectors $(\pm1,\pm2),(\pm2,\pm1)$; a vector (a,b) fits on the board in $(8-|a|)(8-|b|) = 7\times 6 = 42$ positions, identically for all 8 vectors. Ordered moves $= 8\times 42 = 336$, so $E = 168$ edges and $2E = 336$. A corner square has $d = 2$ (from a1: b3 and c2). Hence

$$E[\text{return}] = \frac{336}{2} = 168 \text{ moves.}$$

Self-check (independent degree census). The 8x8 knight-degree histogram: 4 squares of degree 2, 8 of degree 3, 20 of degree 4, 16 of degree 6, 16 of degree 8. Squares: $4+8+20+16+16 = 64$; total degree $8+24+80+96+128 = 336 = 2E$ — agreeing with the vector count. Sanity: a central square (degree 8) has return time $336/8 = 42$, much shorter than the badly-connected corner's 168, as it should be. The reversibility structure did all the work — no linear system in 64 unknowns.

Answer: 168 moves.

P3.9 *St. Petersburg, capped EV*

(a) $\mathbb{E} = \sum_{n\geq1} 2^{-n}\,2^n = \sum_{n\geq1} 1 = \infty$: every level contributes one dollar and there are infinitely many levels.

(b) With the cap, levels $n = 1,\dots,20$ still contribute \$1 each ($2^{-n}\times 2^n$), totalling \$20; the event "no head in the first 20 flips" has probability 2^{-20} and pays the capped 2^{20}, contributing \$1 more:

$$\mathbb{E} = 20 + 2^{-20}\cdot 2^{20} = 21.$$

General pattern: cap at $2^K \Rightarrow$ fair price $K+1$.

(c) Four defensible reasons to bid well under \$21 for a single play:

- **Concave utility:** $\mathbb{E}[\ln(\text{payoff})] = \sum_n 2^{-n} n\ln 2 = 2\ln 2$, so a log agent's certainty equivalent for the raw payoff is $e^{2\ln 2} = \$4$ — Bernoulli's original resolution.
- **One-shot heavy tails:** the median payoff is \$2, and about 97% of plays pay $\leq \$32$; the EV is carried by vanishing tail events, and with tails this heavy the sample mean converges uselessly slowly — "fair price" by EV presumes a repetition you are not getting.
- **Counterparty credit:** any valuation above the cap's contribution assumes the casino can and will pay; the cap *is* the credit limit made explicit.
- **The meta-lesson:** price $\neq$ expected value when distributions are wild — variance, bankroll and utility must enter. This is the whole point of risk management.

Self-check. (b) another way: $\mathbb{E} = \sum_{n=1}^{20} 2^{-n}2^n + P(\text{first 20 all tails})\cdot 2^{20} = 20+1 = 21$, and it matches the notes' anchor "cap $2^{30} \Rightarrow \$31$" via $K+1$ with $K = 30$. The log moment uses the standard $\sum_{n\geq1} n\,2^{-n} = 2$; check small terms: $\frac{1}{2}+\frac{2}{4}+\frac{3}{8}+\frac{4}{16} = 1.5+0.4375 \approx 1.94 \to 2$.

Answer: (a) infinite; (b) \$21 (cap at 2^K gives $K+1$); (c) concave utility (log CE = \$4), one-shot heavy tails, counterparty credit — price is not EV for wild distributions. □

P3.10 *Utility shape, risk aversion* ♦

Increasing: $U'(W) > 0$ — more wealth is preferred to less (non-satiation). Without it the agent is not even trying to make money.

Concave: $U''(W) < 0$ — risk aversion. By Jensen's inequality, for any non-degenerate zero-mean gamble X,

$$\mathbb{E}[U(W+X)] < U(W+\mathbb{E}[X]) = U(W),$$

so the investor declines every fair gamble and demands a positive risk premium to carry risk. The fair \$100 flip: $\frac{1}{2}U(W{+}100) + \frac{1}{2}U(W{-}100) < U(W)$ — the chord lies below a concave curve. Marginal-utility phrasing: because U' is decreasing, the hundred dollars lost hurts more than the hundred gained helps.

Standard examples: $\ln W$, $\sqrt{W}$, $-e^{-aW}$. Connections: maximising expected log growth (Kelly) is exactly behaving as a log-utility investor — concavity is what makes over-betting penalised; and log utility gives the St. Petersburg game a finite certainty equivalent despite infinite EV (Bernoulli's resolution), since $\mathbb{E}[\ln(\text{payoff})] = 2\ln 2 < \infty$.

Self-check (numerical Jensen). $U = \sqrt{W}$, $W = 10{,}000$, fair flip of $\pm\$3{,}600$: $\frac{1}{2}\sqrt{13{,}600}+\frac{1}{2}\sqrt{6{,}400} \approx \frac{1}{2}(116.6) + \frac{1}{2}(80) = 98.3 < 100 = \sqrt{10{,}000}$ — declined. Direction check: a convex (risk-loving) U flips the Jensen inequality and *accepts* fair gambles — the sign of U'' is exactly what controls the decision, confirming concavity as the defining property.

Answer: Increasing and concave: $U' > 0$ (more wealth preferred), $U'' < 0$ (risk aversion).

21.4 Solutions — §4 Statistics I: Estimators and Inference

P4.1 *Second moment of the standard normal* ♦

The fastest route is the decomposition of a second moment into variance plus squared mean:

$$E[X^2] = \mathrm{Var}(X) + (E[X])^2 = 1 + 0^2 = 1.$$

This identity is worth knowing in its general form $E[(X-c)^2] = \mathrm{Var}(X) + (E[X]-c)^2$: it is exactly the bias–variance decomposition of mean squared error, with X playing the estimator and c the target.

Self-check (compute another way). With $\varphi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ we have $\varphi'(x) = -x\,\varphi(x)$, so integrating by parts with $u = x$ and $dv = x\varphi(x)\,dx = -d\varphi$,

$$\int_{-\infty}^{\infty} x^2\varphi(x)\,dx \;=\; \big[-x\varphi(x)\big]_{-\infty}^{\infty} + \int_{-\infty}^{\infty}\varphi(x)\,dx \;=\; 0+1 \;=\; 1,$$

confirming the identity route. A third check: $X^2 \sim \chi^2_1$, whose mean equals its degrees of freedom, 1.

Answer: $E[X^2] = 1$.

P4.2 *Bias-variance-MSE trade-off in variance estimation*

Under normality, $S/\sigma^2 \sim \chi^2_{n-1}$, so with $Q \sim \chi^2_{n-1}$ (mean $n-1$, variance $2(n-1)$):

$$E[S] = (n-1)\sigma^2, \qquad E[S^2] = \sigma^4\big(\mathrm{Var}(Q)+(EQ)^2\big) = \sigma^4(2(n-1)+(n-1)^2) = (n-1)(n+1)\,\sigma^4.$$

Expand the MSE as a function of the divisor c:

$$\mathrm{MSE}(c) = E\Big[\Big(\frac{S}{c}-\sigma^2\Big)^2\Big] = \frac{E[S^2]}{c^2} - \frac{2\sigma^2 E[S]}{c} + \sigma^4 = \sigma^4\left[\frac{(n-1)(n+1)}{c^2} - \frac{2(n-1)}{c} + 1\right].$$

Setting the derivative in c to zero: $-2(n-1)(n+1)/c^3 + 2(n-1)/c^2 = 0$, i.e. $c^* = n+1$. Plugging back in:

$$\mathrm{MSE}(n+1) = \frac{2\sigma^4}{n+1} \;<\; \mathrm{MSE}(n) = \frac{(2n-1)\sigma^4}{n^2} \;<\; \mathrm{MSE}(n-1) = \frac{2\sigma^4}{n-1}.$$

So the MSE ranking is $n+1$ (best), then n (MLE), then $n-1$ (unbiased, worst of the three). Dividing by $n+1$ shrinks the estimate toward zero; the small bias it introduces is more than paid for by the variance reduction — the same logic that makes ridge and shrinkage estimators work.

Self-check (small case $n = 2$). Then $S = (X_1 - X_2)^2/2$ and $S/\sigma^2 \sim \chi^2_1$, so $E[S^2] = 3\sigma^4 = (n-1)(n+1)\sigma^4$ with $n = 2$. The formula gives $c^* = 3$ and $\mathrm{MSE}(3) = \sigma^4(3/9 - 2/3 + 1) = \frac{2}{3}\sigma^4 = 2\sigma^4/(n+1)$. Ordering check at $n = 10$: $2/11 \approx 0.182 < 19/100 = 0.19 < 2/9 \approx 0.222$.

Answer: $c = n + 1$ minimises MSE; ranking $n + 1 \prec n \prec n - 1$. Unbiasedness is a constraint, not a virtue — a slightly biased (shrunk) estimator strictly dominates the unbiased one on MSE. □

P4.3 *Bessel's correction and the bias of s*

(a) Decompose deviations from μ around the sample mean:

$$\sum_i (X_i - \mu)^2 = \sum_i (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2$$

(the cross term $2(\bar{X} - \mu)\sum_i(X_i - \bar{X})$ vanishes identically). Taking expectations, $E[\sum_i(X_i - \mu)^2] = n\sigma^2$ and $E[n(\bar{X} - \mu)^2] = n\,\mathrm{Var}(\bar{X}) = \sigma^2$, hence

$$E\Big[\sum_i (X_i - \bar{X})^2\Big] = n\sigma^2 - \sigma^2 = (n-1)\sigma^2.$$

Intuition: the sample mean chases the data, so deviations measured from $\bar{X}$ are systematically smaller than deviations from μ — by exactly one degree of freedom.

(b) With μ known, $E[\sum_i(X_i - \mu)^2] = n\sigma^2$ directly, so divide by n: no degree of freedom is spent estimating the centre. (Regression generalises this: $\hat{\sigma}^2 = \mathrm{RSS}/(n - p)$ after fitting p parameters.)

(c) No. By Jensen's inequality applied to the strictly concave square root (equivalently: $\mathrm{Var}(s) > 0$ unless s is degenerate),

$$(E[s])^2 \;<\; E[s^2] \;=\; \sigma^2 \quad\Longrightarrow\quad E[s] < \sigma.$$

So even after Bessel's correction, the sample volatility is biased *low*. Unbiasedness is not preserved under nonlinear transformations.

Self-check. For (a), take $n = 2$: $\sum_i(X_i - \bar{X})^2 = (X_1 - X_2)^2/2$, and $E[(X_1 - X_2)^2] = 2\sigma^2$, giving expectation $\sigma^2 = (2-1)\sigma^2$. For (c), compute $E[s]$ exactly at $n = 2$ for normal data: $s = |X_1 - X_2|/\sqrt{2}$ with $X_1 - X_2 \sim \mathcal{N}(0, 2\sigma^2)$, and $E|\mathcal{N}(0, \tau^2)| = \tau\sqrt{2/\pi}$, so $E[s] = \sigma\sqrt{2/\pi} \approx 0.80\,\sigma < \sigma$.

Answer: (a) $E[\sum_i(X_i - \bar{X})^2] = (n-1)\sigma^2$, so divide by $n - 1$; (b) divide by n when μ is known; (c) no — $E[s] < \sigma$ by Jensen, e.g. $E[s] \approx 0.80\,\sigma$ at $n = 2$. □

P4.4 *Sum of i.i.d. normals* ♦

Sums of independent normals are normal, and for independent variables both means and variances add:

$$X_1 + \cdots + X_k \;\sim\; \mathcal{N}(km,\; ks^2).$$

Contrast this with the classic confusion: *scaling* one variable gives $kX_1 \sim \mathcal{N}(km, k^2s^2)$. Summing k independent copies multiplies the standard deviation by $\sqrt{k}$ only — this is exactly why risk grows like $\sqrt{T}$ while drift grows like T under i.i.d. returns.

Self-check (MGF). Each X_i has moment generating function $M(t) = \exp(mt + s^2t^2/2)$; independence makes the MGF of the sum the product,

$$M(t)^k = \exp\left(kmt + ks^2t^2/2\right),$$

which is precisely the MGF of $\mathcal{N}(km, ks^2)$. Dimension check: variance is in squared units, so it must scale linearly in the number of terms while the sd scales as $\sqrt{k}$.

Answer: $\mathcal{N}(km,\; ks^2)$ — mean km, variance ks^2 (sd $s\sqrt{k}$).

P4.5 *Sum of normals without independence* ♦

Trap: independence (or joint normality) was never stated, and without it the question cannot be answered. Two separate failures:

- *Unknown variance even if the sum were normal.* $\mathrm{Var}(X+Y) = 4 + 9 + 2\,\mathrm{Cov}(X,Y)$, and Cauchy–Schwarz only bounds $|\,\mathrm{Cov}\,| \le \sigma_X\sigma_Y = 6$, so the variance can be anywhere in $[1, 25]$.
- *The sum need not be normal at all.* Marginal normality does not imply joint normality. Standard counterexample: let $X \sim \mathcal{N}(0,1)$ and $Y = X$ on $\{|X| \le c\}$, $Y = -X$ on $\{|X| > c\}$. By the symmetry of the normal density, $Y \sim \mathcal{N}(0,1)$ marginally, yet $X + Y$ equals $2X$ when $|X| \le c$ and 0 otherwise — a distribution with a point mass at 0, certainly not normal. Shifting and scaling transplants the same construction to the given means and variances.

If the two variables *are* independent (the intended textbook reading), means and variances add:

$$X + Y \sim \mathcal{N}(5+3,\ 4+9) = \mathcal{N}(8, 13).$$

Self-check. Under independence, the MGF product gives $\exp(8t + 13t^2/2)$, the $\mathcal{N}(8,13)$ MGF. Boundary sanity: perfect positive correlation gives sd $= 2 + 3 = 5$, variance 25, matching the Cauchy–Schwarz upper bound; perfect negative correlation gives sd $= 1$, variance 1, the lower bound.

Answer: Not determined — it depends on the joint dependence; the variance alone ranges over $[1, 25]$ and the sum need not even be normal. If independent: $\mathcal{N}(8, 13)$.

P4.6 *Distribution recognition (binomial)* ♦

Recognition checklist: a *fixed* number of trials ($n = 50$), each a Bernoulli success/failure, *independent*, with *constant* success probability ($p = 0.7$). That is the definition of a binomial:

$$\#\text{fills} \ \sim\ \text{Binomial}(n = 50,\ p = 0.7), \qquad E = np = 35, \quad \mathrm{Var} = np(1-p) = 10.5.$$

Why the distractors fail: geometric / negative binomial count *trials until* a success quota (random n, fixed successes) — here n is fixed and successes are random; Poisson would need p small with np moderate. Worth saying out loud in an interview: in reality fill probability co-moves with queue depth, volatility and toxicity, so independence and constant p are modelling assumptions — exactly the kind you state and then question.

Self-check. Mean 35 is 70% of 50. Variance sanity: $np(1-p)$ is maximised at $p = 0.5$ (would be 12.5); at $p = 0.7$ it must be smaller, and $10.5 < 12.5$. Complement symmetry: unfilled orders $\sim$ Binomial$(50, 0.3)$, and the two means $35 + 15 = 50$.

Answer: Binomial$(n = 50, p = 0.7)$; mean 35, variance 10.5.

P4.7 *Normal approximation to the binomial* ♦

Match the first two moments (de Moivre–Laplace, the CLT for Bernoulli sums):

$$\text{mean} = np = 20, \qquad \text{variance} = np(1-p) = 100 \times 0.2 \times 0.8 = 16,$$

so approximate by $\mathcal{N}(20, 16)$, i.e. sd $= 4$. The standard distractor is $\mathcal{N}(20, 20)$ — forgetting the $(1-p)$ factor (that would be the Poisson limit, where variance equals mean, valid only for small p). Rule-of-thumb validity here: $np = 20 \ge 10$ and $n(1-p) = 80 \ge 10$, so the approximation is respectable; add a continuity correction (± 0.5) for point or boundary probabilities.

Self-check. Variance must be strictly below the mean whenever p is not small: $16 < 20$, and the ratio $16/20 = 0.8 = 1 - p$. Plausibility of the sd: about 95% of outcomes in 20 ± 8, i.e. $[12, 28]$ successes out of 100 at $p = 0.2$ — sensible.

Answer: $\mathcal{N}(\text{mean } 20,\ \text{variance } 16)$ (sd 4); the trap answer is variance 20.

P4.8 *CLT - what it does not say*

The CLT is being asked to certify exactly the things it is silent about.

- *The CLT is an asymptotic statement about the centre, not the far tails at fixed n.* It says the standardised sum converges in distribution; it says nothing at $n = 252$, and convergence is slowest precisely in the tails (Berry–Esseen error degrades with the third moment). A 4-sigma event lives in the large-deviations zone, where the tail of the *sum* is governed by the tails of the *summands*: for power-law-tailed daily P&L, the sum's extreme tail remains power-law (dominated by the single worst day), no matter how bell-shaped the middle looks.
- *Independence fails.* Daily P&L exhibits volatility clustering and serial dependence; the classical CLT needs i.i.d. (or weak-dependence variants with much smaller effective n). Dependent summands can normalise far more slowly or not at all.
- *Finite variance may effectively fail.* In stressed regimes the variance is unstable or practically unbounded; with infinite variance the limit is a heavy-tailed stable law, not a normal.
- *(Bonus) Parameter uncertainty.* Even granting normality, μ and σ are estimates; at 4 sigma the answer is exquisitely sensitive to both.

Direction: all of these fatten the true tail relative to the Gaussian, so the normal-table number *understates* the loss probability — compare the distribution-free Cantelli bound in the Sharpe/loss section, which sits orders of magnitude above $\Phi(-4)$.

Self-check (concrete case). Daily P&L with Student-t_3 tails: any finite sum of 252 such days still has a tail exponent of 3 (heavy tails are preserved under addition), so its 4-sigma exceedance probability is orders of magnitude above $\Phi(-4) \approx 3.2 \times 10^{-5}$ — consistent with the claim that the Gaussian answer is a lower bound in practice.

Answer: CLT is asymptotic and about the centre (silent at fixed n and in far tails, where the summands' tails rule); i.i.d. fails (vol clustering, dependence); finite variance may fail (stable, not normal, limits); plus parameter error at 4 sigma. The normal-table probability is biased low — the true tail is fatter.

P4.9 *Confidence interval: variance vs sd* ♦

First convert variance to standard deviation — the step the question is testing:

$$\sigma = \sqrt{100} = 10.$$

A central 95% interval for a normal is mean $\pm\, 1.96\sigma \approx$ mean $\pm\, 2\sigma$:

$$1000 \pm 2 \times 10 = [980,\ 1020].$$

The trap answer treats the variance 100 as if it were the sd, giving $1000 \pm 200 = [800, 1200]$ — off by a factor of σ itself. (Pedantry worth one sentence if pressed: for a single random variable this is a central *probability* interval; "confidence interval" properly refers to interval estimates of a parameter — the test's wording is the colloquial one.)

Self-check. $P(|Z| \le 2) \approx 95.4\%$, close enough to 95% for an "approximate" answer; the exact 1.96 multiplier gives $[980.4, 1019.6]$, which rounds to the same interval. Width sanity: total width 40 is 4 sds — the standard 95% span.

Answer: $[980, 1020]$ (sd $= 10$, mean ± 2 sd); the distractor $[800, 1200]$ confuses variance with sd.

P4.10 *Hypothesis testing recipe*

Follow the recipe. *Null:* daily signs are i.i.d. fair coin flips, $p = 1/2$ (up days independent of each other). *Statistic:* N_{up}, the number of up days. *Null distribution:* $N_{\text{up}} \sim \text{Binomial}(252, 0.5) \approx \mathcal{N}(126,\ 63)$, sd $= \sqrt{63} \approx 7.94$. *Compute:*

$$z = \frac{140 - 126}{7.94} \approx 1.76, \qquad p_{\text{two-sided}} = 2\big(1 - \Phi(1.76)\big) \approx 2 \times 0.039 \approx 7.8\%.$$

Conclusion: fail to reject at 5% — "obviously better than a coin flip" is not supported at the conventional threshold. A one-sided test gives $p \approx 3.9\%$ and would reject; but choosing the one-sided alternative *after* seeing that the count came out high halves the p-value by construction — a garden-of-forking-paths move. The directional hypothesis must be fixed before looking.

Two further objections worth volunteering: daily signs are not i.i.d. (autocorrelation and volatility clustering shrink the effective n, pushing the true p-value up), and hit rate ignores magnitudes — a 55% hit rate with negatively skewed payoffs can still lose money.

Self-check. Continuity correction: $z = (139.5 - 126)/7.94 \approx 1.70$, $p_{\text{two-sided}} \approx 8.9\%$ — same conclusion. CI duality: the 95% interval for p is $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/252} = 0.556 \pm 0.061 = [0.494, 0.617]$, which contains 0.5, consistent with non-rejection.

Answer: $z \approx 1.76$, two-sided $p \approx 8\%$: not significant at 5%. One-sided $p \approx 3.9\%$ would reject, but the direction must be pre-specified — choosing it after seeing the data is p-hacking.

P4.11 *Sharpe ratio standard error (the humility formula)*

Work at daily frequency, then annualise. Daily Sharpe $\text{SR}_d = 1.2/\sqrt{252} \approx 0.0756$; sample size $n = 3 \times 252 = 756$ days:

$$\text{SE}_d \approx \sqrt{\frac{1 + 0.0756^2/2}{756}} = \sqrt{\frac{1.0029}{756}} \approx 0.0364, \qquad \text{SE}_{\text{ann}} = \sqrt{252}\,\text{SE}_d \approx 0.58.$$

Because $\text{SR}_d^2/2$ is negligible at daily frequency, the clean rule of thumb is $\text{SE}_{\text{ann}} = \sqrt{1.0029/3} \approx 0.58$. Anchor consistency: the notes' benchmark "one year of daily data measures SR to about ± 1" scaled by $1/\sqrt{3} \approx 0.577$ matches.

Answer: $\text{SE}_{\text{ann}} \approx 0.58$. Versus 0: $z \approx 2.1$, just significant at 5%. Versus 1: $z \approx 0.35$, indistinguishable (would take ≈ 96 years). ± 0.25 requires about 16 years of daily data.

P4.12 *Multiple testing / deflated-Sharpe logic*

(a) With $n = 1000$ days, the t-statistic is effectively standard normal under the null. Per test, $P(|t| \geq 2.5) = 2(1 - \Phi(2.5)) \approx 2 \times 0.0062 = 0.0124$. Across 100 independent null tests,

$$P(\max_i |t_i| \geq 2.5) = 1 - (1 - 0.0124)^{100} \approx 1 - e^{-1.24} \approx 71\%.$$

Seeing a 2.5 somewhere among 100 noise signals is the *expected* outcome, not evidence. Equivalently, the expected maximum of N standard normal draws grows like $\sqrt{2 \ln N}$: for the 200 tail events implicit in 100 two-sided tests, $\sqrt{2 \ln 200} \approx 3.3$ — the best-of-100 *under pure noise* typically beats 3.

(b) Bonferroni: per-test level $0.05/100 = 5 \times 10^{-4}$ two-sided, i.e. 2.5×10^{-4} per tail, requiring $|t| \geq \Phi^{-1}(1 - 2.5 \times 10^{-4}) \approx 3.5$. The observed 2.5 is not close.

(c) The recipe: always ask "how many things were tried?"; correct with Bonferroni-style thresholds or the Deflated Sharpe Ratio (which penalises the reported Sharpe by the expected maximum under the null given the number of trials and non-normality); demand genuine out-of-sample or forward evidence; and only then discuss costs, capacity and leakage. Note the trials are rarely independent in practice — correlated variants reduce the effective number of tests, which the DSR framework handles via the variance of trial outcomes.

Self-check. Expected count of null signals clearing 2.5: $100 \times 0.0124 = 1.24 \geq 1$, so by a Poisson approximation $P(\text{at least one}) \approx 1 - e^{-1.24} \approx 71\%$ — matches the direct computation. And the $\sqrt{2 \ln N}$ anchor (≈ 3.3) sits consistently just below the Bonferroni cutoff 3.5.

Answer: (a) $\approx 71\%$ — best-of-100 noise clears $|t| = 2.5$ more often than not; (b) $|t| \approx 3.5$ (Bonferroni at 5%); (c) count the trials, apply multiple-testing corrections / Deflated Sharpe, and demand out-of-sample confirmation.

21.5 Solutions — §5 Regression and Regularisation

P5.1 *OLS univariate slope and R-squared*

(a) With an intercept the slope is $\hat{\beta} = \widehat{\mathrm{Cov}}(X, Y)/\widehat{\mathrm{Var}}(X) = \rho\,\sigma_Y/\sigma_X = 0.4 \times 3/1 = 1.2$.

(b) In univariate regression $R^2 = \rho^2 = 0.16$: only 16% of Y's daily variance is explained.

(c) Predicted move $= \hat{\beta} \times 2\% = 2.4\%$.

(d) The slope carries the volatility ratio $\sigma_Y/\sigma_X = 3$: a modest correlation times a large vol ratio gives $\beta > 1$. R^2 measures tightness of the relationship, β measures magnitude per unit of X — different objects.

Self-check. Recompute the slope the long way: $\mathrm{Cov} = \rho\sigma_X\sigma_Y = 0.4 \times 0.01 \times 0.03 = 1.2 \times 10^{-4}$, $\mathrm{Var}(X) = 10^{-4}$, ratio $= 1.2$. And $R^2 = 0.16 \le 1$ no matter how large β is, confirming (d) is not a contradiction.

Answer: (a) 1.2; (b) 0.16; (c) +2.4%; (d) $\beta = \rho\,\sigma_Y/\sigma_X$ carries the vol ratio 3, so modest correlation still yields a slope above 1.

P5.2 *OLS derivation and projection*

(a) Minimise $L(\beta) = \|y - X\beta\|^2 = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X\beta$. The gradient $\nabla_\beta L = -2X^\top y + 2X^\top X\beta = 0$ gives the normal equations $X^\top X\beta = X^\top y$; full column rank makes $X^\top X$ invertible, so $\hat{\beta} = (X^\top X)^{-1}X^\top y$. The Hessian is $2X^\top X \succ 0$ (for $v \neq 0$, $v^\top X^\top Xv = \|Xv\|^2 > 0$ by full rank), so this is the unique global minimum.

(b) $\hat{\varepsilon} = y - X\hat{\beta}$ gives $X^\top\hat{\varepsilon} = X^\top y - X^\top X(X^\top X)^{-1}X^\top y = 0$: residuals are orthogonal to every regressor. Equivalently $\hat{y} = Hy$ is the orthogonal projection of y onto $\mathrm{col}(X)$, and $y = \hat{y} + \hat{\varepsilon}$ is an orthogonal decomposition.

(c) Symmetry: $H^\top = X\big((X^\top X)^{-1}\big)^\top X^\top = H$ since $(X^\top X)^{-1}$ is symmetric. Idempotence: $H^2 = X(X^\top X)^{-1}(X^\top X)(X^\top X)^{-1}X^\top = H$ — projecting twice is projecting once. Trace: by cyclicity, $\mathrm{tr}(H) = \mathrm{tr}\left((X^\top X)^{-1}X^\top X\right) = \mathrm{tr}(I_p) = p$: the fit spends p degrees of freedom (this is what makes $\hat{\sigma}^2 = \mathrm{RSS}/(n-p)$ the right denominator).

(d) Centred univariate case: $X = x$ (one column, mean removed), so $\hat{\beta} = (x^\top x)^{-1}x^\top y = \sum_i x_i y_i / \sum_i x_i^2 = \widehat{\mathrm{Cov}}(x, y)/\widehat{\mathrm{Var}}(x)$.

Self-check. Orthogonality (b) implies Pythagoras: $\|y\|^2 = \|\hat{y}\|^2 + \|\hat{\varepsilon}\|^2$; equivalently the decomposition behind R^2 — consistent. Independently, differentiate back: plugging $\hat{\beta}$ into the gradient gives $-2X^\top y + 2X^\top X(X^\top X)^{-1}X^\top y = 0$.

Answer: $\hat{\beta}_{\mathrm{OLS}} = (X^\top X)^{-1}X^\top y$; residuals satisfy $X^\top\hat{\varepsilon} = 0$; H is the symmetric idempotent projector onto $\mathrm{col}(X)$ with $\mathrm{tr}(H) = p$ (degrees of freedom used); univariate centred case reduces to Cov/Var. □

P5.3 *Gauss–Markov assumptions and violations*

(a) Assumptions: (1) linearity / correct specification; (2) full column rank of X; (3) strict exogeneity $\mathbb{E}[\varepsilon \mid X] = 0$; (4) spherical errors $\mathrm{Var}(\varepsilon \mid X) = \sigma^2 I$ (homoskedastic *and* uncorrelated). Conclusion: $\hat{\beta}_{\mathrm{OLS}}$ is **BLUE** — among estimators *linear in* y and *unbiased*, it has minimum variance ($\mathrm{Var}(\tilde{\beta}) - \mathrm{Var}(\hat{\beta}) \succeq 0$, hence minimal for every $c^\top\beta$). **Normality is not assumed.**

(b)(i) Leakage breaks strict exogeneity: the regressor is correlated with the error. Consequence: **bias and inconsistency** — more data does not save you; this is the fatal one. Fix: redesign the signal to be point-in-time; in the omitted-variable version, add the missing control or use an instrument.

(ii) Heteroskedasticity breaks spherical errors. OLS stays unbiased but is *inefficient* and its reported standard errors are wrong. Fix: WLS/GLS (whiten, then Gauss–Markov applies to the transformed data, restoring BLUE) or robust standard errors for inference.

(iii) Near-multicollinearity violates *no* assumption while the rank is technically full: OLS is still BLUE — but $\mathrm{Var}(\hat{\beta}) = \sigma^2(X^\top X)^{-1}$ explodes along the near-singular direction, so "best" is still terrible. Fix: ridge, or drop/combine the collinear features.

(iv) Fat tails break nothing in Gauss–Markov (normality was never assumed): OLS remains BLUE. Only exact finite-sample t/F inference is affected. Fix: rely on CLT asymptotics or bootstrap.

Self-check. Order the four by severity: only (i) attacks $\mathbb{E}[\hat{\beta}]$; (ii) and (iii) attack variance/inference; (iv) attacks only the exact sampling distribution. That matches the MSE decomposition bias^2 + variance: bias failures are unfixable by more data, variance failures are merely expensive.

Answer: Linearity, full rank, strict exogeneity, spherical errors $\Rightarrow$ OLS is BLUE; normality NOT assumed. (i) exogeneity: bias + inconsistency (fatal) — redesign; (ii) heteroskedasticity: inefficiency + wrong SEs — WLS/robust; (iii) no assumption broken: variance explodes — ridge/drop; (iv) nothing broken: only exact t/F — asymptotics/bootstrap.

P5.4 *Omitted-variable bias*

(a) The short regression gives $\hat{\beta}_1 = (x_1^\top x_1)^{-1}x_1^\top y$. Substitute $y = x_1\beta_1 + x_2\beta_2 + \varepsilon$:

$$\hat{\beta}_1 = \beta_1 + \beta_2\,\frac{x_1^\top x_2}{x_1^\top x_1} + \frac{x_1^\top \varepsilon}{x_1^\top x_1}.$$

Taking expectations (exogeneity kills the last term):

$$\mathbb{E}[\hat{\beta}_1] = \beta_1 + \beta_2\,\frac{\mathrm{Cov}(x_1, x_2)}{\mathrm{Var}(x_1)}$$

— the true coefficient plus the omitted coefficient times the auxiliary regression slope of x_2 on x_1.

(b) $\mathbb{E}[\hat{\beta}_1] = 0 + 1 \times 0.3 = 0.3$. The backtest reports a "signal loading" of 0.3 that is nothing but market beta leaking through the correlation.

(c) The mistake: omit market exposure while your signal correlates with the market, and you rediscover beta and call it alpha. Fix: include x_2 in the regression, or *residualise* x_1 against the market (and other factors) before testing — after residualisation $\mathrm{Cov}(x_1^\perp, x_2) = 0$, so the bias term vanishes. (This is exactly what factor-residualised returns, e.g. Axioma-style, are for.)

Self-check. Two limits: if $\mathrm{Cov}(x_1, x_2) = 0$ the bias vanishes and the short regression is fine; if $x_1 = x_2$ the auxiliary slope is 1 and the short regression reports $\beta_1 + \beta_2 = 1$, i.e. the full market coefficient — exactly what regressing y on the market alone should give. Sign check: positive correlation and positive β_2 bias the estimate upward, as computed.

Answer: $\mathbb{E}[\hat{\beta}_1] = \beta_1 + \beta_2 \,\mathrm{Cov}(x_1, x_2)/\mathrm{Var}(x_1)$; the backtest reports 0.3 — pure market beta disguised as alpha; fix by including the market or residualising the signal against it first.

P5.5 *Gauss–Markov BLUE proof + ridge follow-up*

(a) Unbiasedness for *every* β forces $\mathbb{E}[\tilde{\beta}] = CX\beta = \beta$, i.e. $CX = I_p$. Write $C = (X^\top X)^{-1}X^\top + D$; then $CX = I_p + DX$, so $DX = 0$. With $\mathrm{Var}(y) = \sigma^2 I$,

$$\mathrm{Var}(\tilde{\beta}) = \sigma^2 CC^\top = \sigma^2\Big[(X^\top X)^{-1} + (X^\top X)^{-1}X^\top D^\top + DX(X^\top X)^{-1} + DD^\top\Big]$$

$$= \sigma^2\big[(X^\top X)^{-1} + DD^\top\big],$$

the cross terms dying because $DX = 0$ (and its transpose $X^\top D^\top = 0$). Hence $\mathrm{Var}(\tilde{\beta}) - \mathrm{Var}(\hat{\beta}) = \sigma^2 DD^\top \succeq 0$ (a Gram matrix), with equality iff $D = 0$, i.e. iff $\tilde{\beta}$ *is* OLS. One idea: any competitor is OLS plus a piece orthogonal to the design, and that piece only adds variance. PSD ordering means $\mathrm{Var}(c^\top\tilde{\beta}) \geq \mathrm{Var}(c^\top\hat{\beta})$ for every linear functional.

(b) No contradiction. The theorem crowns OLS best *among linear unbiased* estimators. Ridge is biased, so it competes outside that class, and MSE = bias^2 + variance lets it win: there always exists some $\lambda > 0$ for which ridge has strictly smaller MSE than OLS (Hoerl–Kennard). Under Gaussian errors OLS is the MLE and attains the Cramér–Rao bound, making it best among *all* unbiased estimators — which still says nothing about biased ones. Unbiasedness is a constraint, not a virtue.

(c) Here $X = \mathbf{1}_n$ and $\beta = \mu$. A linear estimator $\sum_i w_i y_i$ is unbiased iff $\sum_i w_i = 1$, and its variance is $\sigma^2 \sum_i w_i^2$. By Cauchy–Schwarz, $1 = (\sum_i w_i)^2 \le n \sum_i w_i^2$, so $\sum_i w_i^2 \ge 1/n$ with equality iff $w_i = 1/n$: the sample mean. And OLS on a constant regressor is exactly $(\mathbf{1}^\top \mathbf{1})^{-1}\mathbf{1}^\top y = \bar{y}$ — the theorem verified concretely.

Self-check. Part (c) is itself an independent verification of (a) in the simplest design. Additionally, set $D = 0$ in the variance formula: it collapses to $\sigma^2(X^\top X)^{-1}$, the known OLS variance; and $DD^\top$ is $p \times p$ PSD by construction ($v^\top DD^\top v = \|D^\top v\|^2 \ge 0$).

Answer: $\mathrm{Var}(\tilde{\beta}) - \mathrm{Var}(\hat{\beta}) = \sigma^2 DD^\top \succeq 0$ with $DX = 0$; ridge does not contradict the theorem because it is biased (outside the class) and wins on MSE; for the mean, the unique best linear unbiased weights are $w_i = 1/n$ — the sample mean. □

P5.6 *Ridge derivation, invertibility, shrinkage*

(a) $L(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|_2^2$; the gradient $-2X^\top y + 2X^\top X\beta + 2\lambda\beta = 0$ gives $(X^\top X + \lambda I)\beta = X^\top y$, so

$$\hat{\beta}_{\text{Ridge}} = (X^\top X + \lambda I)^{-1} X^\top y.$$

The Hessian $2(X^\top X + \lambda I) \succ 0$ strictly, so the minimiser is unique — even when OLS is not.

(b) $X^\top X \succeq 0$ has eigenvalues $d_i \ge 0$; adding λI shifts every eigenvalue to $d_i + \lambda \ge \lambda > 0$. All eigenvalues strictly positive $\Rightarrow$ invertible, always. This is the mechanical fix for multicollinearity and for $p > n$.

(c) With $X^\top X = I$: $\hat{\beta}_{\text{Ridge}} = (1+\lambda)^{-1}X^\top y = \hat{\beta}_{\text{OLS}}/(1+\lambda)$ — every coefficient shrunk by the same factor, never set exactly to zero. Numerically: $1.2/1.5 = 0.8$.

(d) Put a prior $\beta \sim \mathcal{N}(0, \tau^2 I)$ with Gaussian noise σ^2: the log-posterior is $-\|y - X\beta\|^2/(2\sigma^2) - \|\beta\|^2/(2\tau^2) + \text{const}$, so the MAP estimate is exactly ridge with $\lambda = \sigma^2/\tau^2$. Strong prior (small τ) $\Rightarrow$ large λ $\Rightarrow$ heavy shrinkage.

Self-check. Limits: $\lambda \to 0$ recovers OLS; $\lambda \to \infty$ shrinks everything to 0 (infinitely confident prior at zero) — both sensible. Differentiate back in the 1-D orthonormal case at $b = 0.8$: $2(b - 1.2) + 2\lambda b = 2(-0.4) + 2(0.5)(0.8) = -0.8 + 0.8 = 0$.

Answer: $\hat{\beta}_{\text{Ridge}} = (X^\top X + \lambda I)^{-1}X^\top y$; always invertible because every eigenvalue is at least $\lambda > 0$; orthonormal case shrinks OLS by $1/(1+\lambda)$, so $1.2 \to 0.8$; MAP under a Gaussian prior with $\lambda = \sigma^2/\tau^2$. □

P5.7 *Regularisation menu* ★

The five to lead with:

- **L2 / ridge** — penalise $\|w\|_2^2$: smooth shrinkage toward zero, handles collinearity, Gaussian prior.
- **L1 / lasso** — penalise $\|w\|_1$: exact zeros, so feature selection; Laplace prior.
- **Dropout** — randomly zero units in training: prevents co-adaptation, acts as an implicit ensemble of subnetworks.
- **Early stopping** — halt training when validation loss turns: limits effective capacity along the optimisation path (for linear models, closely related to ridge).
- **Data augmentation / noise injection** — enlarge the training distribution with invariances; in finance: train across assets and regimes rather than one series.

Backups when probed: elastic net (L1 + L2); label smoothing; batch norm's implicit regularising effect; ensembling/bagging; weight decay as distinct from L2 under Adam (AdamW); and the tree-model menu — `max_depth`/number of leaves, `min_data_in_leaf`/`min_child_weight`, L1/L2 penalties on leaf values (`lambda_l1`/`lambda_l2`), `feature_fraction`/`bagging_fraction` subsampling, small learning rate with more trees, early stopping on validation. Finance-specific: covariance *shrinkage* (Ledoit–Wolf) is regularisation of an estimator, not just of a predictive model.

Self-check. Apply the unifying test to each item: regularisation = anything that constrains effective capacity or pulls the fit toward a prior, trading bias for variance. L2/L1 add explicit priors; dropout and augmentation inject noise/invariances that shrink the hypothesis space actually reachable; early stopping bounds the optimisation path; every tree control caps how far a tree can chase noise. All pass; nothing on the list merely "improves optimisation" without constraining fit.

Answer: L2/ridge, L1/lasso, dropout, early stopping, data augmentation/noise injection — with elastic net, label smoothing, batch norm, bagging/ensembling, tree controls (depth, min leaf data, leaf-value L1/L2, feature/row subsampling, small learning rate) and Ledoit–Wolf covariance shrinkage as backups.

P5.8 *Lasso soft-thresholding*

(a) Case $b > 0$: the objective is smooth, derivative $2(b - \hat{\beta}_j) + \lambda = 0$ gives $b = \hat{\beta}_j - \lambda/2$, admissible only if $\hat{\beta}_j > \lambda/2$. Case $b < 0$: symmetrically $b = \hat{\beta}_j + \lambda/2$, admissible if $\hat{\beta}_j < -\lambda/2$. Case $b = 0$: the subdifferential of the objective at 0 is $-2\hat{\beta}_j + \lambda[-1, 1]$; it contains 0 iff $|\hat{\beta}_j| \le \lambda/2$, and the objective is convex, so $b = 0$ is then optimal. Combining:

$$\hat{\beta}_j^{\text{Lasso}} = \text{sign}(\hat{\beta}_j)\big(|\hat{\beta}_j| - \tfrac{\lambda}{2}\big)_+ \qquad \text{(soft-thresholding).}$$

(b) Threshold $\lambda/2 = 0.5$: $2.0 \to 1.5$; $0.6 \to 0.1$; $-0.3 \to 0$ (inside the threshold, killed exactly).

(c) At $b = 0$ the ridge penalty's pull is $\frac{d}{db}\lambda b^2 = 2\lambda b = 0$ — the penalty exerts *no force* at zero, so any nonzero data pull moves the coefficient off zero; ridge only ever multiplies by $1/(1+\lambda)$. The L1 penalty keeps a *constant* force λ (per unit of $|b|$, i.e. $\lambda/2$ against the data pull in this parametrisation) all the way to zero: whenever the data pull $|\hat{\beta}_j|$ is below the threshold, the penalty wins outright and pins the coefficient at exactly zero. Geometrically: the L1 ball has corners on the axes; the L2 ball is round.

Self-check. Verify (b) for $\hat{\beta}_j = 0.6$ by direct evaluation: at $b = 0.1$: $(0.5)^2 + 1(0.1) = 0.35$; at $b = 0$: 0.36; at $b = 0.2$: $0.16 + 0.2 = 0.36$. The claimed optimum 0.35 beats both neighbours. And the formula's two limits: $\lambda \to 0$ returns OLS, $\lambda \ge 2|\hat{\beta}_j|$ kills everything.

Answer: Soft-thresholding $\text{sign}(\hat{\beta})(|\hat{\beta}| - \lambda/2)_+$; with $\lambda = 1$: $(2.0, 0.6, -0.3) \to (1.5, 0.1, 0)$; lasso zeros exist because the L1 penalty keeps a constant force at zero while the ridge penalty's force vanishes there.

P5.9 *Lasso vs ridge in alpha research*

(a) Penalised regression is equivalent to minimising the loss subject to a norm ball constraint. The L1 ball is a diamond (cross-polytope) with corners on the coordinate axes; the elliptical level sets of the quadratic loss typically first touch the ball at a corner — where some coordinates are exactly zero. The L2 ball is round: contact happens at a generic boundary point, so coefficients shrink but essentially never land exactly on zero.

(b) With $p = 500$ approaching $n = 750$ and heavy collinearity, plain OLS variance explodes, so regularisation is mandatory. Lasso's failure mode: within a group of highly correlated signals it tends to pick *one* representative near-arbitrarily and zero the rest, and the choice is unstable — resample the data and a different group member gets selected. For alpha research that instability is costly (turnover in which signals are "on," fragile attribution). Ridge keeps the whole group with weight shared across it (grouping behaviour) and handles the collinearity, at the cost of no selection. Additionally, when $p > n$, lasso can select at most n variables — a structural limit to know.

(c) Elastic net combines both penalties, $\lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2^2$: the L2 part makes the solution unique and spreads weight across correlated groups (grouping effect), the L1 part still zeroes genuinely useless signals. That is precisely the correlated-many-signals regime, hence the default.

Self-check. The duplicate-feature thought experiment: let $x_1 = x_2$ exactly. Ridge's objective is strictly convex (thanks to the penalty), and by symmetry its unique solution splits the coefficient equally — stable. Lasso's fit and penalty are identical for any split $b_1 + b_2 = s$ with $b_1, b_2 \geq 0$: a whole continuum of minimisers, so the selected representative is arbitrary — exactly the instability claimed in (b).

Answer: L1's diamond has corners on the axes (exact zeros), L2's ball is round (shrinkage only); here ridge or elastic net — lasso's selection within correlated groups is arbitrary and resample-unstable (and caps at n selections when $p > n$); elastic net adds L2's grouping and uniqueness to L1's sparsity, the pragmatic default.

P5.10 *Inverse regression*

(a) The two regressions are different projections: Y-on-X minimises *vertical* squared distances, X-on-Y minimises *horizontal* ones. They produce two different lines (unless the data are perfectly collinear), each attenuated toward zero relative to the SD line by a factor ρ. Fitted slopes are not reciprocals.

(b) $\hat{\beta}_{X|Y} = \rho\, \sigma_X/\sigma_Y = 0.5 \times 1 = 0.5$ — the same number as the forward slope here, not 2. The shortcut overstates by $2/0.5 = 4 = 1/\rho^2$.

(c) $\hat{\beta}_{Y|X} = \rho\, \sigma_Y/\sigma_X = B$ and $\hat{\beta}_{X|Y} = \rho\, \sigma_X/\sigma_Y = D$. Multiplying: $BD = \rho^2 \leq 1$, hence $D = \rho^2/B$, which equals $1/B$ only when $|\rho| = 1$ (exactly collinear data). In general naive inversion inflates the answer by the factor $1/\rho^2$.

Self-check. Internal consistency of the given numbers: with $\sigma_X = \sigma_Y$, the forward slope should itself equal $\rho = 0.5$ — and it does, matching the stated 0.5. And the identity: $BD = 0.5 \times 0.5 = 0.25 = \rho^2$. Sanity bound: both slopes have $|\text{slope}| \leq \sigma_{\text{num}}/\sigma_{\text{den}}$, and $2 > \sigma_X/\sigma_Y = 1$ immediately flags the shortcut as impossible.

Answer: Slopes are not reciprocals: $\hat{\beta}_{Y|X}\hat{\beta}_{X|Y} = \rho^2$, so the true X-on-Y slope is $\rho^2/B = 0.5$, not 2 — the shortcut overstates by $1/\rho^2 = 4\times$; inversion is legitimate only when $|\rho| = 1$.

P5.11 *Hedge-ratio inversion, market application*

(a) Per \$1 of S, hedged return $= r_S - h\, r_A$ with h the notional ratio. Variance $= \sigma_S^2 + h^2\sigma_A^2 - 2h\rho\sigma_S\sigma_A$; minimising over h: $2h\sigma_A^2 = 2\rho\sigma_S\sigma_A$, so

$$h^* = \rho\, \frac{\sigma_S}{\sigma_A} = 0.6 \times \frac{2}{1} = 1.2 \quad \Rightarrow \quad \text{short \$1.2M of } A.$$

This is exactly the regression slope of S on A.

(b) The intern's regression gives $\hat{\beta}_{A|S} = \rho\, \sigma_A/\sigma_S = 0.6 \times 0.5 = 0.3$; inverting, $h = 1/0.3 = 3.33$, i.e. *\$3.33M* short. Over-hedge factor: $3.33/1.2 = 2.78 = 1/\rho^2$ — the inverse-regression trap in production.

(c) In $\%^2$ per day (on \$1M):

- Unhedged: $\sigma = 2\%$, i.e. \$20,000/day.
- Correct hedge $h = 1.2$: variance $= 4 + (1.2)^2(1) - 2(1.2)(0.6)(2)(1) = 4 + 1.44 - 2.88 = 2.56$, so $\sigma = 1.6\% =$ \$16,000/day.
- Intern $h = 3.33$: variance $= 4 + (3.33)^2(1) - 2(3.33)(0.6)(2)(1) = 4 + 11.11 - 8.00 = 7.11$, so $\sigma = 2.67\% \approx$ \$26,700/day.

Punchline: the over-hedged book is *riskier than not hedging at all* — the extra \$2.1M of A is unhedged exposure in its own right. Secondary lesson: even the correct hedge only cuts vol from 2.0% to 1.6%, because with $\rho = 0.6$ the residual fraction $\sqrt{1 - \rho^2} = 0.8$ of the risk is unhedgeable by A.

Self-check. The correct-hedge residual should equal $\sigma_S\sqrt{1-\rho^2} = 2 \times 0.8 = 1.6\%$ — matches the direct variance computation. And the two slopes multiply to $1.2 \times 0.3 = 0.36 = \rho^2$.

Answer: (a) \$1.2M; (b) \$3.33M, over-hedged by $1/\rho^2 = 2.78\times$; (c) \$20,000 unhedged, \$16,000 correctly hedged, ≈\$26,700 for the intern — the inverted hedge is riskier than no hedge at all.

P5.12 *Beta vs correlation*

(a) False. $\beta = \rho\,\sigma_Y/\sigma_X$: beta mixes correlation with the *relative volatility.* Counterexample: $\rho = 0.2$ with $\sigma_Y/\sigma_X = 10$ gives $\beta = 2$ — a "high-beta" stock whose returns are barely aligned with the index (think an illiquid small-cap or a crypto-adjacent name against an equity index).

(b) The *beta* sizes the hedge: hedging neutralises P&L magnitude, and magnitude lives in σ_Y/σ_X. The *correlation* (through $R^2 = \rho^2$, i.e. the residual fraction $1-\rho^2$) tells you whether hedging achieves anything: it is the share of variance the hedge can remove. Correlation says whether to hedge, beta says how much.

(c) $\beta = 0.2 \times 10/1 = 2$; explained fraction $= R^2 = \rho^2 = 4\%$. Reconciliation: the index-sized piece of the stock is large per unit of index move (beta 2), yet it is a tiny share of the stock's own huge variance — a beta-2 hedge here would be technically correctly sized and almost pointless, leaving residual vol $10\% \times \sqrt{0.96} \approx 9.8\%$.

Self-check. Bound check: $|\beta| \le \sigma_Y/\sigma_X = 10$, and $2 \le 10$. Consistency: R^2 recomputed from the regression view — explained variance $= \beta^2\sigma_X^2 = 4 \times 1 = 4$ out of $\sigma_Y^2 = 100$, i.e. 4% (matches ρ^2).

Answer: False — e.g. $\rho = 0.2$, $\sigma_Y/\sigma_X = 10$ gives $\beta = 2$; beta sizes the hedge, correlation (via R^2) decides if it is worthwhile; numbers: $\beta = 2$, explained variance 4%, residual vol ≈ 9.8% — correctly sized yet nearly pointless.

P5.13 *Choosing a hedge instrument*

(a) Hedge ratio $h = \rho\,\sigma_Y/\sigma_F$; residual vol $= \sigma_Y\sqrt{1-\rho^2}$ (the part of Y orthogonal to the hedge).

- F_1: $h_1 = 0.9 \times 4/8 = 0.45 \Rightarrow$ short \$450k. Residual $= 4\%\sqrt{1-0.81} = 4\% \times 0.436 = 1.74\%$.
- F_2: $h_2 = 0.6 \times 4/1 = 2.4 \Rightarrow$ short \$2.4M. Residual $= 4\%\sqrt{1-0.36} = 4\% \times 0.8 = 3.2\%$.

(b) Choose F_1. The residual vol of an optimally sized hedge is $\sigma_Y\sqrt{1-\rho^2}$ — it depends *only on the correlation*, not on the hedge instrument's vol or beta. Higher $|\rho| \Rightarrow$ better hedge, full stop.

(c) The rule confuses sizing with quality. F_2 carries the much larger hedge ratio (2.4 vs 0.45) — by the rule it looks like "more hedge" — yet it leaves nearly twice the residual risk (3.2% vs 1.74%). Beta only tells you *how much* to trade once the instrument is chosen; correlation tells you *which* instrument can actually absorb the risk. (Real desks then layer on liquidity and cost — but risk-wise the correlation decides.)

Self-check. Recompute F_1's residual directly: variance $= \sigma_Y^2 + h^2\sigma_F^2 - 2h\rho\sigma_Y\sigma_F = 16 + (0.45)^2(64) - 2(0.45)(0.9)(4)(8) = 16 + 12.96 - 25.92 = 3.04$, and $\sqrt{3.04} = 1.74\%$ — matches $\sigma_Y\sqrt{1-\rho^2} = 4\sqrt{0.19}$. Same check for F_2: $16 + 5.76 - 11.52 = 10.24$, $\sqrt{10.24} = 3.2\%$.

Answer: F_1: short \$450k, residual 1.74%; F_2: short \$2.4M, residual 3.2%. Choose F_1 — residual vol $\sigma_Y\sqrt{1-\rho^2}$ depends only on correlation; the higher-beta rule confuses hedge sizing with hedge quality.

P5.14 *WLS precision weighting*

(a) Take $\hat{\mu} = wx_1 + (1-w)x_2$ (unbiased for any w). Variance $= w^2(1) + (1-w)^2(4)$; setting the derivative to zero: $2w - 8(1-w) = 0$, so $w = 4/5$. Variance $= (16/25)(1) + (1/25)(4) = 20/25 = 0.8 < 1$: yes, strictly better than the good measurement alone — even a noisy second reading adds information when weighted correctly.

(b) This is OLS of the "regression" $x_i = \mu + \varepsilon_i$ on a constant with heteroskedastic noise, done properly: minimise $\sum_i (x_i - \mu)^2/\sigma_i^2$, i.e. *whiten* each observation by dividing by σ_i and run OLS on the transformed data. The first-order condition gives precision weighting: $w_i \propto 1/\sigma_i^2$ (here

$1 : 1/4$, i.e. $4/5$ and $1/5$), with combined variance $\left(\sum_i 1/\sigma_i^2\right)^{-1}$. Gauss–Markov applies to the whitened model, so WLS is BLUE.

(c) Unweighted OLS minimises raw squared residuals, so the crisis months — which have the largest residual variance — dominate the objective while carrying the *least* information per observation; the fit chases the noisiest regime and the reported standard errors are wrong on top. Fixes: WLS with weights $\propto$ inverse conditional variance (e.g. inverse realised variance / vol-scaled returns), which restores efficiency; or, if you only need valid inference with the same point estimates, heteroskedasticity-robust (White) standard errors.

Self-check. Recompute the optimal variance the closed-form way: $(1/1 + 1/4)^{-1} = (5/4)^{-1} = 0.8$ — matches the calculus route. And equal weights would give $(1/4)(1) + (1/4)(4) = 1.25 > 0.8$, confirming the weighting matters.

Answer: $w = 4/5$ on the sd-1 reading, combined variance 0.8 (better than either alone); this is WLS with precision weights $w_i \propto 1/\sigma_i^2$ and combined variance $(\sum 1/\sigma_i^2)^{-1}$; in crisis-heteroskedastic data unweighted OLS lets the noisiest points steer the fit — use inverse-variance WLS (or robust SEs for inference).

P5.15 *Residual weighting vs GLS*

Adjudication. Teammate 3 has the right formula for the wrong problem. Two opposite weighting philosophies are being conflated:

- **WLS/GLS (statistics):** you are estimating *one* model efficiently under heteroskedastic noise; you *down*-weight noisy observations ($w_i = 1/\sigma_i^2$) because they carry less information. Goal: efficiency.
- **Residual weighting (the ML move):** weighting by $|y_i - \hat{y}_{A,i}|$ *up*-weights the points A got wrong so that B *specialises on A's errors.* That is the **boosting** idea in disguise (safe names: boosting-style reweighting, residual fitting, cascading — not "stacking," which trains a meta-learner on model *outputs*). Goal: specialisation, i.e. bias reduction of the ensemble.

So nobody's weights are "upside-down" — they answer different questions. But two genuine fixes to Teammate 2's execution: (1) the cleaner boosting form fits B *to the residuals* $y_i - \hat{y}_{A,i}$ and combines additively, $F = A + \eta B$ with shrinkage $\eta < 1$, rather than refitting y with weights and naively averaging (averaging halves A's calibrated signal and mixes objectives); (2) B should be weak/shallow — a specialist, not a second generalist.

Backfire regime. In financial data, A's largest residuals concentrate in high-volatility regimes, where much of the error is *irreducible noise*, not structure. Up-weighting them makes B specialise on noise — overfitting crisis observations exactly where WLS logic says to down-weight. Guardrails: shrinkage, shallow B, early stopping, or weighting by residuals *after* vol-normalising returns.

Self-check. Limiting cases separate the two philosophies cleanly: if A's residuals are pure i.i.d. noise, the weights are uninformative and B chases noise — the scheme should be abandoned (and WLS logic wins); if A systematically misses one regime (structured residuals), the weights concentrate exactly there and the cascade adds real skill (boosting logic wins). The scheme is valid precisely when residuals contain structure — consistent with the adjudication.

Answer: It is not WLS: GLS/WLS down-weights noisy points ($1/\sigma_i^2$) to fit one model efficiently, while this scheme up-weights hard points so B specialises on A's errors — boosting-style residual fitting; combine as a cascade $F = A + \eta B$ (fit B on residuals, shrink), not a naive average; it backfires in high-vol regimes where large residuals are irreducible noise.

P5.16 *Dart problem, Gaussian, pitchfork*

(a) Write $a = m + u$. Then $S = g(u + d/2) + g(u - d/2)$ with $g(t) = e^{-t^2/(2\sigma^2)}$, which is even — so S is symmetric in u, and a smooth symmetric function has $S'(m) = 0$. (Directly: the two derivative terms at $u = 0$ are equal and opposite.)

(b) Classify the critical point via the second derivative. For one term, $g''(t) = g(t)(t^2/\sigma^4 - 1/\sigma^2)$; at $u = 0$ each term has $t = \pm d/2$, so

$$S''(m) = \frac{2\,e^{-d^2/(8\sigma^2)}}{\sigma^4}\Big(\frac{d^2}{4} - \sigma^2\Big).$$

For $d < 2\sigma$ this is negative: the midpoint is a maximum (indeed the global one — an equal two-component Gaussian mixture is unimodal exactly when the separation is at most 2σ, the classic bimodality threshold). For $d > 2\sigma$ it is *positive*: the midpoint becomes a local *minimum*, and two symmetric maxima split off on either side — a pitchfork bifurcation at $d = 2\sigma$. As $d/\sigma \to \infty$ the two optima migrate onto the targets themselves: you commit to one target rather than split the difference.

(c) A Gaussian log-density is a negative *squared* distance: maximising a Gaussian score is minimising squared loss. Squared loss is minimised by the **mean** — the conditional mean is what L2 regression estimates, and "aim at the mean" is the regression content of the $p = 2$ exponent. Sharp addendum: if the score were *multiplicative* across targets (equivalently, if you summed *log*-densities), the objective would be $-\sum_i (a - x_i)^2/(2\sigma^2)$, maximised at the mean for *every d* — the bifurcation exists only because we add densities, not log-densities. Precision about which objective you are optimising is the whole game.

Self-check. Units: $S''(m)$'s sign depends only on d/σ, as it must (the problem has one dimensionless parameter). Limits: $d \to 0$, the two bumps merge and the midpoint is trivially optimal; $d \gg \sigma$, aiming at a target scores ≈ 1 while the midpoint scores $2e^{-d^2/(8\sigma^2)} \to 0$, so the optimum must indeed leave the midpoint — consistent with the 2σ threshold. Boundary: at $d = 2\sigma$ exactly, $S''(m) = 0$, the degenerate pitchfork point.

Answer: The midpoint is always critical by symmetry; it is the optimum iff $d \le 2\sigma$ (since $S''(m) \propto d^2/4 - \sigma^2$), and past $d = 2\sigma$ a pitchfork sends two maxima toward the targets; the squared exponent is L2 loss — Gaussian score maximisation = squared-loss minimisation = predict the (conditional) mean.

P5.17 *Dart problem, L1, median vs endpoint*

(a) For continuous F, differentiate through the expectation: $g'(c) = \mathbb{E}[\text{sign}(c - X)] = P(X < c) - P(X > c) = 2F(c) - 1$. Setting $g'(c) = 0$ gives $F(c) = 1/2$; since $|x - c|$ is convex in c, g is convex and this stationary point is the global minimum — the median. For the two-point empirical distribution, $g(c) = \frac{1}{2}|x_1 - c| + \frac{1}{2}|x_2 - c|$ is flat and minimal on the whole interval $[x_1, x_2]$: *every* point between the targets is optimal under absolute loss (any point in the interval is a median).

(b) The game flips the conclusion. For a between the targets, write $a = x_1 + t$, $t \in [0, d]$: $S = e^{-t} + e^{-(d-t)}$, a sum of two convex exponentials, hence convex in t — and a convex function on an interval attains its *maximum at an endpoint.* Outside the interval both distances grow, so S strictly decreases; the global maximum is therefore at a target. The margin:

$$S(0) - S(d/2) = \big(1 + e^{-d}\big) - 2e^{-d/2} = \big(1 - e^{-d/2}\big)^2 > 0 \quad \text{for every } d > 0.$$

With Laplace tails you aim *at* a target, never between them. Intuition: e^{-r} rewards being very close much more than squared-exponential-type spreading — the kink at zero pays you for committing.

(c) Mapping: **L2 loss** $\leftrightarrow$ conditional **mean** (outlier-sensitive), **L1 loss** $\leftrightarrow$ conditional **median** (robust) — and part (b) is a warning that "maximise a score" and "minimise a loss" can point to different aims even for the same tail shape (minimising expected absolute distance says anywhere in $[x_1, x_2]$; maximising summed Laplace density says an endpoint). The classic conflation to avoid: *losses* (what functional you predict: mean vs median) versus *penalties* (how you constrain coefficients: ridge shrinks, lasso selects). "L1" appears in both sentences and means different things.

Self-check. Numeric probe at $d = 2$: endpoint $1 + e^{-2} \approx 1.135$ vs midpoint $2e^{-1} \approx 0.736$; difference $\approx 0.399 = (1 - e^{-1})^2 \approx (0.632)^2$. Limits: $d \to 0$ gives margin $(1 - 1)^2 = 0$ — targets coincide, no preference, as it must. Convexity check: $S''(t) = e^{-t} + e^{-(d-t)} > 0$ on $(0, d)$, so no interior maximum exists. And (a) sanity: for symmetric X the median equals the mean, so L1 and L2 point estimates agree there — consistent with both derivations.

Answer: (a) $g'(c) = 2F(c) - 1$ and convexity give the median; with two points, the whole interval $[x_1, x_2]$ is optimal. (b) S is convex between the targets, so the maximum is at an endpoint, beating the midpoint by $(1 - e^{-d/2})^2 > 0$: with Laplace tails, aim at a target. (c) L2 $\leftrightarrow$ mean, L1 $\leftrightarrow$ median; keep losses (what you predict) distinct from penalties (ridge shrinks, lasso selects).

21.6 Solutions — §6 Linear Algebra, PCA and Covariance

P6.1 *Vector norms / reverse triangle inequality* ♦

By the triangle inequality, $|A| = |(A - B) + B| \leq |A - B| + |B|$, so $|A - B| \geq |A| - |B|$; swapping the roles of A and B gives $|A - B| \geq |B| - |A|$. Together these are the *reverse triangle inequality*

$$|A - B| \;\geq\; \big|\,|A| - |B|\,\big| \;=\; |5 - 7| \;=\; 2.$$

Equality requires the triangle formed by A, B, $A - B$ to degenerate: A and B parallel and pointing in the *same* direction. Concretely $B = \frac{7}{5}A$ gives $A - B = -\frac{2}{5}A$, with norm $\frac{2}{5} \cdot 5 = 2$. (For contrast: the maximum of $|A - B|$ is $|A| + |B| = 12$, attained anti-parallel.)

Self-check: law of cosines: $|A - B|^2 = 5^2 + 7^2 - 2 \cdot 5 \cdot 7 \cos\theta = 74 - 70 \cos\theta$, minimised at $\cos\theta = 1$, giving $|A - B|^2 = 4$, i.e. $|A - B| = 2$ — any misalignment strictly increases the distance. One-dimensional sanity case: $A = 5$, $B = 7$ on the real line, $|5 - 7| = 2$.

Answer: 2, attained when A and B are parallel in the same direction (reverse triangle inequality).

P6.2 *Matrix transformations rapid-fire* ♦

(a) SR. Operators act right-to-left on column vectors: first Rx, then $S(Rx) = (SR)x$. RS is the classic wrong-order trap (it shears first). *Self-check:* feed a concrete x through the two steps and through the product — only SR reproduces "rotate, then shear."

(b) The y-axis. The middle row and middle column are e_2, so the y-component passes through untouched, while the (x, z) block is a standard planar rotation. *Self-check:* the matrix maps $(0, 1, 0)^\top$ to itself for every t, and at $t = 0$ it is the identity.

(c) $\det(A) = 0$. Dependent columns mean the image of the unit cube is squashed into a lower-dimensional set — zero volume; equivalently A is singular. *Self-check:* $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ (second column twice the first): $\det = 4 - 4 = 0$.

(d) $B^{-1}A^{-1}$ — inverses reverse order ("socks–shoes"). *Self-check:* $(AB)(B^{-1}A^{-1}) = I$ (the inner pair cancels first).

(e) Areas are multiplied by $|\det| = 2$ and orientation is inverted (the sign flip): every shape doubles in area and is mirrored. *Self-check:* $\mathrm{diag}(2, -1)$ has determinant -2; it stretches x by 2 (unit square $\to$ area 2) and flips the y-axis (orientation reversed).

Answer: (a) SR; (b) the y-axis; (c) $\det(A) = 0$; (d) $B^{-1}A^{-1}$; (e) areas doubled, orientation inverted.

P6.3 *Eigenvectors of a 2D rotation* ♦

The matrix is $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Its characteristic polynomial is $\lambda^2 + 1 = 0$, so $\lambda = \pm i$: no real eigenvalues, hence *no real eigenvectors*. Geometrically that is obvious: every nonzero vector is turned by 90 degrees, so none keeps its direction.

The bait option imports 3D intuition: a rotation of $\mathbb{R}^3$ does have its axis as an eigenvector (eigenvalue 1), but the "axis" of a 2D rotation is not a vector inside $\mathbb{R}^2$ — there is no axis to point along. General fact worth stating: a 2D rotation by angle θ has real eigenvectors only for $\theta = 0$ (identity, $\lambda = 1, 1$) and $\theta = \pi$ ($\lambda = -1, -1$, every vector is an eigenvector).

Self-check: two independent routes. (i) Trace/determinant: $\operatorname{tr} R = 0 = \lambda_1 + \lambda_2$ and $\det R = 1 = \lambda_1\lambda_2$ force $\lambda^2 + 1 = 0$, i.e. $\pm i$. (ii) Geometric: $Rx \perp x$ for every x, so $Rx = \lambda x$ with real $\lambda \neq 0$ would make x orthogonal to itself — impossible for $x \neq 0$ ($\lambda = 0$ is ruled out since $\det R \neq 0$).

Answer: It has no real eigenvectors — eigenvalues are $\pm i$; the "axis of rotation" option describes 3D, not $\mathbb{R}^2$.

P6.4 *Correlation-matrix feasibility*

(a) A valid correlation matrix is the covariance matrix of standardised variables, hence PSD: $x^\top \Sigma x = \operatorname{Var}(x^\top Z) \geq 0$ for all x, so all eigenvalues $\lambda_i \geq 0$ and $\det = \prod \lambda_i \geq 0$. Here, using $\det = 1 + 2abc - a^2 - b^2 - c^2$ for $\begin{pmatrix} 1 & a & b \\ a & 1 & c \\ b & c & 1 \end{pmatrix}$ with $a = 0.9$, $b = -0.9$, $c = 0.8$:

$$\det = 1 + 2(0.9)(-0.9)(0.8) - 0.81 - 0.81 - 0.64 = 1 - 1.296 - 2.26 = -2.556 < 0.$$

A negative determinant means an odd number of negative eigenvalues; since $\operatorname{tr} = 3 > 0$ rules out all three negative, exactly one eigenvalue is negative — the matrix is not PSD, so it is not a correlation matrix. Intuition: B is nearly the same bet as A, and C is nearly the opposite bet, so B and C must be strongly *negatively* correlated; $+0.8$ is geometrically impossible.

(b) Each entry was estimated on a different sample. The one-line PSD proof needs all entries to come from a *single* quadratic form — one dataset, one Σ. Pairwise-complete estimation breaks that structure: there is no single sample whose covariance produces all three numbers simultaneously, so nothing forces the assembled matrix to be PSD. (Same failure mode as asynchronous data across assets.)

(c) Treat standardised variables as unit vectors with ρ = cosine of the angle between them. Feasibility of the triple is $\det \geq 0$:

$$\det(c) = -c^2 - 2(0.81)c + 1 - 0.81 - 0.81 = -c^2 - 1.62c - 0.62 \geq 0 \iff c^2 + 1.62c + 0.62 \leq 0,$$

with roots $c = \frac{-1.62 \pm \sqrt{1.62^2 - 4(0.62)}}{2} = \frac{-1.62 \pm 0.38}{2}$, i.e. $c = -1$ and $c = -0.62$. So $\rho_{BC} \in [-1, -0.62]$: the recorded $+0.8$ is far outside. (Equivalently the general bound $\rho_{BC} \in [ab - \sqrt{(1-a^2)(1-b^2)},\ ab + \sqrt{(1-a^2)(1-b^2)}] = [-0.81 - 0.19, -0.81 + 0.19]$.) Repairs: (i) eigenvalue clipping — floor negative eigenvalues at 0, rescale the diagonal back to 1; (ii) shrinkage toward the identity, $(1-\alpha)\Sigma + \alpha I$ (the same $+\lambda I$ medicine as ridge for near-singular $X^\top X$); (iii) impose factor structure (low-rank plus diagonal).

Self-check: the two routes agree — the quadratic's roots (-1 and -0.62; Vieta: product 0.62, sum -1.62, both match) coincide with the cosine bound -0.81 ± 0.19; and evaluating det at $c = 0.8$ by cofactor expansion also gives $0.36 - 1.458 - 1.458 = -2.556$, matching the formula above.

Answer: (a) $\det = -2.556 < 0$, so one eigenvalue is negative — not PSD, not a valid correlation matrix; (b) pairwise estimation on different samples breaks the single-quadratic-form structure that guarantees PSD; (c) $\rho_{BC} \in [-1, -0.62]$; repair by eigenvalue clipping, shrinkage toward I, or factor structure.

P6.5 *PCA derivation and explained variance*

(a) The projection of the data onto w has variance $w^\top \Sigma w$. Maximise subject to $w^\top w = 1$ with a Lagrangian: $L = w^\top \Sigma w - \lambda(w^\top w - 1)$, and $\nabla_w L = 2\Sigma w - 2\lambda w = 0$ gives $\Sigma w = \lambda w$ — stationary points are *eigenvectors* of Σ. At an eigenvector the objective equals $w^\top \Sigma w = \lambda$, so the maximiser is the eigenvector of the largest eigenvalue λ_1, and the maximised variance is λ_1. Explained fraction of PC$_i$: $\lambda_i/\sum_j \lambda_j$.

(b) For $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: eigenvalues 3 (eigenvector $(1,1)^\top/\sqrt{2}$) and 1 (eigenvector $(1,-1)^\top/\sqrt{2}$). PC1 explains $\frac{3}{3+1} = \frac{3}{4} = 75\%$.

(c) Without centring, $X^\top X = n\,\bar{x}\bar{x}^\top +$ (centred scatter): the rank-one mean term is added to the true scatter, and with a large offset it dominates — the leading "component" chases the direction of the *mean vector*, not the direction of maximal variance, and its explained share is inflated. Centre first (PCA is defined on the covariance, not the raw second moment).

Self-check (b): trace $= 4 = 3+1$ and det $= 3 = 3 \cdot 1$, both consistent; direct verification $\Sigma(1,1)^\top = (3,3)^\top = 3\,(1,1)^\top$; and the shortcut for symmetric 2×2 with equal diagonal d and off-diagonal c — eigenvalues $d \pm c = 2 \pm 1$ with eigenvectors $(1, \pm 1)^\top$ — agrees.

Answer: (a) $w =$ top eigenvector of Σ, maximised variance $= \lambda_1$; (b) directions $(1,1)^\top/\sqrt{2}$ and $(1,-1)^\top/\sqrt{2}$, PC1 explains 75%; (c) it chases the mean direction, not the maximal-variance direction. □

P6.6 *Equicorrelated covariance / PC1 as market mode*

(a) $\mathbf{1}\mathbf{1}^\top$ is rank one with eigenvector $\mathbf{1}$ (eigenvalue n) and eigenvalue 0 on the $(n-1)$-dimensional complement $\{x : \sum_i x_i = 0\}$. Since Σ is a linear combination of I and $\mathbf{1}\mathbf{1}^\top$, it has the same eigenvectors:

$$\text{on } \mathbf{1}: \quad \lambda_1 = \sigma^2\big[(1-\rho) + \rho n\big] = \sigma^2\big(1 + (n-1)\rho\big);$$

$$\text{on the complement}: \quad \lambda = \sigma^2(1-\rho), \text{ multiplicity } n-1.$$

With $n = 100$, $\rho = 0.3$: $\lambda_1 = \sigma^2(1 + 29.7) = 30.7\,\sigma^2$ and $\lambda = 0.7\,\sigma^2$ (99 times).

(b) PC1 is the equal-weight direction $\mathbf{1}/\sqrt{n}$ — long everything, equally. Explained fraction $= \lambda_1/\operatorname{tr}\Sigma = \sigma^2(1+(n-1)\rho)/(n\sigma^2) = \frac{1+99(0.3)}{100} = 30.7\%$. Note the limit: as $n \to \infty$ the share tends to ρ — with $\rho = 0.3$, roughly a third of all variance in a large equicorrelated book is one single direction.

(c) PC1 has same-sign (here equal) weights on every asset: it *is* the broad-market portfolio, and any average pairwise correlation $\rho > 0$ makes it the dominant eigenvalue. A signal S with $\langle S, \mathbf{1}\rangle \neq 0$ (net long or net short) has a component along PC1, so a large slice of its P&L variance is market beta rather than selection — which is exactly why one beta-neutralises: project S off the market mode (next subsection's recipe) before sizing.

Self-check: the eigenvalues must sum to the trace: $\sigma^2(1+(n-1)\rho) + (n-1)\sigma^2(1-\rho) = \sigma^2[1 + (n-1)\rho + (n-1) - (n-1)\rho] = n\sigma^2 = \operatorname{tr}\Sigma$. Small case $n = 2$, $\sigma^2 = 2$, $\rho = 0.5$: formulas give $\lambda = 2(1.5) = 3$ and $2(0.5) = 1$ — exactly the worked staple $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

Answer: (a) $\lambda_1 = \sigma^2(1+(n-1)\rho)$ once (eigenvector $\mathbf{1}$), $\lambda = \sigma^2(1-\rho)$ with multiplicity $n-1$; (b) PC1 = equal weights, explaining $\frac{1+(n-1)\rho}{n} = 30.7\%$; (c) same-sign dominant eigenvector = market mode; net-directional signals carry market beta and must be projected off it.

P6.7 *PCA via the SVD / conditioning*

(a) $X^\top X = (USV^\top)^\top(USV^\top) = VSU^\top USV^\top = VS^2V^\top$ (using $U^\top U = I$). So the eigenvectors of the scatter matrix $X^\top X$ are the columns of V with eigenvalues s_i^2. The sample covariance is $C = X^\top X/(n-1) = V\frac{S^2}{n-1}V^\top$, hence the principal directions are the columns of V and the sample variance along v_i is $s_i^2/(n-1)$. (Under the $1/n$ convention it is s_i^2/n — pick one and stay consistent.)

(b) $n - 1 = 4$: variances $16/4 = 4$ and $4/4 = 1$. PC1 share $= \frac{4}{4+1} = 80\%$ (equivalently $s_1^2/(s_1^2 + s_2^2) = 16/20$ — the $(n-1)$ cancels in the ratio).

(c) Call the SVD of X. Forming $X^\top X$ explicitly *squares the condition number* ($\kappa(X^\top X) = \kappa(X)^2$), so directions belonging to small singular values drown in floating-point error, and rounding can even return tiny negative eigenvalues for what should be a PSD matrix. The SVD works on X directly and is backward stable; it also never materialises the $p \times p$ product.

Self-check: total variance two ways: sum of per-PC variances $4 + 1 = 5$ must equal $\mathrm{tr}(X^\top X)/(n-1) = (s_1^2 + s_2^2)/4 = 20/4 = 5$; shares sum to $80\% + 20\% = 100\%$. Dimension check: $X^\top X$ is $p \times p$ and V's columns live in $\mathbb{R}^p$, the space where directions should live.

Answer: (a) directions = columns of V; variance along v_i is $s_i^2/(n-1)$; (b) variances 4 and 1, PC1 explains 80%; (c) the SVD of X — forming $X^\top X$ squares the condition number. ☐

P6.8 *Gram-Schmidt / orthogonalising correlated signals*

(a) Projection (Gram–Schmidt step):

$$B_\perp = B - \frac{\langle A, B\rangle}{\langle A, A\rangle} A = B - \frac{\mathrm{Cov}(A,B)}{\mathrm{Var}(A)} A = B - 0.8\,A,$$

since both are demeaned with unit variance, so $\mathrm{Cov}(A, B) = \rho = 0.8$ and $\mathrm{Var}(A) = 1$. Then $\mathrm{Cov}(A, B_\perp) = \mathrm{Cov}(A, B) - 0.8\,\mathrm{Var}(A) = 0.8 - 0.8 = 0$, hence zero correlation. (This is exactly the OLS residual of B regressed on A — residuals are orthogonal to regressors.)

(b) $\mathrm{Var}(B_\perp) = \mathrm{Var}(B) - 2(0.8)\,\mathrm{Cov}(A, B) + 0.8^2\,\mathrm{Var}(A) = 1 - 1.28 + 0.64 = 0.36 = 1 - \rho^2$. So $\rho^2 = 64\%$ of B's variance was already in A — only about a third of B is new.

(c) Zero raw inner product equals zero covariance only when the means are zero: $\mathrm{Cov}(A, B) = \mathbb{E}[AB] - \mathbb{E}[A]\mathbb{E}[B]$, so with nonzero means two orthogonal vectors can be correlated and vice versa. Demean first, or equivalently include an intercept in the regression (residuals are then orthogonal to the constant column too).

(d) $B_\perp$ is the *incremental information* in B net of what A already trades. It is the right object when A is a frozen black box you must not double-count against: the combined book $A + B_\perp$ has additive variances (no covariance term), so risk aggregates cleanly and B gets credit only for what is genuinely new. If instead you were free to re-optimise the weights of A and B jointly, feeding raw B into the combiner is equivalent — orthogonalisation earns its keep precisely when the incumbent is untouchable.

Self-check: regression route: regressing unit-variance B on A has $R^2 = \rho^2 = 0.64$, so the residual variance is $1 - R^2 = 0.36$ — matches the direct expansion. Boundary sanity: $\rho = 0$ gives $B_\perp = B$ (nothing removed), $\rho = 1$ gives $\mathrm{Var}(B_\perp) = 0$ (nothing left).

Answer: (a) $B_\perp = B - 0.8A$, uncorrelated with A by construction; (b) $\mathrm{Var}(B_\perp) = 1 - \rho^2 = 0.36$; 64% of B was already in A; (c) orthogonality = zero correlation only for demeaned signals; (d) $B_\perp$ is B's incremental information — trade it when the overlapping incumbent is frozen.

P6.9 *Beta neutralisation via PC1*

(a) $\langle S, M\rangle = 2 - 1 + 1 = 2$ and $\langle M, M\rangle = 3$, so

$$S' = S - \tfrac{2}{3}(1,1,1)^\top = \left(\tfrac{4}{3},\, -\tfrac{5}{3},\, \tfrac{1}{3}\right)^\top.$$

(b) $\langle S', M\rangle = \frac{4}{3} - \frac{5}{3} + \frac{1}{3} = 0$. The weights sum to zero: the position is long–short balanced against the equal-weight market portfolio, so a broad move that lifts all three assets equally produces zero P&L to first order — the signal no longer loads on the dominant risk direction PC1. It is beta-neutral *by construction*, not by regression estimate.

(c) With $M = \mathbf{1}$ the projection term is $\frac{\langle S, \mathbf{1}\rangle}{n}\mathbf{1} = \bar{S}\,\mathbf{1}$: subtracting it is exactly *cross-sectional demeaning* of the signal. A real estimated PC1 only approximately has equal weights — high-beta names carry larger loadings — so projecting off the true PC1 removes a *beta-weighted* average rather than the plain average. Demeaning is the cheap first cut; PC1 projection is the refinement that respects each name's actual market loading. (Caveat worth saying aloud: in a mixed universe or a stressed regime, PC1 need not be a clean "market" — inspect the loadings' signs before calling it that.)

Self-check: Pythagoras — projection splits norms: $\|S\|^2 = 4 + 1 + 1 = 6$ should equal $\|S'\|^2 + \frac{\langle S,M\rangle^2}{\|M\|^2} = (\frac{16}{9} + \frac{25}{9} + \frac{1}{9}) + \frac{4}{3} = \frac{42}{9} + \frac{12}{9} = \frac{54}{9} = 6$. And the demeaning view: $\bar{S} = 2/3$, so $S' = (2,-1,1)^\top - \frac{2}{3}(1,1,1)^\top$ — the same vector.

Answer: (a) $S' = (4/3, -5/3, 1/3)^\top$; (b) $\langle S', M\rangle = 0$ and the weights sum to zero — market-neutral by construction; (c) with equal-weight M the projection is exactly cross-sectional demeaning; an estimated PC1 instead removes a beta-weighted average.

P6.10 *A-orthogonal matrices* ♦

First translate the definition into one matrix equation: $\langle Qx, Qy\rangle_A = x^\top(Q^\top AQ)y$ equals $x^\top Ay$ for *all* x, y iff

$$\boxed{Q^\top AQ = A}$$

(ordinary orthogonality is the $A = I$ case). Now test each candidate against $A = \mathrm{diag}(2,3,5)$:

- Swap $e_1 \leftrightarrow e_3$: $Q^\top AQ = \mathrm{diag}(5,3,2) \neq A$. A permutation *reorders* the weights — it can only work if the swapped weights are equal.
- $Q = \mathrm{diag}(1,-1,1)$: $Q^\top AQ = \mathrm{diag}(1^2 \cdot 2,\ (-1)^2 \cdot 3,\ 1^2 \cdot 5) = A$. Sign flips square away — axis reflections are always A-orthogonal for diagonal A.
- Swap $e_1 \leftrightarrow e_2$: $\mathrm{diag}(3,2,5) \neq A$.
- Rotation in the x–y plane: the upper 2×2 block of $Q^\top AQ$ is $R^\top \mathrm{diag}(2,3)R$, which for generic t has off-diagonal entry $\cos t \sin t \neq 0$ (and even at $t = \pi/2$ it becomes $\mathrm{diag}(3,2)$). Rotations *mix* unequal weights — they preserve $\langle\cdot,\cdot\rangle_A$ only on blocks where A is a multiple of the identity.

The concept in one line: A-orthogonality is congruence-invariance of a weighted geometry; with distinct diagonal weights, the only surviving axis symmetries are the sign flips. □

Answer: (B) $Q = \mathrm{diag}(1,-1,1)$.

21.7 Solutions — §7 Calculus for Quant Interviews

P7.1 *Derivatives toolkit: power rule* ♦

Write $\sqrt{x} = x^{1/2}$ and apply the power rule $(x^n)' = nx^{n-1}$:

$$\frac{d}{dx}\, x^{1/2} \;=\; \tfrac{1}{2}\, x^{-1/2} \;=\; \frac{1}{2\sqrt{x}}.$$

Self-check (independent route, implicit differentiation): let $y = \sqrt{x}$, so $y^2 = x$. Differentiating both sides: $2y\, y' = 1$, hence $y' = 1/(2y) = 1/(2\sqrt{x})$ — same result. *Numerical spot-check:* at $x = 4$, the difference quotient $(\sqrt{4.01} - \sqrt{4})/0.01 \approx (2.00250 - 2)/0.01 = 0.250$, matching $1/(2 \cdot 2) = 0.25$.

Answer: $\dfrac{1}{2\sqrt{x}}$.

P7.2 *Derivatives toolkit: chain-rule cycle* ♦

Each differentiation applies the chain rule (a factor of 4) and advances the trig cycle $\sin \to \cos \to -\sin \to -\cos \to \sin$:

$$4\cos(4x) \;\to\; -16\sin(4x) \;\to\; -64\cos(4x) \;\to\; 256\sin(4x).$$

Sine returns to itself after exactly four derivatives, and the chain rule contributes 4 each time, so the fourth derivative is $4^4 \sin(4x) = 256\sin(4x)$. *Self-check (closed form):* $\frac{d^n}{dx^n}\sin(kx) =$

$k^n \sin(kx + \frac{n\pi}{2})$; with $n = 4$, $k = 4$ this gives $4^4 \sin(4x + 2\pi) = 256 \sin(4x)$ — the phase shift of a full period confirms we are back at sine. Sign sanity: the two minus signs picked up at steps 2 and 3 cancel as $(-1)^2$, so the positive coefficient is right. **Answer:** $256 \sin(4x)$.

P7.3 *Derivatives toolkit: logistic identity*

(a) Differentiate $p(x) = (1 + e^{-x})^{-1}$ by the chain rule:

$$p'(x) = \frac{e^{-x}}{(1+e^{-x})^2} = \underbrace{\frac{1}{1+e^{-x}}}_{p} \cdot \underbrace{\frac{e^{-x}}{1+e^{-x}}}_{1-p} = p(1-p),$$

using $1 - p = \dfrac{e^{-x}}{1+e^{-x}}$. **(b)** At $p = 0.80$: sensitivity $= p(1-p) = 0.8 \times 0.2 = 0.16$, i.e. about 16 cents of price move per unit of log-odds ($\Delta p \approx 0.16\,\Delta x$ for small Δx). **(c)** $p(1-p)$ is a downward parabola in p, maximised at $p = \frac{1}{2}$ with value $\frac{1}{4}$. A 50-cent contract moves the most per unit of news; contracts near 0 or 1 barely react. Practically: the same headline should reprice mid-priced contracts far more than long shots, which matters for sizing, for deciding where quoted markets are most at risk of going stale, and it is the exact analogue of an at-the-money digital having the most concentrated sensitivity. *Self-check (numerical):* $p = 0.8$ corresponds to $x = \ln 4 \approx 1.386$. At $x = 1.486$: $e^{-1.486} \approx 0.25 \times 0.905 = 0.226$, so $p \approx 1/1.226 = 0.8155$, giving slope $\approx 0.0155/0.1 = 0.155$ — consistent with 0.16 (slightly below because $p(1-p)$ decreases as p rises past 0.8). Bound sanity: $p(1-p) \le \frac{1}{4}$ always, and $0.16 < 0.25$. **Answer:** (a) $p' = p(1-p)$; (b) 0.16 per unit of log-odds; (c) at $p = \frac{1}{2}$ (sensitivity $\frac{1}{4}$), where a contract reprices most per unit of news.

P7.4 *Derivatives toolkit: optimisation recipe*

Step 1: critical point. By the product rule,

$$f'(s) = e^{-s/2} + s \cdot \big(-\tfrac{1}{2}\big)e^{-s/2} = e^{-s/2}\big(1 - \tfrac{s}{2}\big),$$

which vanishes iff $s^* = 2$ (the exponential never vanishes). **Step 2: justify the maximum** (the justification sentence is graded). Since $e^{-s/2} > 0$, the sign of f' is the sign of $1 - s/2$: positive on $(0, 2)$, negative on $(2, \infty)$, so f rises then falls — a unique interior maximum. Equivalently $f''(s) = e^{-s/2}(\frac{s}{4} - 1)$ gives $f''(2) = -\frac{1}{2e} < 0$; and the boundary behaviour $f(0) = 0$, $f(s) \to 0$ as $s \to \infty$ with a single interior critical point forces a maximum there. **Step 3: report both s^* and $f(s^*)$.** $f(2) = 2e^{-1} = 2/e \approx 0.736$ ticks per quote. *Self-check (bracketing):* $f(1) = e^{-1/2} \approx 0.607$, $f(2) \approx 0.736$, $f(3) = 3e^{-3/2} \approx 0.669$ — the middle value is the largest of the three, consistent with a peak at $s = 2$. Trading sanity: quoting too tight ($s \to 0$) earns nothing per fill, quoting too wide never fills; an interior optimum is exactly what the economics demands. **Answer:** $s^* = 2$ ticks, with maximum expected profit $f(2) = 2/e \approx 0.736$ ticks per quote.

P7.5 *Integration by parts* ♦

Integration by parts with $u = \ln x$ (by LIATE the log is always u: it dies when differentiated) and $dv = dx$, so $du = \frac{1}{x}dx$, $v = x$:

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x}\, dx = x \ln x - \int 1 \, dx = x \ln x - x + C.$$

Self-check (differentiate back — ten seconds, catches most sign errors): $\frac{d}{dx}(x \ln x - x) = \ln x + x \cdot \frac{1}{x} - 1 = \ln x + 1 - 1 = \ln x$. *Definite sanity:* $\int_1^e \ln x \, dx = (e \cdot 1 - e) - (1 \cdot 0 - 1) = 1$; plausible, since the integrand rises from 0 to 1 over an interval of length $e - 1 \approx 1.72$, giving an average value $\approx 0.58 \in (0, 1)$. **Answer:** $x \ln x - x + C$.

P7.6 *Integration: substitution to Gamma + symmetry caveat*

(a) The screaming tell is a factor $x\,dx$ next to functions of x^2: substitute $u = x^2$, $du = 2x\,dx$, transforming the limits immediately ($0 \to 0$, $\infty \to \infty$). Since $x^3 dx = x^2 \cdot x\,dx = u \cdot \frac{du}{2}$,

$$\int_0^\infty x^3 e^{-x^2} dx = \frac{1}{2}\int_0^\infty u\,e^{-u} du = \frac{1}{2}\,\Gamma(2) = \frac{1!}{2} = \frac{1}{2},$$

using the named Gamma result $\int_0^\infty u^n e^{-u} du = n!$. **(b)** The integrand is odd ($(-x)^3 e^{-(-x)^2} = -x^3 e^{-x^2}$) over a symmetric interval, so the integral is 0 — and the argument is legitimate because the integrand is *absolutely integrable*: by (a), $\int_{-\infty}^\infty |x^3 e^{-x^2}| dx = 2 \times \frac{1}{2} = 1 < \infty$, so each half-line contributes a finite value and the two cancel exactly. For $\int_{-\infty}^\infty x\,dx$ both halves diverge, so the improper integral simply does not exist; only the symmetric principal value $\lim_{R\to\infty} \int_{-R}^R x\,dx$ is 0, which is a weaker statement — "odd so zero" needs the absolute-integrability caveat said aloud. *Self-check for (a) (differentiate back):* the antiderivative $F(x) = -\frac{1}{2}(x^2+1)e^{-x^2}$ satisfies $F'(x) = -\frac{1}{2}\left[2x\,e^{-x^2} + (x^2+1)(-2x)e^{-x^2}\right] = xe^{-x^2}(x^2+1-1) = x^3 e^{-x^2}$, and $F(\infty) - F(0) = 0 - (-\frac{1}{2}) = \frac{1}{2}$. Positivity sanity: the integrand on $(0,\infty)$ is strictly positive, and $\frac{1}{2} > 0$. **Answer:** (a) $\frac{1}{2}$; (b) 0, valid because the integrand is odd *and* absolutely integrable — for $\int x\,dx$ both halves diverge, so the integral is undefined (only its principal value is 0). □

P7.7 *Improper integrals: p-tests, comparison, log refinement*

(a) Converges. Tail at ∞ with $p = \frac{3}{2} > 1$. (Explicitly: antiderivative $-2x^{-1/2}$ gives value 2.) **(b) Converges.** The only problem point is the finite endpoint $x = 1$; substituting $u = 1 - x$ gives $\int_0^1 u^{-1/2} du$, a finite-endpoint blow-up with $p = \frac{1}{2} < 1$. (Value: 2.) **(c) Diverges.** Log refinement: $u = \ln x$ turns it into $\int_{\ln 2}^\infty \frac{du}{u}$, the $p = 1$ boundary case — log divergence. **(d) Diverges.** Comparison: $2 + \sin x \geq 1$, so the integrand is $\geq \frac{1}{x} \geq 0$, and $\int_1^\infty \frac{dx}{x}$ diverges. (Note the integrand $\to 0$; that proves nothing — the question is how fast.) **(e) Converges.** Exponentials beat polynomials: any $\int^\infty x^k e^{-cx} dx$ with $c > 0$ converges. (Value via $u = x/10$: $10^8 \int_0^\infty u^7 e^{-u} du = 10^8 \cdot 7!$.) *Self-check:* every verdict is backed by an explicit antiderivative or an explicit bound, not just a rule name; the two p-test convergents (a), (b) produce positive finite values; and the boundary logic is consistent — at ∞ we needed fast decay ($p > 1$: (a) passes, (c) sits exactly on the failing boundary), at a finite endpoint we can afford a mild blow-up ($p < 1$: (b) passes). $1/x$ fails at both ends, which is exactly why (c) and (d) both reduce to it and diverge. **Answer:** (a) converges ($= 2$); (b) converges ($= 2$); (c) diverges (log); (d) diverges (compare $1/x$); (e) converges ($= 10^8 \cdot 7!$).

P7.8 *Improper integrals: interior singularity*

Two independent red flags, either of which should stop you:

- **Sign:** the integrand $1/x^2$ is strictly positive, so no negative answer is possible — a positive integrand can never produce a negative or zero integral.
- **Illegal FTC:** the Fundamental Theorem of Calculus requires an antiderivative valid on the whole interval, but $1/x^2$ has an *interior* singularity at $x = 0$; $-1/x$ is not an antiderivative across it. Never apply FTC across an interior singularity — find every problem point and split there.

Correct procedure: split at 0 and require *every* piece to converge. Near the finite endpoint 0, $\int_0^1 x^{-2} dx$ has $p = 2 \geq 1$, which diverges; explicitly $\int_\varepsilon^1 x^{-2} dx = \frac{1}{\varepsilon} - 1 \to \infty$. The left half diverges identically by symmetry. Hence the integral diverges (to $+\infty$). *Self-check (bound):* on $[-1, 1]$ we have $|x| \leq 1$, so $1/x^2 \geq 1$; even if the integral existed it would be ≥ 2, flatly contradicting -2 — and the explicit ε-computation confirms $+\infty$. **Answer:** The FTC was applied across the interior singularity at $x = 0$; splitting there shows each half diverges ($p = 2$ at a finite endpoint), so the integral diverges to $+\infty$ — and the negative sign was the instant giveaway.

P7.9 *Improper integrals: combined p-tests via limit comparison*

Problem points: $x = 0$ (possible blow-up from x^{-p} when $p > 0$) and $x = \infty$. Split at any interior point, say $x = 1$, and require both pieces to converge. **Near 0:** $1 + x \to 1$, so limit-compare with x^{-p}: the ratio $\frac{1/(x^p(1+x))}{1/x^p} = \frac{1}{1+x} \to 1 \in (0, \infty)$, so $\int_0^1$ behaves like $\int_0^1 x^{-p} dx$, which converges iff $p < 1$ (finite-endpoint p-test: mild blow-up allowed). **Near ∞:** $x^p(1 + x) \sim x^{p+1}$, and the ratio against $x^{-(p+1)}$ tends to 1, so $\int_1^\infty$ behaves like $\int_1^\infty x^{-(p+1)} dx$, which converges iff $p + 1 > 1$, i.e. $p > 0$ (tail p-test: need fast decay). Both conditions must hold simultaneously, so the range of convergence is $0 < p < 1$. **Boundary sanity:** $p = 0$: $\int_0^\infty \frac{dx}{1+x} = \ln(1 + x)\big|_0^\infty$ diverges (log at ∞) excluded. $p = 1$: near 0 the integrand $\sim \frac{1}{x}$, log-divergent excluded. As always, whenever an effective $p = 1$ appears, the verdict is log divergence. *Self-check (explicit interior value):* at $p = \frac{1}{2}$, substitute $u = \sqrt{x}$ ($x = u^2$, $dx = 2u\,du$):

$$\int_0^\infty \frac{2u\,du}{u\,(1 + u^2)} = 2\int_0^\infty \frac{du}{1 + u^2} = 2 \cdot \frac{\pi}{2} = \pi,$$

finite — confirming convergence strictly inside the claimed range. **Answer:** $0 < p < 1$. □

P7.10 *Moments as integrals: normalisation via Gamma*

Normalisation questions are improper-integral evaluations in probability clothing. The one workhorse needed: substituting $u = x/\theta$,

$$\int_0^\infty x^n e^{-x/\theta} dx = \theta^{n+1} \int_0^\infty u^n e^{-u} du = \theta^{n+1}\, n!.$$

Find c: with $\theta = 2$, $n = 1$: $\int_0^\infty xe^{-x/2} dx = 2^2 \cdot 1! = 4$, so $c = \frac{1}{4}$. **Mean (LOTUS):** $\mathbb{E}[X] = \frac{1}{4}\int_0^\infty x^2 e^{-x/2} dx = \frac{1}{4} \cdot 2^3 \cdot 2! = \frac{16}{4} = 4$. **Second moment and variance:** $\mathbb{E}[X^2] = \frac{1}{4}\int_0^\infty x^3 e^{-x/2} dx = \frac{1}{4} \cdot 2^4 \cdot 3! = \frac{96}{4} = 24$, hence $\mathrm{Var}(X) = 24 - 4^2 = 8$. *Self-check (recognise the distribution):* f is the Gamma density with shape 2 and scale 2 — the sum of two independent exponentials, each with mean 2 (rate $\lambda = \frac{1}{2}$, moments $n!/\lambda^n$). So the mean is $2 + 2 = 4$ and the variance is $4 + 4 = 8$ (each exponential has variance $1/\lambda^2 = 4$). Scale sanity: standard deviation $\sqrt{8} \approx 2.8$, the same order as the mean, as expected for a mildly right-skewed waiting-time distribution. **Answer:** $c = \frac{1}{4}$, $\mathbb{E}[X] = 4$, $\mathrm{Var}(X) = 8$. □

P7.11 *Moments as integrals: Gaussian fourth moment*

(a) Standardise: $Z = X/\sigma$, so $\mathbb{E}[X^4] = \sigma^4\,\mathbb{E}[Z^4]$ and we need $\int_{-\infty}^\infty z^4\varphi(z)\,dz$ with $\varphi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$. The key structural fact is $\varphi'(z) = -z\,\varphi(z)$, so $z\varphi(z)\,dz$ integrates to $-\varphi(z)$. Integrate by parts with $u = z^3$, $dv = z\,\varphi(z)\,dz$, $v = -\varphi(z)$:

$$\int_{-\infty}^\infty z^4\varphi(z)\,dz = \Big[-z^3\varphi(z)\Big]_{-\infty}^\infty + 3\int_{-\infty}^\infty z^2\varphi(z)\,dz = 0 + 3 \cdot 1 = 3,$$

since the boundary term dies (exponentials beat polynomials) and $\int z^2\varphi = \mathrm{Var}(Z) = 1$. Hence $\mathbb{E}[X^4] = 3\sigma^4$: Gaussian kurtosis is exactly 3. **(b)** A ratio of 9 against a Gaussian's 3 means the return distribution has *fat tails* (excess kurtosis 6): extreme moves are far more common than the model admits. One practical consequence (any one suffices): σ-based limits or VaR calibrated on the Normal will be breached much more often than advertised — under the Normal a 4σ day has probability $\approx 6 \times 10^{-5}$ (roughly once in 60 years of trading days), yet fat-tailed markets deliver them routinely. Standard fixes: heavier-tailed models (Student-t), or recognising volatility clustering — a mixture of Gaussians with time-varying σ has kurtosis above 3 even if each regime is Gaussian, so conditioning on the current vol regime removes much of the excess. *Self-check:* the same parts argument with general even power gives the recursion $\mathbb{E}[Z^{2m}] = (2m-1)\,\mathbb{E}[Z^{2m-2}]$, i.e. the double factorial $1, 3, 15, \ldots$; at $m = 1$ it returns $\mathbb{E}[Z^2] = 1\cdot 1 = 1$, and at $m = 2$ it gives $3\cdot 1 = 3$, agreeing with the direct computation. Symmetry sanity: all odd moments vanish, so kurtosis is the first shape parameter beyond variance — exactly why desks track it. **Answer:** (a) $\mathbb{E}[X^4] = 3\sigma^4$

(kurtosis 3); (b) sample kurtosis ≈ 9 means fat tails / vol clustering — Normal-calibrated risk limits materially understate the frequency of extreme moves. □

P7.12 *Taylor micro-kit: limit*

Taylor first (faster, shows more): $e^x = 1 + x + \frac{x^2}{2} + O(x^3)$, so the numerator is $\frac{x^2}{2} + O(x^3)$ and

$$\frac{e^x - 1 - x}{x^2} = \frac{1}{2} + O(x) \longrightarrow \frac{1}{2}.$$

Self-check 1 (L'Hopital, checking the form each time): at $x = 0$ the expression is $\frac{0}{0}$, so differentiate: $\frac{e^x-1}{2x}$, again $\frac{0}{0}$; differentiate once more: $\frac{e^x}{2} \to \frac{1}{2}$. *Self-check 2 (numerical):* at $x = 0.1$: $(1.105171 - 1.1)/0.01 = 0.5171$; the next Taylor term predicts $\frac{1}{2} + \frac{x}{6} = 0.5167$ — the numerics match the expansion to the next order, not just the limit. **Answer:** $\frac{1}{2}$.

P7.13 *Taylor micro-kit: compounding estimate*

Route: through the exponential. $(1.02)^{30} = e^{30\ln(1.02)}$. The crude first-order kit $(1+\frac{a}{n})^n \approx e^a$ with $a = 0.6$ gives $e^{0.6} \approx 1.822$. Refine with the second-order log expansion $\ln(1+x) \approx x - \frac{x^2}{2}$:

$$\ln(1.02) \approx 0.02 - 0.0002 = 0.0198, \qquad 30 \times 0.0198 = 0.594,$$

$$e^{0.594} = e^{0.6} \cdot e^{-0.006} \approx 1.822 \times 0.994 \approx 1.811.$$

So $(1.02)^{30} \approx 1.81$ — the second-order term pulls the naive $e^{0.6}$ down by about 1%. *Self-check 1 (rule of 72):* at 2% per period, doubling takes about $72/2 = 36$ periods; after only 30 periods we should sit noticeably below 2.0 — and 1.81 does. *Self-check 2 (convexity bound):* compounding must beat the linear estimate $1 + 30 \times 0.02 = 1.60$, and $1.81 > 1.60$. The true value is 1.8114, so the two-term estimate is accurate to about 0.03% — the pattern to remember: convert powers to $e^{n\ln(1+x)}$ and keep two terms of the log. **Answer:** $(1.02)^{30} \approx 1.81$ (true value 1.8114).

P7.14 *Change of variables in an ODE* ♦

Pure chain rule. From $\tau = 1/t$ we get $t = 1/\tau$ and

$$\frac{dt}{d\tau} = -\frac{1}{\tau^2}, \qquad \frac{dx}{d\tau} = \frac{dx}{dt}\cdot\frac{dt}{d\tau} = x \cdot \left(-\frac{1}{\tau^2}\right) = -\frac{x}{\tau^2}.$$

Check by explicit solution: $x(t) = Ce^t = Ce^{1/\tau}$, so $\dfrac{dx}{d\tau} = Ce^{1/\tau} \cdot \left(-\frac{1}{\tau^2}\right) = -x/\tau^2$ — matches.

Answer: (C) $dx/d\tau = -x/\tau^2$.

P7.15 *Differentiate a parameter-dependent integral* ♦

The fastest route: x is a constant with respect to s, so evaluate the integral in closed form first, then differentiate:

$$F(x) = x\int_x^{2x} s\,ds = x \cdot \frac{(2x)^2 - x^2}{2} = x \cdot \frac{3x^2}{2} = \frac{3}{2}x^3 \qquad \Rightarrow \qquad F'(x) = \frac{9}{2}x^2.$$

Cross-check with the full Leibniz rule (limits and integrand both depend on x):

$$F'(x) = \underbrace{2\cdot(2x)x}_{\text{upper limit}} - \underbrace{1\cdot x\cdot x}_{\text{lower limit}} + \underbrace{\int_x^{2x} s\,ds}_{\partial/\partial x\text{ of integrand}} = 4x^2 - x^2 + \frac{3x^2}{2} = \frac{9}{2}x^2.$$

Both routes agree — and under a clock, evaluate-then-differentiate is the safer one whenever the integral is elementary.

Answer: (B) $\frac{9}{2}x^2$.

P7.16 *Probability by integration* ♦

Condition on X: given $X = x$, $P(Y < x^2) = x^2$ (uniform CDF). Averaging over X,

$$P(X^2 > Y) = \mathbb{E}\big[P(Y < X^2 \mid X)\big] = \mathbb{E}[X^2] = \int_0^1 x^2\,dx = \frac{1}{3}.$$

Equivalently: the probability is the area under the parabola $y = x^2$ inside the unit square.

Sanity check: $P(X^2 > Y) < P(X > Y) = \frac{1}{2}$ since $X^2 < X$ on $(0,1)$ — and $\frac{1}{3} < \frac{1}{2}$. The trap options bait a symmetry guess (1/2) or a quarter-circle association ($\pi/4$, which belongs to $P(X^2 + Y^2 < 1)$, a different question).

Answer: (C) 1/3.

P7.17 *Newton's method: the slowest root* ♦

Find the roots. $\tanh(u) = 0$ iff $u = 0$, so the roots solve $x^4 - x^2 = x^2(x^2 - 1) = 0$: $x \in \{-1, 0, 1\}$.

Determine multiplicities. Near $x = 0$: $x^4 - x^2 \approx -x^2$ and $\tanh u \approx u$, so $f(x) \approx -x^2$ — a root of *multiplicity 2.* At $x = \pm 1$: $(x^4 - x^2)' = 4x^3 - 2x = \pm 2 \neq 0$, so both are *simple* roots.

Convergence rates. Newton is quadratic at simple roots, but only *linear* at a root of multiplicity m, with error ratio $(m-1)/m$ per step. Derivation for the model case $f = (x - r)^m$:

$$x_{k+1} - r = x_k - r - \frac{(x_k - r)^m}{m(x_k - r)^{m-1}} = (x_k - r)\left(1 - \frac{1}{m}\right),$$

so with $m = 2$ the error only halves each iteration — linear, ratio $\frac{1}{2}$ — while at ± 1 the digits double per step. The slowest root is $x = 0$: the one with the smallest *absolute value.*

The distractor: "the smallest one" is -1 — a simple root with quadratic convergence. The option pair (smallest vs smallest absolute value) is engineered to punish reading fast. And (B) is the textbook overstatement: quadratic convergence *requires* a simple root ($f'(r) \neq 0$).

Bonus (worth saying in an interview): the fix at a known multiplicity is the modified step $x_{k+1} = x_k - m\,f/f'$, which restores quadratic convergence. □

Answer: (D) the root with the smallest absolute value ($x = 0$, multiplicity 2, linear convergence).

21.8 Solutions — §8 ML Theory: Trees, Ensembles, Bias–Variance

P8.1 *Bias-variance decomposition*

Drop the argument x_0. Insert and subtract f and $\mathbb{E}_D[\hat{f}]$:

$$y - \hat{f} = \underbrace{\varepsilon}_{\text{test noise}} + \underbrace{(f - \mathbb{E}[\hat{f}])}_{\text{a deterministic constant}} + \underbrace{(\mathbb{E}[\hat{f}] - \hat{f})}_{\text{mean zero over } D}.$$

Square and take $\mathbb{E}_{D,\varepsilon}$. The squared terms give σ^2, $\big(f - \mathbb{E}[\hat{f}]\big)^2 = \text{Bias}^2$, and $\mathbb{E}[(\hat{f} - \mathbb{E}[\hat{f}])^2] = \text{Var}(\hat{f})$. Cross terms: (i) both terms containing ε factor as $\mathbb{E}[\varepsilon]\cdot\mathbb{E}[\cdot] = 0$ because the test noise is independent of the training randomness and has mean zero; (ii) the remaining term is $2(f - \mathbb{E}[\hat{f}])\,\mathbb{E}[\mathbb{E}[\hat{f}] - \hat{f}] = 0$ because the first factor is a constant and the second factor has mean zero by construction. Hence

$$\mathbb{E}[(y - \hat{f})^2] = \big(f - \mathbb{E}[\hat{f}]\big)^2 + \text{Var}(\hat{f}) + \sigma^2.$$

More data, same model class: the *variance* term shrinks (the fitted model concentrates around its own expectation); the *bias* term converges to the approximation error of the model class and does not vanish; σ^2 is irreducible by definition.

Self-check (constant predictor). Take $\hat{f} \equiv c$, ignoring the data. Directly: $\mathbb{E}[(y - c)^2] = \mathbb{E}[(f - c + \varepsilon)^2] = (f - c)^2 + \sigma^2$. The formula: $\text{bias}^2 = (f - c)^2$, variance $= 0$, plus σ^2 — identical.

Answer: $\mathbb{E}[(y-\hat{f})^2] = (f - \mathbb{E}[\hat{f}])^2 + \mathrm{Var}(\hat{f}) + \sigma^2$; the ε cross terms die by independence and $\mathbb{E}[\varepsilon] = 0$, the third because $\hat{f} - \mathbb{E}[\hat{f}]$ has mean zero against a constant. More data shrinks only the variance term. □

P8.2 *Decision trees: Gini and information gain*

(a) Parent: $G = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = \frac{1}{2}$. Each child has proportions $\left(\frac{3}{4}, \frac{1}{4}\right)$: $G_{\text{child}} = 1 - \frac{9}{16} - \frac{1}{16} = \frac{6}{16} = \frac{3}{8}$. Both children have 8 of the 16 samples, so the weighted Gini is $\frac{1}{2} \cdot \frac{3}{8} + \frac{1}{2} \cdot \frac{3}{8} = \frac{3}{8} = 0.375$, an impurity decrease of $\frac{1}{2} - \frac{3}{8} = \frac{1}{8} = 0.125$.

(b) Parent entropy = 1 bit (a fair 50/50). Child entropy $= -\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4} = \frac{3}{4}(2 - \log_2 3) + \frac{1}{4} \cdot 2 = 0.75 \times 0.415 + 0.5 \approx 0.811$ bits. Information gain $= 1 - 0.811 \approx 0.189$ bits.

(c) None of them. A tree splits on thresholds, i.e. on the *rank order* of the feature; a strictly monotone transform preserves exactly which partitions of the samples are achievable by a threshold, so the same split (at the transformed threshold) produces the same child sets and the same impurity numbers. This is why scaling and monotone transforms are wasted effort for tree models.

Self-check. Bounds: binary Gini lies in $[0, \frac{1}{2}]$ and binary entropy in $[0, 1]$ bit — all computed values respect them; both children are mirror images so the weighted value must equal the single-child value, which it does; and $\log_2 \frac{4}{3} = 2 - \log_2 3 = 2 - 1.585 = 0.415$ confirms the entropy arithmetic.

Answer: Gini $0.5 \to 0.375$, decrease $= 0.125$; information gain ≈ 0.189 bits; nothing changes under a monotone feature transform.

P8.3 *Trees cannot extrapolate*

A tree partitions feature space into rectangles and predicts a *constant* on each leaf. VIX = 80 falls into the open-ended rightmost cell, whose rule is roughly "VIX > (largest threshold learned, high 20s)", so the model outputs approximately the average target of the highest-VIX training days — about what it predicts for VIX ≈ 30. The prediction function is flat beyond the training range: trees (and sums of trees) *cannot extrapolate*, so realised vol is severely underpredicted exactly when it matters. A linear regression extrapolates along its fitted slope — directionally more sensible here, though extrapolation reliability is its own risk.

Fixes: (1) make the feature *stationary/relative* so extreme regimes map into the seen range — ratios (VIX over its trailing median), log-changes, rolling z-scores or ranks instead of raw levels; (2) add explicit interaction/spread features or hybridise with a linear component (fit trees on the residuals of a linear baseline, or linear-in-leaf variants such as LightGBM's `linear_tree`). A monotonic constraint ("higher VIX ⇒ higher predicted vol") is a further regularising guard: it cannot restore the missing slope but guarantees the saturation is at least monotone.

Self-check (degenerate case). A depth-0 tree predicts the global training mean for every input — the "constant beyond the training range" behaviour is just the general form of this, so the mechanism is right. And the direction matches reality: March 2020 realised vol exceeded anything in 2015–2019, so a saturated forecast is an underprediction, as claimed.

Answer: the model outputs roughly its VIX-30 prediction (top-leaf mean) because tree predictions are piecewise constant and cannot extrapolate; a linear model would extend along its slope; fixes: stationary/ratio/z-scored features and explicit interactions or linear-hybrid designs (plus monotonic constraints).

P8.4 *Ensemble variance formula and the correlation floor*

(a) $\mathrm{Var}\left(\sum_i X_i\right) = N\sigma^2 + N(N-1)\rho\sigma^2$ (there are $N(N-1)$ ordered covariance pairs, each $\rho\sigma^2$). Dividing by N^2:

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{N} + \frac{N-1}{N}\rho\sigma^2 = \rho\sigma^2 + \frac{1-\rho}{N}\sigma^2.$$

(b) As $N \to \infty$ the second term dies and $\rho\sigma^2$ survives: *correlation between learners is the floor on ensemble variance* — averaging more trees cannot cross it. Bagging alone leaves trees highly correlated (they all chase the same strong features), so random forests additionally sample a random feature subset at every split: this attacks ρ itself, accepting slightly worse individual trees, because lowering the floor beats polishing the $1/N$ term.

(c) Validity requires the equicorrelated covariance matrix to be PSD: $\mathrm{Var}\left(\sum X_i\right) = N\sigma^2[1+(N-1)\rho] \geq 0$, i.e. $\rho \geq -\frac{1}{N-1}$. At the boundary $\rho = -\frac{1}{N-1}$:

$$\mathrm{Var}(\bar{X}) = \sigma^2\Big[-\frac{1}{N-1} + \frac{1}{N}\Big(1+\frac{1}{N-1}\Big)\Big] = \sigma^2\Big[-\frac{1}{N-1} + \frac{1}{N}\cdot\frac{N}{N-1}\Big] = 0.$$

Maximally anti-correlated predictors cancel exactly — the diversification ideal (in book terms: negatively correlated strategies are worth more than uncorrelated ones).

(d) Need $\frac{(1-\rho)\sigma^2}{N} \leq 0.05 \iff \frac{0.8}{N} \leq 0.05 \iff N \geq 16$.

Self-check. Boundary cases of (a): $\rho = 1 \Rightarrow \mathrm{Var} = \sigma^2$ for all N (identical models, averaging useless); $\rho = 0 \Rightarrow \sigma^2/N$ (i.i.d. rate); $N = 1 \Rightarrow \sigma^2$. For (d): $N = 16$ gives $0.2 + 0.8/16 = 0.25$, exactly the floor 0.2 plus 0.05.

Answer: (a) $\rho\sigma^2 + \frac{1-\rho}{N}\sigma^2$; (b) limit $\rho\sigma^2$ — the correlation floor, which feature-subsampling attacks directly; (c) $\rho \geq -\frac{1}{N-1}$, where the ensemble variance is exactly 0; (d) $N = 16$. ☐

P8.5 *Bootstrap and out-of-bag*

(a) Each draw misses the point with probability $1 - \frac{1}{n}$, and draws are independent:

$$P(\text{never drawn}) = \Big(1 - \frac{1}{n}\Big)^n \xrightarrow{n\to\infty} e^{-1} \approx 0.368.$$

(b) So each tree sees on average $1 - e^{-1} \approx 63.2\%$ of the distinct points. (c) The *out-of-bag (OOB) error*: every training point is predicted using only the $\approx 36.8\%$ of trees that never saw it, which yields an honest generalisation estimate — effectively cross-validation for free, no held-out set needed — and drives OOB-based feature importances.

Self-check. Small cases: $n = 2$: $(1/2)^2 = 0.25$; $n = 5$: $(4/5)^5 = 1024/3125 \approx 0.328$; $n = 100 \approx 0.366$ — increasing towards 0.368 from below. Analytically, $\ln(1-1/n)^n = n\ln(1-1/n) \approx n(-\frac{1}{n} - \frac{1}{2n^2}) = -1 - \frac{1}{2n} \to -1$.

Answer: (a) $(1 - 1/n)^n \to 1/e \approx 36.8\%$; (b) $\approx 63.2\%$; (c) the out-of-bag error estimate (free cross-validation).

P8.6 *Boosting as functional gradient descent*

(a) Treat the model's values $F(x_i)$ at the n training points as free coordinates. Then $\frac{\partial L}{\partial F(x_i)} = -(y_i - F(x_i))$, so the negative gradient at F_{t-1} is the residual vector $y_i - F_{t-1}(x_i)$. The tree h_t is fit to approximate this vector (a projection of the gradient onto the class of trees), and $F_t = F_{t-1} + \eta h_t$ is a gradient step of length η. Residual fitting *is* steepest descent on squared loss.

(b) $-\frac{\partial}{\partial F}|y - F| = \mathrm{sign}(y - F)$: each tree is fit to the *sign* of the residual (LAD boosting; optimal leaf values become medians of the in-leaf residuals) — outlier-robust, exactly the L1-vs-L2 mean/median story.

(c) Boosting attacks *bias*: each round fits what the current ensemble still misses, so shallow high-bias trees suffice. Its overfitting knobs: learning rate η (shrinkage), number of rounds with early stopping, tree depth/leaf count, row/feature subsampling, L1/L2 penalties on leaf weights, minimum child samples. More rounds *do* eventually overfit (training error keeps falling, validation error turns). Bagging attacks *variance* of deep low-bias trees fit in parallel; adding more bagged trees never increases expected test error — the average just converges (to the $\rho\sigma^2$ floor), so "more trees" is safe in a forest but a tuning decision in boosting.

Self-check (one concrete run). Single observation, $y = 10$, $F_0 = 0$, $\eta = 0.1$; each tree can fit its target exactly. Residual recursion: $r_t = (1-\eta)^t \cdot 10$ — geometric decay to zero, exactly gradient descent with step η on a quadratic. After 2 rounds: $F_2 = 1 + 0.9 = 1.9 = 10(1 - 0.81)$. Units also check: the gradient carries units of y, matching what a regression tree predicts.

Answer: (a) $-\partial L/\partial F|_{F_{t-1}} = y - F_{t-1}$, the residual; (b) $\operatorname{sign}(y - F)$ with median leaf values; (c) boosting cuts bias (knobs: η, rounds/early stopping, depth, subsampling, leaf penalties) and can overfit with rounds; bagging cuts variance and more trees never hurt. □

P8.7 *Good features for tree models* ★

Organise the answer around what tree splits can and cannot do.

- *Splits are thresholds on rank order.* Monotone transforms and standardisation are irrelevant (the same partitions remain reachable), and robustness to outliers comes free. Do not spend effort scaling; spend it below.
- *Trees cannot extrapolate.* Predictions are constant beyond the training range, so prefer *stationary* constructions over raw levels: returns rather than prices, ratios to rolling baselines, cross-sectional ranks and z-scores. A feature whose live range drifts outside the training range silently saturates.
- *Splits are axis-aligned.* A tree cannot natively express relations *between* columns ($x_1 > x_2$ needs a staircase of splits), so engineer interactions explicitly: ratios, spreads, differences, imbalances (bid-ask imbalance, price over moving average, book-to-price).
- *Missing values are handled natively* (LightGBM learns a default split direction), so avoid distorting imputations.
- *Dangers:* high-cardinality ID-like categoricals are overfit fuel (the tree memorises identities); leakage — construct everything point-in-time; and where you *know* the sign of an effect, monotonic constraints encode it as regularisation.

Self-check (invariance claim on a toy). Samples $x = 1, 10, 100$ with labels $0, 0, 1$: best split "$x > 55$" isolates the third point. After $\log_{10}$: $0, 1, 2$ and split "> 1.5" produces the identical partition with identical impurity — confirming split-invariance to monotone maps. The extrapolation point is the P8.3 mechanism, verified there.

Answer: good tree features are stationary (returns, ratios, ranks/z-scores, not levels), interaction-explicit (ratios/spreads/imbalances, since splits are axis-aligned), and rank-meaningful; scaling and monotone transforms do not matter; handle missing natively; beware high-cardinality IDs and leakage; encode known signs via monotonic constraints.

21.9 Solutions — §9 Deep Learning

P9.1 *Backprop mechanics and vanishing gradients*

(a) Backprop is the chain rule organised back-to-front: the sensitivity $\delta_\ell = \partial a_L/\partial a_\ell$ is computed once per layer and *reused* for every parameter feeding that layer, so the whole gradient costs a small constant times one forward pass (backward $\approx 2\times$ forward flops), independent of parameter count. Finite differences need one perturbed forward pass per parameter: $10^7 \times 10\,\text{ms} = 10^5\,\text{s} \approx 28$ hours per gradient step, versus ~ 30 ms for forward+backward — a factor of about 10^7 (i.e. exactly the parameter count).

(b) With $z_\ell = w_\ell a_{\ell-1}$, the chain rule gives

$$\frac{\partial a_L}{\partial w_1} = \Big[\prod_{\ell=2}^{L} \sigma'(z_\ell)\, w_\ell\Big]\, \sigma'(z_1)\, a_0.$$

The sigmoid satisfies $\sigma' = \sigma(1-\sigma) \le \frac{1}{4}$ (max at $\sigma = \frac{1}{2}$). With $|w_\ell| \le 1$ and $|a_0| \le 1$, the magnitude is at most $(1/4)^L = 4^{-20} = 2^{-40} \approx 9 \times 10^{-13}$ for $L = 20$: the early layers receive essentially zero signal. Any product of factors bounded below 1 decays exponentially — that is the vanishing-gradient mechanism (factors above 1 give the mirror-image exploding case).

(c) *ReLU:* $\sigma' = 1$ on the active half — the 1/4 shrink factor becomes 1. *Residual connections:* $\frac{\partial}{\partial x}(x + f(x)) = 1 + f'$ — an identity term keeps a product-free gradient path through depth. *LSTM gating:* the cell state updates additively through gates near 1, so the through-time product becomes a sum-like highway. *Batch norm:* keeps pre-activations z in the non-saturated region, so $\sigma'(z)$ is not driven to ≈ 0.

Self-check. $4^{20} = 2^{40} = (2^{10})^4 \approx (10^3)^4 = 10^{12}$, so $4^{-20} \approx 10^{-12}$ (more precisely 9.1×10^{-13}); the finite-difference estimate 10^5 s $= 10^5/3600 \approx 27.8$ h; and the sigmoid bound: $\sigma'(0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, indeed the maximum of $p(1-p)$.

Answer: (a) backprop reuses per-layer sensitivities — cost $\approx$ a constant times one forward pass, vs ≈ 28 hours per step for finite differences on 10^7 parameters; (b) the gradient is a product of L factors, each at most $\frac{1}{4}$ in magnitude here, so at $L = 20$ it is bounded by $4^{-20} \approx 10^{-12}$; (c) ReLU makes the derivative factor 1, residuals add an identity path, LSTM gates make time propagation additive, batch norm keeps activations unsaturated. □

P9.2 *Self-attention and the sqrt(dk) scaling*

(a) $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$. Each term has mean 0 and variance $\mathbb{E}[q_i^2 k_i^2] = \mathbb{E}[q_i^2]\,\mathbb{E}[k_i^2] = 1$ (independence), and distinct terms are uncorrelated, so $\mathrm{Var}(q \cdot k) = d_k$.

(b) Standard deviation $\sqrt{64} = 8$. Two independent logits $\sim$ scale 8 differ by about $8\sqrt{2} \approx 11$ in a typical draw. Softmax of $[11, 0]$: weight $\frac{1}{1+e^{-11}}$ with $e^{-11} \approx 1.7 \times 10^{-5}$ — effectively one-hot. Attention starts out frozen onto single random tokens.

(c) The softmax Jacobian is $\partial p_i / \partial z_j = p_i(\delta_{ij} - p_j)$. At a one-hot p, every entry is ≈ 0 (either $p_i \approx 0$ or $1 - p_j \approx 0$): the layer is saturated exactly like a sigmoid in its tails, and no gradient flows to the scores — the attention pattern cannot learn its way out.

(d) Dividing by $\sqrt{d_k}$ makes the score variance 1 *independent of head dimension*, so softmax operates in its sensitive regime at initialisation: attention distributions start diffuse and sharpen only as training gives them reason to.

Self-check. $d_k = 1$: formula gives $\mathrm{Var}(q_1 k_1) = 1$, matching the product-of-independents computation directly. The tail number: $e^{-11} = e^{-8} e^{-3} \approx 3.4 \times 10^{-4} \times 5.0 \times 10^{-2} \approx 1.7 \times 10^{-5}$. And the Jacobian check at uniform $p = (\frac{1}{2}, \frac{1}{2})$ gives entries $\pm\frac{1}{4} \neq 0$ — gradients flow when softmax is *not* saturated, isolating saturation as the culprit.

Answer: (a) $\mathrm{Var}(q \cdot k) = d_k$; (b) std $= 8$ at $d_k = 64$; typical logit gaps ≈ 11 make softmax one-hot to within $\sim 2 \times 10^{-5}$; (c) the softmax Jacobian $p_i(\delta_{ij} - p_j)$ vanishes at one-hot outputs — saturated, no learning signal; (d) unit-variance scores for any d_k, keeping softmax un-saturated at init. □

P9.3 *Transformers on tick data*

(a) $L^2 = (5 \times 10^5)^2 = 2.5 \times 10^{11}$ entries; at 2 bytes each that is 5×10^{11} bytes = 500 GB *per head per layer* — an order of magnitude beyond an entire A100 (40–80 GB) before counting any other activations. Naive full-day tick attention is not merely slow; it does not fit.

(b) FlashAttention never materialises the $L \times L$ matrix: it streams tiles through on-chip SRAM and recomputes what it needs, cutting attention *memory* to $O(L)$. But every query–key pair is still computed, so *compute remains* $O(L^2)$ — training over years of 500k-tick days stays quadratic and brutal. Say both halves: memory fixed, compute not.

(c) Transformer with a KV cache: appending one tick costs one query against all L cached keys, i.e. $O(L)$ per layer per new tick, and the cache itself grows with the session — cost and memory both scale with the window. SSM/recurrent models: a fixed-size state updated in $O(1)$ per tick, constant memory, no re-attending — the right computational shape for streaming inference in live trading (S4 trains in $O(L \log L)$, Mamba in $O(L)$; both step recurrently at inference).

(d) Aggregate ticks before modelling: time/volume/dollar bars or order-flow features (OFI, imbalance, microprice deltas) cut L by 2–4 orders of magnitude while *adding* prior knowledge, after which any sequence model (or LightGBM on features) is on the table — or run an SSM on ticks if the raw stream truly carries the alpha.

Self-check. Arithmetic: $(5 \times 10^5)^2 = 25 \times 10^{10} = 2.5 \times 10^{11}$; $\times 2$ B $= 5 \times 10^{11}$ B $= 500$ GB at 10^9 B/GB. Shape check on (c): the KV cache holds L key/value vectors, so one new query costs L dot products — $O(L)$, consistent; an RNN-style state is size-fixed by construction — $O(1)$.

Answer: (a) 2.5×10^{11} entries ≈ 500 GB per head per layer — infeasible; (b) FlashAttention fixes memory ($O(L)$), not compute (still $O(L^2)$); (c) transformer: $O(L)$ per new tick with a growing KV cache; SSM: $O(1)$ per tick with fixed state; (d) aggregate ticks into bars/flow features first (or use an SSM on the raw stream).

P9.4 *The model stopped working*

The five hypotheses, each with its test:

- *Regime change / non-stationarity* — the relationship weakened as the world changed. Diagnostics: rolling IC and rolling Sharpe through time; feature-distribution drift by year (KS/PSI); refit by sub-period and compare coefficients.
- *Overfit all along* — 2005–2018 was in-sample luck. Diagnostics: strict walk-forward re-run with the current codebase; deflated Sharpe accounting for how many variants were ever tried; sensitivity of P&L to small hyperparameter perturbations (a real edge is not knife-edge).
- *Alpha decay / crowding* — the edge was real and got arbitraged away. Diagnostics: return per unit of turnover trending to zero; rising implementation shortfall and impact; correlation of the strategy's P&L with published factors and factor ETFs after the idea became public.
- *Look-ahead / leakage in the original backtest* — the historical P&L was never achievable. Diagnostics: rebuild data point-in-time (as-of joins, restatement-safe fundamentals, correct timestamps); compare early live fills against backtest assumptions (slippage, availability).
- *Structural change* — regulation, tick-size regimes, new participant mix, market microstructure shifts. Diagnostics: align the performance break date with known structural events; test the same signal in markets the change did not touch.

Check *leakage* and *overfitting* first: they answer the prior question of whether the edge ever existed. They are also the cheapest to test from your own artefacts (no new data), and every other conclusion changes if they fail — a regime analysis of a never-real edge is wasted work.

Self-check (coverage). The five split cleanly into "the edge was never real" (overfit, leakage) versus "real, then gone" (regime, crowding, structure) — both branches covered, no overlap; and the diagnostics discriminate: crowding predicts *gradual* decay with rising costs, structural change predicts a *sharp* break at a datable event, leakage predicts live-vs-backtest divergence from day one. Distinct fingerprints, so the tests can actually separate the hypotheses.

Answer: (1) regime change, (2) overfitting, (3) alpha decay/crowding, (4) look-ahead/leakage, (5) structural change — with the paired diagnostics above; test leakage and overfitting first because they determine whether the edge ever existed and are cheapest to verify.

P9.5 *Numpy vs lists, and why PyTorch exists* ★

Lists vs numpy. A Python list is an array of *pointers* to boxed PyObjects: every element carries a type tag and reference count, lives wherever the heap allocator put it, and every operation on it dispatches dynamically through the interpreter. A numpy array is *one contiguous typed buffer*; a vectorised operation runs a compiled C loop with SIMD over cache-friendly memory — no boxing, no pointer chasing, no per-element interpreter tick, no per-element refcounting. One vectorised op is typically 10–100× a Python loop before any parallelism enters. (MCQ bait to dodge: this is *not* a GIL story — the win is single-threaded, from memory layout and compiled loops.)

Why PyTorch does not use numpy. numpy is CPU-only and has no autograd — the two things a deep-learning framework cannot live without. A `torch.Tensor` is a numpy-like buffer *plus*: (1) autograd metadata — every op records a `grad_fn` node so backprop can run; (2) a device abstraction — the same API executes CUDA kernels with asynchronous streams and a caching GPU allocator; (3) deep-learning dtypes (fp16/bf16) and fused kernels; (4) distributed primitives. Interop is zero-copy on CPU: `torch.from_numpy` shares the same memory.

Self-check (do the magnitudes cohere?). Summing 10^7 Python floats in a loop at ∼50–100 ns of dispatch per element ≈ 0.5–1 s; `np.sum` on the same data is memory-bandwidth bound: 8×10^7 bytes at $\sim 2 \times 10^{10}$ B/s ≈ 4 ms — a ∼100–250× ratio, consistent with the 10–100× claim. And zero-copy `from_numpy` confirms the storage layout is identical — the difference really is autograd + device machinery, not the buffer.

Answer: lists store pointers to boxed objects with per-element interpreter dispatch; numpy stores one contiguous typed buffer processed by compiled C loops with SIMD (not a GIL effect) — 10–100× per op. PyTorch adds what numpy lacks: autograd (grad_fn graph) and GPU/device execution (plus fp16/bf16, fused kernels, distributed); `torch.from_numpy` is zero-copy on CPU.

P9.6 *GPU vs CPU speedup* ★

Headline: **one to two orders of magnitude (10–100×)** for dense batched linear algebra. Derive it from hardware ratios rather than folklore:

- *Compute-bound* (big matmuls): A100 ≈ 19.5 TFLOPs FP32, ≈ 312 TFLOPs FP16 on tensor cores, versus a good multi-core CPU at ∼1–2 TFLOPs ⇒ ∼10–100×, more with mixed precision.
- *Memory-bound* ops (elementwise, reductions): the bandwidth ratio rules — ∼2 TB/s HBM vs ∼100–200 GB/s DDR ⇒ ∼10–20×.

Caveats to volunteer before being asked: *PCIe transfers can eat the entire win* (shipping data to the card costs more than computing on it — keep data resident); *small, sequential, or branchy workloads* can be *slower* on GPU (kernel-launch overhead, poor occupancy, no latency-hiding); and versus naive single-threaded Python the observed gap is 10^3–10^4×, but that credits the GPU for vectorisation a CPU could also have delivered — compare against a well-vectorised CPU baseline to keep the claim honest.

Self-check (two independent routes agree). FLOPs route: 19.5/1.5 ≈ 13× (FP32) up to 312/1.5 ≈ 200× (FP16 tensor cores); bandwidth route: 2000/150 ≈ 13×. Both land in or bracket the 10–100× band; and the everyday observation — a CPU training epoch of hours becoming minutes on one GPU — sits at ∼30–60×, inside the same band.

Answer: ∼10–100× (1–2 orders of magnitude) for dense batched linear algebra — FLOPs ratio when compute-bound, ∼10–20× bandwidth ratio when memory-bound; the advantage collapses for small/sequential/transfer-dominated workloads and inflates to 10^3–10^4× only against naive single-threaded Python.

P9.7 *Python class variables* ⧫

A class variable is defined on the class object itself, so there is *one shared binding*: every instance that has not shadowed it sees the same value, because attribute lookup goes instance dict → class dict. Rebinding on the class (`C.x = 7`) changes what all such instances see; mutating a *mutable* class variable in place (e.g. appending to a class-level list) is likewise visible everywhere — the classic shared-mutable-default bug. The intended MCQ answer: **it is shared by the class — the same value for all objects**. Pedantic caveat worth one sentence in interview: assigning *through an instance* (`obj.x = 5`) creates a per-instance attribute that shadows the class variable for that object only.

Self-check (mental trace). `class C: x = 0; a, b = C(), C(); C.x = 7` ⇒ `a.x == b.x == 7` (both look ups fall through to the class); then `a.x = 1` ⇒ `b.x` still 7, confirming the shadowing caveat rather than contradicting the sharing rule.

Answer: it is shared by the class — all instances see the same value (until an instance assignment shadows it for that object alone).

P9.8 *Training bottleneck diagnosis* ♦

The job is **I/O bound** (disk-bound): throughput is limited by data delivery, not by compute — the GPU starves waiting on the input pipeline. Worth volunteering the fix list: parallel data loading with prefetch (`DataLoader` workers), local NVMe or caching instead of network storage, sharded binary formats (Parquet/TFRecord/WebDataset) instead of millions of small files, decode/augment offline or on GPU (DALI). Contrast terms to have ready: *compute-bound* (GPU pegged at 100%), *memory-bandwidth-bound*, *CPU-preprocessing-bound* (CPU pegged, disk idle).

Self-check (the utilisation signature is discriminating). If compute were the bottleneck, GPU utilisation would be high; if CPU preprocessing were, the CPU would be pegged with the disk quiet; here the *only* saturated resource is the disk, and the bottleneck is by definition the saturated resource — so I/O bound is not just consistent but uniquely identified.

Answer: I/O bound — the data pipeline (disk reads), not compute, is the bottleneck.

P9.9 *Fail fast vs fail silent* ♦

The **fail-fast principle**: an unexpected NaN in a critical column means the pipeline's output will likely be invalid, so the remaining ∼71 hours of compute would be spent manufacturing a corrupt artefact. Crashing at hour one (i) saves the compute, and (ii) — more important in a research context — prevents *silent corruption from propagating*: a model trained on quietly-skipped data does not announce the problem, it just misleads, and silently dropped rows also shift distributions and make runs unreproducible (which rows vanished this time?). An unexpected NaN is usually a symptom of an upstream join or vendor problem that affects more than the visible rows, so surface it, fix the cause, rerun. (In the MCQ, the distractor options described benefits that do not generally happen.)

Self-check (cost asymmetry). Crash-now cost: 1 hour of compute plus a fix. Continue cost: 72 hours of compute *plus* debugging a subtly wrong model later — and if the output feeds trading decisions, the cost of quietly-wrong is unbounded relative to the cost of loudly-absent. The asymmetry independently justifies the principle without appeal to authority.

Answer: to avoid wasting the remaining 3 days of compute on output that will likely be invalid, and to stop silent corruption propagating downstream — the fail-fast principle.

P9.10 *Immutability and concurrency* ♦

It **eliminates race conditions**. A data race requires a write to shared state; if data cannot be modified after creation, threads cannot overwrite what another thread is reading — there are no read-modify-write hazards, no torn reads, and *nothing to lock.* Any number of threads can share the same structure safely; an "update" builds a new version, so every thread holds a consistent snapshot. Downstream benefits: no locks means no deadlocks from lock-ordering, and code becomes referentially transparent and easy to test. The cost — allocation and copying — is what persistent data structures with structural sharing exist to amortise.

Self-check (contrapositive on the canonical race). The textbook race is two threads executing `x += 1` and losing an increment: it requires reading, modifying, and writing back *shared mutable* `x`. Make `x` immutable and that program cannot be written — each thread can only produce a *new* value, and combining them forces an explicit, visible coordination point. The hazard is not merely mitigated; it is unrepresentable.

Answer: immutability eliminates race conditions — threads cannot overwrite shared state, so there are no read-modify-write hazards and nothing to lock.

P9.11 *Shared-node OOM forensics* ♦

The job is **consuming most of the node's available RAM**. When the node exhausts memory, the kernel's OOM-killer terminates processes to free it — and its victim selection (an `oom_score` heuristic favouring large, recent, low-priority processes, adjustable via `oom_score_adj` and cgroup limits) is *not obliged to pick the culprit.* So the hog survives while its neighbours are sacrificed: "runs without issue" is precisely what the offender looks like. Everyday SLURM reality: request memory honestly (`-mem`/`-mem-per-cpu`); with cgroup enforcement configured, the kill is contained to the offending job's cgroup and this failure mode disappears. Diagnostics: `dmesg` OOM-killer lines naming sacrificed PIDs, and `sacct` MaxRSS showing the survivor's footprint.

Self-check (eliminate the alternatives). CPU oversubscription slows neighbours but does not OOM-kill them; disk/network saturation causes stalls and timeouts, not OOM; a bug in the neighbours' code would not correlate with sharing a node with this particular job. Only memory exhaustion — with kernel victim selection decoupled from blame — explains "offender healthy, neighbours OOM-killed".

Answer: the job is eating the node's RAM, triggering the OS OOM-killer, which kills other processes (not necessarily the offender) to free memory.

21.10 Solutions — §10 Sharpe, Loss Probabilities and Model Combination

P10.1 *P(loss) = Phi(-SR) and time scaling*

(a) With $R \sim \mathcal{N}(\mu, \sigma^2)$ and $\mathrm{SR} = \mu/\sigma$,

$$P(R<0) = P\Big(Z < -\frac{\mu}{\sigma}\Big) = \Phi(-\mathrm{SR}) = \Phi(-1.5) \approx 6.7\%.$$

(b) Under i.i.d. returns the mean scales with the horizon T and the sd with $\sqrt{T}$, so $\mathrm{SR}_T = \mathrm{SR}\sqrt{T}$. Monthly: $\mathrm{SR}_{1/12} = 1.5/\sqrt{12} \approx 0.433$, hence

$$P(\text{losing month}) = \Phi(-0.433) \approx 33\%.$$

(c) Shrinking the horizon shrinks drift linearly but noise only as the square root, so signal-to-noise degrades as $\sqrt{T}$: at short horizons the drift is invisible inside the noise and the loss frequency drifts toward 1/2. Same strategy, different clock — this is also why staring at daily P&L makes a good strategy feel bad (roughly $\Phi(-1.5/\sqrt{252}) \approx 46\%$ of days lose).

Self-check. Anchor consistency: SR 1 gives $\Phi(-1) \approx 15.9\%$ losing years; our higher SR 1.5 must lose less often, and $6.7\% < 15.9\%$. Limits: as $T \to \infty$, $\Phi(-\mathrm{SR}\sqrt{T}) \to 0$; as $T \to 0$, $\to \Phi(0) = 50\%$; the monthly 33% sits between the annual 6.7% and 50% as it must.

Answer: (a) $\Phi(-1.5) \approx 6.7\%$; (b) $\Phi(-1.5/\sqrt{12}) \approx 33\%$; (c) SR scales as $\sqrt{T}$, so shorter horizons are noise-dominated and loss frequency tends to 50%.

P10.2 *Sharpe to loss probability with assumption demolition* ★

Assumptions (state them before computing). (i) Annual P&L X is normal; (ii) the parameters are the *true* ones and constant; (iii) years are i.i.d.; (iv) horizon is one year and "losing \$100M" means *terminal* annual P&L $\le -\$100$M (not touching $-\$100$M intra-year). Justification: normality is the maximum-entropy default given only mean and vol, and annual P&L aggregates many shorter-period P&Ls (a CLT gesture — with known weaknesses in the tails); i.i.d. and known parameters are what the given numbers implicitly assert.

Computation. Mean P&L $\mu = \mathrm{SR} \times \sigma = 2 \times \$50\text{M} = \$100\text{M}$. A $-\$100$M year sits at

$$z = \frac{-100-100}{50} = -4 \qquad \Longrightarrow \qquad P = \Phi(-4) \approx 3.2 \times 10^{-5} \approx 0.003\%.$$

Direction: higher in reality. Every assumption breaks the same way:

- *Fat tails* — real P&L distributions have excess kurtosis; Φ underprices 4-sigma events badly.
- *Vol clustering / non-constant σ* — even if each regime is conditionally normal, a scale mixture of normals is unconditionally fat-tailed.
- *Negative skew* — many strategies have short-vol-like payoff profiles: frequent small gains, rare large losses.
- *Estimated vs true Sharpe* — a reported SR 2 is typically selected (best strategy, survivor); the deflated-Sharpe argument says true μ is lower, moving the loss threshold closer.
- *Path vs terminal* — if "losing \$100M" means the drawdown ever touching $-\$100$M during the year, the probability is strictly larger (for driftless Brownian motion, exactly $2\times$ by the reflection principle; with positive drift, between $1\times$ and $2\times$).
- *Regime change, crowding, correlations to one in crises* — the years are not exchangeable draws.

The distribution-free anchor (the interviewer's own follow-up). Cantelli's one-sided inequality needs only the mean and variance: $P(X - \mu \leq -k\sigma) \leq 1/(1+k^2)$. Here $k = 4$: bound $= 1/17 \approx 5.9\%$. The truth lives between 0.003% and 5.9% — a gap of over three orders of magnitude that is *entirely* the Gaussian tail assumption. Naming that gap is the point of the question.

Self-check. Recompute z: loss of \$100M means $X = -100$; $(-100 - 100)/50 = -4$. Anchor table: $\Phi(-4) \approx 3.2 \times 10^{-5}$. Monotonicity sanity: merely losing *any* money is a 2-sigma event, $\Phi(-2) \approx 2.3\%$; losing \$100M must be far rarer, and $0.003\% \ll 2.3\%$.

Answer: Under normal, i.i.d., known-parameter, terminal-P&L assumptions: $P = \Phi(-4) \approx 3.2 \times 10^{-5} \approx 0.003\%$. In reality *higher* (fat tails, vol clustering, negative skew, estimated Sharpe, path-vs-terminal, regime/crowding risk); distribution-free ceiling: Cantelli gives $\leq 1/17 \approx 5.9\%$.

P10.3 *Cantelli's inequality: proof, application, tightness*

(a) Let $Y = \mu - X$, so $E[Y] = 0$, $\mathrm{Var}(Y) = \sigma^2$, and the event is $\{Y \geq k\sigma\}$. For any $u \geq 0$: if $Y \geq k\sigma$ then $Y + u \geq k\sigma + u > 0$, so $(Y+u)^2 \geq (k\sigma+u)^2$. Markov's inequality on the nonnegative variable $(Y+u)^2$ gives

$$P(Y \geq k\sigma) \;\leq\; \frac{E[(Y+u)^2]}{(k\sigma+u)^2} \;=\; \frac{\sigma^2+u^2}{(k\sigma+u)^2}.$$

Optimise the free parameter: minimising over u (set the derivative to zero, or just try $u = \sigma/k$):

$$\frac{\sigma^2+\sigma^2/k^2}{(k\sigma+\sigma/k)^2} = \frac{\sigma^2(k^2+1)/k^2}{\sigma^2(k^2+1)^2/k^2} = \frac{1}{1+k^2}. \qquad \blacksquare$$

(b) Losing at least \$100M means $X \leq -100 = \mu - 4\sigma$ (since $(100+100)/50 = 4$), so $k = 4$:

$$P \;\leq\; \frac{1}{1+16} = \frac{1}{17} \approx 5.9\%.$$

The Gaussian assumption gives $\Phi(-4) \approx 3.2 \times 10^{-5}$; the bound is larger by a factor of about 1800 — three-plus orders of magnitude. Neither number is "the answer": $\Phi(-4)$ is the best case of a specific thin-tailed model, 1/17 is the worst case over *all* distributions with that mean and vol, and reality sits in between depending on tail assumptions.

(c) Take the two-point distribution

$$X = \mu + \frac{\sigma}{k} \text{ w.p. } \frac{k^2}{1+k^2}, \qquad X = \mu - k\sigma \text{ w.p. } \frac{1}{1+k^2}.$$

Mean: $\frac{\sigma}{k} \cdot \frac{k^2}{1+k^2} - k\sigma \cdot \frac{1}{1+k^2} = \frac{\sigma k - \sigma k}{1+k^2} = 0$ above μ. Variance: $\frac{\sigma^2}{k^2} \cdot \frac{k^2}{1+k^2} + k^2\sigma^2 \cdot \frac{1}{1+k^2} = \sigma^2 \frac{1+k^2}{1+k^2} = \sigma^2$. And $P(X - \mu \le -k\sigma) = \frac{1}{1+k^2}$ exactly — equality holds, so no better distribution-free bound exists.

Self-check. At $k = 1$ the bound is $1/2$, strictly better than the vacuous two-sided Chebyshev ($1/k^2 = 1$), and safely above the Gaussian $\Phi(-1) \approx 15.9\%$. At $k = 4$: $1/17 \approx 5.88\%$ and $\Phi(-4) \approx 0.0032\%$ — bound respected with room to spare.

Answer: (a) Markov on $(\mu - X + u)^2$ optimised at $u = \sigma/k$ gives $1/(1 + k^2)$; (b) $1/17 \approx 5.9\%$ versus Gaussian 3.2×10^{-5} — an $\approx 1800\times$ model-risk gap; (c) the two-point law at $\mu + \sigma/k$ and $\mu - k\sigma$ with masses $k^2/(1 + k^2)$, $1/(1 + k^2)$ attains it. □

P10.4 *Sharpe vs EV at maximum capacity* ♦

Compute the daily Sharpe (EV/sd) of each: (A) $5000/1 = 5000$; (B) 0.67; (C) 0.5; (D) 0.25. Annualised ($\times\sqrt{252} \approx 15.9$): (A) astronomical; (B) ≈ 10.6; (C) ≈ 7.9; (D) ≈ 4.0. Option (A) is the bait.

The crux is the phrase *at maximum capacity.* Sharpe is the right ranking criterion only because a scalable strategy can be levered: any high-Sharpe edge can be turned into any desired expected P&L at chosen risk. The capacity clause removes exactly that move — you must compare the strategies *as they stand.* A large, diversified, well-capitalised trading firm is effectively risk-neutral at these dollar magnitudes, so it simply maximises expected value: **(D)**, \$100k/day ≈ \$25.2M/year.

Why the risk in (D) does not matter: annualised, D earns \$25.2M with a standard deviation of $400\text{k} \times \sqrt{252} \approx \$6.35\text{M}$ — a standalone annual Sharpe of ≈ 4, so a merely *flat* year is a 4σ event ($\Phi(-4) \approx 3 \times 10^{-5}$). Even a 2σ-bad year still makes ≈ \$12.5M. The scary-looking "40% of days lose" is daily noise, irrelevant at the horizon a firm cares about. Meanwhile (A), for all its Sharpe of 5000, earns \$1.26M/year — a rounding error at firm scale. (You run (A) too, of course — it is free money — but the question asks which you *most prefer*, and at capacity preference goes by EV.)

Self-check via certainty equivalents: for a mean–variance agent to prefer (A), risk aversion would have to be so extreme that the firm would also reject a coin flip winning \$1.0M vs losing \$0.84M — no trading firm behaves that way. Cross-check of the rule itself: if capacity were *unlimited*, lever (A) $\times 10^4$ and it dominates everything — confirming that the capacity clause is precisely what flips the answer from Sharpe to EV.

Answer: (D) — at maximum capacity you cannot lever, so a well-capitalised firm maximises expected value: \$100k/day (≈ \$25M/yr, standalone annual Sharpe still ≈ 4). Picking the Sharpe-5000 strategy (A) — \$1.26M/yr — is the trap the capacity clause is designed to spring. (Note: at the live sitting the recorded answer was (A); on adjudicated review the canonical trading-firm answer is (D).)

P10.5 *Var(A+B) and the Sharpe of combined strategies*

Throughout, $\mathrm{Var}(A + B) = \mathrm{Var}(A) + \mathrm{Var}(B) + 2\,\mathrm{Cov}(A, B) = 100 + 100 + 2\rho \cdot 100 = 200(1 + \rho)$ (in $\$\mathrm{M}^2$), and the mean is always $10 + 10 = 20$.

(a) $\rho = 0.5$: Var = 300, vol = $\sqrt{300} \approx \$17.3\text{M}$, SR = $20/17.3 \approx 1.15$. Positive correlation is redundancy: risk is superadditive relative to the diversified case, and the Sharpe improvement over a single leg is modest.

(b) $\rho = 0$: Var = 200, vol ≈ \$14.1M, SR = $20/14.1 = \sqrt{2} \approx 1.41$ — the classic $\sqrt{2}$ diversification gain.

(c) $\rho = -0.3$: Var = 140, vol ≈ \$11.8M, SR = $20/11.8 \approx 1.69$. Negative correlation is hedging: combined risk falls below the sum and can fall below either leg alone.

(d) N uncorrelated strategies, each mean μ and vol σ: the sum has mean $N\mu$ and vol $\sigma\sqrt{N}$, so $\mathrm{SR}_N = \sqrt{N}\,\mathrm{SR}_1 = \sqrt{\sum_i \mathrm{SR}_i^2}$ (all equal). The quadrature formula in general, however, presumes each strategy contributes equal *risk.* A raw dollar sum weights legs by their vols: a leg with Sharpe 1 but tiny vol adds almost nothing to a raw sum's P&L, while a high-vol leg dominates the book's risk. Scaling each leg to equal vol (weights $\propto 1/\sigma_i$, "risk-balanced") restores the symmetric case and delivers $\sqrt{\sum \mathrm{SR}_i^2}$ for uncorrelated legs.

Self-check. (b) via quadrature: $\sqrt{1^2 + 1^2} = 1.414$, matching the direct computation. Limiting cases of the formula $\mathrm{SR}(\rho) = 20/\sqrt{200(1+\rho)}$: $\rho = 1$ gives vol 20, SR = 1 (no benefit — same strategy twice); $\rho \to -1$ gives vol $\to 0$, SR $\to \infty$ (perfect hedge with positive carry) — both limits sensible. Monotonicity: $1.15 < 1.41 < 1.69$ as ρ falls.

Answer: (a) vol $\approx$ \$17.3M, SR ≈ 1.15; (b) vol $\approx$ \$14.1M, SR $= \sqrt{2} \approx 1.41$; (c) vol $\approx$ \$11.8M, SR ≈ 1.69; (d) $\mathrm{SR}_N = \sqrt{N}\,\mathrm{SR}_1$, and the quadrature rule needs equal-risk (vol-scaled) weights, not a raw dollar sum. ☐

P10.6 *Decorrelating a new model from a black box*

(a) Two independent failures:

- *It discards genuinely predictive features.* A feature can be correlated with A's output and still carry incremental information A does not use; deleting it throws away alpha wholesale rather than removing only the overlap.
- *It does not even achieve the goal.* Pairwise feature-to-A correlations do not bound the correlation of the *trained model's output* with A. Concrete counterexample: suppose A's output effectively equals $X_1 + \cdots + X_{10}$ for i.i.d. features X_i. Each single feature has $\mathrm{Corr}(X_i, A) = 1/\sqrt{10} \approx 0.32$ — below any reasonable filter threshold — yet a model trained on all ten can reproduce A exactly (correlation 1).

(b) *Exact one-liner (post-hoc):* regress B's output on A's and trade the residual, $B_\perp = B - \hat{\beta}A$ with $\hat{\beta} = \mathrm{Cov}(A,B)/\mathrm{Var}(A)$. By construction $\mathrm{Cov}(B_\perp, A) = \mathrm{Cov}(B,A) - \hat{\beta}\,\mathrm{Var}(A) = 0$: you trade exactly B's incremental information. Trade-off: the part of B's alpha that overlapped A is stripped out — but that part was already being monetised by A, so nothing real is lost at the book level; B's standalone Sharpe falls even as the book improves.

Training-time alternative: add a correlation penalty to the loss, $\mathcal{L} = \mathrm{MSE} + \lambda\,\mathrm{Corr}(A,B)^2$, and tune λ. Trade-off: a continuous dial between B's standalone accuracy and orthogonality; costs some fit, requires A's outputs at training time, and gives approximate rather than exact orthogonality.

Why bother at all: with Corr = 0.7, $\mathrm{Var}(A+B) = \mathrm{Var}\,A + \mathrm{Var}\,B + 2 \times 0.7\,\sigma_A\sigma_B$ — heavily superadditive risk for mostly redundant signal. After orthogonalisation the cross term is zero and (with risk-balanced weights) the book's Sharpe approaches $\sqrt{\mathrm{SR}_A^2 + \mathrm{SR}_{B_\perp}^2}$.

Self-check. The residual method's exactness: $\mathrm{Cov}(B - \beta A, A) = \mathrm{Cov}(B,A) - \beta\,\mathrm{Var}(A)$, which vanishes precisely at the OLS slope $\beta = \mathrm{Cov}(B,A)/\mathrm{Var}(A)$. The counterexample in (a): $\mathrm{Corr}(X_i, \sum_j X_j) = \sigma^2/(\sigma \cdot \sigma\sqrt{10}) = 1/\sqrt{10}$ — ten individually "weak" features jointly reconstruct A.

Answer: (a) feature-dropping discards predictive information, and pairwise feature correlations do not bound the trained output's correlation with A; (b) regress B on A and trade the residual (exact, loses only the already-monetised overlap), or penalise $\mathrm{Corr}(A,B)^2$ in the training loss (tunable, approximate).

21.11 Solutions — §12 Market Making I: the Economics of the Spread

P12.1 *Spread decomposition*

The spread decomposes into four components:

- **Adverse selection** — the flow that chooses to trade against your quote is biased against you (informed pickoffs); the spread is the insurance premium.

- **Inventory risk** — the price can move while you hold the position a fill gives you.
- **Order-processing costs and fees** — infrastructure, exchange fees, clearing.
- **Margin** — the maker's profit.

Volatility widens the first two: when prices move more, informed traders' private information is worth more (their pickoff profit per fill grows), and any inventory you are handed is riskier to carry. Competition compresses the margin (and squeezes operational slack out of processing costs) but cannot push the spread below the actuarial cost of the first two components — a maker quoting inside that cost loses money in expectation and exits.

Self-check (sanity bound): the Glosten–Milgrom miniature confirms the floor independently: with zero fees, zero inventory cost and zero profit margin, the breakeven spread is still $2\pi > 0$ whenever a fraction $\pi > 0$ of flow is informed. So "competition drives the spread to zero" fails exactly when adverse selection exists, which is consistent with the decomposition.

Answer: spread = adverse selection + inventory risk + order-processing/fees + margin; volatility widens the first two; competition compresses the margin.

P12.2 *Glosten-Milgrom zero-profit quotes*

(a) Condition on the side that trades. An informed trader buys only when $V = 101$; noise buys half the time regardless:

$$P(\text{buy} \mid V{=}101) = \pi + \tfrac{1-\pi}{2} = 0.65, \qquad P(\text{buy} \mid V{=}99) = \tfrac{1-\pi}{2} = 0.35, \qquad P(\text{buy}) = \tfrac{1}{2}.$$

By Bayes, $P(V{=}101 \mid \text{buy}) = \frac{0.65\times 0.5}{0.5} = 0.65$. The zero-profit ask is the value *conditional on being lifted*:

$$a = \mathbb{E}[V \mid \text{buy}] = 99 + 2(0.65) = 100.30,$$

and by symmetry the bid is $b = \mathbb{E}[V \mid \text{sell}] = 99.70$.

(b) Breakeven spread $= a - b = 2\pi = 0.60$. The friend's error: *unconditional* flow is indeed 50/50, but the flow that trades against a *fixed* quote is conditionally biased against it — your resting ask is lifted disproportionately in the states where $V = 101$. Quotes must be set at the expected value conditional on being traded with, not at the unconditional mean. Any spread below 0.60 loses in expectation even with zero fees, zero inventory cost, and no profit margin.

Self-check (direct P&L): at ask a, conditional on being lifted, expected value delivered is $101 \times 0.65 + 99 \times 0.35 = 100.30$, so expected profit per lifted offer is $a - 100.30$: zero exactly at $a = 100.30$, negative below. Small-case check: at $\pi = 0$ the formula gives spread 0 — the friend's claim is precisely the no-informed-flow special case.

Answer: ask 100.30, bid 99.70; breakeven spread $2\pi = 0.60$. The claim ignores that the flow hitting a fixed quote is conditionally biased against it — quote the value conditional on being hit. □

P12.3 *Glosten-Milgrom with asymmetric prior*

(a) Buy side: $P(\text{buy} \mid 101) = \pi + \frac{1-\pi}{2} = 0.6$ and $P(\text{buy} \mid 99) = \frac{1-\pi}{2} = 0.4$, so

$$P(\text{buy}) = 0.6(0.6) + 0.4(0.4) = 0.52, \qquad P(101 \mid \text{buy}) = \frac{0.6 \times 0.6}{0.52} = \frac{9}{13} \approx 0.692.$$

Ask $= \mathbb{E}[V \mid \text{buy}] = 99 + 2 \cdot \frac{9}{13} = 100 + \frac{5}{13} \approx 100.385$. Sell side: $P(\text{sell} \mid 101) = 0.4$, $P(\text{sell} \mid 99) = 0.6$, so $P(\text{sell}) = 0.6(0.4) + 0.4(0.6) = 0.48$ and $P(101 \mid \text{sell}) = \frac{0.24}{0.48} = \frac{1}{2}$: bid $= 99 + 2(\frac{1}{2}) = 100.00$ exactly.

(b) Spread $= \frac{5}{13} \approx 0.385 < 2\pi = 0.4$: the asymmetric prior has *less* uncertainty (variance $4p(1-p)$ peaks at $p = \frac{1}{2}$), so there is less for the informed to know that you do not, and the insurance premium shrinks. The quote midpoint is $100 + \frac{5}{26} \approx 100.192$, *not* the prior mean 100.20. The bid updates further (0.20 down vs 0.185 up for the ask): a *sell* against the bullish prior is the more surprising event, so it moves the posterior more ($0.6 \to 0.5$ on a sell vs $0.6 \to 0.692$ on a buy).

(c) $p \to 1$: both posteriors $\to 1$, so ask = bid = 101 and the spread $\to 0$ — no private information left to protect against. $\pi \to 0$: posteriors collapse to the prior, ask = bid = 100.2, spread $\to 0$ — no informed flow. Both limits behave.

Self-check (recompute the spread another way): spread $= 2\,[P(101 \mid \text{buy}) - P(101 \mid \text{sell})] = 2\left(\frac{9}{13} - \frac{1}{2}\right) = 2\cdot\frac{5}{26} = \frac{5}{13}$, matching ask − bid $= 100\frac{5}{13} - 100$. Also $P(\text{buy}) + P(\text{sell}) = 0.52 + 0.48 = 1$.

Answer: ask $= 100 + \frac{5}{13} \approx 100.38$, bid $= 100.00$; spread $\frac{5}{13} \approx 0.385 < 2\pi$; midpoint $100.19 \neq$ prior mean 100.20, and the bid side updates further because a sell is the more surprising trade under a bullish prior. □

P12.4 *Mark-outs*

(a) **Mark-outs**: the mid-price change t seconds after your fill, signed so that negative means the market moved against your trade, bucketed by counterparty, venue, and size. The average mark-out on your fills *is* realized adverse selection — an estimator of the informed-flow premium (the π of the Glosten–Milgrom model) computed from your own trades. A mark-outs-by-counterparty dashboard panel is the concrete, operational answer.

(b) Counterparty X's flow is toxic: consistently negative mark-outs mean X trades when you are about to be wrong. Widen or pull *for X specifically* if the venue lets you differentiate (or against the patterns that precede X's fills); cap the size you show X. If X's flow cannot be distinguished from the rest, you must widen for everyone — indistinguishable toxic flow raises the average toxicity of all flow.

(c) Volatility doubled: **widen**. Both vol-sensitive spread components (adverse selection and inventory risk) scale up; the flow itself has not become more informed, so you keep quoting, just at a spread that re-prices the new risk.

(d) Imminent unscheduled news: **pull first**, re-quote wide once the news is out, tighten as the picture clears. Around an announcement, the informed/faster fraction of flow spikes and the price can gap through your quote; being on the wrong side of one gap costs many spreads.

Self-check (map each action to the spread component that changed): (b) adverse selection rose *for one bucket* → widen/pull for that bucket; (c) σ rose → the two vol-driven components rise → widen; (d) momentarily $\pi \to$ large, so the breakeven spread 2π can exceed any spread that would still get filled — quoting nothing *is* the zero-profit quote. Each action is exactly the decomposition's prescription.

Answer: (a) mark-outs (post-fill mid drift) by counterparty/venue/size; (b) widen/pull/size-cap against X, or widen for all if unidentifiable; (c) widen; (d) pull, re-quote wide, tighten as the picture clears.

21.12 Solutions — §13 Market Making II: Inventory and Skewing

P13.1 *Reservation price computation*

(a) The inventory shift is $q\gamma\sigma^2(T-t) = 60 \times 0.001 \times 2.5 \times 0.4 = 0.06$, so

$$r = 100 - 0.06 = 99.94.$$

(b) Quotes: bid 99.84, ask 100.04. Compared with mid-centered quotes (99.90/100.10), *both* are shifted down by 0.06: the ask undercuts, so your offer is more likely to be lifted (you sell the long more eagerly); the bid backs off, so you are less likely to be hit into an even bigger long. The book leans toward shedding inventory while still quoting two-sided. (c) With $q = -60$ the shift flips sign: $r = 100.06$, quotes 99.96 bid at 100.16 ask — now leaning to buy the short back.

Self-check: recompute the shift stepwise: $60 \times 0.001 = 0.06$; $\times 2.5 = 0.15$; $\times 0.4 = 0.06$. Sign logic: long ($q > 0$) must give $r < s$ (a long wants to sell) — it does; $q = 0$ recovers mid-centered quotes; flipping q flips the shift exactly, by linearity.

Answer: (a) $r = 99.94$; (b) 99.84 bid at 100.04 ask — skewed to sell down the long; (c) $r = 100.06$, quotes 99.96 at 100.16.

P13.2 *Reservation-price anatomy, skew vs widen*

(a) The reservation price is the price at which you are indifferent between holding and shedding a marginal unit of inventory. Carrying q units to the horizon adds P&L variance $q^2\sigma^2(T-t)$; under exponential (mean-variance) utility the *marginal* risk cost of one more unit is proportional to $q\gamma\sigma^2(T-t)$. So: q sets the direction and urgency (more inventory, bigger concession); γ converts variance into the price concession you are willing to pay; σ^2 is the rate at which risk accrues; $(T-t)$ is how long you must carry it.

(b) As $T - t \to 0$ the skew vanishes: with no time left to hold the position, inventory carries (in the model) no further risk, so you quote around mid. Internally consistent — though real desks add an explicit terminal liquidation penalty (you cannot magically flatten at the bell for free), which keeps end-of-horizon skews alive in practice.

(c) Shifting both quotes sheds inventory through *both* fill channels at once: a lower ask raises the probability your offer is lifted (you sell), and a lower bid reduces the probability your bid is hit (you avoid buying more) — while you remain two-sided, keeping the flow, the queue position, and the information that comes with being in the market. Improving only the ask gives away edge on one side and does nothing to slow accumulation on the other.

(d) **Widening** prices *uncertainty* — adverse selection and volatility — and is symmetric: it addresses not knowing. **Skewing** manages a *known* directional position and is asymmetric: it addresses knowing exactly what you hold. A vol spike widens; an inventory build skews; toxic flow may widen or pull.

Self-check (limits and dimensions): $\gamma \to 0$ (risk-neutral) gives $r = s$ — no concession, consistent with (a); $\sigma \to 0$ likewise kills the skew. Dimensionally, $q \cdot \gamma \cdot \sigma^2 \cdot (T-t)$ carries [units] $\times$ [1/(price $\times$ units)] $\times$ [price^2/time] $\times$ [time] = [price], as a price correction must.

Answer: the correction is the marginal risk cost of carrying q to horizon; skew $\to 0$ as $T - t \to 0$ (no time left to bear risk); both quotes move so inventory sheds through both fill probabilities while staying two-sided; widening prices uncertainty, skewing manages a known position. ☐

P13.3 *Fermi estimation + interval calibration* ★

The point estimate is a rate $\times$ time chain said fast: 6 blocks/hour $\times 24 = 144$/day; $\times 365 \approx$ 52,600/year; $\times 14$ years (2012 to 2026) $\approx$ 735,000. The 15-second version: 50k/yr $\times 14 =$ 700k. Any answer of the form "about 700–750k" is full marks on the point.

The interval is the actual test. On a number you produced in 15 seconds from a possibly-misremembered rate, an honest 95% interval must be *embarrassingly wide* — order-of-magnitude, e.g. 200k to 2M (point estimate $\times/\div$ 3 is a good default when any link in the chain is soft). What it must *not* be is a memorized-fact interval: at the actual interview the candidate said $\approx$ 100k with interval 98k–120k, and the truth ($\approx$ 735k) fell far outside — that is the graded failure, because a 95% interval that misses is a direct demonstration of miscalibration. The one wrong move is refusing to give an interval at all: the interval *is* the statement of your uncertainty. Where should the width come from? The 10-minute rate is handed to you and the 14-year window is known, so the width prices your own arithmetic under pressure and any soft facts (e.g. if you are not sure whether halvings change the block rate — they do not, difficulty retargets to hold 10 minutes — your interval must cover being wrong about that).

The drill (how to train this): pick any quantity, produce point + 95% interval in 15 seconds, repeat many times, and score *coverage*: about 19/20 intervals should contain the truth. Missing half the time means far too narrow; never missing means too wide (you are giving away edge).

Self-check (independent recomputation via minutes): $14 \times 525{,}600 \approx 7.36$M minutes; divide by $10 \Rightarrow 736$k — matches the per-day chain. Coverage check: $[200\text{k}, 2\text{M}]$ contains 736k comfortably, while $[98\text{k}, 120\text{k}]$ does not — the calibration lesson in one line.

Answer: $\approx 735{,}000$ blocks (144/day $\approx$ 52.6k/yr $\times 14$); a calibrated 95% interval is order-of-magnitude wide, e.g. 200k–2M — and refusing to give one is the only losing move.

P13.4 *Make-a-market game: position and P&L accounting* ★

Bookkeeping convention first (this is where the game is won or lost): the counterparty's BUY lifts your *offer* — you SOLD one unit at your ask; their SELL hits your *bid* — you BOUGHT one unit at your bid. So the tape reads: sold at 200k, sold at 600k, sold at 1M, bought at 4M.

Position: 3 sold $-$ 1 bought $=$ **short 2 units**. (At the live interview the candidate answered "short 2 million" — *short* and *2* were right, but "million" conflated position size with price level. The clean answer is "short 2 units.")

Cash (millions of block-units): $+0.2 + 0.6 + 1.0 - 4.0 = -2.2$M.

Mark-to-market at the final mid 2.5M: equity $=$ cash $+q\times$ mid $= -2.2 + (-2)(2.5) = -7.2$M — the book is down badly. (Marked at the true value ≈ 0.735M instead: $-2.2 + (-2)(0.735) \approx -3.7$M; down either way — three sales went off far below value before the buy printed far above it.)

The flow read: repeated buys mean your offer is cheap — jump the market in multiples, not ticks, until the flow flips (the candidate did move 100–200 $\rightarrow$ 400–600 $\rightarrow$ 900k–1M $\rightarrow$ 4–6M, which is the right behavior). The flip is the information payoff: they *bought* at your 1M offer and later *sold* at your 4M bid, so they are acting as if the value lies in $(1\text{M}, 4\text{M})$ — and the final quote 2M–3M correctly sits inside that bracket. The graded checklist: (1) width = confidence; (2) move with the flow, never repeat an offer; (3) always know your inventory — the closing question is the whole point; (4) know your rough P&L. The game grades *process*, not proximity to the true number.

Self-check (recount two independent ways): by side: offers lifted at 200k, 600k, 1M (3 sales), bid hit at 4M (1 buy), net -2. Cash: received $0.2 + 0.6 + 1.0 = 1.8$M, paid 4.0M, net -2.2M. Equity by liquidation thought-experiment: buying back 2 units at the final mid costs $2 \times 2.5 = 5.0$M, leaving $-2.2 - 5.0 = -7.2$M with a flat book — matches cash $+q\times$ mid.

Answer: short 2 *units*; cash -2.2M; mark-to-market ≈ -7.2M at the final mid 2.5M; the buy-then-sell flip brackets the counterparty's implied value inside $(1\text{M}, 4\text{M})$.

P13.5 *Pre-trade risk controls / position limits* ⧫

(C). Hard, automatically enforced pre-trade position limits are the standard control: a risk layer rejects any order that would push the position beyond its limit, mechanically bounding worst-case inventory *no matter what* the upstream logic does — bug, bad data, or runaway loop. (A) and (B) are detective controls with human latency: in fast markets a runaway algorithm builds a catastrophic position in seconds, and a dashboard or alert only tells you it already happened. (D) reduces the probability of a bug but cannot bound the loss when one slips through — and one always can. Note the design detail inside (C): block only the *breaching* side; the algorithm must remain free to trade the position back down, otherwise the limit locks in the very risk it was meant to cap.

Self-check (control-theory sanity test): a control that bounds the bad state *ex ante* dominates one that reports it *ex post* whenever the state can move faster than the responder — algorithmic trading is exactly that regime, and (C) is the only ex-ante binding option. Consistent, too, with inventory economics: the loss tail scales with $|q|$, so capping $|q|$ caps the tail.

Answer: (C) — hard position limits that automatically block further trading on the breaching side.

P13.6 *Inventory hedging with perps*

Hedge externally: short the perpetual future on the same or another venue B against the long spot on A. Your delta is now approximately flat, so you can *keep quoting* — keep earning the spread and keep seeing the flow — while carrying basis instead of direction. The perp costs (or earns) the funding rate rather than directional exposure, and you can unwind the pair slowly: route future sell flow to close the spot leg while buying back the perp, or let both legs run off.

Residual risks:

- *Basis risk* — the perp-spot spread can move against the pair.
- *Funding cost* — if funding runs against your side, the hedge bleeds carry while it is on.
- *Venue/counterparty risk* — collateral now sits on venue B; where inventory and margin live overnight is a first-class risk decision.
- *Margin/liquidation risk on the short perp leg* — a violent spike can stop the hedge out exactly when it is most needed.
- *Execution cost* of putting the hedge on (spread, fees, impact on B).

The colleague's proposal fails twice: pulling quotes forfeits the franchise (flow, queue position, information) *and* does nothing about the inventory you already hold. The larger lesson: at size, inventory management becomes a *routing and financing* problem — where to hedge, what carry to pay, where collateral sits — not just a quoting problem.

Self-check (P&L decomposition): long spot + short perp = basis position + funding leg; the directional term cancels by construction (both legs carry ≈ unit delta), so the remaining exposures are exactly the residuals listed — nothing directional left unaccounted. Comparison bound: dumping Q into your own book costs on the order of the impact ($\propto \sqrt{Q}$, large at size) *and* signals your position; the hedge caps the cost at spread + fees + basis.

Answer: short perps against the long spot (same or another venue): stay quoting, carry funding/basis instead of direction; residual risks are basis moves, funding cost, venue/counterparty exposure, margin/liquidation on the hedge leg, and hedge execution cost.

21.13 Solutions — §14 Market Microstructure

P14.1 *Square-root impact law by dimensional analysis*

Write the units equation:

$$\text{price} \;=\; \text{shares}^{\alpha}\left(\frac{\text{shares}}{\text{day}}\right)^{\beta}\left(\frac{\text{price}}{\text{day}^{1/2}}\right)^{\gamma} \;=\; \text{shares}^{\alpha+\beta}\,\text{day}^{-\beta-\gamma/2}\,\text{price}^{\gamma}.$$

Matching exponents on each unit: price $\Rightarrow \gamma = 1$; shares $\Rightarrow \alpha+\beta = 0$; time $\Rightarrow -\beta - \frac{\gamma}{2} = 0 \Rightarrow \beta = -\frac{1}{2}$, hence $\alpha = \frac{1}{2}$. So

$$I = Y\,\sigma\,\sqrt{\frac{Q}{V}},$$

the square-root law, with Y dimensionless.

What the ansatz sneaked in: the entire argument rests on the premise that Q, V, σ are the *only* relevant variables (no tick size, no spread, no execution duration or participation rate as an independent quantity) and that the law is a pure power law with a universal dimensionless Y. Dimensional analysis then shows the square root is the *only* consistent power law in those three variables — but the variable-selection premise is doing the heavy lifting; it is the empirics (roughly $Y \approx 1$ across markets and epochs) that justify it.

Self-check (plug back and scale): units of $\sigma\sqrt{Q/V}$: $(\text{price}\cdot\text{day}^{-1/2}) \times (\text{shares}/(\text{shares/day}))^{1/2} = \text{price}\cdot\text{day}^{-1/2}\cdot\text{day}^{1/2} = \text{price}$. Scaling sanity: doubling both Q and V leaves impact unchanged — impact depends on your size *relative to the market*, as intuition demands.

Answer: $\gamma = 1$, $\beta = -\frac{1}{2}$, $\alpha = \frac{1}{2}$, i.e. $\Delta P = Y\sigma\sqrt{Q/V}$; the hidden assumption is that Q, V, σ (in power-law form, universal Y) are the only variables that matter. □

P14.2 *Impact vs alpha capture*

(a) $Q/V = 250\text{k}/10\text{M} = 0.025$; $\sqrt{0.025} \approx 0.158$; so

$$\Delta P \approx 2\% \times 0.158 \approx 0.32\% = 32 \text{ bp}.$$

(b) **No.** Your own impact (≈ 32 bp) exceeds the 25 bp alpha — at this size the trade destroys its own edge; even the *average* execution cost, a substantial fraction of the peak impact, leaves nothing after spread and fees. Since impact scales as $\sqrt{Q}$ while the per-share alpha does not depend on your size, shrinking the order restores positive expectancy: e.g. 25k shares gives $\sqrt{0.0025} = 0.05$, impact ≈ 10 bp against 25 bp of alpha. (c) The Almgren–Chriss trade-off. Slicing over five days does cut the impact, but (i) a *two-hour* alpha is long gone by day five — you would pay real costs to acquire a position whose forecast has expired; and (ii) slower execution bears more risk of adverse price moves while you are still un-executed. Too big eats the alpha, too small misses it, too slow risks the move.

Self-check (recompute (a) differently): $Q/V = 1/40$, $\sqrt{1/40} = 1/\sqrt{40} \approx 1/6.32 \approx 0.158$, so $2\% \times 0.158 = 31.6$ bp. Robustness bound for (b): even using the common refinement cost $\approx \frac{2}{3}\times$ peak impact ≈ 21 bp, the margin over 25 bp alpha is ≈ 4 bp before spread and fees — effectively zero, so the conclusion stands.

Answer: (a) ≈ 32 bp; (b) no — impact is larger than the 25 bp alpha at this size (trade smaller, e.g. 25k shares for ≈ 10 bp impact); (c) the alpha decays long before the slices execute, and slower execution adds adverse-move risk.

P14.3 *Kyle's lambda and reconciliation with concave impact*

(a) λ is Kyle's lambda: the price move per unit of signed order flow — a direct measure of *illiquidity* (the inverse of depth). Uses: comparing liquidity across assets, venues, and time; pre-trade cost models and sizing (an impact budget); flagging liquidity droughts when λ spikes.

(b) (i) Linear: $4\times$ the flow $\Rightarrow 4 \times 2 = 8$ bp. (ii) Square root: $\sqrt{4} = 2\times \Rightarrow 4$ bp.

(c) Empirically the local slope is not constant in trade size: fitted per size bucket, $\lambda(Q) \propto Q^{-1/2}$, so total impact $\lambda(Q)\,Q \propto \sqrt{Q}$ — concave. A single linear regression estimates an *average* slope over whatever size distribution you sampled; extrapolating it linearly to $4\times$ the typical size overstates impact. The square-root prediction (4 bp) is the one supported by metaorder studies. Economically, large orders move the price less per share because they are deliberately worked where and when liquidity is available, and convey proportionally less information per share.

Self-check (consistency identity): if total impact is $I(Q) = c\sqrt{Q}$, then the local slope is $I(Q)/Q = c\,Q^{-1/2}$ — exactly the stated empirical scaling $\lambda(Q) \propto Q^{-1/2}$, so the two statements are one law. And both models agree at the calibration size by construction ($Q = 10\text{k} \Rightarrow 2$ bp), diverging only out of sample — which is why in-sample regression fit cannot pick between them; size-bucketed fits can.

Answer: (a) $\lambda =$ illiquidity: price move per unit signed volume; (b) 8 bp linear vs 4 bp square-root; (c) empirically $\lambda(Q) \propto Q^{-1/2}$ so total impact $\propto \sqrt{Q}$ — trust the concave, 4 bp prediction. □

P14.4 *OBI and VAMP computation*

(a) $\text{OBI} = \frac{1000-100}{1100} = \frac{900}{1100} \approx +0.82$ (strong buying pressure);

$$P_{\text{VAMP}} = \frac{100 \times 1000 + 99 \times 100}{1100} = \frac{109{,}900}{1100} \approx 99.91.$$

(b) The 1,000-share bid wall means sellers must chew through enormous resting demand to push the price down, while a mere 100 shares of buying lifts the whole ask: the next move is far more likely up than down, so the "true" price sits near the ask. Weighting the *ask price* by the *bid size* encodes exactly this — a heavy bid pins value toward the ask. The plain mid (99.50) ignores the pressure; VAMP ($\approx$ 99.91) reads it. (The naive same-side weighting would do the opposite and put the price near the crowded side — backwards.) (c) Precise answer: this is the *volume-weighted (imbalance-weighted) mid*, a first-order proxy. Stoikov's formal *micro-price* is defined as the limit of expected future mids conditional on the book state; the weighted mid is its first-order approximation. Making that naming distinction out loud is worth doing at an HFT-adjacent shop.

Self-check (boundary cases): $q_b = q_a$ gives OBI = 0 and VAMP = plain mid — correct neutral limit; $q_b \gg q_a$ sends VAMP $\to P_a$, matching the near-certain uptick logic of (b); and VAMP is a convex combination of P_b and P_a (weights $\frac{1000}{1100} + \frac{100}{1100} = 1$), so it always lies inside the quoted spread.

Answer: OBI $\approx +0.82$; $P_{\text{VAMP}} \approx 99.91$, pinned near the ask; it is the weighted mid — a first-order proxy for Stoikov's micro-price (expected future mid), not the formal micro-price itself.

P14.5 *OBI signal monetisation*

Three independent killers:

- **The economics do not clear the spread.** A 58% hit rate on a one-tick move is an expected edge of $(0.58 - 0.42) \times 1 = 0.16$ ticks per trade, while crossing the spread costs a full spread (≥ 1 tick) plus taker fees, *twice* per round trip. Cost exceeds edge by construction — short-horizon book signals are usually worth a fraction of the spread.
- **Latency, queues, and adverse fill timing.** By the time you observe OBI = 0.8 and your order arrives, faster participants have repriced; the thin ask you meant to lift is gone precisely when the signal was right. You get filled preferentially in the states where the move already happened or the book refilled — the live fill distribution is strictly worse than the backtest's, which assumed you trade the quote you saw.
- **Displayed depth is not commitment, and the signal double-counts.** Resting size can be flickered or spoofed, inflating backtest OBI; and OBI is highly autocorrelated, so "58% of ticks" largely re-samples the same persistent episodes rather than 58% of independent bets. Worse, the payoff is asymmetric: the occasions when the thin side is swept *through* — exactly when you buy the top — are the expensive ones.

How a maker monetises it: passively. Use OBI/VAMP as a quote-*placement* input: lean (skew) quotes toward the imbalance — quote tighter/earlier on the side the price is drifting toward, back away on the other side — so the drift is captured while *earning* the spread instead of paying it; and use extreme imbalance as a pull/fade trigger to avoid being picked off on the stale side. The general principle: a signal smaller than the spread pays the passive, not the aggressive.

Self-check (net-expectancy bound): aggressive round trip: +0.16 ticks of edge $-\ \geq 1$ tick of spread $-2\times$ taker fees < 0, robustly negative *granting* the 58%; passive use collects spread + drift-conditional fill quality, the standard maker P&L on flow-anticipating signals — signs opposite, same signal, which is the whole point.

Answer: taking pays ≥ 1 tick + fees for ≈ 0.16 ticks of edge; live fills arrive after faster players reprice (worse than backtest by selection); displayed imbalance is spoofable/flickery and autocorrelated with asymmetric downside. Monetise it passively: skew quote placement with the imbalance and pull on extremes — earn the spread, don't pay it.

21.14 Solutions — §15 Crypto Market Structure

P15.1 *Bitcoin mechanics* ★

Answer in the four beats the question asks for.

(1) Getting a transaction included. A user signs a transfer with their private key and broadcasts it; pending transactions sit in every node's *mempool.* Each transaction attaches a fee, and block space is scarce, so miners sort candidates by fee rate — attaching a higher fee is how a user buys priority. In congestion, fees spike and low-fee transactions wait.

(2) Mining a block. Miners assemble mempool transactions into a candidate block and search for a nonce such that the SHA-256 hash of the block header falls below a difficulty target (proof of work). Finding one is a lottery with win rate proportional to hash power; the difficulty retargets every 2016 blocks so that blocks keep arriving roughly every 10 minutes — about 144 per day.

(3) The chain. Every block header contains the hash of the *previous* block header, so the blocks form a hash-linked chain: changing any historical transaction changes that block's hash and breaks every later link, so rewriting history means redoing the proof of work for the entire suffix. Nodes adopt the chain with the most cumulative work; that is why a transaction buried under several blocks (the 6-confirmation convention) is treated as final.

(4) Miner pay. Two sources: the *block subsidy* of newly minted bitcoin (halving every 210,000 blocks, roughly 4 years: $50 \to 25 \to 12.5 \to 6.25 \to 3.125\ldots$), plus the sum of the fees of the transactions included.

Self-check (numbers hang together): total supply $= 210{,}000 \times 50 \times (1+\frac{1}{2}+\frac{1}{4}+\cdots) = 210{,}000 \times 100 =$ 21M — the famous cap drops out of the halving schedule. Cadence: $6/\text{hr} \times 24 = 144$ blocks/day, the same anchor as the blocks-since-2012 Fermi estimate (≈ 52.6k/yr).

Trading relevance (worth one closing sentence): the 10-minute cadence plus the fee market means on-chain transfers have stochastic latency — exactly the friction cross-venue crypto arbitrage has to cross, and why inventory logistics matter as much as quoting.

Answer: Blocks are mined by a proof-of-work lottery roughly every 10 minutes; each block hashes the previous one into a chain and the most-work chain wins; users attach fees to get transactions included; miners are paid the block subsidy of newly minted coins (halving every $\sim$4 years) plus transaction fees.

P15.2 *Cross-exchange arbitrage, leg risk* ♦

Go through the options against the economics of the trade, whose edge is a few basis points per round trip.

- *Fees (a):* known constants, priced in *ex ante* — if fees exceeded the discrepancy you would simply not fire. They cannot surprise you after the fact.
- *Latency (b):* missing an opportunity is an opportunity cost with zero realized loss. Annoying, not dangerous.
- *Withdrawal times (d):* a real friction, but it is a capital-efficiency and inventory-logistics problem spread across many trades (pre-position inventory on both venues), not the dominant per-trade risk.
- *Leg risk (c):* you intended to be flat; if only the buy (or only the sell) fills, you hold a naked directional position in a volatile asset until you chase the missing leg or unwind — at whatever price the market has moved to. This is the tail that eats the strategy.

Self-check by magnitudes: suppose the edge is 3 bp and the coin runs 80%/yr volatility. Five minutes of vol is $0.80/\sqrt{365 \times 288} \approx 0.80/324 \approx 25$ bp — a one-sigma move while you scramble to complete the leg is nearly ten times the edge, so a single bad leg-out erases dozens of successful arbs. Fees, by contrast, are bounded by construction below the edge.

Answer: (c) — leg risk: a partial fill leaves a large unhedged delta position; fees and latency are secondary distractors.

P15.3 *The market-maker mandate* ♦

Neither — and recognizing that the question is a trap is the point.

A designated market maker's contract is about *liquidity*, not levels or activity: presence in the book, a maximum spread, a minimum quoted depth, an uptime percentage, across named venues. Consider what promising each forbidden thing would mean:

- *Higher prices:* committing to support a price level means buying whatever supply arrives at that level — economically, writing an uncapped put on the token for a fee. No market-neutral liquidity business can carry that, and deliberately propping a price is market manipulation (painting a floor).
- *Higher volumes:* volume is generated by other participants; the only way a single firm can *guarantee* volume is to trade with itself — wash trading, again manipulation.

What the project actually buys is *tradability*: tight spreads and real depth lower the cost for anyone to enter or exit, which may well make the token more attractive — but price and volume are *outcomes* of the market, never deliverables of the market maker. This is the compliance-correct framing and the one a trading firm wants to hear you give unprompted.

If the format forces a choice between the two (as the live MCQ's binary phrasing did): pick *higher volumes* — liquidity provision legitimately supports tradability and hence volume, while any price promise is manipulation outright — and then immediately reframe to the true deliverables (spread, depth, uptime).

Self-check via incentives: both forbidden promises reduce to the same absurdity — either unbounded inventory risk (price support) or self-dealing (volume support). A promise a rational, compliant firm cannot structure is not the promise being made.

Answer: Neither. A market maker promises liquidity — maximum spread, minimum depth, uptime — never price levels or volumes; promising either would border on market manipulation.

P15.4 *Perpetual funding arithmetic*

(a) Direction. The perp has no expiry, so funding is the cash flow that ties it to spot: the rate is (a clamp of) the perp's premium over the spot index. Perp above index $\Rightarrow$ positive funding $\Rightarrow$ *longs pay shorts* every interval. Paying to stay long pushes longs to close and makes shorting attractive, pressuring the perp back *down* toward the index.

(b) Annualisation. Three intervals per day:

$$0.015\% \times 3 = 0.045\%/\text{day}, \qquad 0.045\% \times 365 \approx 16.4\%/\text{yr (simple)}.$$

Compounded: $(1.00015)^{3\times365} = e^{1095\ln(1.00015)} \approx e^{0.1642} \approx 1.179$, i.e. about 17.9%/yr.

(c) Weekly cost. 21 intervals $\times$ 0.015% = 0.315% of notional; on \$2M that is

$$2{,}000{,}000 \times 0.00315 \approx \$6{,}300 \text{ per week.}$$

Self-check: against the standard anchor 0.01%/8h $\approx$ 11%/yr: our rate is 1.5$\times$ that, and $1.5 \times 10.95\% = 16.4\%$. And the weekly number back-solves: \$6,300/\$2M = 0.315%, times 52 $\approx$ 16.4%/yr.

Answer: (a) Longs pay shorts; pressure pushes the perp down toward the index. (b) 0.045%/day $\approx$ 16.4%/yr simple, $\approx$ 17.9% compounded. (c) About \$6,300 per week on \$2M.

P15.5 *Cash-and-carry*

The trade (strongest version). This is the crypto cash-and-carry: buy spot, short an equal notional of the perp *on the same venue with the spot posted as collateral* (cross-margin), so the book is coin-for-coin delta-neutral and a rally increases collateral value in step with the short's mark-to-market loss. Collect funding every 8 hours while it is positive.

Why it is not free money:

- **Funding floats.** The rate resets every interval as a clamp of the perp premium; 30% is a two-week snapshot, not a lock. When crowded longs de-lever, the rate compresses or flips negative — and then *you* are the one paying to hold the position.
- **Basis risk.** There is no expiry forcing convergence at any horizon. You enter with the perp rich, but an early unwind at an adverse basis can surrender months of carry in a single mark.
- **Venue / counterparty risk.** Both legs and the collateral sit on an exchange; post-FTX, *where inventory sits* is a first-class risk decision. A large slice of the 30% is compensation for exactly this.
- **Margin / liquidation risk on the short leg.** If the legs sit on different venues (or the collateral is stablecoin), a sharp rally generates margin calls on the short faster than spot gains can be moved; a forced buy-in at the top converts a hedged book into a realised loss.
- **Frictions.** Fees and impact on four executions (into and out of both legs), funding leakage during exit, and the capital cost of fully-funded spot.

Positioning read. Persistent positive funding means leveraged longs are paying a fat premium to stay long: *crowded* positioning. At extremes it is a contrarian gauge — and a warning that the carry trade itself is crowded, so its unwind correlates with liquidation cascades.

Self-check by magnitudes: 30% of \$10M $\approx$ \$58k per week of carry; one 5% adverse basis/unwind mark is \$500k $\approx$ nine weeks of carry. A yield that one bad mark can erase is a risk premium, not an arbitrage.

Answer: Cash-and-carry (long spot, short perp, same-venue cross-margin) collects the funding, but the 30% is a floating risk premium, not free money: funding can compress or flip, the basis can move against an unwind, the venue/counterparty can fail, the short leg carries margin/liquidation risk in rallies, and frictions bite; persistent positive funding signals crowded leveraged longs.

P15.6 *Liquidation price and cascades*

(a) Per coin of position: initial margin $= P_0/L = 50{,}000/10 = 5{,}000$; equity at price P is $5{,}000 + (P - 50{,}000)$; the maintenance requirement is $0.005P$. Liquidation when

$$5{,}000 + P - 50{,}000 = 0.005P \quad\Longrightarrow\quad 0.995\,P = 45{,}000 \quad\Longrightarrow\quad P \approx \$45{,}226,$$

about 9.55% below entry. In general, with leverage L and maintenance fraction m,

$$P_{\text{liq}} = P_0\,\frac{1-1/L}{1-m}, \qquad \text{adverse move} = \frac{1/L - m}{1-m} \approx \frac{1}{L} - m.$$

Rule of thumb: a $1/L$ move against you wipes the margin; the maintenance buffer triggers liquidation slightly *earlier* (at a higher price for a long).

Self-check: plug back: at $P = 45{,}226$, equity per coin $= 5{,}000 - 4{,}774 = 226$ and requirement $= 0.005 \times 45{,}226 \approx 226$. And the crude $1/L$ rule gives 45,000, correctly just below the exact answer.

(b) The cascade. A liquidation is a *forced market sell* of the position. Price trading into the cluster triggers the first tranche; those sells push price down further, triggering the next tranche — mechanical positive feedback *in the direction of the move*, producing one-way flow and gaps. For a maker this flow is dangerous: it is relentless, all same-side, and marks against every fill in the short run. The standard response, in order: *widen or pull* quotes first (the flow is toxic in the mark-out sense even though it is forced rather than informed), keep inventory discipline (no knife-catching at size, ladder bids rather than one large level), then *re-provide liquidity at wide levels*: because the flow is forced and information-free about fundamentals, cascades overshoot and revert, and the liquidity premium available to whoever has risk capacity at the bottom is exactly the compensation for standing in front of it.

Answer: (a) $P_{\text{liq}} = 0.9 \times 50{,}000/0.995 \approx \$45{,}226$, a $\sim 9.5\%$ drop; in general an adverse move of about $1/L$ (less the maintenance buffer). (b) Forced sells trigger further liquidations — a cascade with the move; widen or pull first, then supply liquidity wide into the overshoot.

P15.7 *Constant-product AMM pricing and impact*

(a) $p = y/x = 200{,}000/100 = \$2{,}000$ per ETH, and $k = xy = 2 \times 10^7$.

(b) After the buy the pool holds $x' = 90$ ETH, so it must hold $y' = k/x' = 2 \times 10^7/90 = 222{,}222.22$ USDC. The trader pays

$$\Delta y = \frac{k}{x - \Delta x} - y = 222{,}222.22 - 200{,}000 = 22{,}222.22 \text{ USDC},$$

an average price of $22{,}222.22/10 = \$2{,}222.22$ per ETH. The new marginal price is $y'/x' = 222{,}222.22/90 = \$2{,}469.14$. Closed form worth knowing: buying a fraction f of reserves gives average price $p/(1-f)$ and marginal price $p/(1-f)^2$; here $f = 0.1$ gives 2,222.22 and 2,469.14 — impact is built into the curve and super-linear in size.

Self-checks: invariant: $90 \times 222{,}222.22 = 2.0 \times 10^7$. The average price must lie between the pre- and post-trade marginal prices: $2{,}000 < 2{,}222 < 2{,}469$; indeed it is exactly their geometric mean, $\sqrt{2{,}000 \times 2{,}469.14} = 2{,}222.2$.

(c) The pool now quotes 2,469 while the CEX trades at 2,000: arbitrageurs buy ETH on the CEX and sell it into the pool until the pool price returns to $\approx 2{,}000$, pocketing the difference. The pool has no view and no oracle — *external arbitrage is its price discovery.* That keeps AMM prices honest, but note who pays: the arbitrageurs' profit is extracted from the passive LPs, which is precisely the adverse-selection cost that surfaces as impermanent loss.

Answer: (a) \$2,000. (b) Pays 22,222.22 USDC, average \$2,222.22/ETH, new pool price \$2,469.14. (c) Arbitrageurs sell ETH into the pool until it re-prices to ≈\$2,000; arbitrage is the mechanism that ties the AMM to fair value, at the LPs' expense.

P15.8 *MEV sandwich attack*

Track the pool state (x, y) through the three trades, with $k = 2 \times 10^7$.

(a) The three legs.

- **Front-run** (attacker buys 10): pays $k/90 - 200{,}000 = 22{,}222.22$; pool $\to (90,\ 222{,}222.22)$.
- **Victim buys 10**: pays $k/80 - 222{,}222.22 = 250{,}000 - 222{,}222.22 = 27{,}777.78$; pool $\to (80,\ 250{,}000)$.
- **Back-run** (attacker sells 10): receives $250{,}000 - k/90 = 27{,}777.78$; pool $\to (90,\ 222{,}222.22)$.

Attacker profit $= 27{,}777.78 - 22{,}222.22 = \$5{,}555.56$ (about 25% of the front-run outlay).

(b) Flat, and whose money. Attacker ETH: $+10 - 10 = 0$. The back-run exactly retraces the victim's segment of the curve (from $x = 80$ back to $x = 90$), so the attacker *receives precisely what the victim paid.* Compare the victim's counterfactual: unsandwiched, they would have paid 22,222.22; sandwiched, they paid 27,777.78 — worse by exactly 5,555.56. The attack is a pure zero-sum transfer of the victim's extra slippage to the attacker.

Self-check by conservation: final pool $(90,\ 222{,}222.22)$ is *identical* to the unsandwiched final state, so the LPs are unaffected; net USDC into the pool $= 22{,}222.22 + 27{,}777.78 - 27{,}777.78 = 22{,}222.22$, matching $200{,}000 \to 222{,}222.22$; net ETH out = 10, the victim's coins. Every account balances.

(c) Limits and the lesson. The victim's *slippage tolerance* caps their worst execution price, hence caps the front-run size (a competent attacker sizes to push the victim exactly to tolerance); priority-fee auctions among competing searchers hand much of the profit to validators/builders; pool fees ($3 \times 0.3\%$ here would eat a large slice) and gas set a minimum viable victim size. Mitigations: tight slippage settings, private order flow (private RPCs, order-flow auctions), batch auctions. The implication: a public mempool *broadcasts your intention to adversaries who can order transactions around yours* — on public chains, execution is an adversarial game, and MEV is its price.

Answer: (a) Profit $= 27{,}777.78 - 22{,}222.22 = \$5{,}555.56$. (b) ETH-flat; the profit is exactly the victim's extra slippage (LPs end unaffected). (c) Bounded by the victim's slippage tolerance, gas/priority-fee auctions, and pool fees; a public mempool shows your order to adversaries before it executes.

P15.9 *Impermanent loss*

(a) Holdings track price. The pool constraints are $xy = k$ and marginal price $p = y/x$. Solving: $x = \sqrt{k/p}$, $y = \sqrt{kp}$. At deposit, price p_0: $x_0 = \sqrt{k/p_0}$, $y_0 = \sqrt{kp_0}$, and the equal-value condition holds automatically since $p_0 x_0 = \sqrt{kp_0} = y_0$. Initial value $V_0 = 2\sqrt{kp_0}$. After the move to $p_1 = rp_0$, arbitrageurs rebalance the pool to

$$x_1 = \sqrt{k/p_1} = \frac{x_0}{\sqrt{r}}, \qquad y_1 = \sqrt{kp_1} = y_0\sqrt{r},$$

so $V_{\text{LP}} = p_1 x_1 + y_1 = 2\sqrt{kp_1} = V_0\sqrt{r}$: *the LP's value grows like* $\sqrt{r}$*, not* r.

(b) Against the HODL benchmark. Holding the deposited coins instead: $V_{\text{HODL}} = p_1 x_0 + y_0 = y_0(r+1) = V_0\frac{1+r}{2}$. Hence

$$\frac{V_{\text{LP}}}{V_{\text{HODL}}} = \frac{2\sqrt{r}}{1+r}, \qquad \text{IL}(r) = \frac{2\sqrt{r}}{1+r} - 1 \;\le\; 0$$

by AM–GM ($\sqrt{r} \le \frac{1+r}{2}$, equality iff $r = 1$). Values: $r = 2$: $2\sqrt{2}/3 - 1 = 0.9428 - 1 = -5.72\%$; $r = 4$: $4/5 - 1 = -20\%$.

(c) Symmetry and the small-move law. Replace $r \to 1/r$: $\frac{2/\sqrt{r}}{1+1/r} = \frac{2\sqrt{r}}{r+1}$ — identical, so $\text{IL}(r) = \text{IL}(1/r)$: a halving hurts exactly as much as a doubling (in relative terms). For $r = 1+\varepsilon$: $\sqrt{r} \approx 1 + \frac{\varepsilon}{2} - \frac{\varepsilon^2}{8}$, so

$$\frac{2\sqrt{r}}{1+r} \approx \frac{2+\varepsilon-\varepsilon^2/4}{2+\varepsilon} \approx 1 - \frac{\varepsilon^2}{8} \qquad \Longrightarrow \qquad \text{IL} \approx -\frac{\varepsilon^2}{8}.$$

A loss *quadratic in the move*, whichever direction, is the signature of a **short gamma** position: the pool mechanically sells the appreciating asset on the way up and buys the depreciating one on the way down. The LP has sold an option; trading fees are the premium collected. Rational iff expected fees (plus incentives) exceed the expected give-away — "LP profits iff fees beat the short-gamma cost." (And "impermanent" is a misnomer: withdraw at $r \ne 1$ and the loss is permanent.)

Self-checks: $\text{IL}(1) = 2/2 - 1 = 0$; the symmetry proved in (c) is itself a consistency check; and testing the Taylor law at $r = 1.1$: exact $2\sqrt{1.1}/2.1 - 1 = -0.113\%$ vs $-\varepsilon^2/8 = -0.125\%$ (right order, slight overestimate as expected for a truncation).

Answer: $V_{\text{LP}} = V_0\sqrt{r}$, $V_{\text{LP}}/V_{\text{HODL}} = 2\sqrt{r}/(1+r)$; $\text{IL}(2) = -5.72\%$, $\text{IL}(4) = -20\%$; $\text{IL}(r) = \text{IL}(1/r)$ and $\text{IL} \approx -\varepsilon^2/8$ for small moves — the LP is short gamma and should provide liquidity only when expected fees exceed the option value given away. □

P15.10 *Impermanent-loss benchmark*

The error: wrong benchmark. "Up 41%" compares against the deposit, but the deposit was half ETH — and ETH doubled. The relevant counterfactual is *holding the deposited assets*:

$$V_{\text{HODL}} = \underbrace{5{,}000}_{\text{USDC}} + \underbrace{5{,}000 \times 2}_{\text{2.5 ETH at \$4,000}} = \$15{,}000.$$

The LP position at \$14,142 lags by \$858, i.e. $858/15{,}000 = 5.72\%$ — exactly $\text{IL}(2) = 1 - 2\sqrt{2}/3$. LPing made money in absolute terms while *costing* \$858 relative to doing nothing.

Verify the pool arithmetic from scratch: deposit $x_0 = 2.5$ ETH, $y_0 = 5{,}000$ USDC, $k = 12{,}500$. Price doubles ($r = 2$), arbitrage rebalances to $x_1 = 2.5/\sqrt{2} = 1.7678$ ETH, $y_1 = 5{,}000\sqrt{2} = 7{,}071.07$ USDC; value $= 1.7678 \times 4{,}000 + 7{,}071.07 = 7{,}071.07 + 7{,}071.07 = \$14{,}142$ — matching both the quoted figure and the closed form $V_{\mathrm{LP}} = V_0\sqrt{r} = 10{,}000\sqrt{2}$. Note the pool now holds *fewer* ETH: it sold ETH all the way up — the short-gamma mechanics in action.

What "winning" requires. The LP genuinely beat holding iff accumulated trading fees (plus any liquidity incentives) over the period exceeded \$858. That is the entire LP business in one line: fees are the premium on the option the pool sold for you.

Two closing precisions: (i) "compared to what" matters — versus holding \$10,000 of pure USDC the LP did great; versus pure ETH (\$20,000) it did terribly; the honest benchmark is the assets actually deposited. (ii) "Impermanent" is a misnomer: withdraw now and the \$858 is permanent.

Self-check: $\mathrm{IL}(2) \times V_{\mathrm{HODL}} = 0.0572 \times 15{,}000 = 858$ — formula and first-principles pool computation agree.

Answer: The benchmark is wrong: holding the deposited assets would be worth \$15,000, so LPing cost \$858 = 5.72% (the $r = 2$ impermanent loss); the LP position genuinely won only if fees earned over the period exceeded \$858.

21.15 Solutions — §16 Options and Volatility

P16.1 *Black-76 digital, prediction-market link*

(a) The formula. Under Black-76 with $r \approx 0$ the digital is worth $e^{-rT}N(d_2) \approx N(d_2)$, "the risk-neutral probability, discounted," with

$$d_2 = \frac{\ln(F/K) - \sigma^2 T/2}{\sigma\sqrt{T}}.$$

Pieces: $T = 30/365 = 0.0822$, $\sqrt{T} = 0.2867$, so $\sigma\sqrt{T} = 0.6 \times 0.2867 = 0.172$ (a 17.2% one-month vol); $\ln(F/K) = -\ln(7/6) \approx -0.154$; $\sigma^2 T/2 = 0.36 \times 0.0822/2 = 0.0148$. Then

$$d_2 = \frac{-0.154 - 0.015}{0.172} \approx -0.98, \qquad N(d_2) = 1 - \Phi(0.98) \approx 1 - 0.837 = 0.163.$$

The digital is worth about *16 cents*: the options market prices roughly a 16% risk-neutral probability of settling above 70k.

Self-check: the required log-move is 0.154 against a monthly vol of 0.172 — about 0.9σ; adding the $-\sigma^2 T/2$ drift of the risk-neutral log-price pushes it to 0.98σ, and a $\sim 1\sigma$ upper tail should be in the mid-teens. Bounds: price in $(0, 1)$ and well below the 50 cents an at-the-money digital would command.

(b) The gap. The prediction market implies 25%; the options market implies 16%. If you trust the options surface, the binary is rich: sell it and hedge with a call spread around 70k (Breeden–Litzenberger: differentiating the strike curve prices a digital at *any* strike — the options-venue-to-prediction-market pipeline). Caveats before firing:

- *Risk-neutral vs real-world:* $N(d_2)$ is a $\mathbb{Q}$-probability; with $r \approx 0$ and one month of horizon the drift adjustment is small, but crypto risk premia are not automatically negligible — some gap can be legitimate.
- *Skew:* using a single implied vol ignores $\partial\sigma/\partial K$; the exact digital is the slope of the *smile-consistent* call curve, which adds a vega $\times$ skew correction to $N(d_2)$. Differentiate the actual strike curve rather than plugging one vol.
- *Vol clock:* strong intraday/weekly vol seasonality in crypto deforms short-dated pricing; calendar T is an approximation.
- *Plumbing:* settlement definitions (which index, what time), fees, capital lockup and counterparty/venue risk on both legs can eat a 9-cent edge.

Answer: (a) $d_2 \approx -0.98$, digital $\approx N(-0.98) \approx 0.163$, i.e. about 16 cents. (b) The prediction market's 25 cents implies a higher probability than options' 16%; check risk-neutral-vs-real-world drift, the skew correction to $N(d_2)$, crypto vol seasonality, and settlement/fee/venue frictions before trading the gap.

P16.2 *Breeden-Litzenberger*

(a) Differentiate under the integral. Write $C(K) = e^{-rT} \int_K^\infty (S - K)\, f_{\mathbb{Q}}(S)\, dS$. By Leibniz, the boundary term vanishes (the integrand is $(K - K)f(K) = 0$ at $S = K$) and the integrand's K-derivative is -1:

$$\frac{\partial C}{\partial K} = -e^{-rT} \int_K^\infty f_{\mathbb{Q}}(S)\, dS = -e^{-rT}\, \mathbb{Q}(S_T > K).$$

So the digital is the negative slope of the call curve: $\mathrm{Dig}(K) = e^{-rT}\, \mathbb{Q}(S_T > K) = -\partial C/\partial K$. Differentiating once more,

$$\frac{\partial^2 C}{\partial K^2} = e^{-rT} f_{\mathbb{Q}}(K) \qquad \Longrightarrow \qquad f_{\mathbb{Q}}(K) = e^{rT}\, \frac{\partial^2 C}{\partial K^2}.$$

(b) Finite differences with $h = 5$, $r = 0$:

$$\mathrm{Dig}(100) \approx -\frac{C(105) - C(95)}{2h} = -\frac{2 - 7}{10} = 0.50,$$

$$f_{\mathbb{Q}}(100) \approx \frac{C(95) - 2C(100) + C(105)}{h^2} = \frac{7 - 8 + 2}{25} = 0.04$$

per unit of price. The two one-sided slope estimates are $(7 - 4)/5 = 0.60$ and $(4 - 2)/5 = 0.40$; they bracket the centered estimate, which is second-order accurate — quote 0.50 and know the bracket.

Self-check by reconstruction: fit the unique quadratic through the three points: $C(K) = 4 - 0.5\,(K - 100) + 0.02\,(K - 100)^2$. Check: $C(95) = 4 + 2.5 + 0.5 = 7$, $C(105) = 4 - 2.5 + 0.5 = 2$. Then $-C'(100) = 0.50$ and $C''(100) = 0.04$ — differentiation recovers exactly the finite-difference answers. Sanity: digital $\in [0, 1]$ and C decreasing in K $(7 > 4 > 2)$.

(c) No-arbitrage and uses. The numerator $C(95) - 2C(100) + C(105) = 1 > 0$ is the price of the 95/100/105 *butterfly*; butterflies have non-negative payoffs, so no-arbitrage forces their price ≥ 0 — which is precisely convexity of C in K, hence a non-negative density. (Similarly the call-spread bounds $0 \leq -\Delta C/\Delta K \leq e^{-rT}$ keep the digital in $[0, 1]$: here 0.6 and 0.4 both qualify.) In practice: differentiate a liquid strike curve (e.g. a crypto options venue) to extract the full risk-neutral density and price *any* European payoff at *any* strike — digitals matching a prediction-market contract (the options-venue-to-Polymarket pipeline), tail probabilities, moments — and to run no-arbitrage checks on a vol surface.

Answer: Digital $= -\partial C/\partial K = e^{-rT}\mathbb{Q}(S_T > K)$; $f_{\mathbb{Q}}(K) = e^{rT}\partial^2 C/\partial K^2$. Numerically $\mathrm{Dig}(100) \approx 0.50$ and $f_{\mathbb{Q}}(100) \approx 0.04$; non-negativity is the butterfly no-arbitrage (convexity) condition; used to extract risk-neutral densities from strike curves and price digitals at arbitrary strikes. □

P16.3 *Sqrt-t vol scaling*

(a) Scaling down. With 252 trading days, $\sqrt{252} \approx 15.87 \approx 16$:

$$\sigma_{\text{daily}} = \frac{20\%}{\sqrt{252}} \approx 1.26\%, \qquad \sigma_{\text{weekly}} = \frac{20\%}{\sqrt{252/5}} = \frac{20\%}{\sqrt{50.4}} \approx 2.82\%.$$

Hence the trader's anchor: "16% annual ≈ 1% a day."

(b) The trap. *Variance* is additive across uncorrelated increments, not volatility: $\mathrm{Var}(\sum R_i) = \sum \mathrm{Var}(R_i)$, so variance scales with t and vol with $\sqrt{t}$. Annualising multiplies daily vol by $\sqrt{252} \approx 15.9$, giving back 20%; multiplying by 252 scales the *standard deviation* by the factor that belongs to the *variance*, effectively double-counting time — 318% would be the answer only if every day moved in the same direction with certainty.

(c) Assumption and violation. The rule needs (at least) uncorrelated increments with stable variance. Violations: serial correlation — momentum (positive autocorrelation) makes long-horizon vol *larger* than the $\sqrt{t}$ extrapolation, mean reversion makes it smaller (the variance-ratio test measures exactly this); and volatility clustering means any point-in-time daily estimate scales poorly to long horizons.

Self-check: invert: $1.26\% \times 15.87 = 20.0\%$; and the weekly figure via the daily one: $1.26\% \times \sqrt{5} = 1.26 \times 2.236 = 2.82\%$ — both routes agree.

Answer: (a) Daily $\approx 1.26\%$, weekly $\approx 2.82\%$. (b) Variance is additive, not vol: annualise vol with $\times\sqrt{252} \approx 15.9$, not $\times 252$. (c) Assumes uncorrelated, stable-variance increments; autocorrelation (momentum or mean reversion) and vol clustering break it.

P16.4 *Variance risk premium, delta-hedged P&L*

(a) Gamma rent vs theta rent. Long option, delta-hedged, rates ≈ 0. Over one day with return $R = \Delta S/S$, a second-order expansion of the hedged position's value gives a gamma gain $\frac{1}{2}\Gamma S^2 R^2$ and a theta decay $\Theta\,\Delta t$. The Black–Scholes equation for a delta-hedged book at $r \approx 0$ says $\Theta = -\frac{1}{2}\Gamma S^2 \sigma_i^2$ (theta is exactly the rent on the gamma, priced at implied vol σ_i). So

$$\text{daily P\&L} \;\approx\; \tfrac{1}{2}\,\Gamma S^2\left(R^2 - \sigma_i^2\,\Delta t\right), \qquad \text{total} \;\approx\; \sum_t \tfrac{1}{2}\,\Gamma_t S_t^2\left(R_t^2 - \sigma_i^2 \Delta t\right):$$

a *gamma-weighted* (realised^2 − implied^2) — the one-line meaning of "trading vol." With realised 20% against implied 25%, the daily R^2 falls short of the theta bill on average: **you lose**. (Caveat worth saying: the ΓS^2 weights make it path-dependent — vol realised while spot is near the strike counts most, so a 20-vol month realised mostly far from the strike loses even more.)

(b) Size of the loss. The straddle premium $\approx 0.8\,S\,\sigma_i\sqrt{T}$ is *linear in σ*. Had the market priced 20 instead of 25, the straddle would cost $0.8\,S\,\sigma_r\sqrt{T}$; to first order the hedged long realises that difference:

$$\text{loss} \approx 0.8\,S\sqrt{T}\,(\sigma_i - \sigma_r) = \text{premium} \times \left(1 - \frac{20}{25}\right) = 20\% \text{ of premium.}$$

Concretely with $S = 100$: premium $\approx 0.8 \times 100 \times 0.25 \times 0.289 \approx 5.8$; loss ≈ 1.15, about 1.2% of spot.

Self-check via vega: straddle vega $= 2S\varphi(d_1)\sqrt{T} \approx 2 \times 100 \times 0.399 \times 0.289 \approx 23$ per unit of vol $= 0.23$ per vol point; times 5 points $= 1.15$ — matches the premium-ratio route. Consistency of (a): if $\sigma_r = \sigma_i$ the expression sums to ≈ 0, recovering the Black–Scholes replication result.

(c) The variance risk premium. Implied vol systematically exceeds subsequently realised vol (equity indices: a few vol points on average), so delta-hedged long-option positions lose on average — that gap is the *variance risk premium.* It persists because options are insurance: hedgers pay up for protection; the short side's losses arrive precisely in high-vol crisis states (when marginal utility is highest), jump risk cannot be hedged away by discrete rebalancing, and limits to arbitrage (margin calls, gamma losses when vol spikes) cap how hard sellers can lean in. The trader on the other side — short vol — earns a steady stream of small carry profits punctuated by rare, violent drawdowns: "picking up nickels in front of a steamroller," or, sized honestly, being paid an insurance premium to bear catastrophe states.

Answer: (a) You lose: daily P&L $\approx \frac{1}{2}\Gamma S^2(R^2 - \sigma_i^2\Delta t)$, i.e. gamma-weighted realised-minus-implied variance, negative when realised $20 <$ implied 25. (b) Roughly $(1 - 20/25) = 20\%$ of the premium ($\approx 1.2\%$ of spot). (c) The variance risk premium: implied exceeds realised on average because sellers are paid to insure crash states; the short-vol counterparty earns steady carry with rare large losses. □

21.16 Solutions — §17 Algorithmic Cost Models

P17.1 *Matmul operation count* ★

Each of the mp entries of the product is a dot product of length n: n multiplications and $n-1$ additions. Totals: mnp multiplications and $mp(n-1)$ additions — "mnp multiply-adds" is the interview-grade summary. Square case: n^3 multiplications and $n^3 - n^2$ additions, so $O(n^3)$. Convention: quote the *multiplication* count — multiplications historically cost more CPU cycles than additions, and the two counts agree to leading order anyway.

Below cubic: Strassen's algorithm recursively multiplies 2×2 blocks with 7 multiplications instead of 8, giving $O(n^{\log_2 7}) \approx O(n^{2.81})$ — practical only for large n (bigger constants, more memory traffic, weaker numerical stability). The theoretical frontier is $\approx O(n^{2.37})$, but those algorithms are "galactic" — constants so large they never win at feasible sizes. Practical BLAS remains cubic: the real-world wins are cache blocking, SIMD, and tensor cores, not asymptotics.

Self-check: smallest case $m = n = p = 2$: formula says $2\cdot2\cdot2 = 8$ multiplications and $mp(n-1) = 4$ additions; writing out a 2×2 product confirms four length-2 dot products = 8 mults, 4 adds — and Strassen's whole point is doing that block product in 7, seeding the recursion.

Answer: $(m \times n)(n \times p)$ costs mnp multiplications (plus $mp(n-1)$ additions); square is $O(n^3)$; multiplications are the conventional count (costlier cycles); Strassen $O(n^{2.81})$ is real but large-n only, $O(n^{2.37})$ is galactic, and practical speed comes from cache/SIMD/tensor cores.

P17.2 *Why count multiplications* ⧫

Because multiplications generally cost more CPU cycles than additions — historically by a wide margin — so the multiply count dominated the running-time estimate and the convention stuck. Two supporting observations: (i) in an $(m \times n)(n \times p)$ product the two counts are nearly equal anyway (mnp vs $mp(n-1)$, identical to leading order), so tracking one number loses nothing asymptotically; (ii) on modern hardware a fused multiply-add executes both in one instruction, which is why "FLOPs" today is often quoted as $2mnp$ — but the reason the *multiplication* became the unit is its historically higher cycle cost.

Self-check: the difference between the two counts is $mnp - mp(n-1) = mp$, a lower-order term — confirming that choosing which to count cannot change the complexity class, only the convention.

Answer: Multiplications cost more CPU cycles than additions, so their count dominates the time estimate (and the two counts agree to leading order anyway).

P17.3 *Low-rank decomposition crossover* ★

(a) NPR multiplications — each of the NR output entries is a length-P dot product.

(b) Association order is the hidden half of the question.

- $(UV)B$: forming UV costs NKP (an $N \times K$ by $K \times P$ product), producing the dense $N \times P$ matrix, then NPR for the product with B. Total $NKP + NPR$ — *strictly worse than the direct NPR for any $K \geq 1$*: re-multiplying the factors together first recreates the dense cost and adds the assembly bill. Never do this.
- $U(VB)$: VB costs KPR (result is $K \times R$ — small), then $U(VB)$ costs NKR. Total $KPR + NKR = KR(N + P)$.

Factored beats dense iff

$$KR(N+P) < NPR \iff K(N+P) < NP \iff \boxed{K < \frac{NP}{N+P}}.$$

Note R cancels: the threshold does not depend on how much data you push through. $NP/(N+P)$ is half the harmonic mean of N and P; for square $N = P$ it reads $K < N/2$. This is the LoRA economics — rank-K adapters with $K \ll \min(N, P)$ make the factored path far cheaper, provided you always multiply the data into V first.

Self-check: boundary cases. $N = P = 2$, $K = 1$, $R = 1$: dense $= 4$; factored $= 1 \cdot 1 \cdot (2 + 2) = 4$ — exact break-even, and indeed the threshold $NP/(N + P) = 1$ sits right at $K = 1$. Full rank $K = P$ (square): $KR(N + P) = PR(N + P) > NPR$ always — a factorisation with no rank reduction can never win. Units check: all three costs scale linearly in R, so the comparison is per-column of data, as it should be.

Answer: (a) NPR multiplications; (b) $(UV)B$ costs $NKP + NPR$ (always worse — never reassemble the dense matrix), $U(VB)$ costs $KR(N+P)$; the factored form wins iff $K < NP/(N+P)$ (half the harmonic mean; $K < N/2$ when square). ☐

P17.4 *Low-rank speedup, numeric*

(a) Dense matrix–vector product: $NP = 4096 \times 4096 = 16{,}777{,}216$ multiplications. Factored, with the correct association $U(Vx)$: $K(N + P) = 64 \times (4096 + 4096) = 64 \times 8192 = 524{,}288$. Ratio: $16{,}777{,}216/524{,}288 = 32$ — the claim is exactly right.

(b) Break-even at $K = \frac{NP}{N+P} = \frac{4096^2}{8192} = 2048$ (half of 4096, as always for square layers); any $K < 2048$ is a win in raw multiplications.

(c) Approximation error: the rank-64 layer equals the dense layer only if the dense weights truly had rank ≤ 64. In general it is a lossy compression, and the accuracy given up is the real price — the flop count is silent about it. (Second-order practicalities: two kernel launches and different memory-access patterns can eat part of the theoretical 32×.)

Self-check: compute the speedup a second way: ratio $= \frac{NP}{K(N+P)} = \frac{NP/(N+P)}{K} = \frac{2048}{64} = 32$ — consistent with (b): the speedup is (break-even rank)/(actual rank).

Answer: (a) verified — exactly 32× fewer multiplications (16.78M vs 524K); (b) break-even rank 2048; (c) the low-rank approximation error (fidelity), which flop counting ignores.

P17.5 *Membership: scan vs sort vs hash* ★

Costs:

- **Scan:** $O(N)$ per query, $O(qN)$ total. No preprocessing, no extra memory.
- **Sort + binary search:** $O(N \log N)$ once, then $O(\log N)$ per query: $O((N + q) \log N)$ total.
- **Hash set:** $O(N)$ expected build, $O(1)$ expected per query: $O(N + q)$ *expected* total.

Crossover (scan vs sort): sorting pays off when

$$qN > N \log N + q \log N \iff q(N - \log N) > N \log N \iff q > \frac{N \log N}{N - \log N} \approx \log N \quad (N \text{ large}).$$

So after roughly $\log N$ queries, pre-sorting wins — a strikingly small number.

When each wins. *Scan:* one or two queries, tiny N, or data you stream past exactly once — zero setup and perfect cache behaviour make it hard to beat at small scale. *Sort:* many queries when you also want *order* — range queries, predecessor/nearest lookups, sorted output — or deterministic worst-case guarantees. *Hash:* many pure membership queries; asymptotically the best at $O(N + q)$ expected, and the phrase to say is “expected $O(1)$ lookups, amortised build”

— paying for it with extra memory, loss of ordering information, and worst-case degradation (collisions/adversarial keys).

Self-check: plug $N = 1024$: crossover $q > \frac{1024\times 10}{1024-10} \approx 10.1$, and $\log_2 N = 10$ — the approximation is tight. Endpoints: $q = 1$: scan N beats sort $N \log N$; $q \to \infty$ with N fixed: hash $O(q)$ beats binary search $O(q \log N)$ beats scan $O(qN)$. House pattern: write both cost expressions, set them equal, solve for the variable asked about.

Answer: scan $O(qN)$; sort-then-search $O((N+q)\log N)$, paying off once $q \gtrsim \log N$; hash set $O(N+q)$ expected — best for many queries, at the price of memory, no ordering, and worst-case degradation.

P17.6 *CS fundamentals rapid-fire* ⧫

(a) *Memoization* — caching a pure function's outputs keyed by input. "Dynamic programming" is the bait: memoization is an ingredient of DP (DP = recursion over overlapping subproblems + memoization/tabulation), but caching by itself, with no subproblem structure, is just memoization. *Self-check:* nothing in the setup mentions subproblems or recurrences — only repeat calls.

(b) It roughly *doubles*: one comparison per record is $O(N)$ per query, so $2N$ records cost $2\times$ the time. *Self-check:* linear cost model — constant work per record, no early exit assumed.

(c) Trace it: after Push(10), Push(20) the stack is $[10, 20]$; val = Pop() = 20, stack $[10]$; Push(40), Push(30) gives $[10, 40, 30]$; Push(val) pushes 20 back: $[10, 40, 30, 20]$; Pop() returns 20; Pop() returns 30. Last popped: **30**. *Self-check:* conservation count — 5 pushes (10, 20, 40, 30, val) and 3 pops in total, so $5 - 3 = 2$ elements must remain, and indeed the final stack is $[10, 40]$.

(d) $O(N)$ space: each active call keeps a frame (locals, return address) on the call stack, and without tail-call optimisation frames cannot be reused even for a tail-position call. (CPython never performs TCO — deep recursion hits the recursion limit around depth 10^3 by default.) *Self-check:* at maximum depth exactly N frames are simultaneously live — linear, not constant.

(e) Left bracket, by distributivity of $\vee$ over $\wedge$: $(A \vee B) \wedge (A \vee \neg B) = A \vee (B \wedge \neg B) = A \vee \text{false} = A$. Right bracket: $(C \wedge D) \wedge (C \wedge \neg D) = C \wedge (D \wedge \neg D) = C \wedge \text{false} = \text{false}$. Whole expression: $A \vee \text{false} = A$. *Self-check:* truth-table spots — A false, B true: left = (T) $\wedge$ (F) = F = A; A false, B false: (F) $\wedge$ (T) = F; A true: T $\wedge$ T = T; the right bracket needs D and $\neg D$ simultaneously — always false.

Answer: (a) memoization; (b) it roughly doubles ($O(N)$); (c) 30; (d) $O(N)$ stack space; (e) A.

21.17 Solutions — §18 Forecasting Returns: the Open-Ended Question

P18.1 *Open-ended process question*

Answer with a *process*, not a model name. A model answer hits five stages, in order:

- **Define the target.** Log vs simple returns; the horizon (next-day here) and why it matters — signal decay, tradability, and noise level all change with horizon. Also pin down what the forecast is scored on: point forecast, direction, or distribution.
- **Explore, then build features in layers.** EDA first: stationarity, outliers, regime breaks, the correlation matrix, PCA for factor structure. Feature layers: endogenous (lagged returns, realised vol, volume, spreads); exogenous lead-lag (the US close leads the European open; overnight futures, rates, FX, commodities); composition-specific (sector weights make particular macro series relevant — e.g. a heavy energy weight makes Brent a feature); sentiment/news if the horizon supports it.
- **Model: baseline first, then a fork.** Start with a trivial baseline (zero / historical mean) and a Ridge regression on standardised features; escalate to LightGBM or sequence models only if the linear information coefficient justifies the added variance. Name the

direct-vs-factorial fork explicitly: model the index series directly, or model constituents and aggregate (often cleaner — see P18.3).

- **Validate honestly.** Walk-forward / time-series splits, never shuffled; metrics that respect the use case: IC, hit rate, and the Sharpe of the implied strategy *after costs* — raw MSE is direction-blind.
- **Name the risks unprompted.** Look-ahead bias, overfitting ("a too-beautiful Sharpe is a bug"), transaction costs, non-stationarity/regime change. Volunteering these is what separates a process answer from a model-name answer.

Self-check (rubric): re-read your narrated answer and test three gates — (i) was the target defined before any model was named? (ii) is there a baseline and an out-of-sample protocol? (iii) did costs and look-ahead appear without prompting? Independently, sanity-check any performance you would claim: daily direction on a broad index sits near 50–55% for good pipelines, so a backtest printing far above that should be treated as a leakage bug, not a triumph.

Answer: A five-stage process — define the target; layered EDA/features (endogenous, lead-lag, composition, sentiment); baseline-then-model with the direct-vs-factorial fork stated; walk-forward validation on IC/hit-rate/net Sharpe; and unprompted naming of look-ahead, overfitting, costs, and non-stationarity.

P18.2 *Backtest critique* ♦

Two answers deserve the label "main issue", and a complete response gives both. Read literally, "the main issue in *executing*" points at extbftransaction costs; the deeper *validity* critique is that the backtest's profit is *beta, not timing alpha.*

extbfValidity: the profit is beta. The strategy is long ETH ten hours out of every twenty-four, over a seven-year sample in which ETH rose by an order of magnitude. Any always-long-by-day rule mechanically inherits a large share of that drift; the specific 9am–7pm window need carry no information at all. Diagnosis tests: (i) benchmark against buy-and-hold scaled to the same average exposure; (ii) randomise the entry/exit clock times — if random ten-hour windows backtest about as well, the timing content is zero; (iii) sub-period analysis: does it survive the drawdown years, or is the P&L concentrated in bull regimes?

extbfExecution: the cost drag is enormous. About 365 round trips per year is roughly 730 fee-paying sides; at 5–10 bp per side in taker fee plus spread plus slippage, that is on the order of 50% of notional per year in friction — enough to destroy a modest edge on its own. Also worth a sentence: a fixed clock-time execution is crowd-visible and easy to front-run.

Self-check (quantitative): ETH went from a few hundred dollars in 2019 to thousands in 2026, i.e. a 10×-order move; a rule long roughly 10/24 of each day rides a substantial fraction of that log drift with no skill, so "backtests very well" is exactly what pure beta predicts. Fee drag arithmetic: $730 \times 7.5\,\text{bp} \approx 5{,}500\,\text{bp} \approx 55\%$ per year — consistent with the cost concern being material, not cosmetic. Both checks point the same way: the burden of proof is on the timing, and the backtest as described has not met it.

Answer: The performance is long-ETH beta harvested over a seven-year bull sample, not timing alpha; the secondary issue is fee/slippage drag from ~365 round trips a year. Test against buy-and-hold and randomised windows before believing the timing.

P18.3 *Open-ended design choice: direct vs factorial*

Case for the factorial route.

- *Cross-section multiplies data:* 40 related series instead of 1 gives far more observations of the shared structure per unit of calendar time.
- *Decomposition isolates the forecastable part:* write $R_i = \beta_i R_{\text{mkt}} + \alpha_i$. The market component is the hardest thing in finance to predict; the beta-neutralised residuals carry cleaner, slower

structure (lead-lag between related names, residual mean-reversion) and are where most systematic edge lives.

- *Information enters at the right level:* Brent belongs in the TotalEnergies model, China sentiment in the luxury names — at stock level these are sharp features; averaged into an index feature they dilute.

Where PCA enters. Eigendecompose the constituent correlation matrix: PC1 $\approx$ the market mode, the next few components $\approx$ sector modes. Uses: (a) build the market factor for residualisation; (b) compress 40 correlated series into a few orthogonal drivers to forecast; (c) check the factor structure is *stable* across subperiods before trusting anything built on it.

Failure modes of the aggregation.

- *Cap-weighted aggregation cancels relative alpha.* The index is dominated by PC1; forecast skill on the α_i largely nets out in the weighted sum. You can build an excellent long-short stock model and a nearly useless index model from the same forecasts.
- *Correlated errors do not diversify.* The variance of the weighted error sum carries the full covariance matrix; if the common-factor forecast is wrong, all 40 stock errors point the same way and aggregation buys you nothing.
- *Plumbing risk:* weight and composition drift, corporate actions, and asynchronous closes create silent look-ahead in the aggregation step.

Self-check (toy case): two stocks, equal weights, $R_i = R_m + \alpha_i$ with $\alpha_2 = -\alpha_1$. The index return is exactly R_m: a *perfect* alpha model contributes zero index forecasting power — confirming the cancellation failure concretely. Conversely it shows where the factorial route genuinely helps an index mandate: a better-estimated market mode and risk model, not the relative alphas.

Answer: The constituent route wins on data volume and on forecasting beta-neutralised structure, with PCA supplying the market/sector factors; but cap-weighted aggregation cancels relative alpha and correlated forecast errors do not diversify — for a pure index target the market-factor forecast still does the heavy lifting.

21.18 Solutions — Puzzle Classics

PZ.1 *Monty Hall*

Switch. Your initial pick holds the car with probability $\frac{1}{3}$, a goat with probability $\frac{2}{3}$ — and nothing the host does changes the probability that your *initial* pick was right, because he can *always* open a goat door regardless of where the car is. Switching wins exactly when the initial pick was wrong:

$$P(\text{win} \mid \text{switch}) = P(\text{initial pick wrong}) = \tfrac{2}{3}.$$

The trick: the host's reveal is *informative about the other door, not about yours.* He never opens the car door, so his choice funnels the remaining $\frac{2}{3}$ of probability onto the one door he pointedly did not open. An equivalent bookkeeping: over the three equally likely car positions, list what switching does — it loses in the one world where you started on the car and wins in the other two.

The protocol matters (the second question). If the host opens one of the two other doors *uniformly at random* and it happens to show a goat, the answer changes: now the event "a goat was revealed" is evidence about your own door too. Bayes: the random host shows a goat with probability 1 when your pick is the car and probability $\frac{1}{2}$ otherwise, so the posterior odds on your pick being right are $\frac{1}{3} \cdot 1 : \frac{2}{3} \cdot \frac{1}{2} = 1{:}1$ — switching is then worth exactly $\frac{1}{2}$, no better than staying. Saying this aloud — *the answer depends on the host's protocol, not just on what you saw* — is what separates a memorized answer from an understood one.

Answer: Switch: $P(\text{win}) = \frac{2}{3}$ under the standard always-reveals-a-goat host; with a random host who happened to show a goat it drops to $\frac{1}{2}$. □

PZ.2 *Two ropes, 45 minutes*

Light rope A at *both* ends and rope B at *one* end, simultaneously. Rope A's two flames meet after exactly 30 minutes (wherever the meeting point is), at which moment rope B has 30 minutes of burn time left. Now light rope B's *other* end: its remaining burn time halves, so it finishes 15 minutes later. Total: $30 + 15 = 45$ minutes.

The trick: lighting both ends halves the remaining *burn time*, never the length — non-uniformity is irrelevant because the two flames jointly consume rope at twice the one-flame rate, wherever the fast and slow stretches lie. One rope therefore gives you 30 or 60; a second rope, re-lit at the right moment, lets you halve any *remaining* time. (The same idea generates 15, 22.5, ... — each extra both-ends ignition is another halving.)

Answer: Light A both ends and B one end; when A dies (30 min), light B's other end; B dies at 45 min.

PZ.3 *Clock-hands angle* ♦

The hour hand moves 30° per hour plus 0.5° per minute: at 09:30 it sits at $9 \times 30 + 30 \times 0.5 = 270 + 15 = 285°$. The minute hand at 30 minutes sits at 180°. Difference: $285 - 180 = 105°$ (already $\leq 180°$, so no reflex correction needed). The classic error is forgetting that the hour hand has moved half way from 9 towards 10, which gives the wrong answer 90°.

Self-check: one-line formula $|30H - 5.5M| = |30 \times 9 - 5.5 \times 30| = |270 - 165| = 105$. Sanity: at 9:00 the (minor) angle is 90° and over the next 30 minutes the minute hand gains 5.5°/min on the hour hand, sweeping past alignment territory — 105° on the other side is plausible in size.

Answer: 105°.

PZ.4 *100 lockers*

Locker n is toggled once by every divisor of n, so it ends *open* iff n has an *odd* number of divisors.

The trick: divisors come in pairs $d \leftrightarrow n/d$, which makes the divisor count even — *except* when some divisor pairs with itself, $d = n/d$, i.e. $n = d^2$. So the divisor count is odd exactly for perfect squares.

The perfect squares up to 100 are $1, 4, 9, 16, 25, 36, 49, 64, 81, 100$ — ten of them.

Small-case check: locker 12 has divisors $1, 2, 3, 4, 6, 12$ (six toggles, ends closed); locker 9 has $1, 3, 9$ (three toggles, ends open) — the pairing $1{\cdot}12, 2{\cdot}6, 3{\cdot}4$ versus $1{\cdot}9, 3{\cdot}3$ shows the mechanism directly.

Answer: 10 lockers open — exactly the perfect squares $1, 4, 9, \ldots, 100$. □

PZ.5 *25 horses, 5 lanes*

Seven races suffice. Run five heats of five (races 1–5), then race the five heat winners (race 6). Label the heats A–E by their winner's finish in race 6, so $A_1 > B_1 > C_1 > D_1 > E_1$ (subscripts = position within the heat).

Race 6's winner A_1 is the fastest horse overall — it beat its own heat and the other heats' best. Now prune who can still be second or third: a horse is eliminated if three horses are already provably faster. D_1, E_1 and their heats are out (three winners beat them); C_2 and below are out (behind C_1, who is behind B_1 and A_1); B_3 and below are out (behind B_2, B_1, A_1); A_4, A_5 are out. Exactly five candidates remain: A_2, A_3, B_1, B_2, C_1. Race them (race 7); the first two across the line are the overall second and third.

Why six races cannot work: after the five heats, ranking information exists only *within* heats; an adversary argument closes it. Every horse must race at least once (an unraced horse could be placed anywhere), so with 25 horses and 5 lanes the first five races are forced to be a perfect partition into disjoint heats. The sixth race must then contain every heat winner: a missing winner has never been compared outside its heat, so it could consistently be the overall fastest or merely fourth — indeterminate either way. That fixes race 6 as exactly the five winners, and after it the second place overall is still unresolved: B_1 and A_2 have never met and no chain of results orders them. Six races cannot pin down the podium, so 7 is minimal.

The trick: heats + a winners' race eliminates whole *tail groups* at once (transitivity does the work), and the candidate count for places 2–3 collapses to exactly one final race.

Answer: 7 races. □

PZ.6 *Sixteen gamers, four per game* ♦

Lower bound. A game eliminates at most 3 players from contention (the losers); to crown 1 champion you must eliminate 15, so at least $\lceil 15/3 \rceil = 5$ games are needed.

Construction achieving it. Four first-round games of 4 produce 4 winners; one final among the winners decides it. Because the best player always wins, the champion is guaranteed to win their heat and then the final — 5 games suffice.

Self-check: small case: 4 players need $(4-1)/3 = 1$ game; the general formula, games $= (n-1)/(g-1)$ for n entrants and g per game, gives $15/3 = 5$. Contrast with the 25-horses classic: there you need the *top 3* and only relative order per race, which is why horses cost 7 races; here the deterministic-winner assumption and the "best only" demand make the naive knockout optimal.

Answer: 5 games.

PZ.7 *Prisoners' hats in a line*

$n-1$ **guaranteed; the first speaker is a coin flip.**

Strategy: agree beforehand that the first caller (backmost) announces the *parity* of the black hats they see — say "black" for odd, "white" for even. That caller is right about their own hat only by luck. Every subsequent prisoner maintains a running parity: they know the parity announced, they can see all black hats in front of them, and they have heard every hat color already (correctly) revealed behind them — so the parity of black hats *excluding their own head* is computable, and comparing it with the announced parity reveals their own color exactly. By induction each of the remaining $n-1$ calls is correct.

The trick: the whole line needs only *one shared bit* of information — a parity check — agreed in advance; every later prisoner then solves one equation in one unknown. It is a one-bit error-correcting scheme dressed as a puzzle.

Generalisation worth saying aloud: with k hat colors, announce the sum of the visible colors mod k (colors encoded $0, \ldots, k-1$); again everyone after the first is saved — the structure is a checksum, not anything specific to two colors.

Answer: $n-1$ guaranteed (the backmost announces the parity of black hats seen; everyone else deduces their own); the first speaker survives with probability $\frac{1}{2}$. □

PZ.8 *Hats: guess or pass*

Strategy. If you see two hats of the *same* colour, name the *opposite* colour; if you see one of each, pass.

Winning probability. The 8 hat configurations are equally likely.

- All-same (RRR, BBB — 2 configurations): every player sees a matching pair and guesses the opposite of it; all three guess, all three are wrong. Team loses.
- Mixed (6 configurations): exactly one player wears the minority colour. That player sees two same-coloured hats and names the opposite — which is their own colour: correct. The other two players each see one hat of each colour and pass. Team wins.

So $P(\text{win}) = 6/8 = 3/4$.

Optimality. Whatever the strategy, any particular guess is right with probability exactly 1/2: your own hat is independent of what you see, and your action depends only on what you see. So, summed over the 8 configurations, the number of correct guess-instances equals the number of wrong guess-instances, say $R = W$. Each winning configuration uses at least one correct instance, so $R \geq w$ where w is the number of winning configurations; every wrong instance must sit in a losing configuration (a winning configuration has zero wrong guesses by definition), and a configuration can host at most 3 wrong guesses, so $W \leq 3(8 - w)$. Combining: $w \leq R = W \leq 3(8 - w)$, hence $4w \leq 24$, i.e. $w \leq 6$ and $P \leq 3/4$. You cannot make guesses *righter* — you can only pile the inevitable wrongness into few configurations (here, the two all-same configurations; for 7 players the same idea via the Hamming code gives 7/8).

Self-check: count the instances in our strategy: the 2 losing configurations absorb $3 \times 2 = 6$ wrong guesses; the 6 winning ones use $1 \times 6 = 6$ correct guesses — $R = W = 6$ exactly, so the counting bound is tight, confirming both the 3/4 computation and that no slack remains.

Answer: See-a-pair-guess-the-opposite wins with probability 3/4, and 3/4 is optimal for three players.

PZ.9 *Broken stick triangle*

Let the stick have length 1 and the break points be $X, Y \sim \mathrm{U}(0,1)$ independent. The three pieces form a triangle iff every piece is shorter than $\frac{1}{2}$ (a piece of length $\geq \frac{1}{2}$ is at least as long as the other two combined, violating the triangle inequality; conversely all pieces $< \frac{1}{2}$ satisfies all three inequalities).

The trick: draw the unit square of (X, Y). Take the case $X < Y$ (probability $\frac{1}{2}$, the triangle above the diagonal); the pieces are X, $Y - X$, $1 - Y$. The conditions

$$X < \tfrac{1}{2}, \qquad Y - X < \tfrac{1}{2}, \qquad Y > \tfrac{1}{2}$$

carve out, inside that half-square triangle, a smaller triangle with vertices $(0, \frac{1}{2}), (\frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, 1)$ of area $\frac{1}{8}$. By symmetry the $X > Y$ case contributes another $\frac{1}{8}$, so

$$P(\text{triangle}) = \tfrac{1}{8} + \tfrac{1}{8} = \tfrac{1}{4}.$$

Sanity check: the answer must be well below $\frac{1}{2}$ — a uniform second break point frequently leaves one long piece; and the complementary events "piece i is too long" each have probability $\frac{1}{4}$ and are disjoint, giving $1 - 3 \cdot \frac{1}{4} = \frac{1}{4}$ directly — a second derivation that agrees.

Answer: $\frac{1}{4}$. □

PZ.10 *Four points on a quarter arc*

Witness decomposition, exactly as in the semicircle classic. For each point i let A_i be the event that all other $n - 1$ points lie in the arc of length f (as a fraction of the circumference) running *clockwise from point* i. If the points fit in some arc of length f, the "first" point of the cluster is such a witness, so $\{\text{all in some } f\text{-arc}\} = \bigcup_i A_i$.

Disjointness (this is where $f \leq 1/2$ is used). If A_i and A_j both held for $i \neq j$, then j lies within clockwise-distance f of i and i within clockwise-distance f of j; those two distances sum to the whole circle, giving $1 \leq 2f$ — impossible for $f < 1/2$, and a probability-zero tie at $f = 1/2$. So the A_i are (almost surely) disjoint.

Probability. Each other point independently lands in the clockwise f-arc with probability f, so $P(A_i) = f^{\,n-1}$ and

$$P(\text{all in some } f\text{-arc}) = \sum_{i=1}^{n} P(A_i) = n f^{\,n-1}.$$

For $n = 4$, $f = 1/4$: $4 \times (1/4)^3 = 4/64 = 1/16$.

Self-check: two independent limits. (i) $f = 1/2$ recovers the semicircle classic $n/2^{n-1}$ — for $n = 4$ that is 1/2, the known answer. (ii) $n = 2$, $f = 1/4$: the formula gives $2 \times 1/4 = 1/2$; directly, the clockwise gap from one point to the other is uniform on $[0, 1)$ and the two points share a quarter-arc iff that gap is $\leq 1/4$ or $\geq 3/4$ — probability $1/2$.

Answer: $1/16$ for four points in a quarter-circle arc; in general $nf^{\,n-1}$ for $f \leq 1/2$. □

PZ.11 *Random walk on a solid*

Collapse states by distance to the target, exactly as for the cube. Let a be the expected steps from the start (the target's antipode) and b from any of the 4 "middle" vertices (each adjacent to both start and target).

From the start, all 4 neighbours are middle vertices:

$$a = 1 + b.$$

From a middle vertex, the 4 neighbours are: the target (absorb), the start, and 2 other middle vertices:

$$b = 1 + \tfrac{1}{4} \cdot 0 + \tfrac{1}{4}a + \tfrac{2}{4}b \quad \Longrightarrow \quad \tfrac{1}{2}b = 1 + \tfrac{1}{4}a \quad \Longrightarrow \quad b = 2 + \tfrac{a}{2}.$$

Substituting: $a = 3 + a/2$, so $a = 6$ and $b = 5$.

Self-check (independent route, via return times): the walk on a regular graph has uniform stationary distribution, so the long-run fraction of time at the target is $1/6$ and the expected return time to it is 6. A return trip is one step to a uniformly random neighbour of the target — always a middle vertex — followed by the hitting time back: $6 = 1 + b$, giving $b = 5$, and then $a = 1 + b = 6$. Sanity against the cube: the cube (8 vertices, degree 3, diameter 3) needs 10 steps; the octahedron is smaller, better connected, and shorter in diameter, so $6 < 10$ is the right direction.

Answer: 6 steps in expectation (5 from a middle vertex). □

PZ.12 *Bold vs timid play*

(a) Bold play. One bet: $P = 18/38 \approx 0.474$.

(b) Timid play. Gambler's ruin with $i = 100$, $N = 200$, $r = q/p = (20/38)/(18/38) = 10/9$:

$$P = \frac{1 - r^{100}}{1 - r^{200}} = \frac{1 - r^{100}}{(1 - r^{100})(1 + r^{100})} = \frac{1}{1 + r^{100}},$$

the clean form because the target is exactly double the stake. Now $r^{100} = (10/9)^{100} = e^{100 \ln(10/9)} \approx e^{10.54} \approx 3.8 \times 10^4$, so

$$P \approx \frac{1}{38{,}000} \approx 2.7 \times 10^{-5}.$$

Bold play is roughly 18,000 times likelier to succeed.

Lesson. With the edge against you, every additional bet gives the edge another bite; bold play minimises total money staked, timid play grinds you into the house edge with near certainty. Flip the sign and the prescription flips: with a *positive* edge you bet small and often, and your success probability tends to 1.

Trading analogue. Crossing the spread as a taker is a negative-edge bet — trade as rarely as your objective allows and in size when you do. A market maker earning a small positive edge per fill wants the opposite: maximum trade count, small size, and the ruin formula is exactly why a small *negative* edge on a high-frequency book means near-certain ruin — the sign of the per-trade edge, not its size, decides your fate at scale.

Self-check: (i) fair-coin limit: as $p \to 1/2$, $r \to 1$ and $1/(1 + r^{100}) \to 1/2$, matching bold play's $1/2$ and the fair-ruin answer $i/N = 100/200$. (ii) Monotonicity in bet size: with \$50 bets ($i = 2$, $N = 4$), $P = 1/(1 + r^2) = 81/181 \approx 0.448$ — between the \$1 and \$100 answers and below the single bold bet, as it must be. (iii) magnitude: $(9/10)^{100} = e^{-100 \times 0.10536} = e^{-10.54} \approx 2.7 \times 10^{-5}$, consistent.

Answer: Bold: $18/38 \approx 0.474$. Timid: $1/(1+(10/9)^{100}) \approx 2.7 \times 10^{-5}$ — about 18,000 times worse. Negative edge: bet boldly and rarely; positive edge: grind small and often.

PZ.13 *Fair coin to probability 1/3* ★

Construction (rejection sampling). Flip the coin twice. HH $\to$ the event occurs; HT or TH $\to$ it does not; TT $\to$ discard the pair and flip twice again. Conditional on the pair not being TT, the three surviving outcomes are equally likely, so the event has probability exactly 1/3.

Explicitly: $P(\text{HH is the first non-TT pair}) = \sum_{k\geq 0}(1/4)^k \cdot (1/4) = \frac{1/4}{1-1/4} = \frac{1}{3}$.

Expected flips. The number of two-flip rounds is geometric with success probability 3/4, so $\mathbb{E}[\text{rounds}] = 4/3$ and, by Wald, $\mathbb{E}[\text{flips}] = 2 \times 4/3 = 8/3 \approx 2.67$.

Fixed number of flips: impossible — say this aloud. With n predetermined flips, every event is a union of atoms of probability 2^{-n}, so its probability is $k/2^n$ for an integer k. Then $k/2^n = 1/3$ would require $2^n = 3k$, but 2^n is never divisible by 3. Exactness *requires* an unbounded (rejection) scheme; volunteering the impossibility argument is what turns a correct answer into a strong one.

Same family, worth having loaded: (i) three flips with target set {HHT, HTH, THH} and the other five outcomes rejected also yields exactly 1/3 per surviving outcome, but at $3 \times 8/3 = 8$ expected flips — the two-flip scheme dominates (in the live interview the candidate reached the construction via this 3-flip, 3/8 hint path); (ii) von Neumann's biased-to-fair trick: flip twice, HT $\to$ H, TH $\to$ T, HH/TT $\to$ repeat — works for any unknown bias since $P(\text{HT}) = P(\text{TH}) = pq$; (iii) simulating a die with a coin: 3 flips give 8 outcomes, reject 2, giving 6 equally likely.

Self-check (symmetry, no series needed): each round produces HH, HT, TH with equal probability and the procedure treats rounds identically, so conditioned on eventual acceptance the three outcomes are exchangeable — each must get probability exactly 1/3; the geometric series above confirms it numerically. And 8/3 flips sits sensibly above the trivial lower bound of 2 (a single fixed round can never carve out exactly 1/3, by the divisibility argument, so some rejection overhead is unavoidable).

Answer: Flip twice: HH = the event, HT/TH = not, TT = reject and repeat — probability exactly 1/3; expected flips 8/3; a fixed number of flips can never do it since $3 \nmid 2^n$.